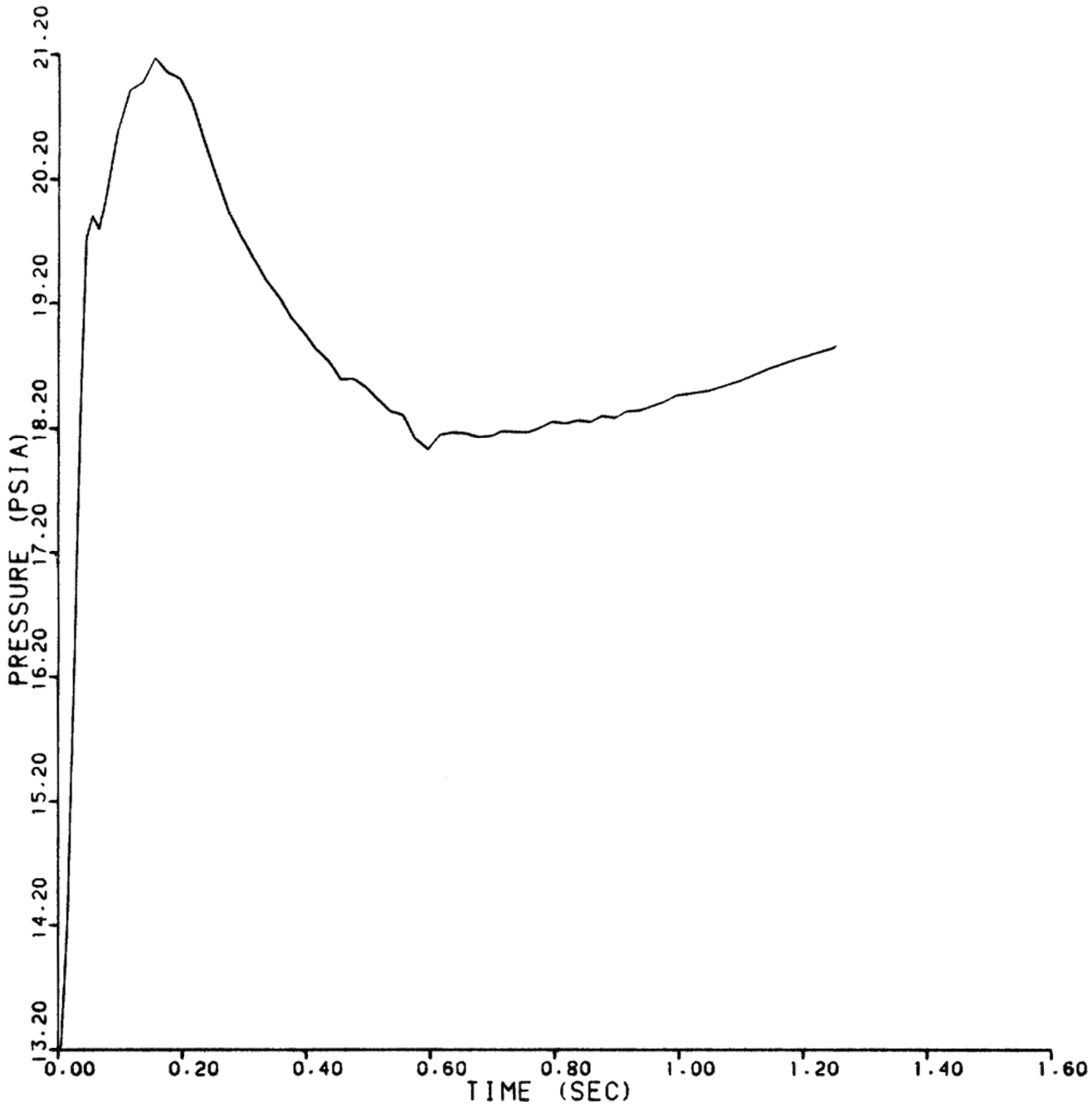

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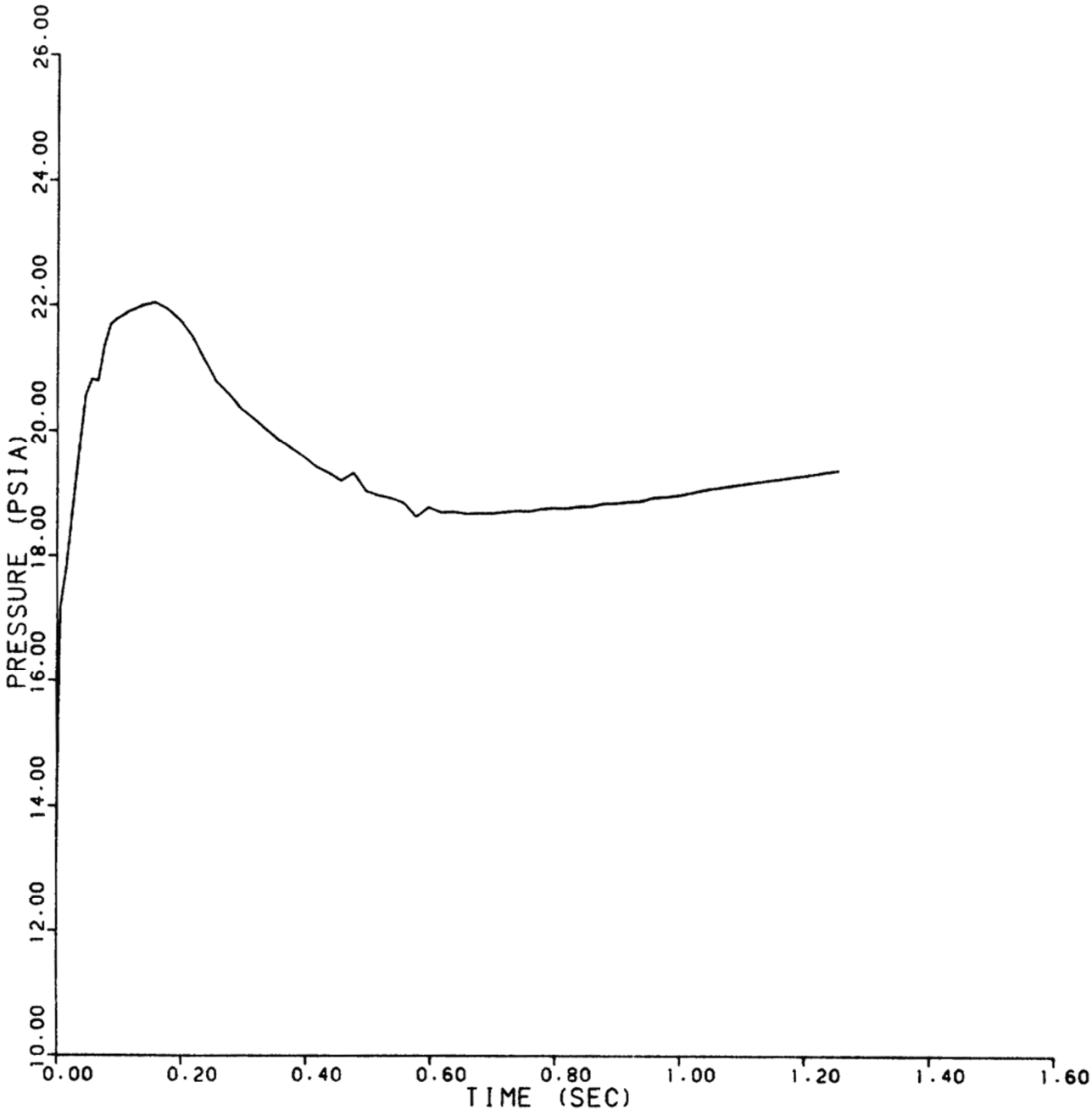
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E20
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 20 OF 74)



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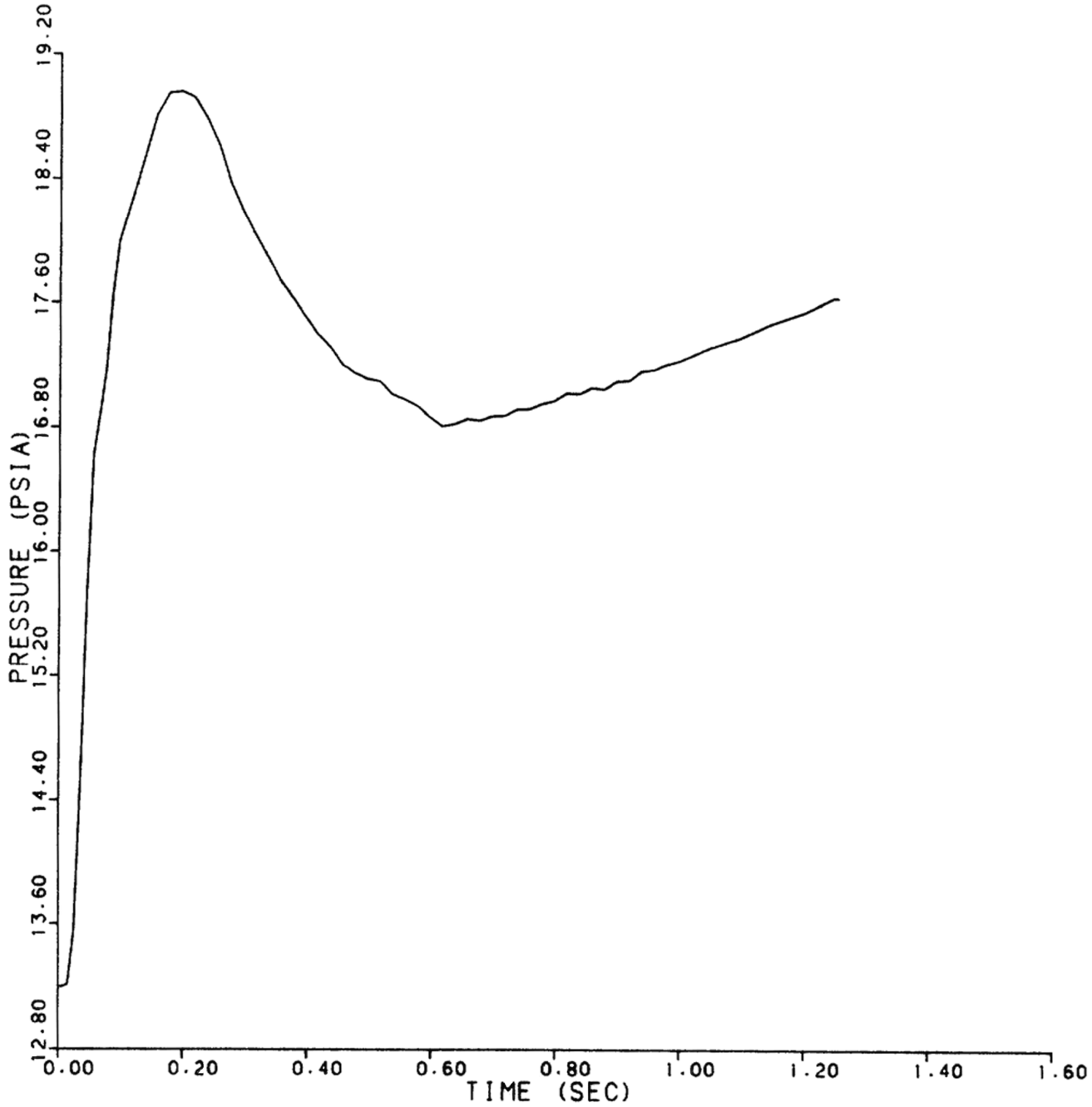
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E21
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 21 OF 74)



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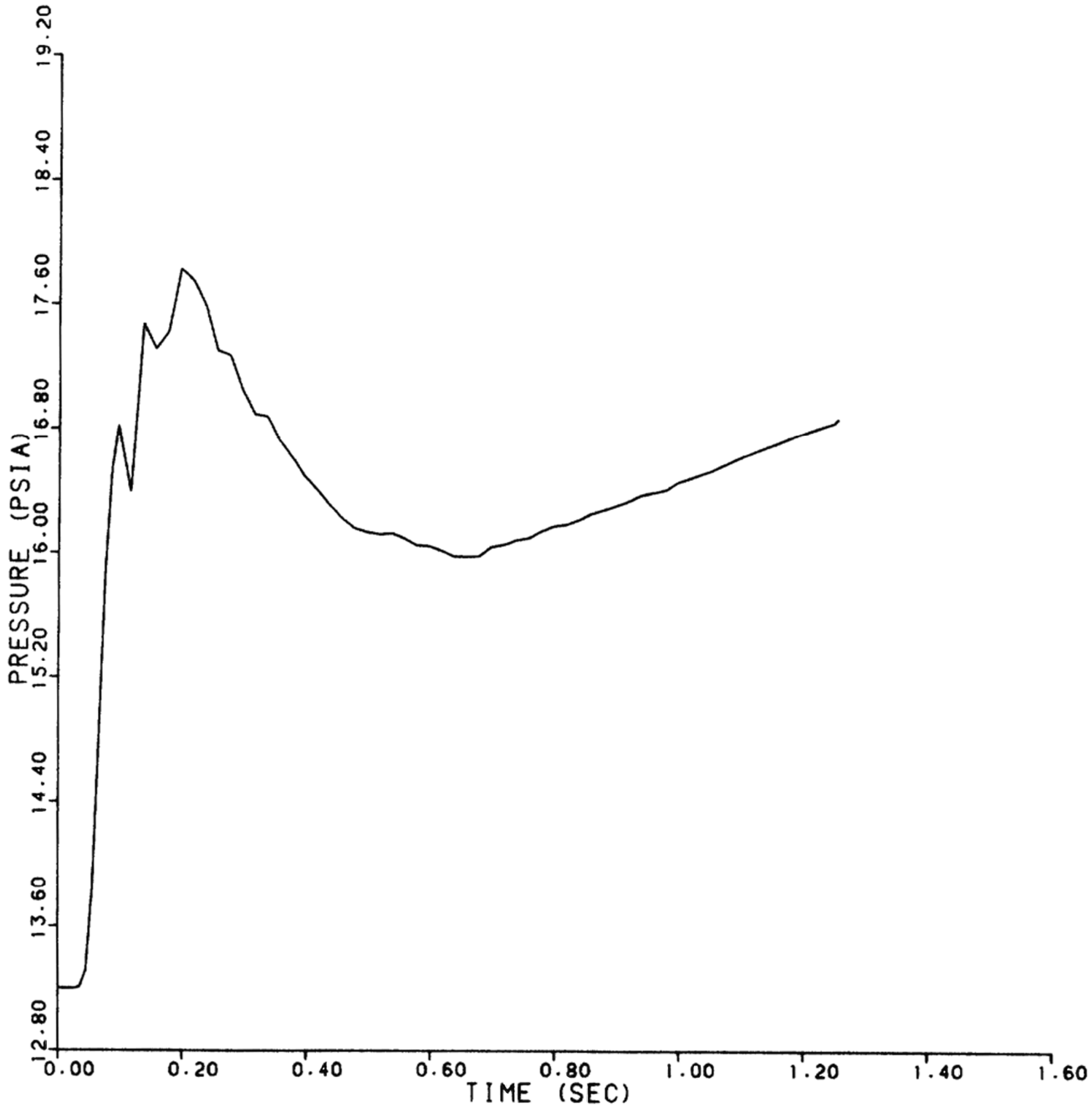
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E22
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 22 OF 74)



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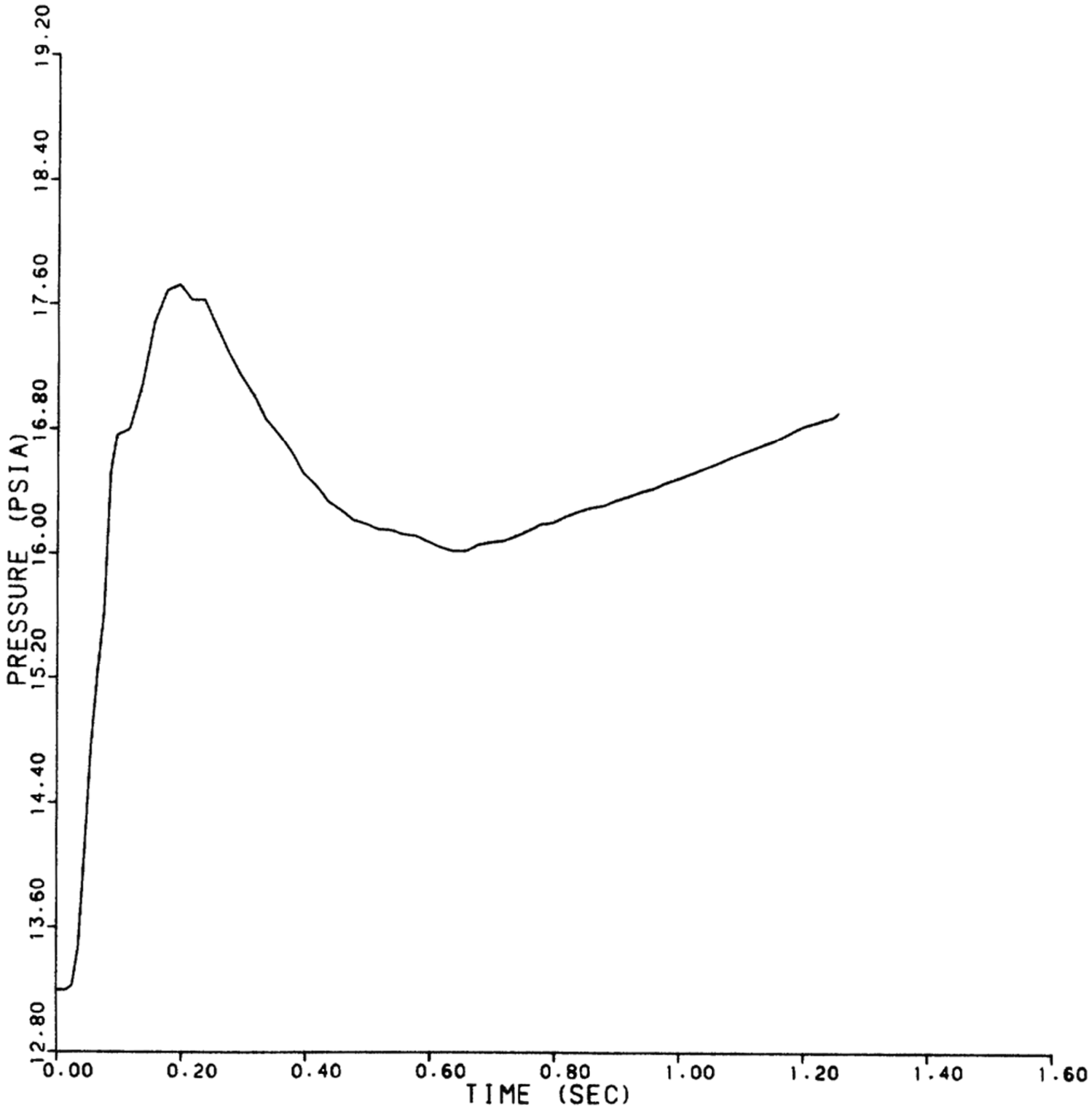
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E23
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 23 OF 74)



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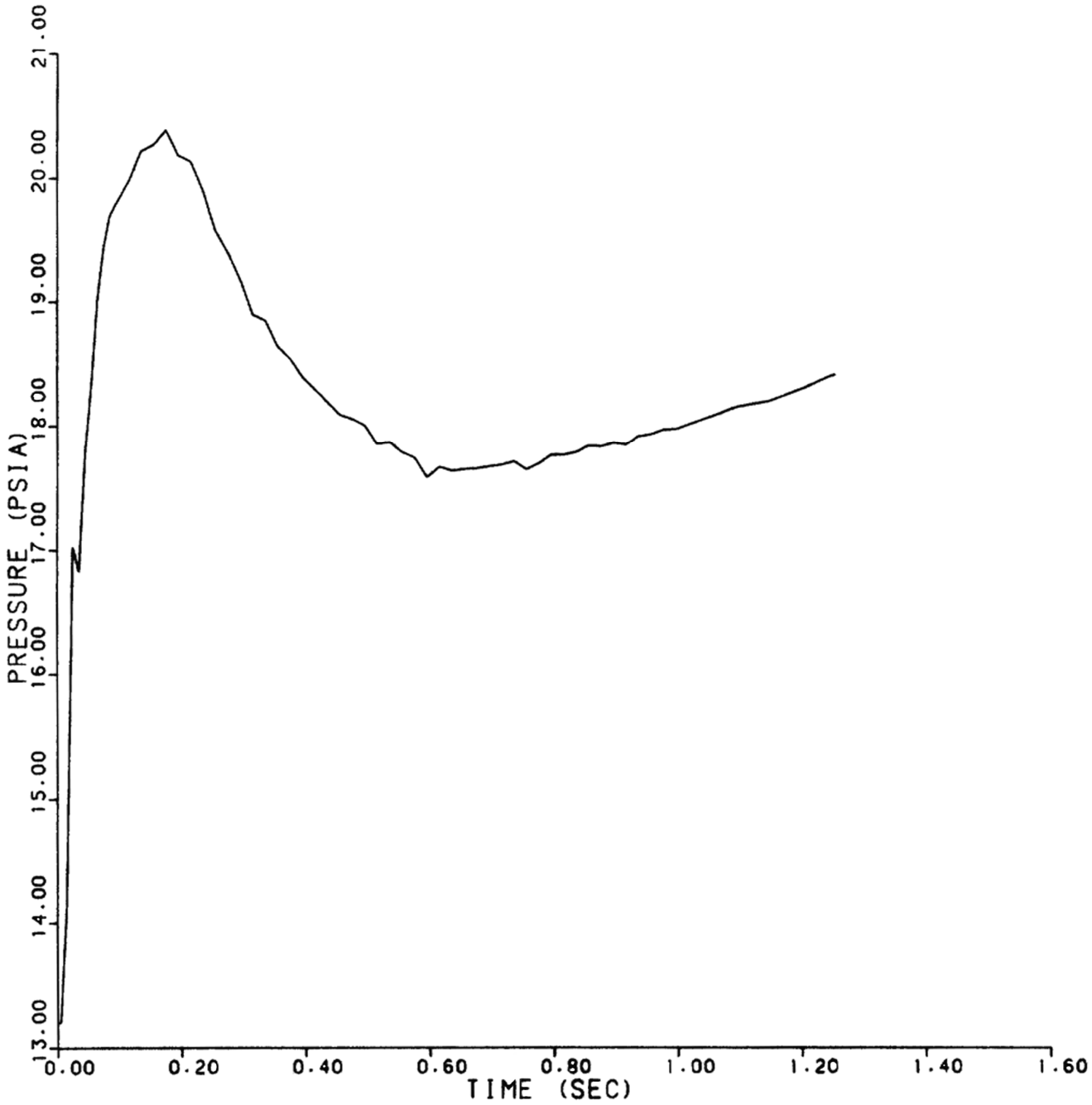
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E24
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 24 OF 74)



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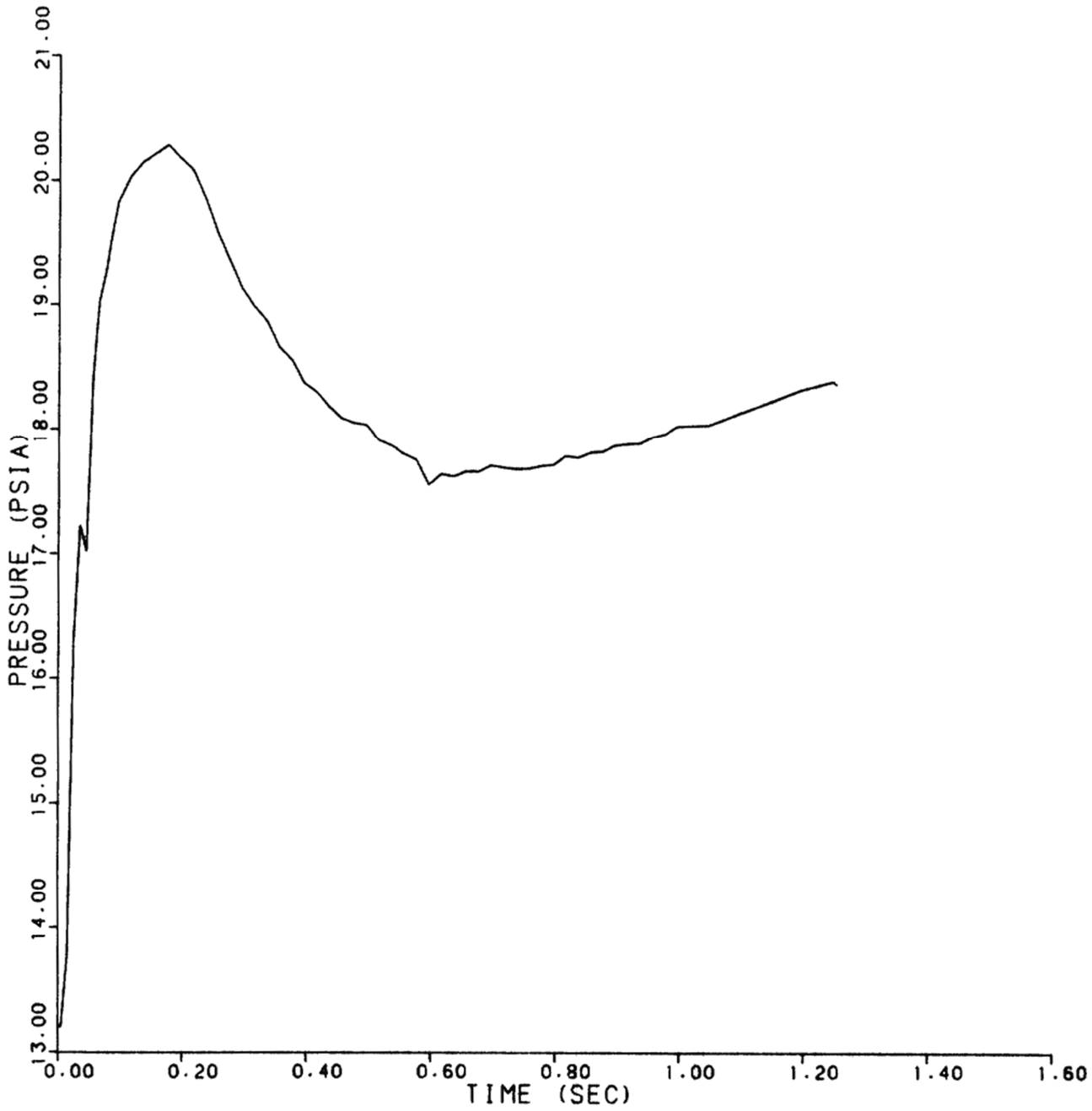
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E25
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 25 OF 74)



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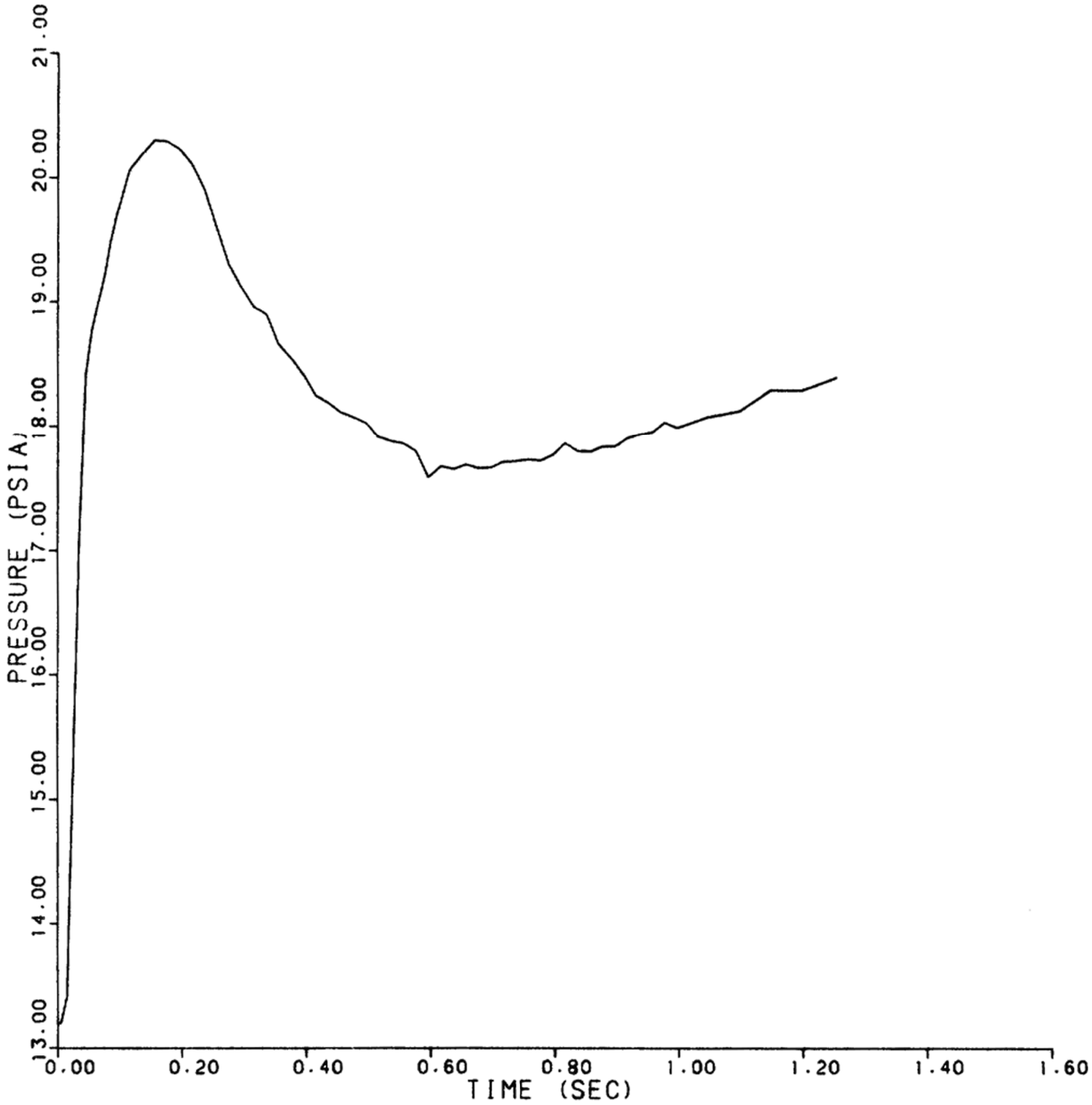
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E26
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 26 OF 74)



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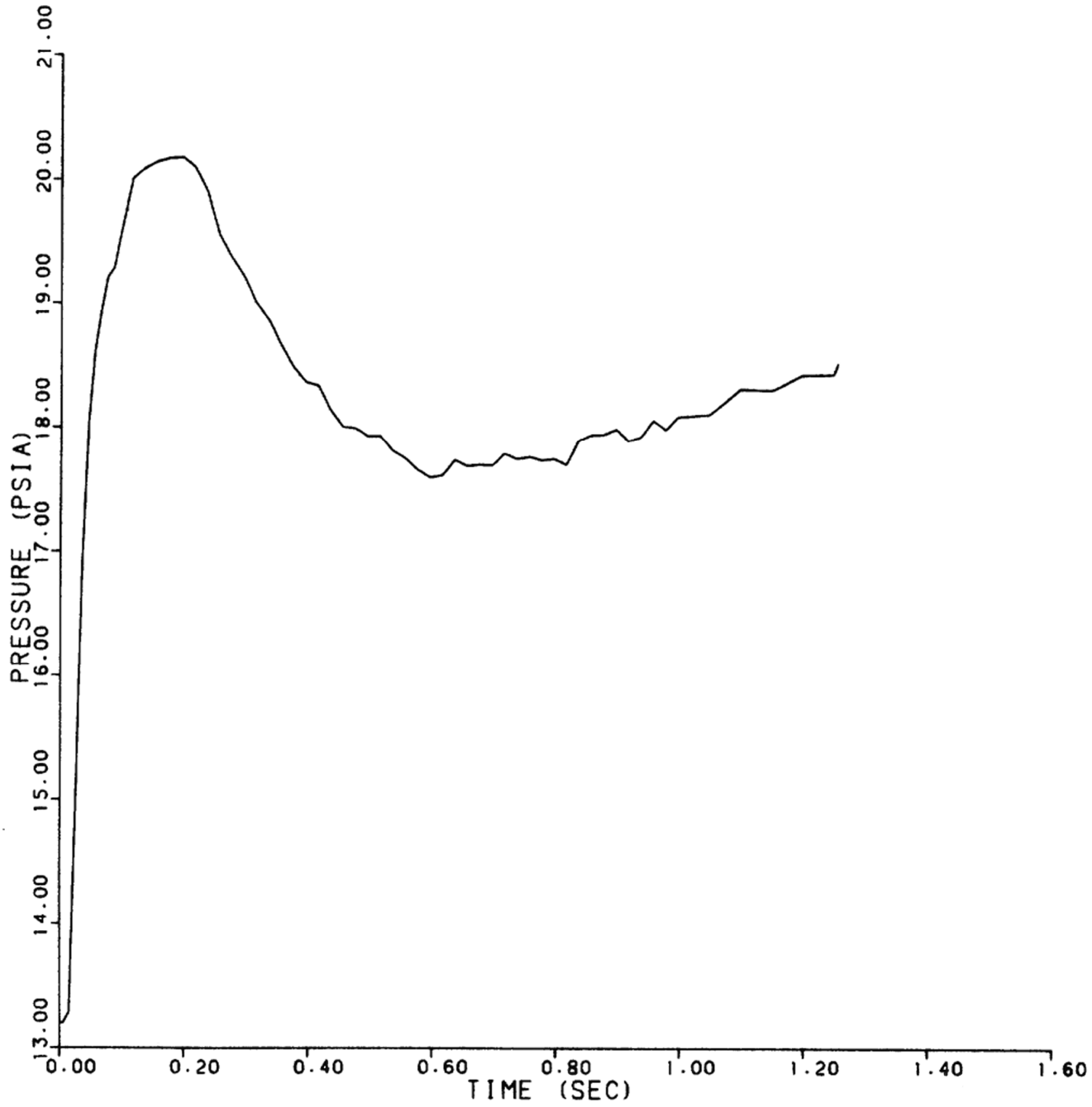
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E27
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 27 OF 74)



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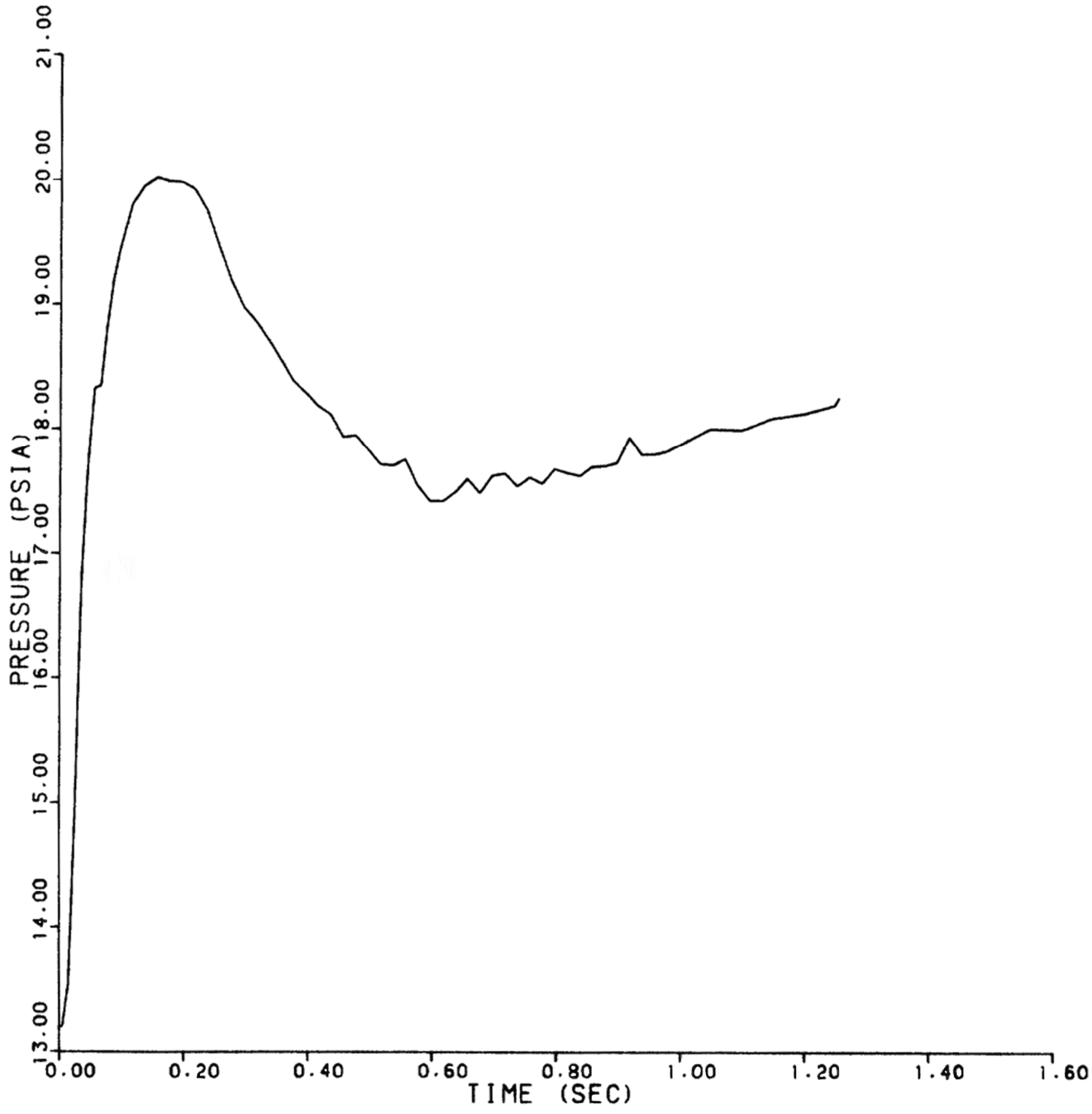
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E28
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 28 OF 74)



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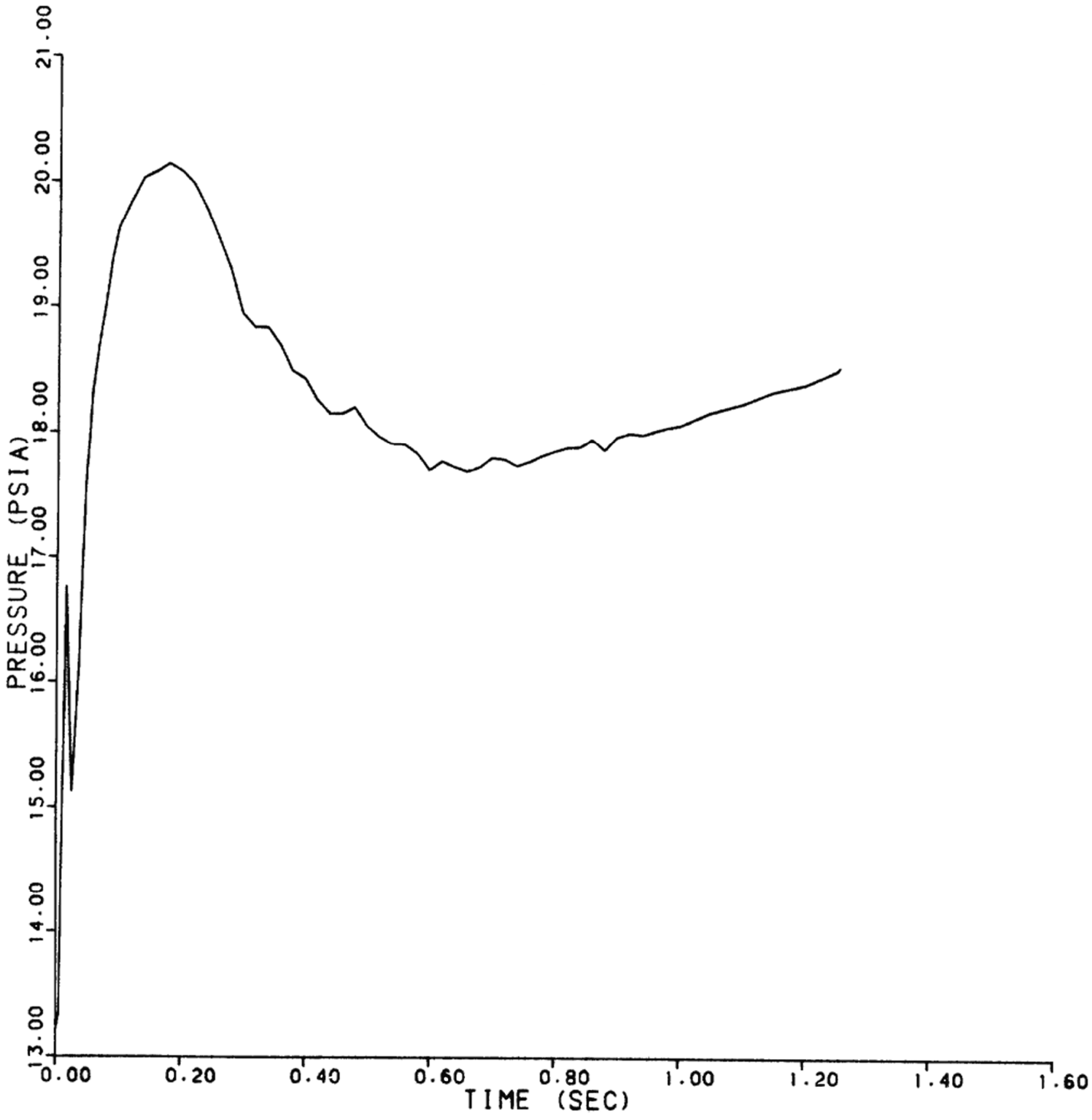
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E29
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 29 OF 74)



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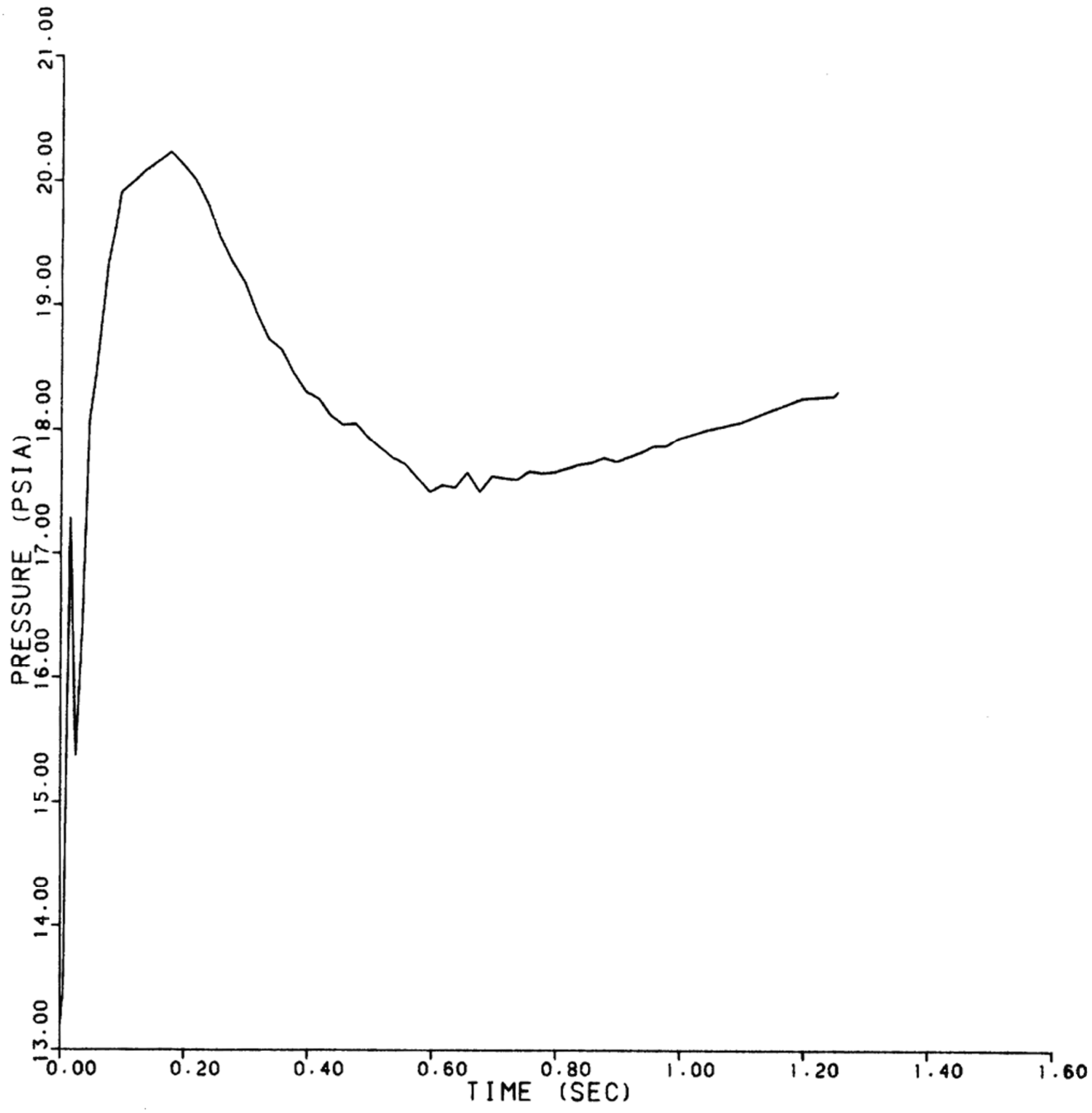
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E30 (306–in.^{2} BREAK AREA) FIGURE 6.2.1–19 (SHEET 30 OF 74)



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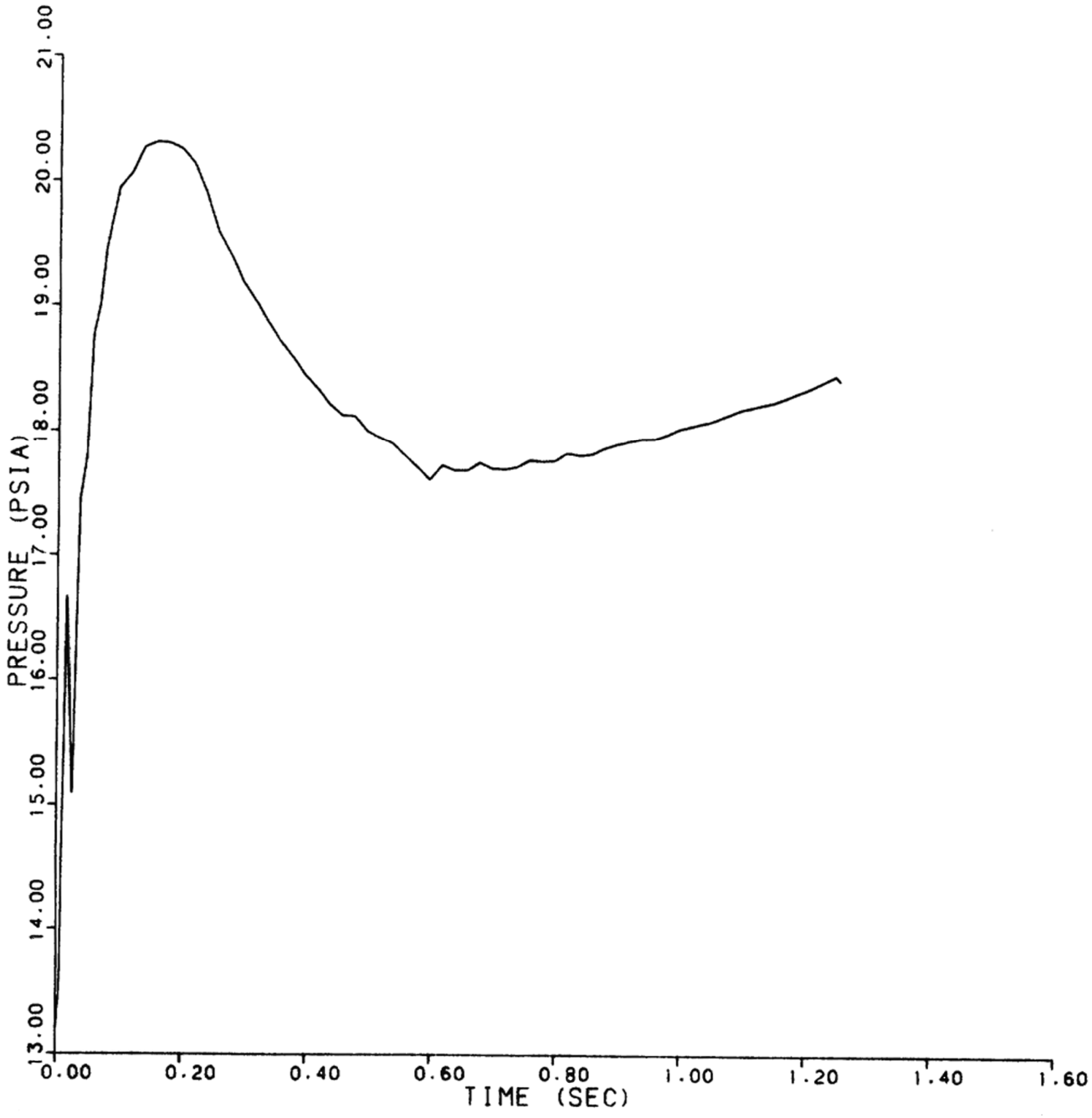
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E31 (306–in.^{2} BREAK AREA) FIGURE 6.2.1–19 (SHEET 31 OF 74)



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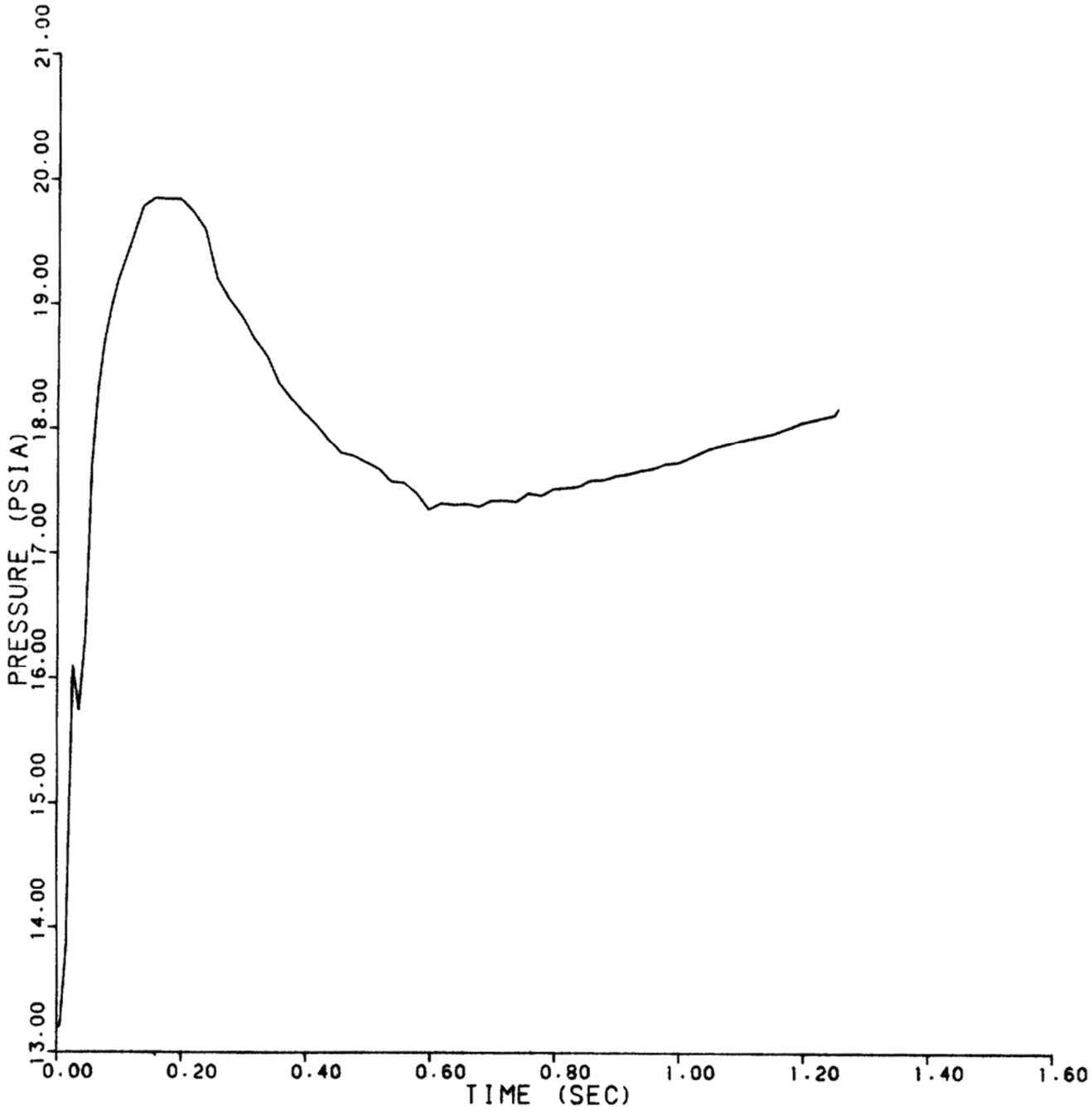
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E32
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 32 OF 74)



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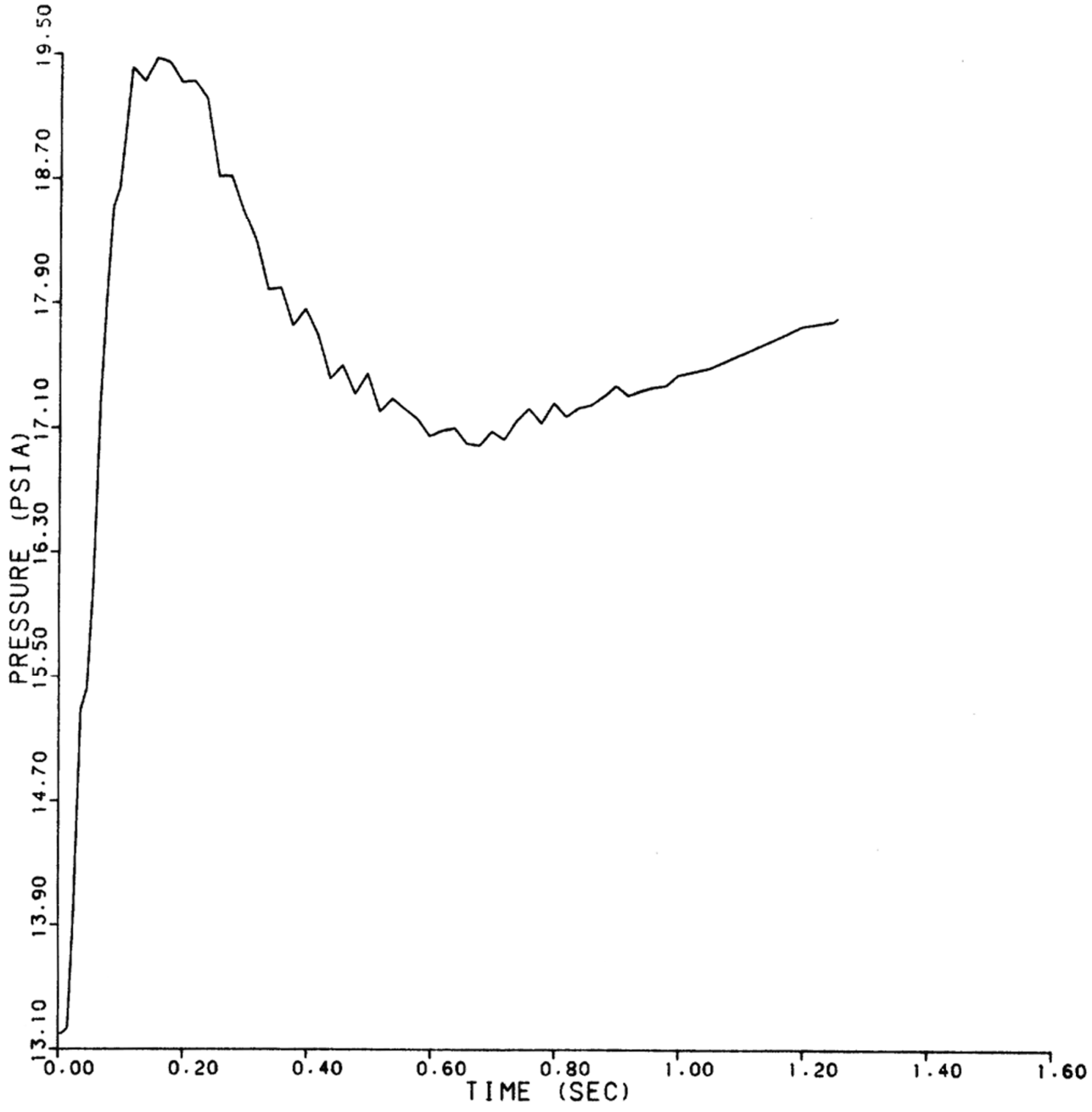
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E33 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 33 OF 74)



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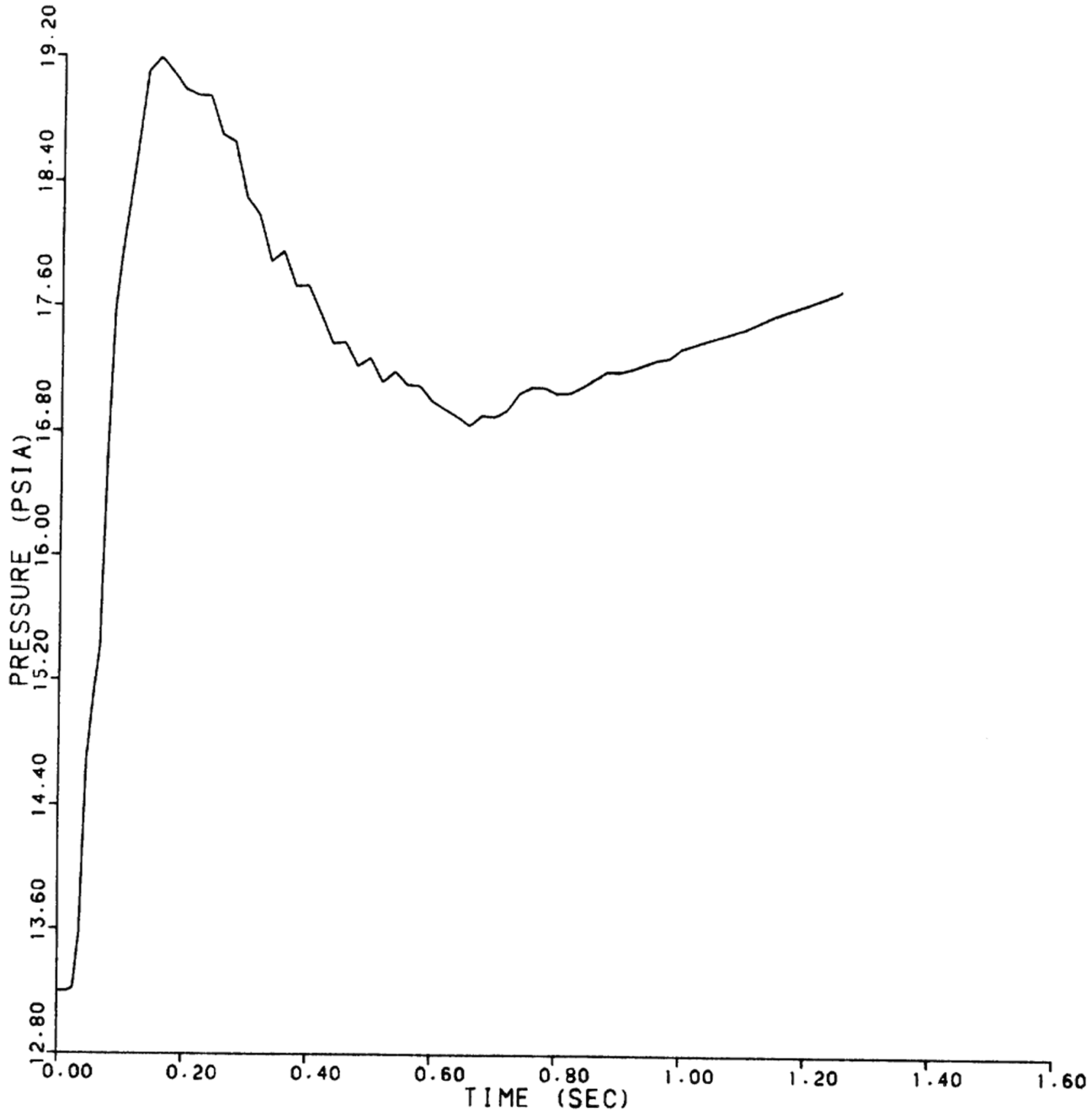
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E34
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 34 OF 74)



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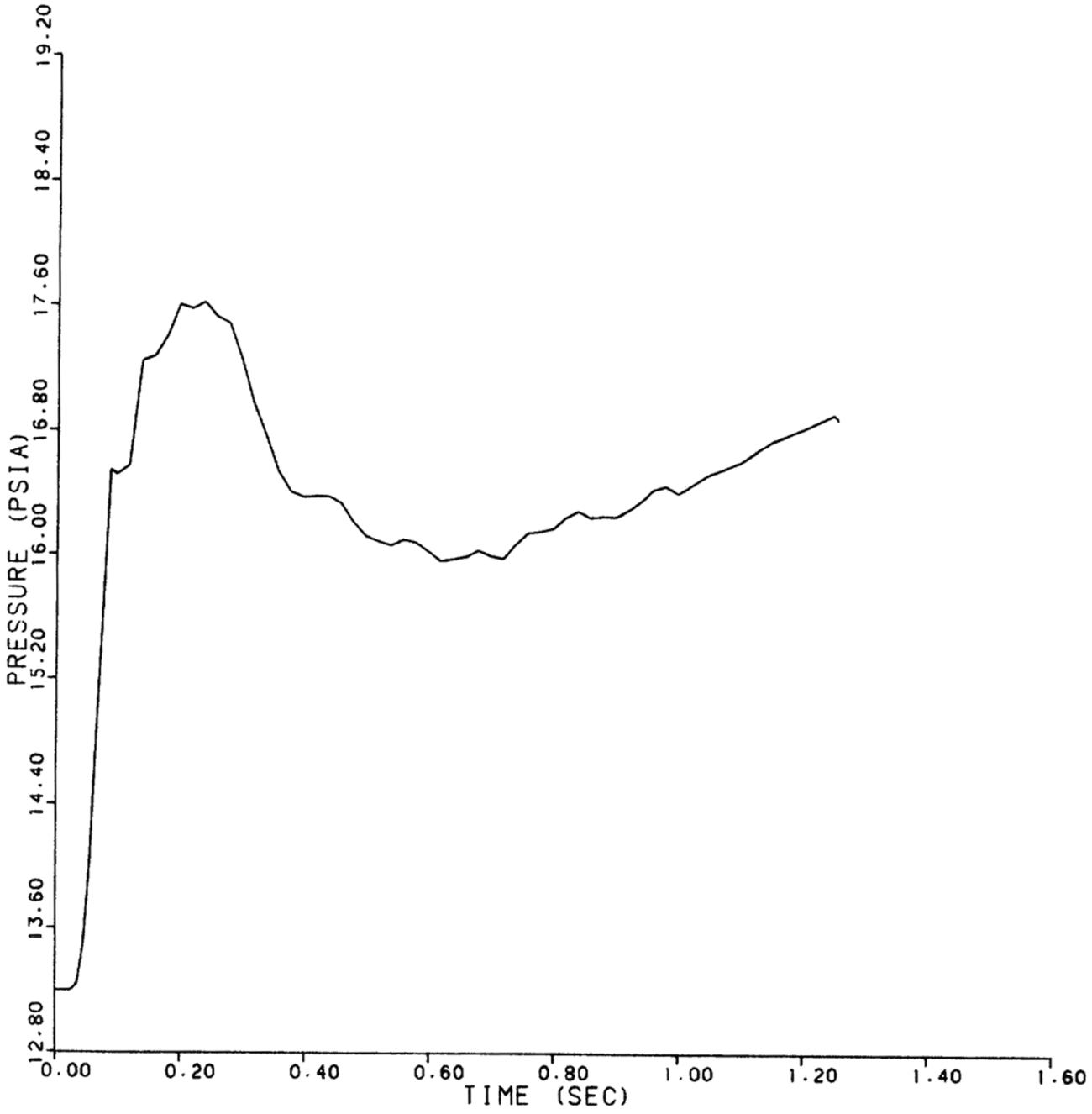
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E35
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 35 OF 74)



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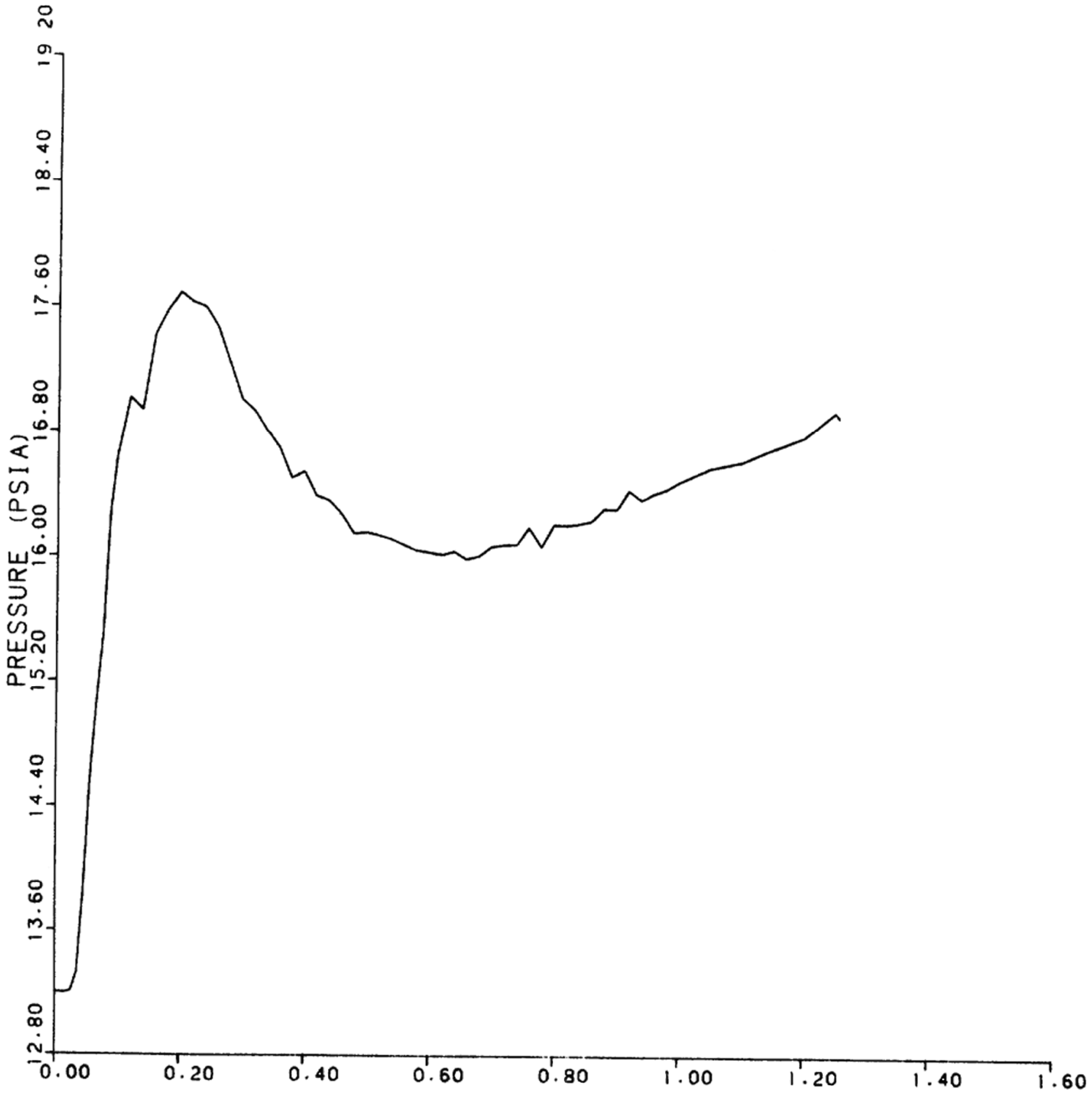
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E36 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 36 OF 74)



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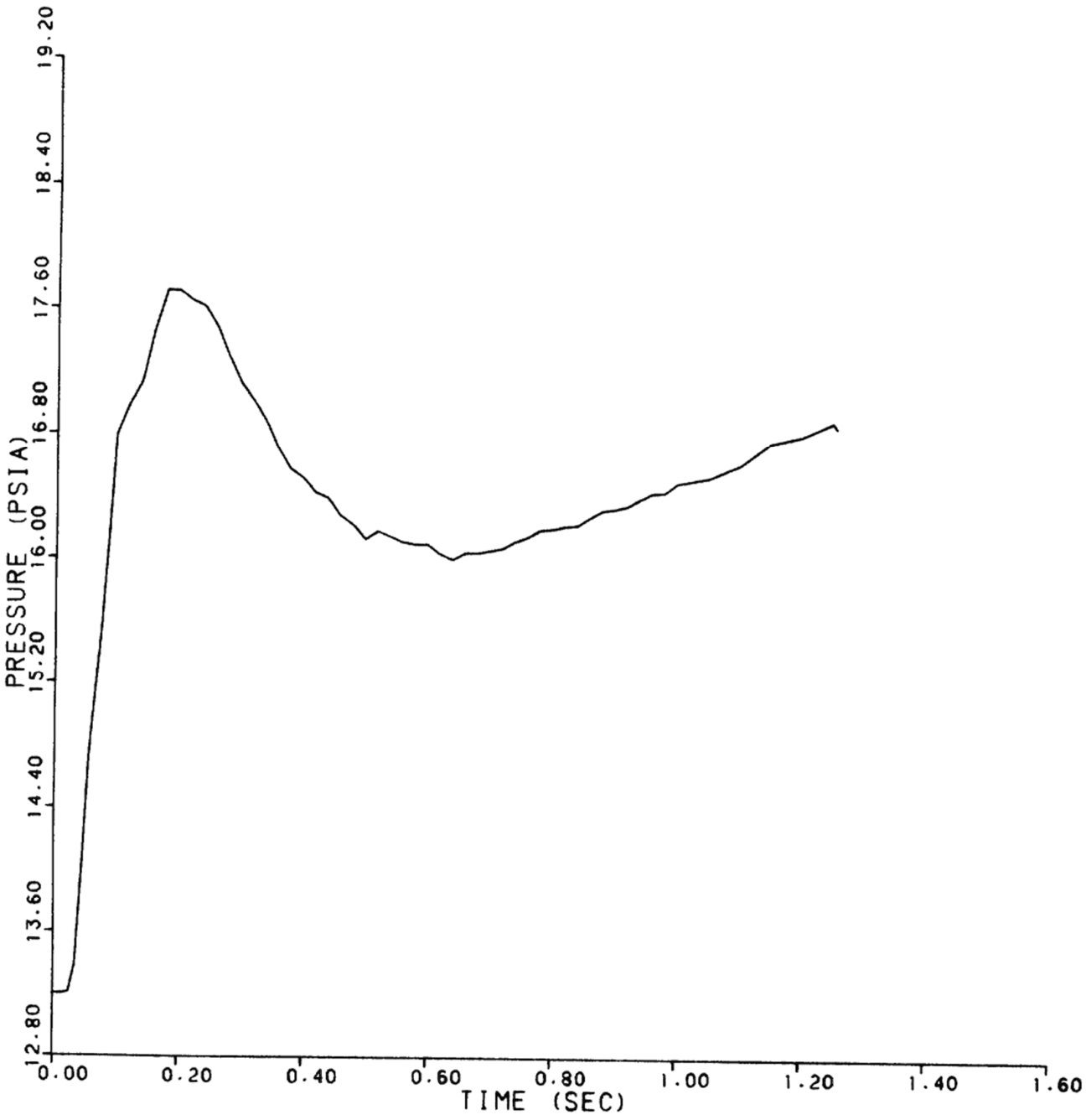
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E37
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 37 OF 74)



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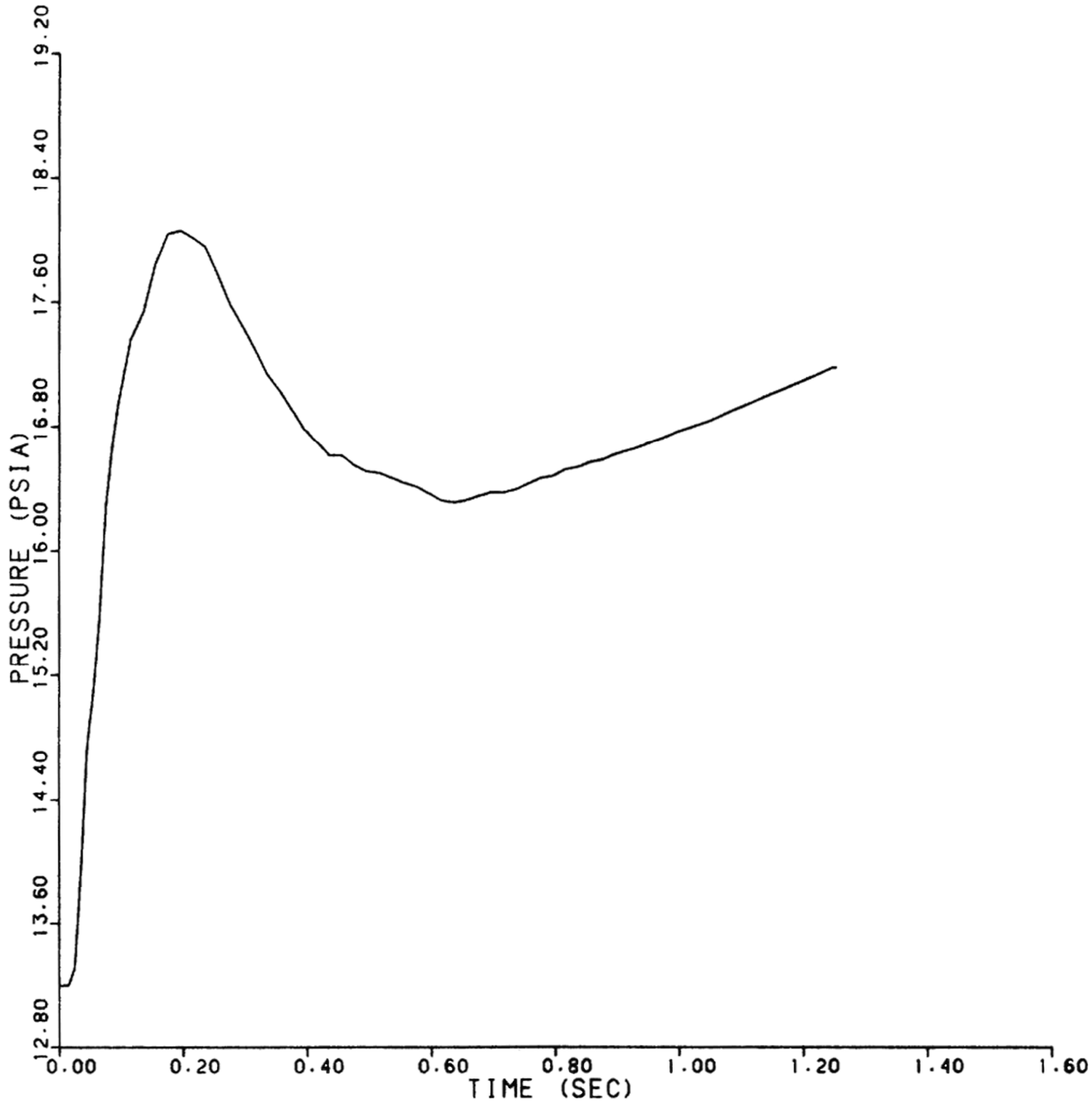
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E38 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 38 OF 74)



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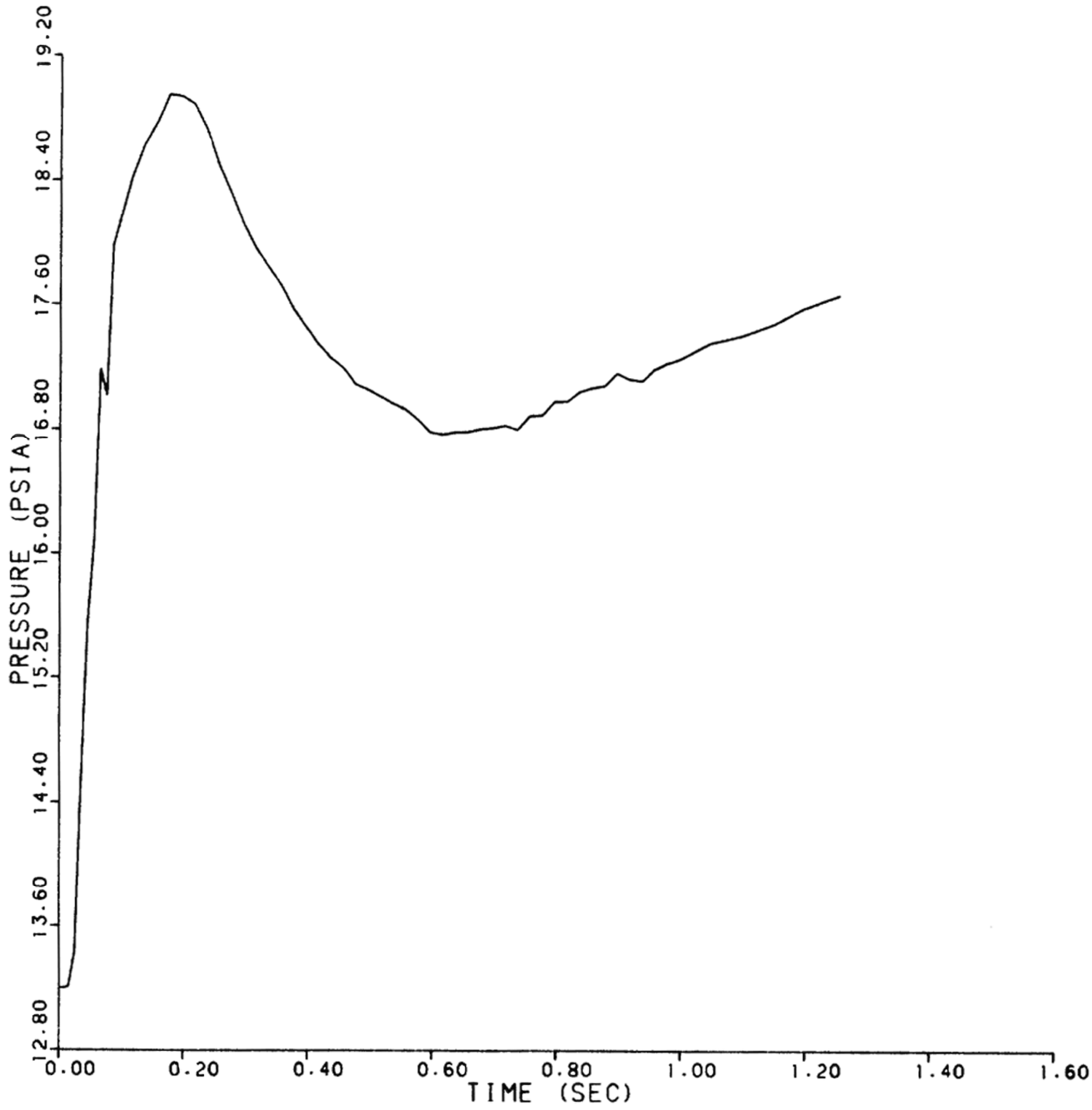
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E39
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 39 OF 74)



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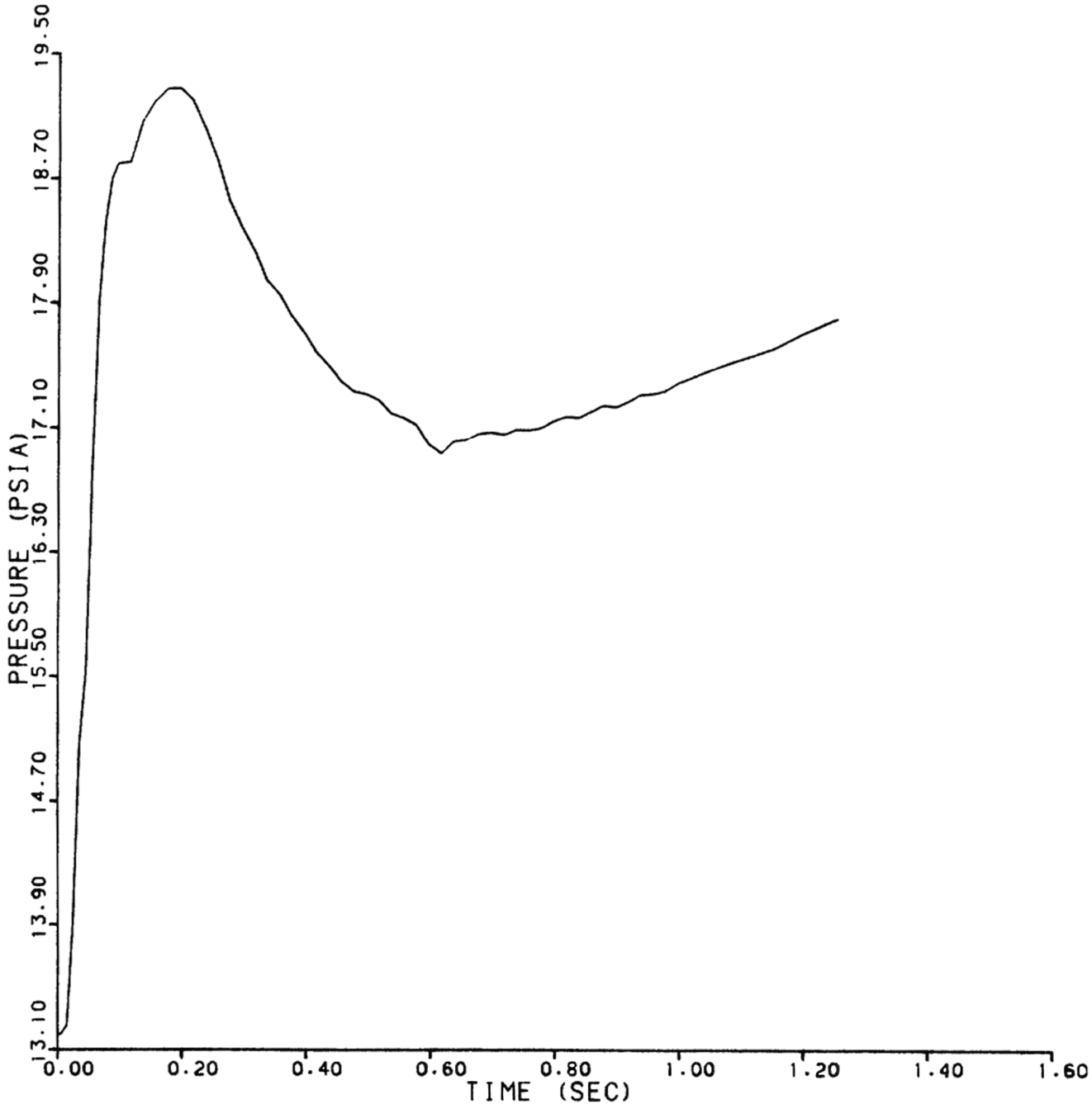
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E40
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 40 OF 74)



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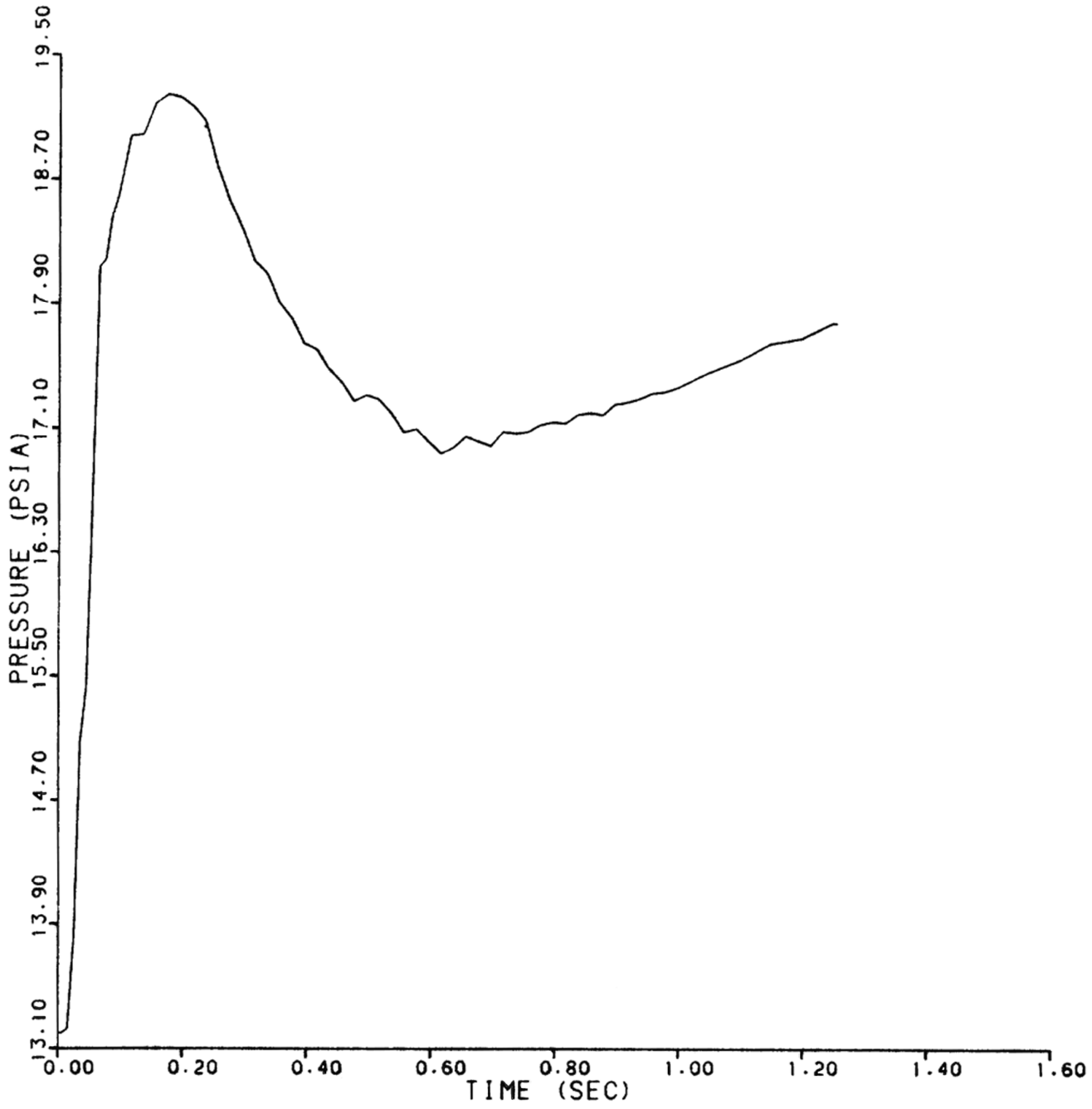
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E41
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 41 OF 74)



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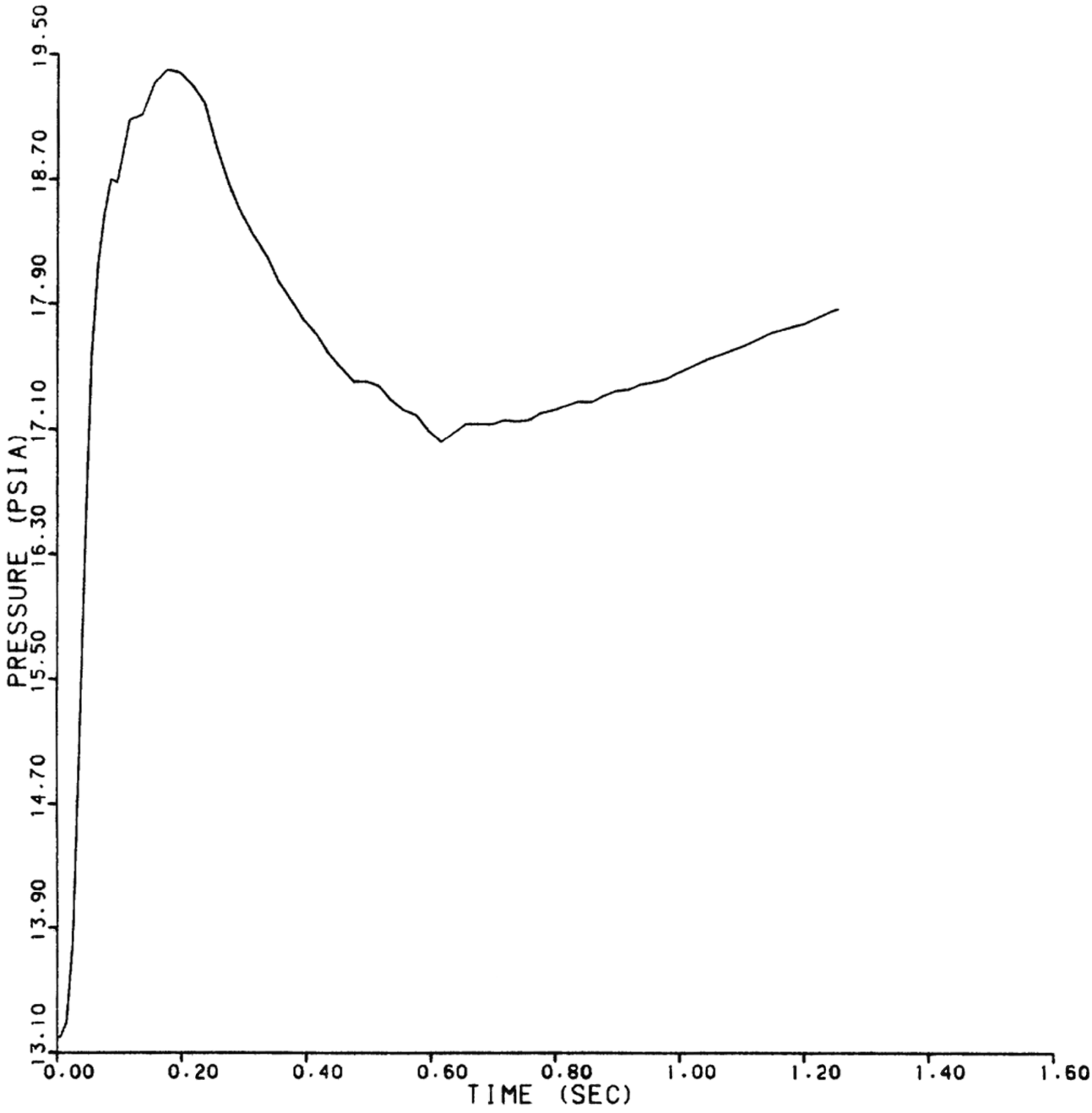
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E42
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 42 OF 74)



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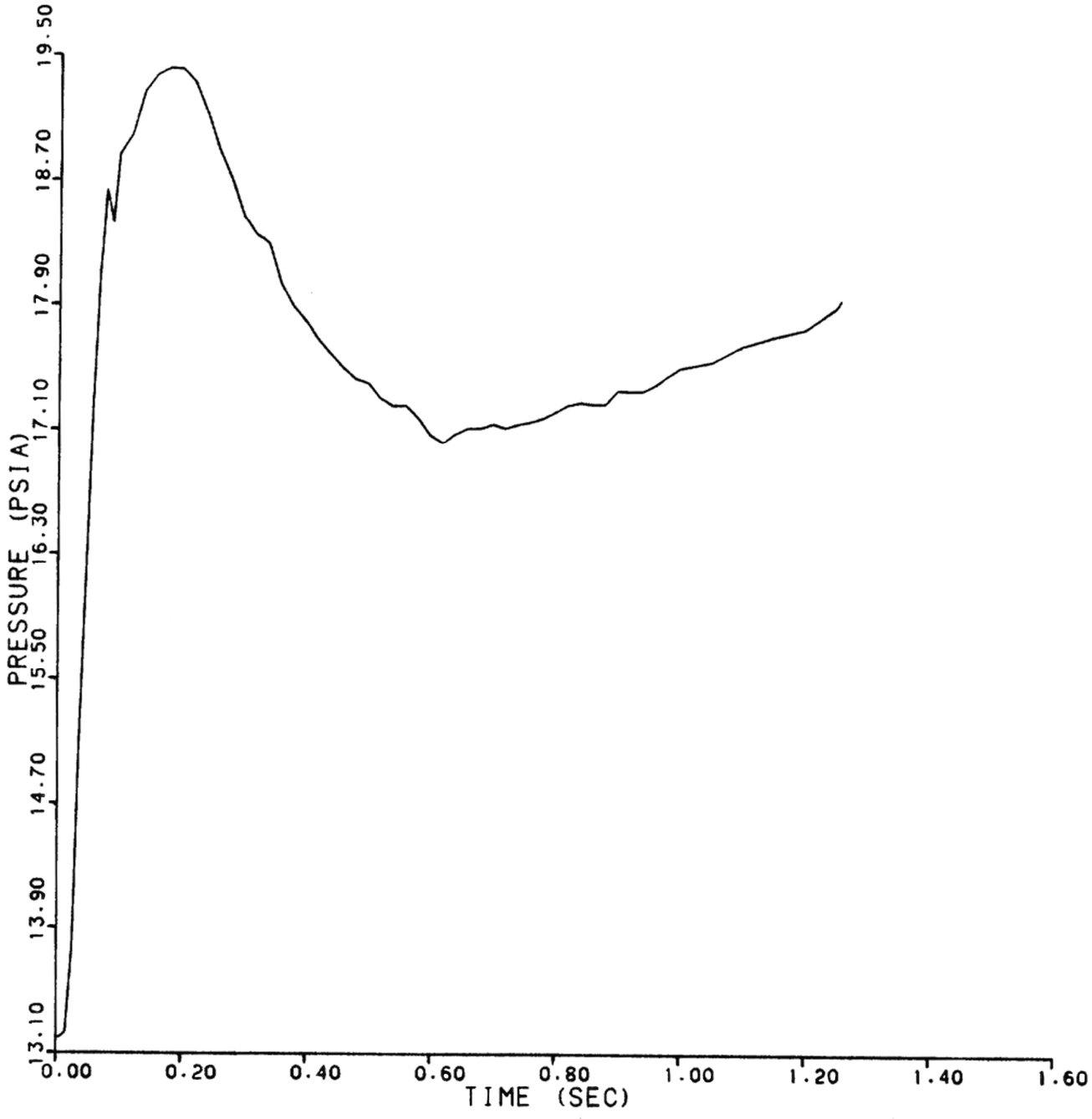
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E43
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 43 OF 74)



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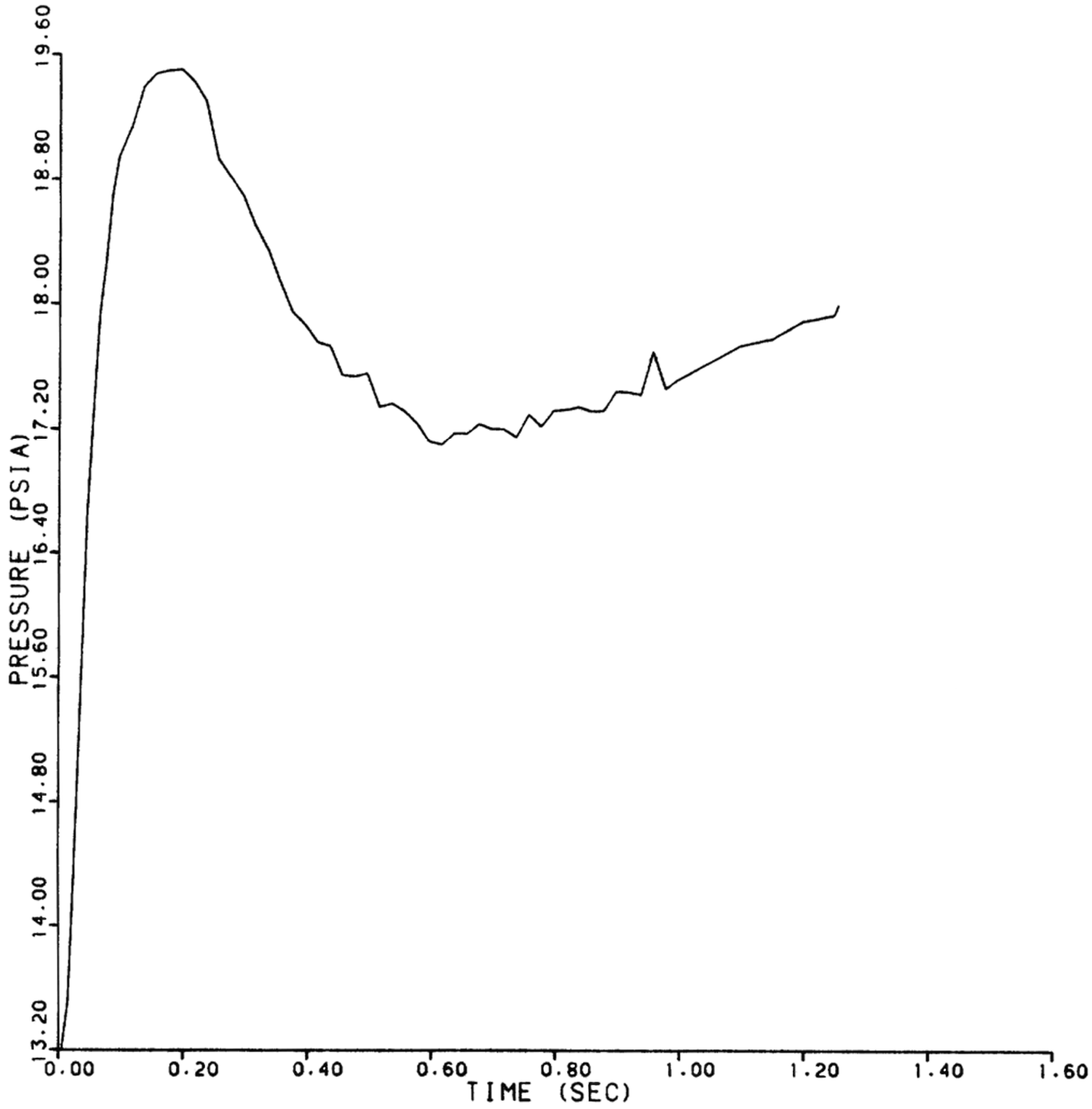
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E44
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 44 OF 74)



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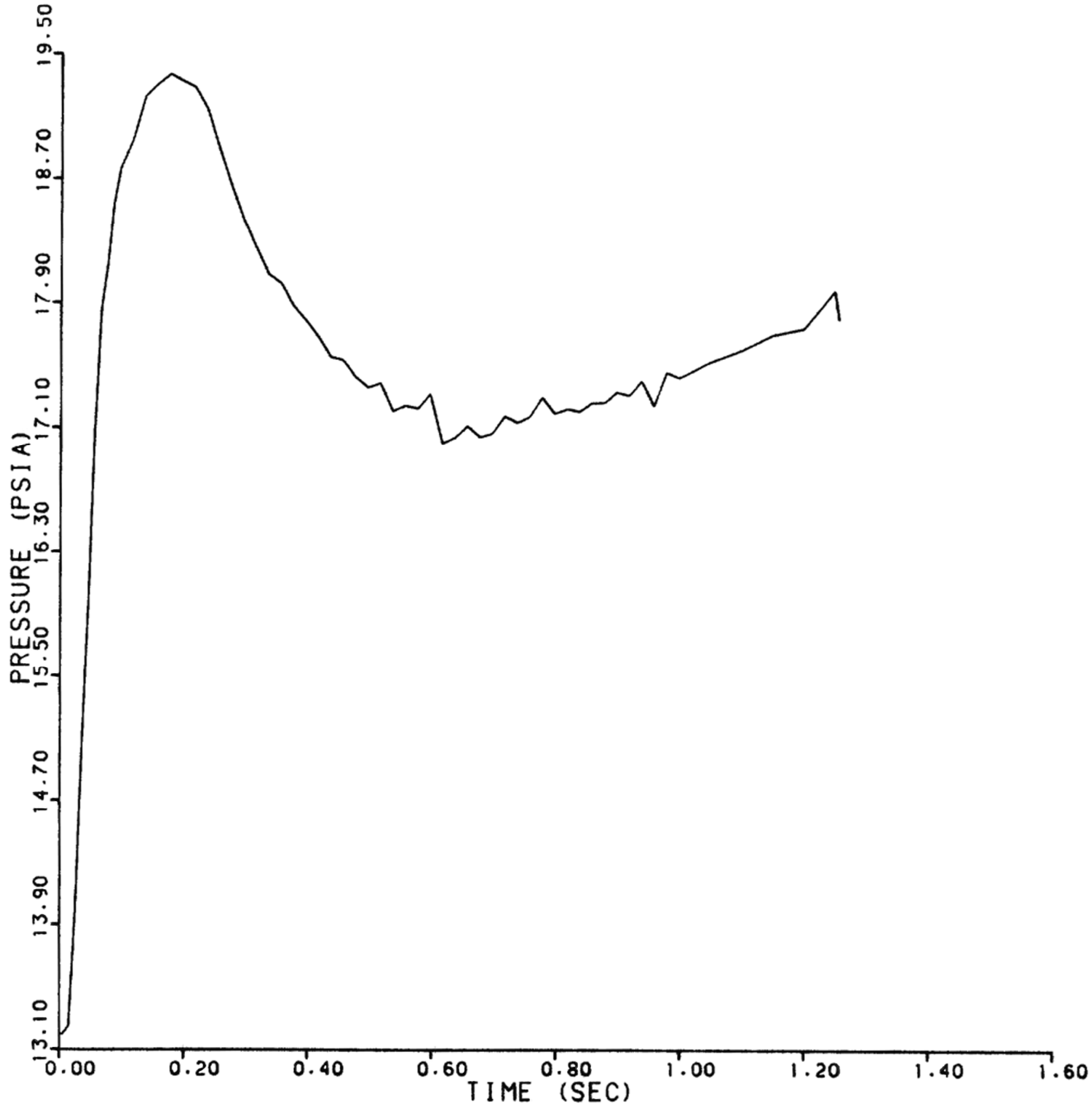
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E45
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 45 OF 74)



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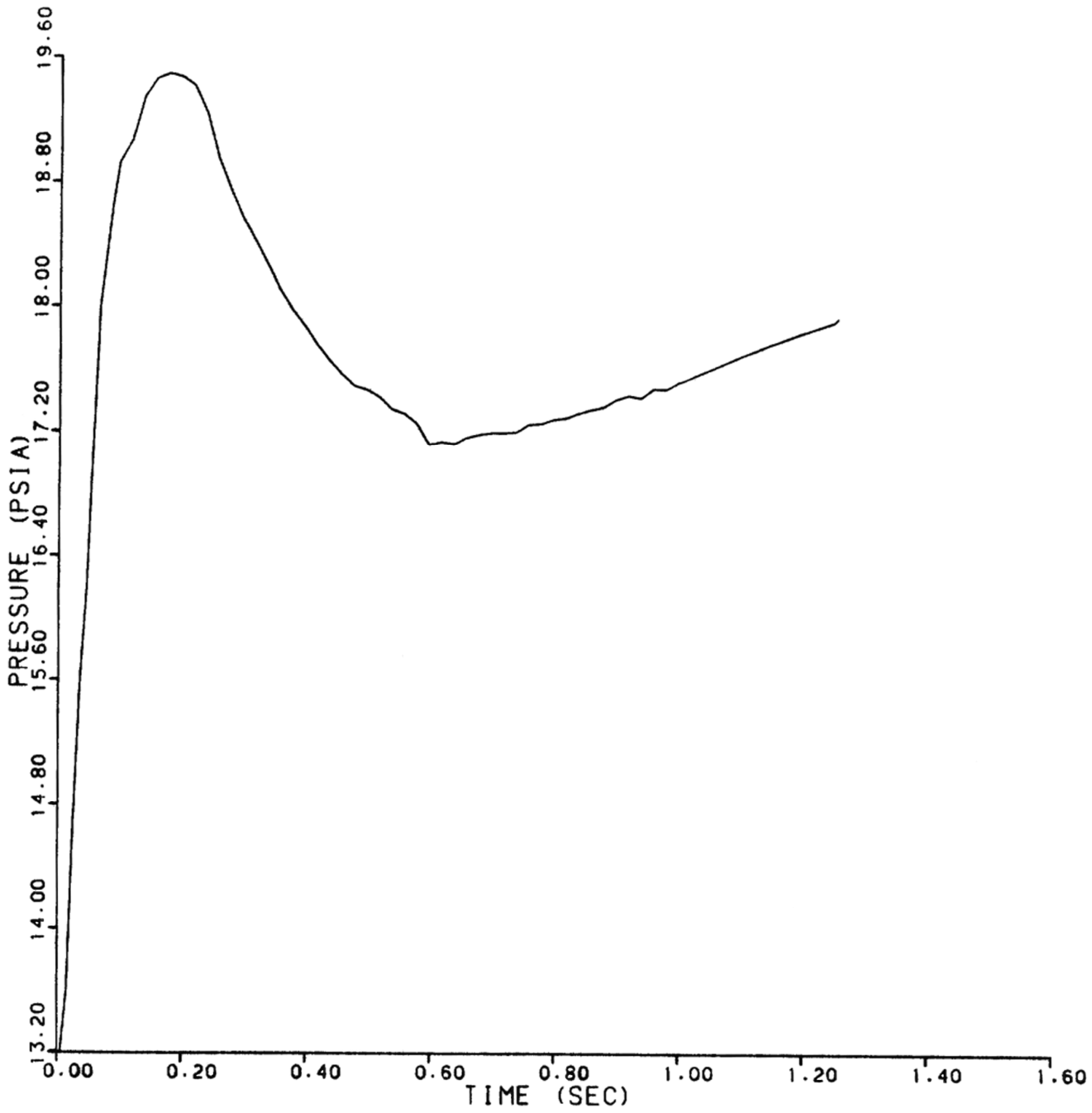
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E46
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 46 OF 74)



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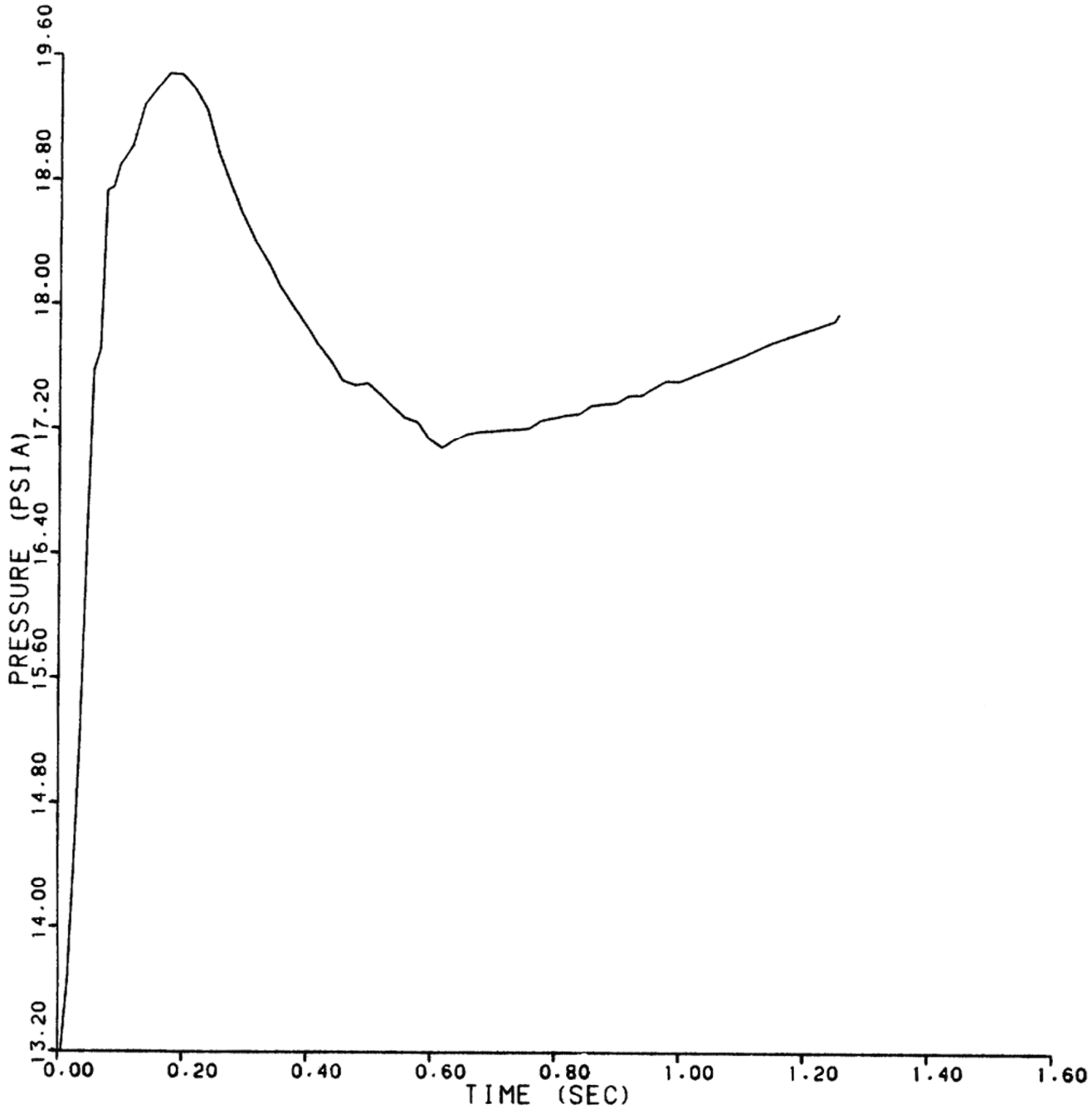
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E47
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 47 OF 74)



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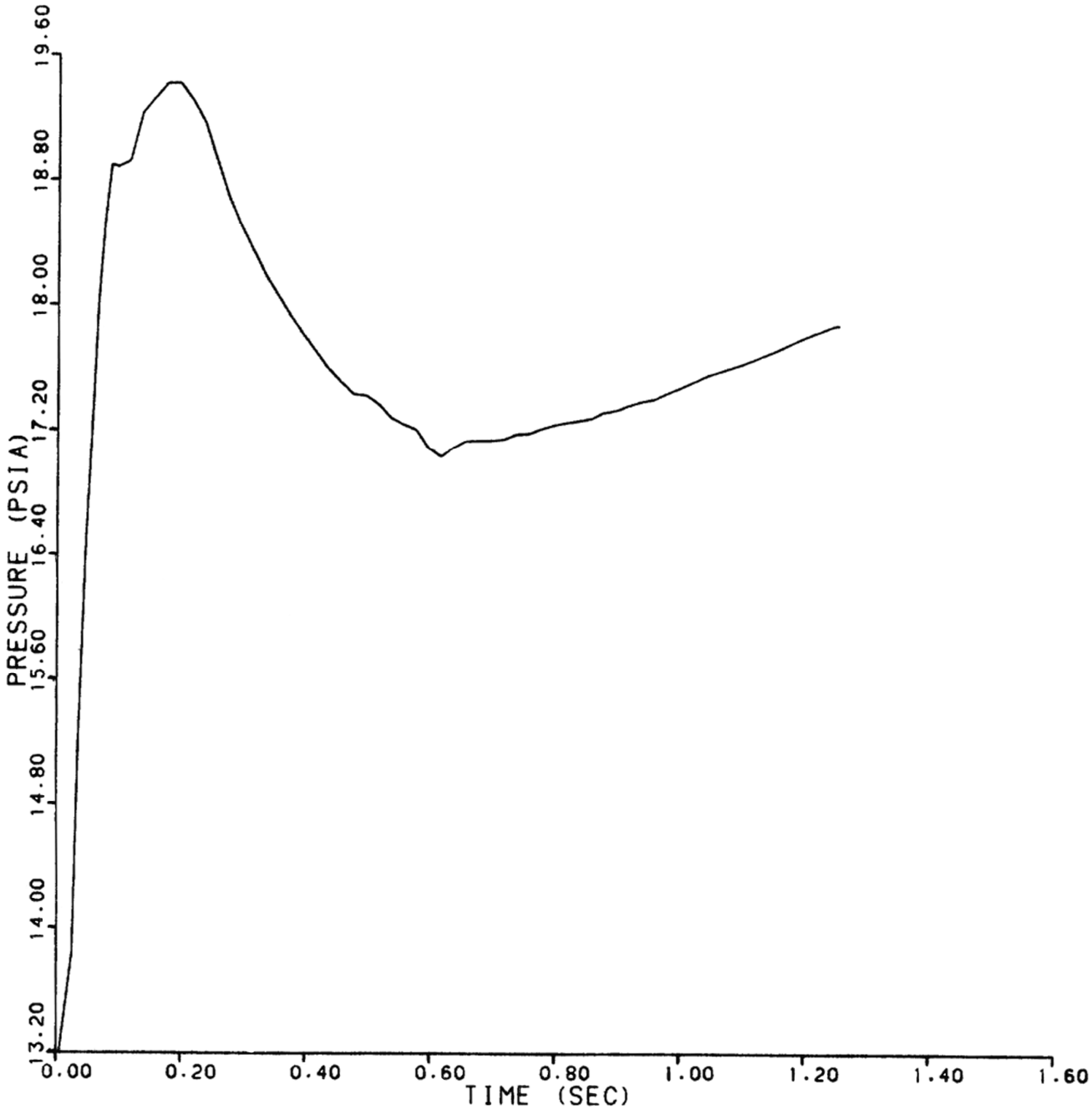
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E48
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 48 OF 74)



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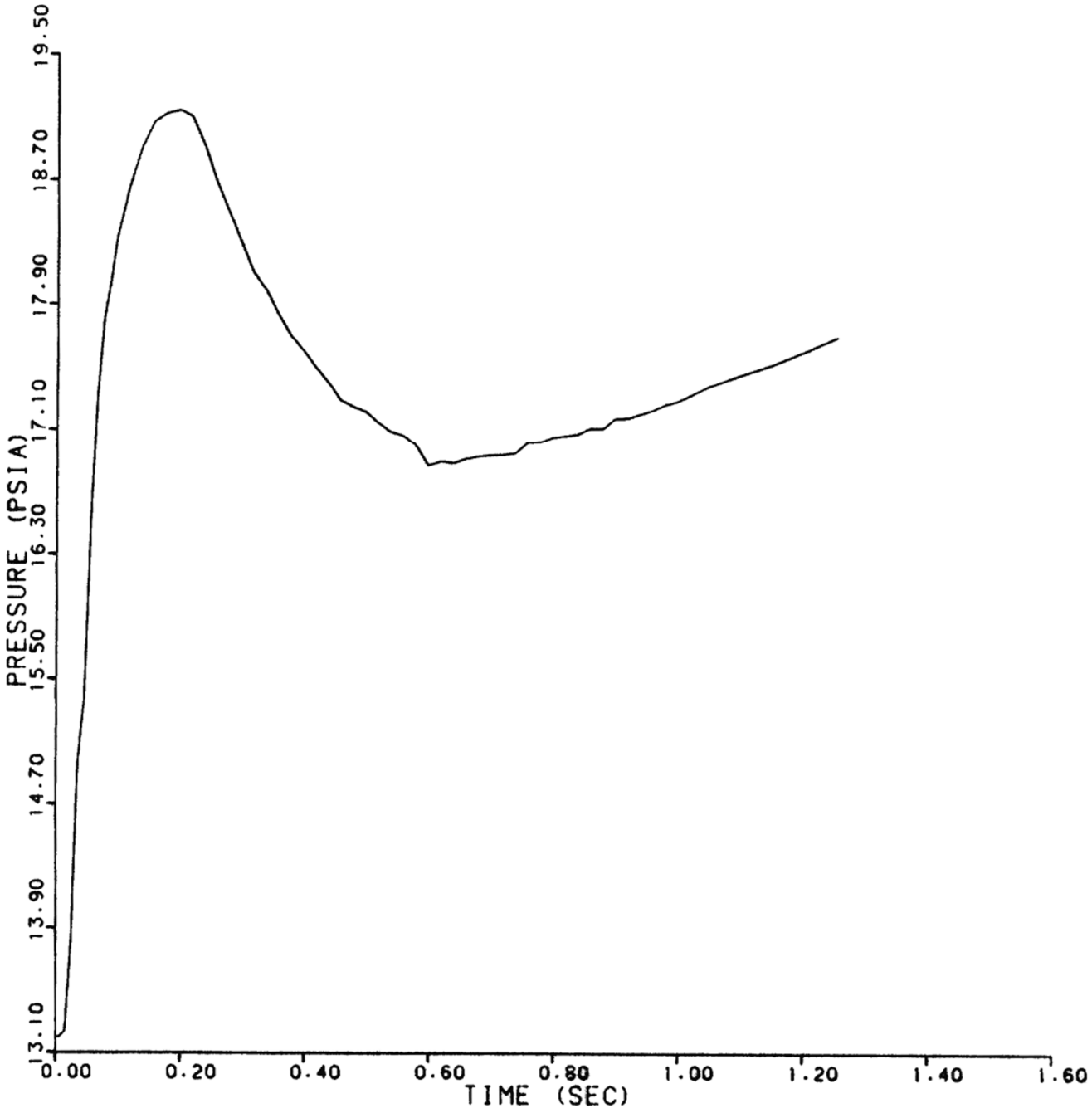
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E49
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 49 OF 74)



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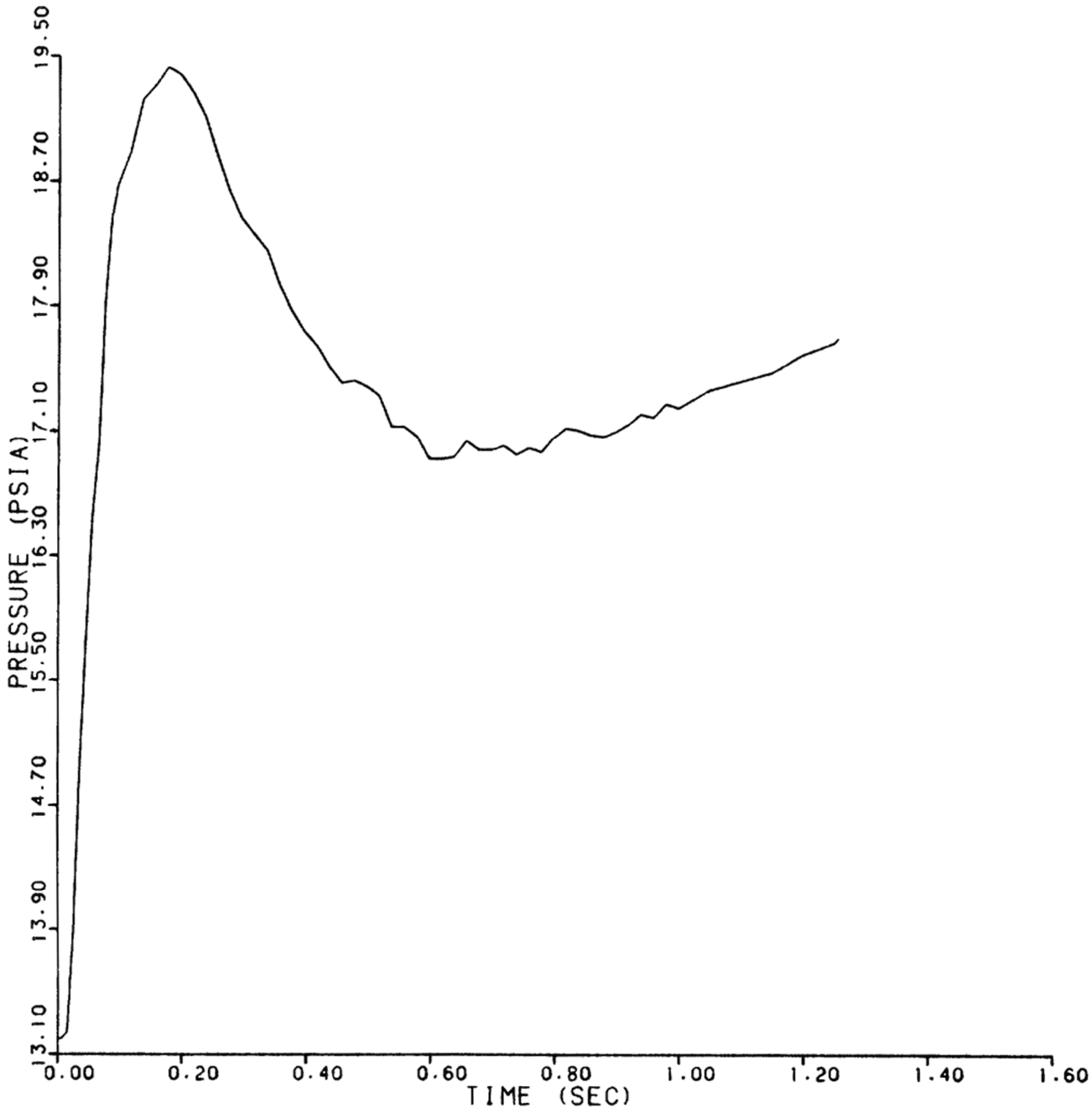
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E50
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 50 OF 74)



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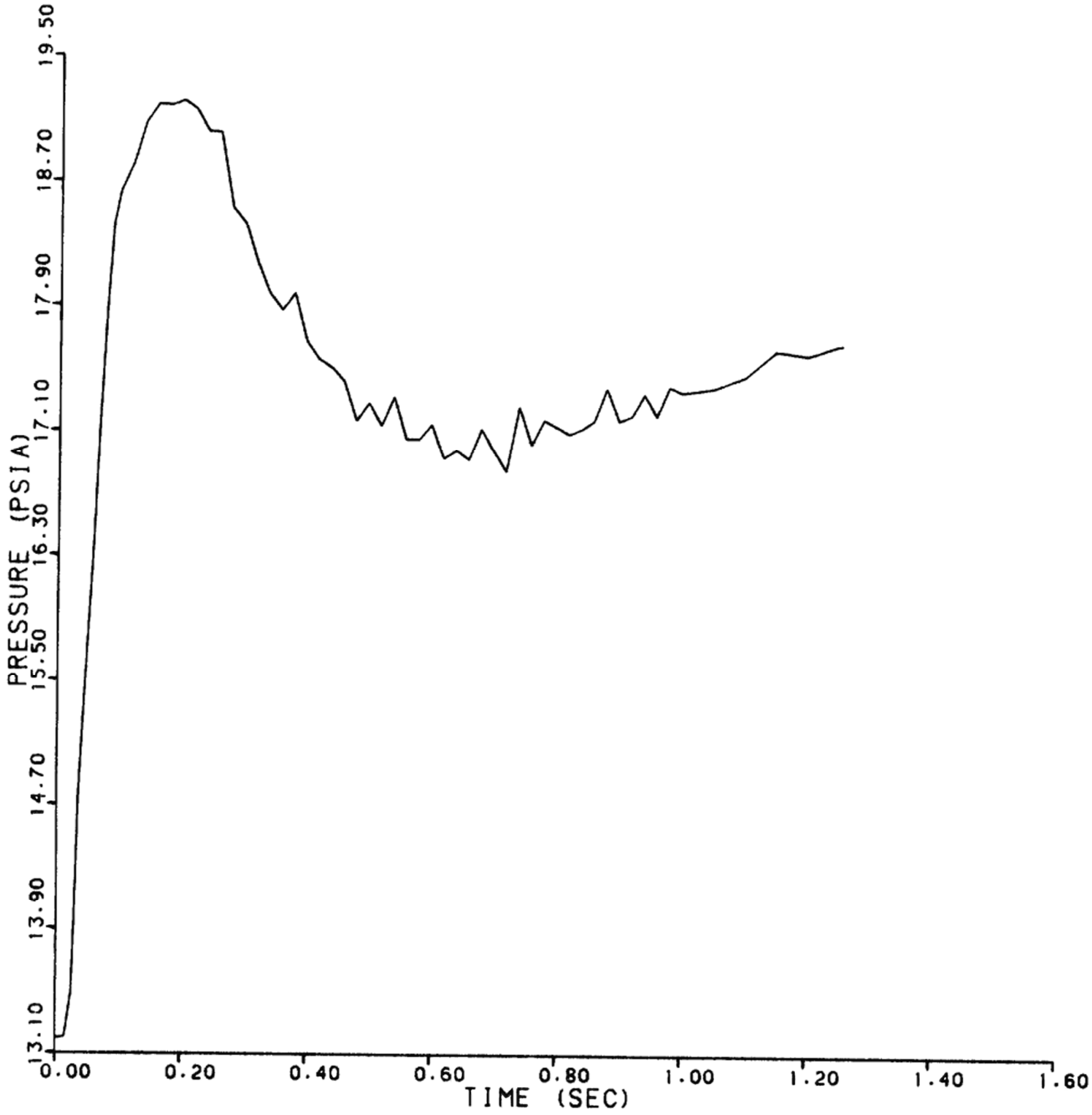
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E51 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 51 OF 74)



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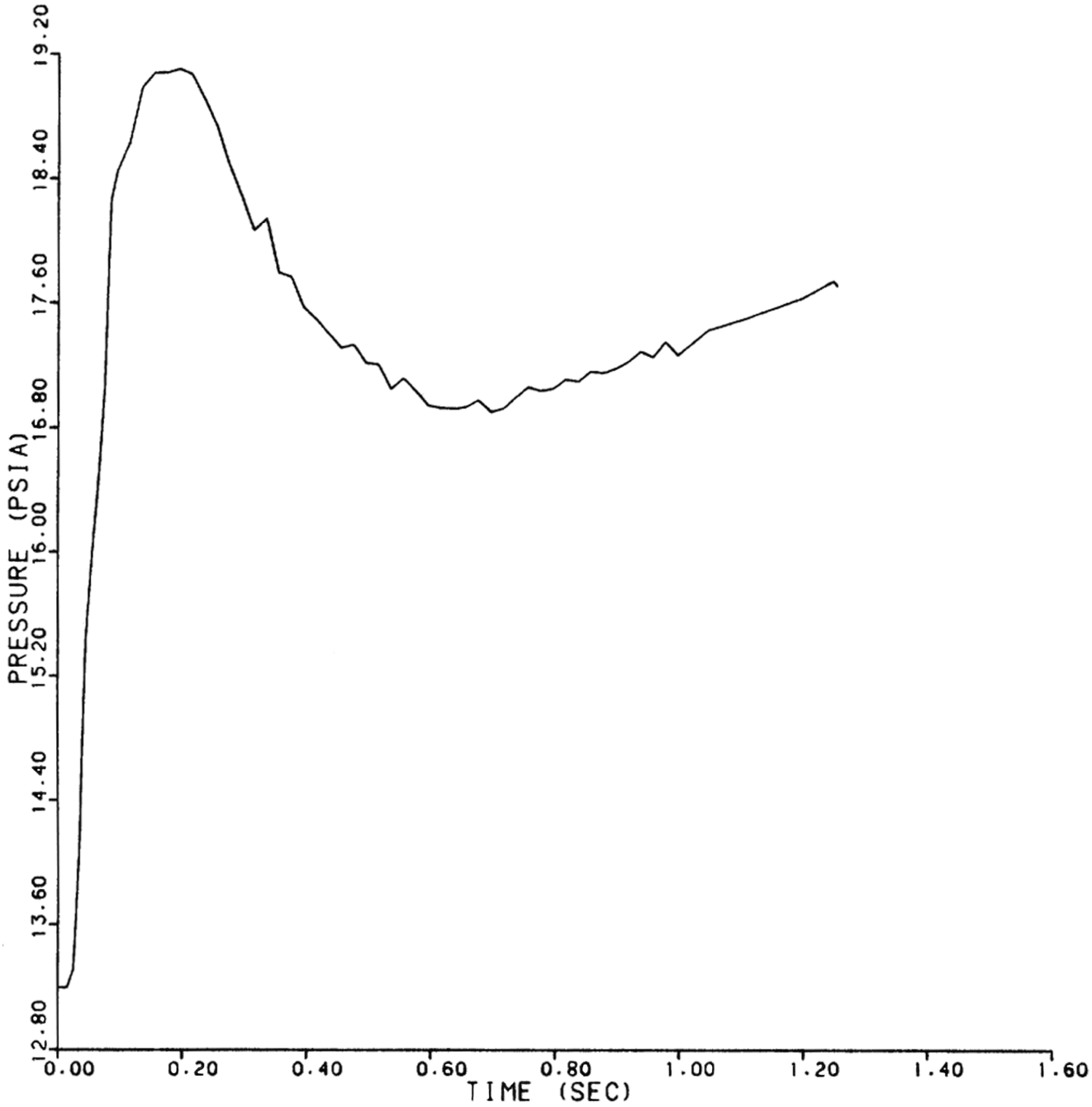
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E52
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 52 OF 74)



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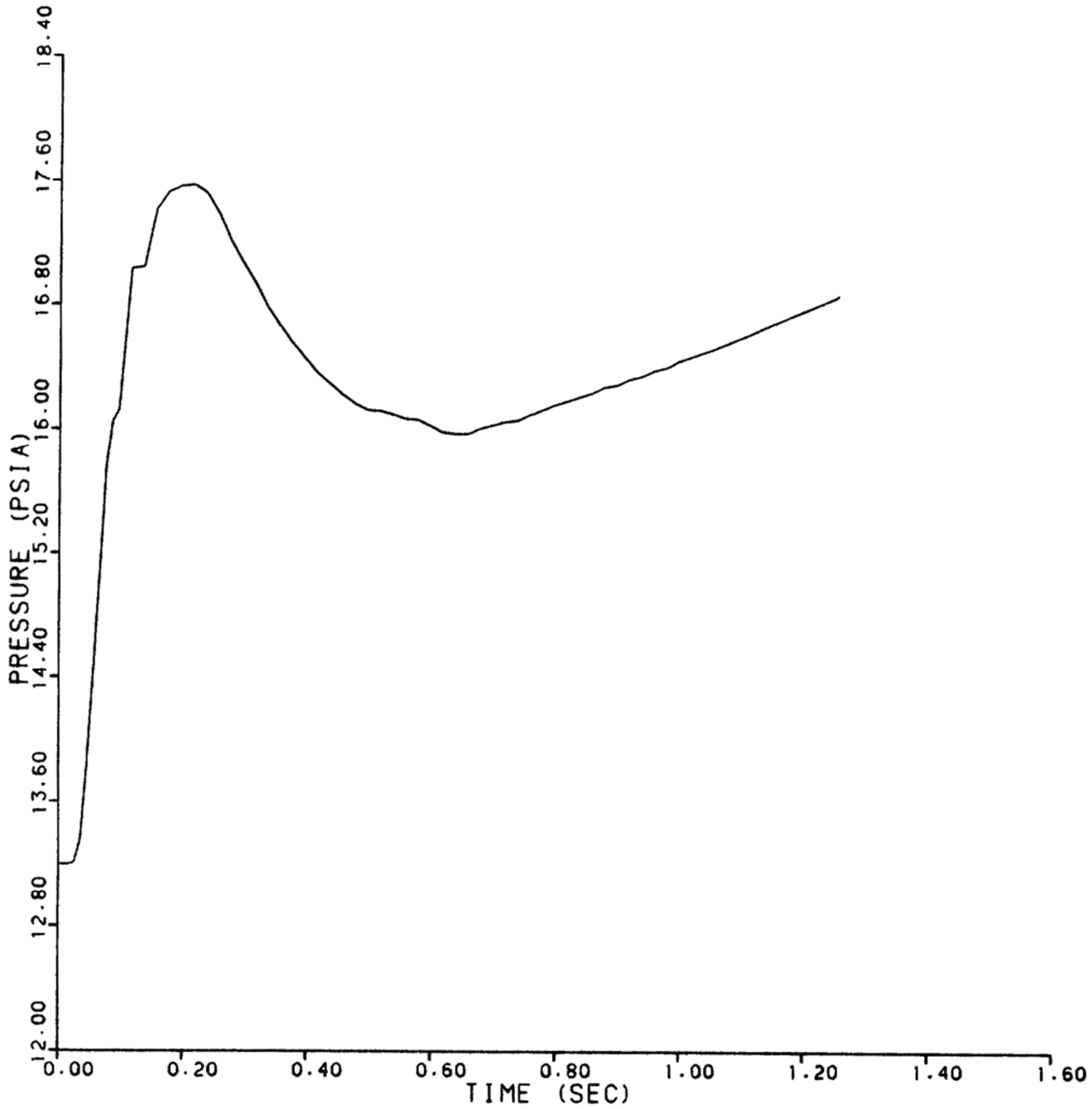
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E53
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 53 OF 74)



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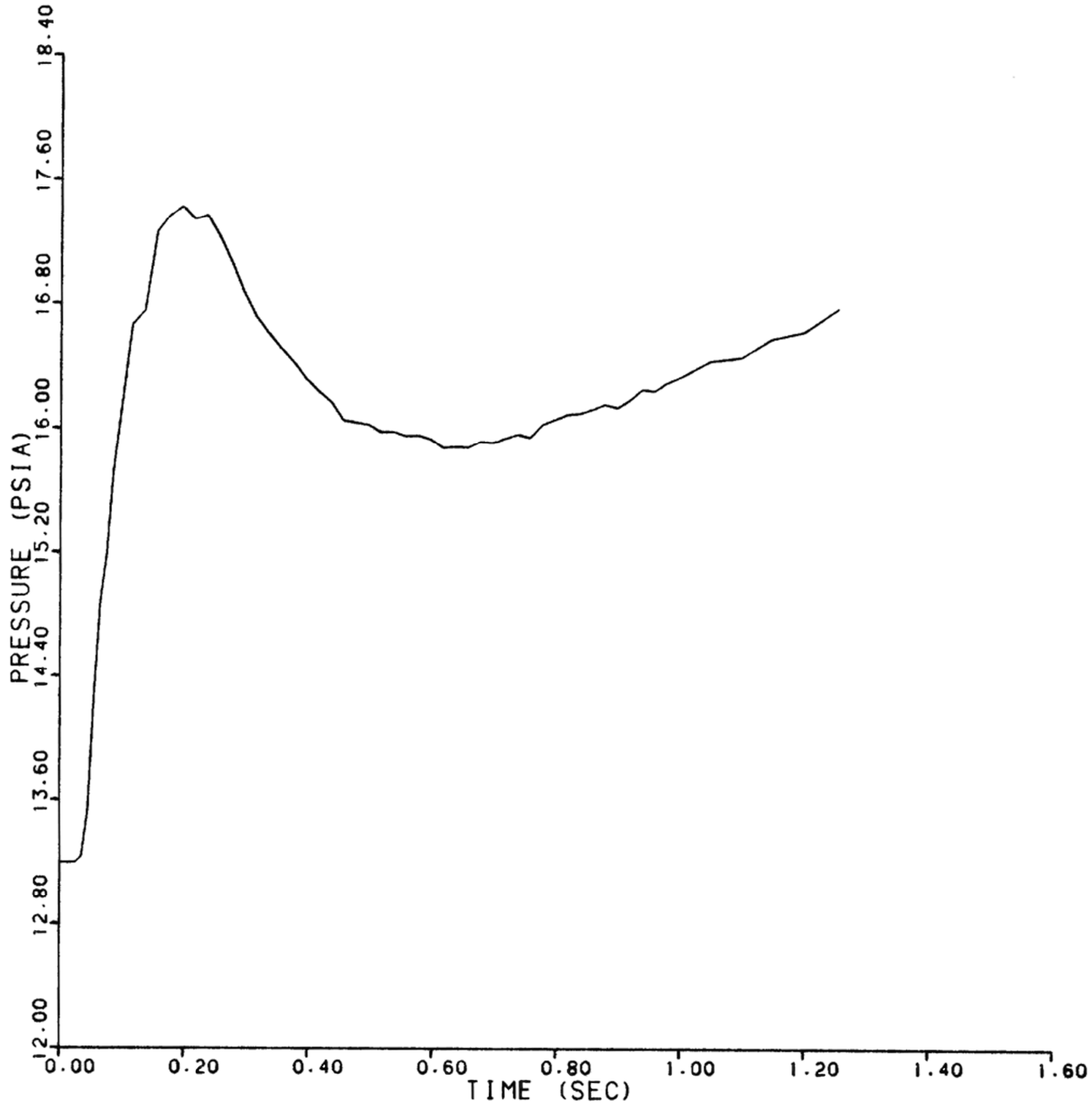
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E54
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 54 OF 74)



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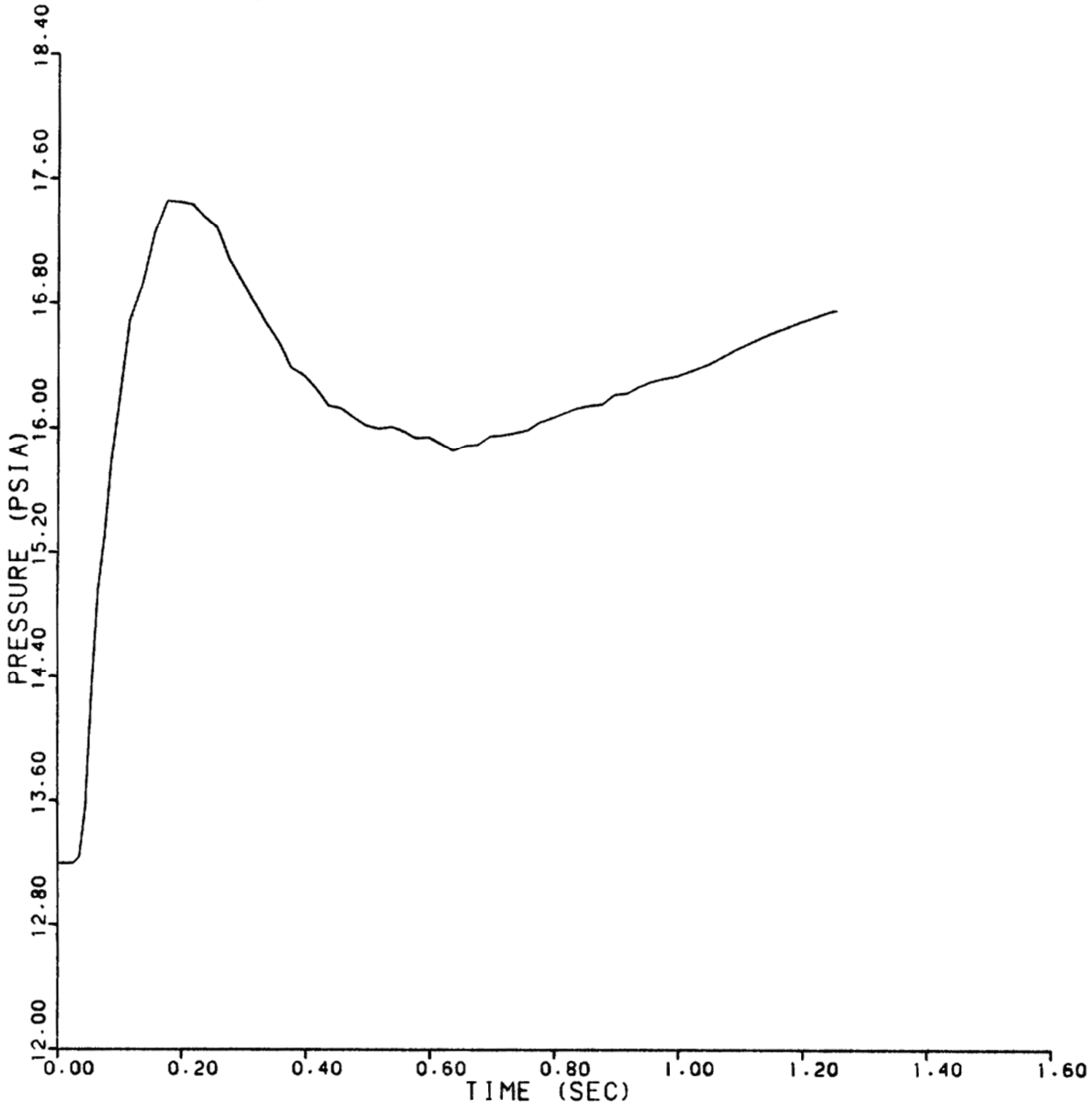
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E55
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 55 OF 74)



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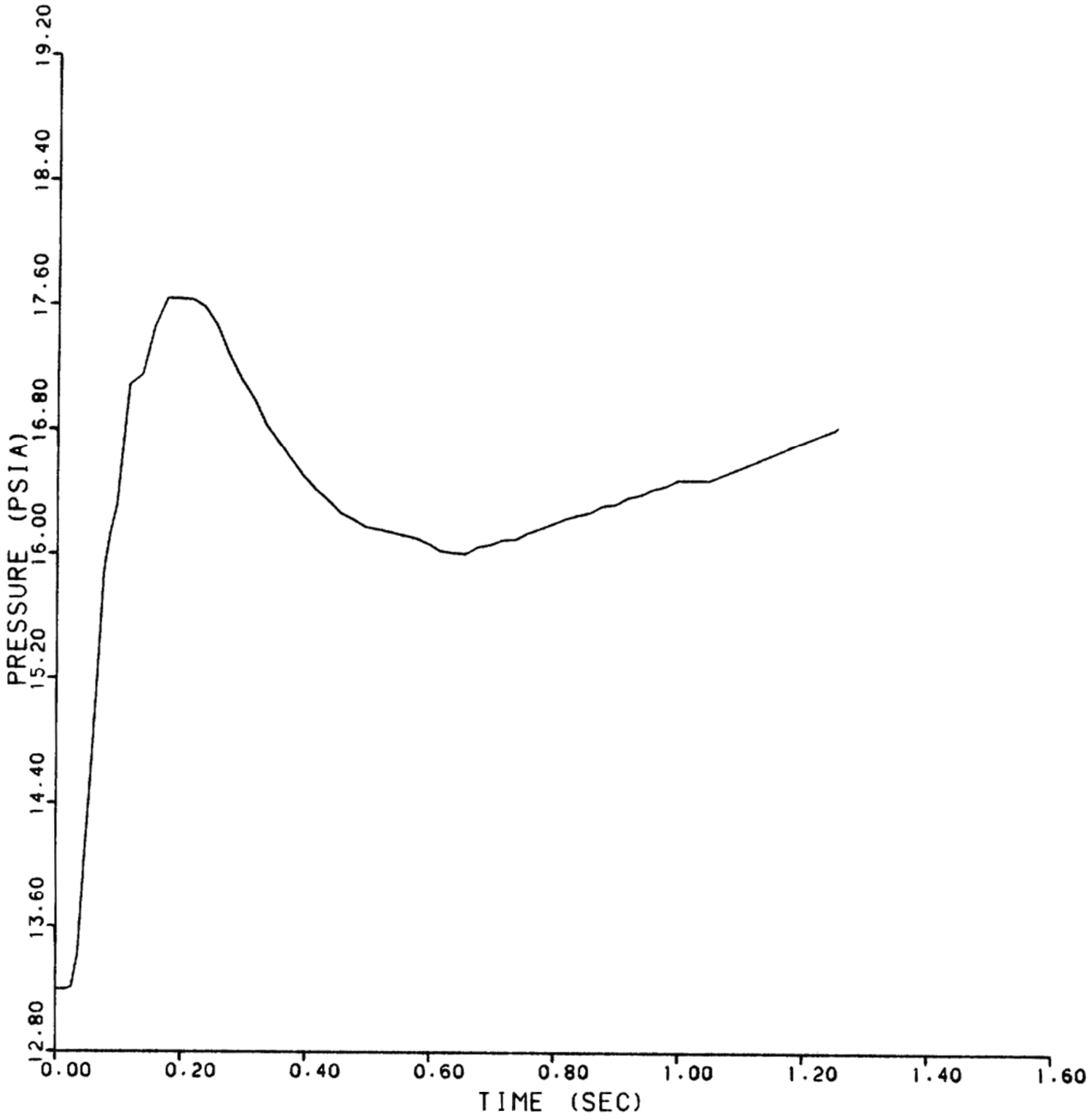
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E56 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 56 OF 74)



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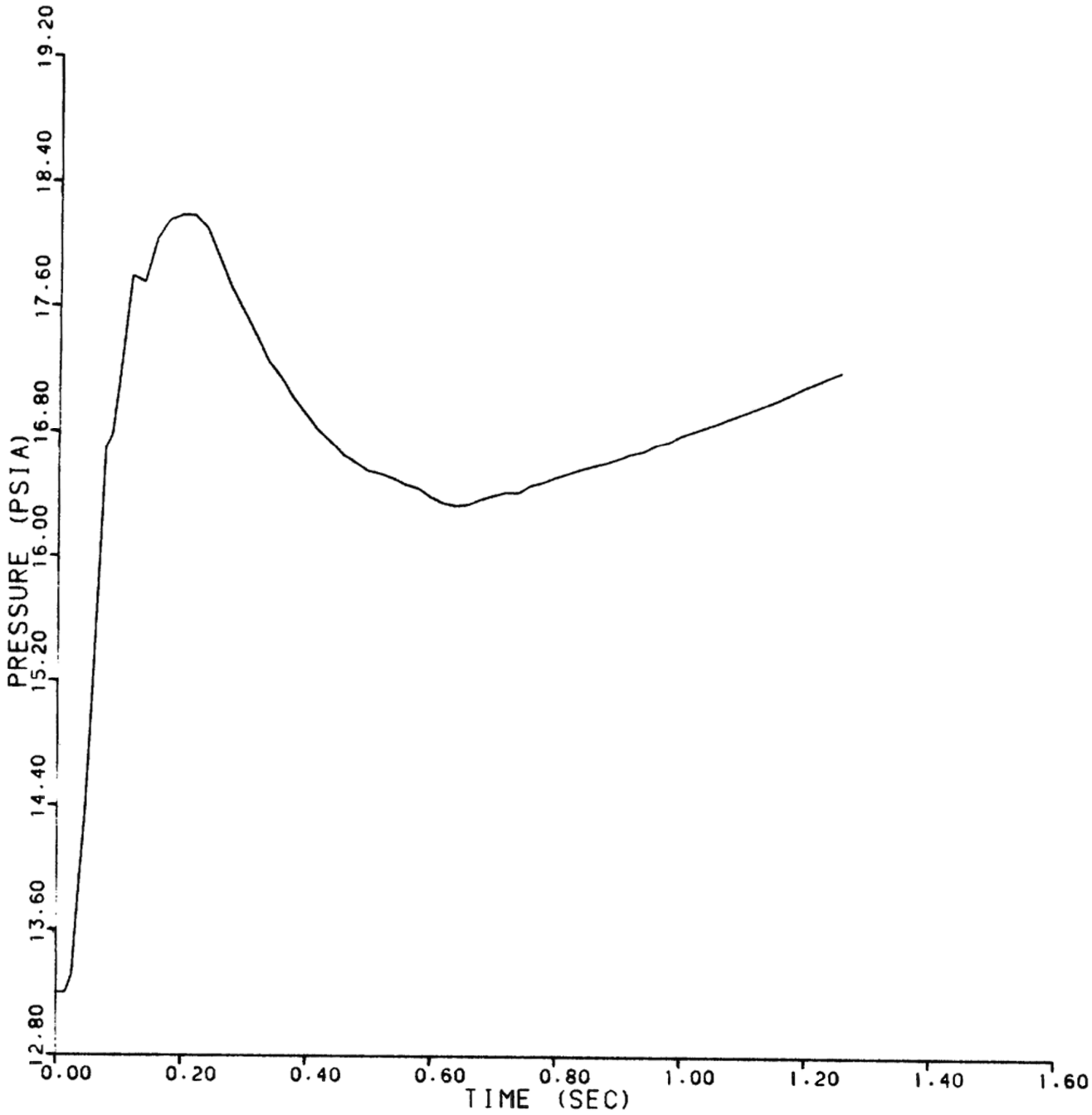
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E57
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 57 OF 74)



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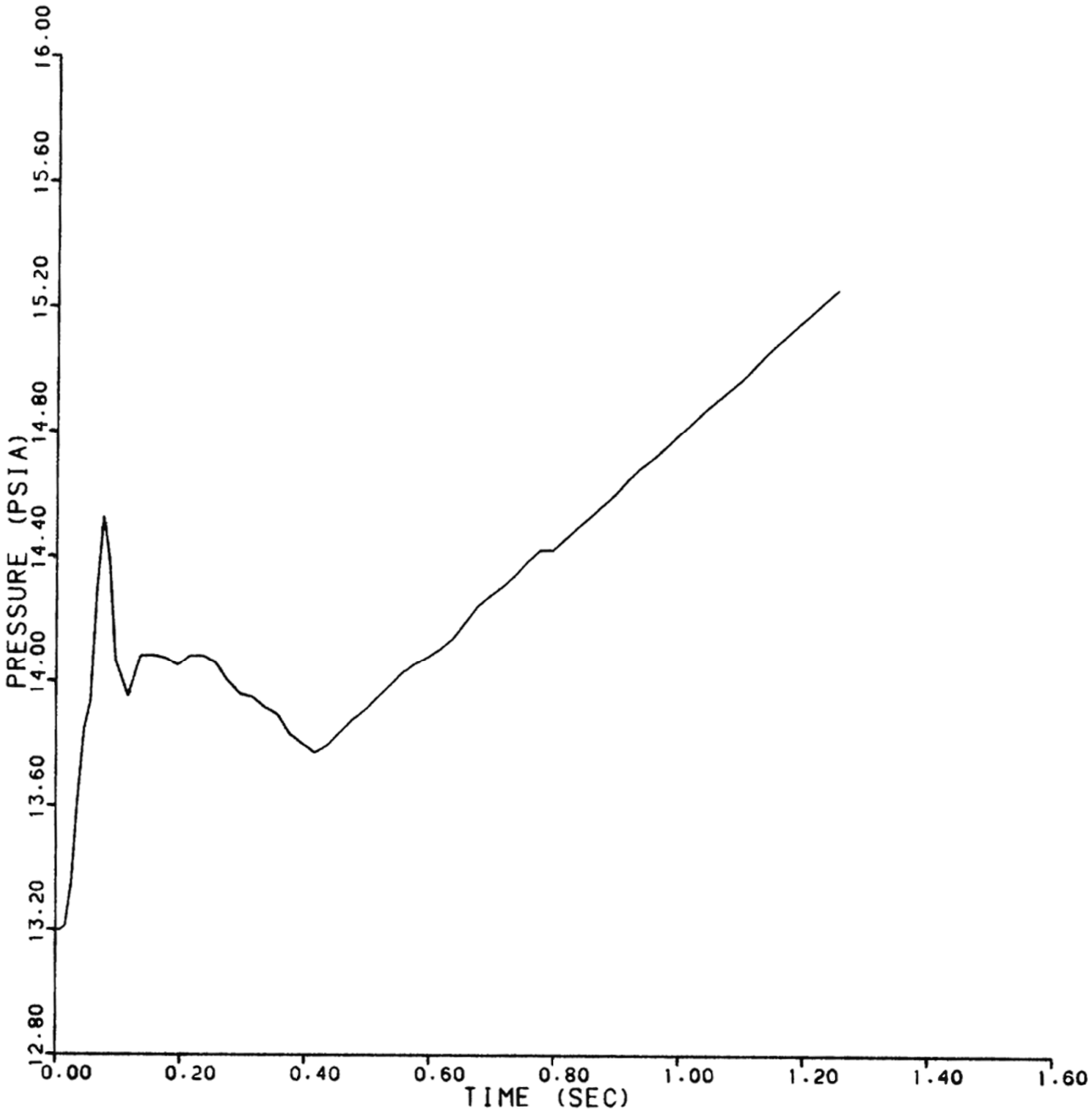
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E58 (306–in.^{2} BREAK AREA) FIGURE 6.2.1–19 (SHEET 58 OF 74)



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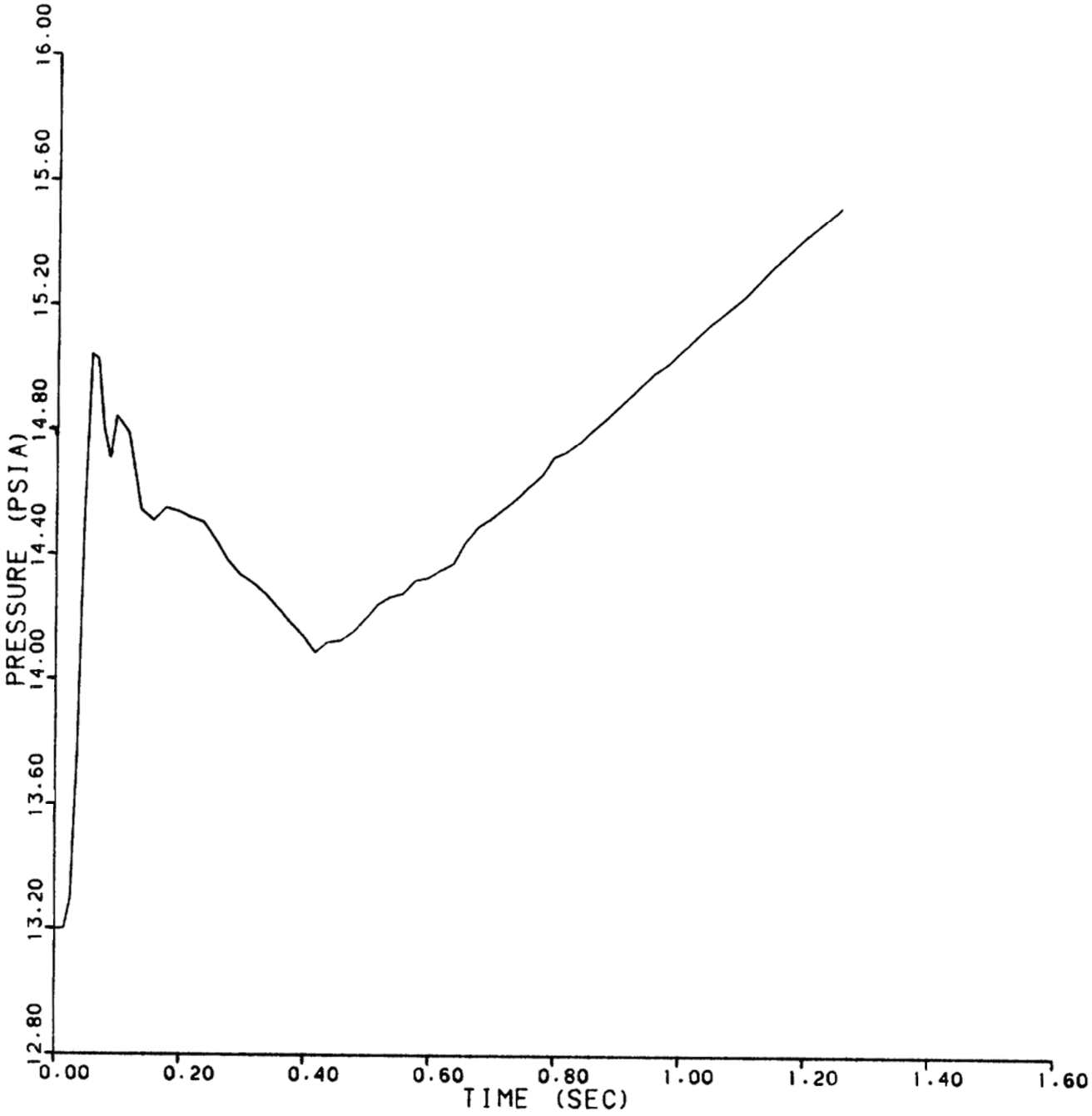
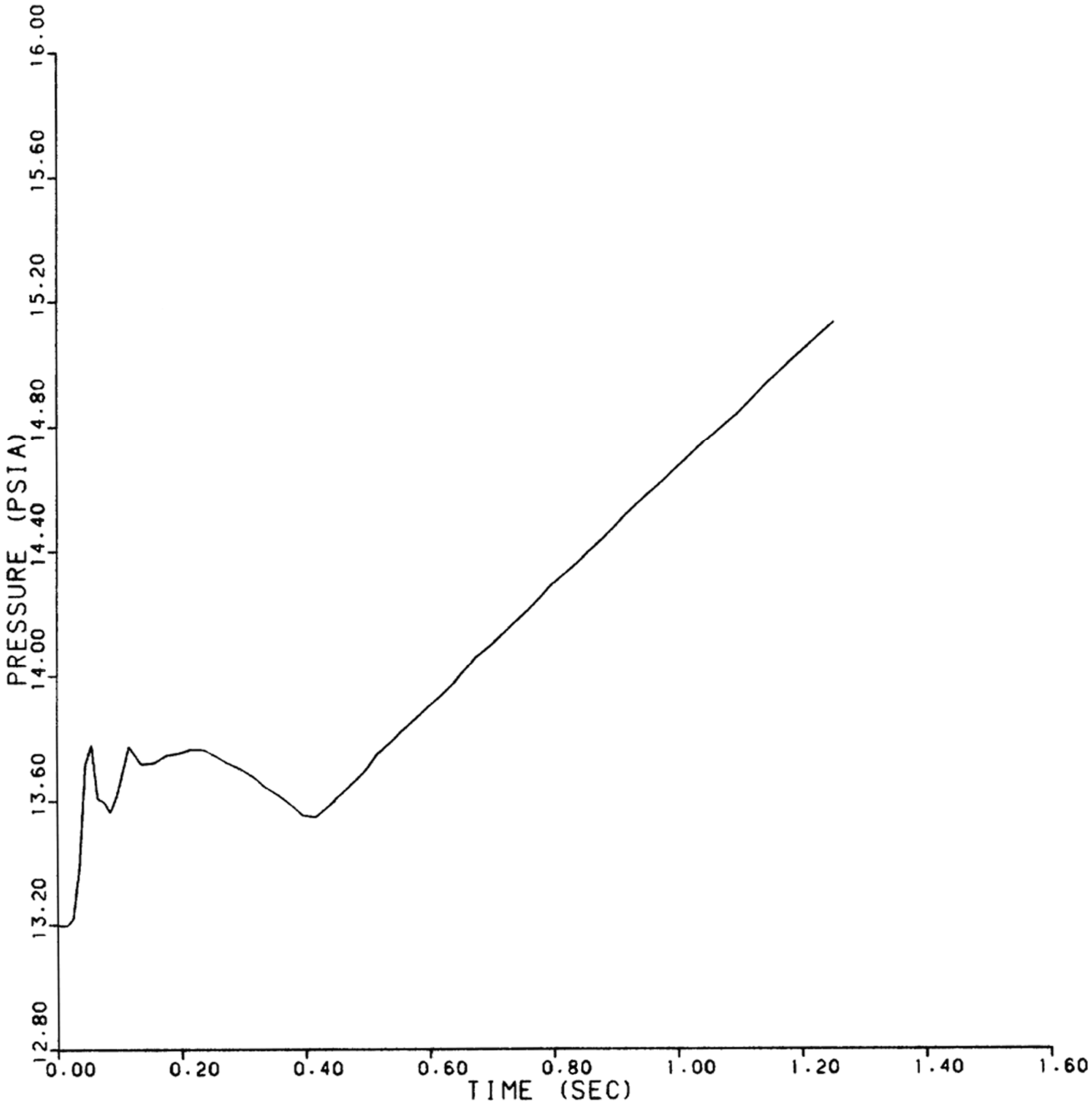
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E59 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 59 OF 74)



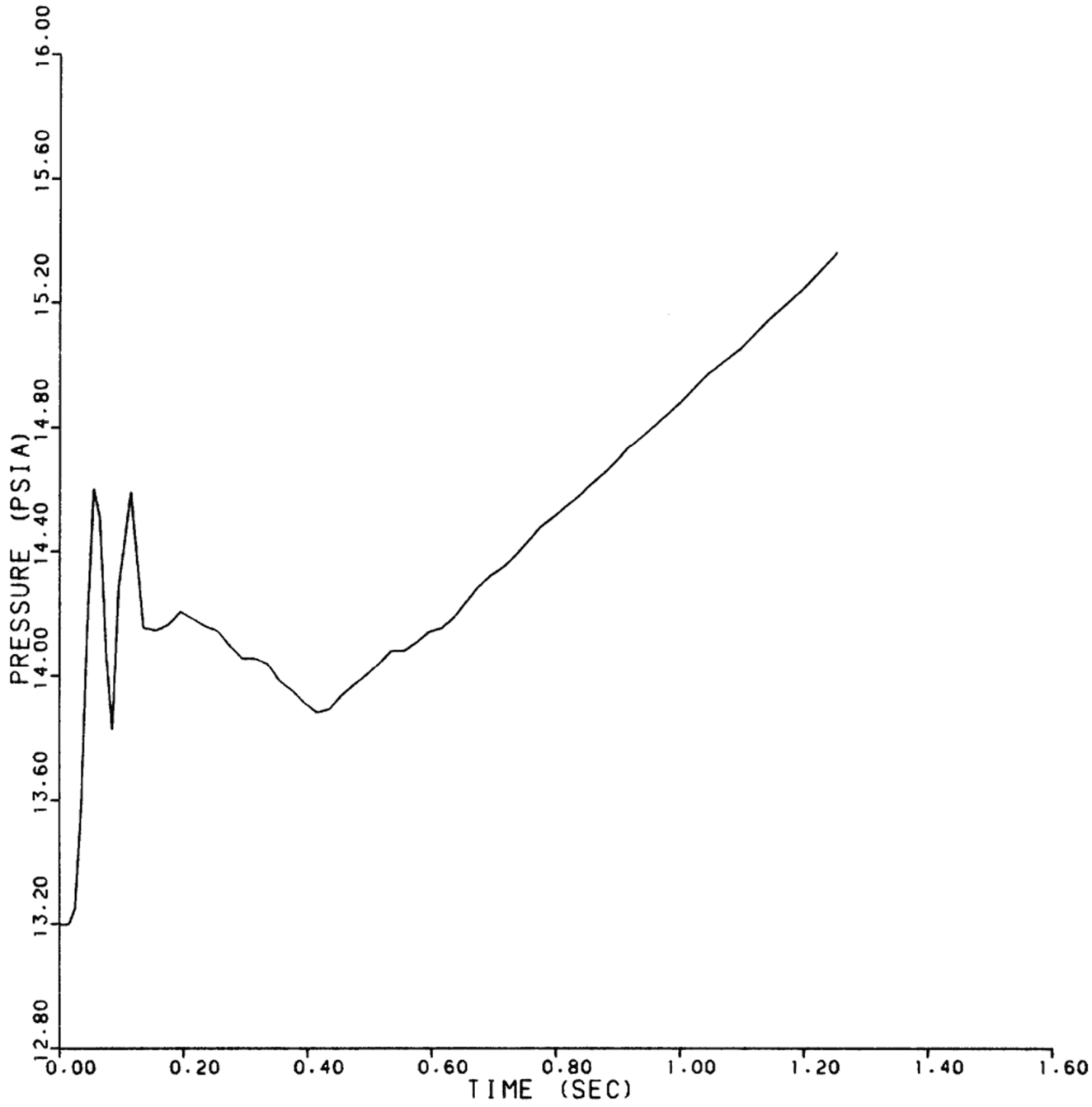
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E60 (306–in.^{2} BREAK AREA) FIGURE 6.2.1–19 (SHEET 60 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E61 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 61 OF 74)



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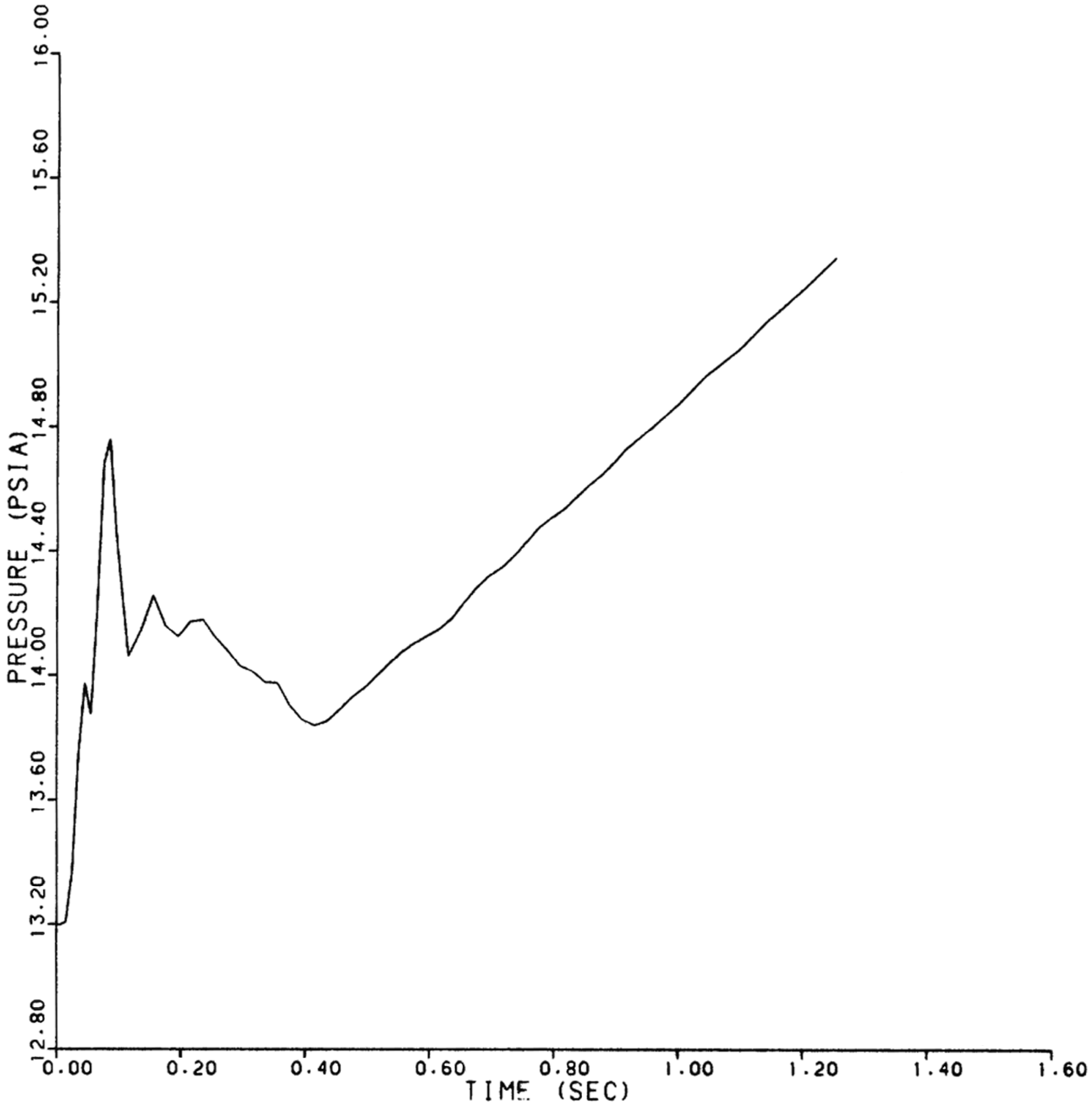
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E62 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 62 OF 74)



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SOUTHERN COMPANY
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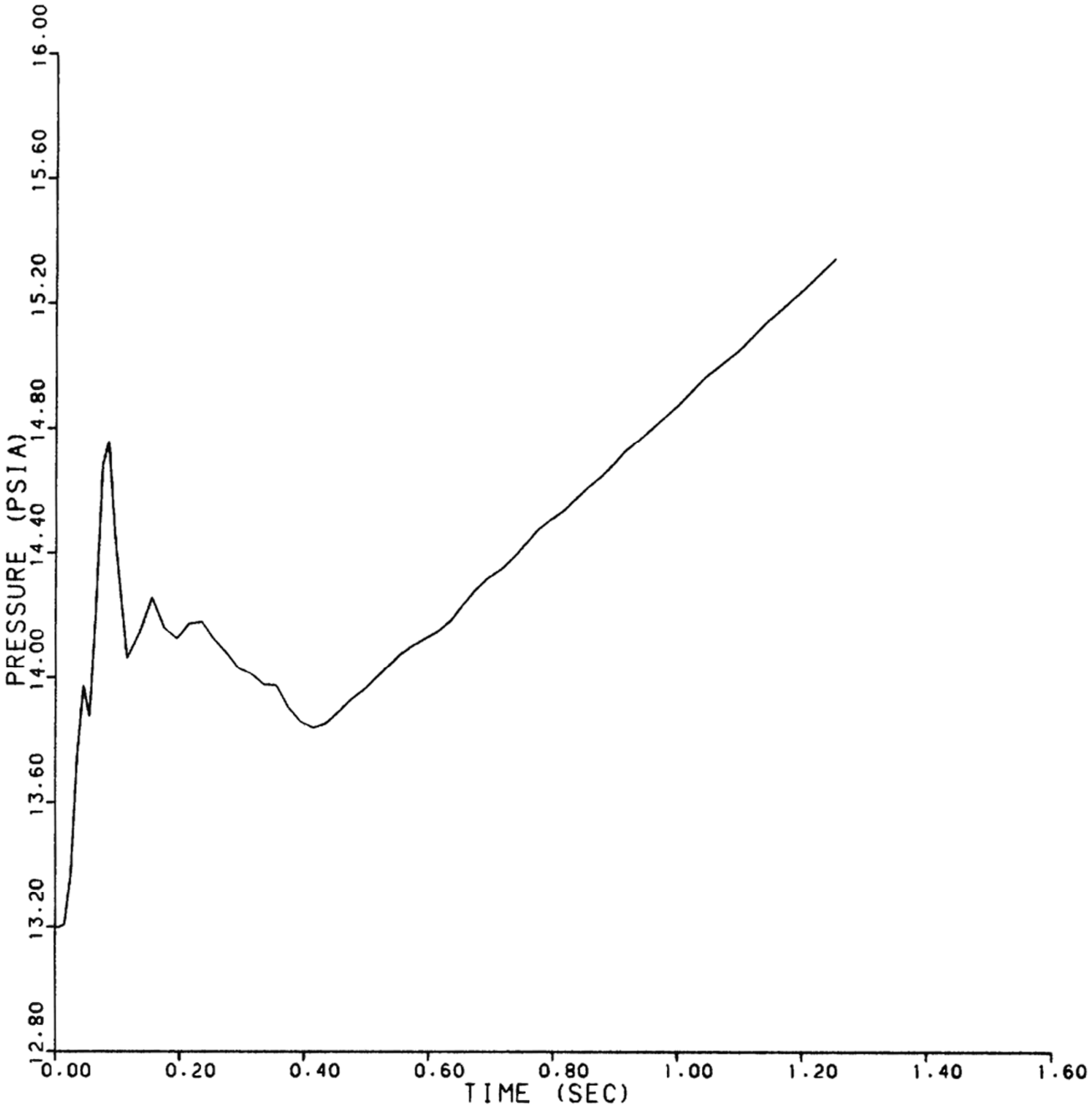
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E63
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 63 OF 74)



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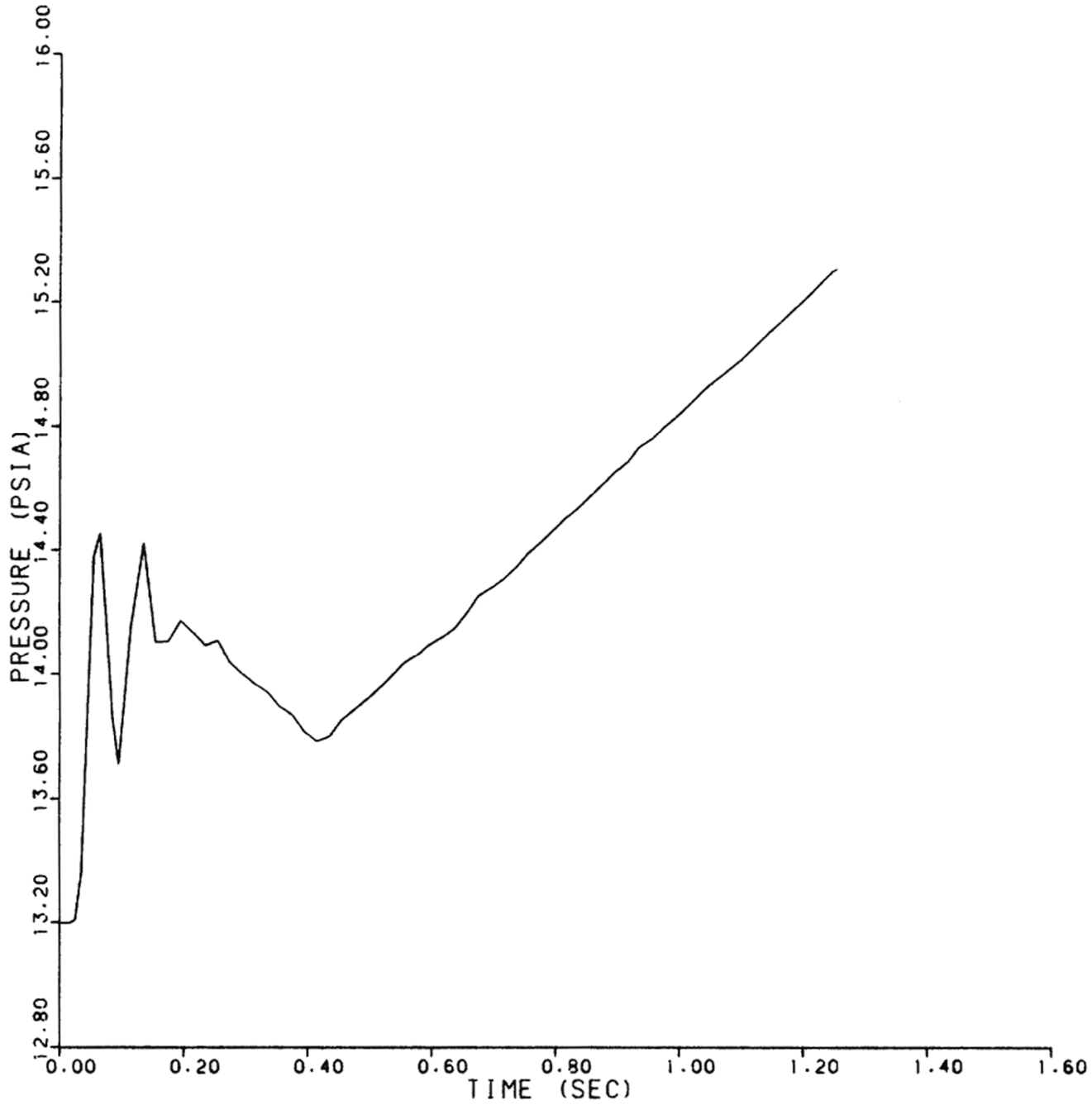
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E64 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 64 OF 74)



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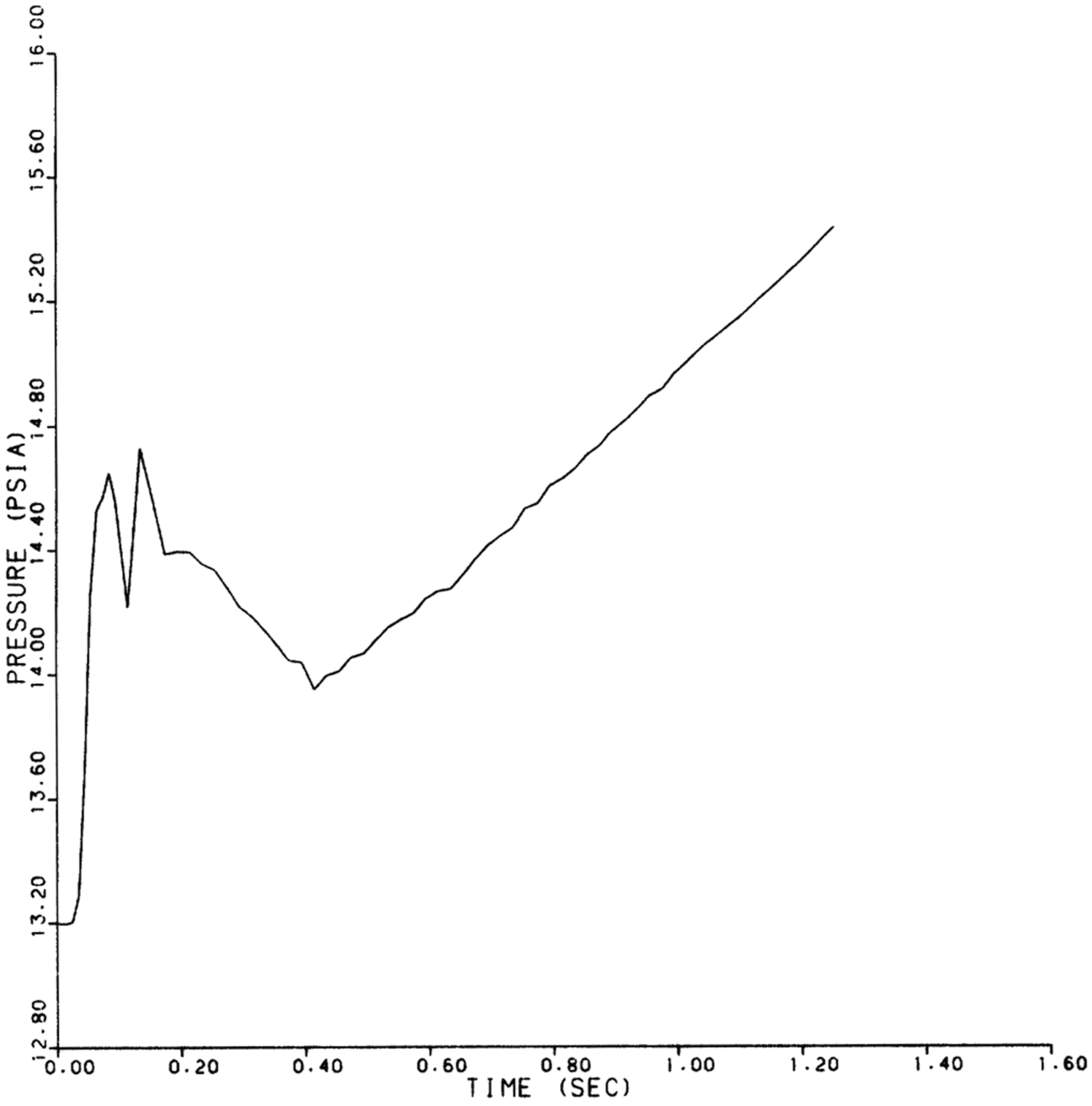
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E65 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 65 OF 74)



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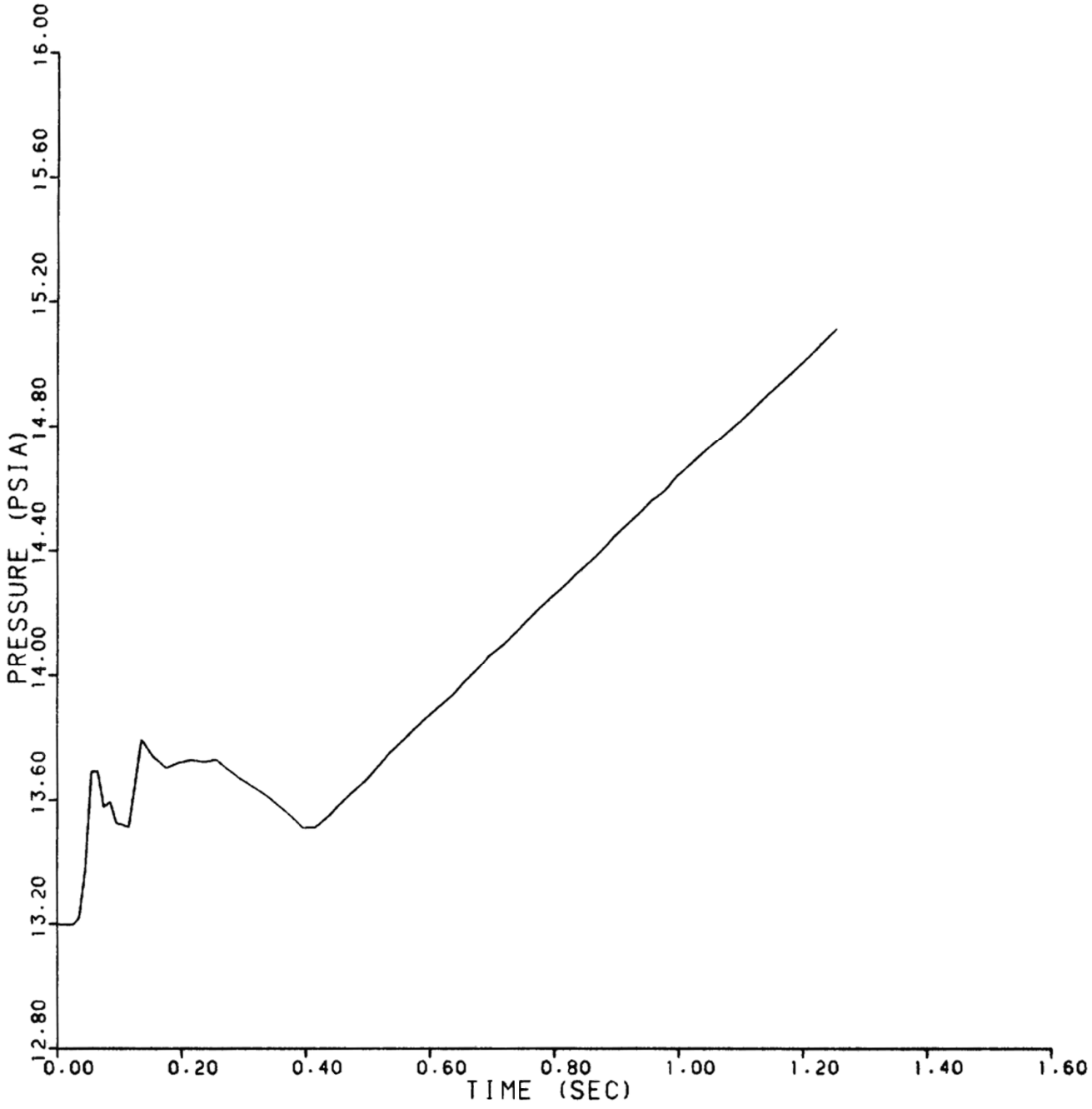
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E66 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 66 OF 74)



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SOUTHERN COMPANY
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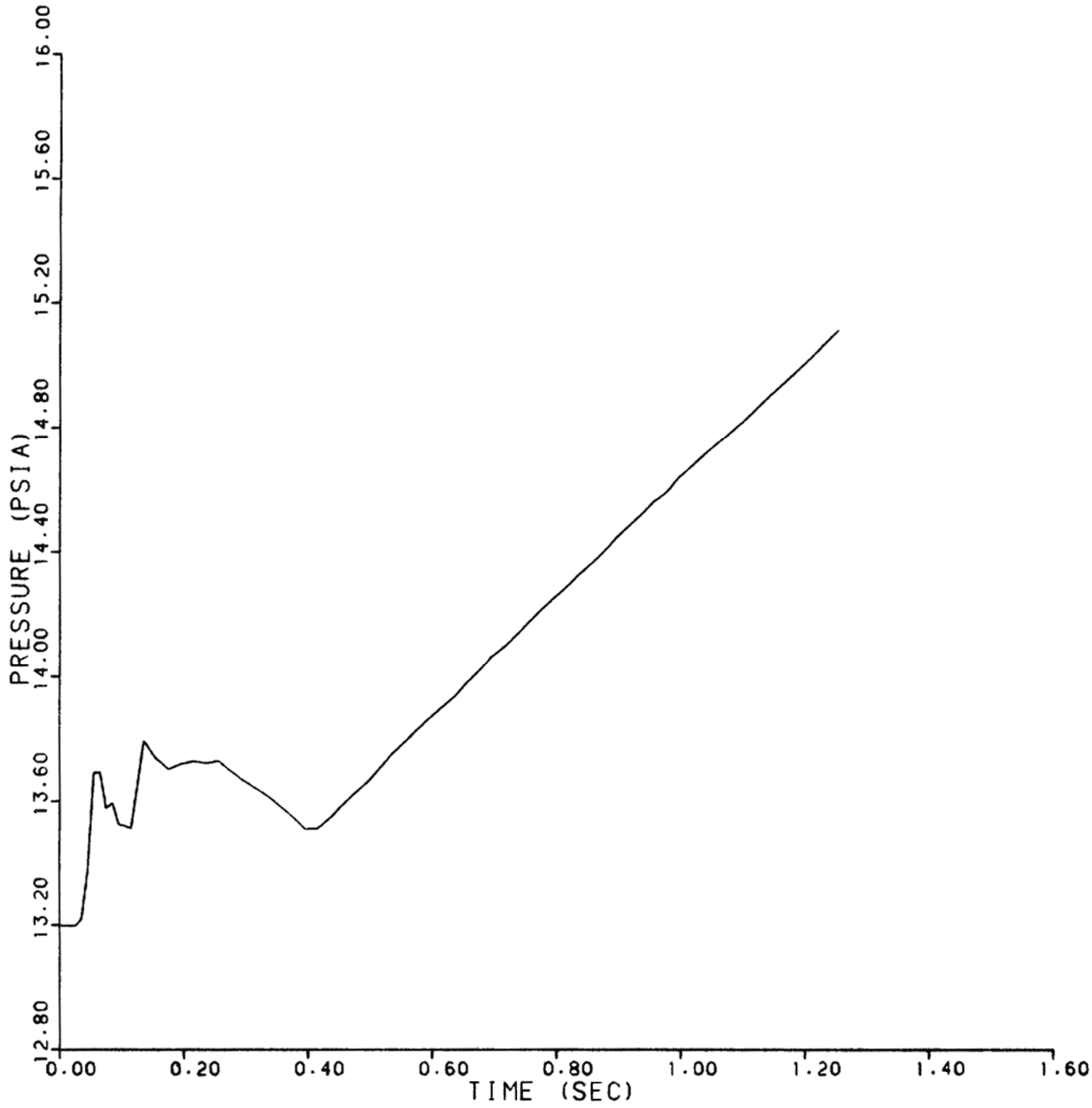
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E67
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 67 OF 74)



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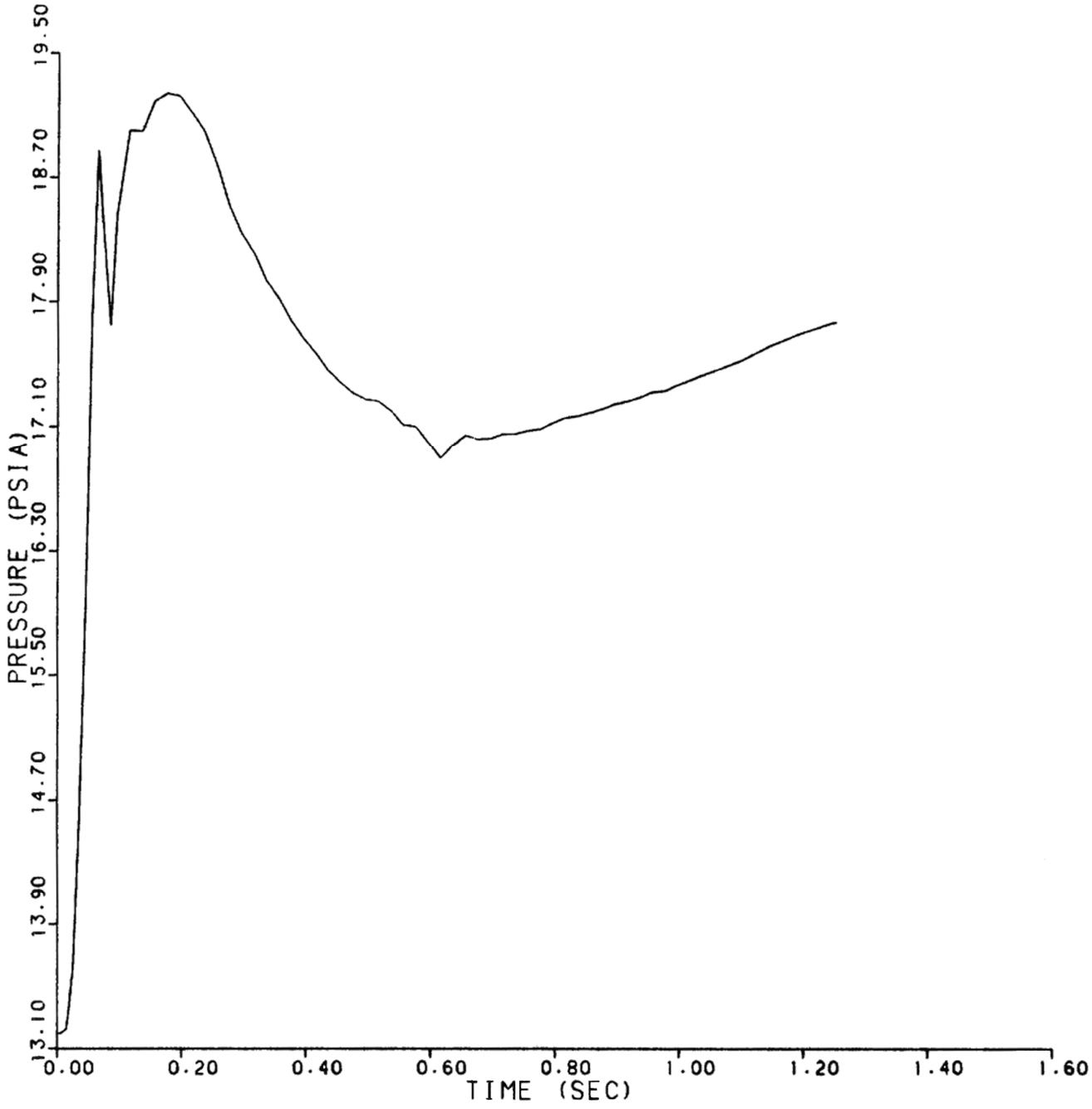
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E68 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 68 OF 74)



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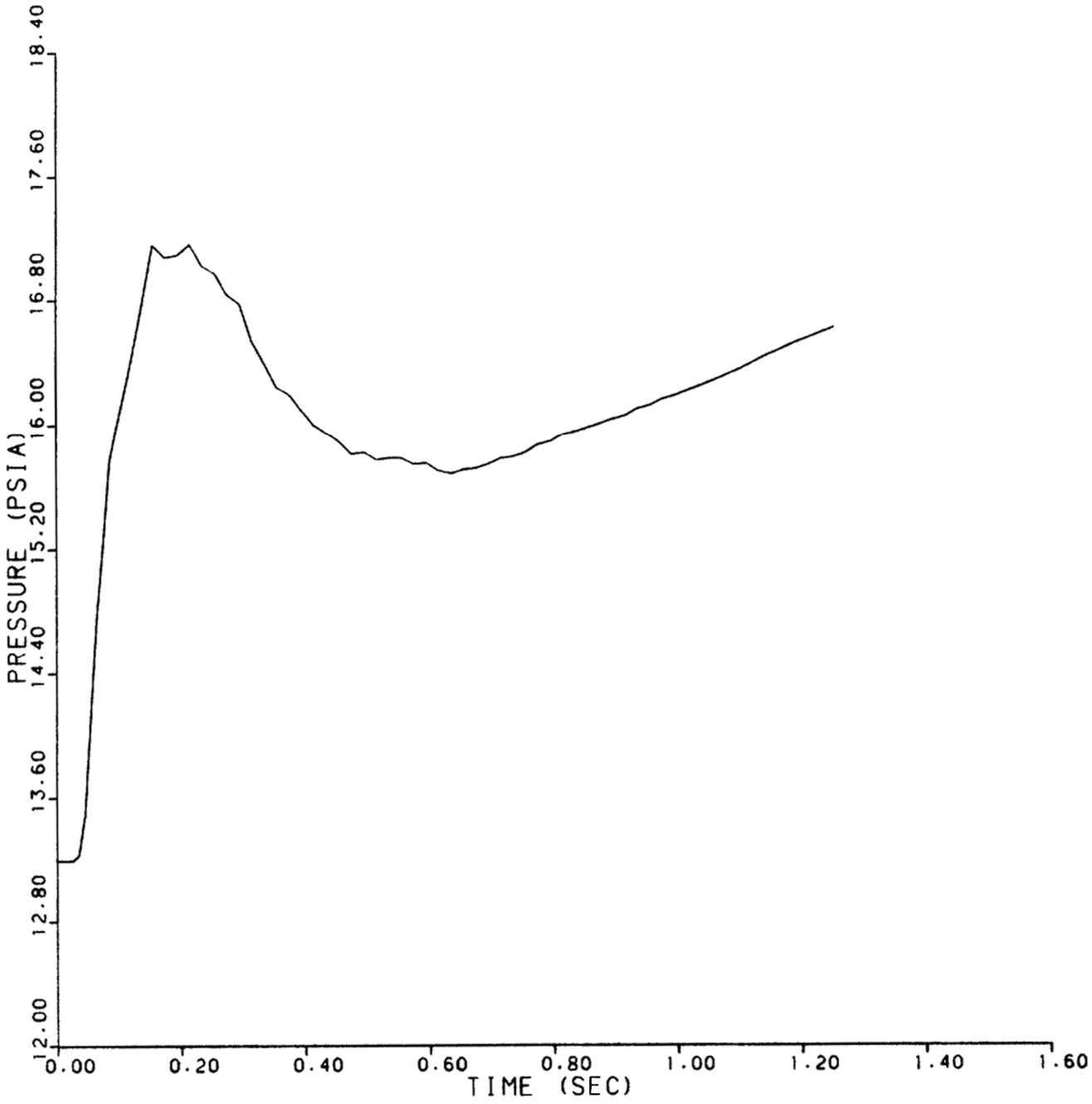
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E69 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 69 OF 74)



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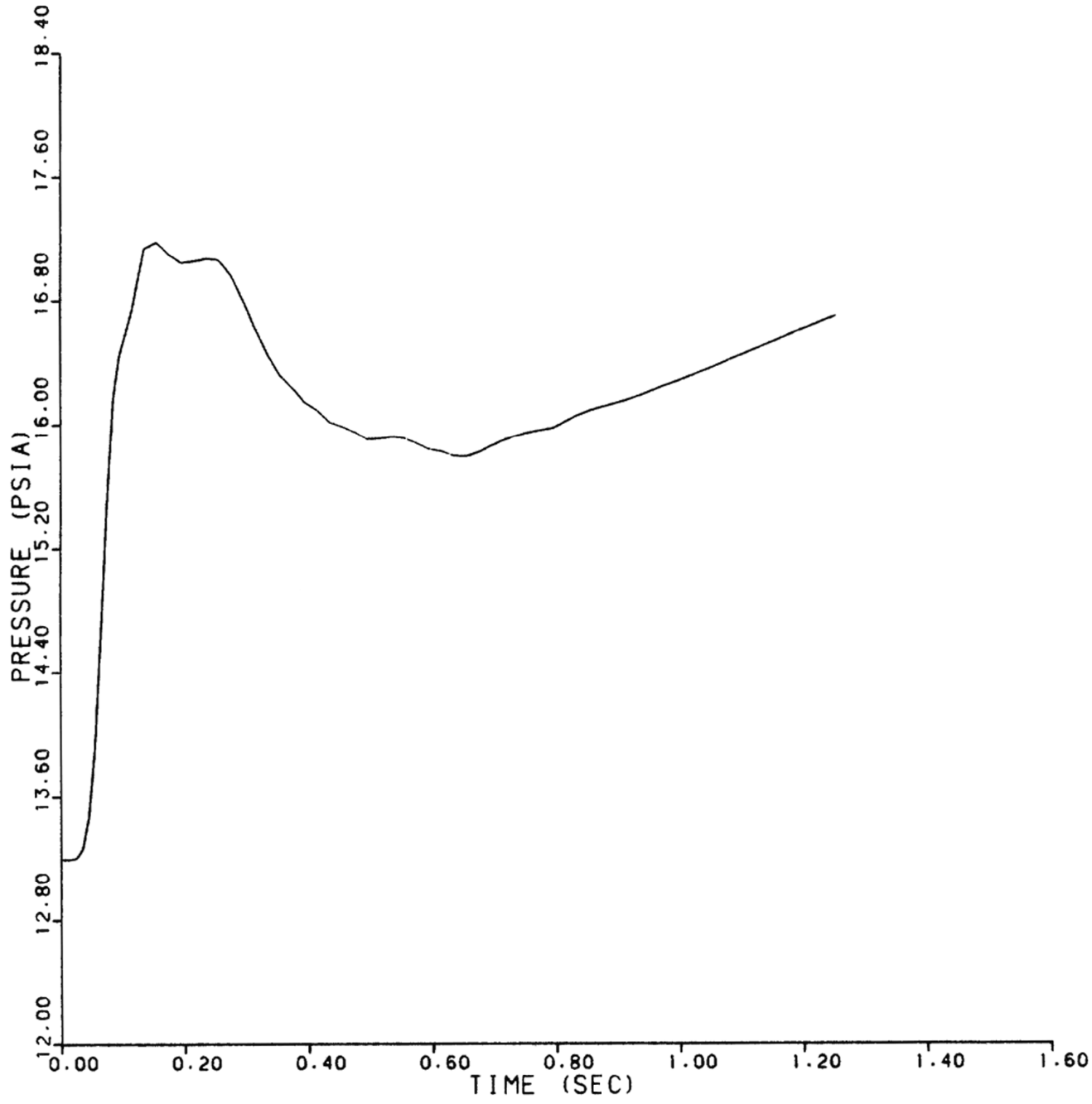
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET NOZZLE NODE – E70 (306–in.^{2} BREAK AREA)
	FIGURE 6.2.1–19 (SHEET 70 OF 74)



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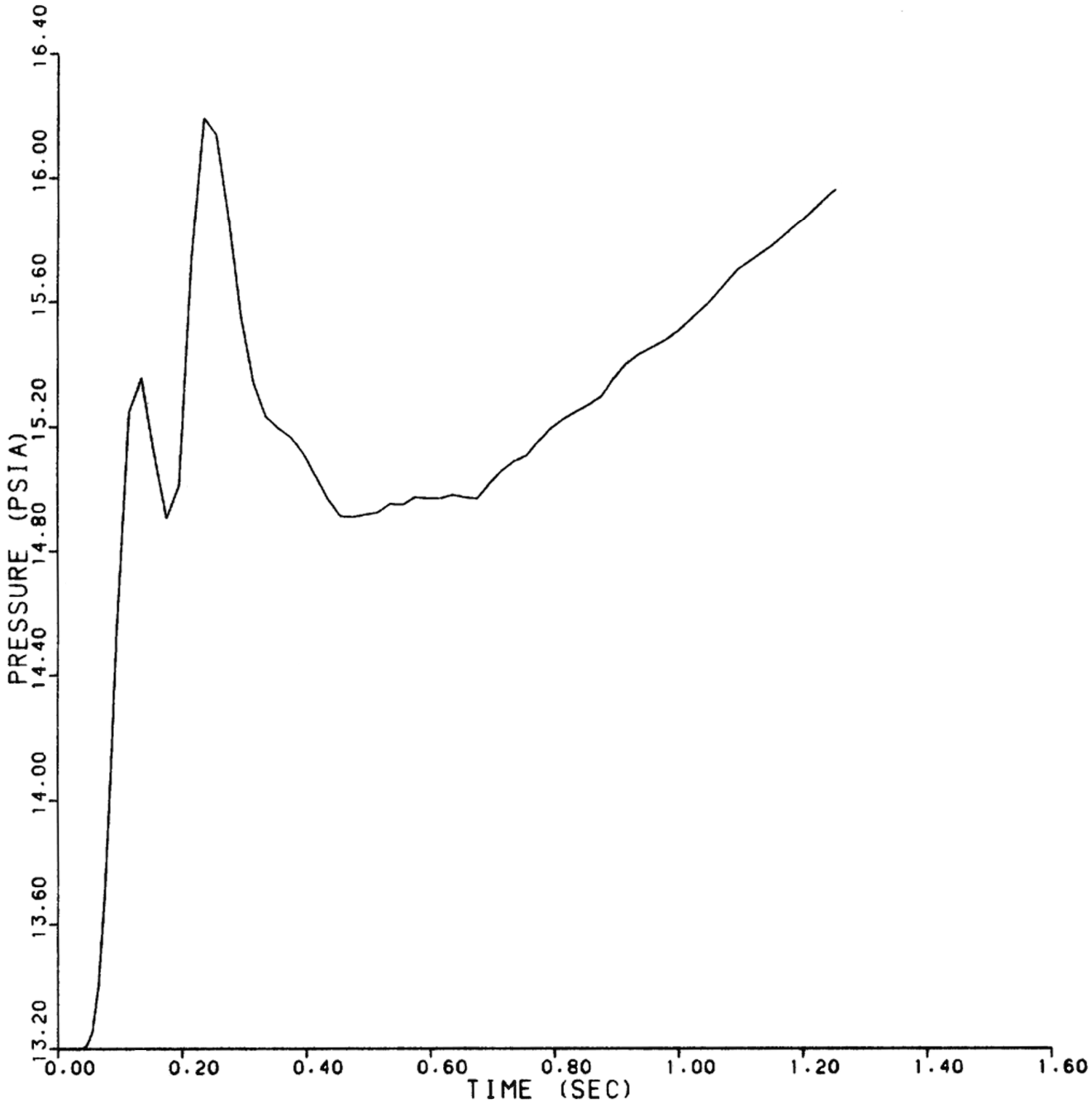
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E71
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 71 OF 74)



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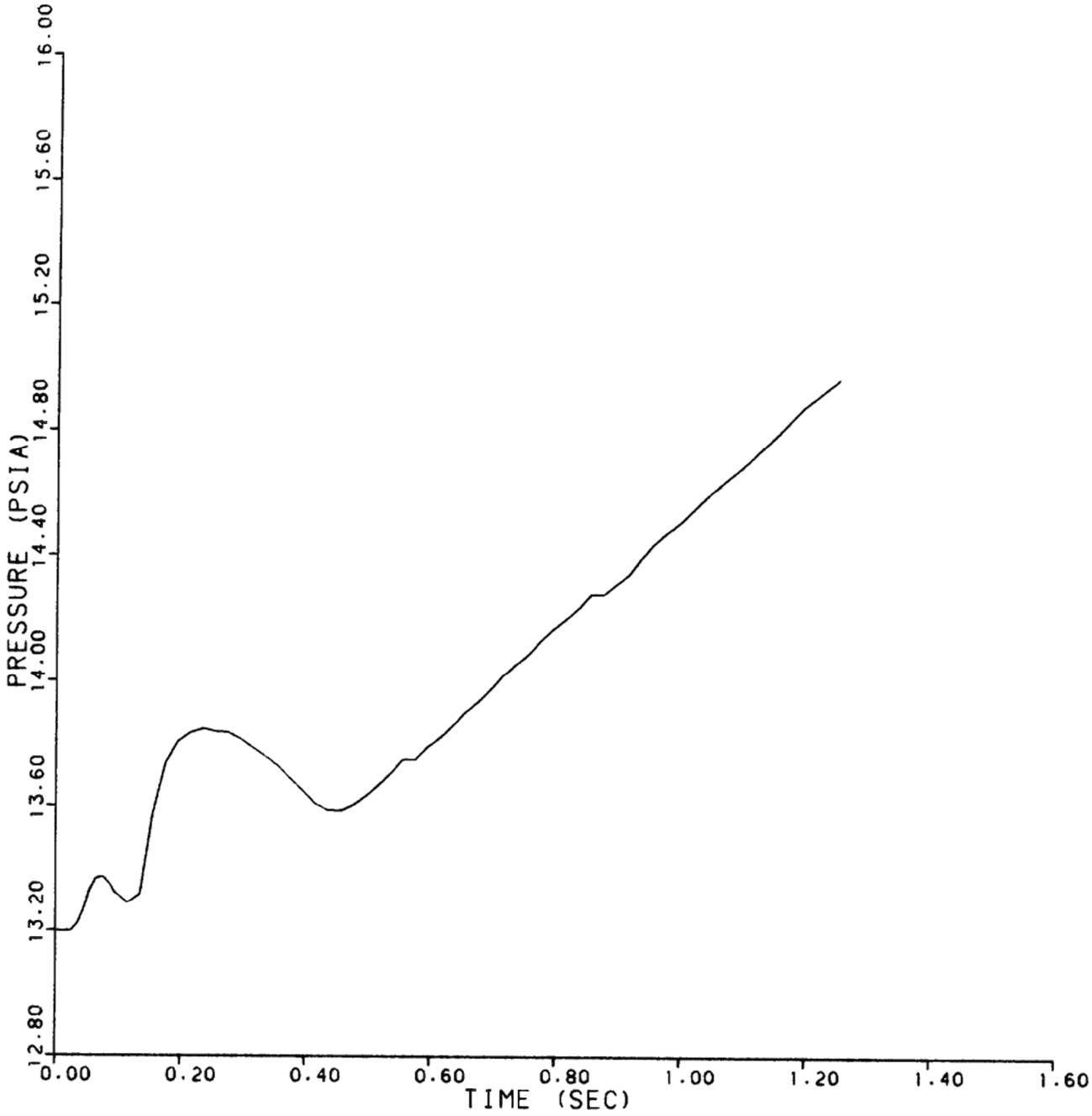
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E72
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 72 OF 74)



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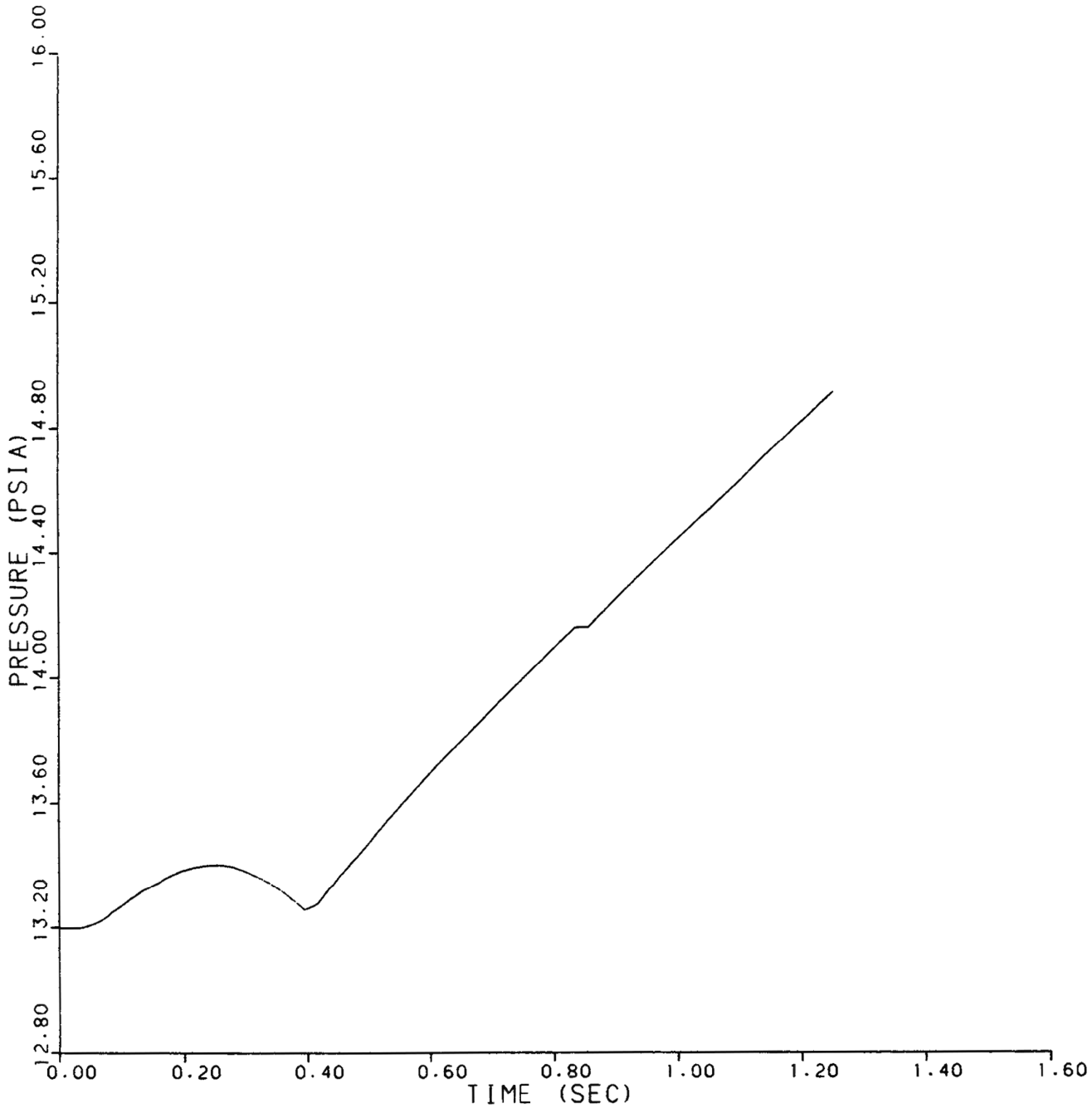
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E73
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 73 OF 74)



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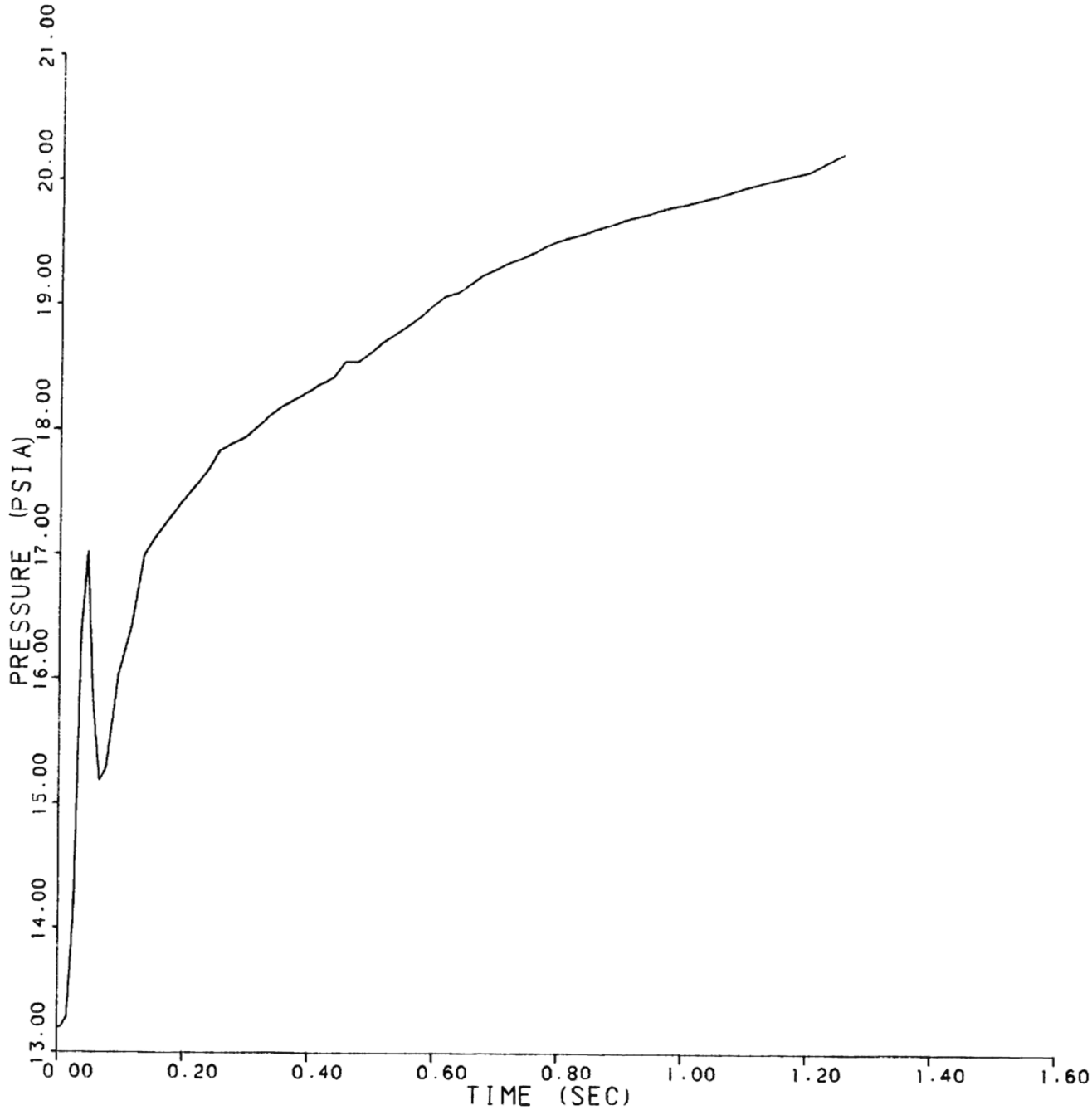
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET NOZZLE
NODE – E74
(306–in.^{2} BREAK AREA)

FIGURE 6.2.1–19 (SHEET 74 OF 74)



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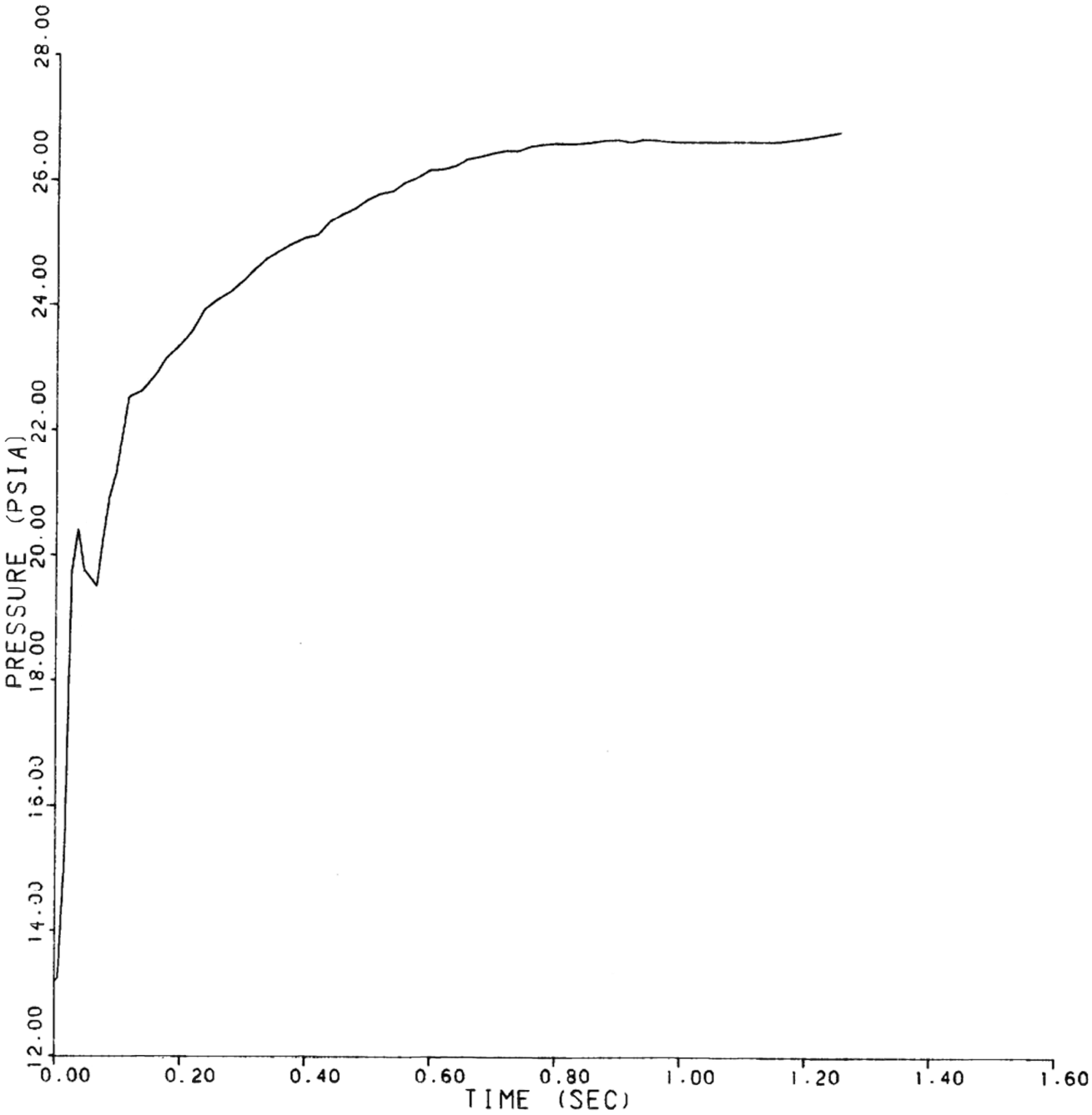
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E1
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 1 OF 74)



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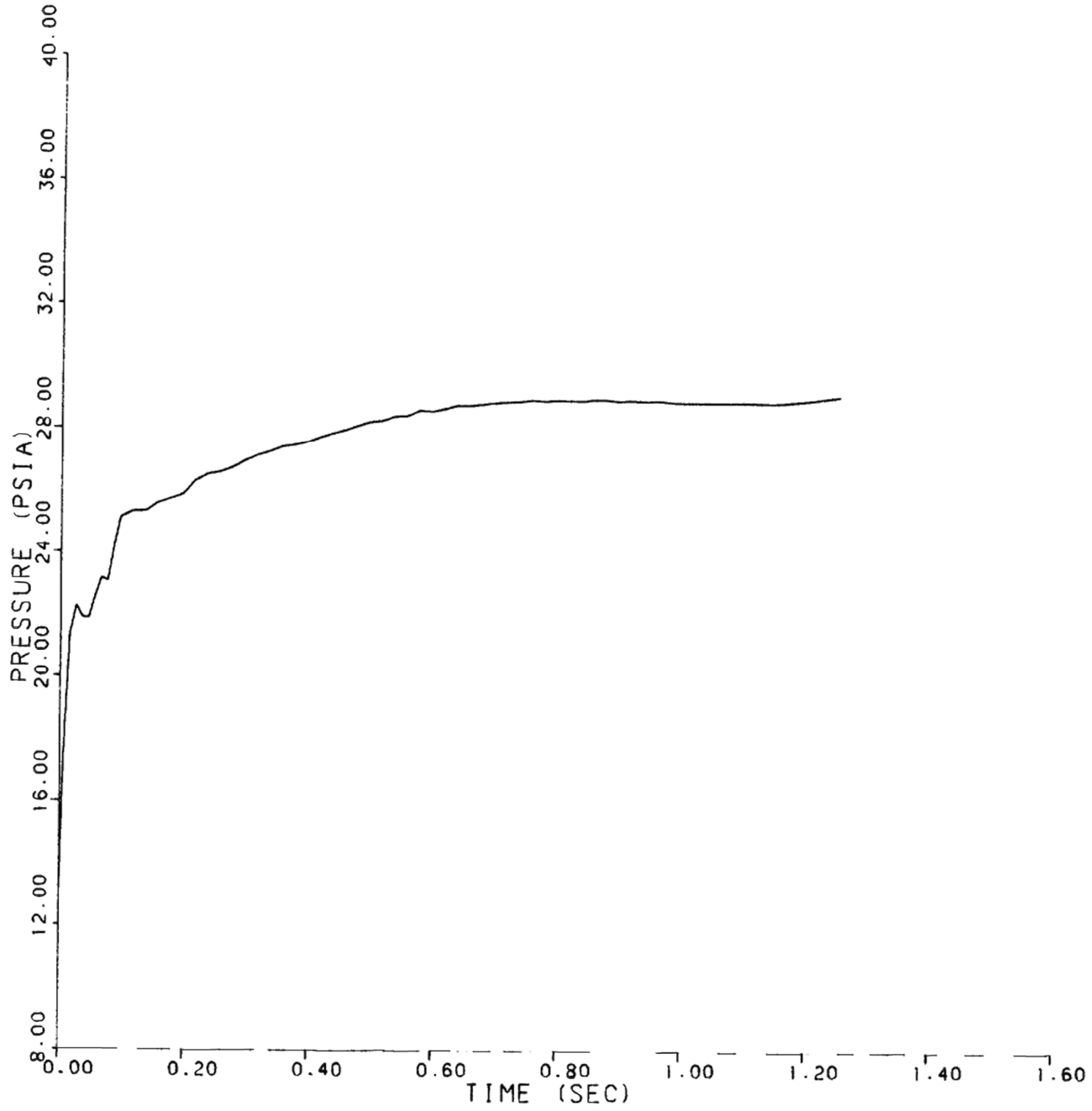
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E2 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 2 OF 74)



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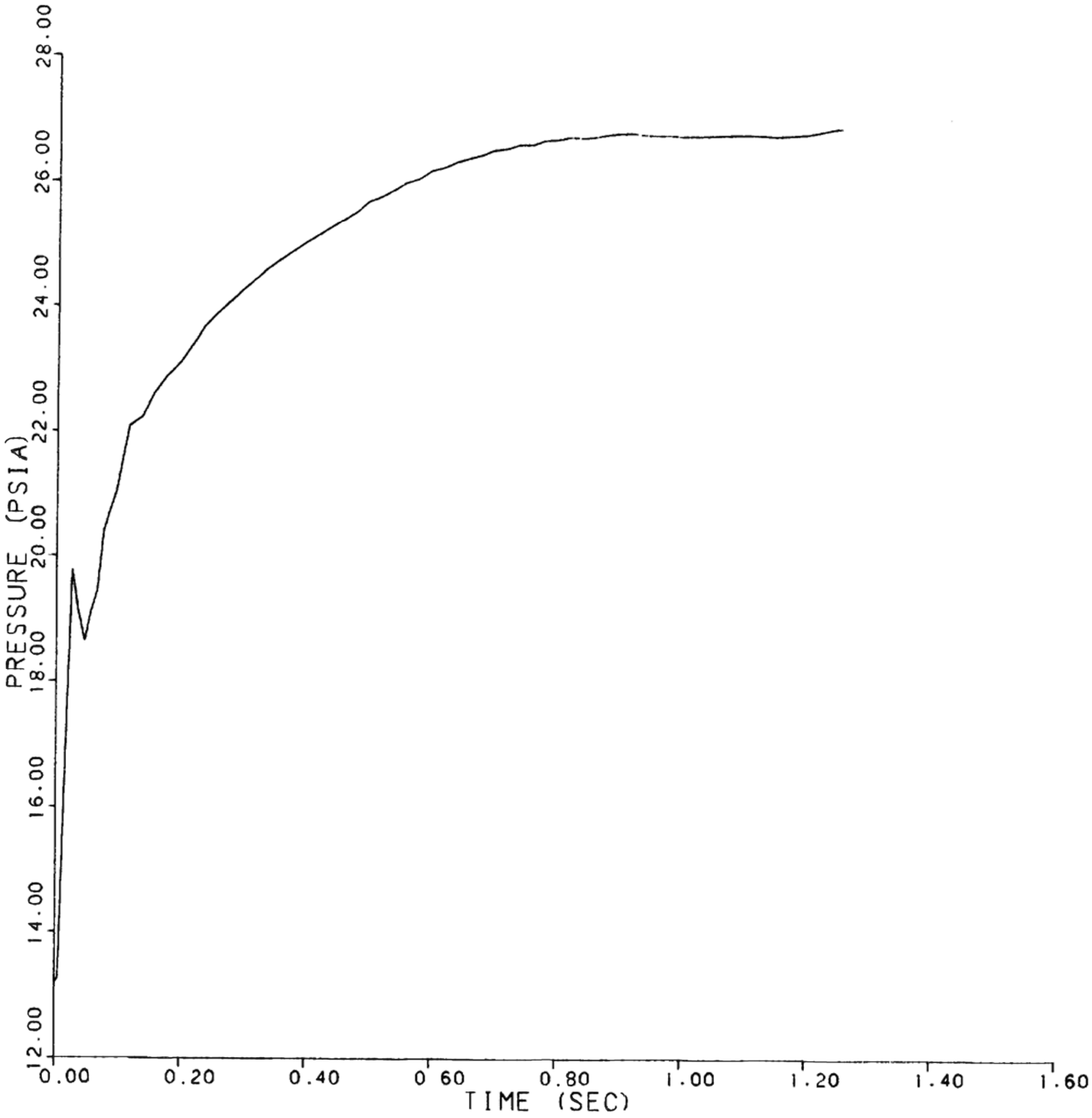
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E3 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 3 OF 74)



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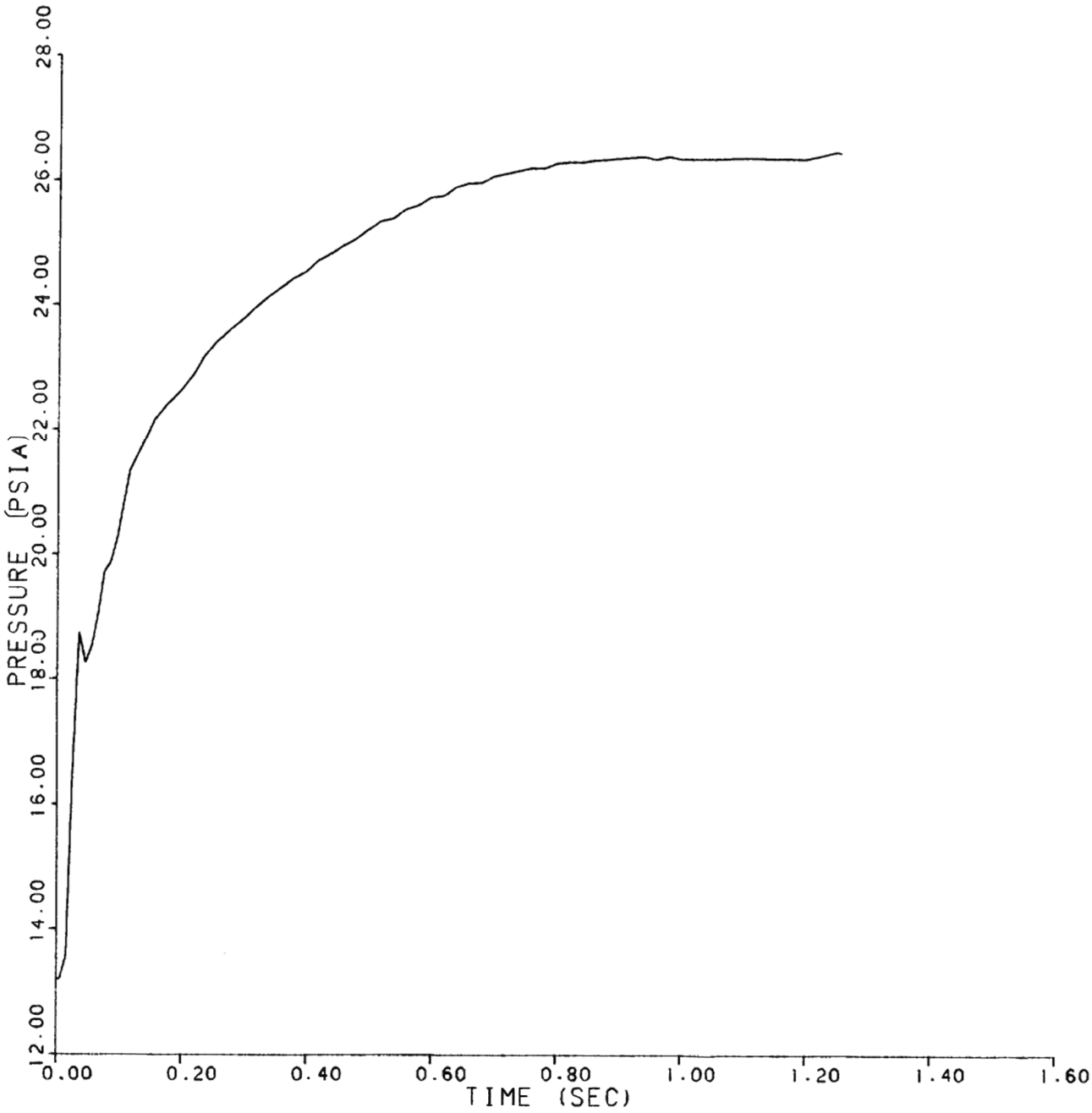
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E4
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 4 OF 74)



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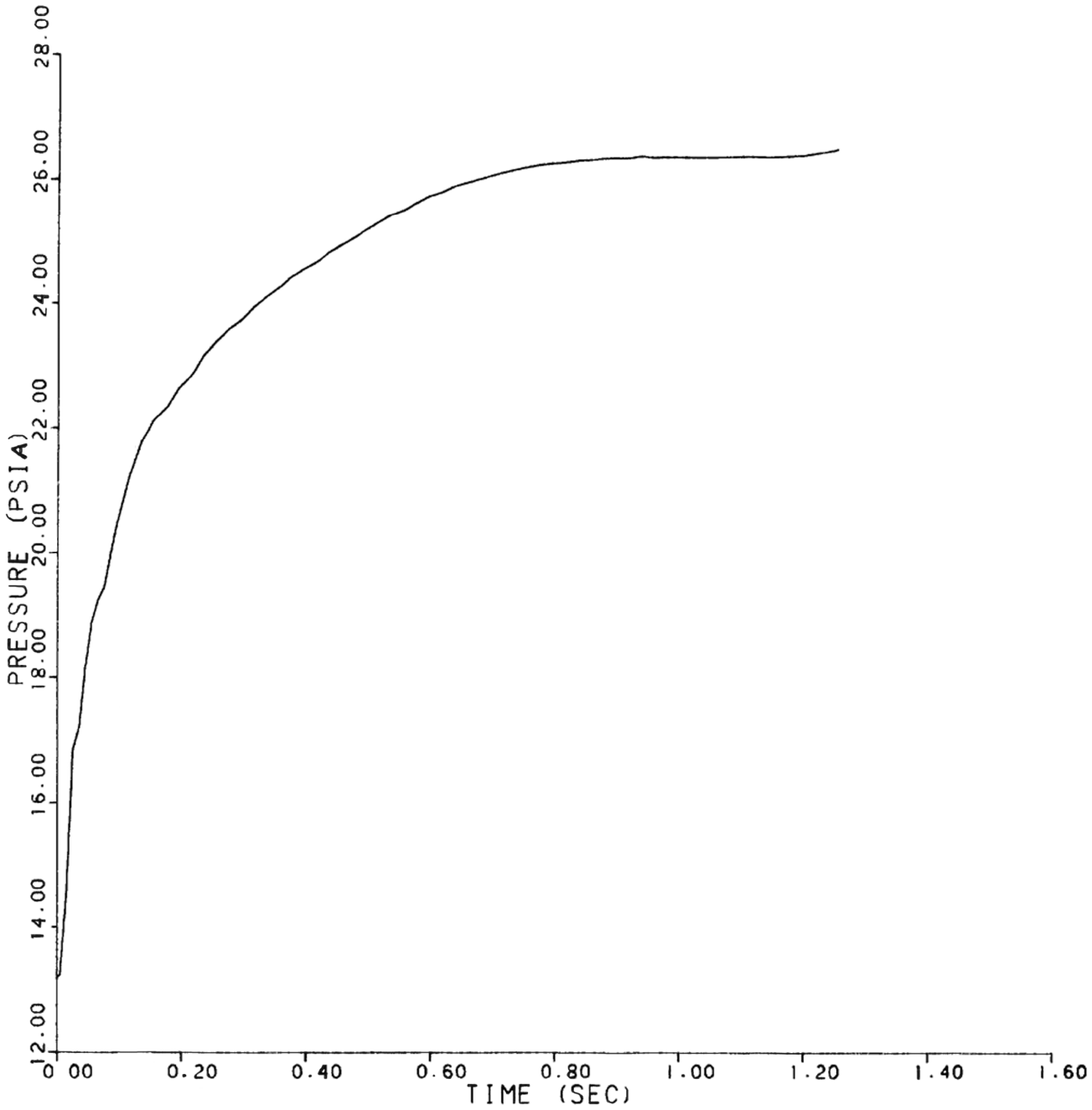
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E5 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 5 OF 74)



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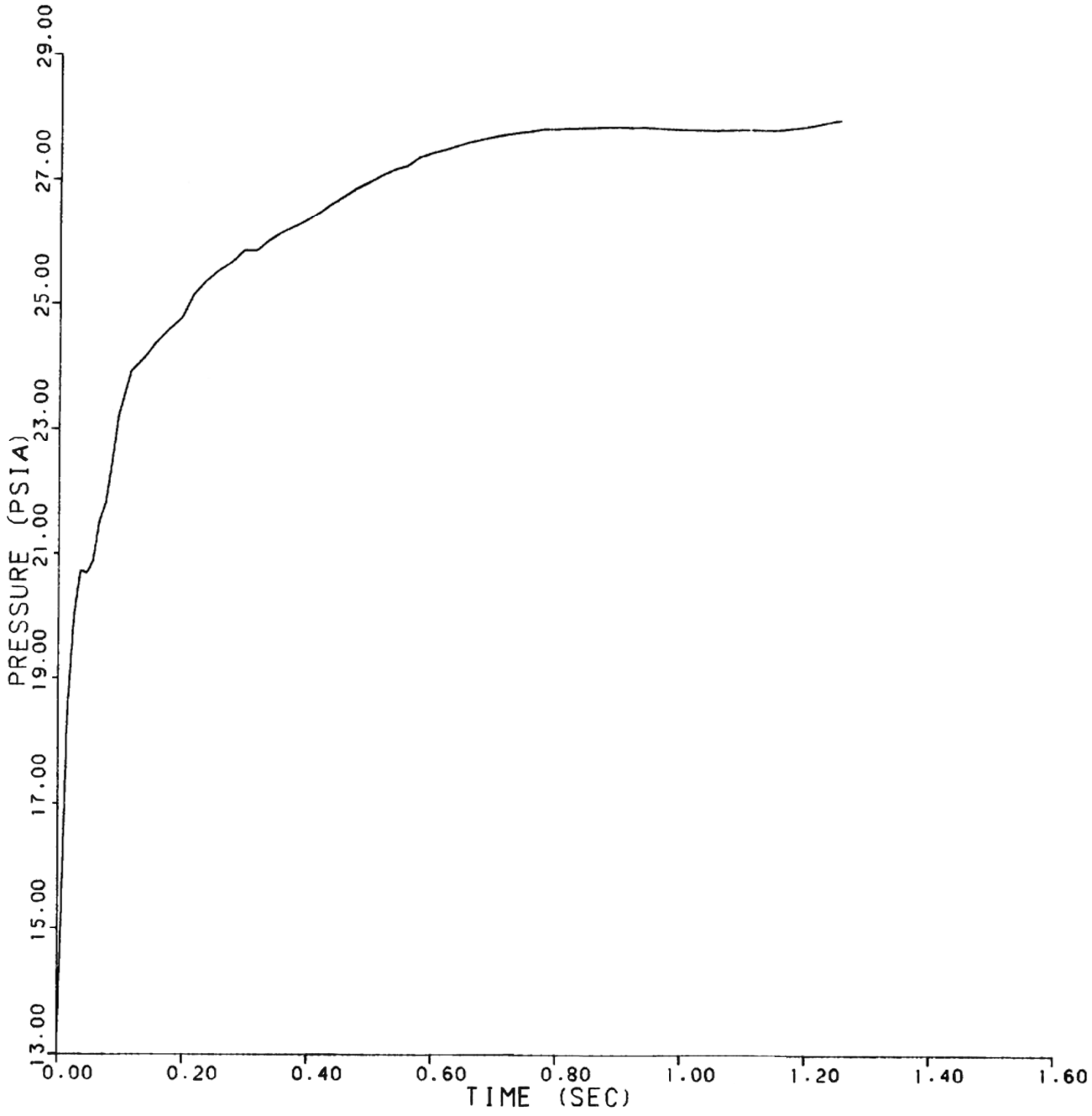
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E6 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 6 OF 74)



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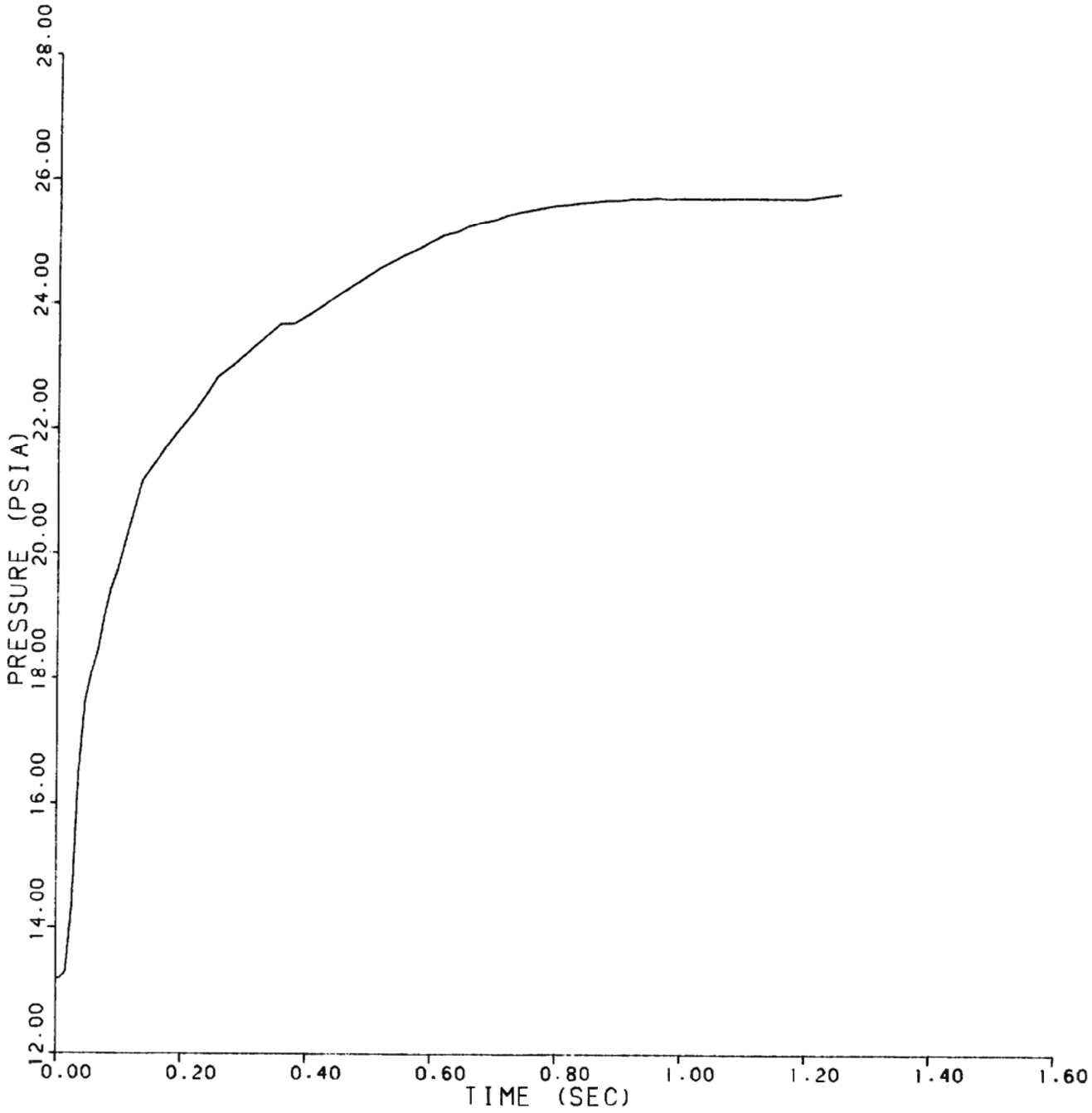
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E7
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 7 OF 74)



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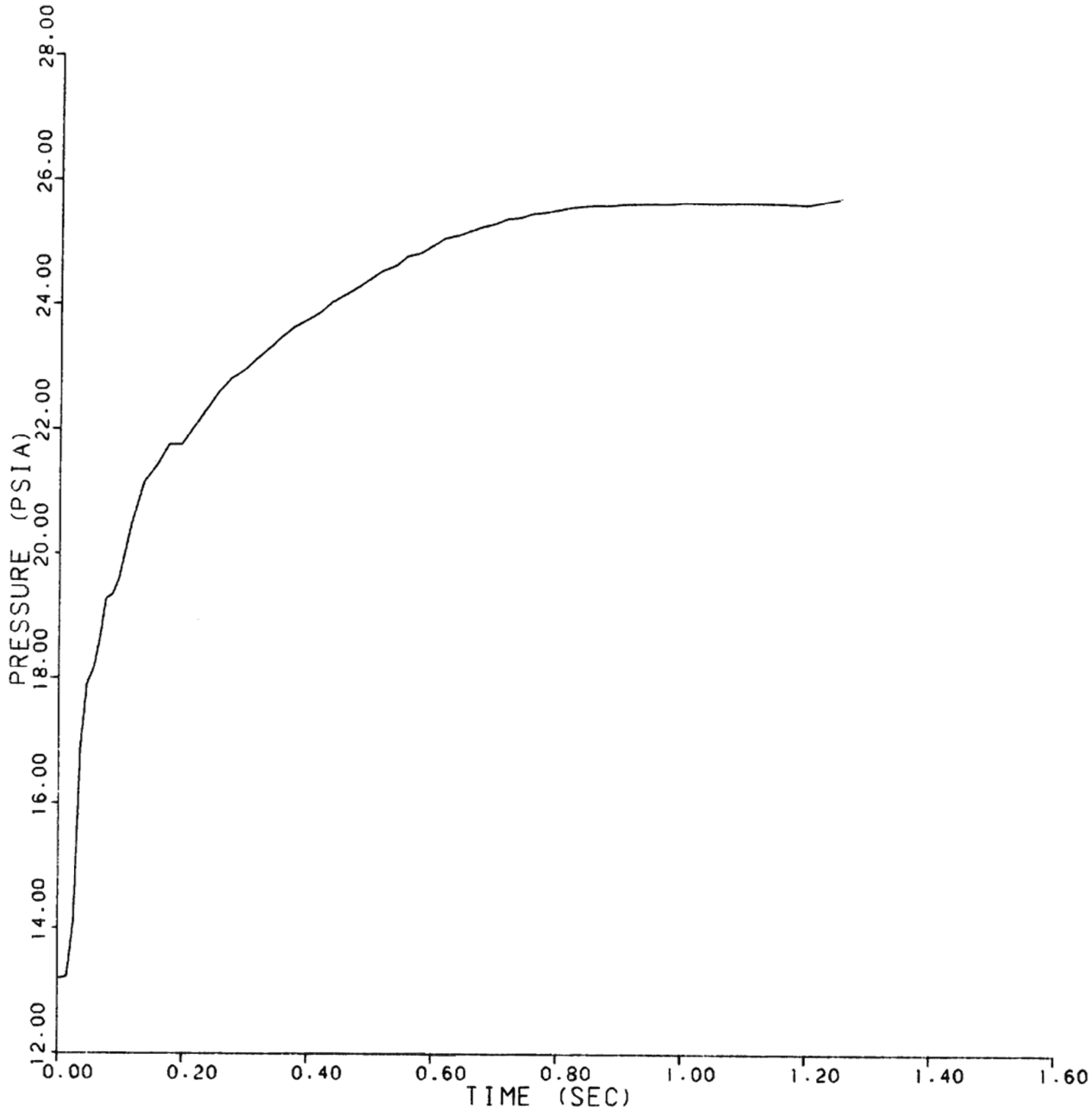
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E8
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 8 OF 74)



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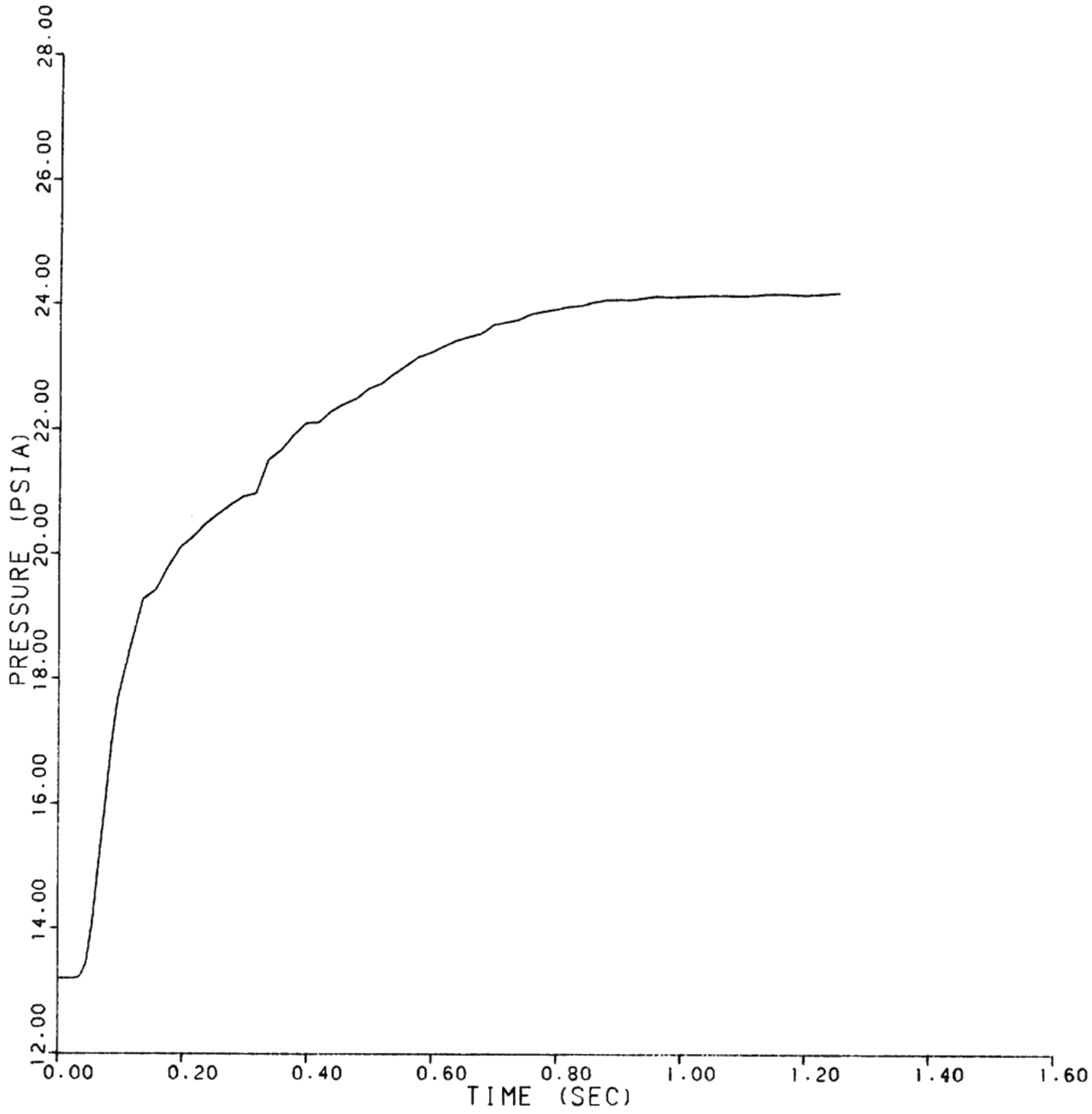
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E9
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 9 OF 74)



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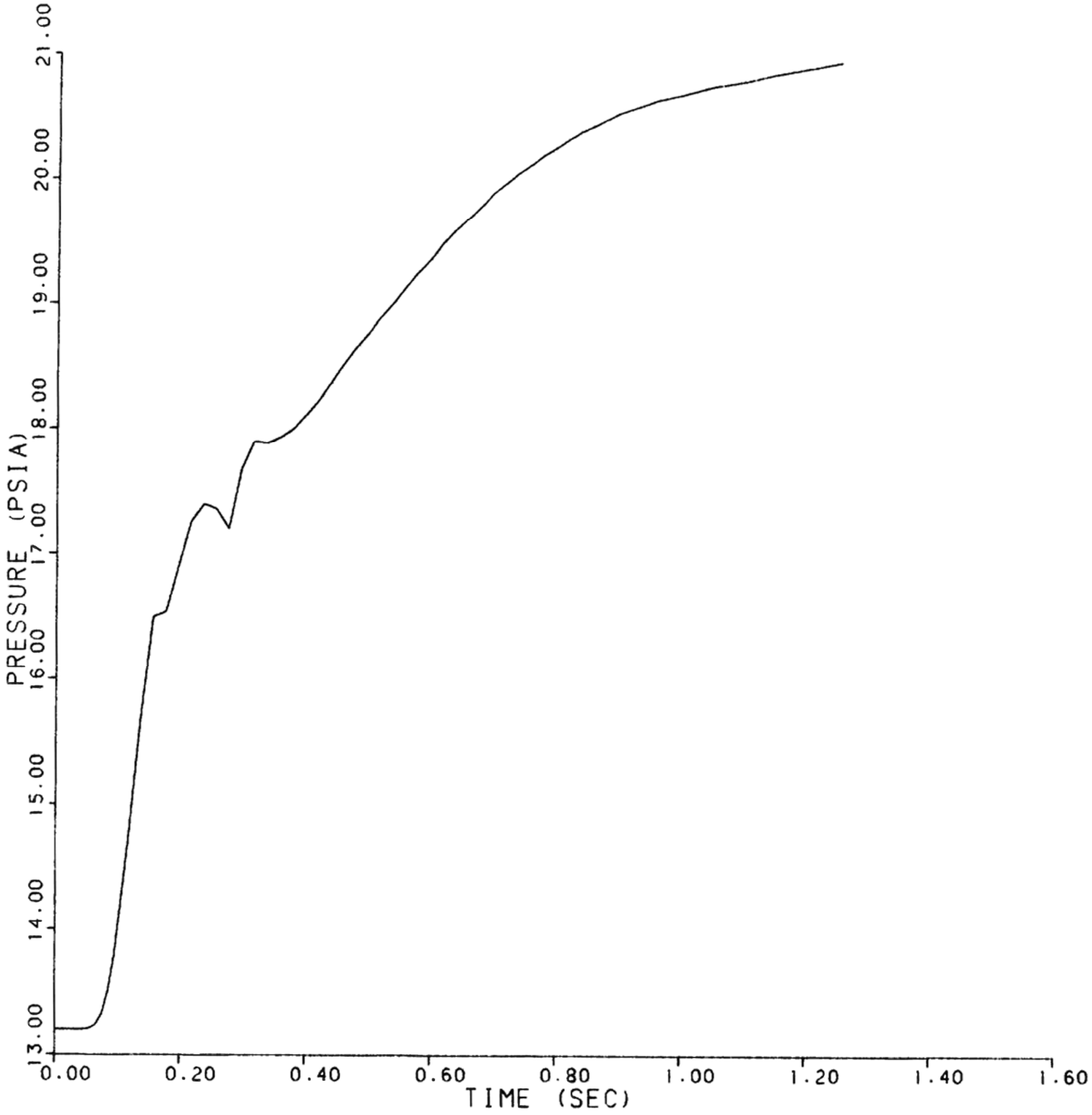
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E10
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 10 OF 74)



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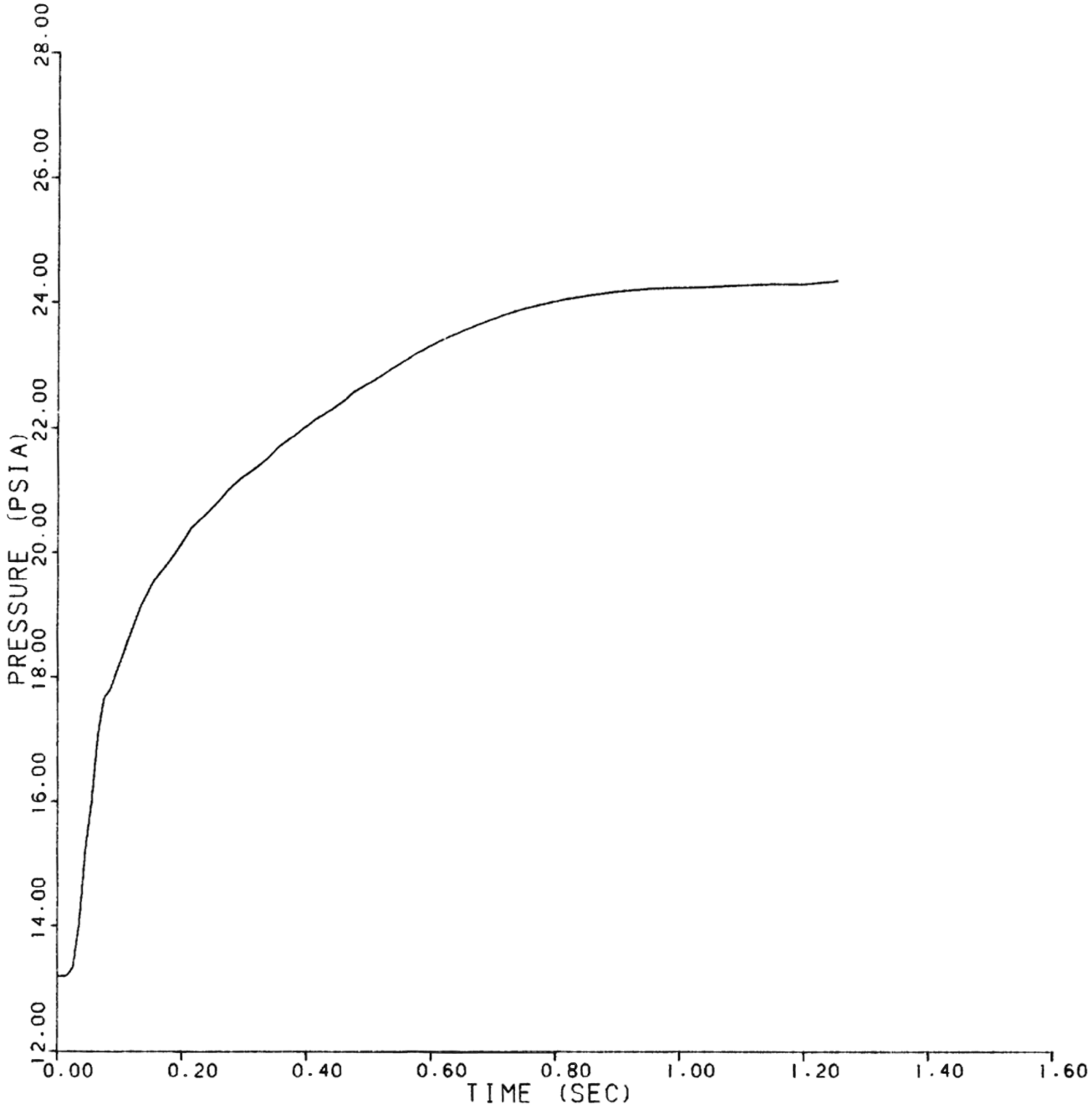
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E11
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 11 OF 74)



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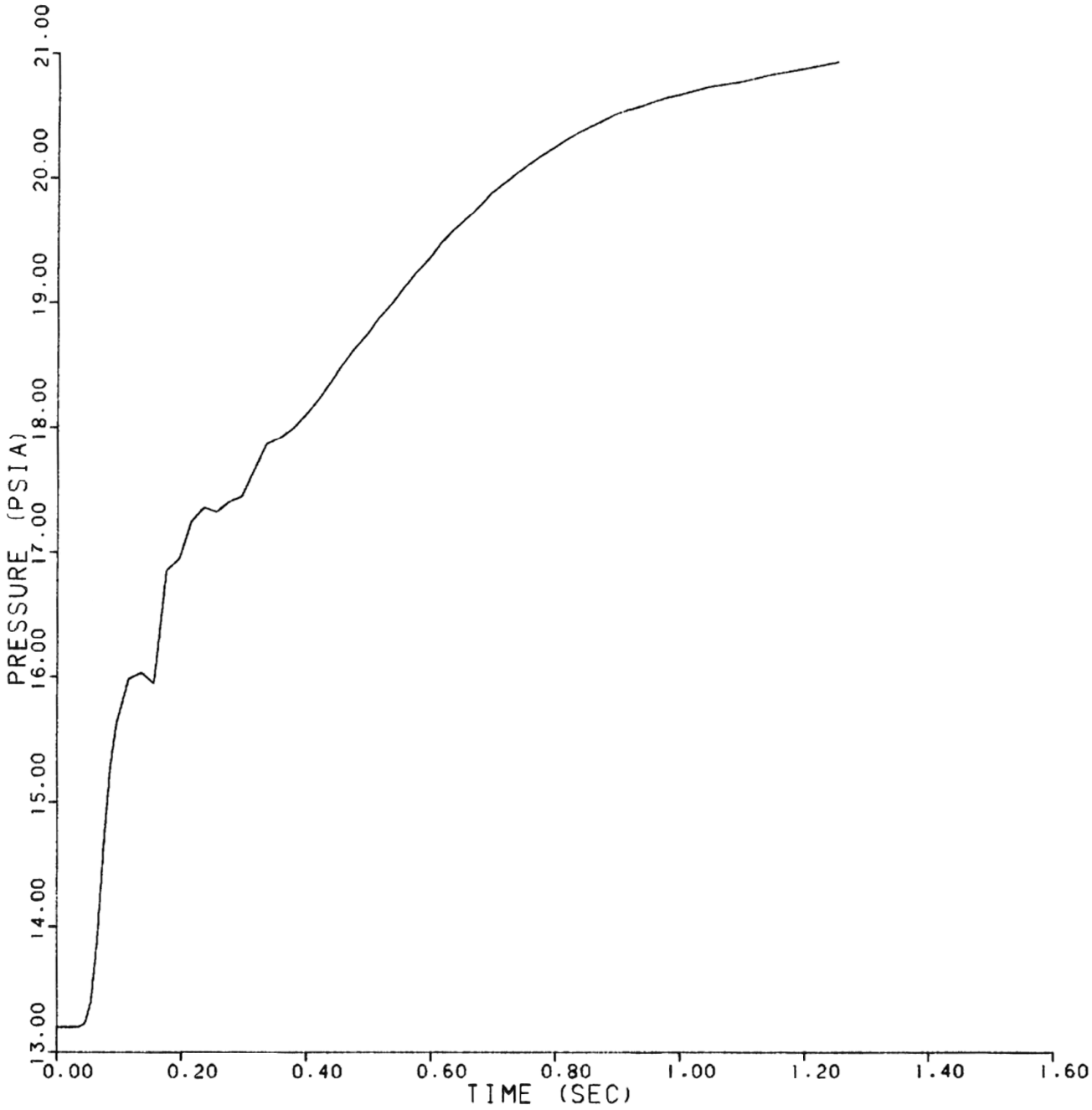
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E12 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 12 OF 74)



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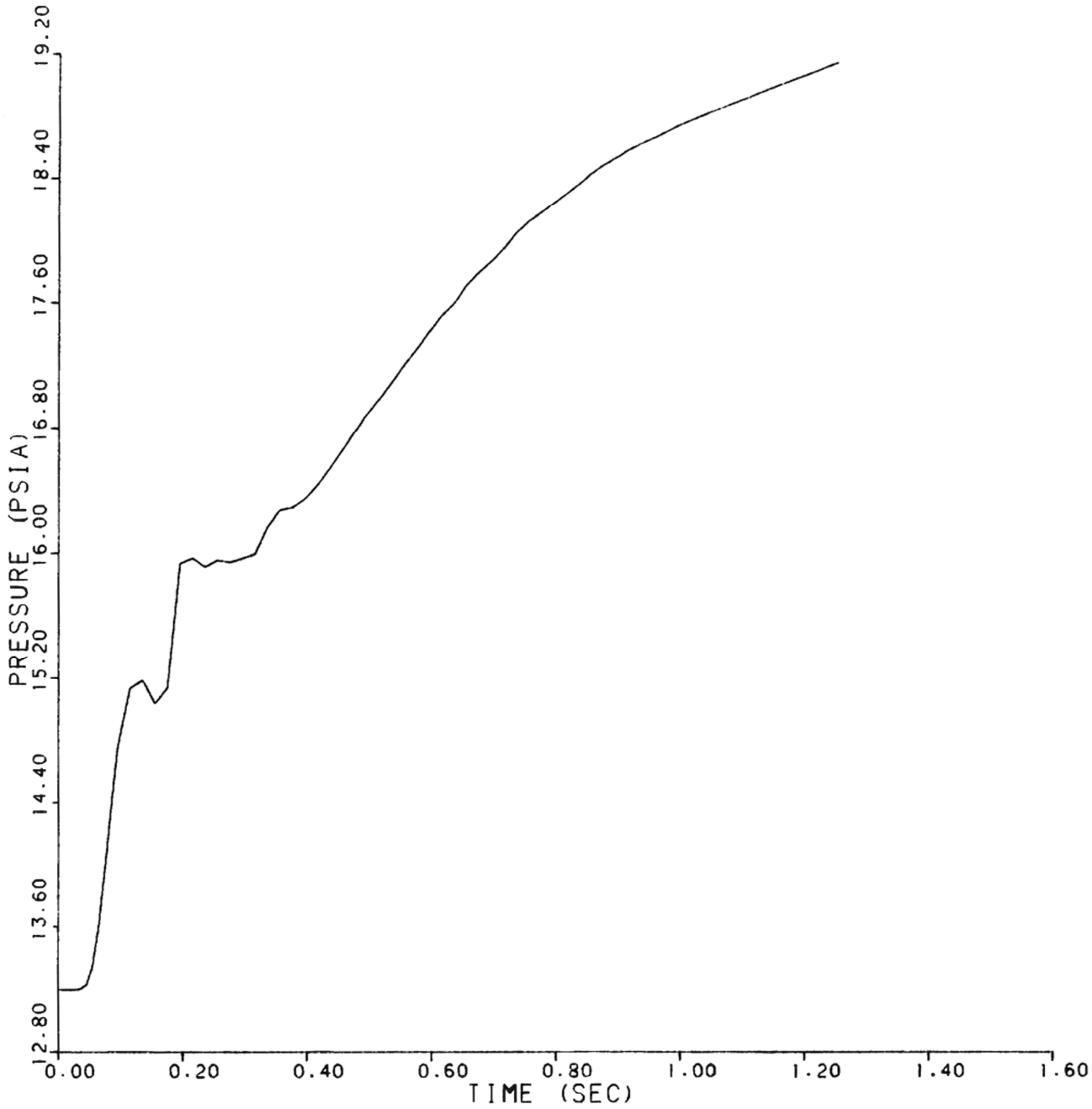
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E13
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 13 OF 74)



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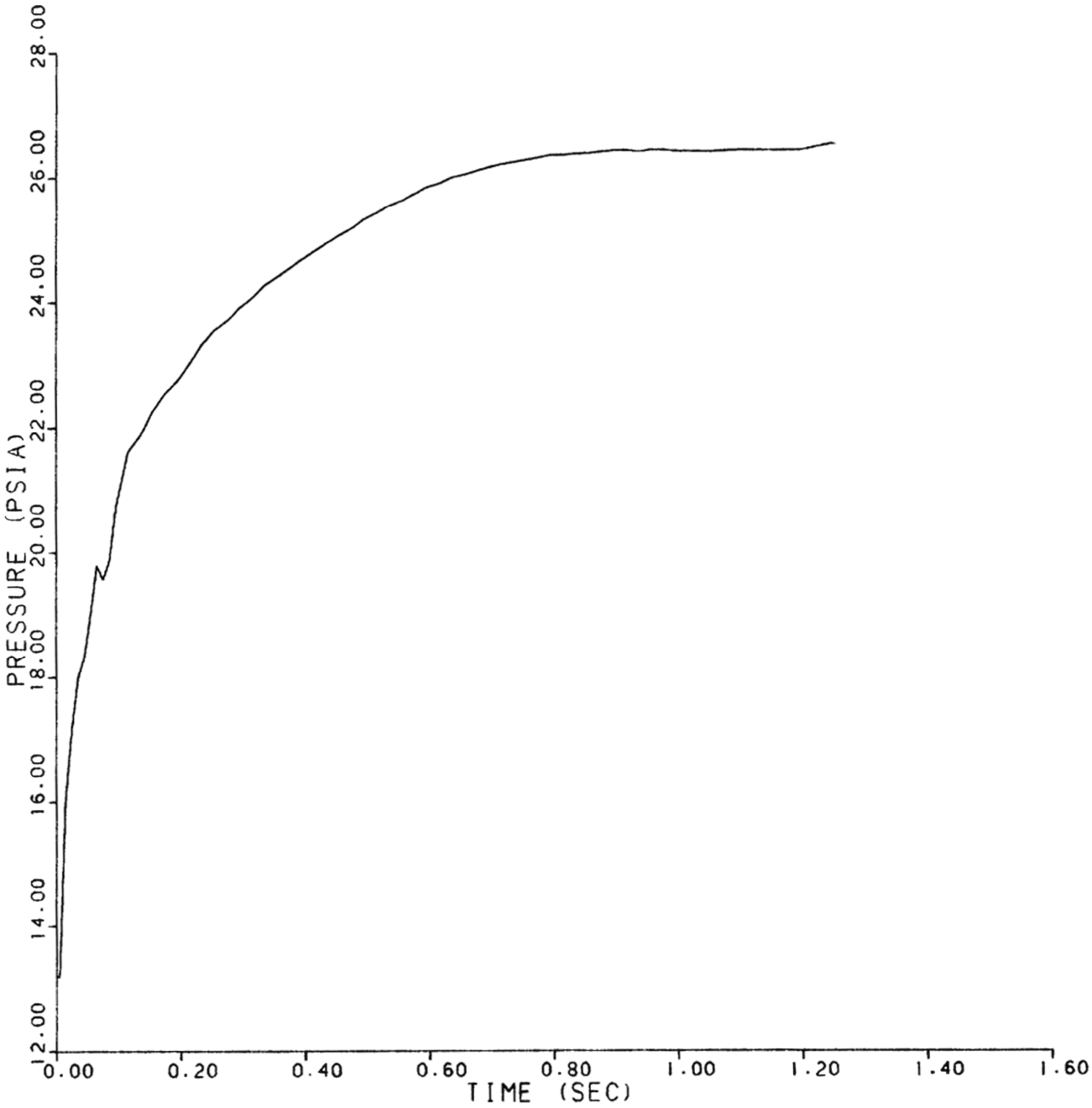
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E14
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 14 OF 74)



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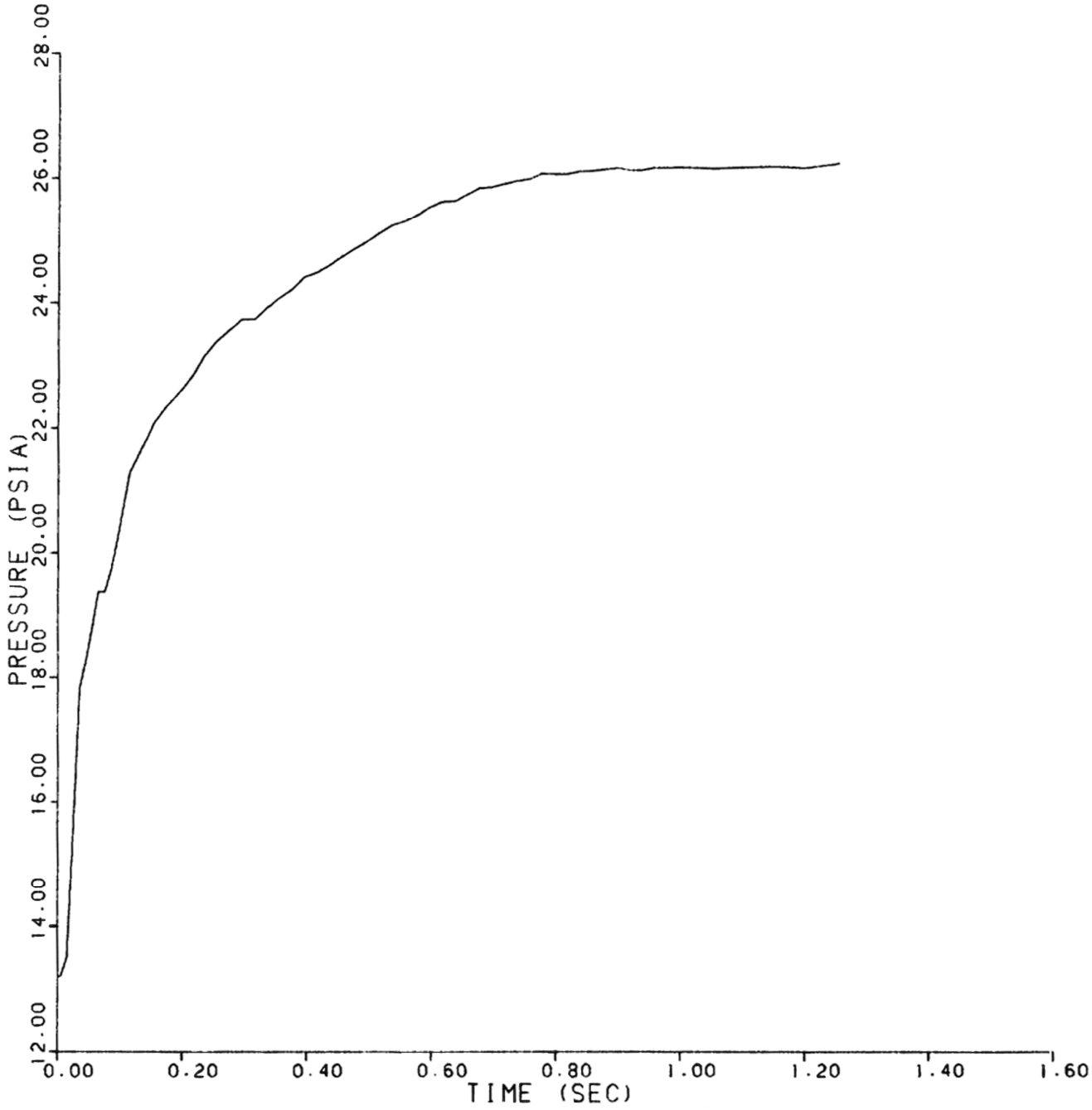
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E15
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 15 OF 74)



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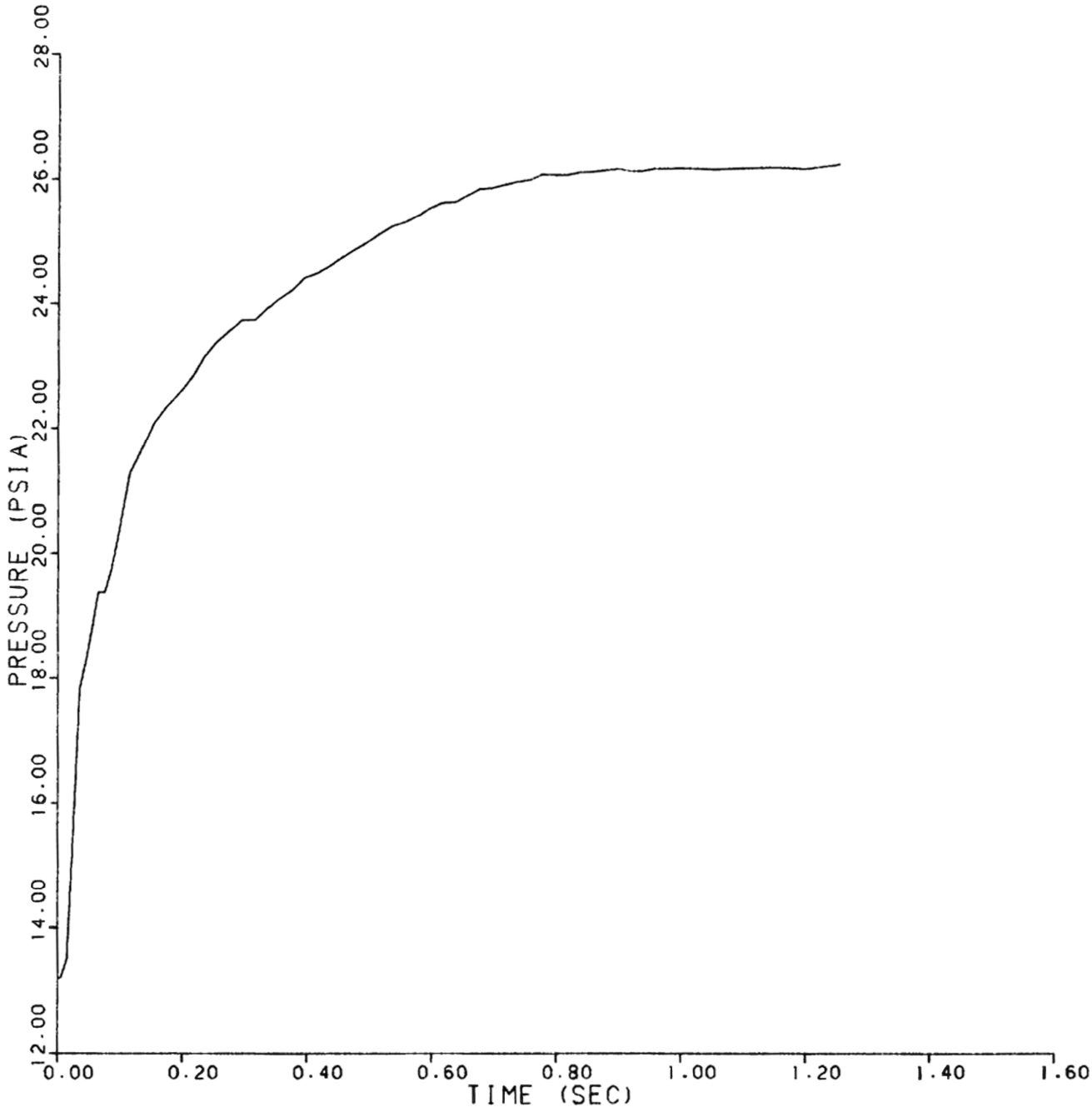
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E16 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 16 OF 74)



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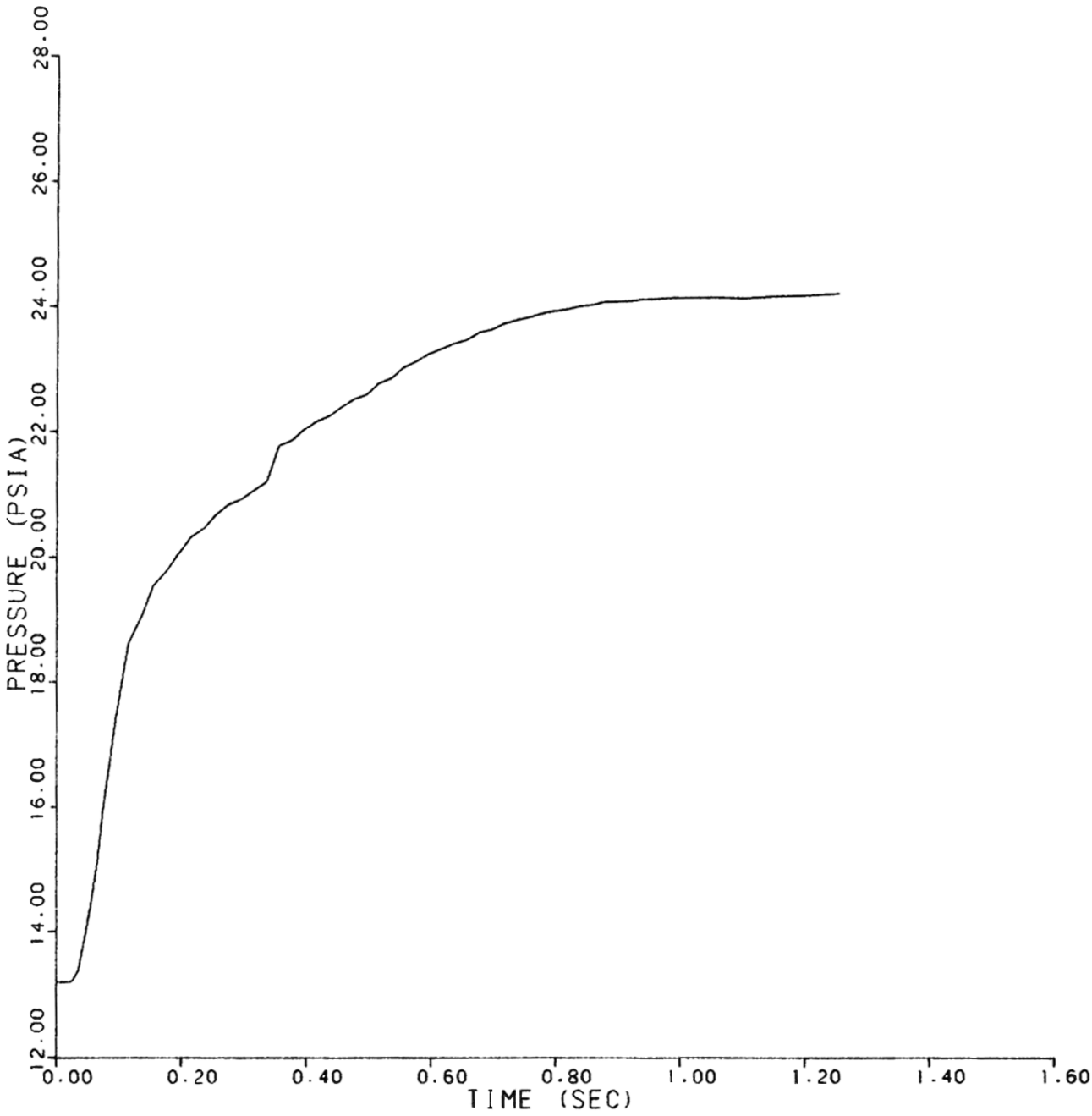
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E17
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 17 OF 74)



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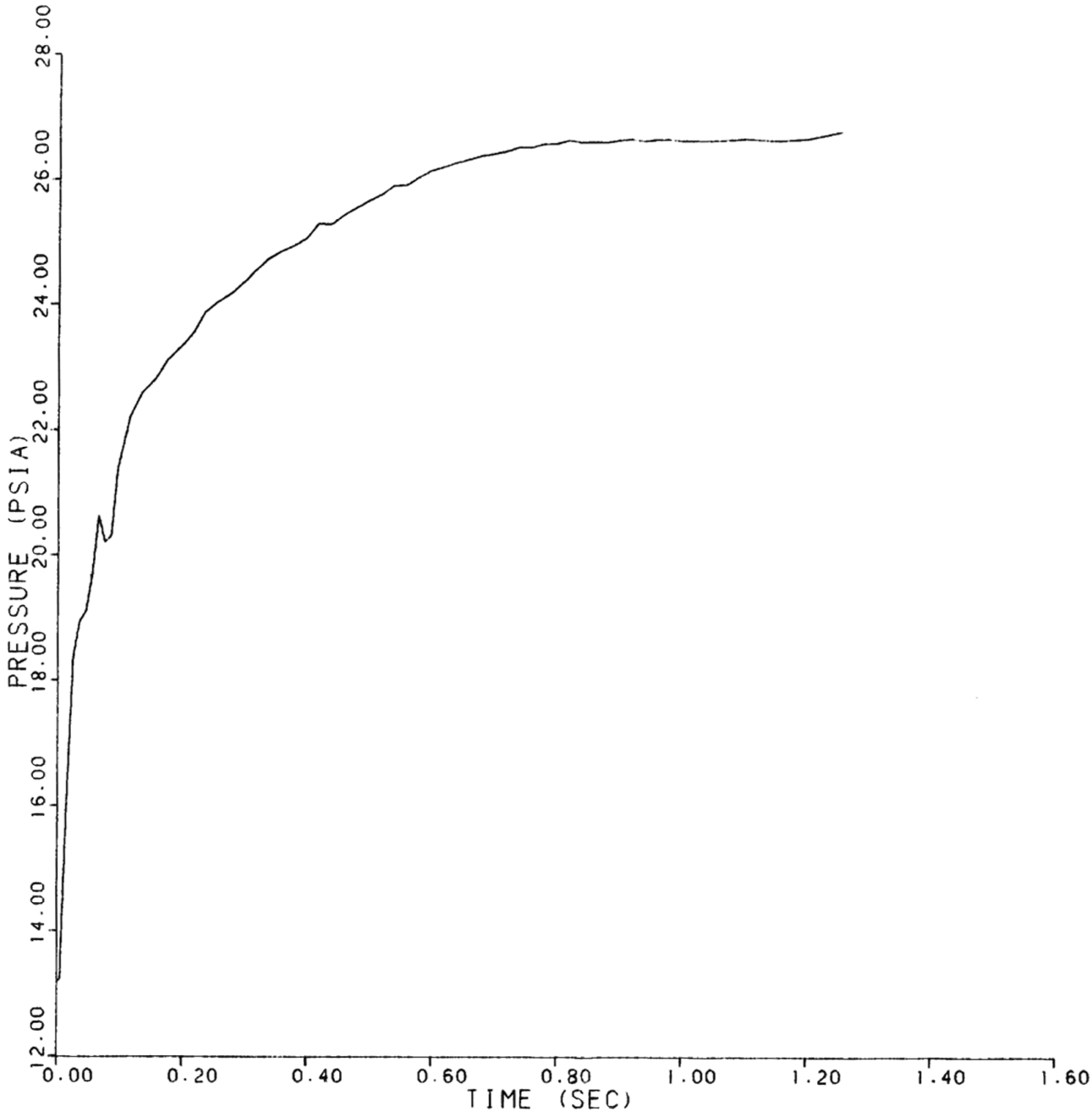
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E18
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 18 OF 74)



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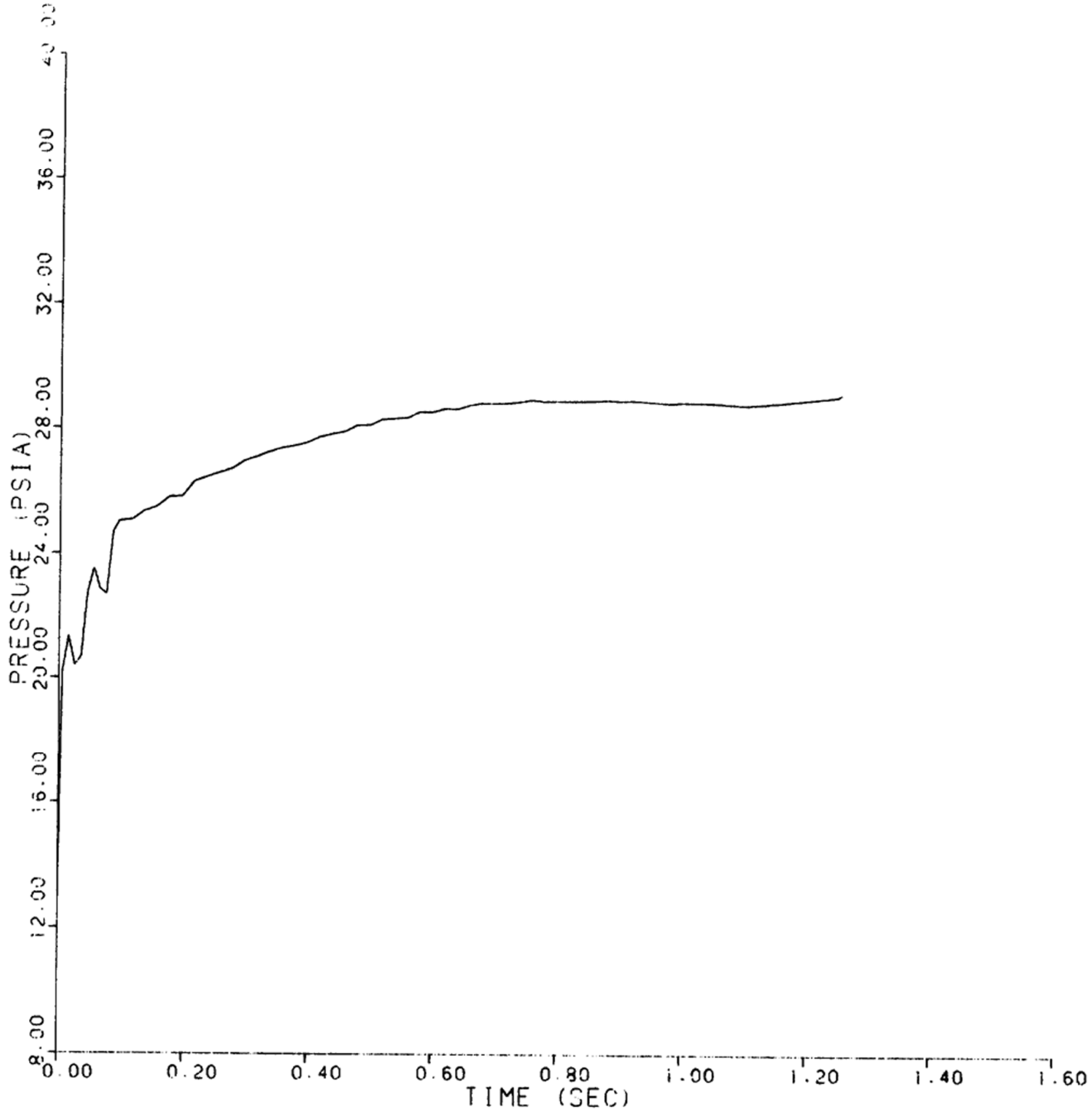
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E19
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 19 OF 74)



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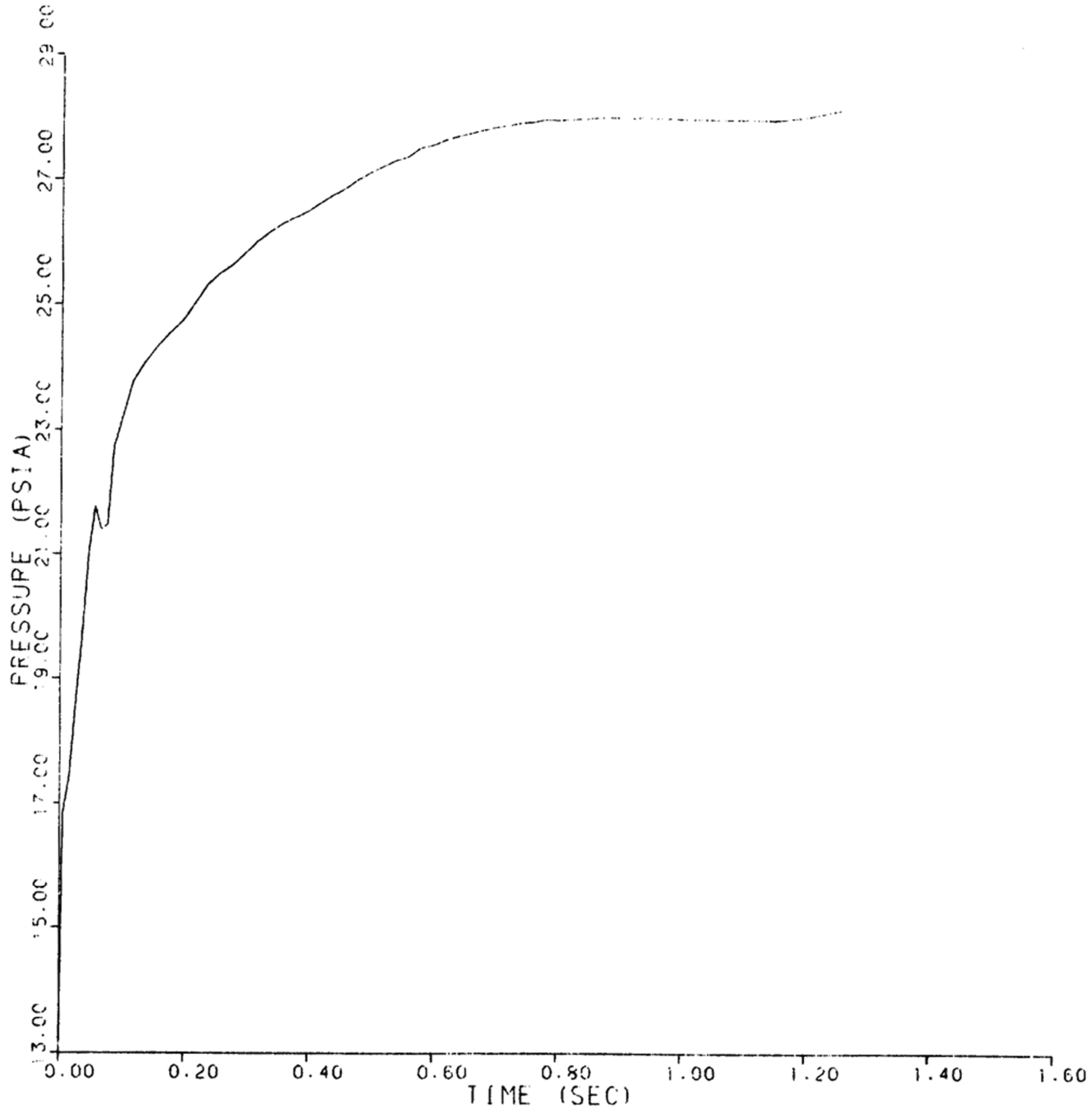
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E20
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 20 OF 74)



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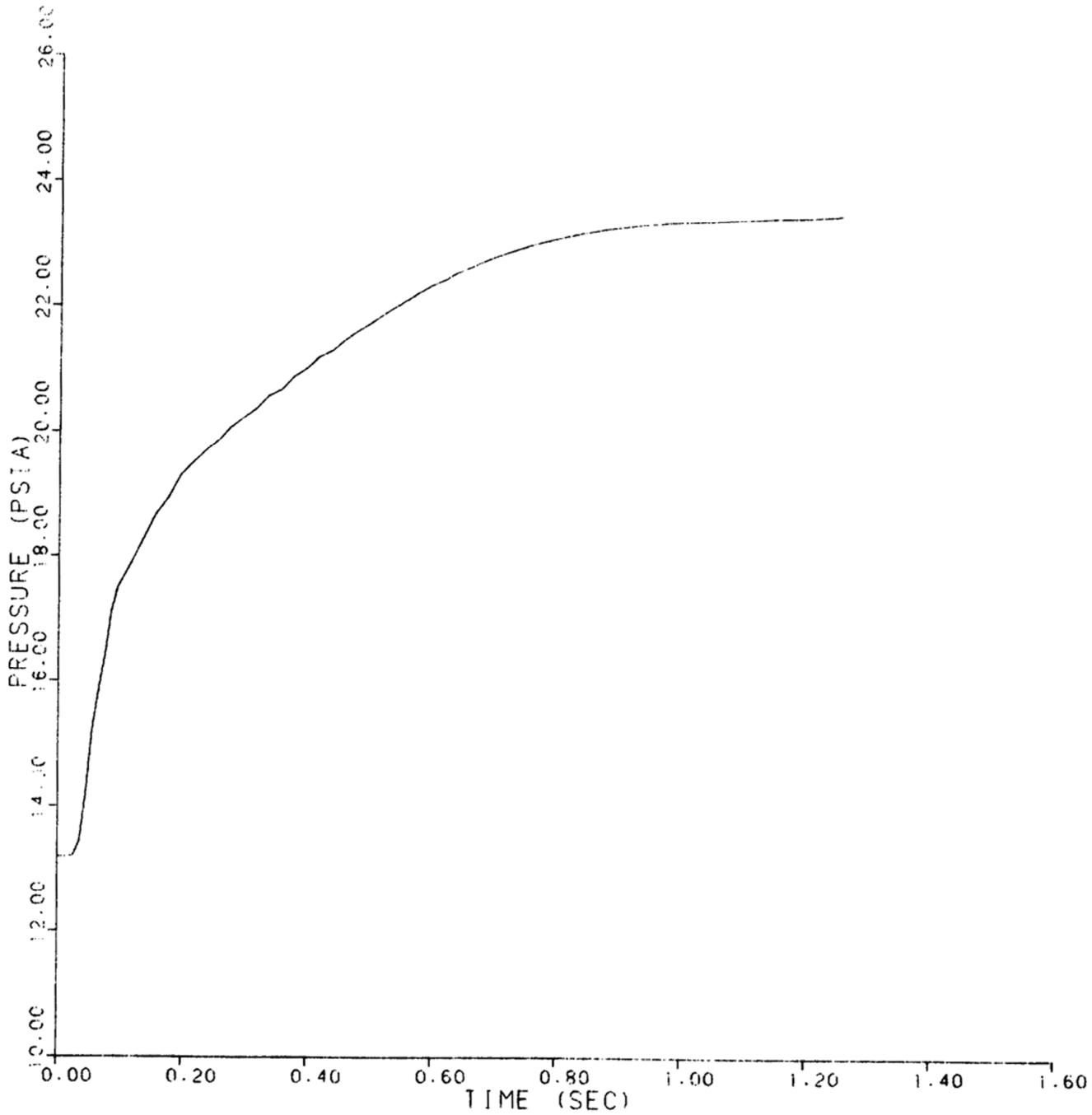
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E21
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 21 OF 74)



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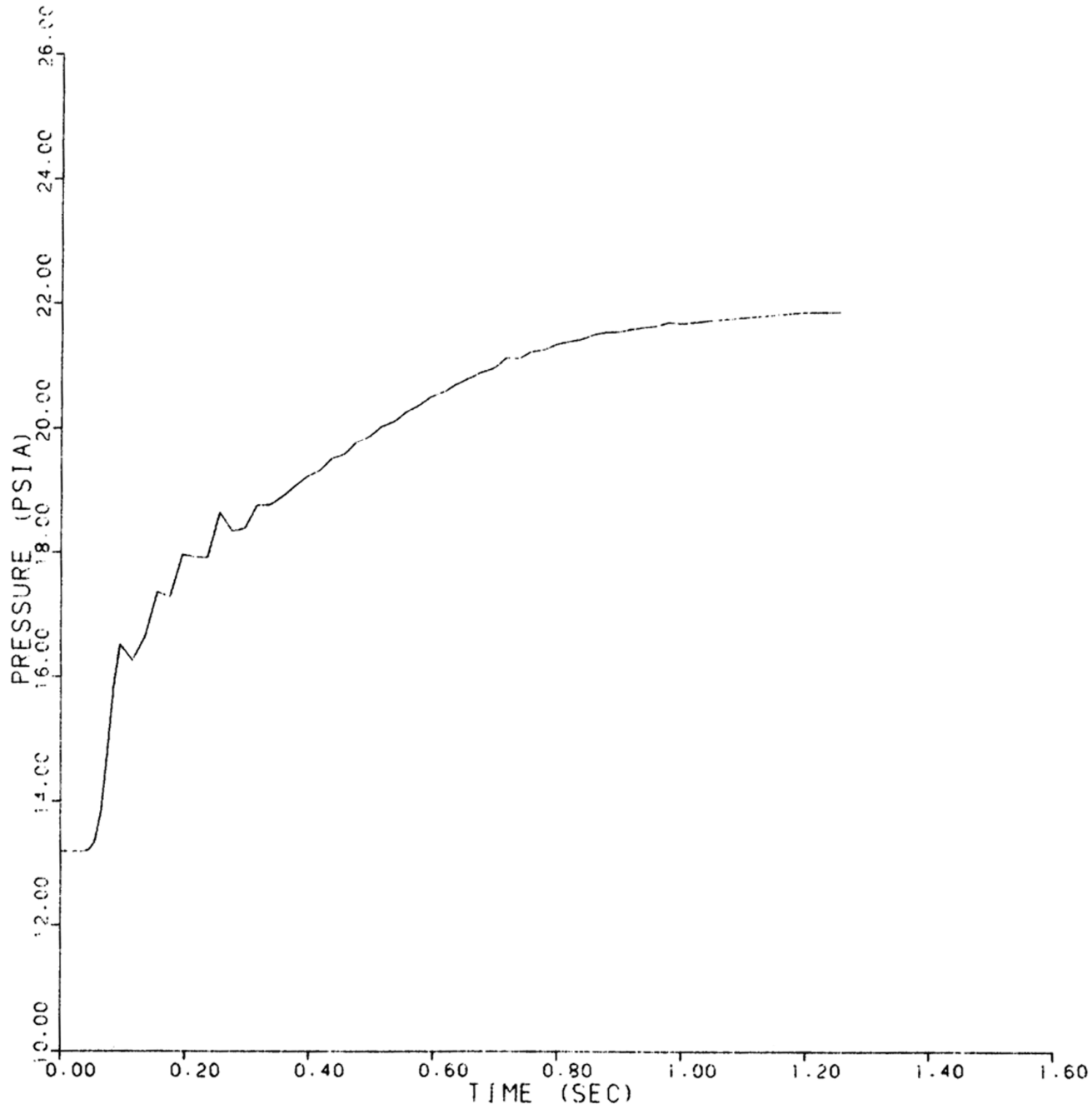
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E22
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 22 OF 74)



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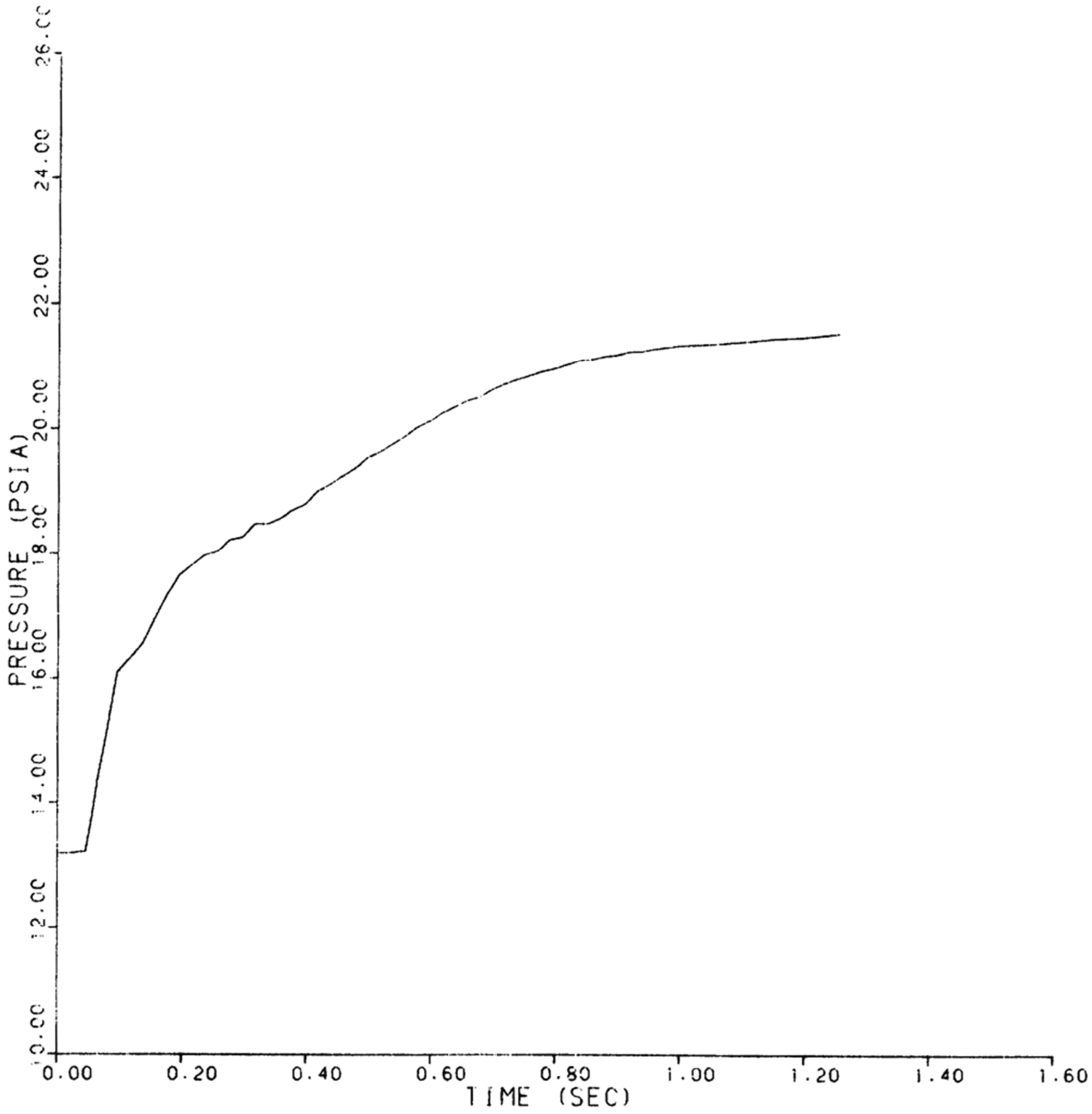
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E23
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 23 OF 74)



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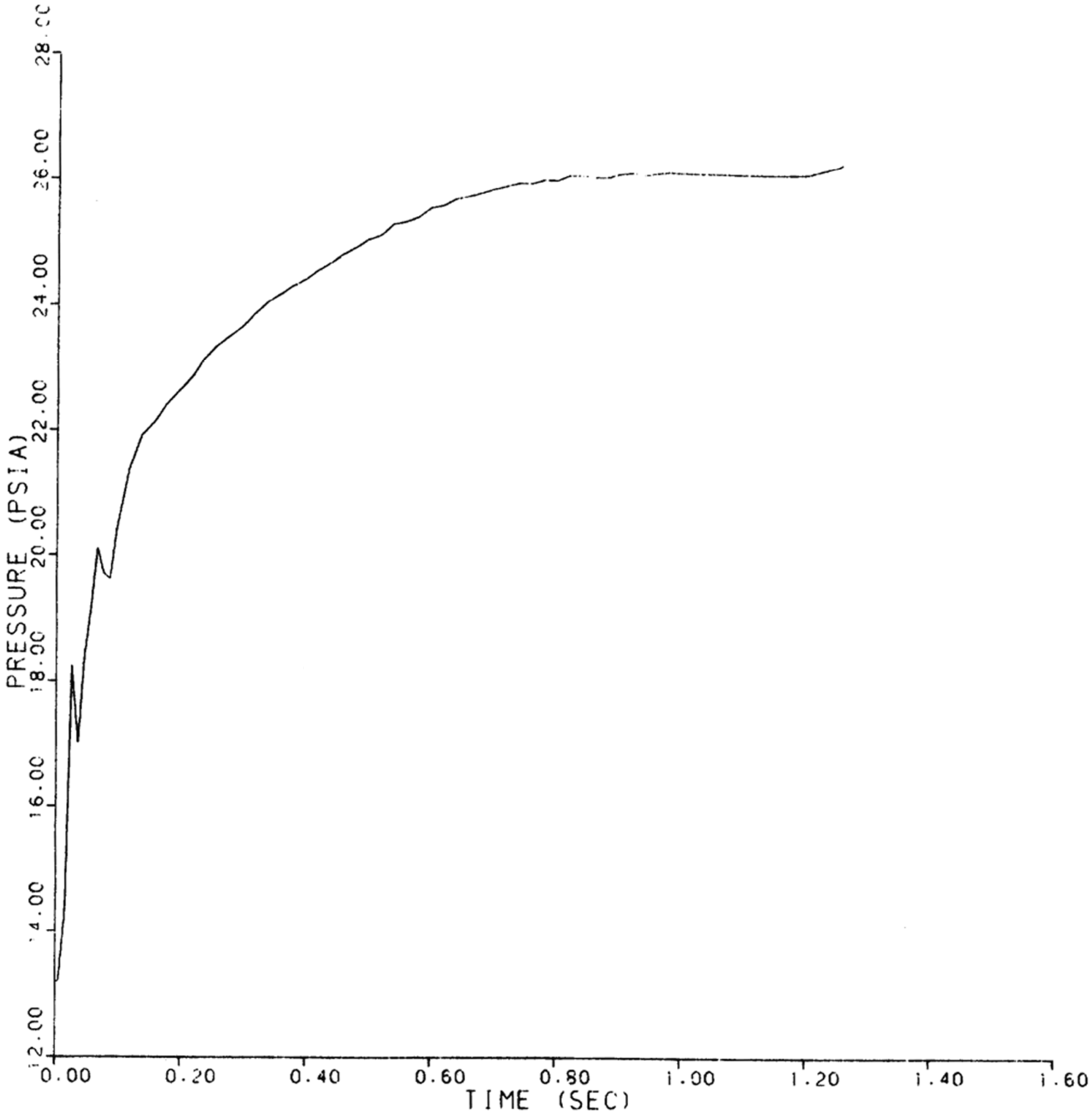
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E24
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 24 OF 74)



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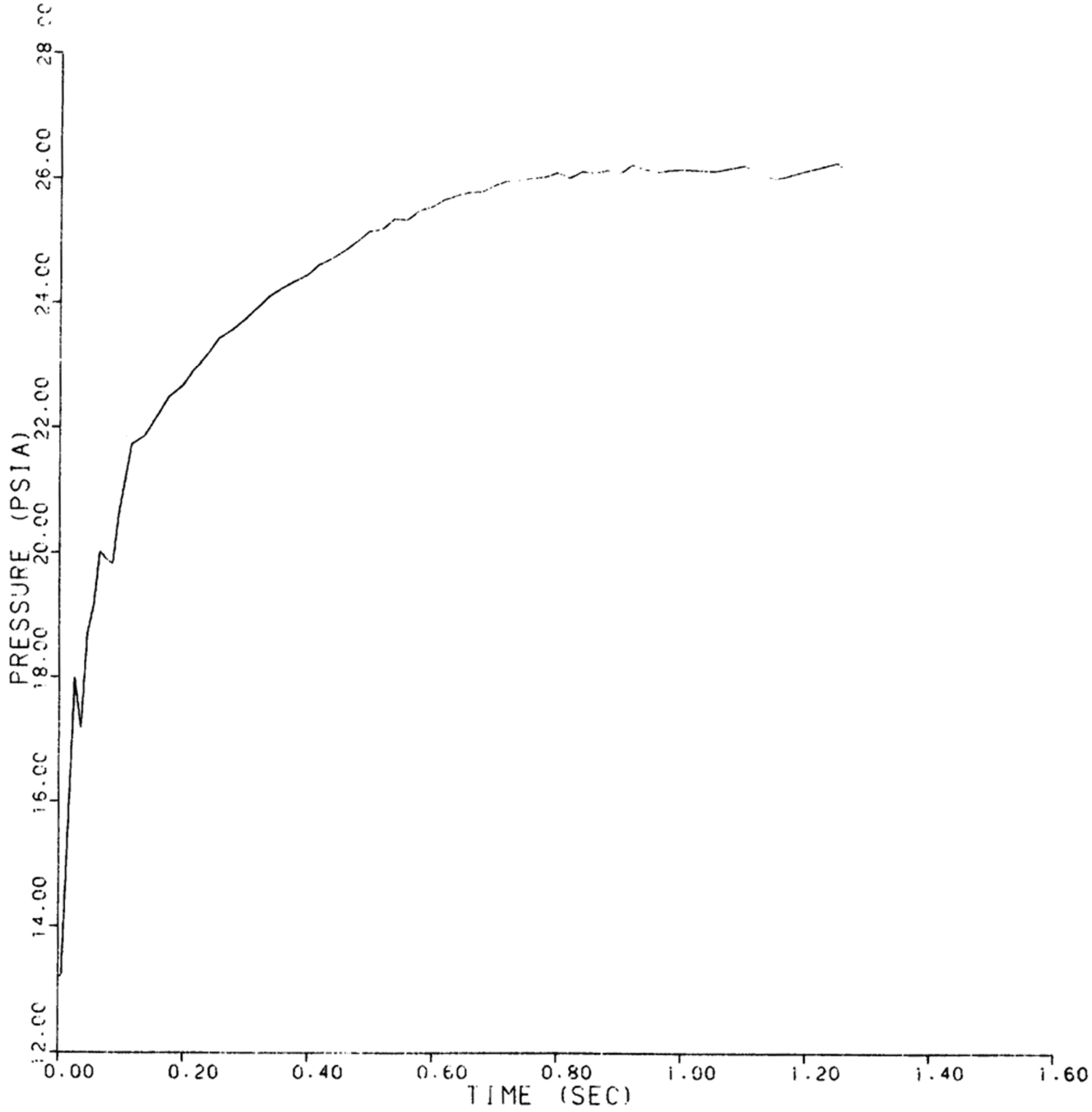
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E25 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 25 OF 74)



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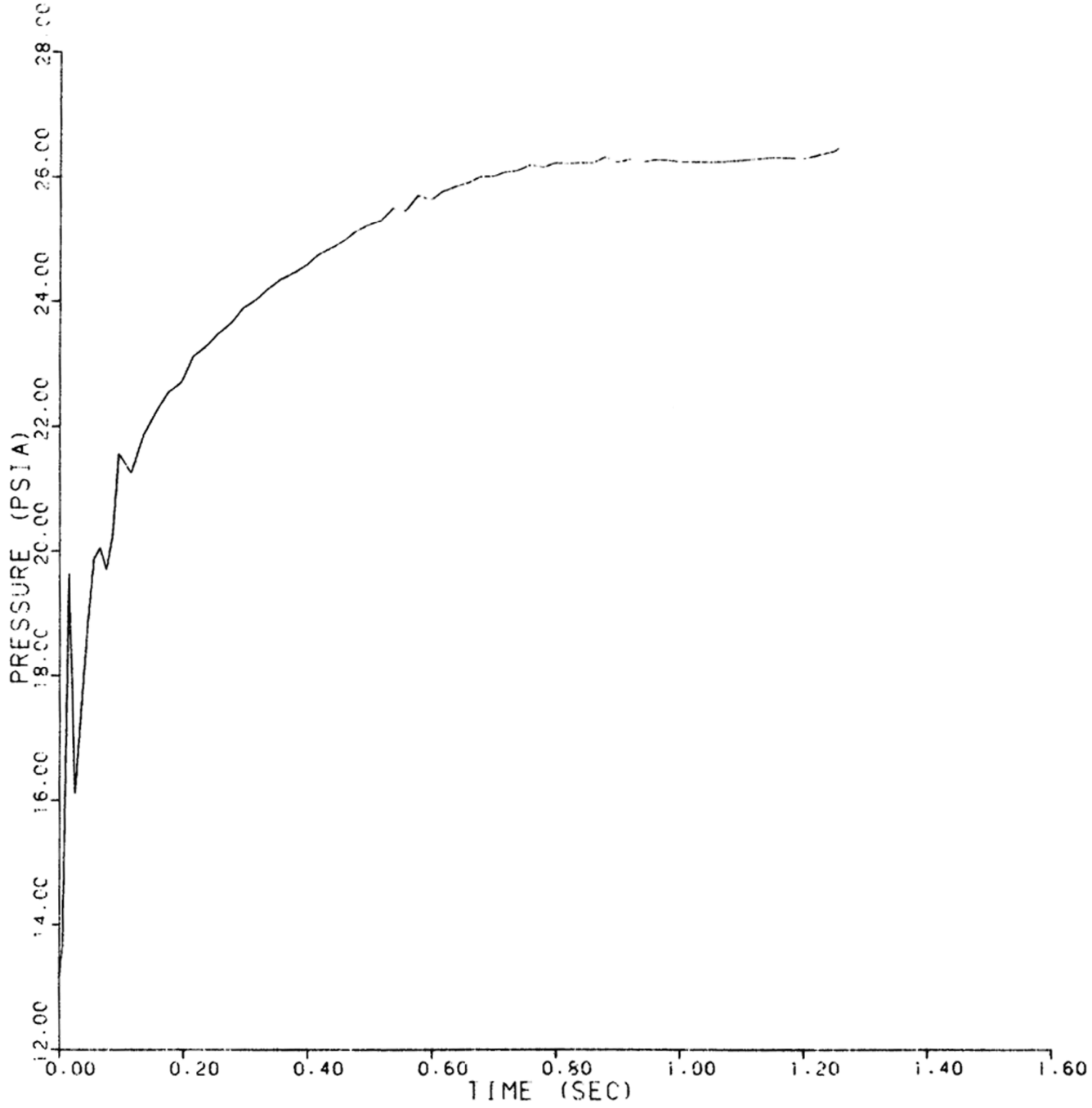
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E26 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 26 OF 74)



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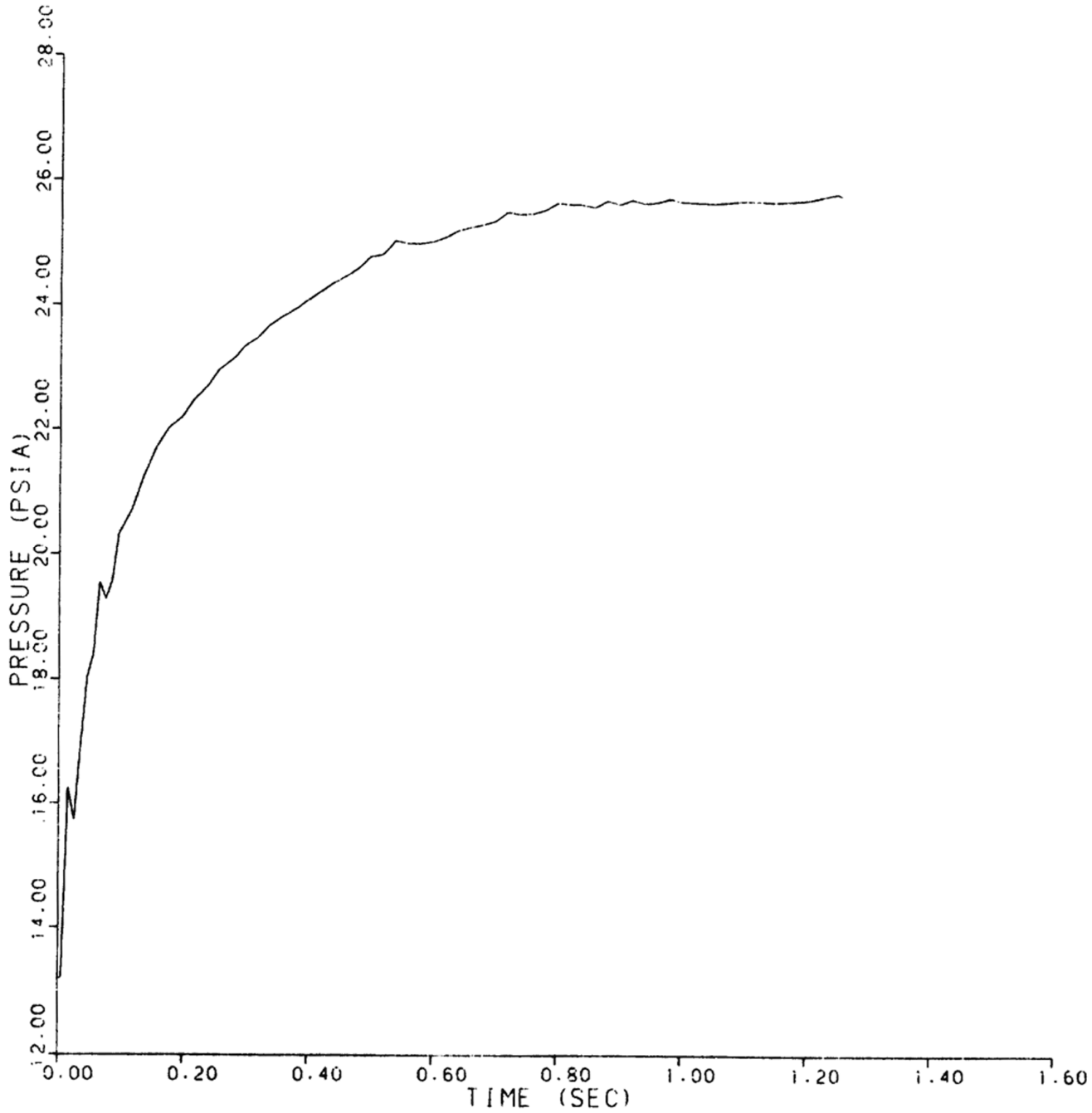
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E27
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 27 OF 74)



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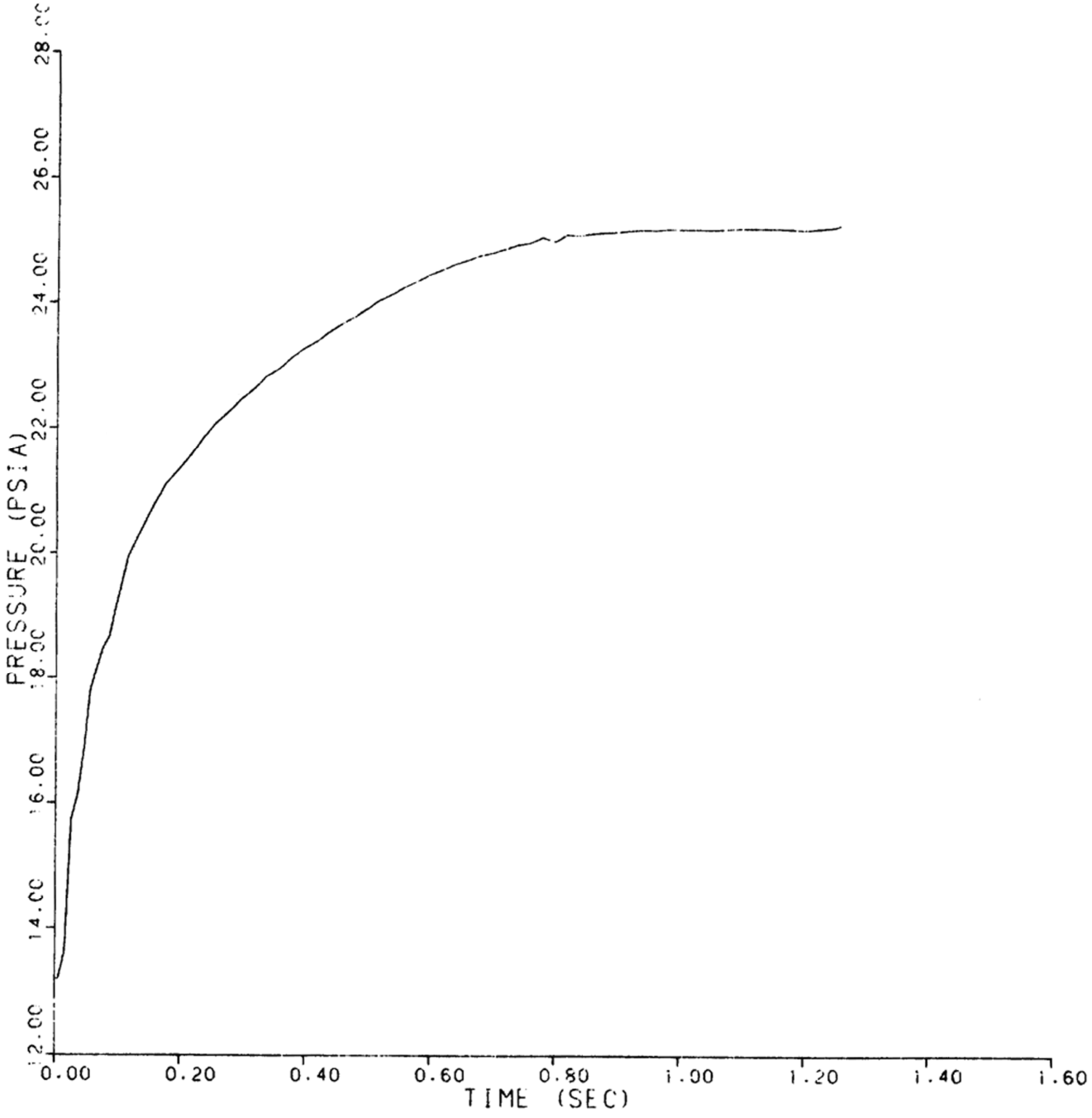
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E28 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 28 OF 74)



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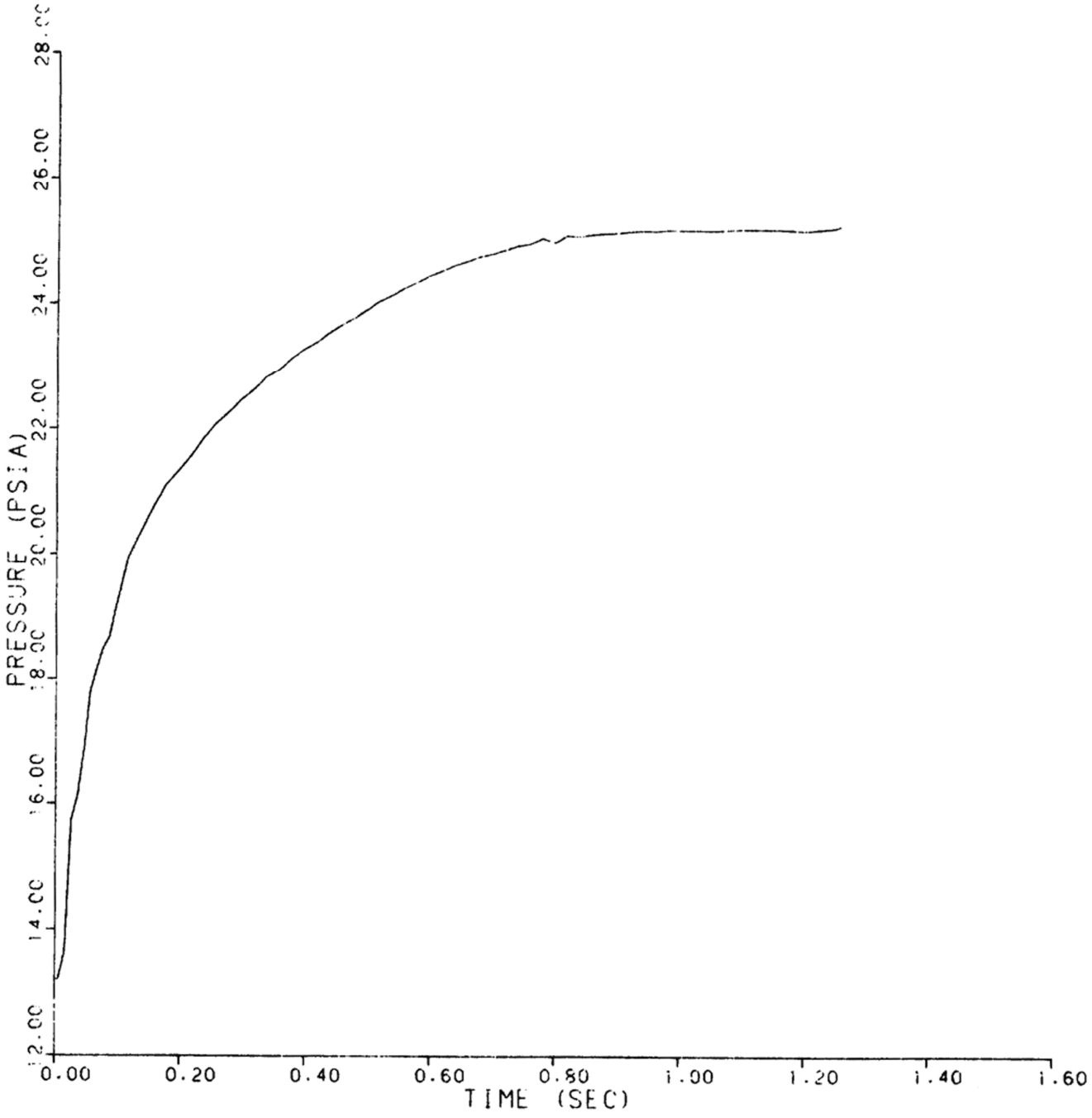
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E29
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 29 OF 74)



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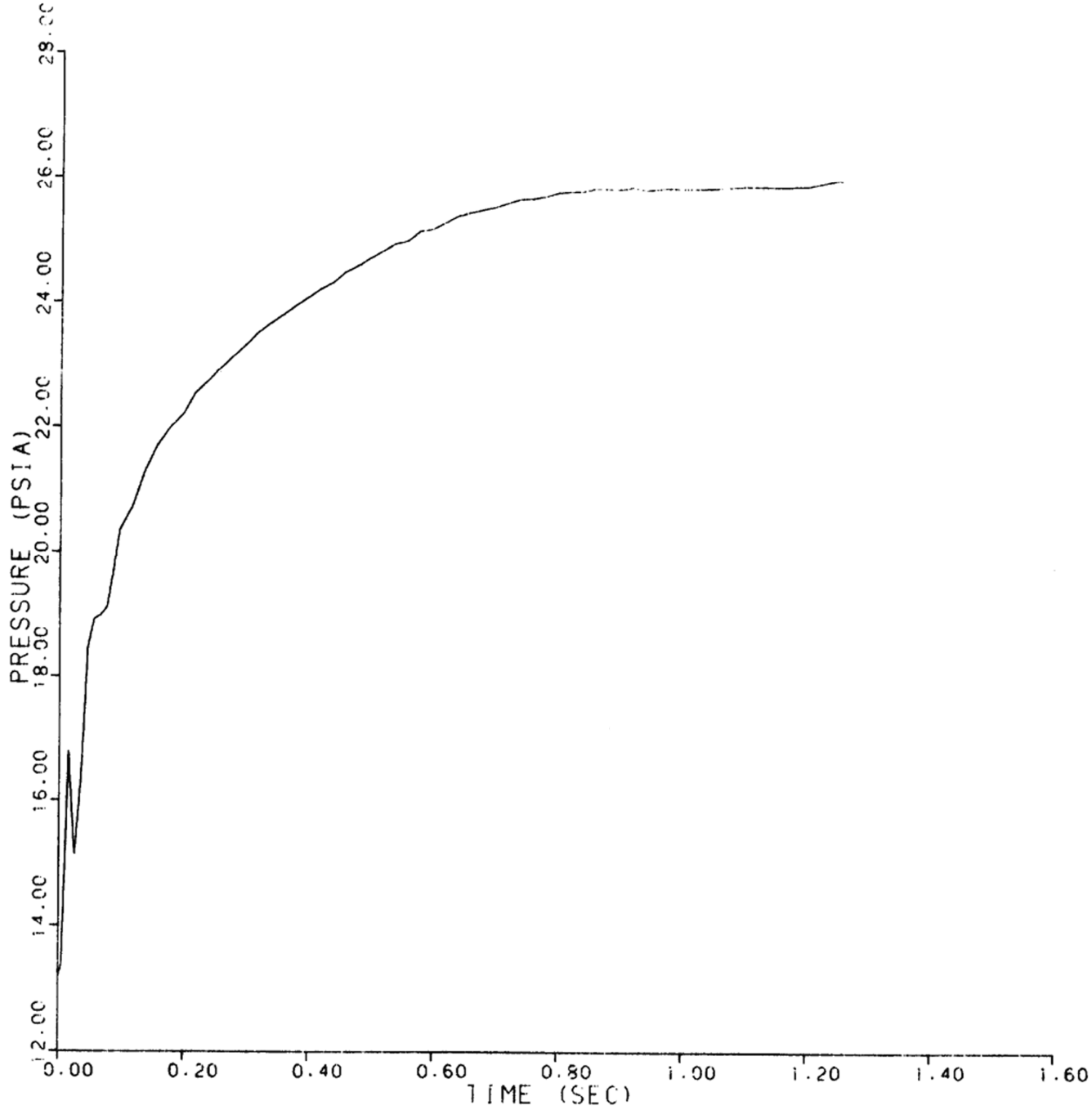
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E30
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 30 OF 74)



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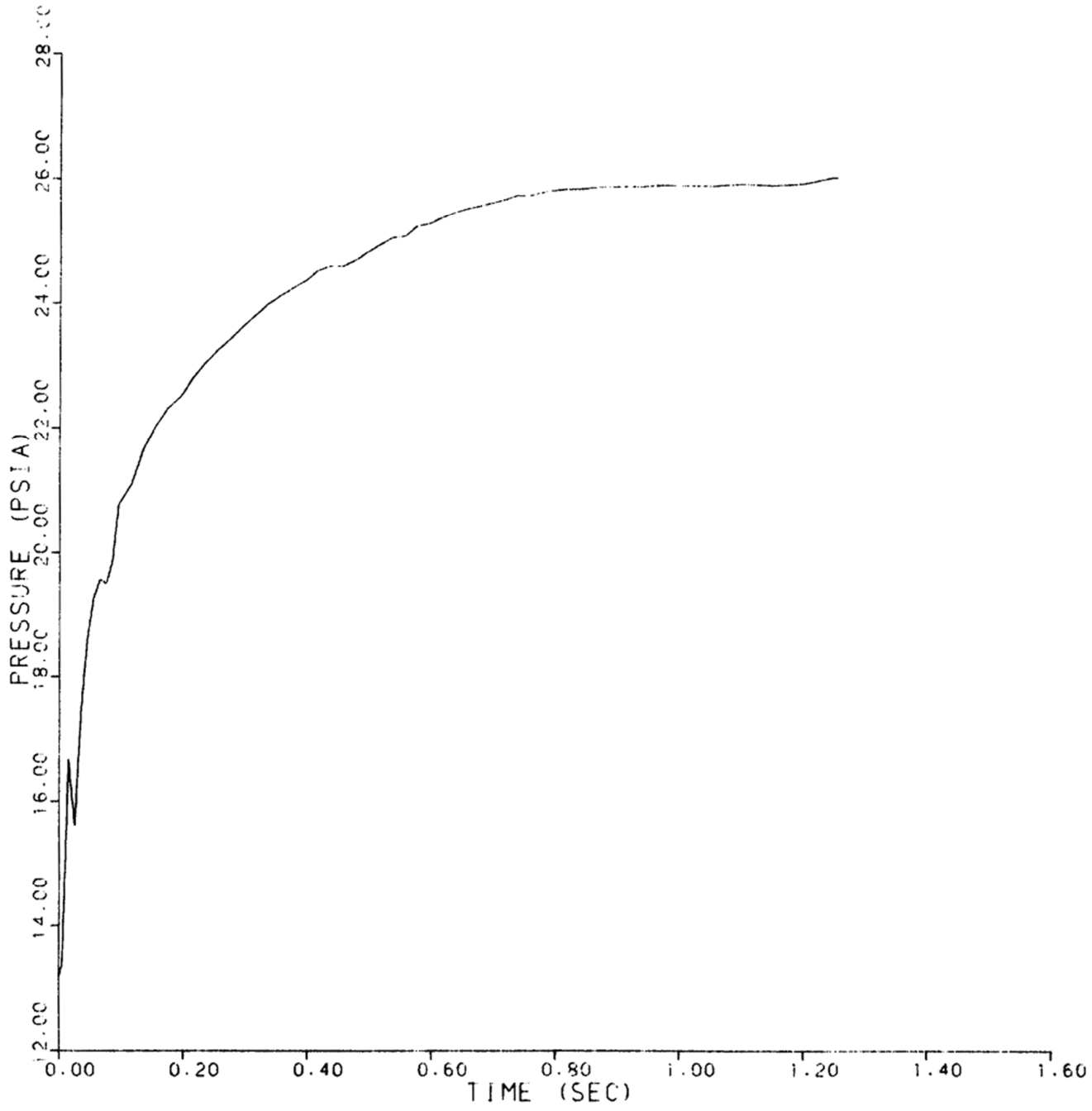
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E31
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 31 OF 74)



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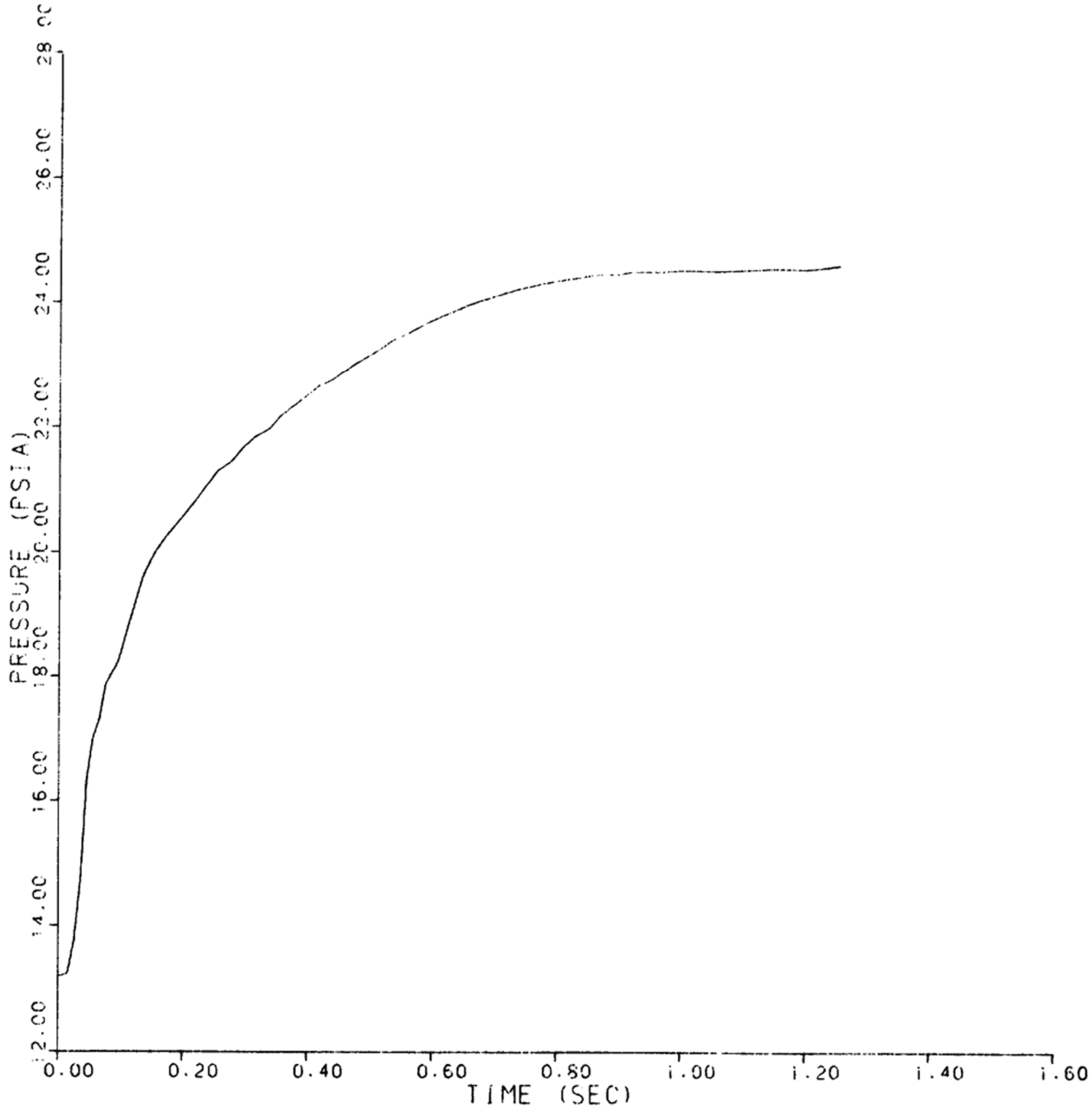
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E32
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 32 OF 74)



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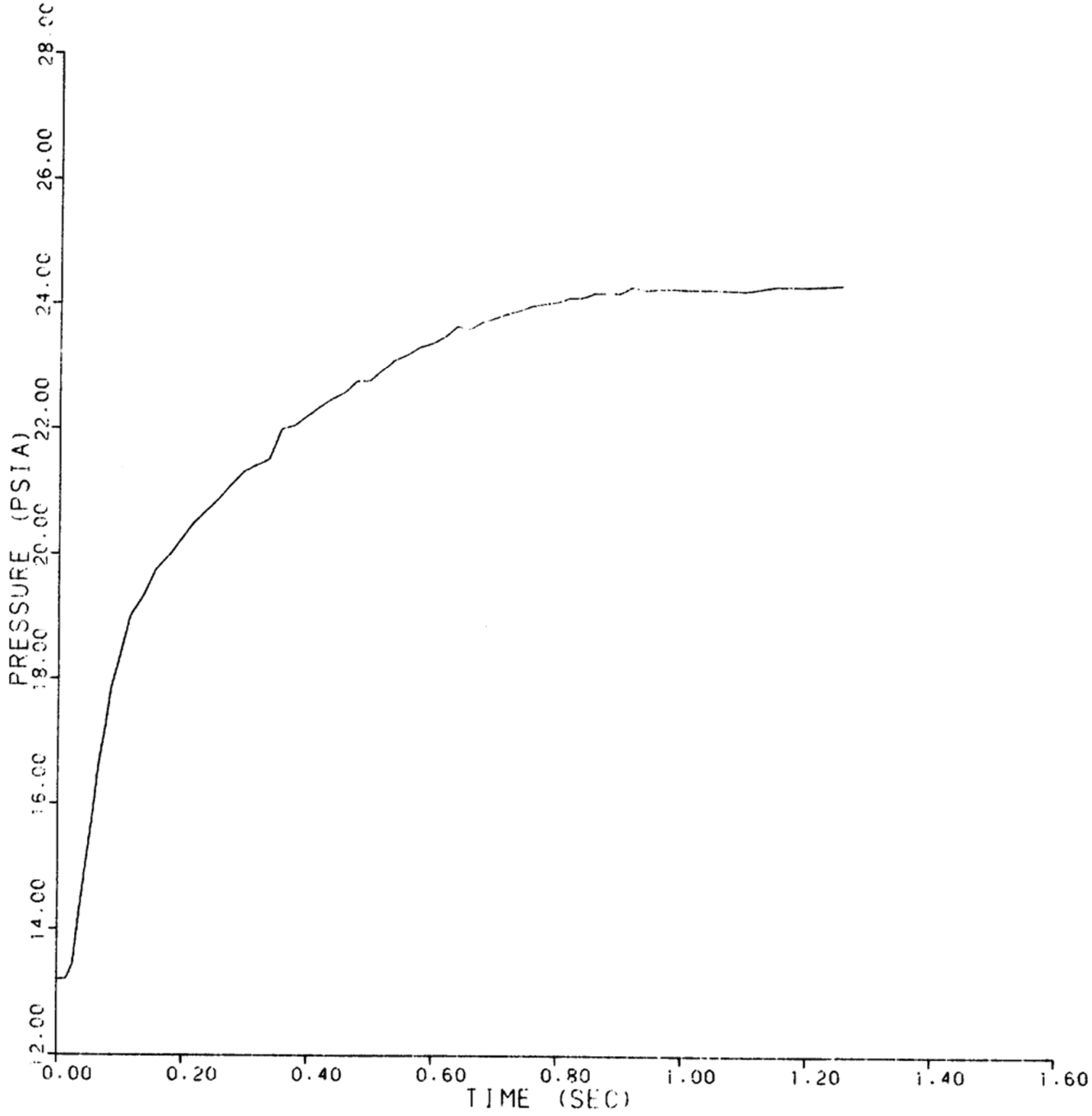
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E33
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 33 OF 74)



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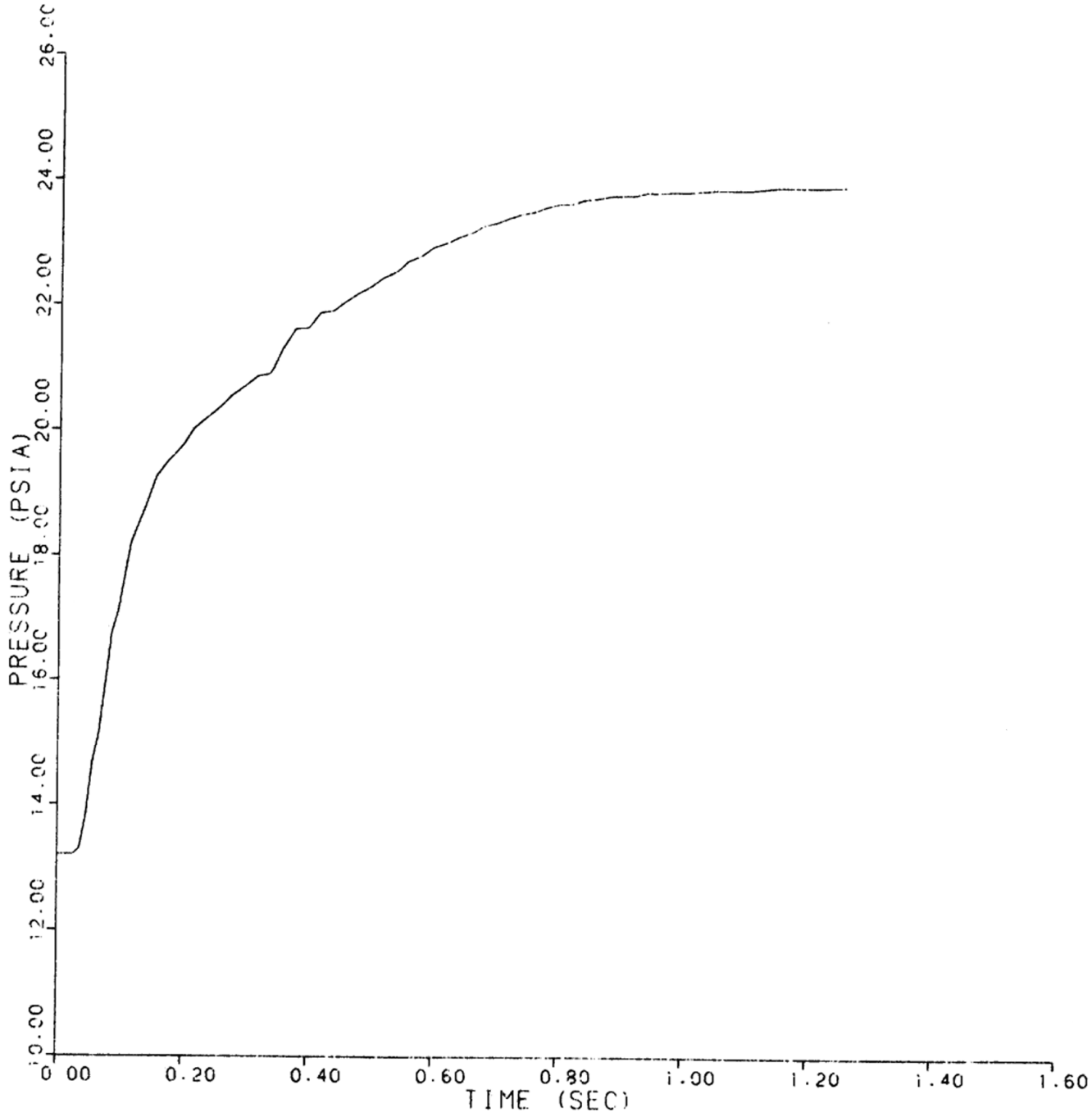
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E34 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 34 OF 74)



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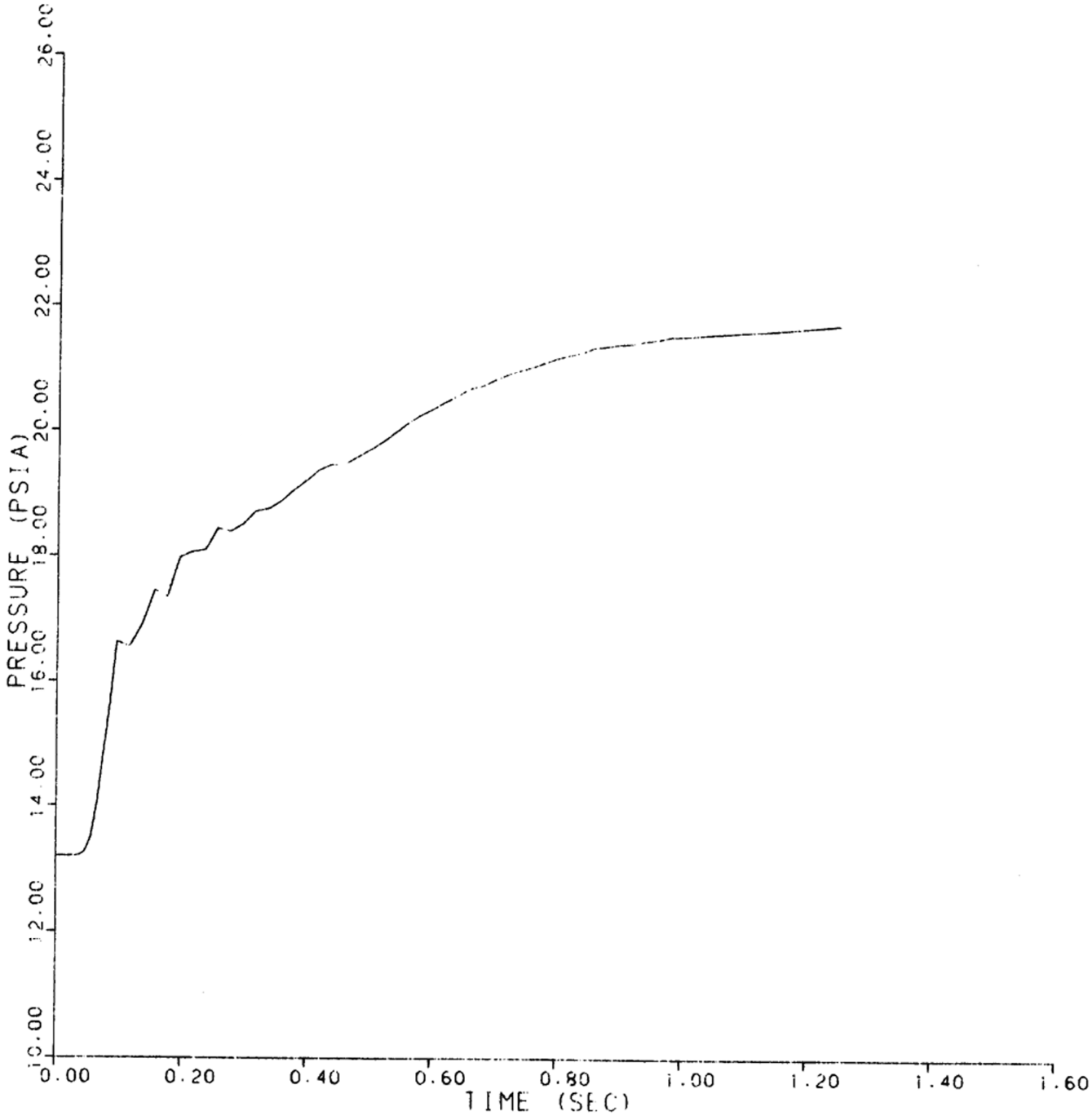
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E35
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 35 OF 74)



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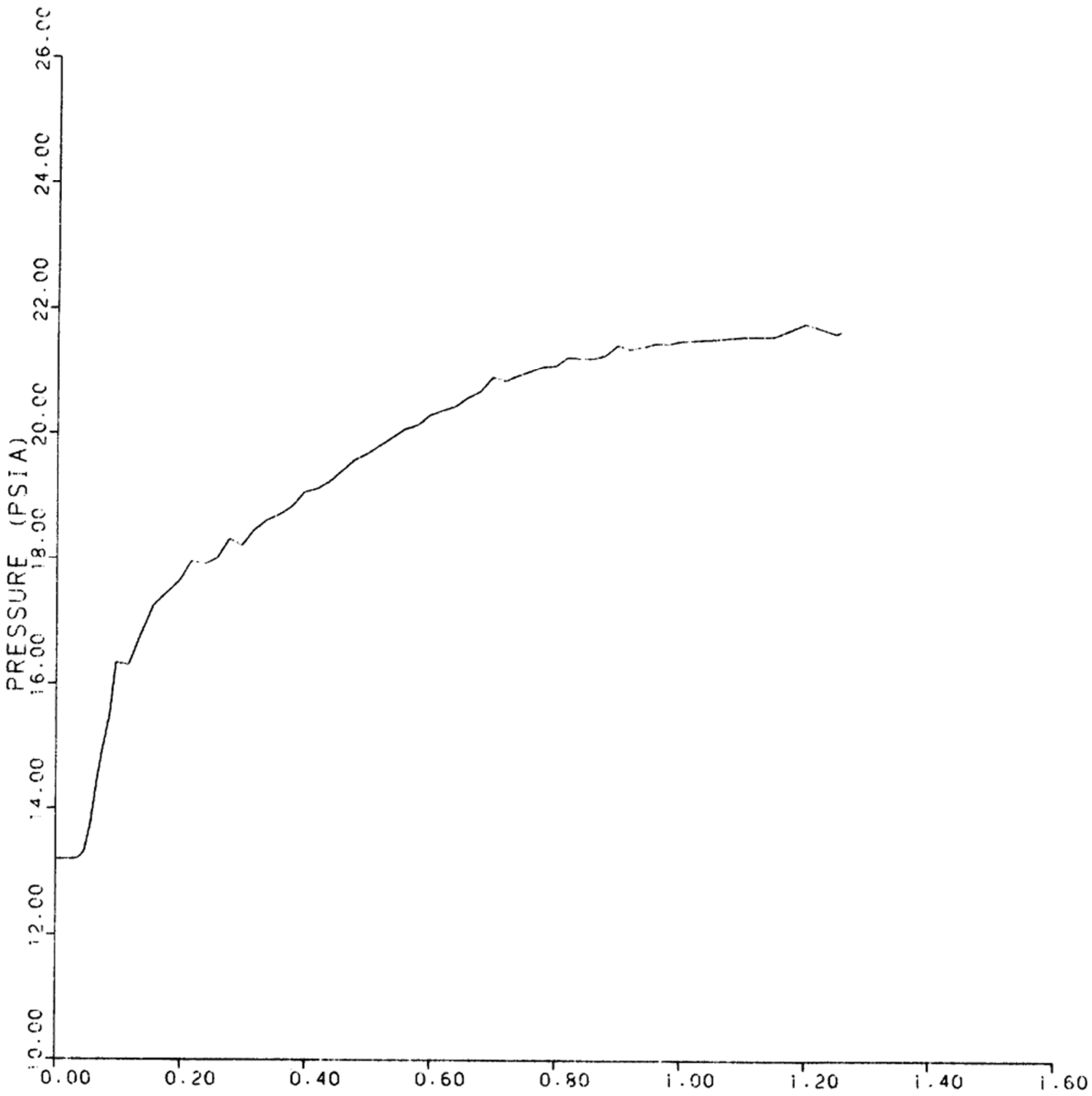
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E36 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 36 OF 74)



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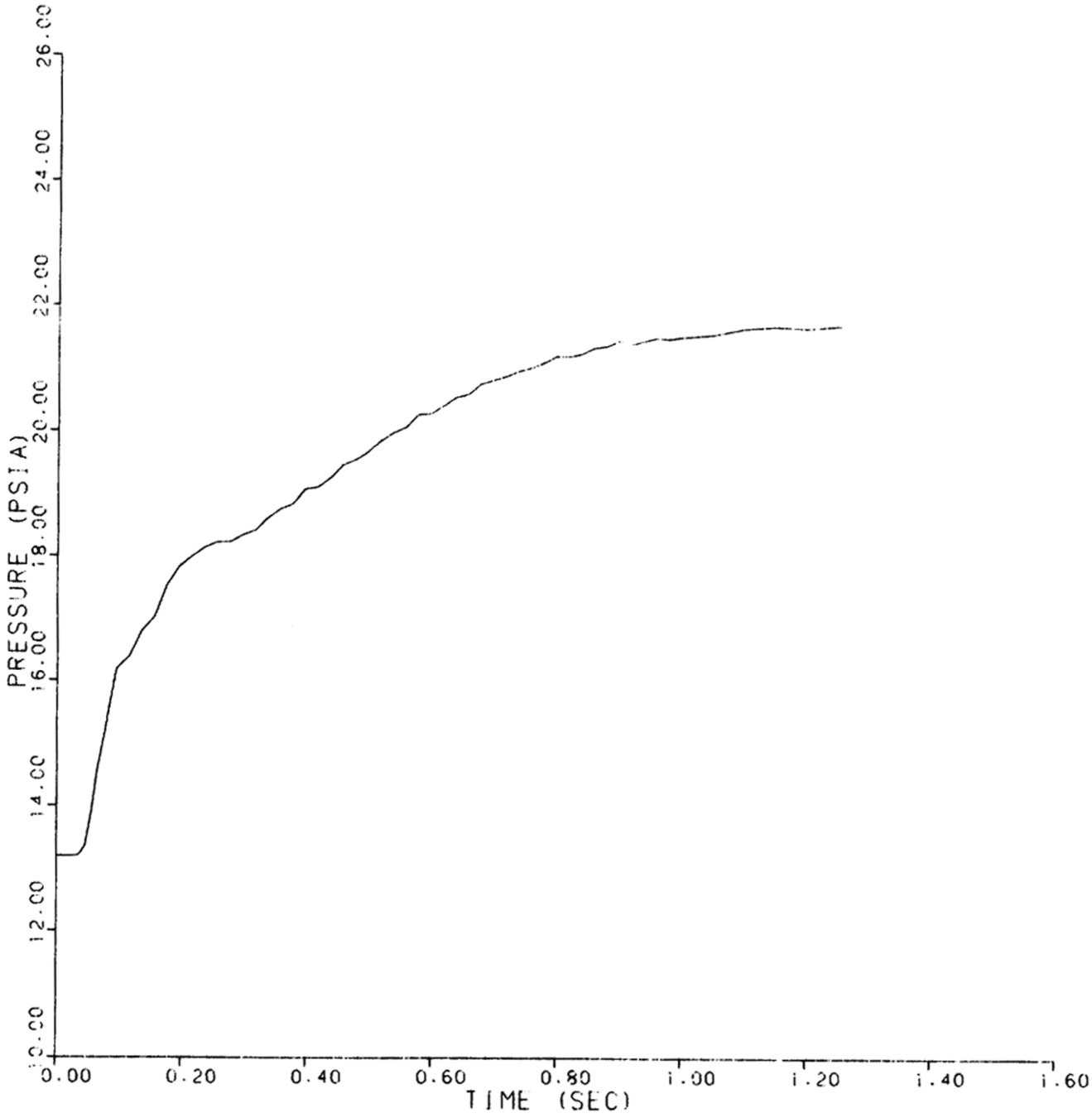
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E37
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 37 OF 74)



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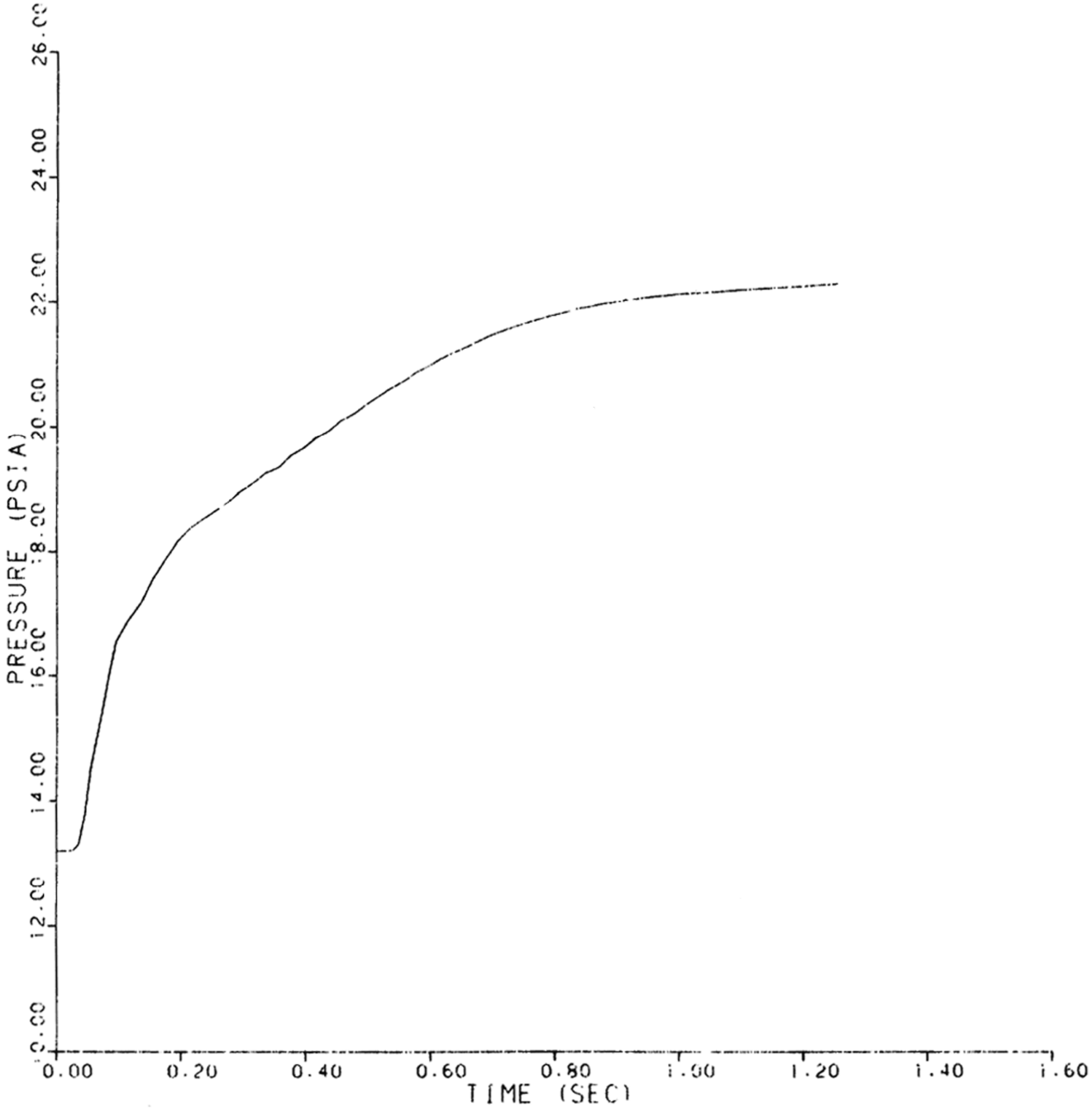
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E38
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 38 OF 74)



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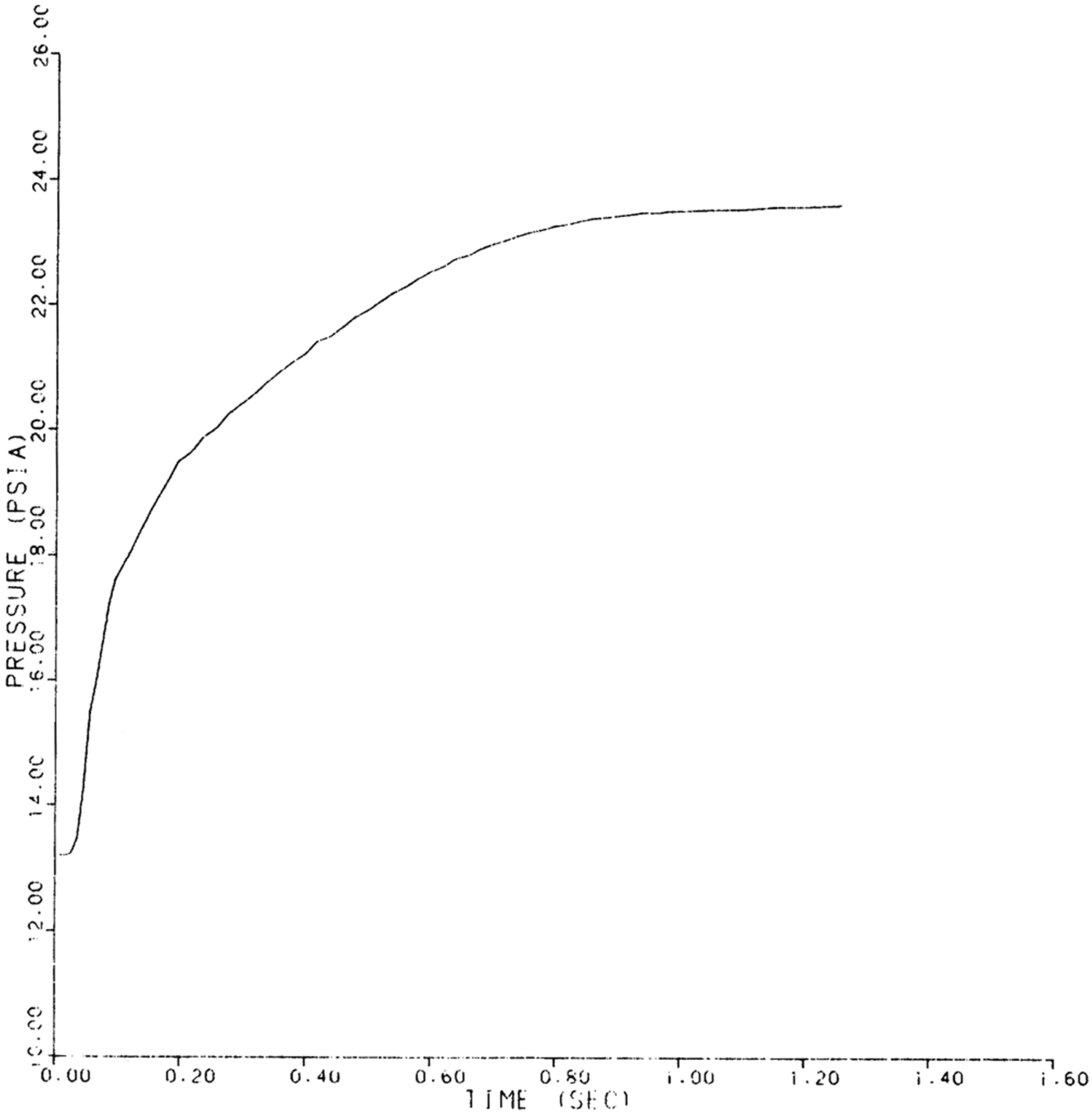
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E39 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 39 OF 74)



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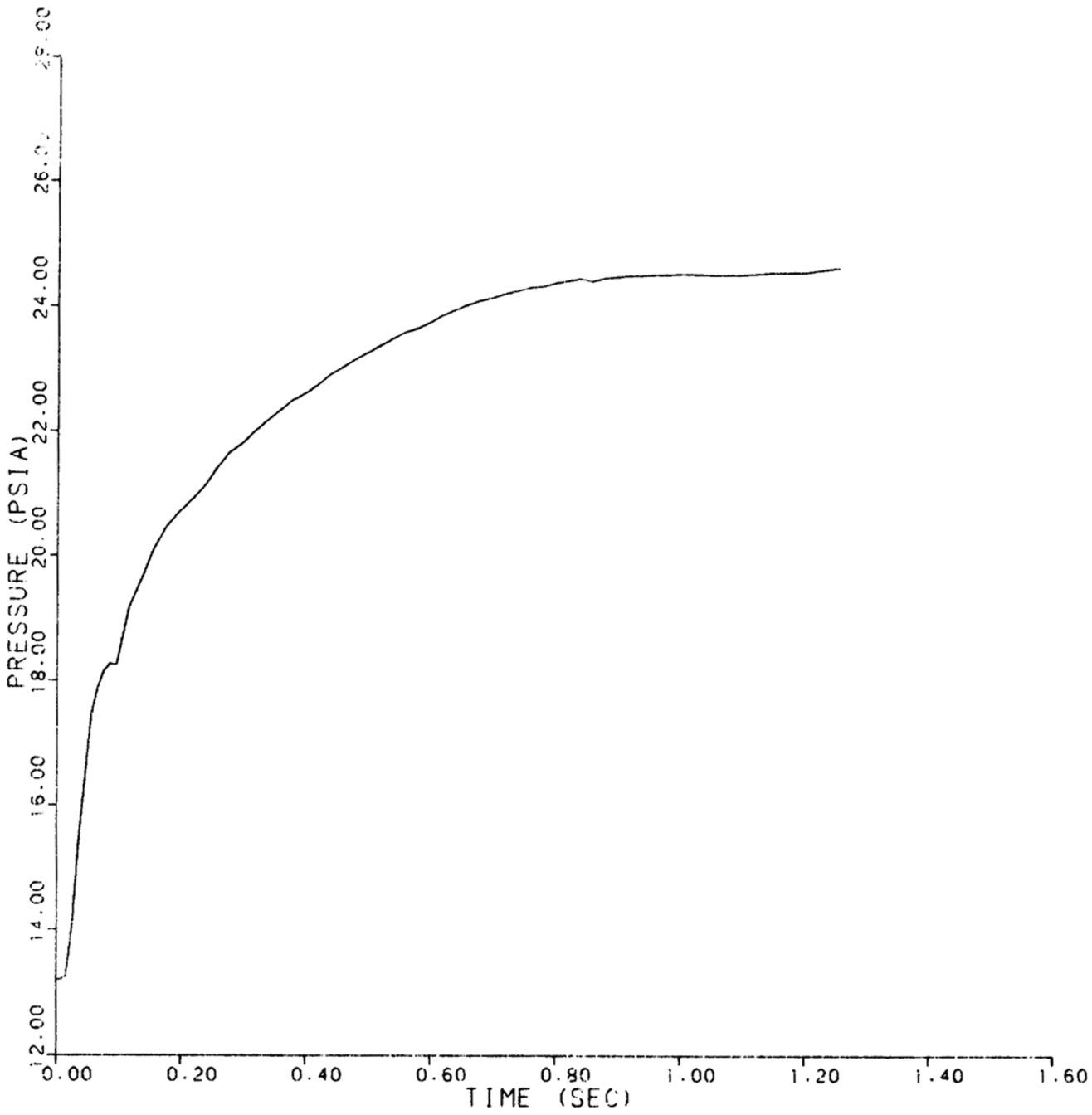
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E40
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 40 OF 74)



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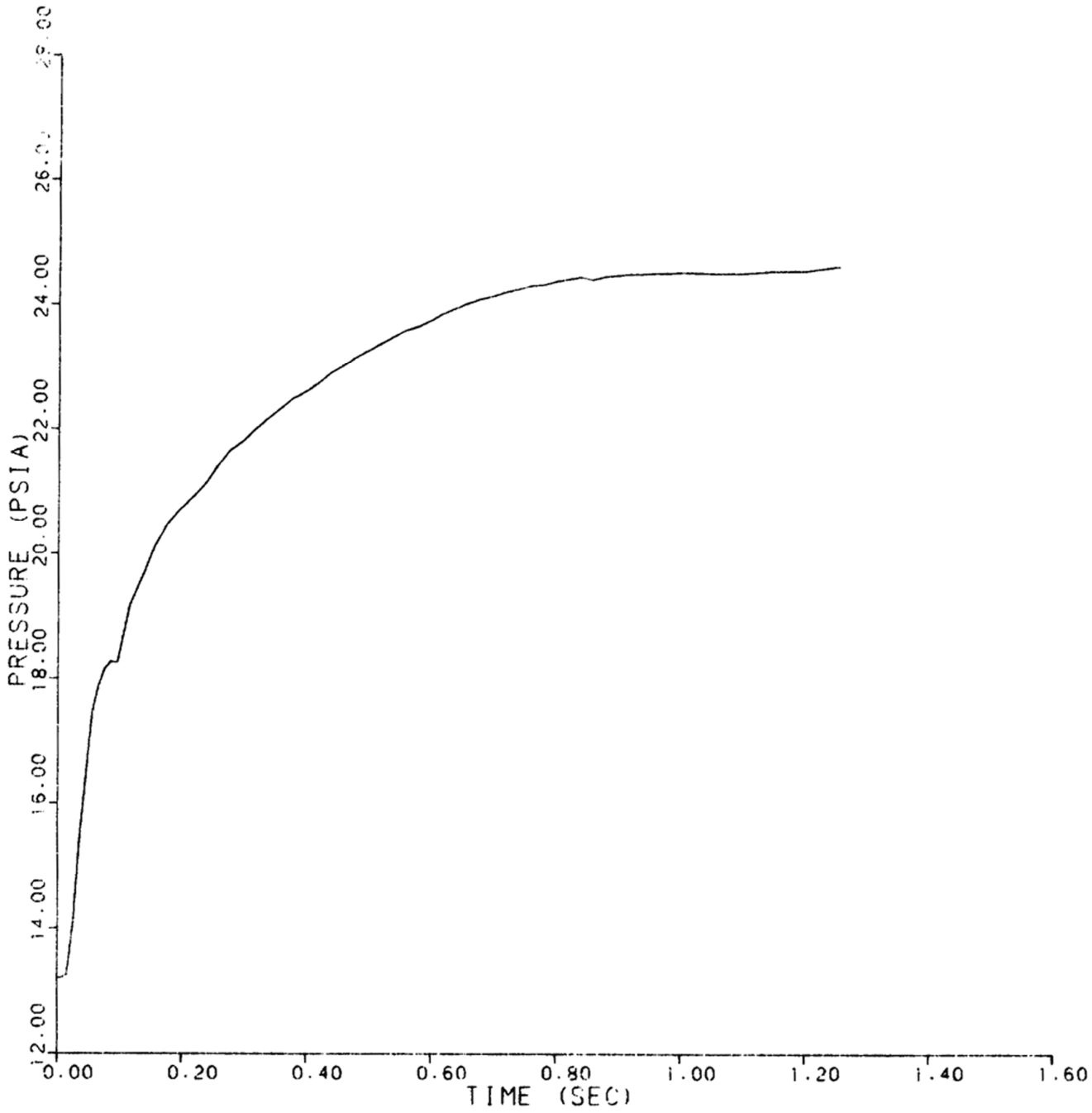
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E41
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 41 OF 74)



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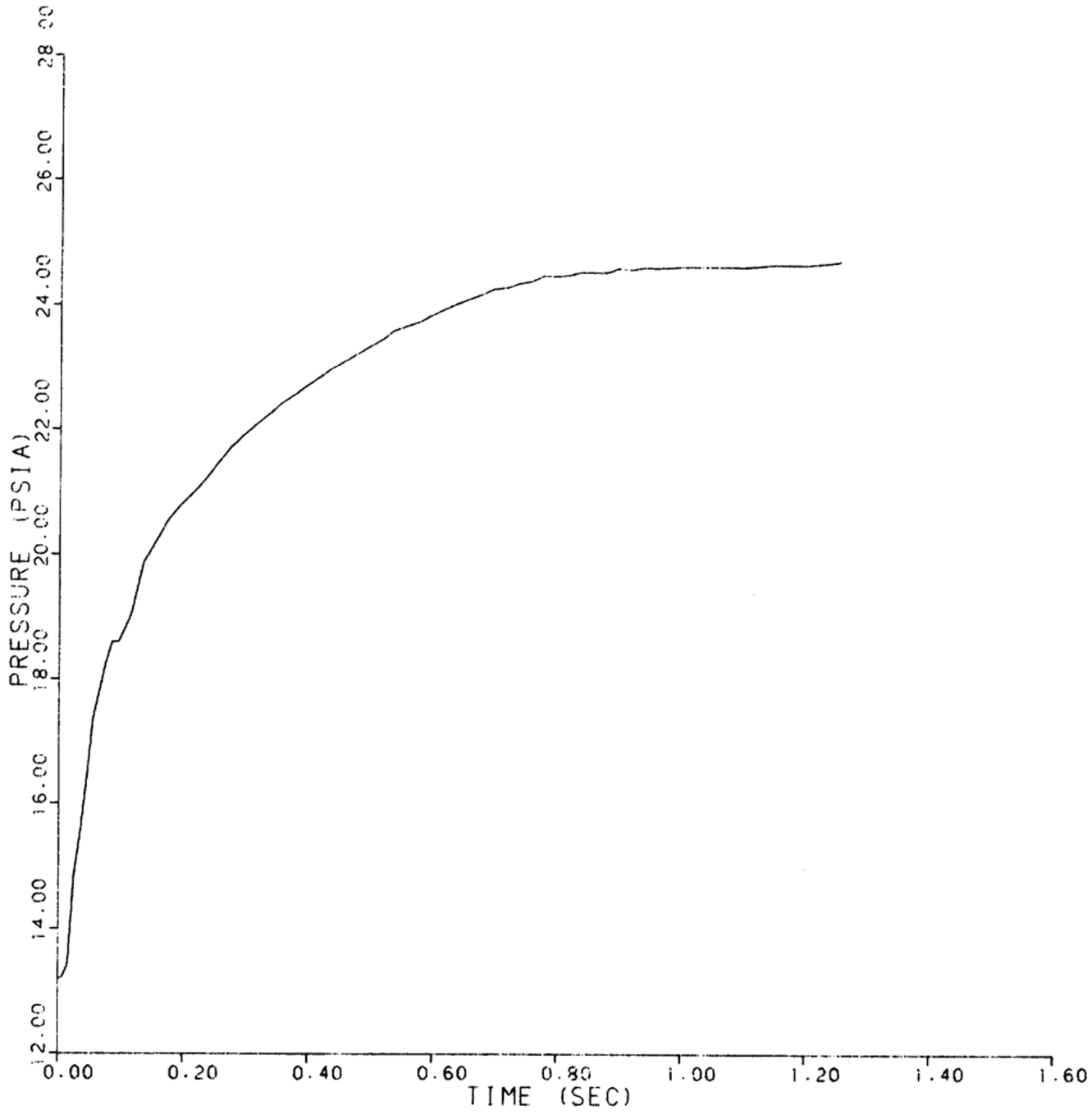
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E42 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 42 OF 74)



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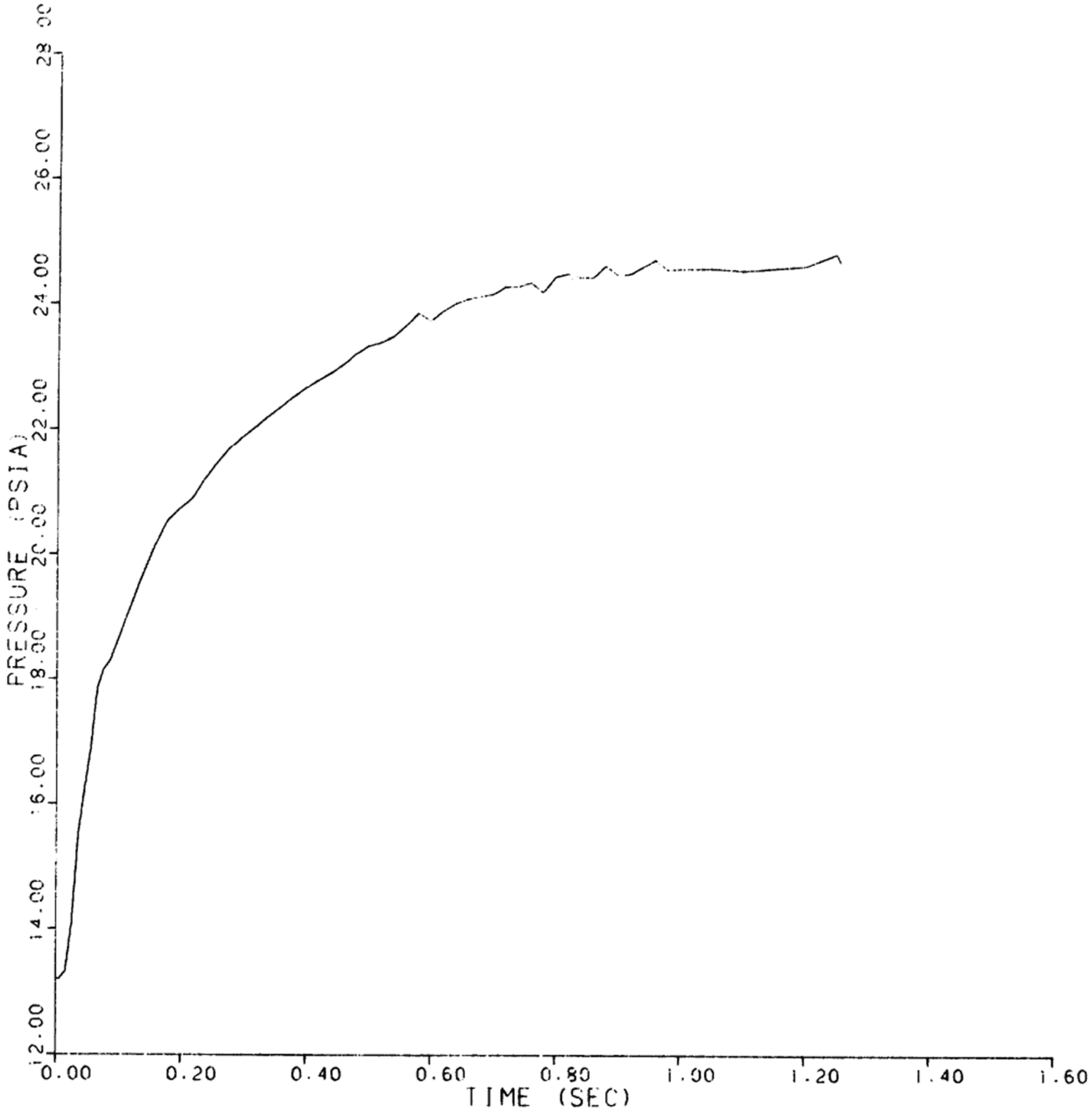
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E43
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 43 OF 74)



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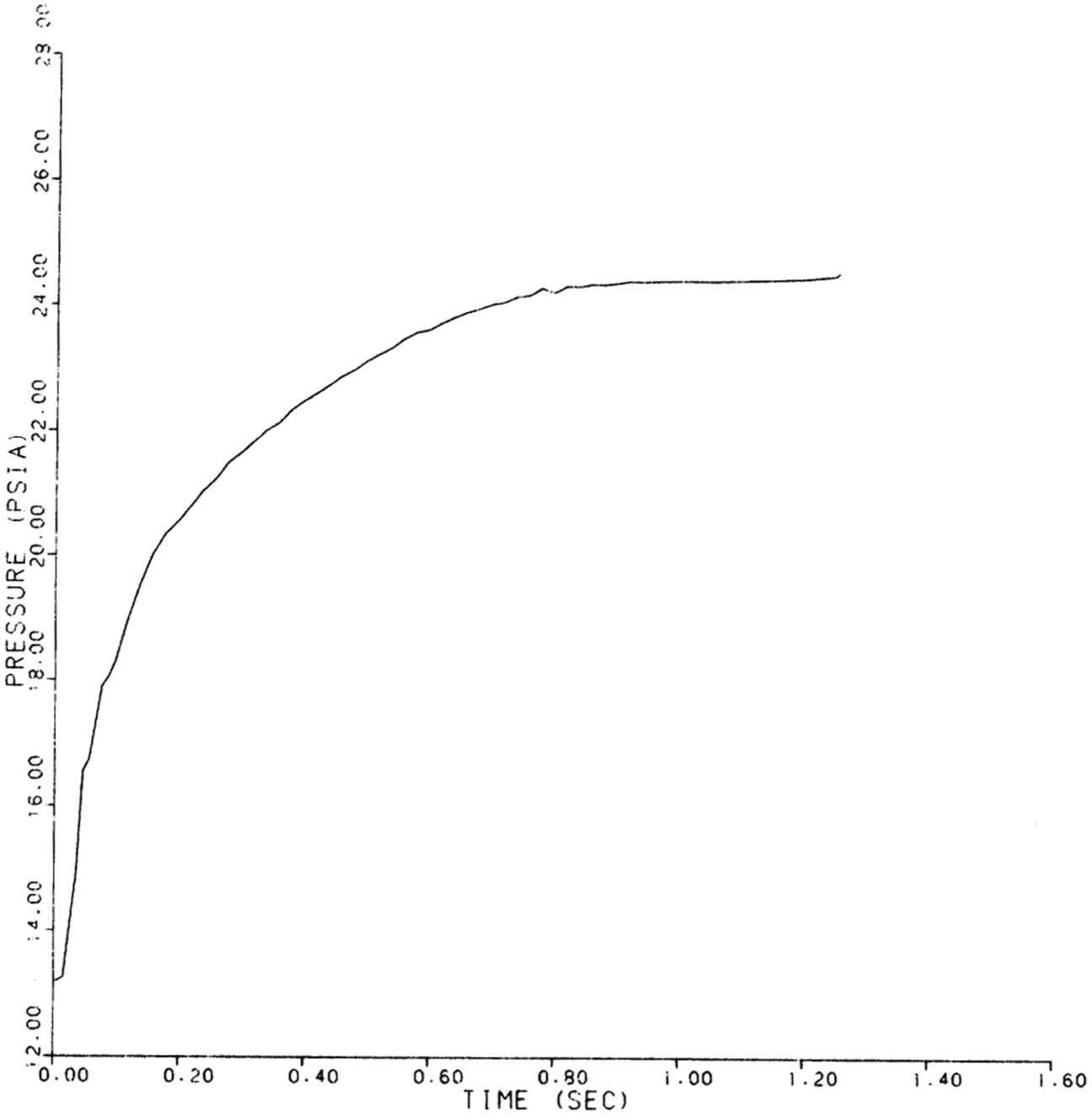
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E44
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 44 OF 74)



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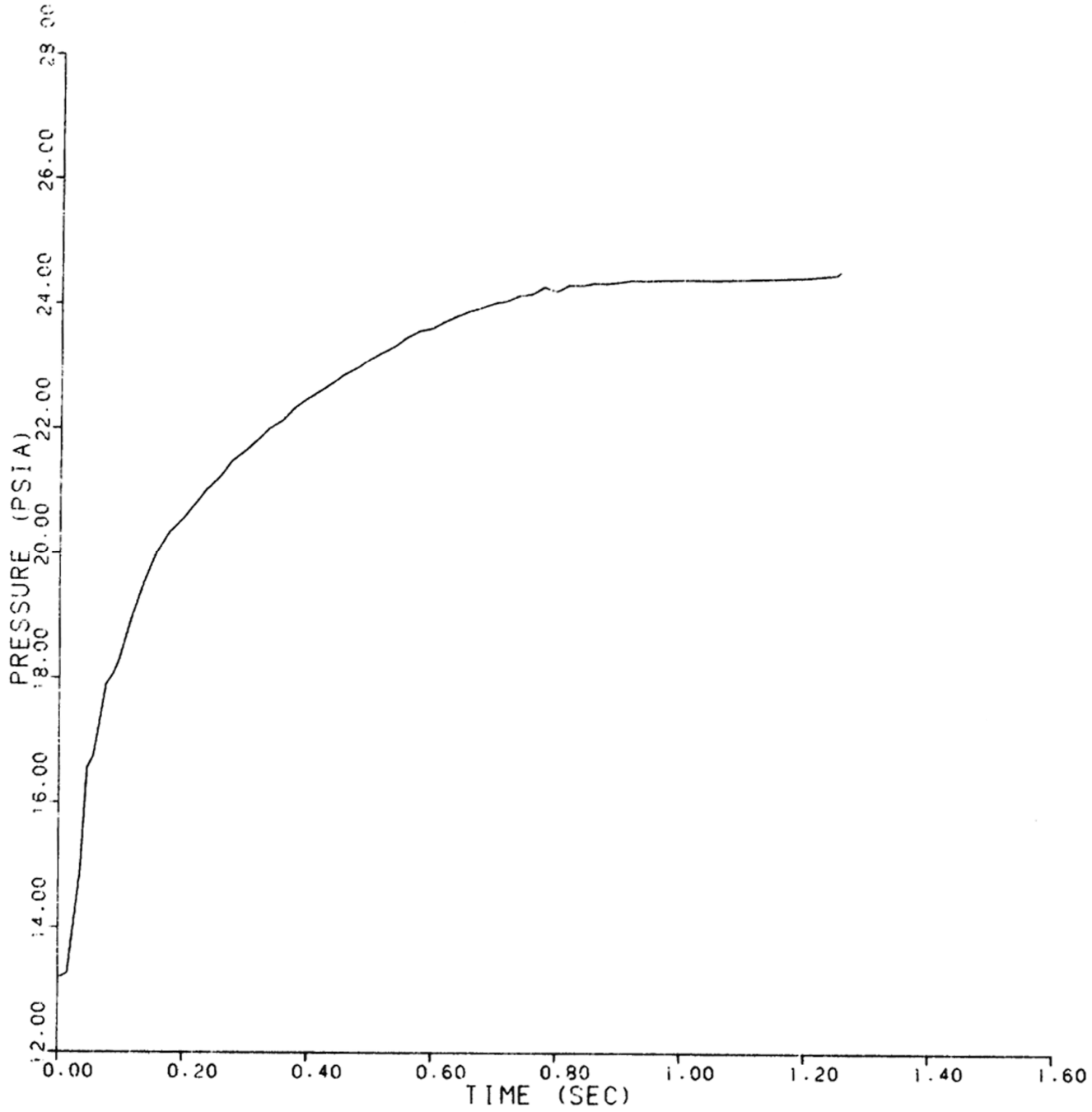
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E45
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 45 OF 74)



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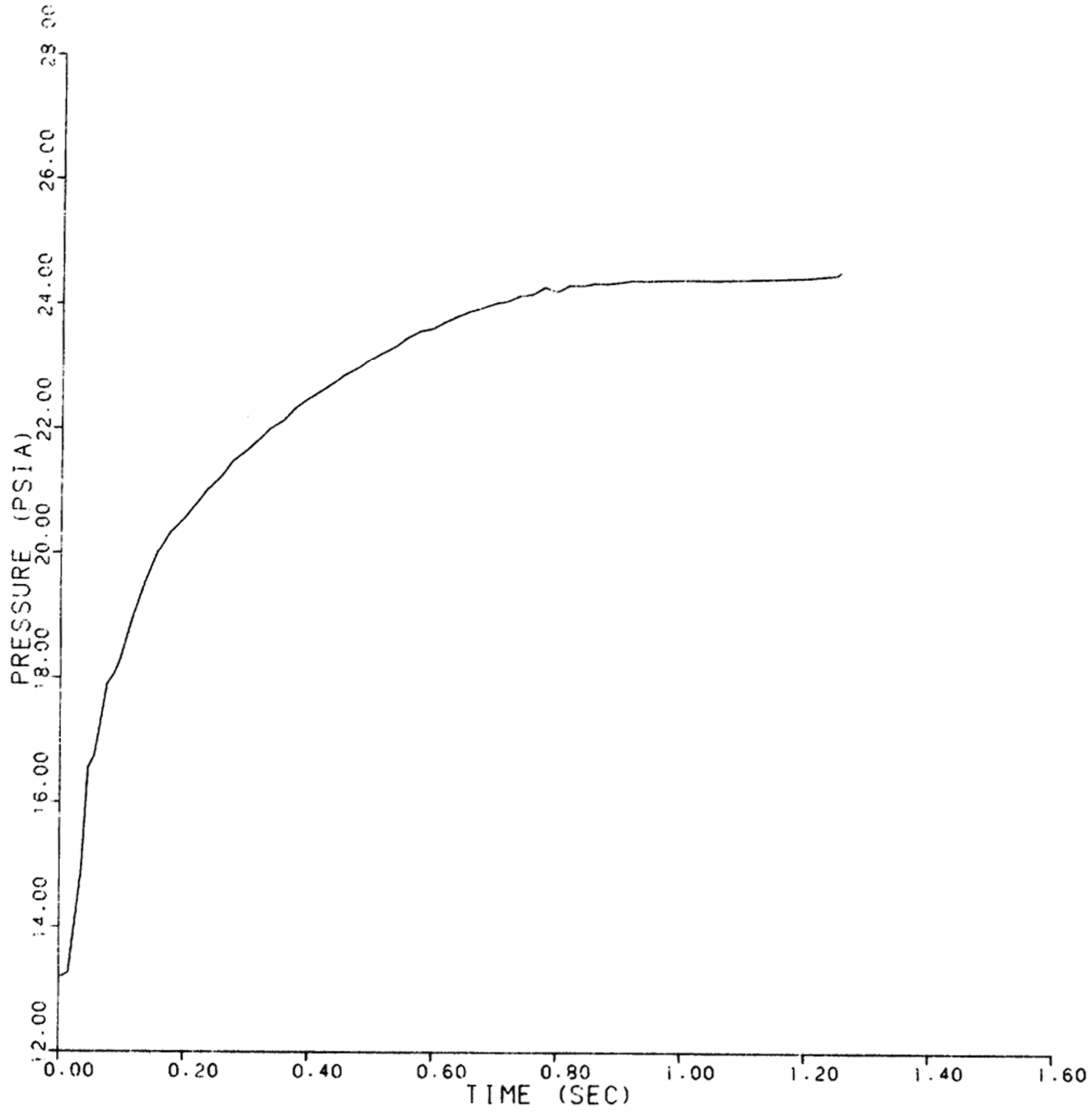
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E46
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 46 OF 74)



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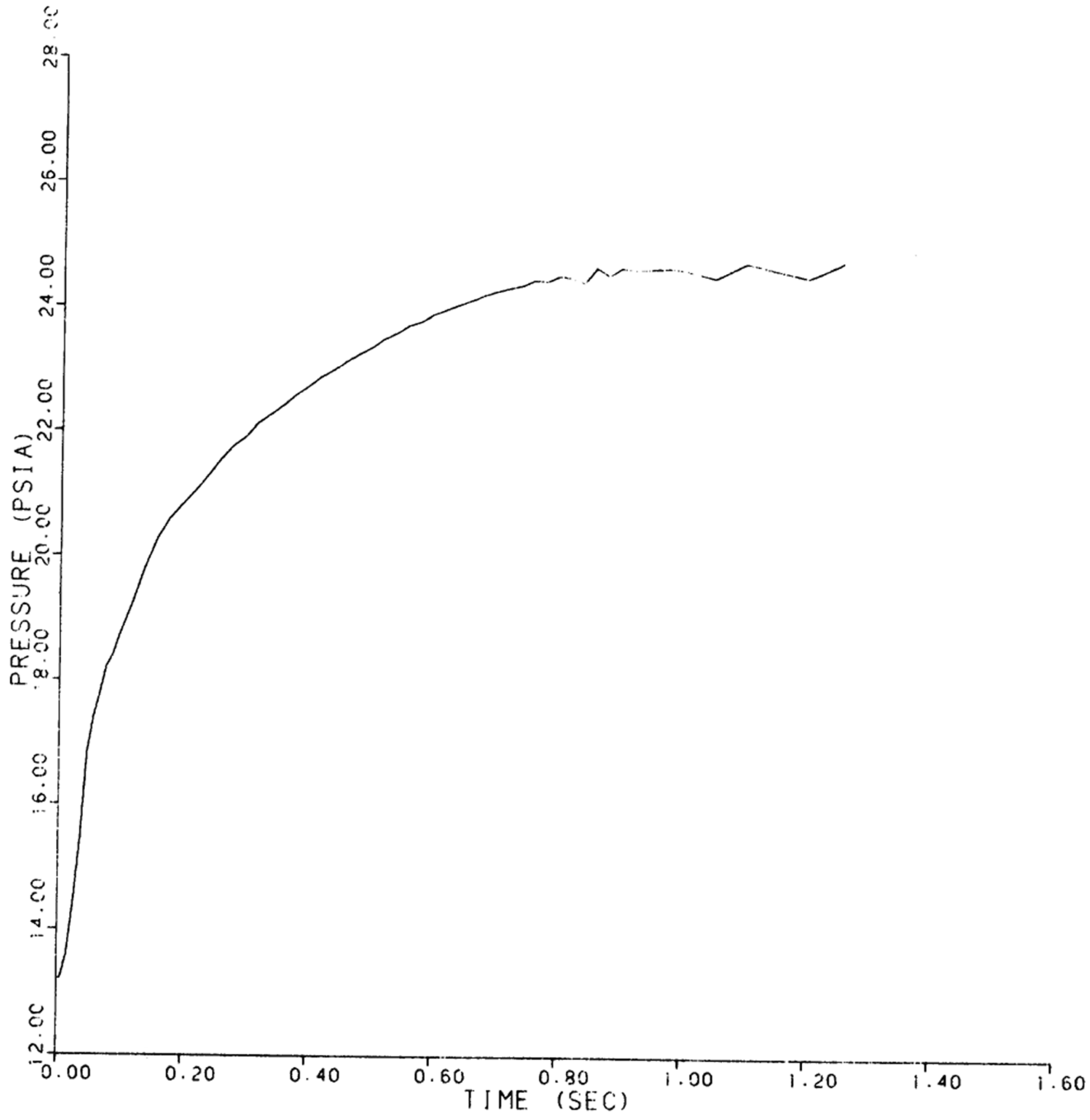
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E47
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 47 OF 74)



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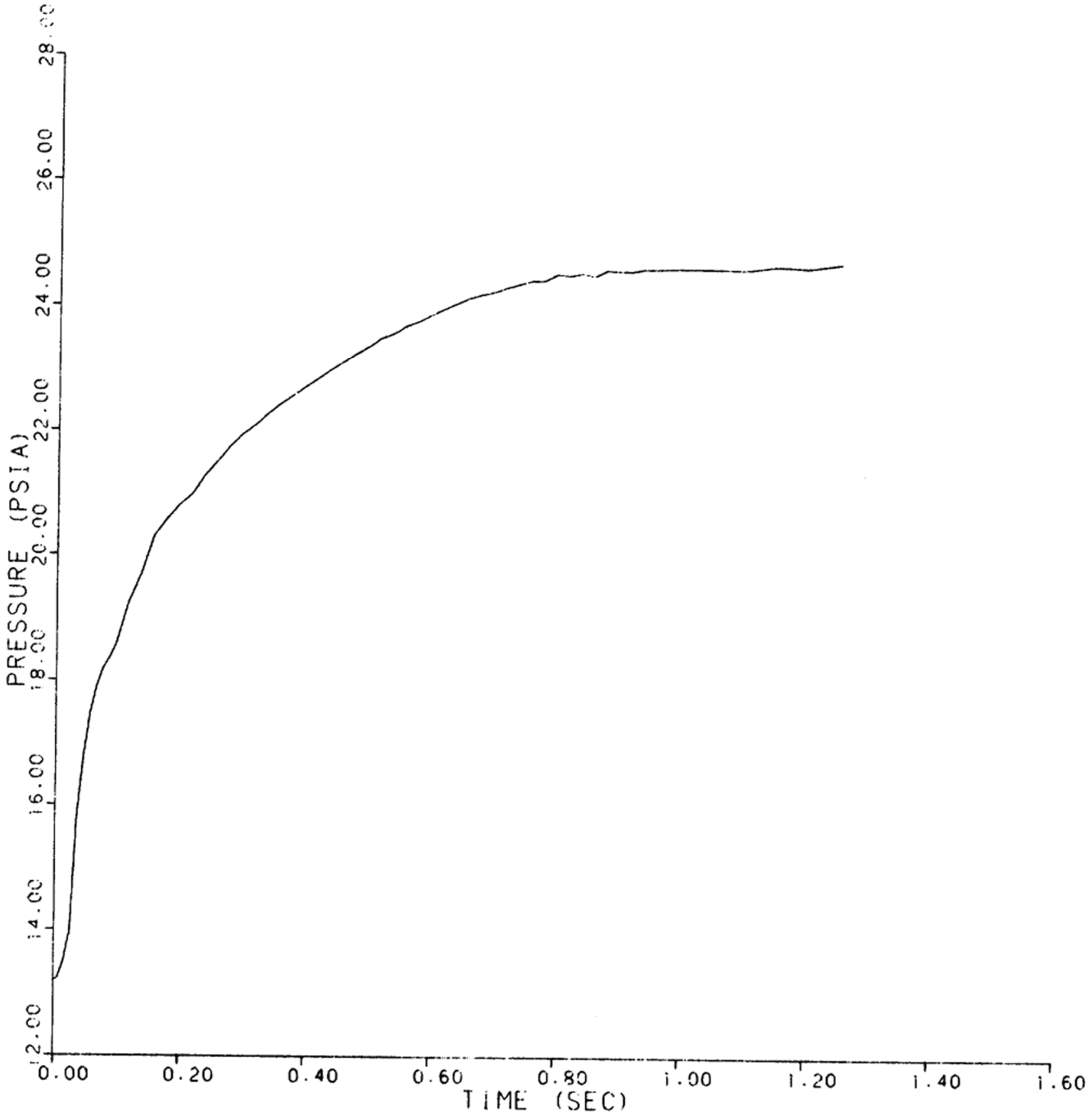
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E48
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 48 OF 74)



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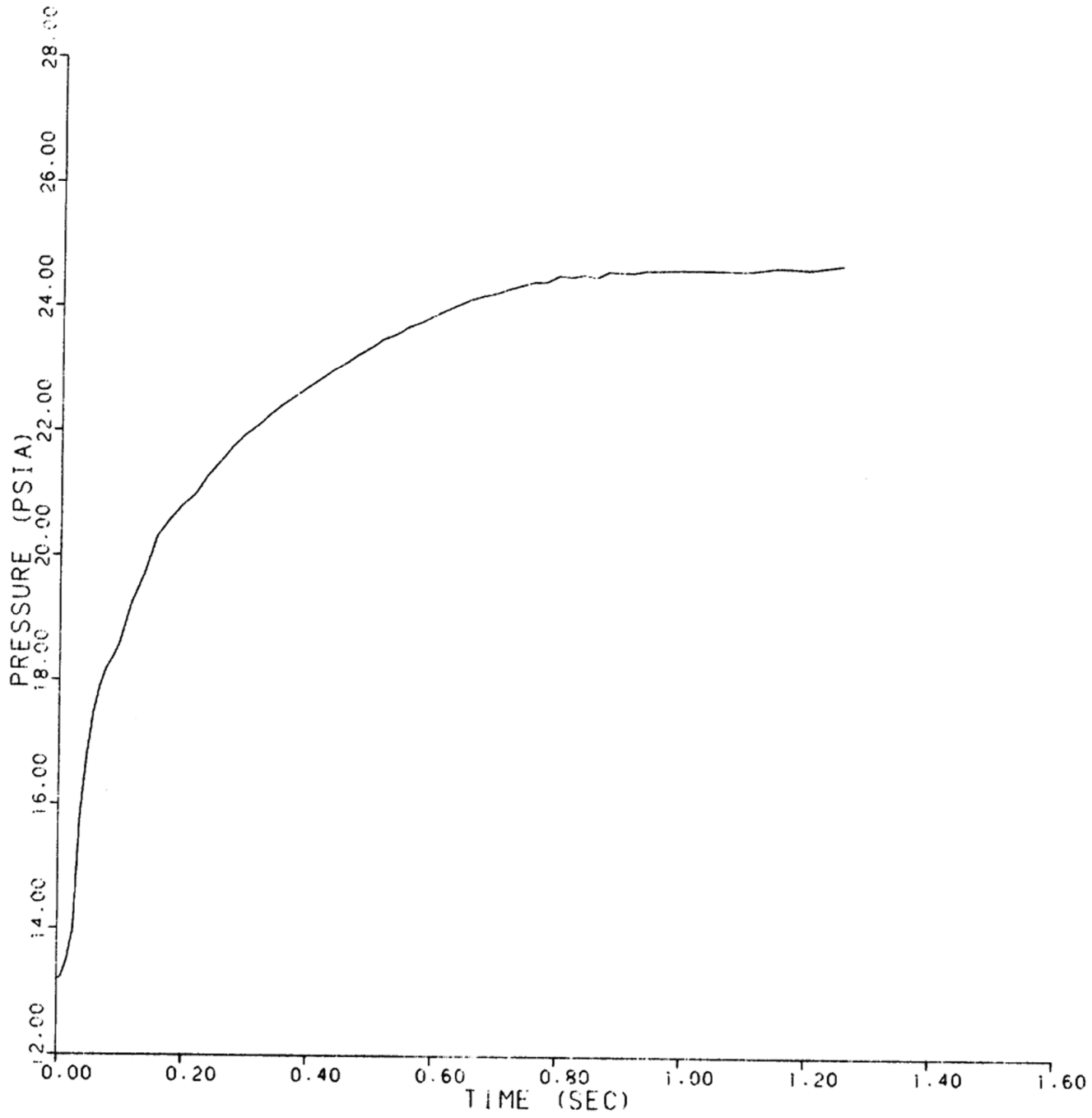
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E49
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 49 OF 74)



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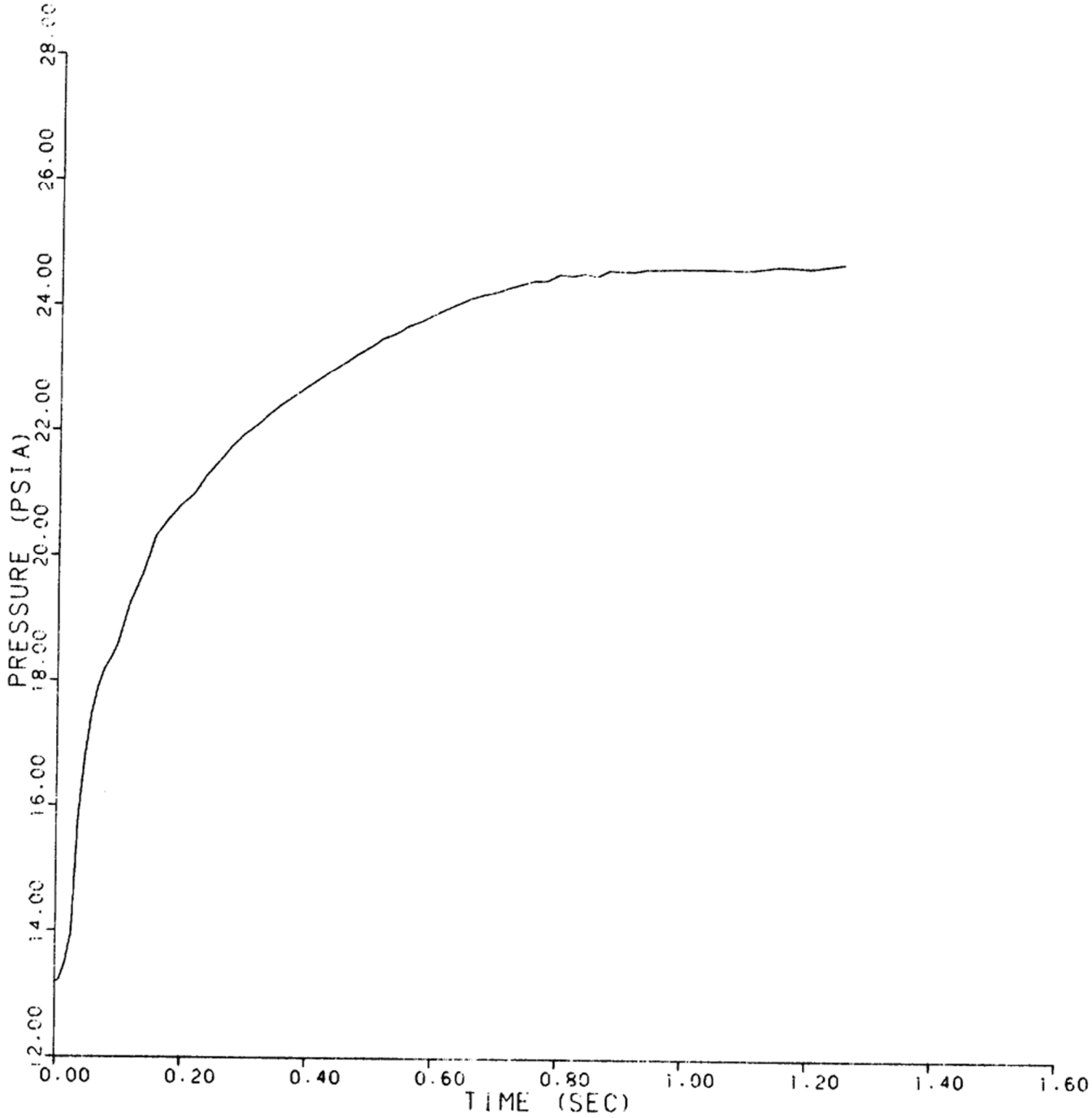
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E50
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 50 OF 74)



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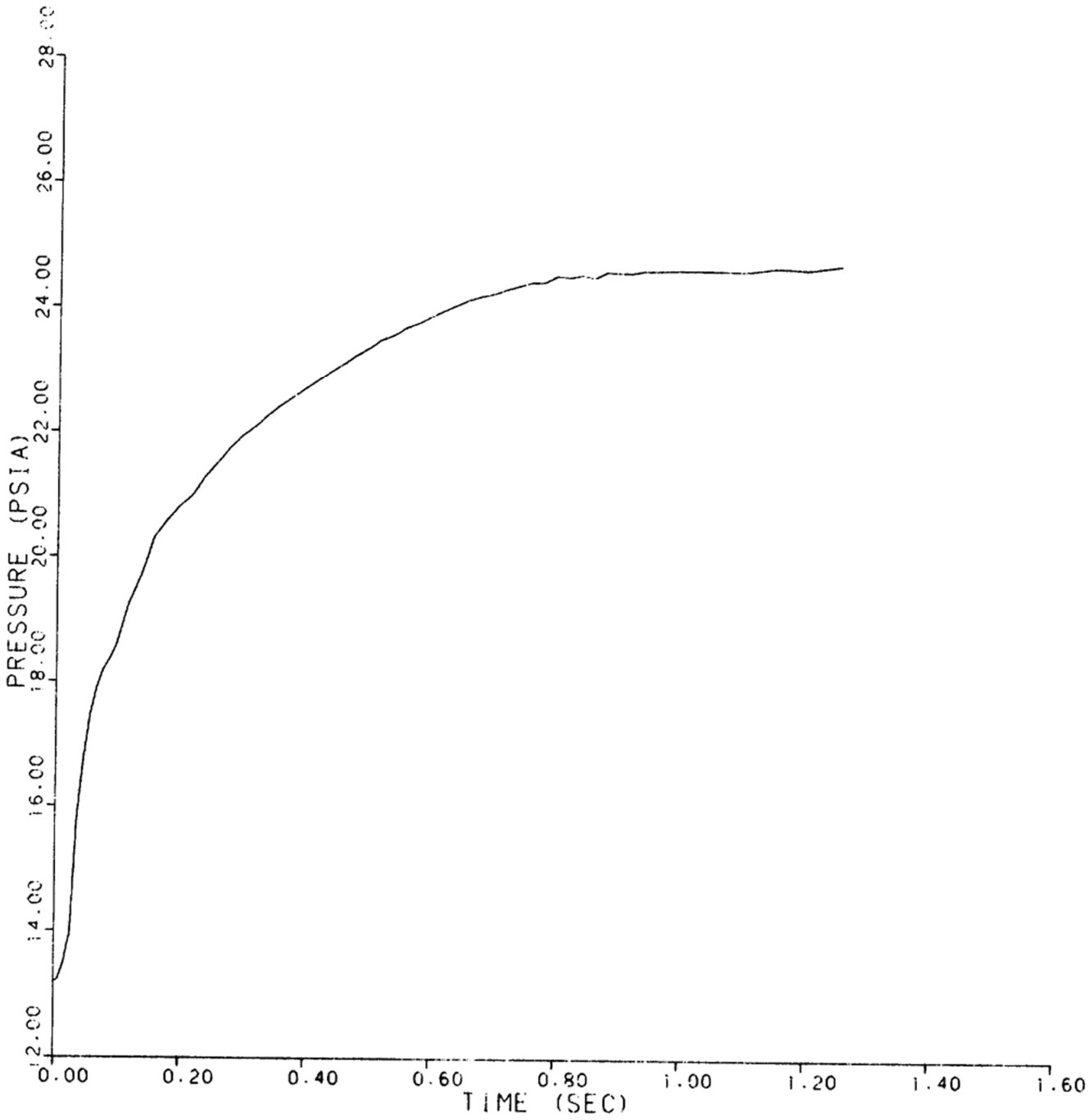
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E51 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 51 OF 74)



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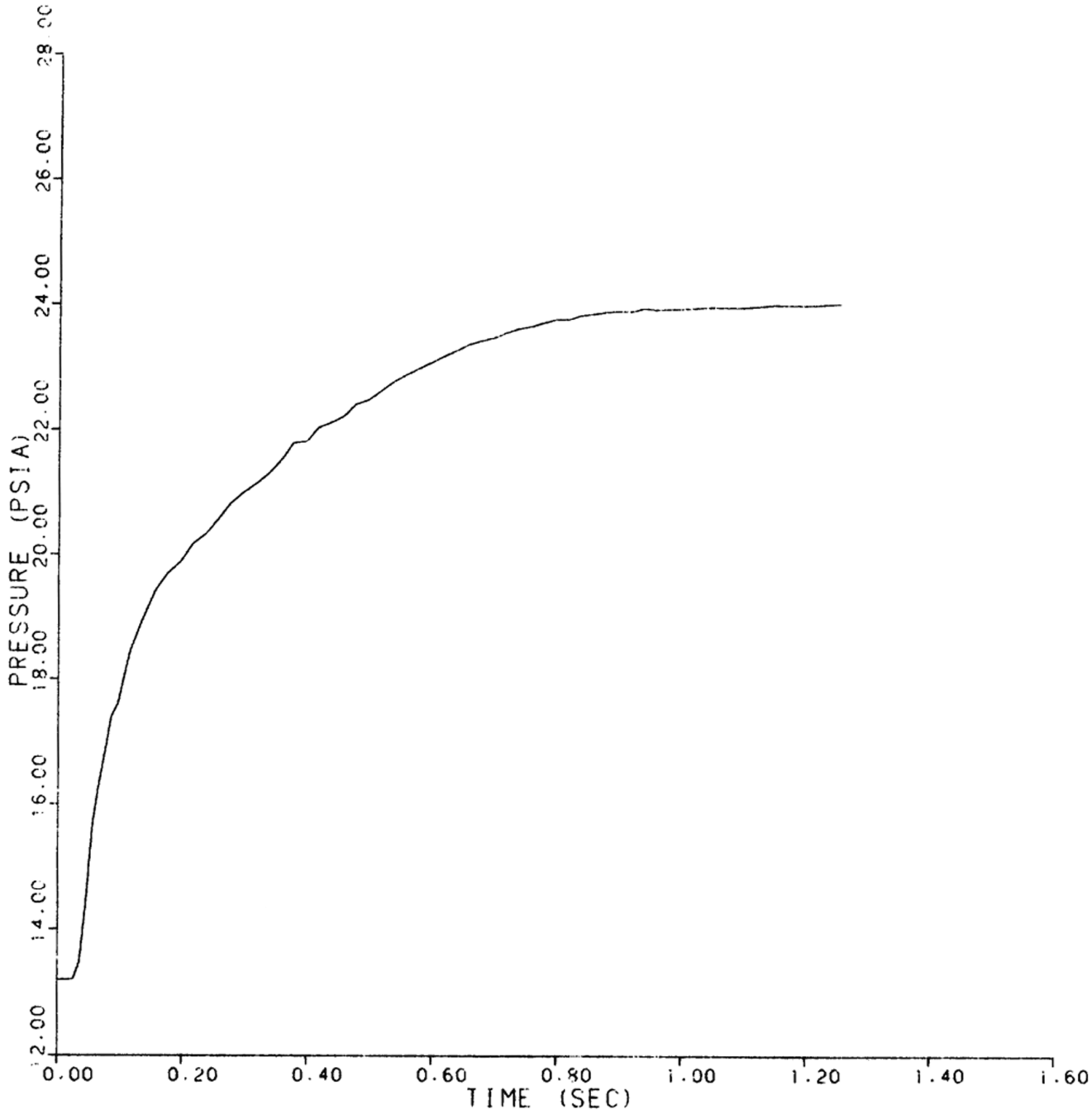
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E52
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 52 OF 74)



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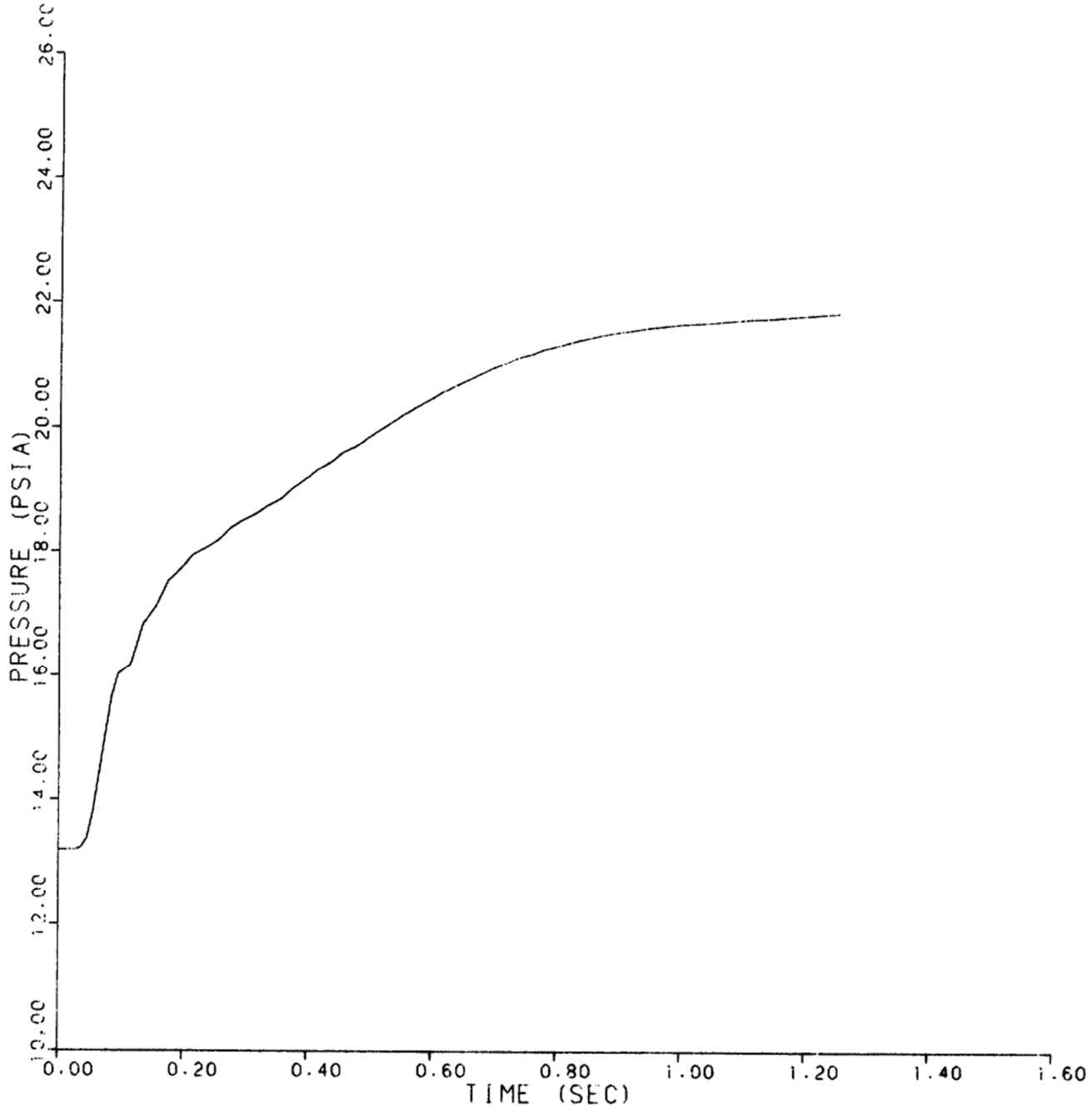
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E53 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 53 OF 74)



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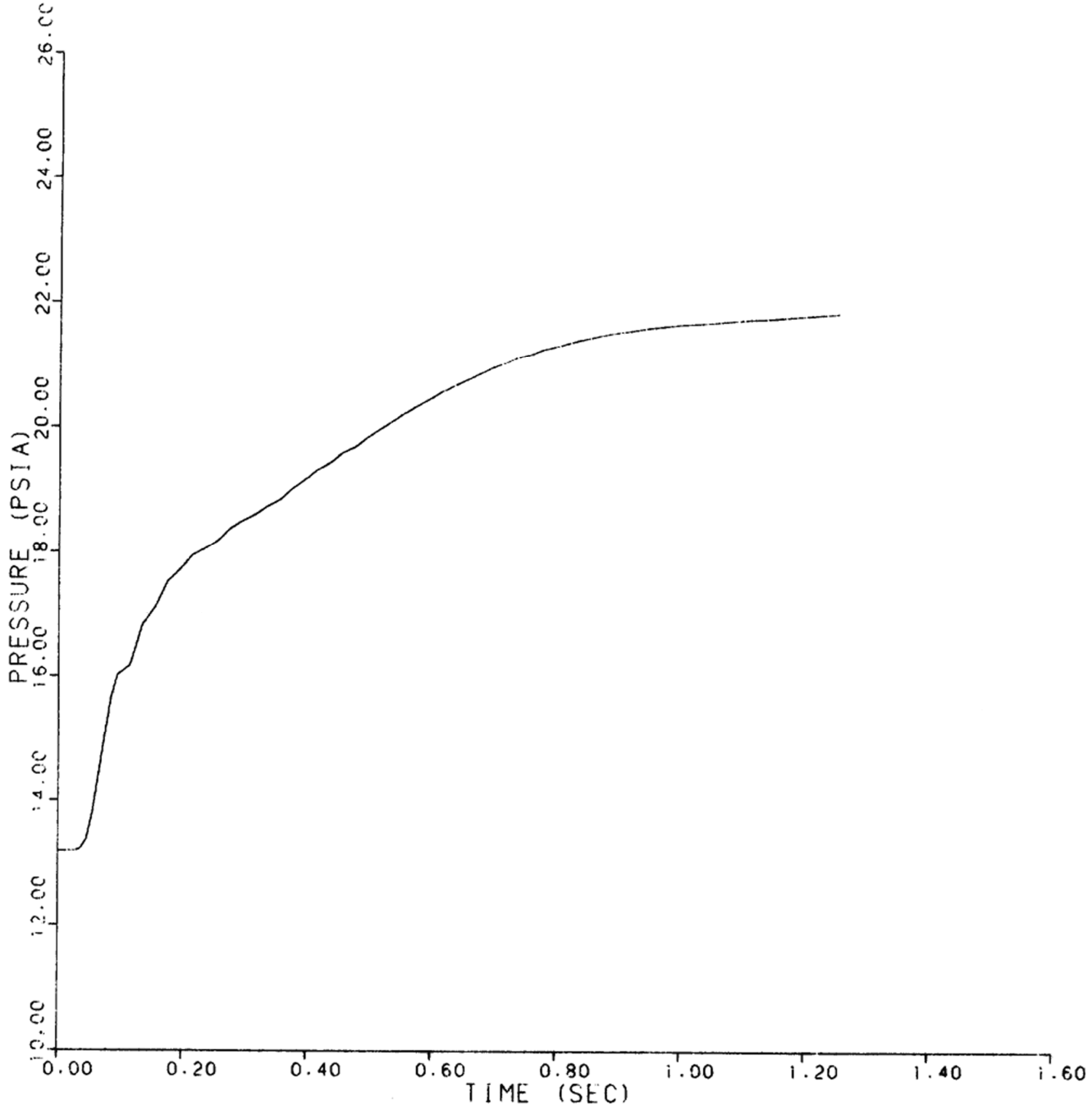
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E54
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 54 OF 74)



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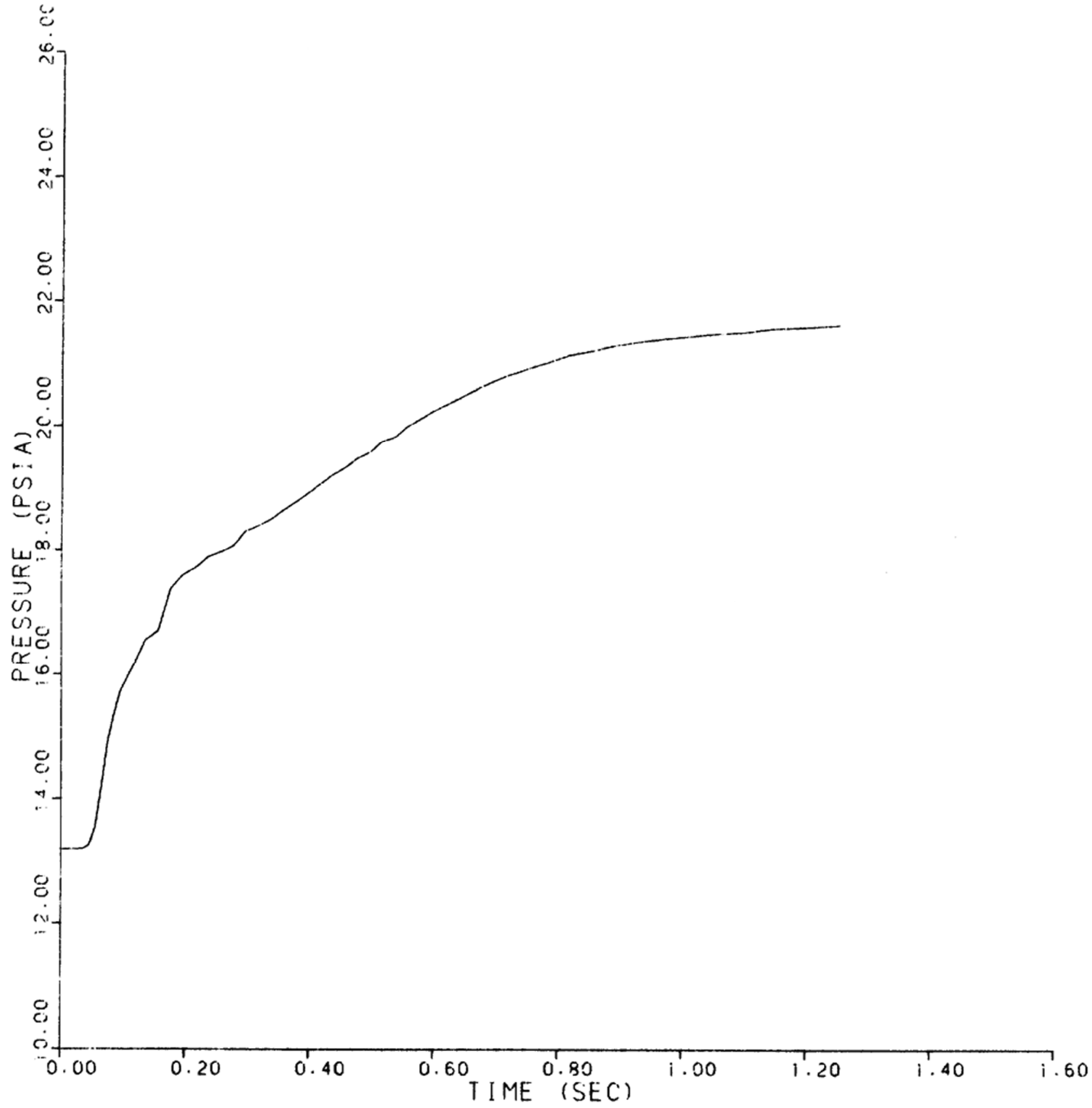
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E55
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 55 OF 74)



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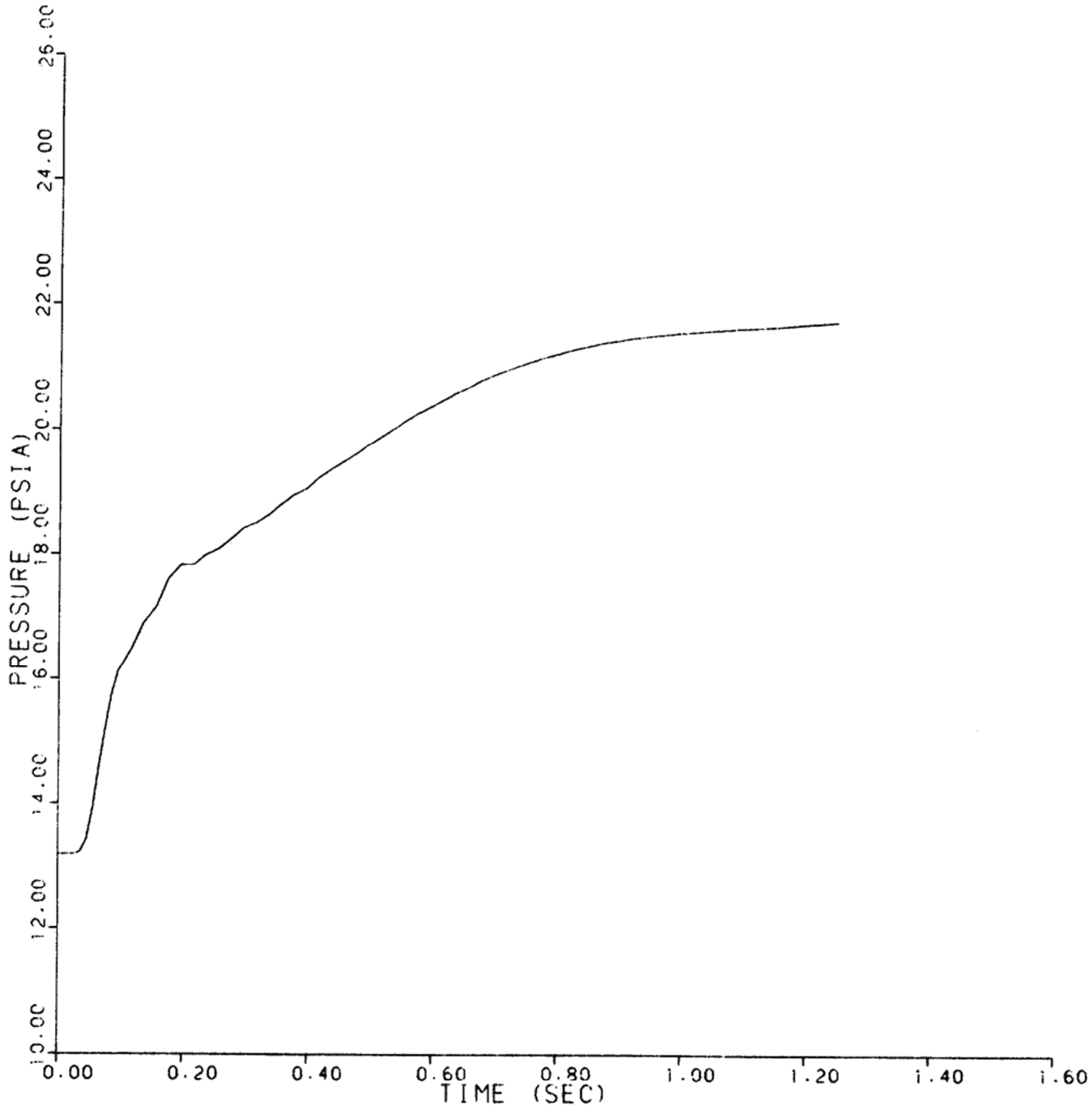
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E56 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 56 OF 74)



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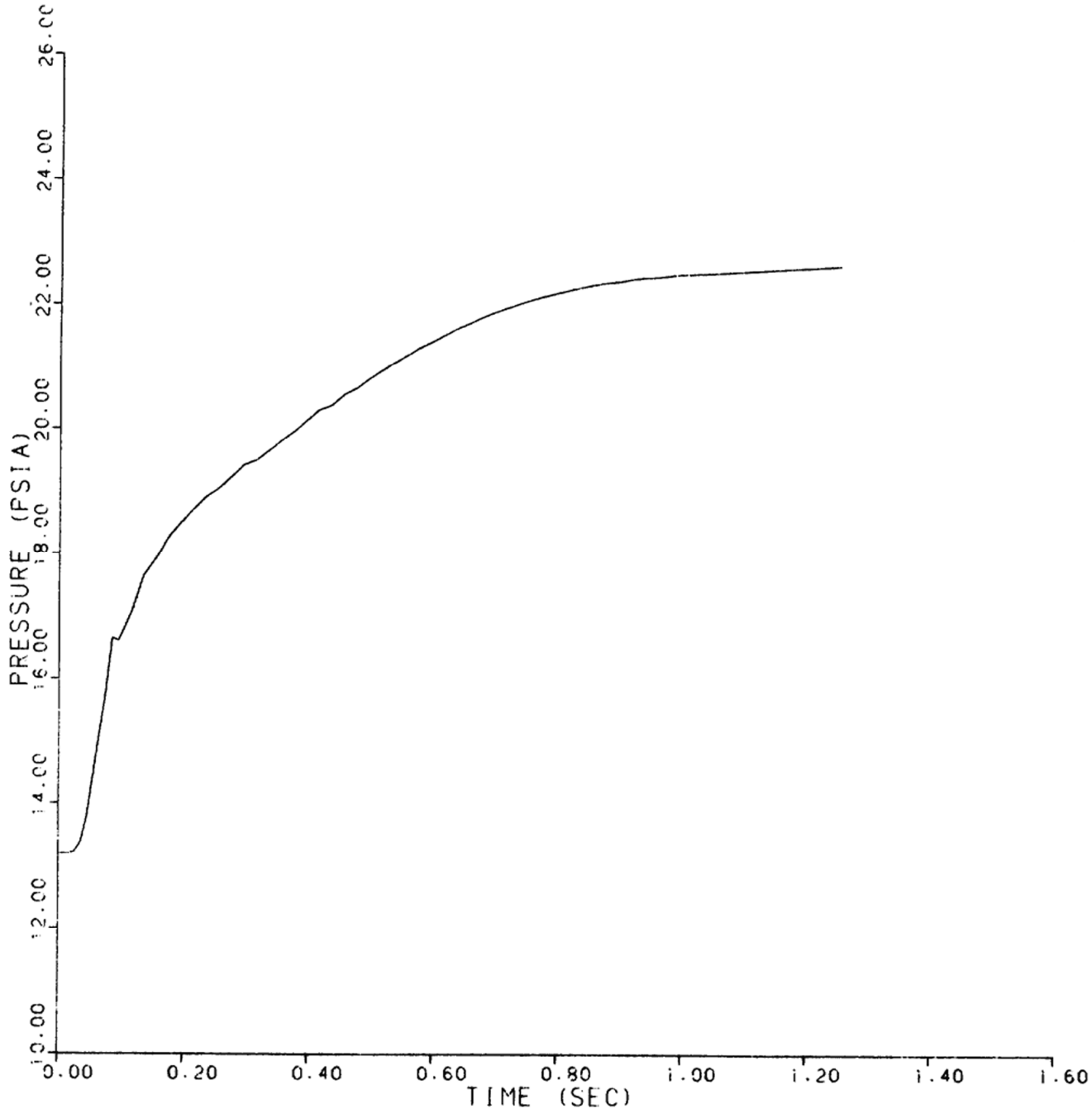
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E57
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 57 OF 74)



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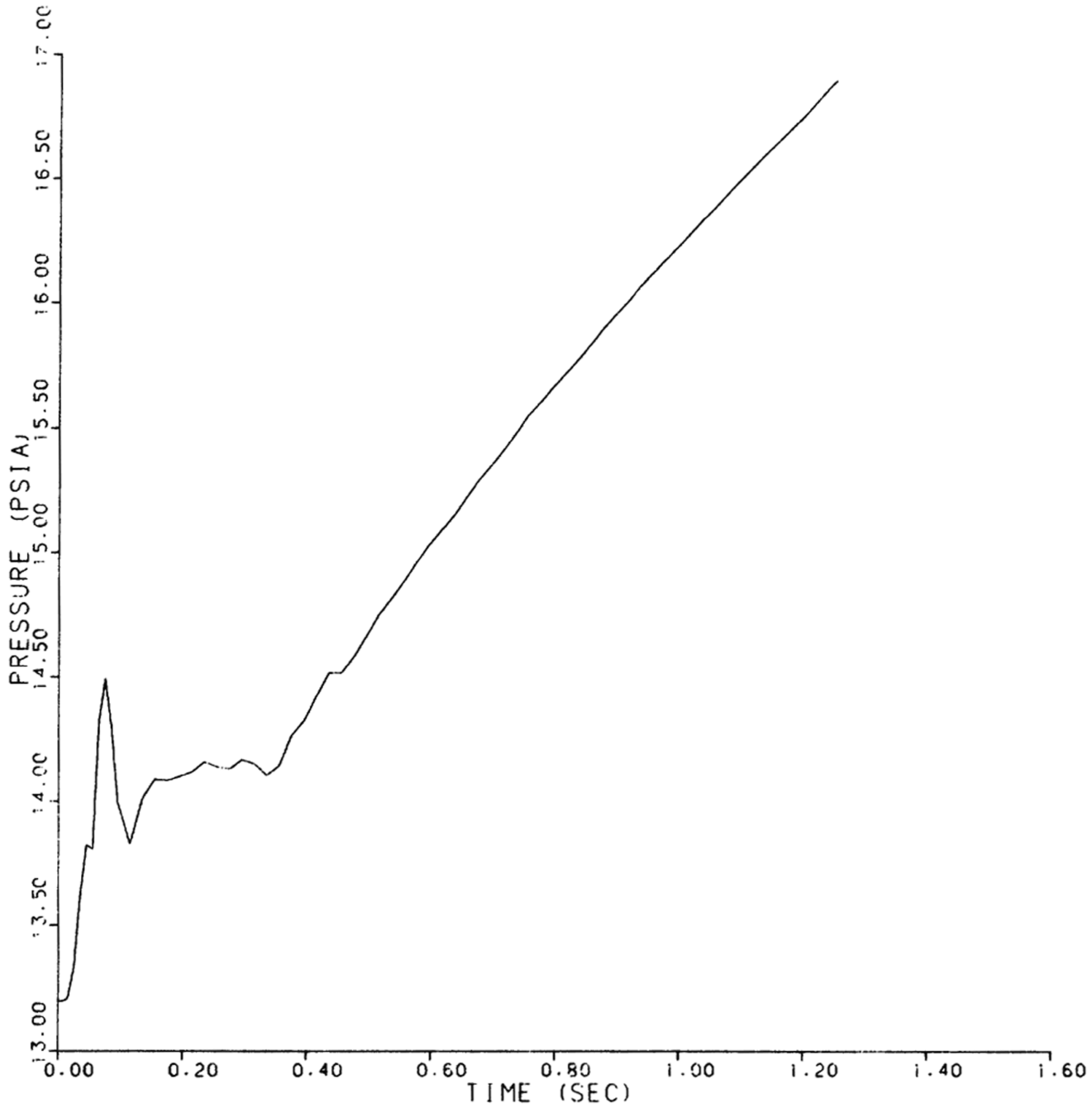
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E58
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 58 OF 74)



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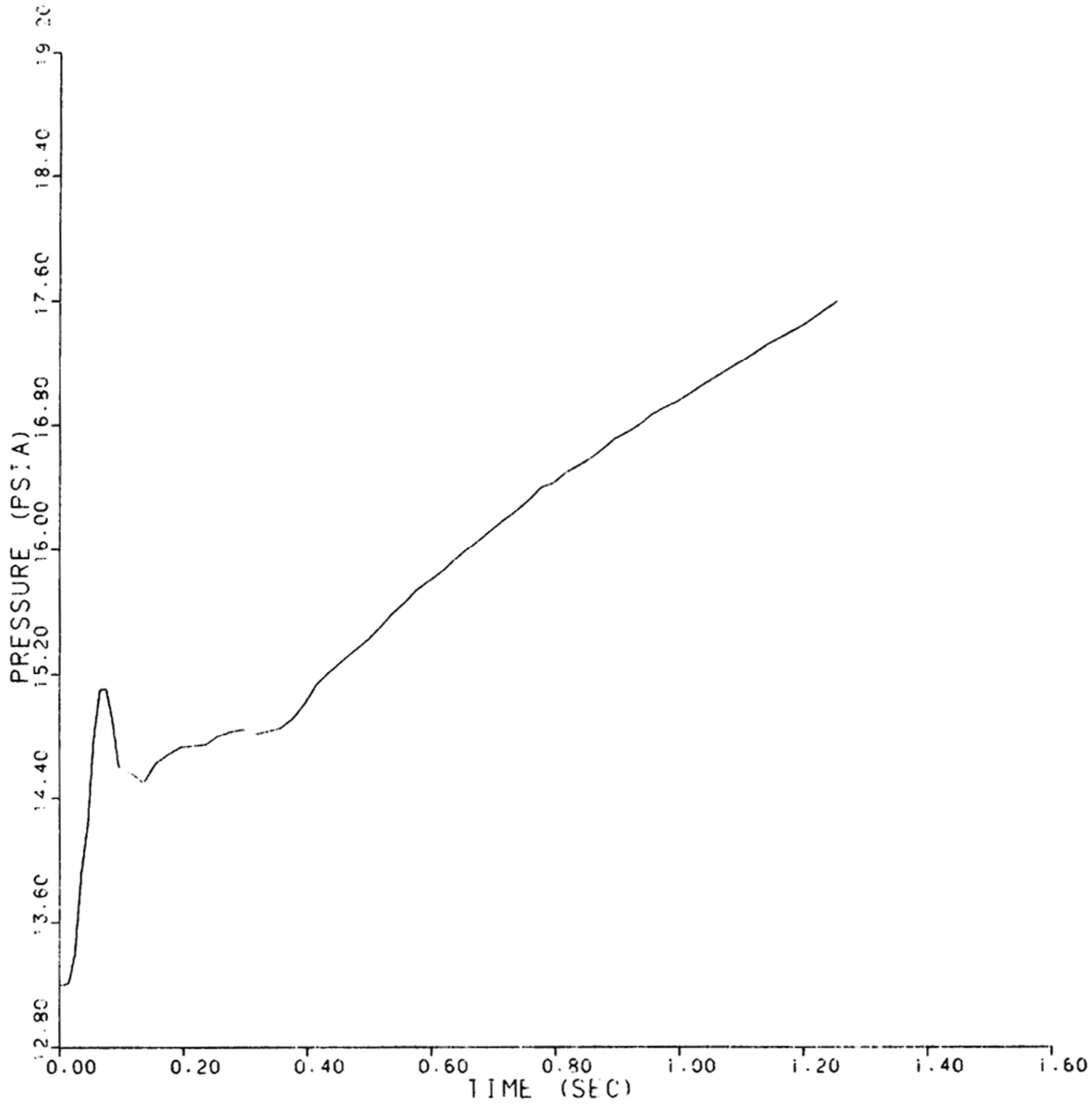
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E59 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 59 OF 74)



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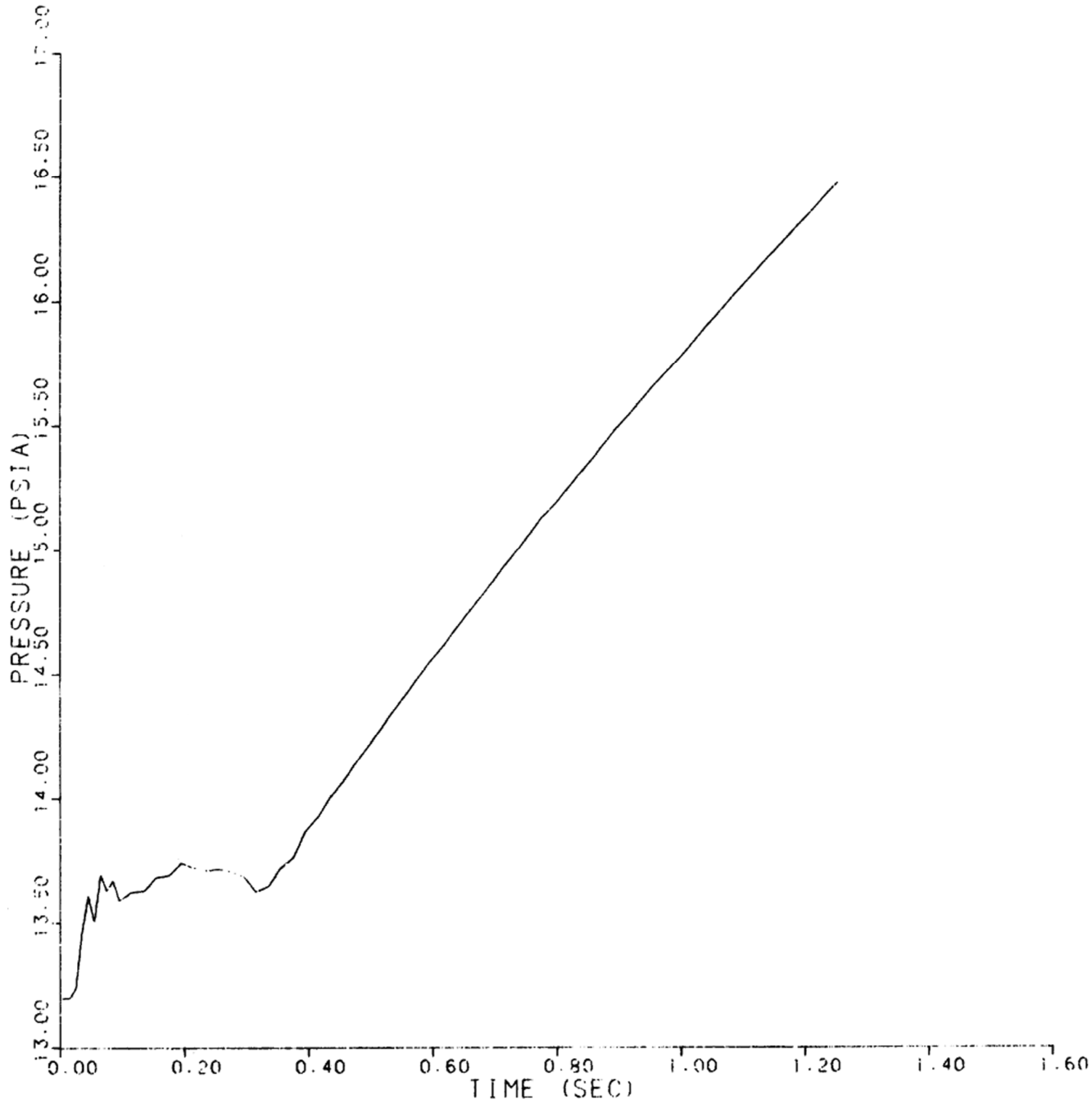
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E60 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 60 OF 74)



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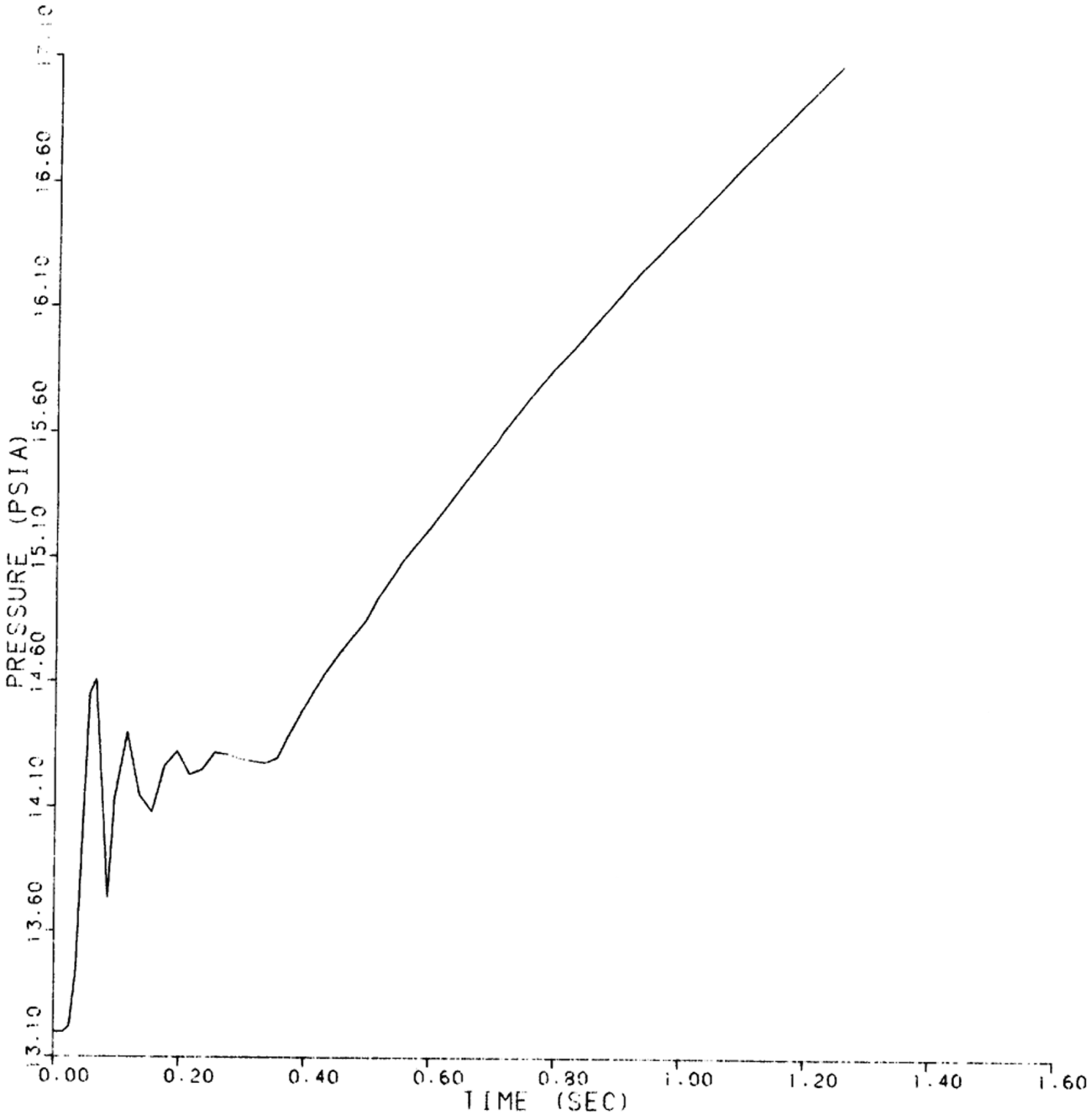
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E61 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 61 OF 74)



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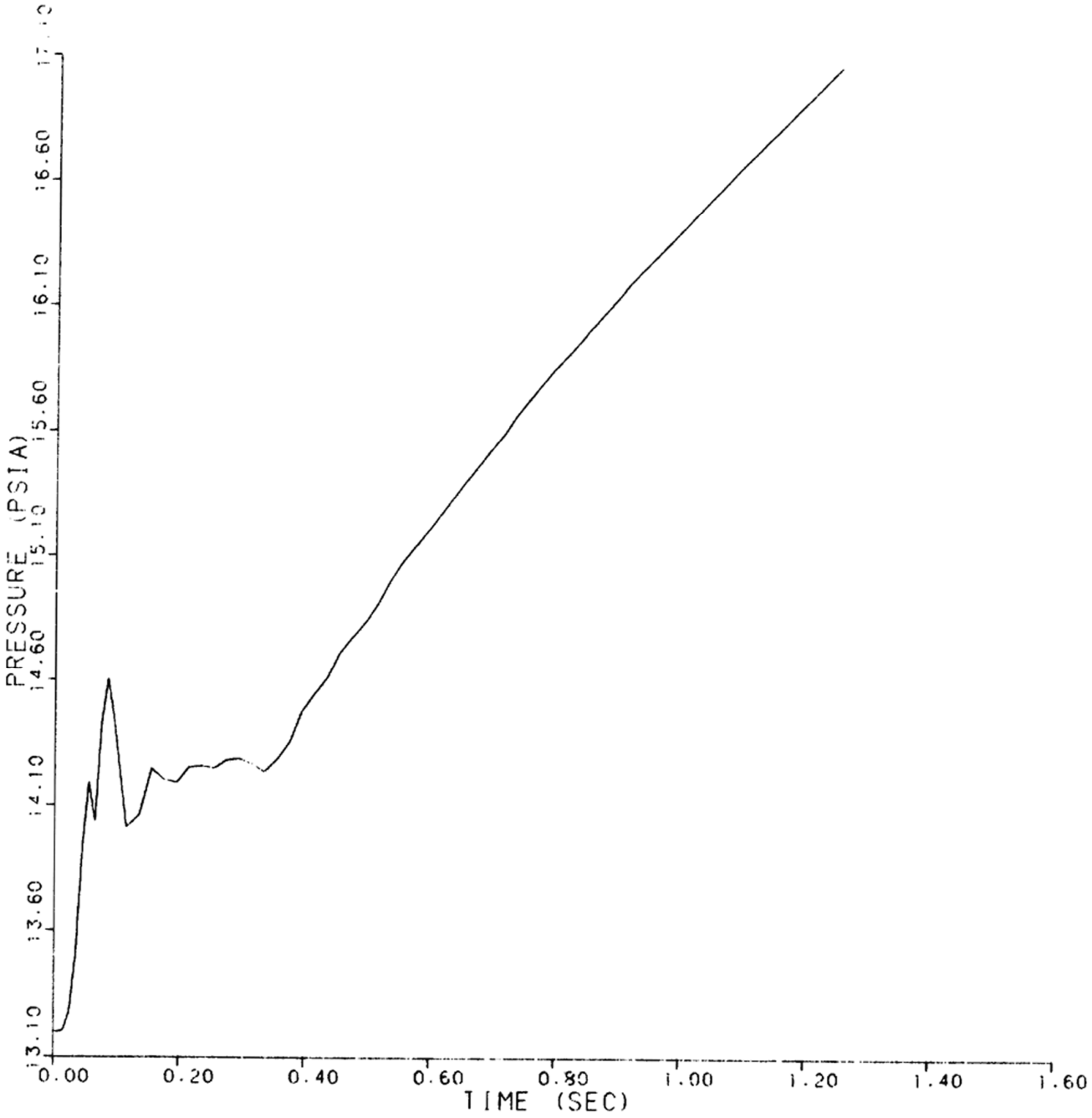
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E62 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 62 OF 74)



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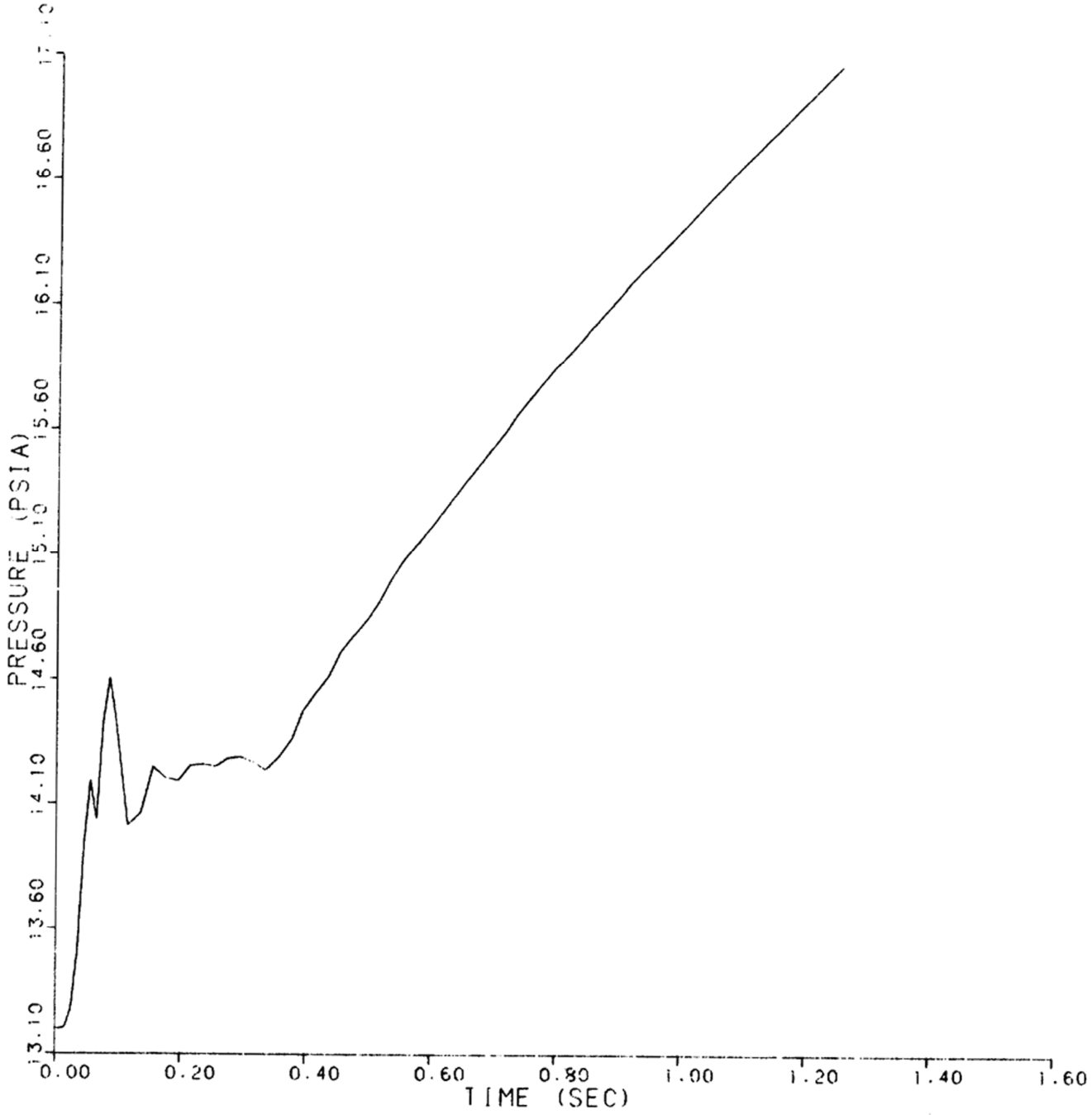
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E63 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 63 OF 74)



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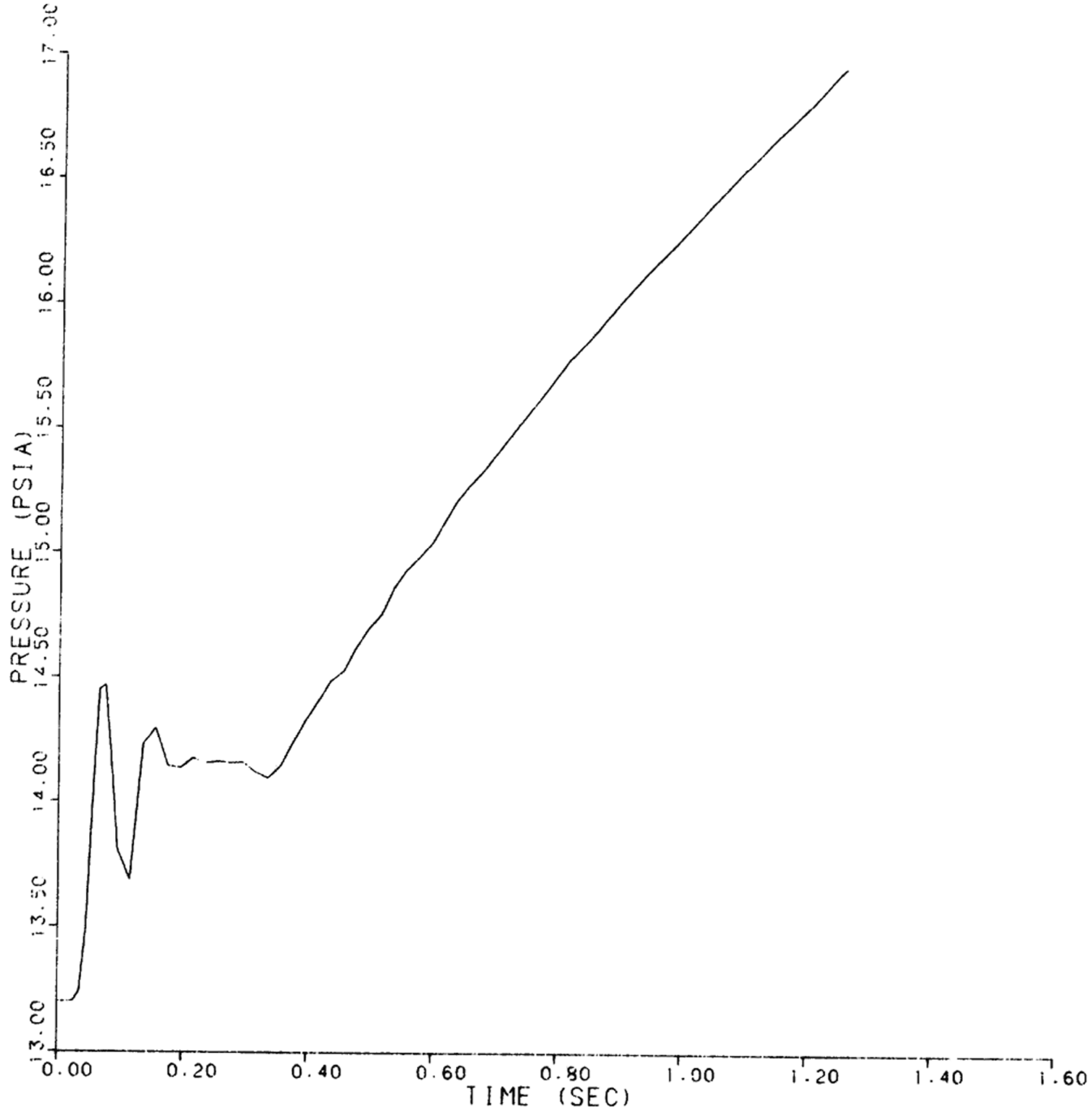
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E64
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 64 OF 74)



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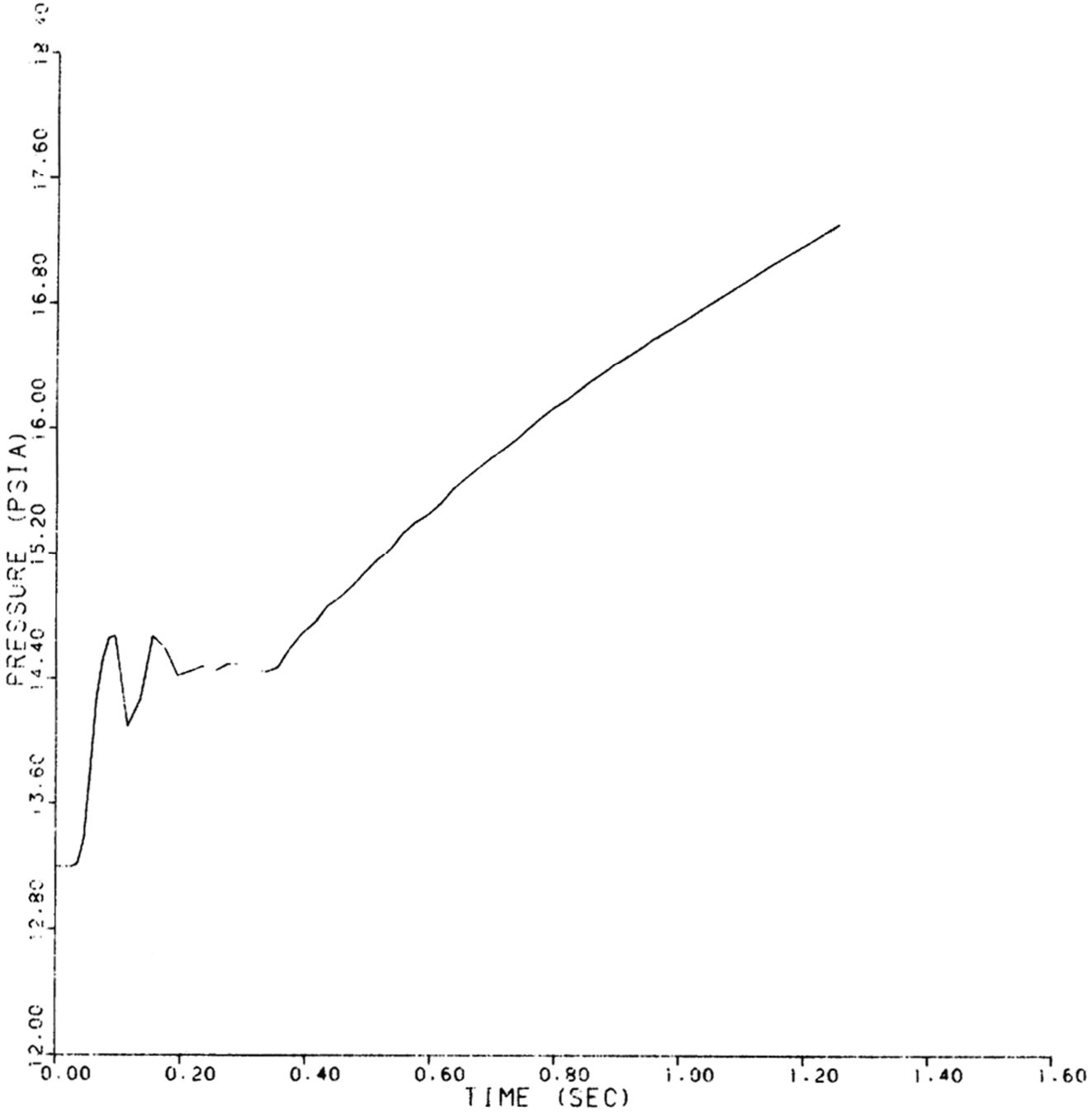
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E65 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 65 OF 74)



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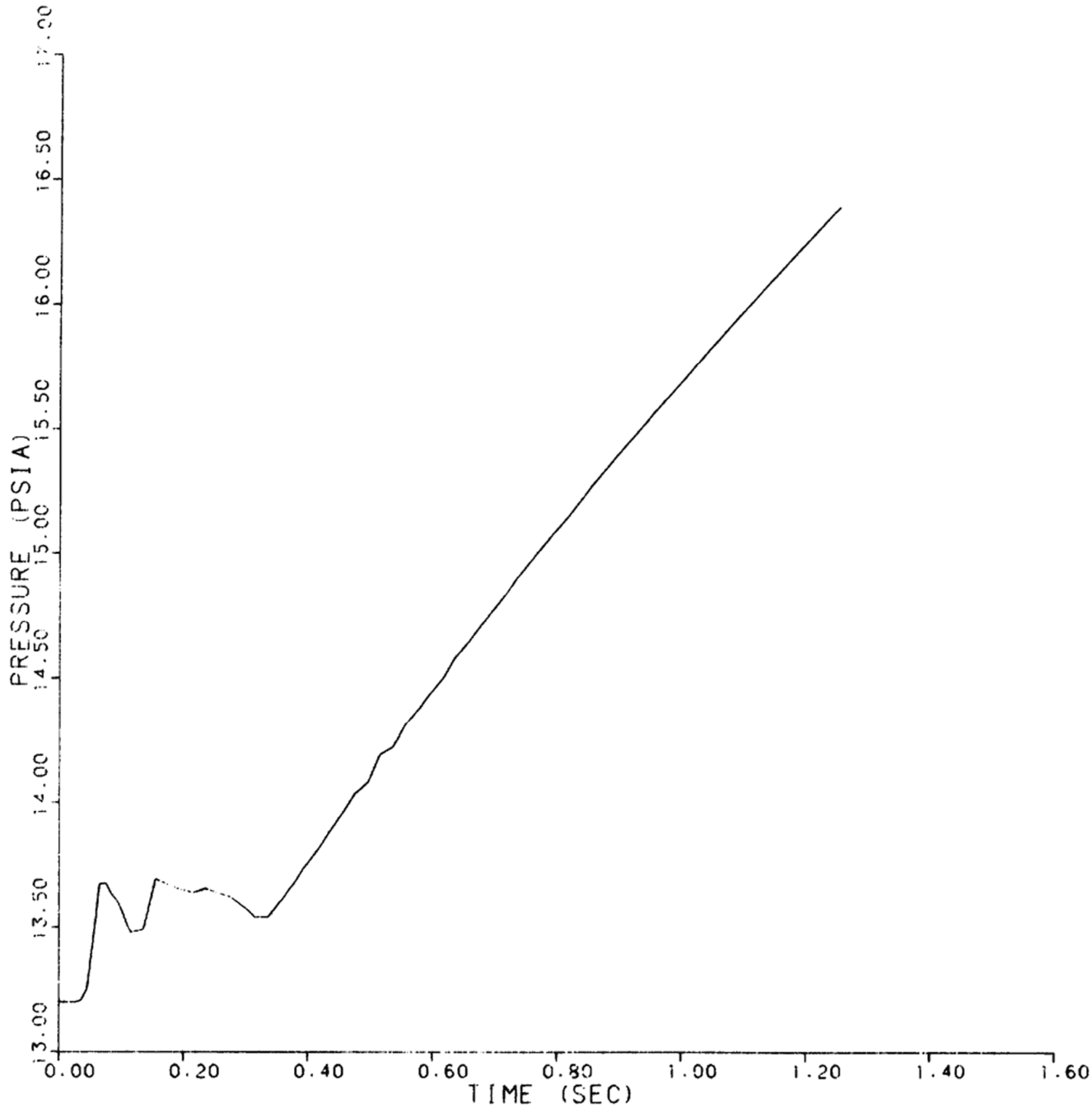
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E66
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 66 OF 74)



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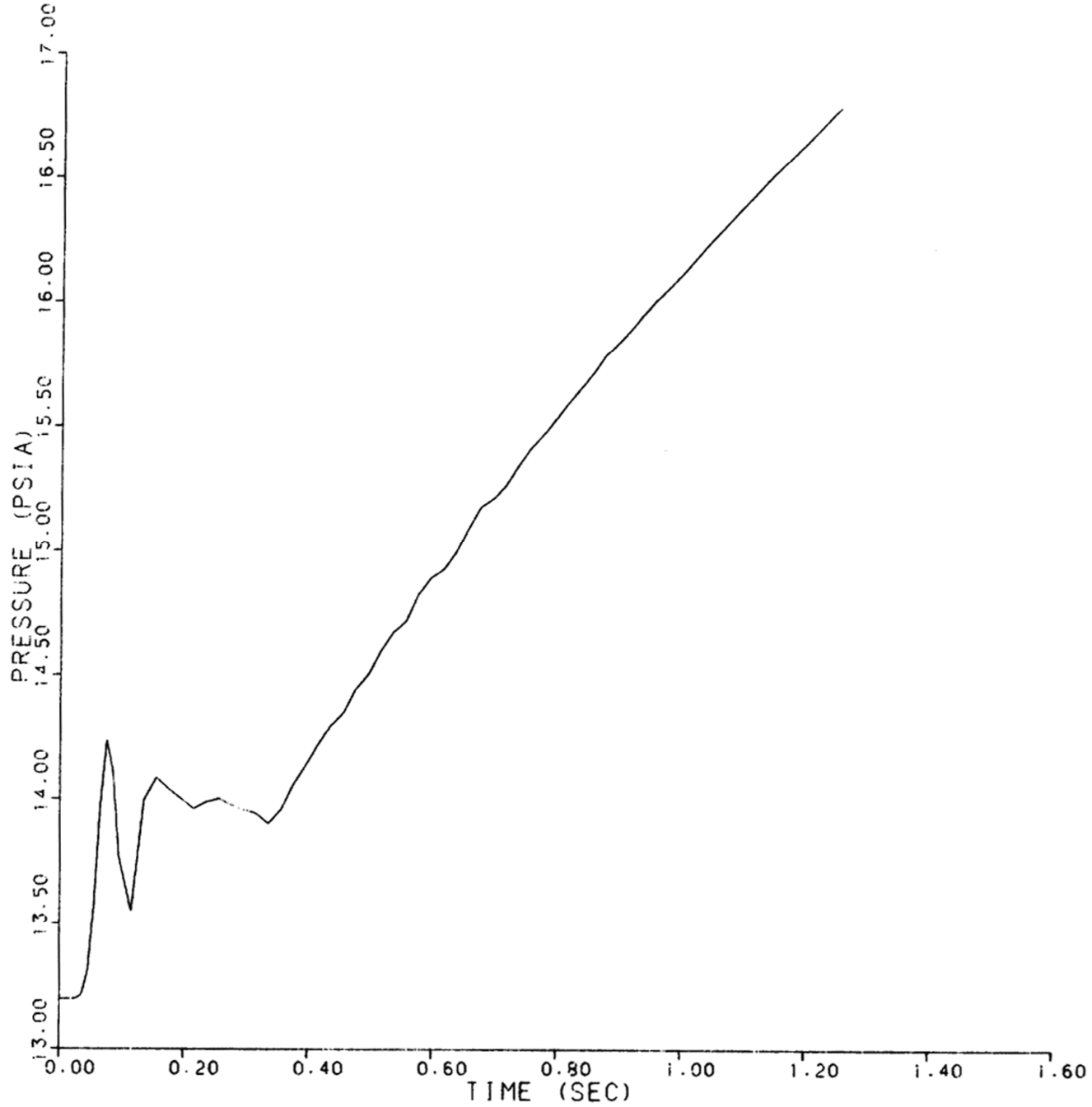
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E67
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 67 OF 74)



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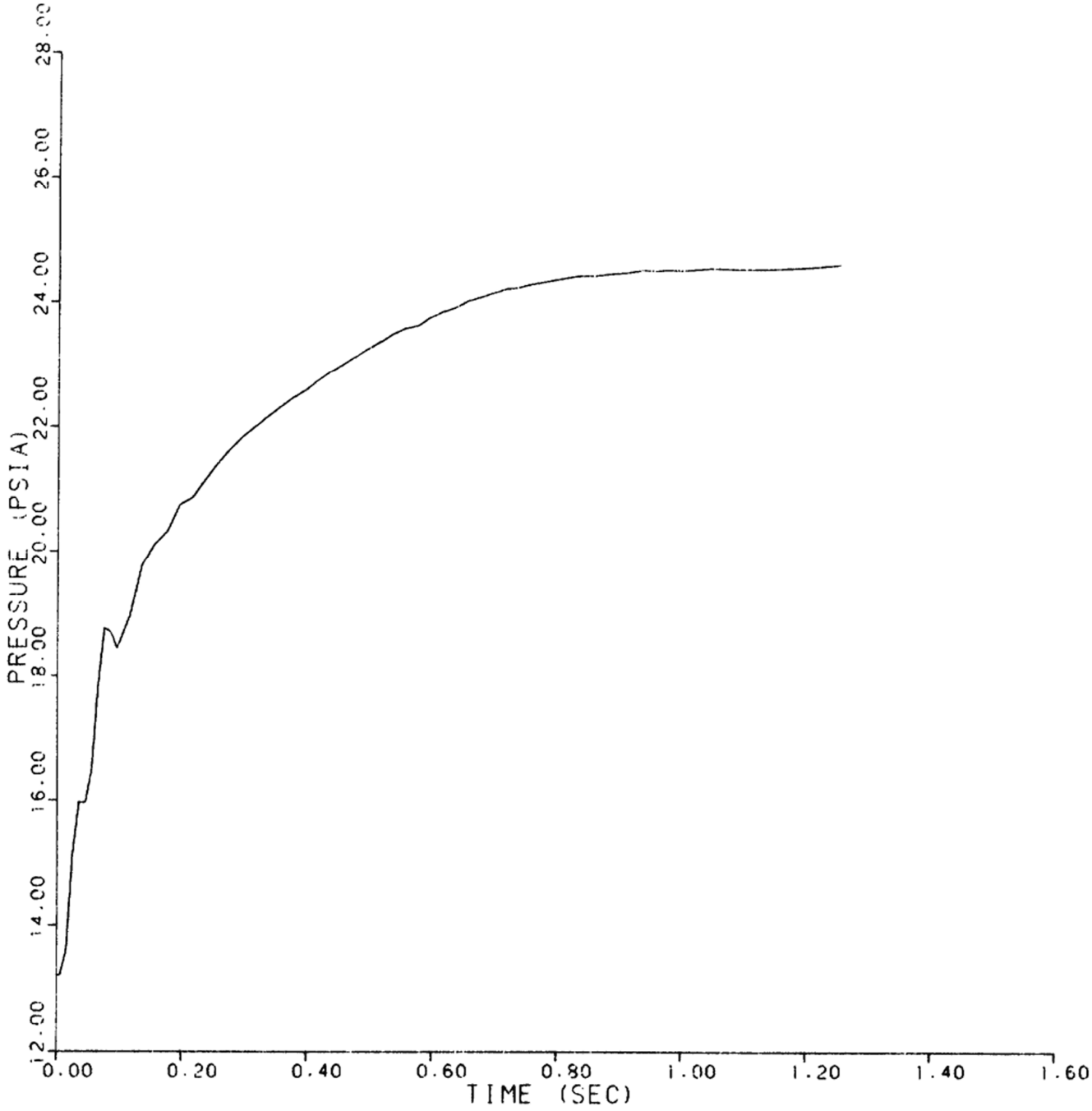
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E68 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 68 OF 74)



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SOUTHERN COMPANY
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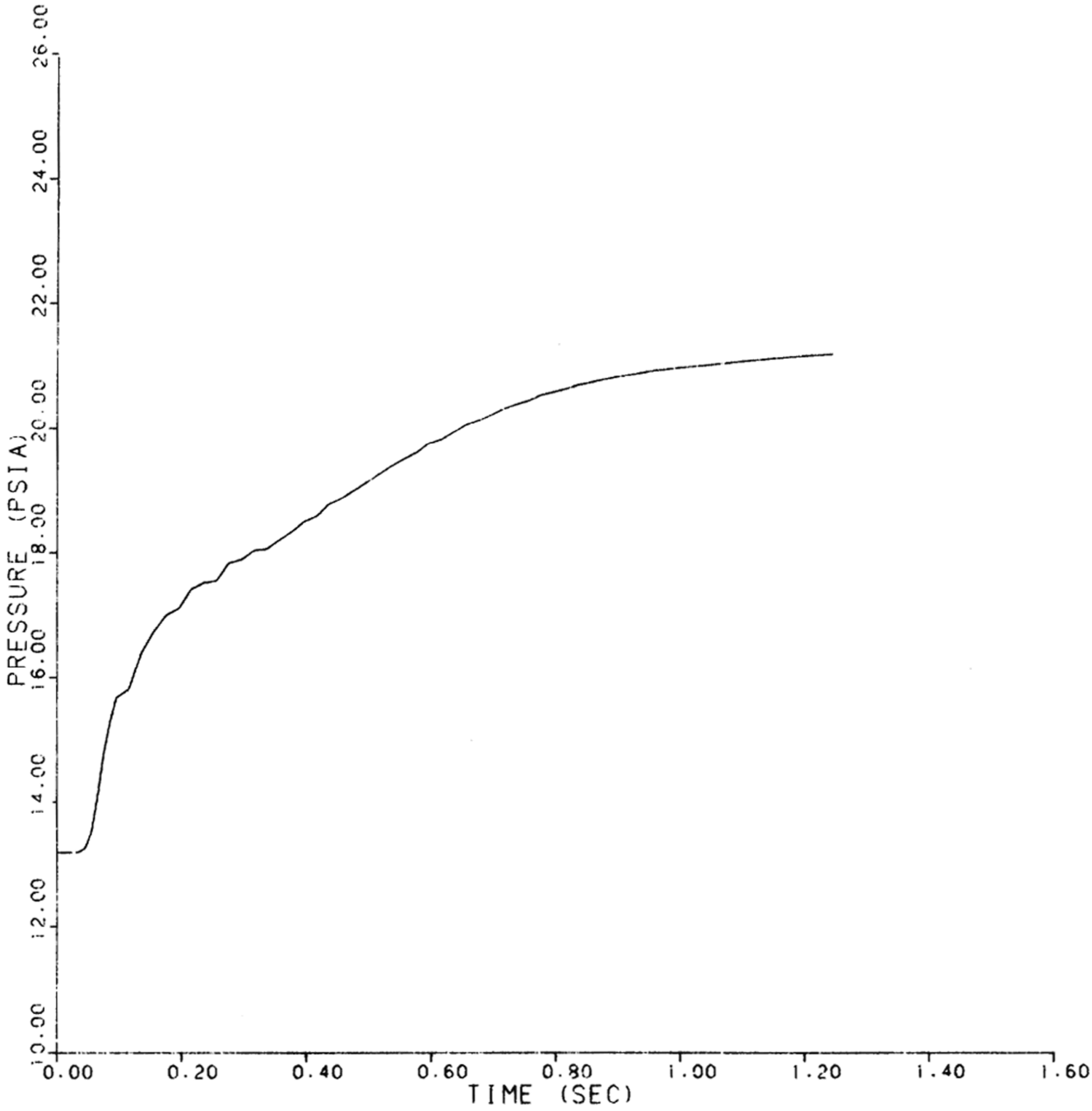
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E69
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 69 OF 74)



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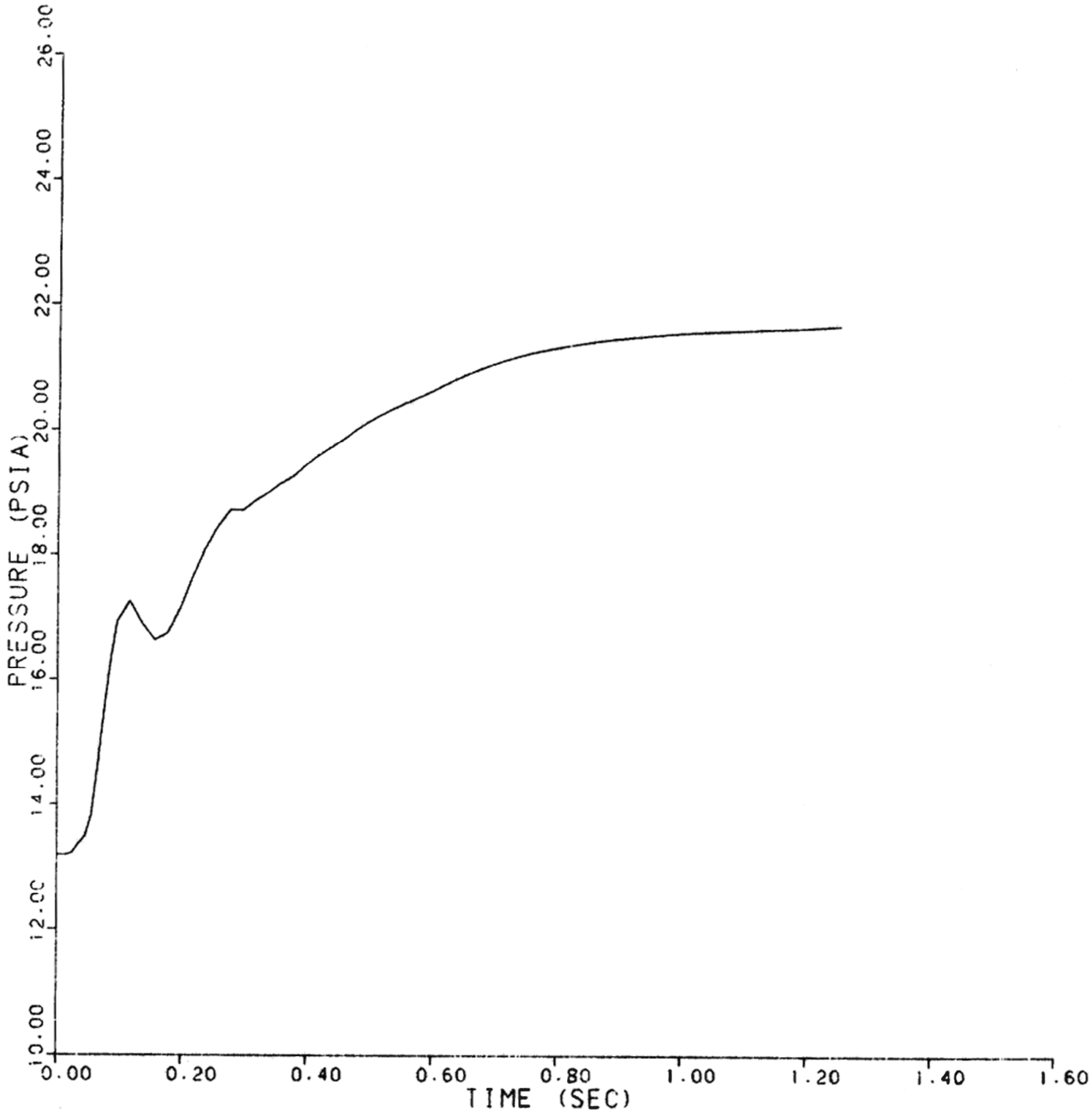
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E70 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 70 OF 74)



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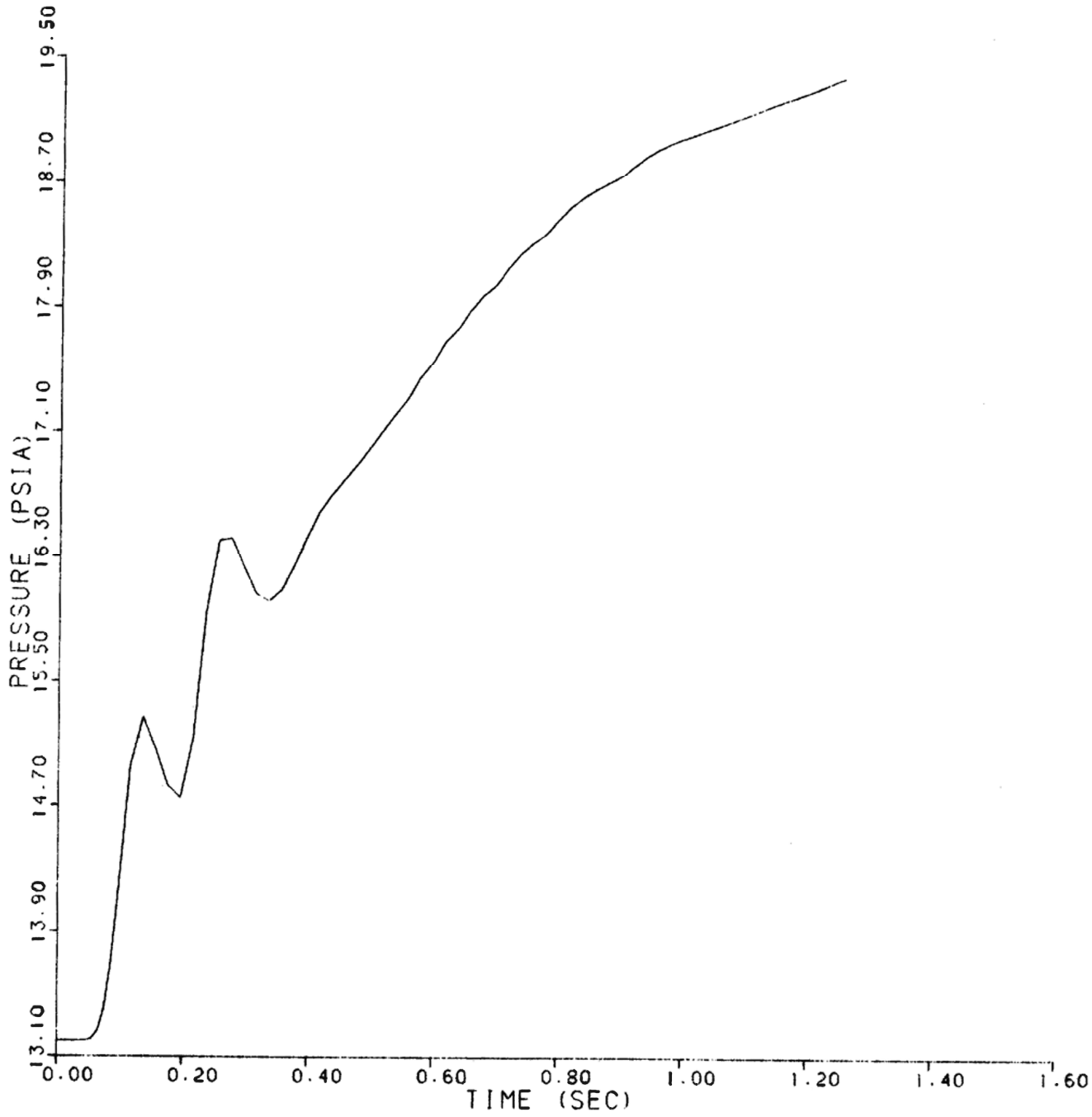
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E71 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 71 OF 74)



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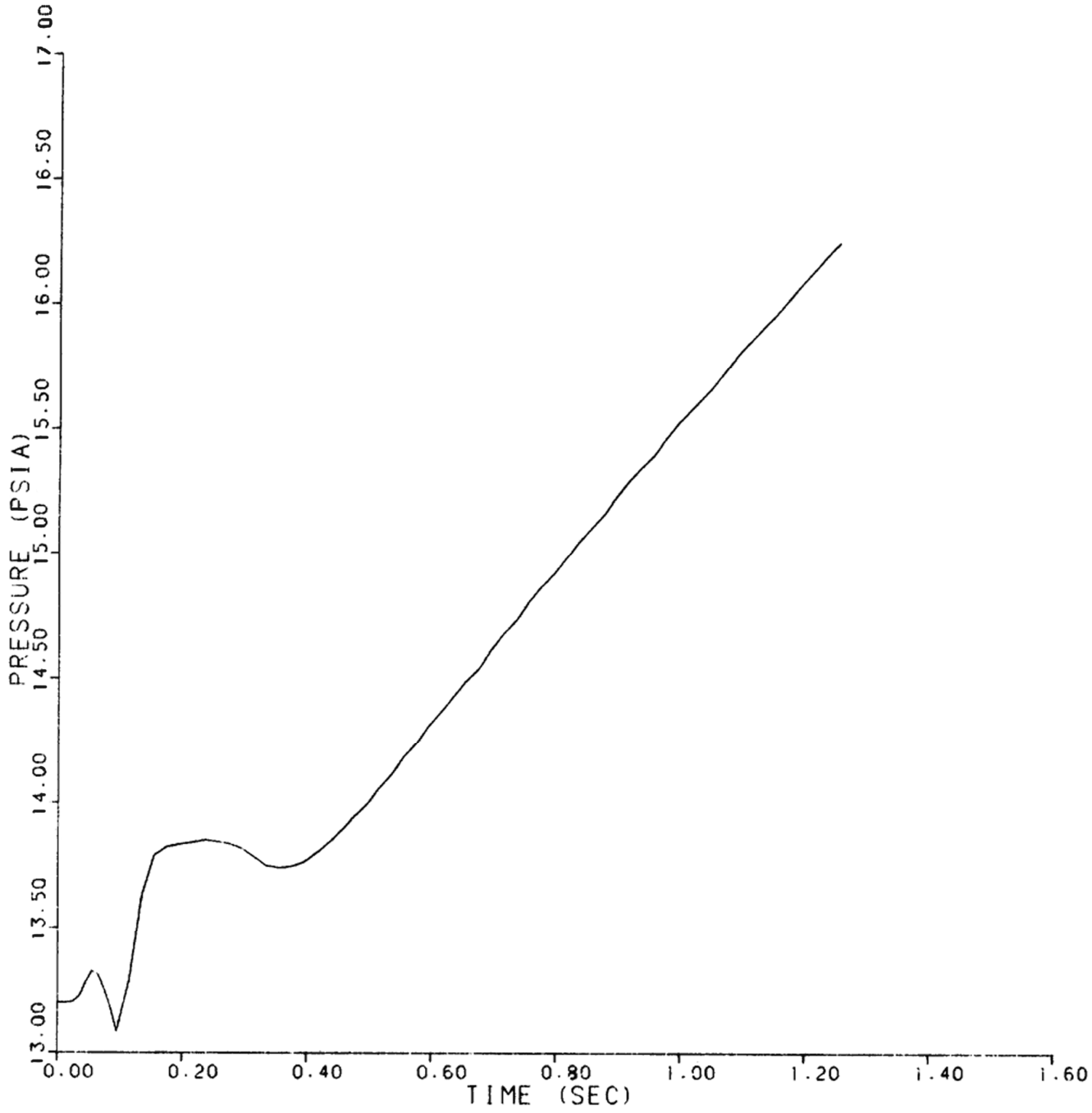
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

REACTOR COOLANT PUMP
INLET NOZZLE – NODE E72
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–20 (SHEET 72 OF 74)



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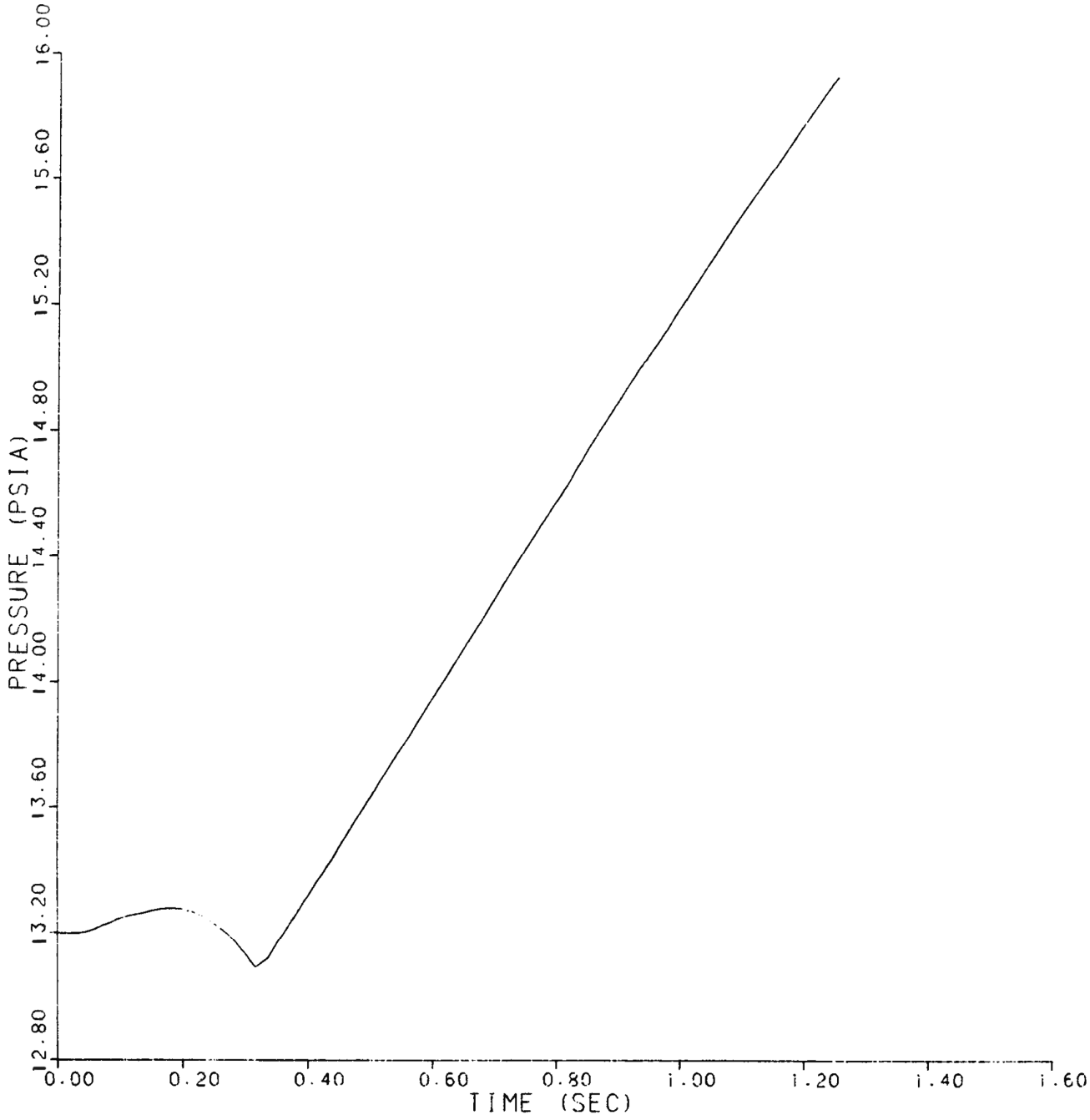
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E73 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 73 OF 74)



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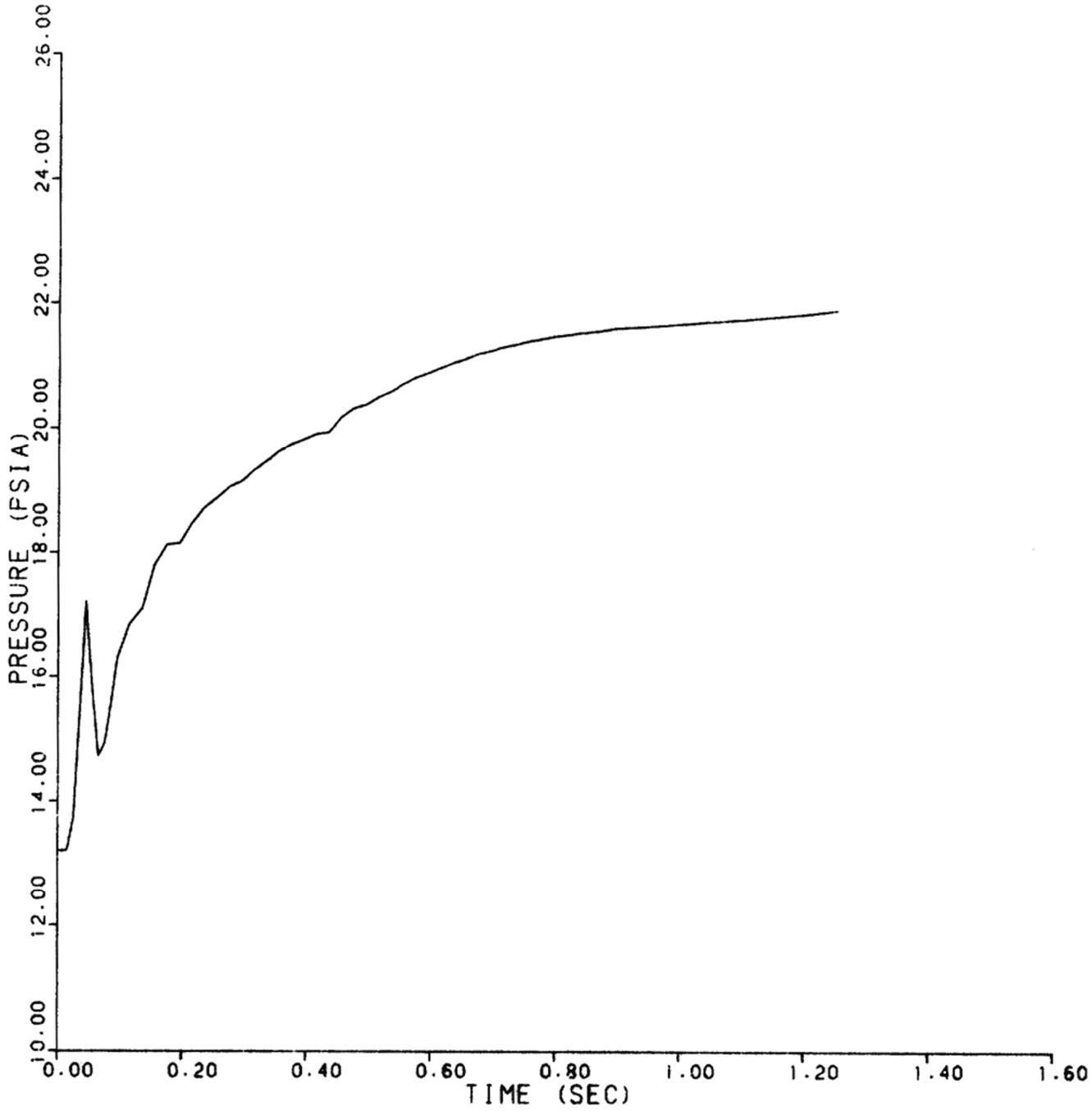
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	REACTOR COOLANT PUMP INLET NOZZLE – NODE E74 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–20 (SHEET 74 OF 74)



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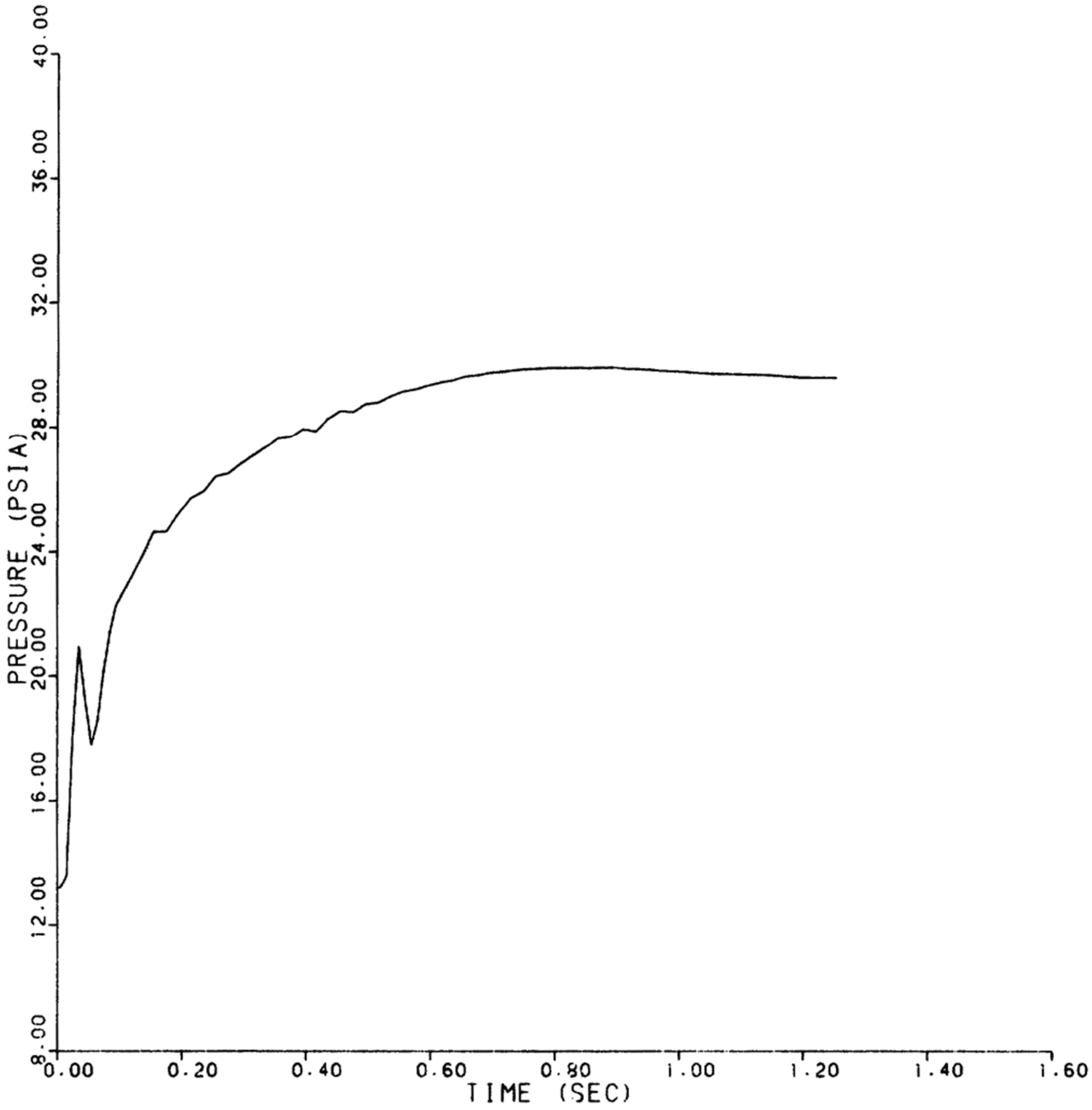
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E1
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 1 OF 74)



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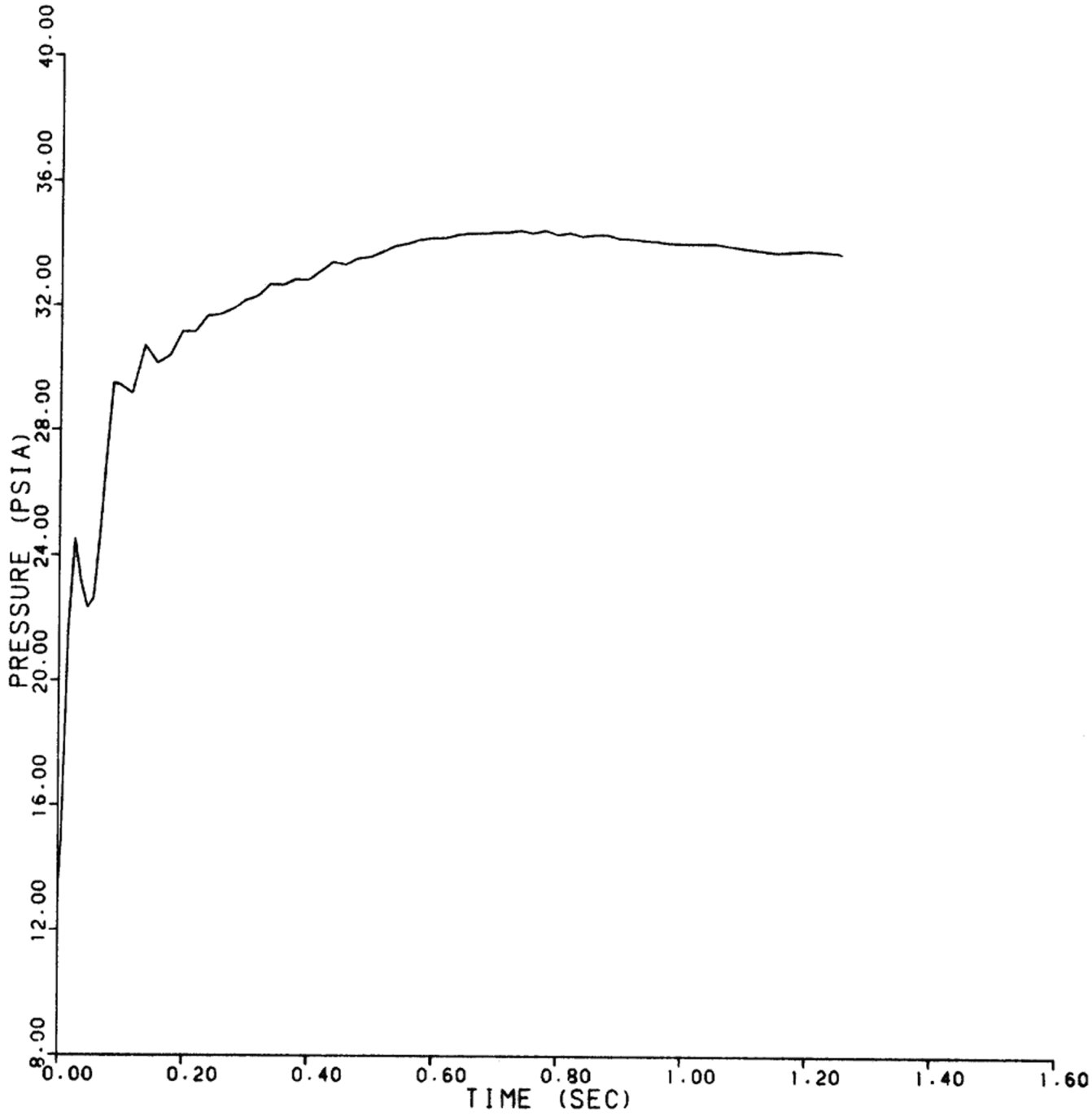
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E2
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 2 OF 74)



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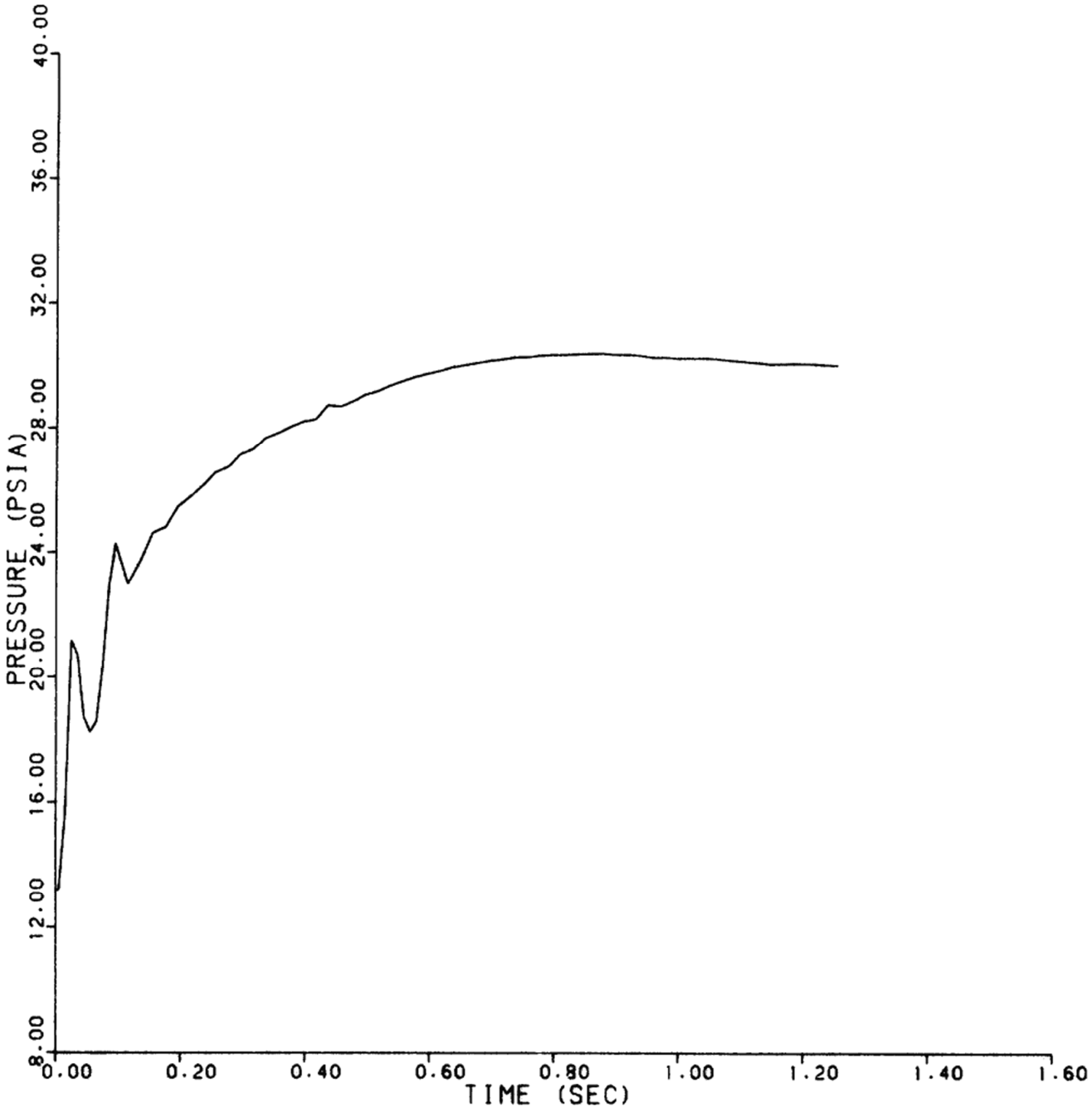
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E3 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 3 OF 74)



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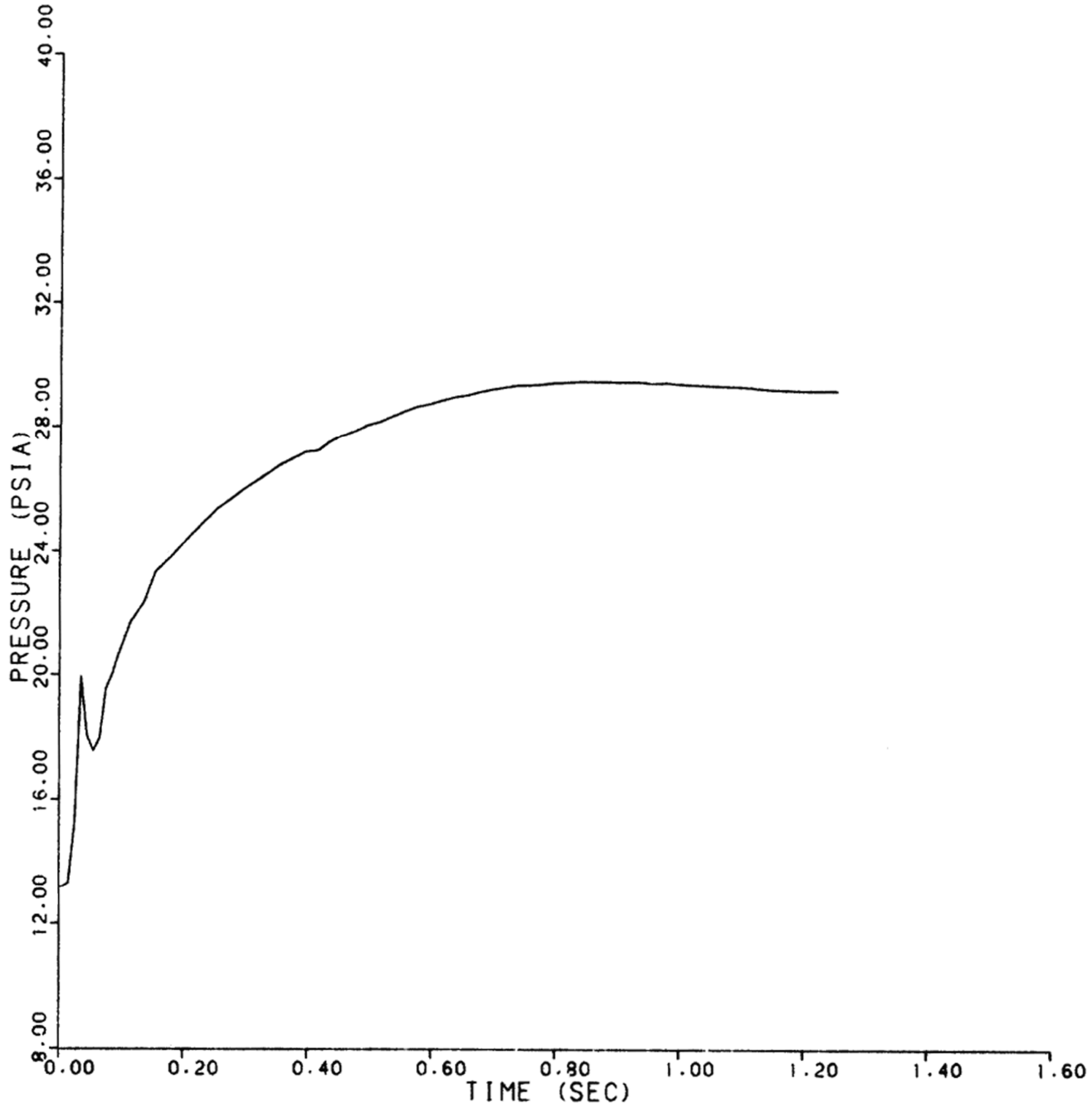
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E4 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 4 OF 74)



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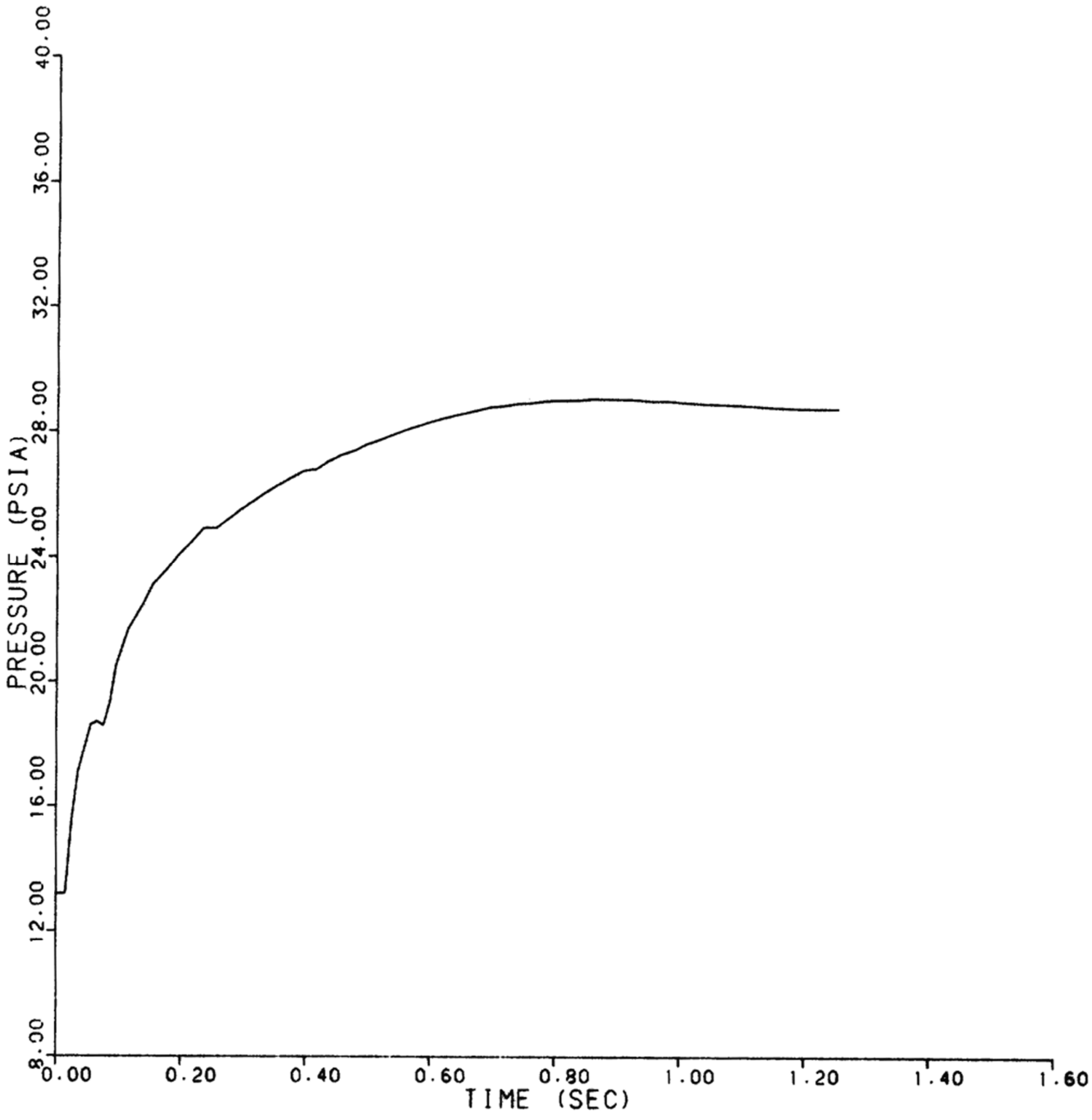
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E5 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 5 OF 74)



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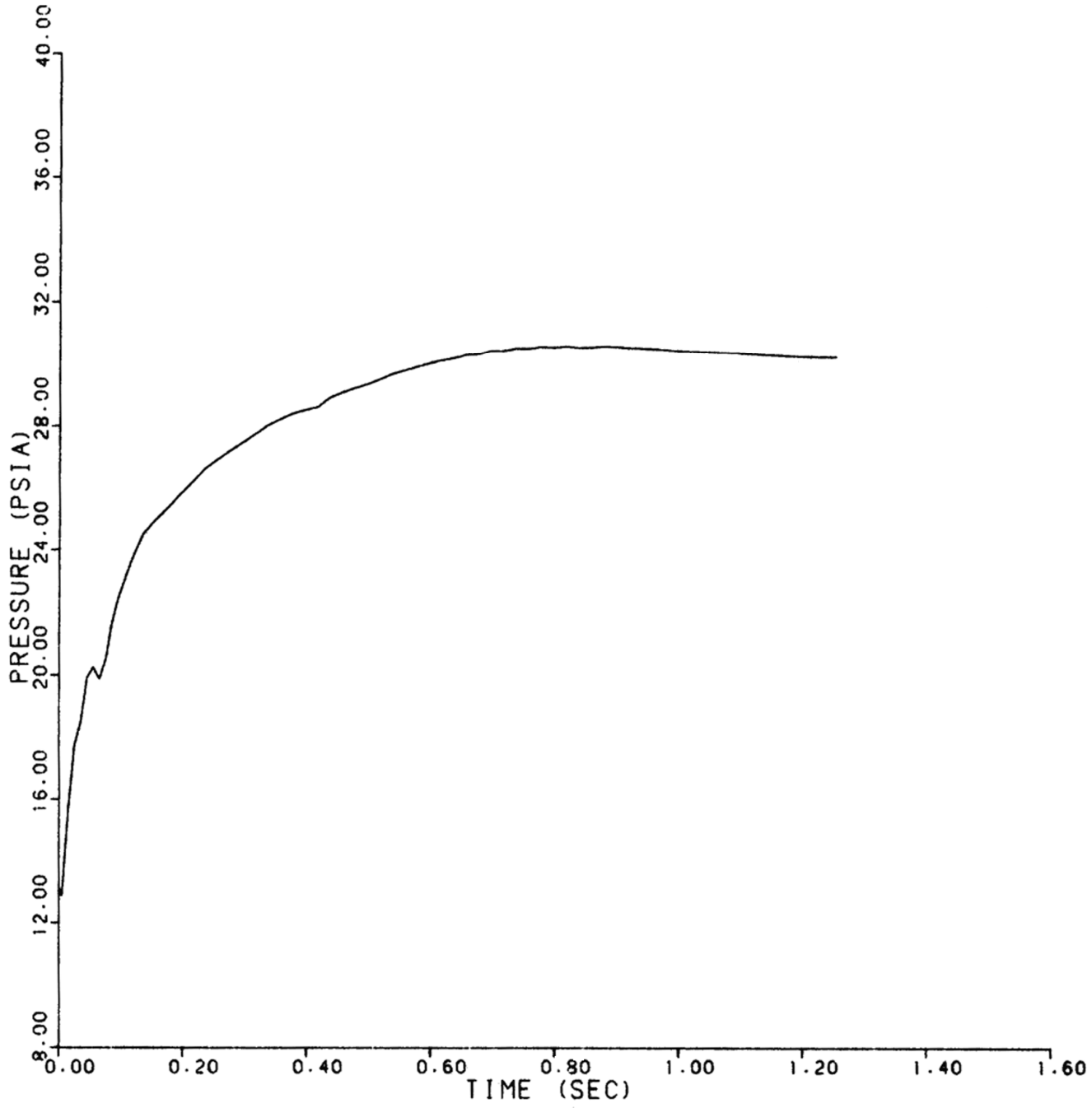
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E6 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 6 OF 74)



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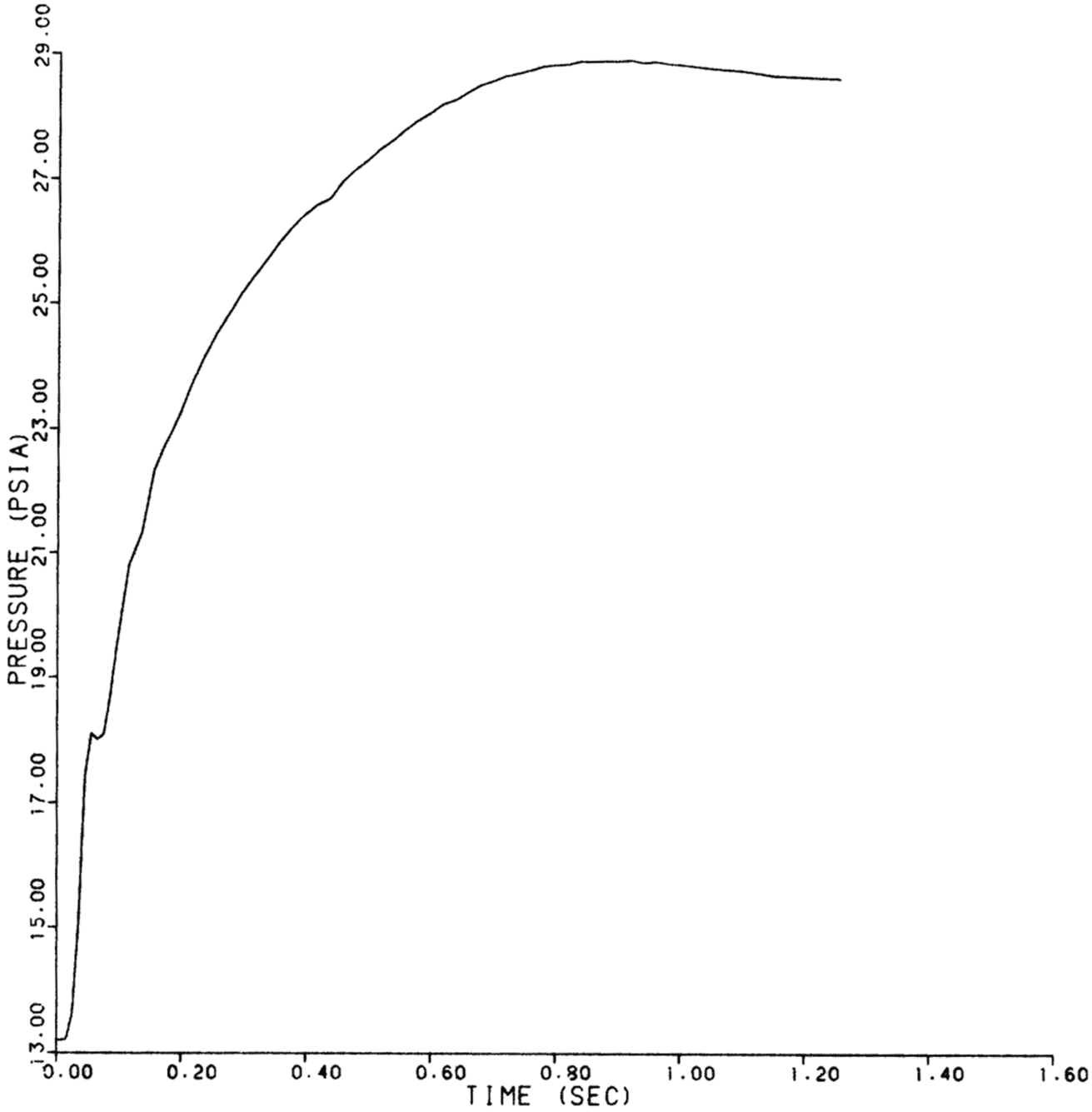
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E7
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 7 OF 74)



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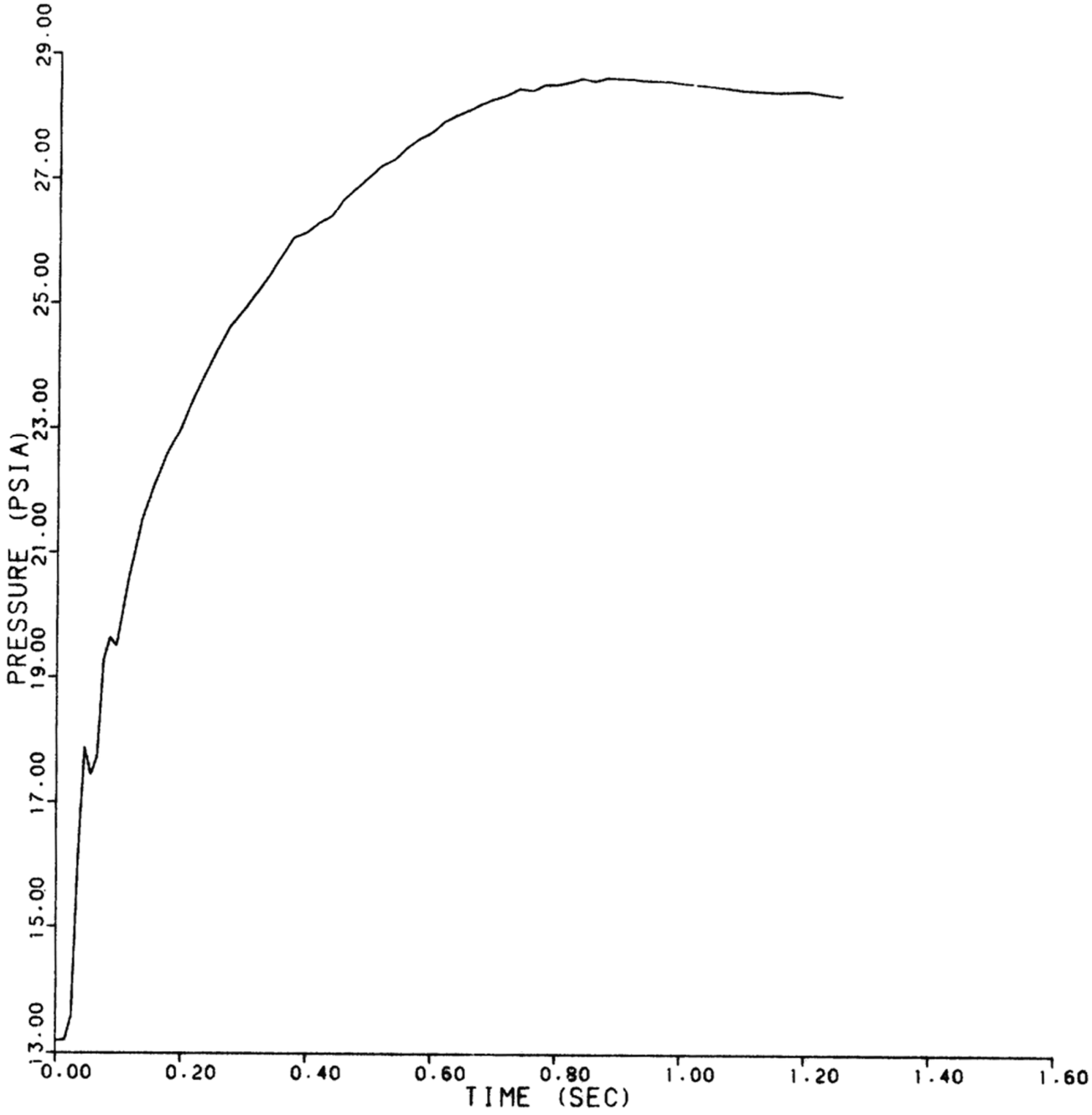
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E8
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 8 OF 74)



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SOUTHERN COMPANY
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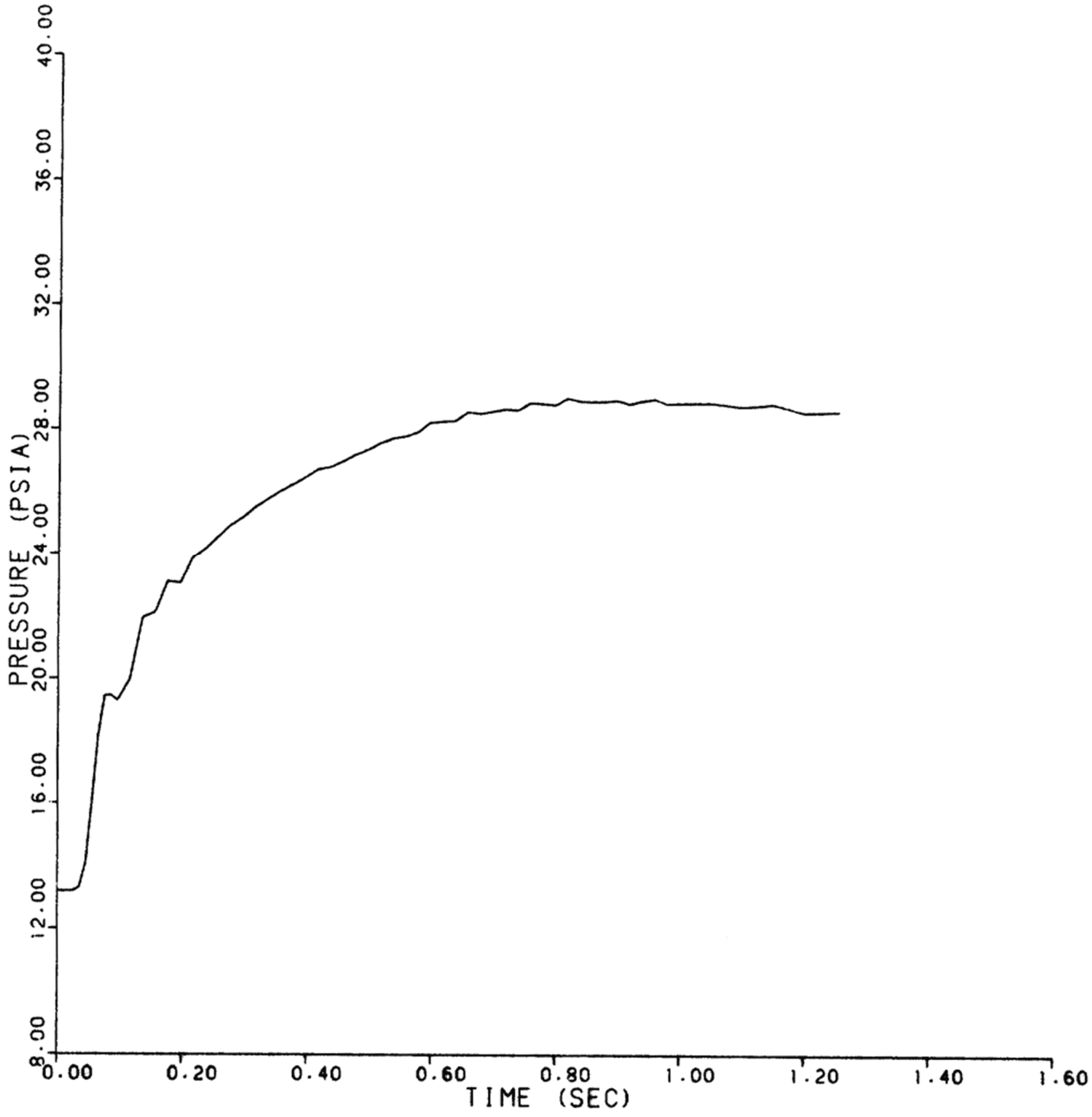
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E9
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 9 OF 74)



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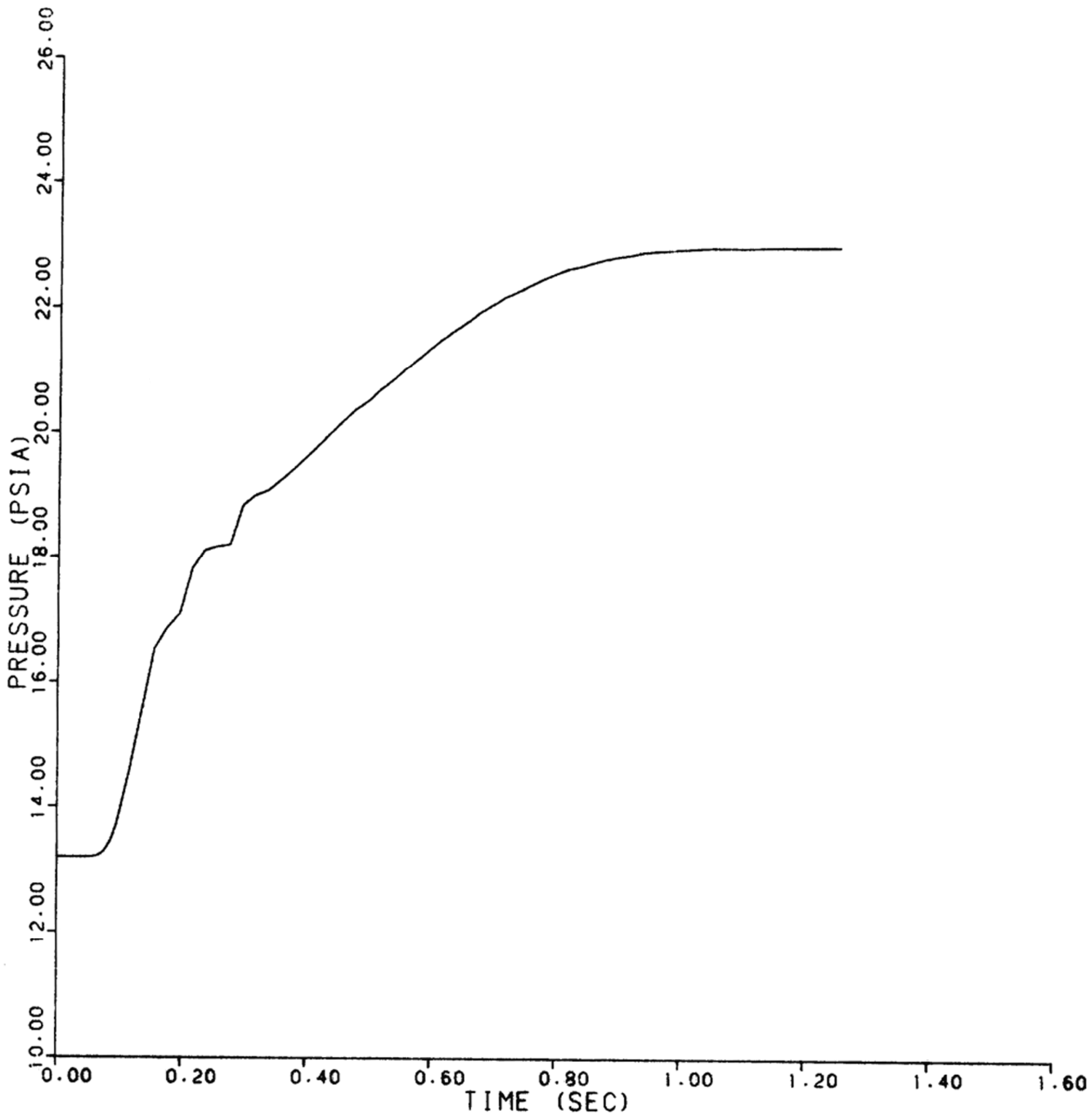
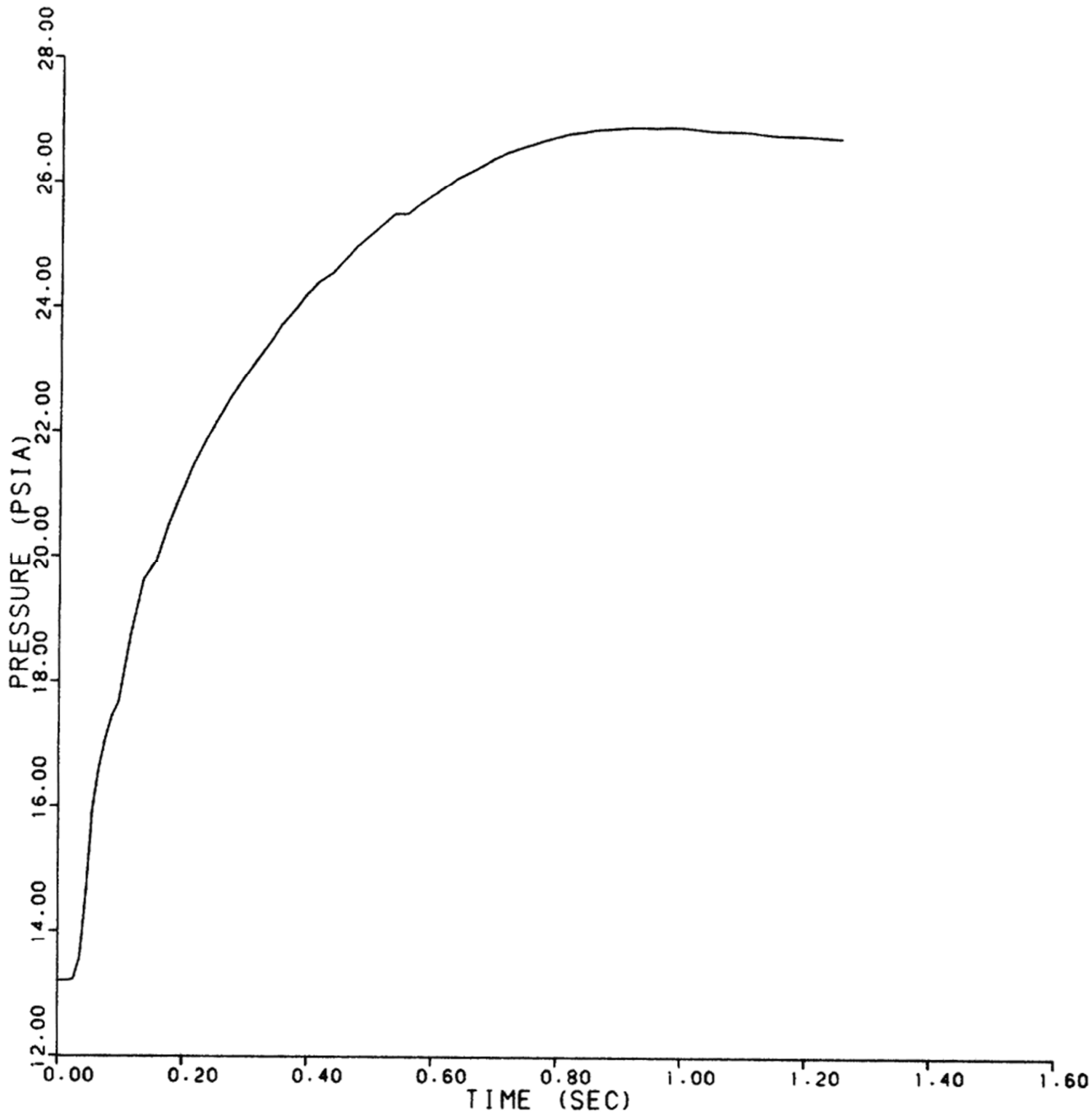
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E10 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 10 OF 74)



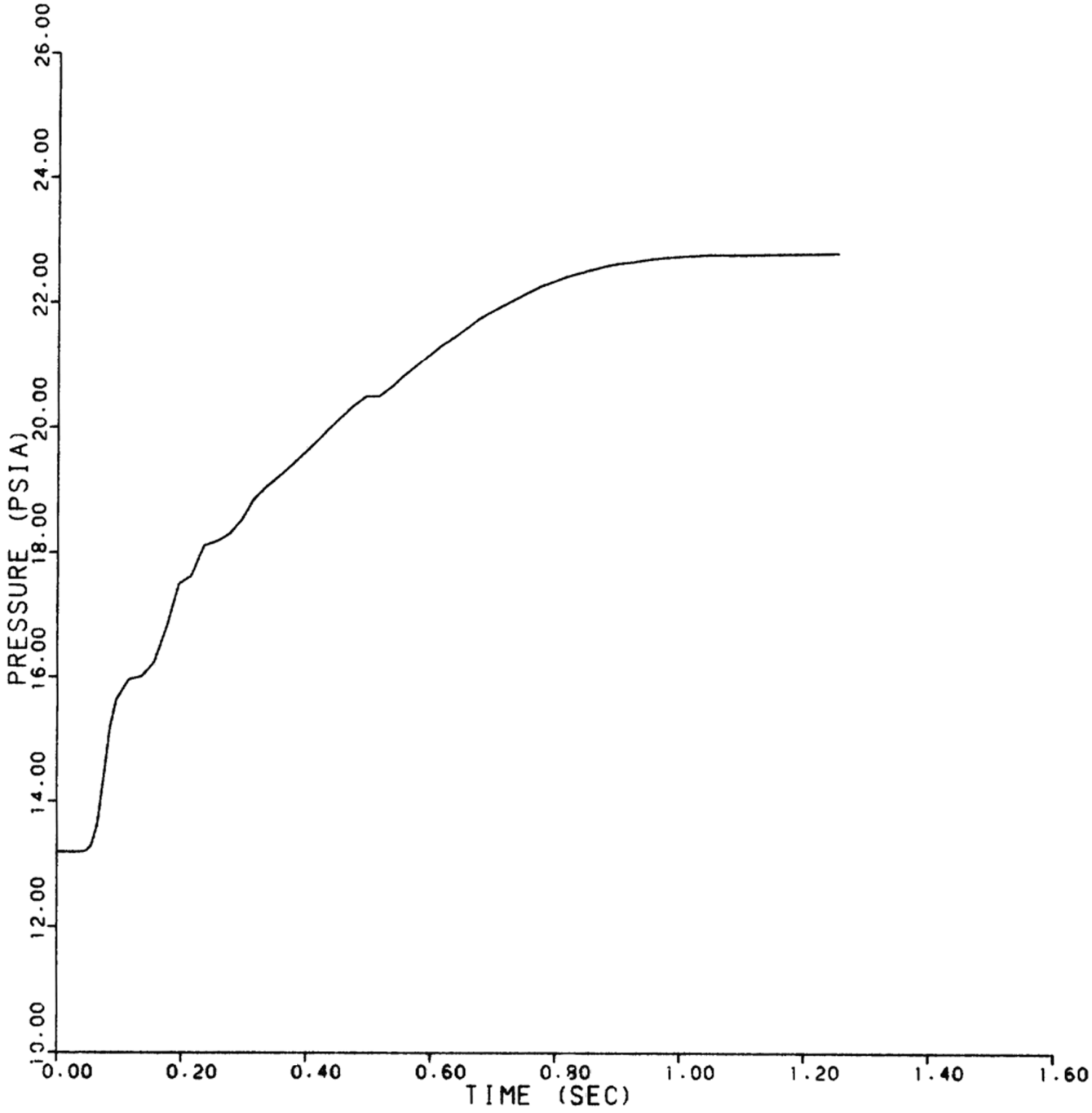
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E11 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 11 OF 74)



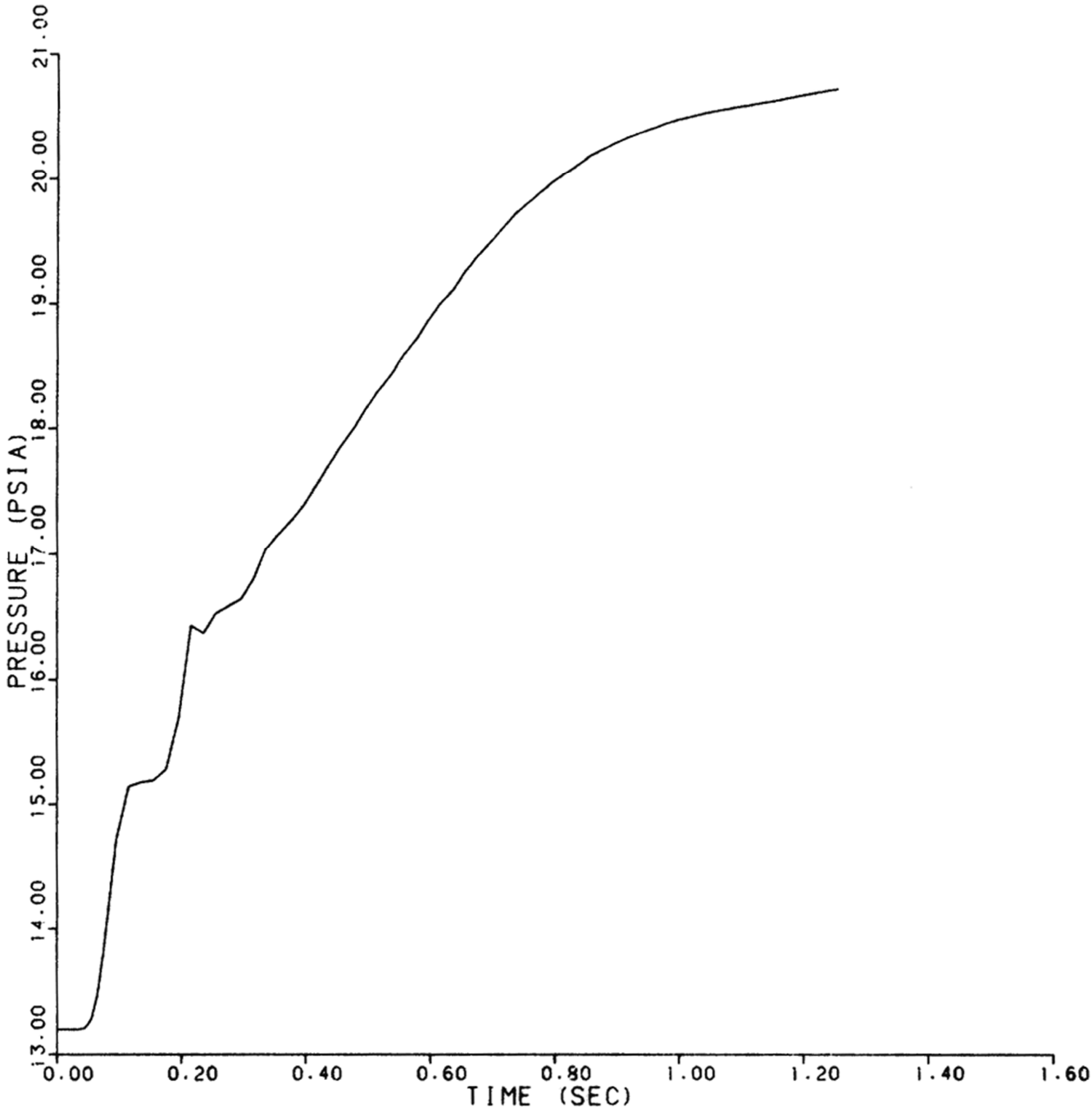
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E12 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 12 OF 74)



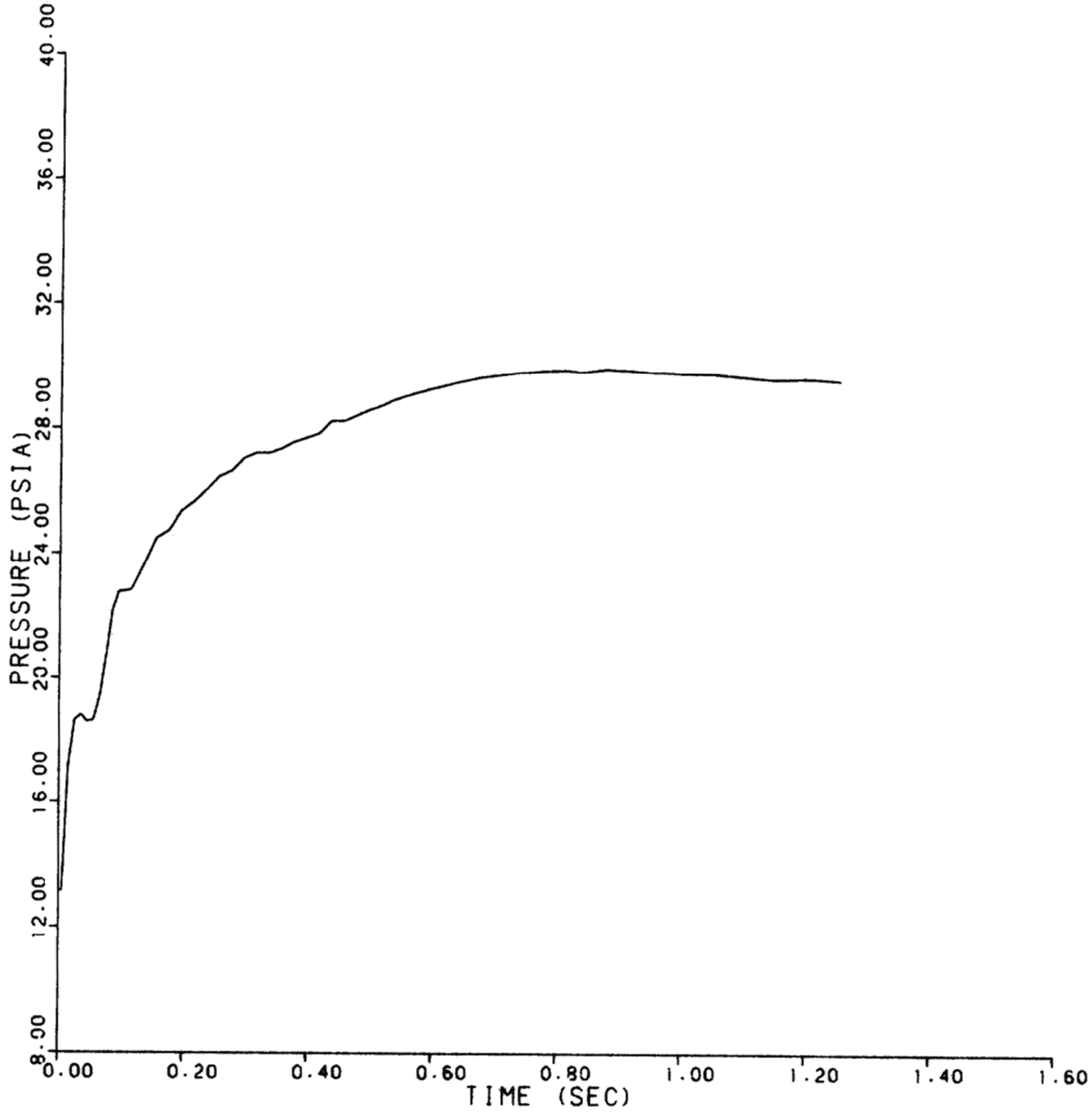
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E13 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 13 OF 74)



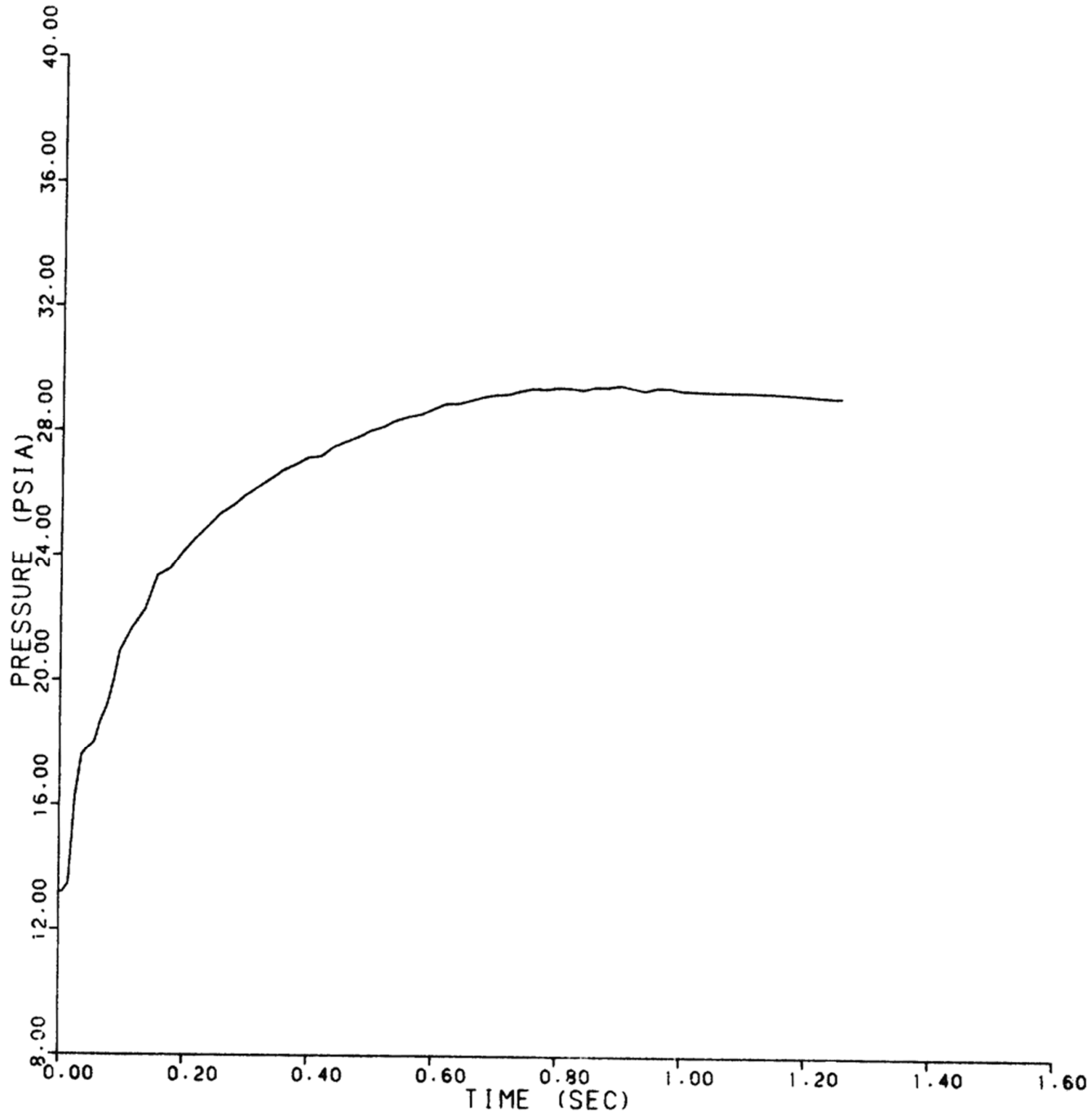
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E14 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 14 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E15 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 15 OF 74)



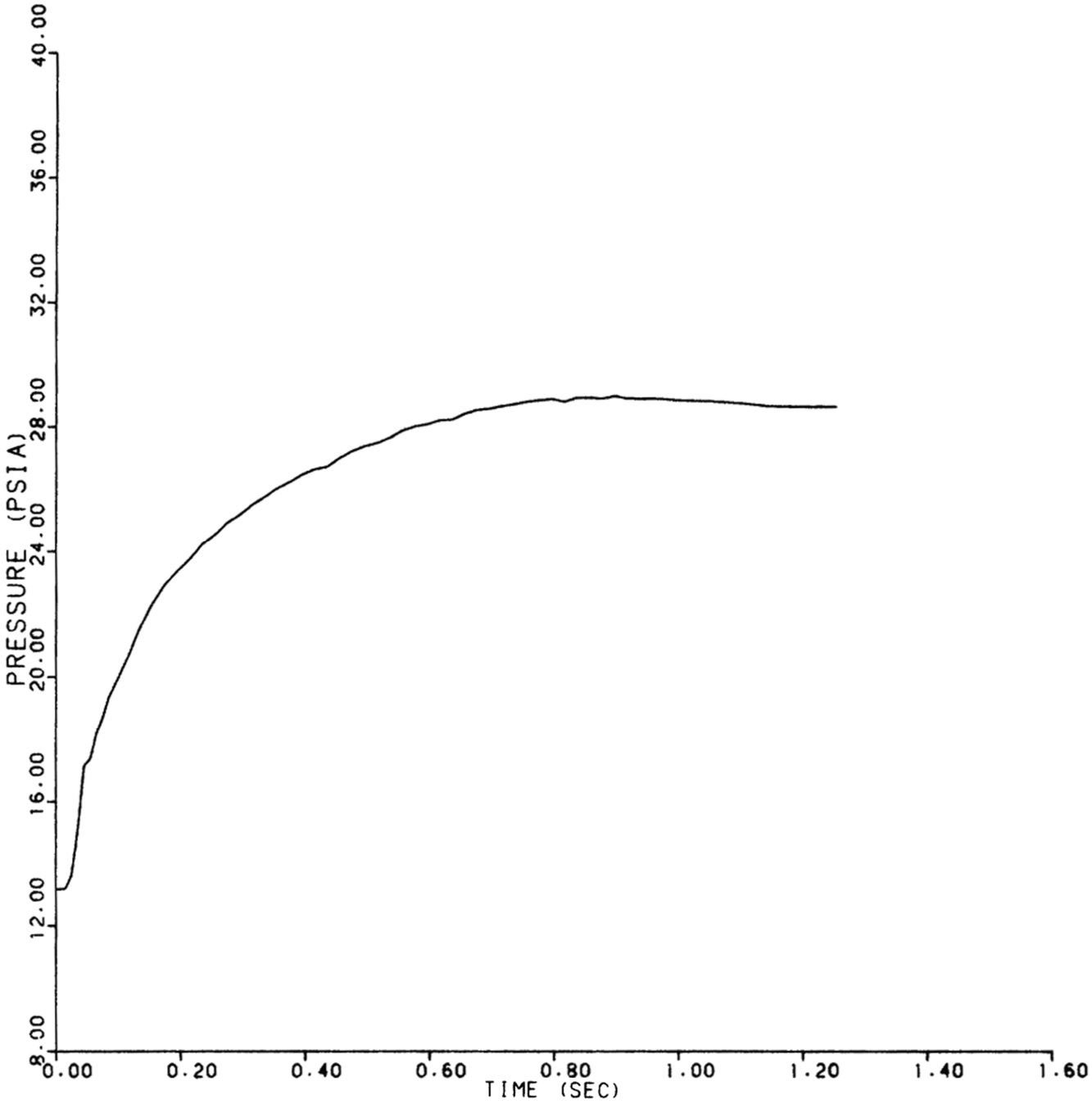
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

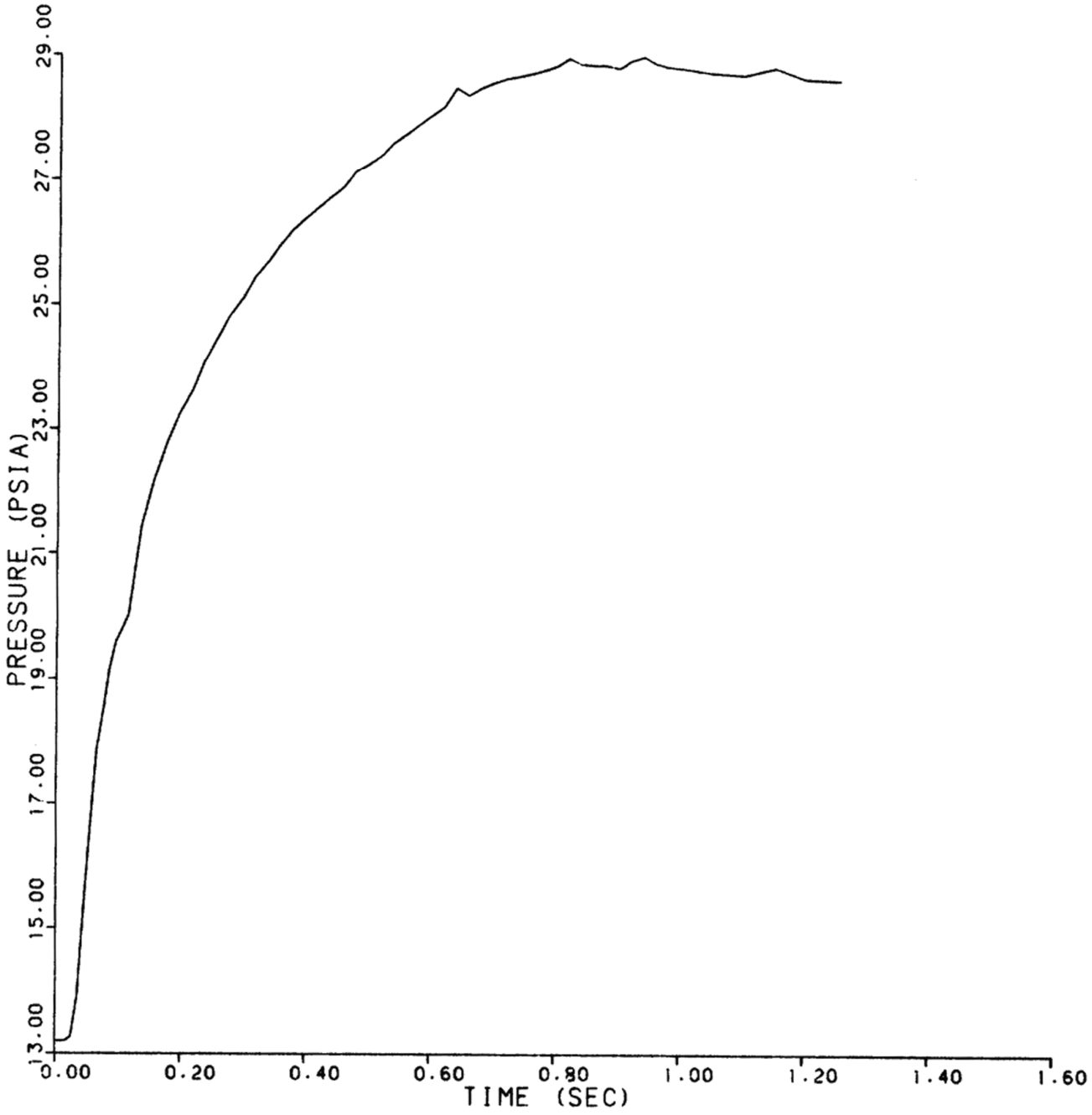
STEAM GENERATOR OUTLET NOZZLE
NODE – E16
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 16 OF 74)



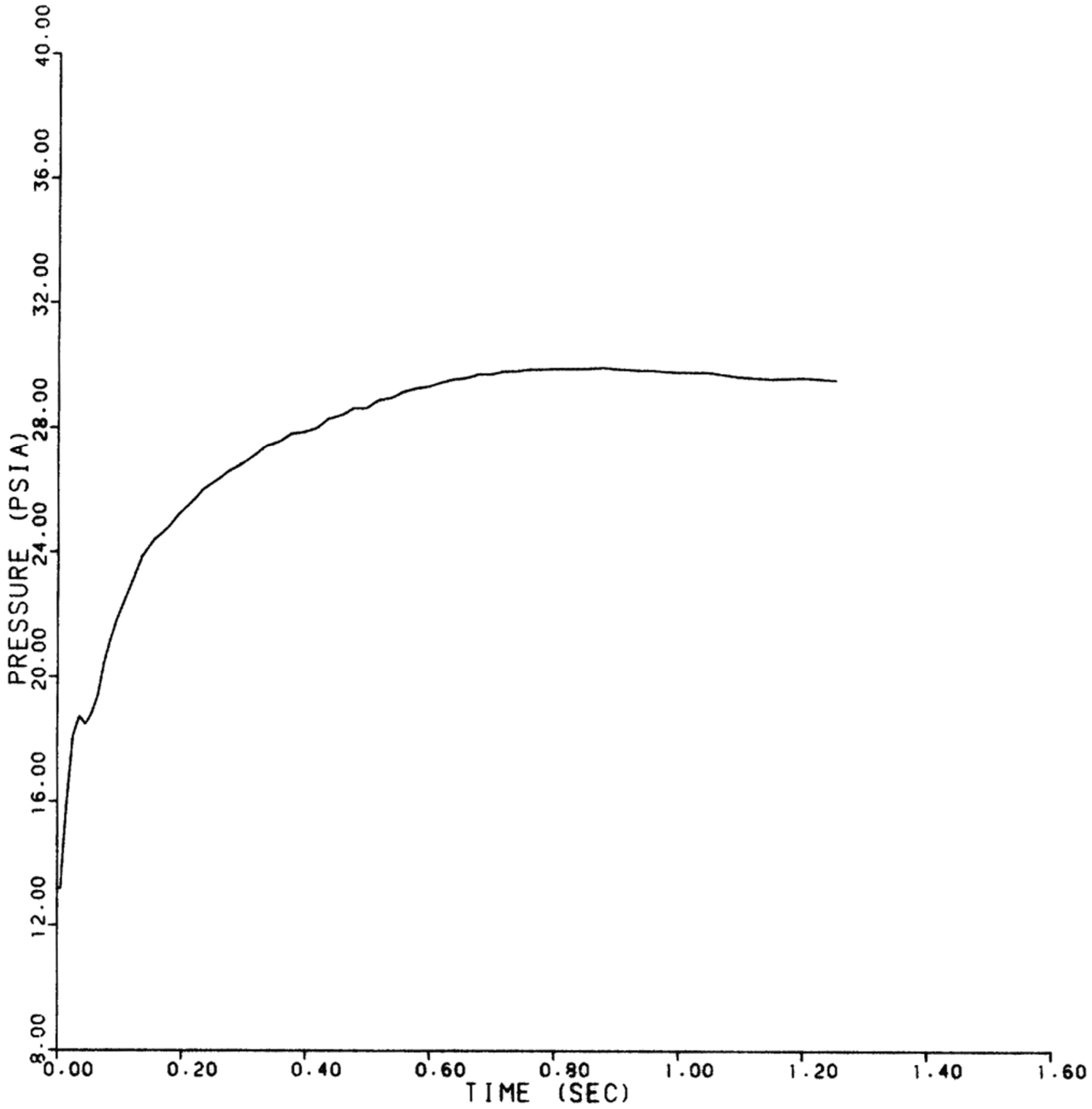
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E17 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 17 OF 74)



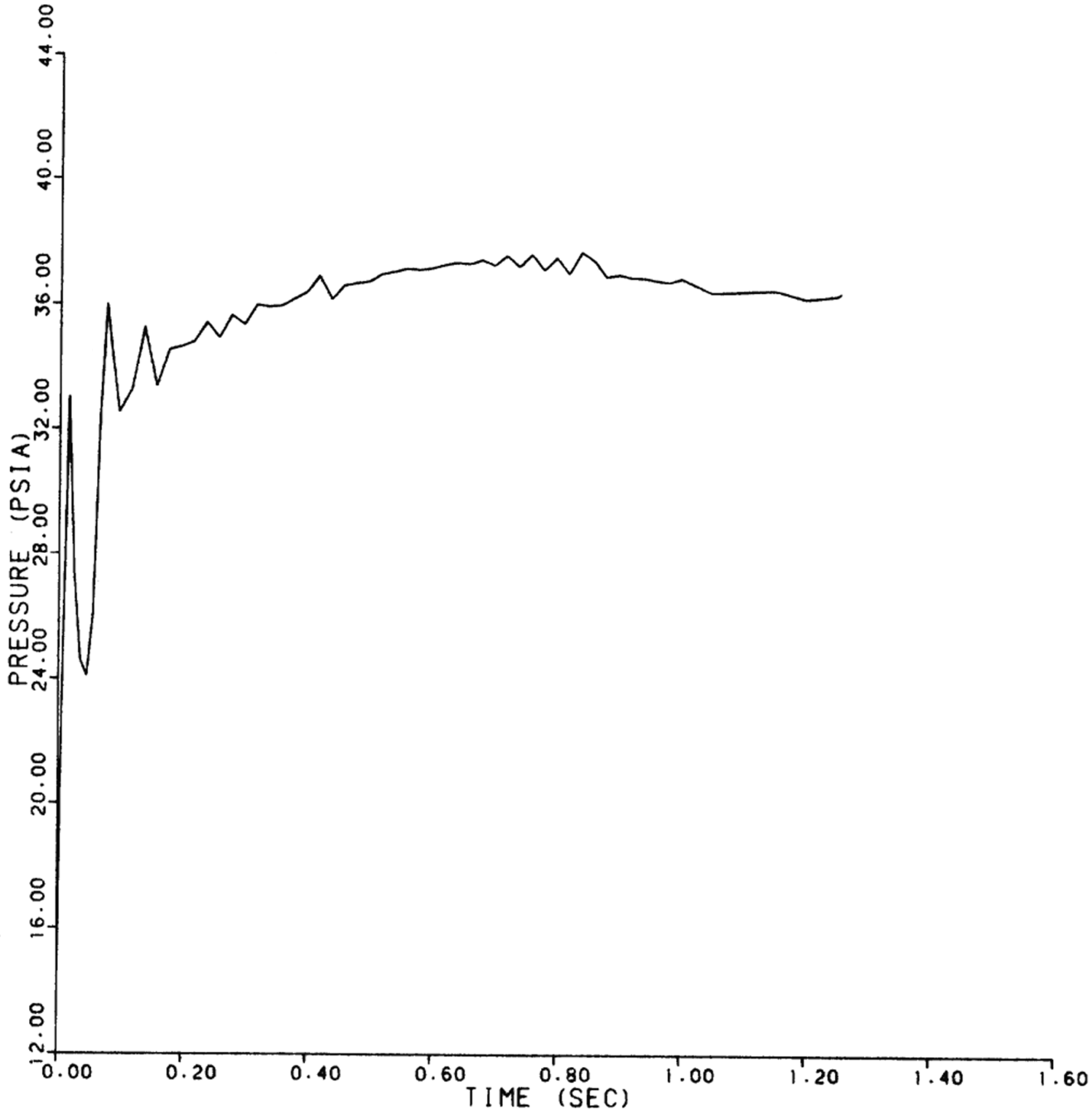
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E18 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 18 OF 74)



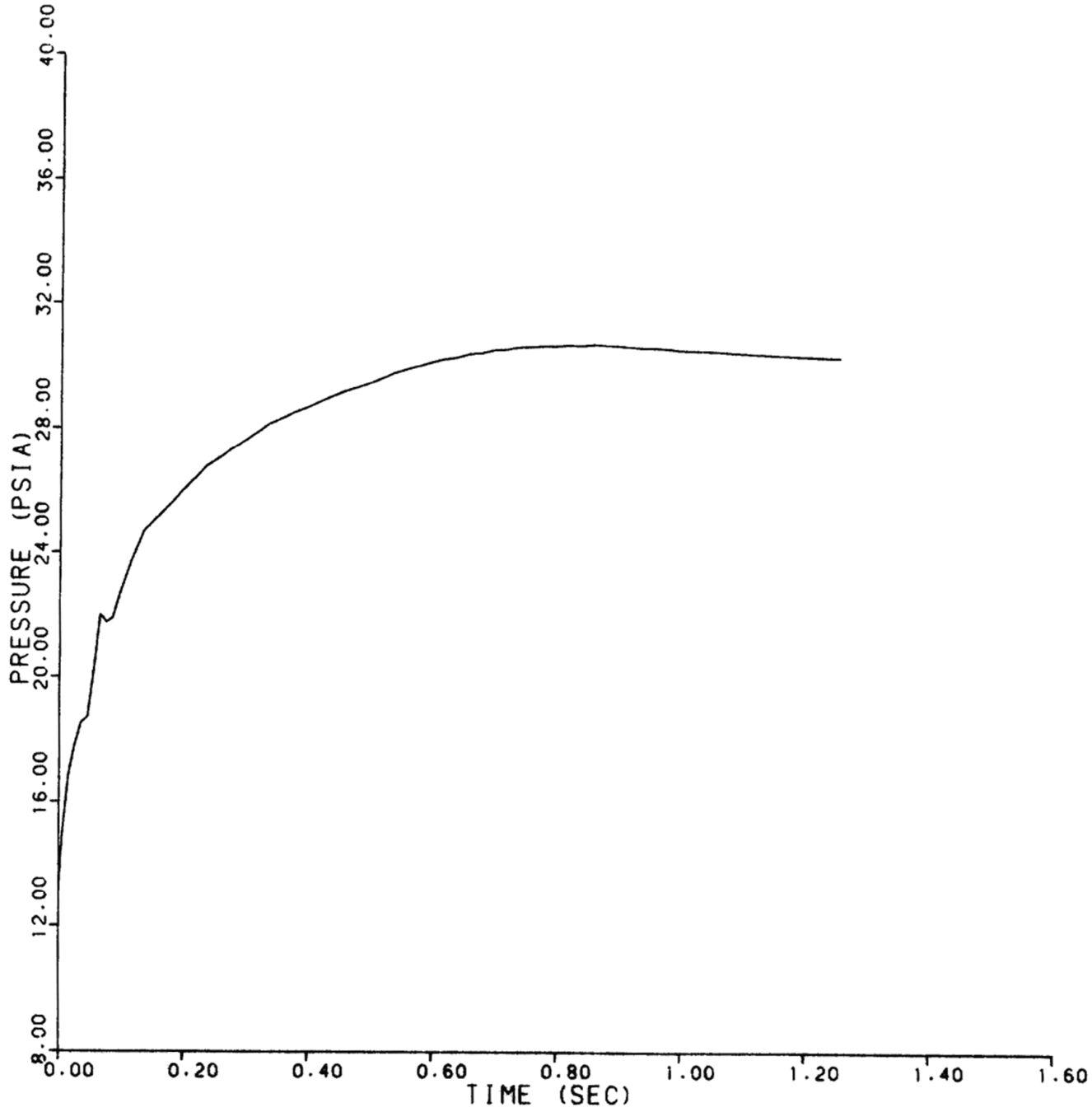
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E19 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 19 OF 74)



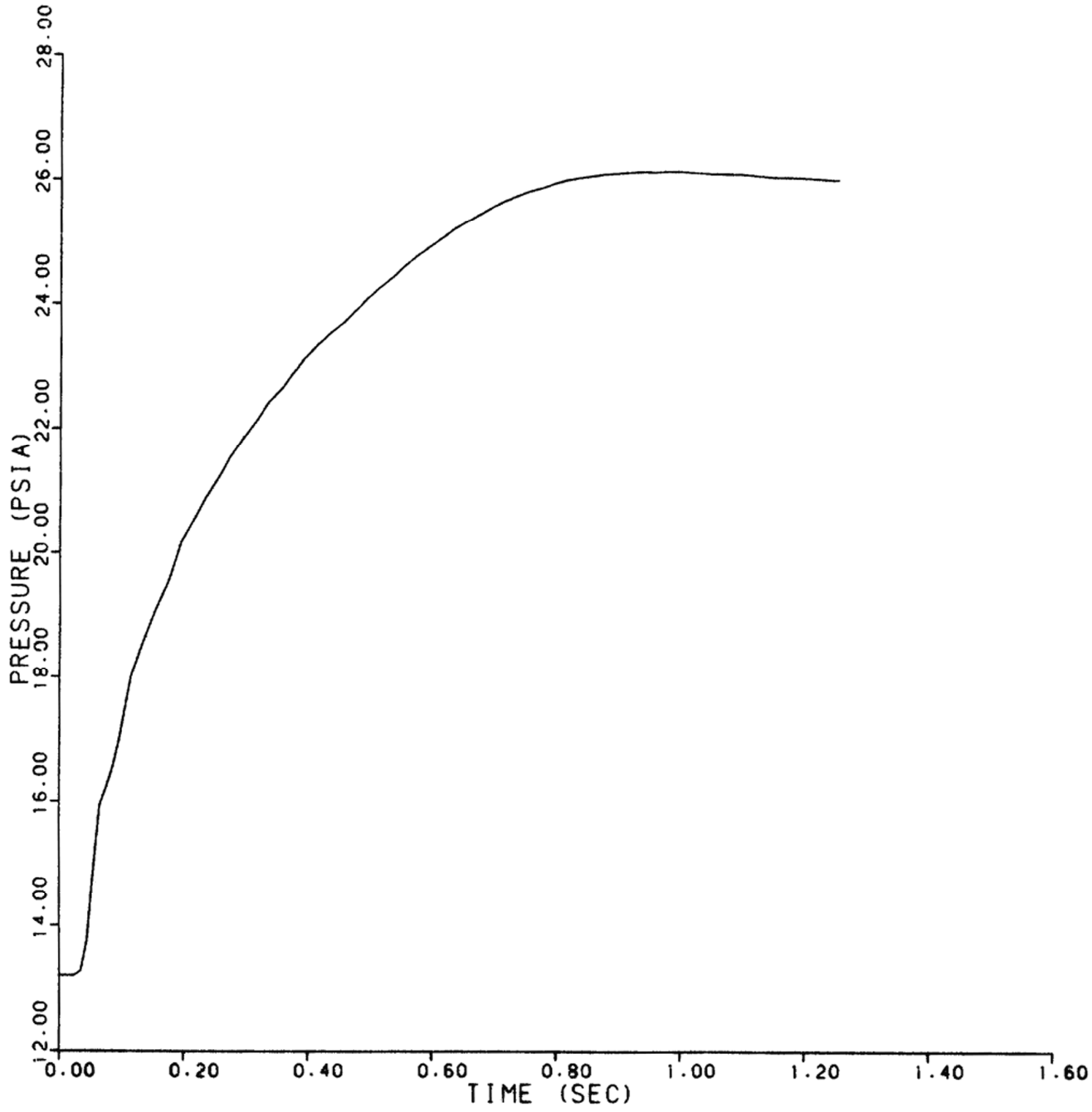
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E20 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 20 OF 74)



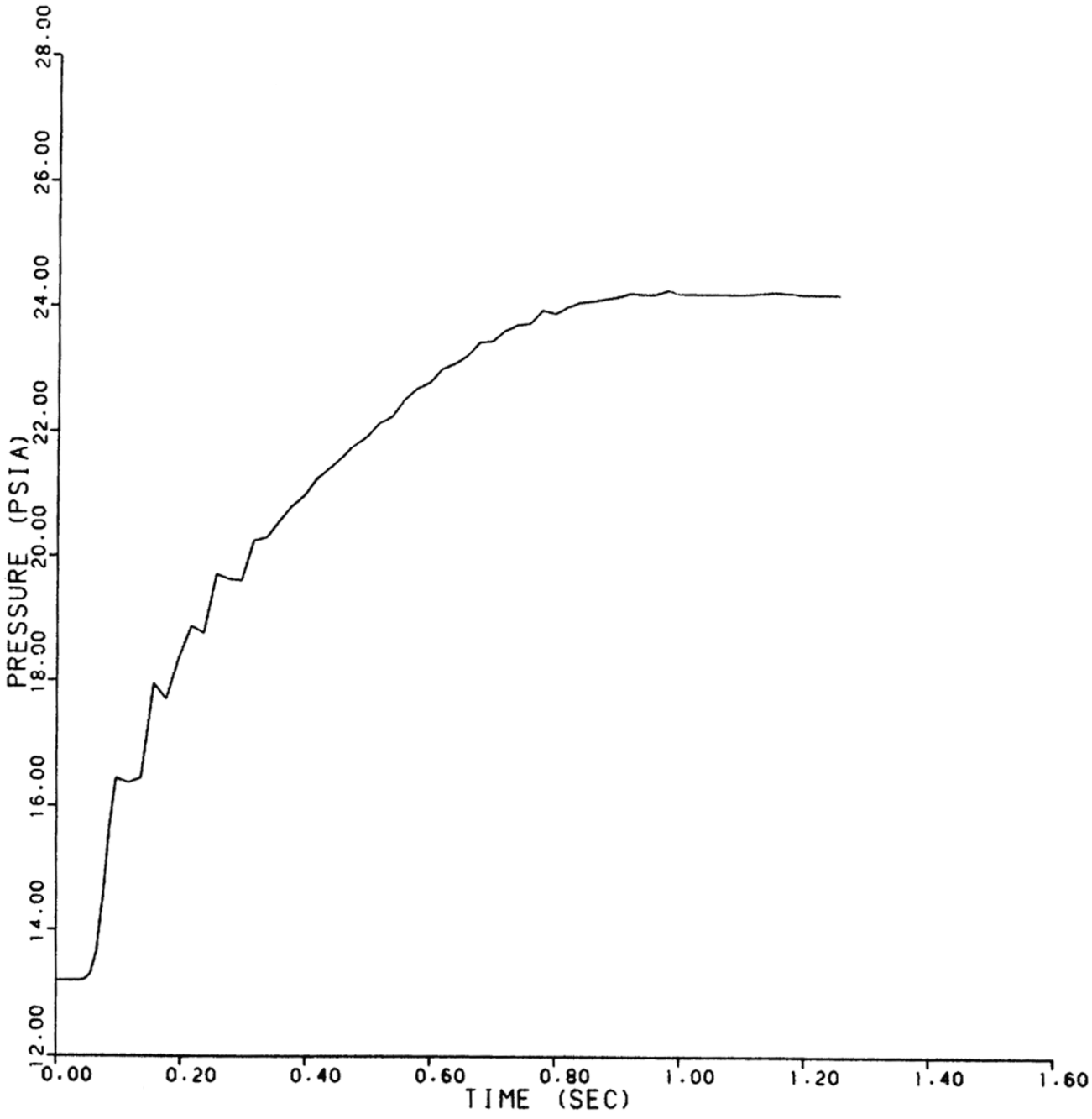
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E21 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 21 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E22 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 22 OF 74)



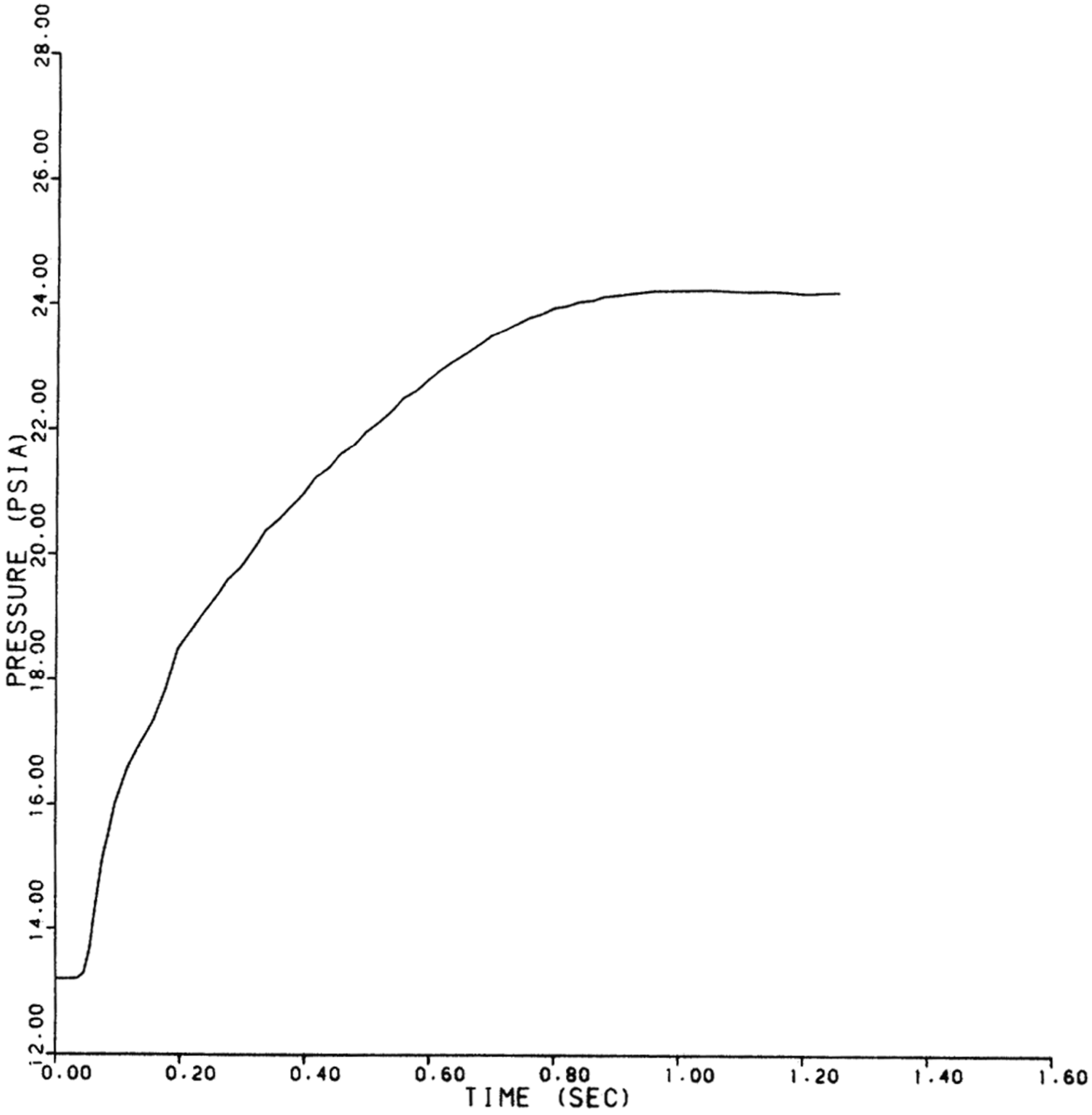
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E23
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 23 OF 74)



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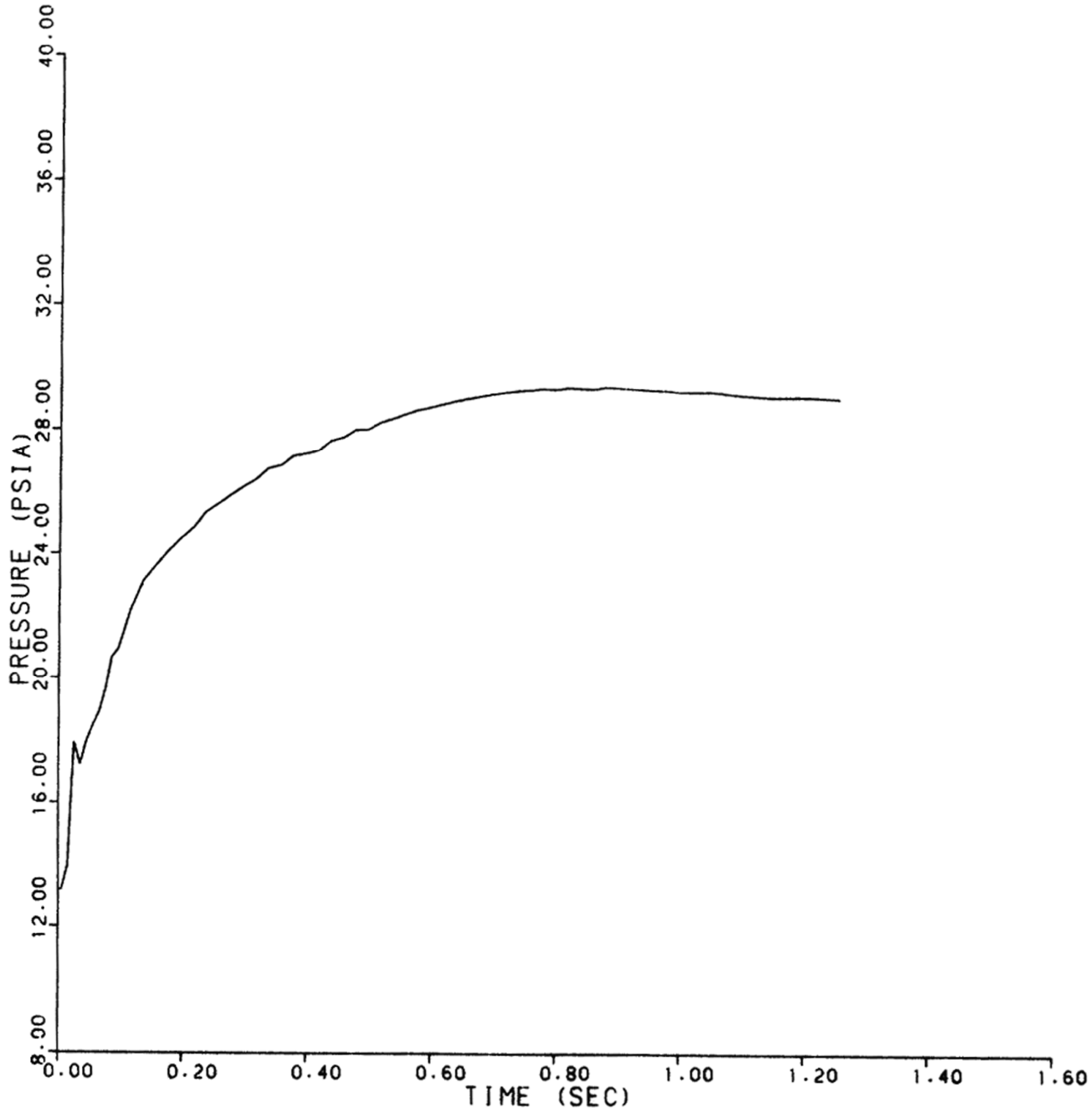
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E24 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 24 OF 74)



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SOUTHERN COMPANY
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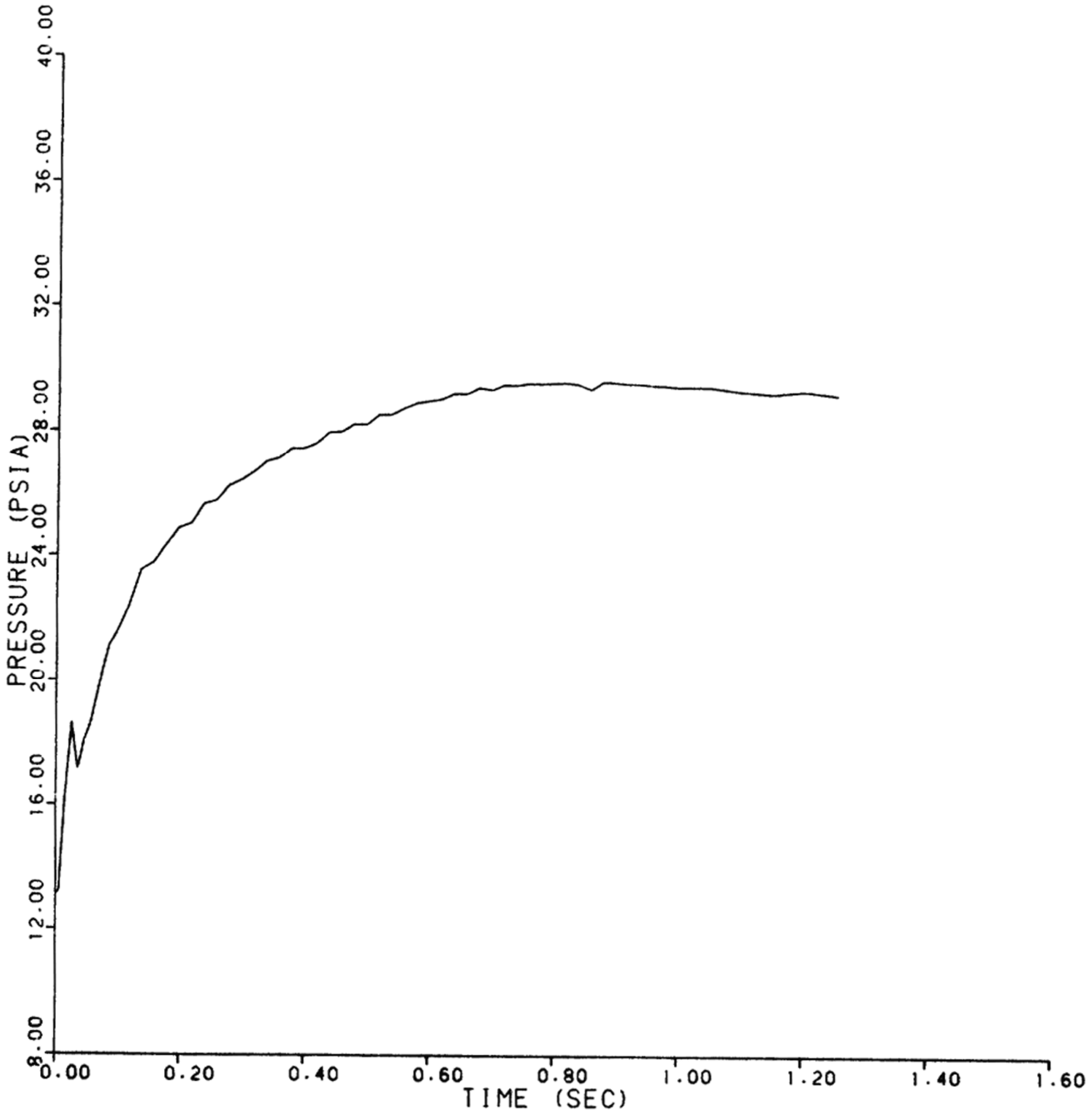
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E25
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 25 OF 74)



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SOUTHERN COMPANY
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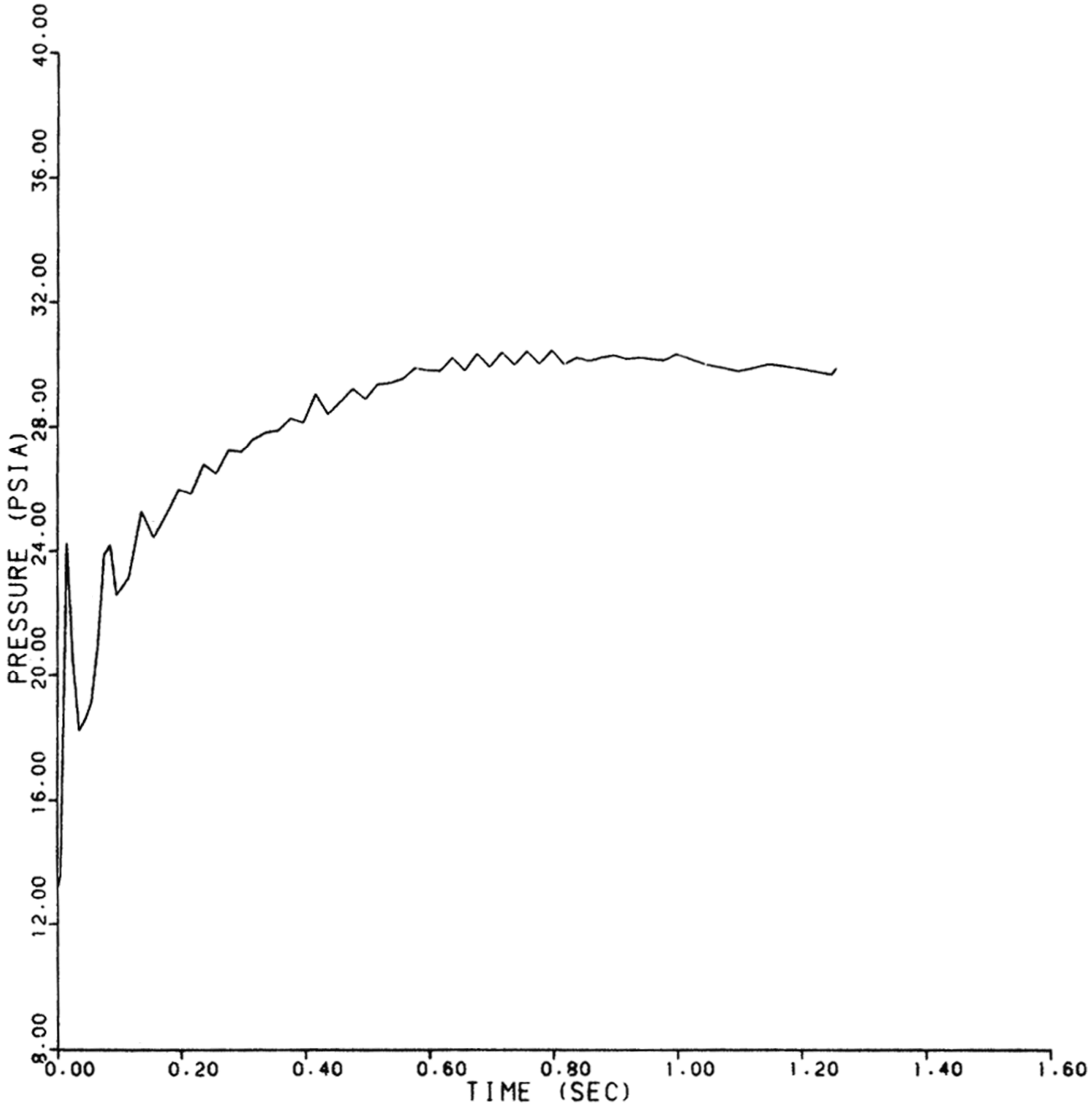
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E26
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 26 OF 74)



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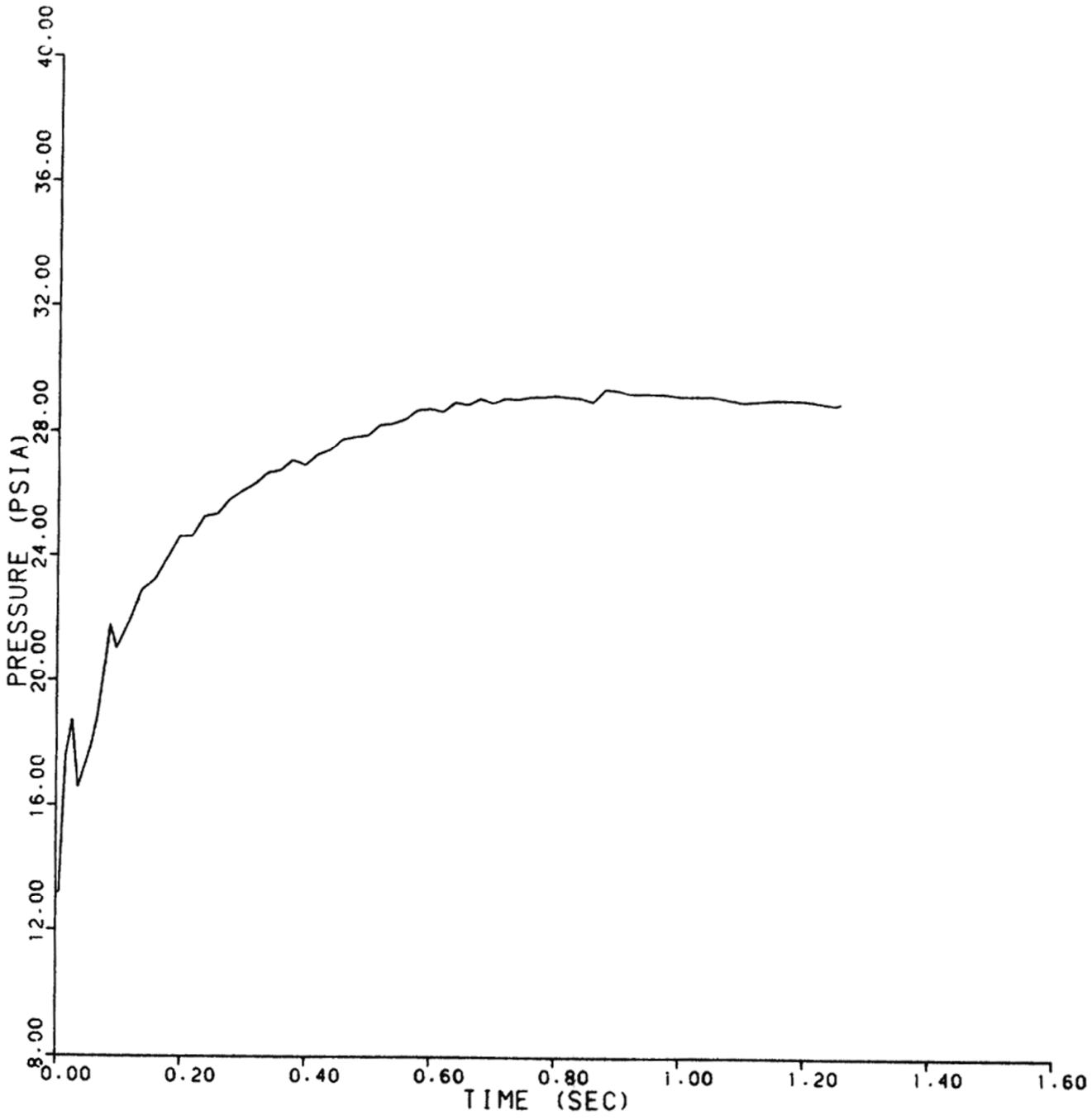
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

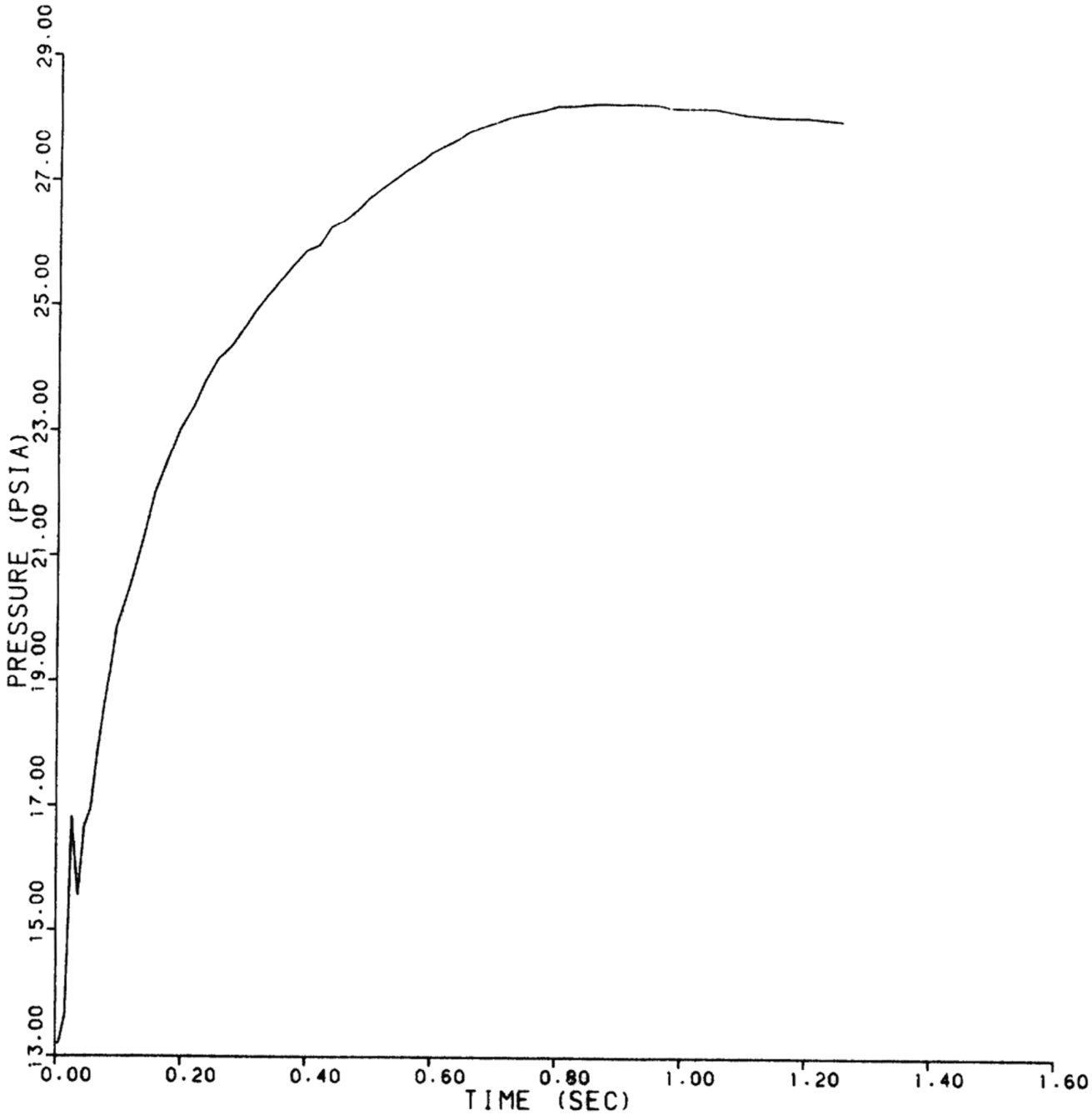
STEAM GENERATOR OUTLET NOZZLE
NODE – E27
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 27 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E28 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 28 OF 74)



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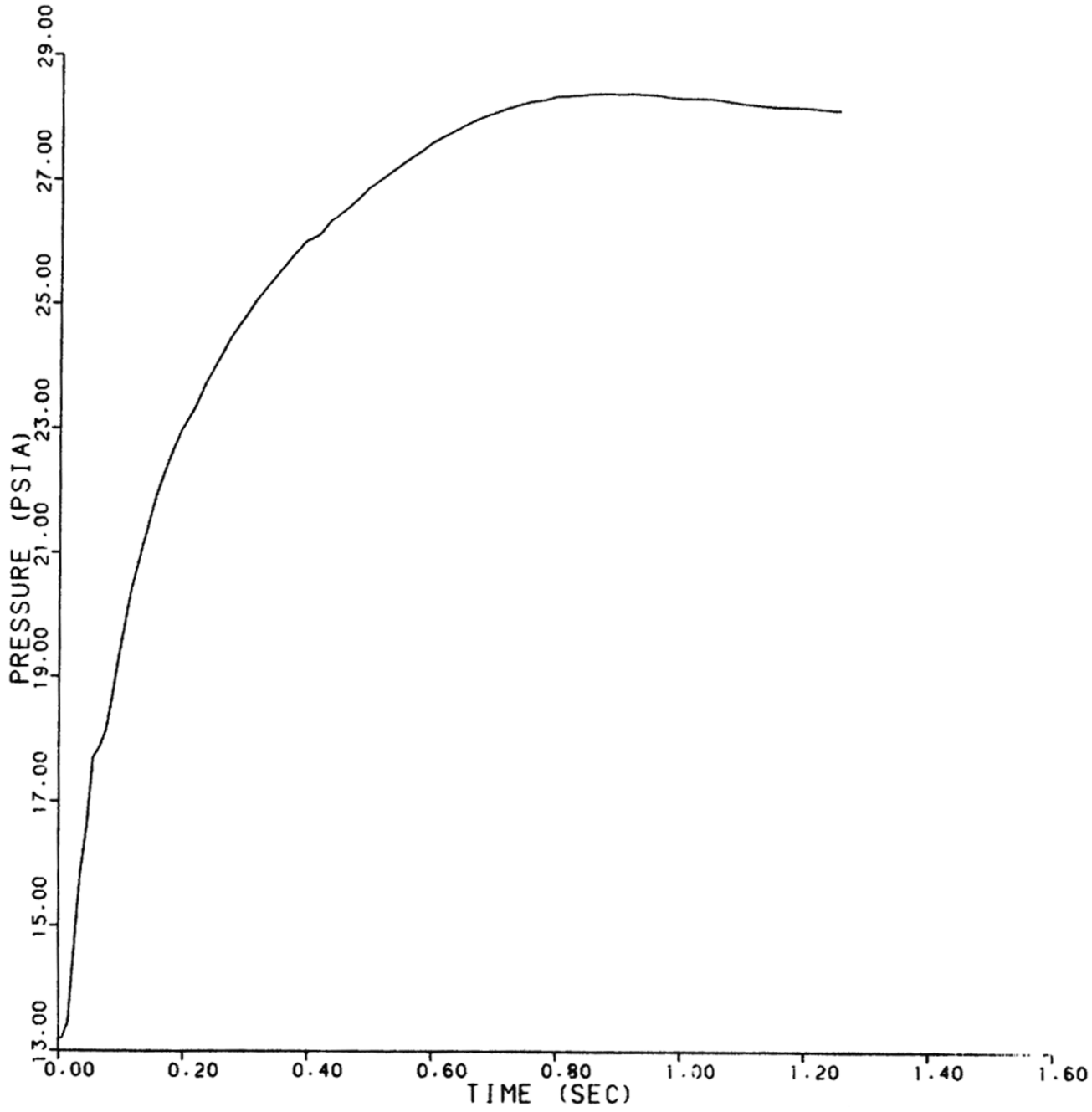
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E29
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 29 OF 74)



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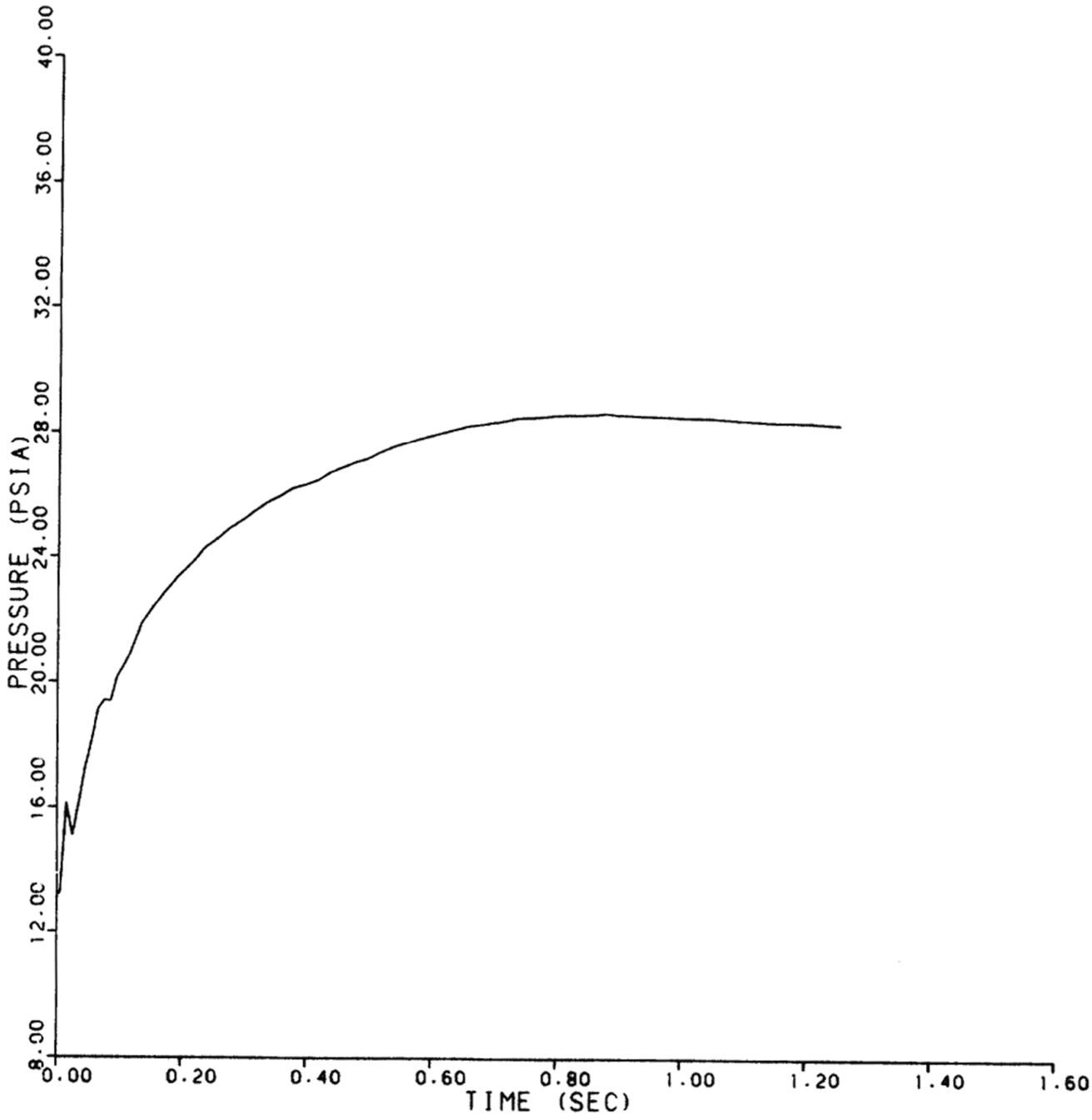
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

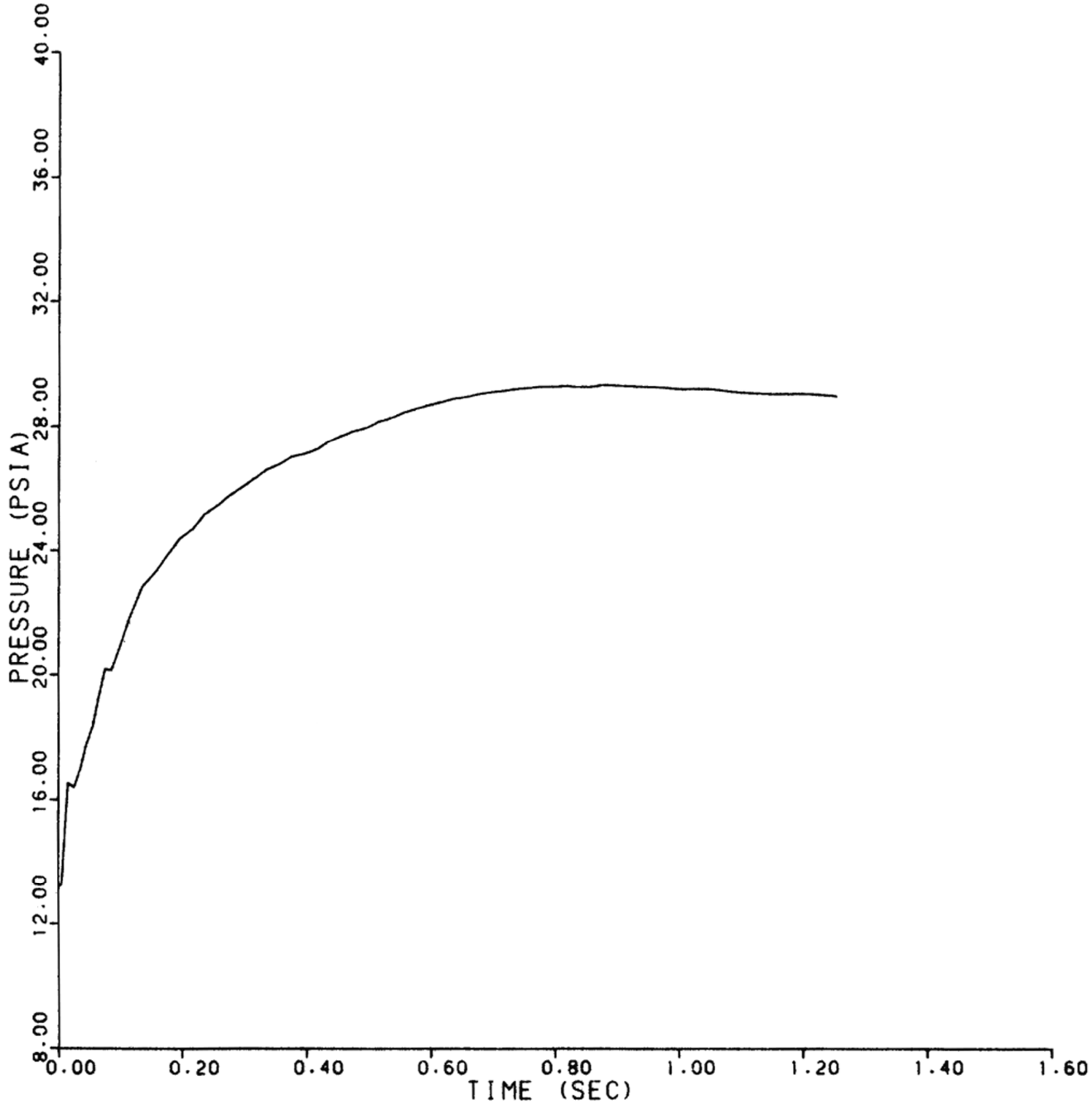
STEAM GENERATOR OUTLET NOZZLE
NODE – E30
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 30 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E31 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 31 OF 74)



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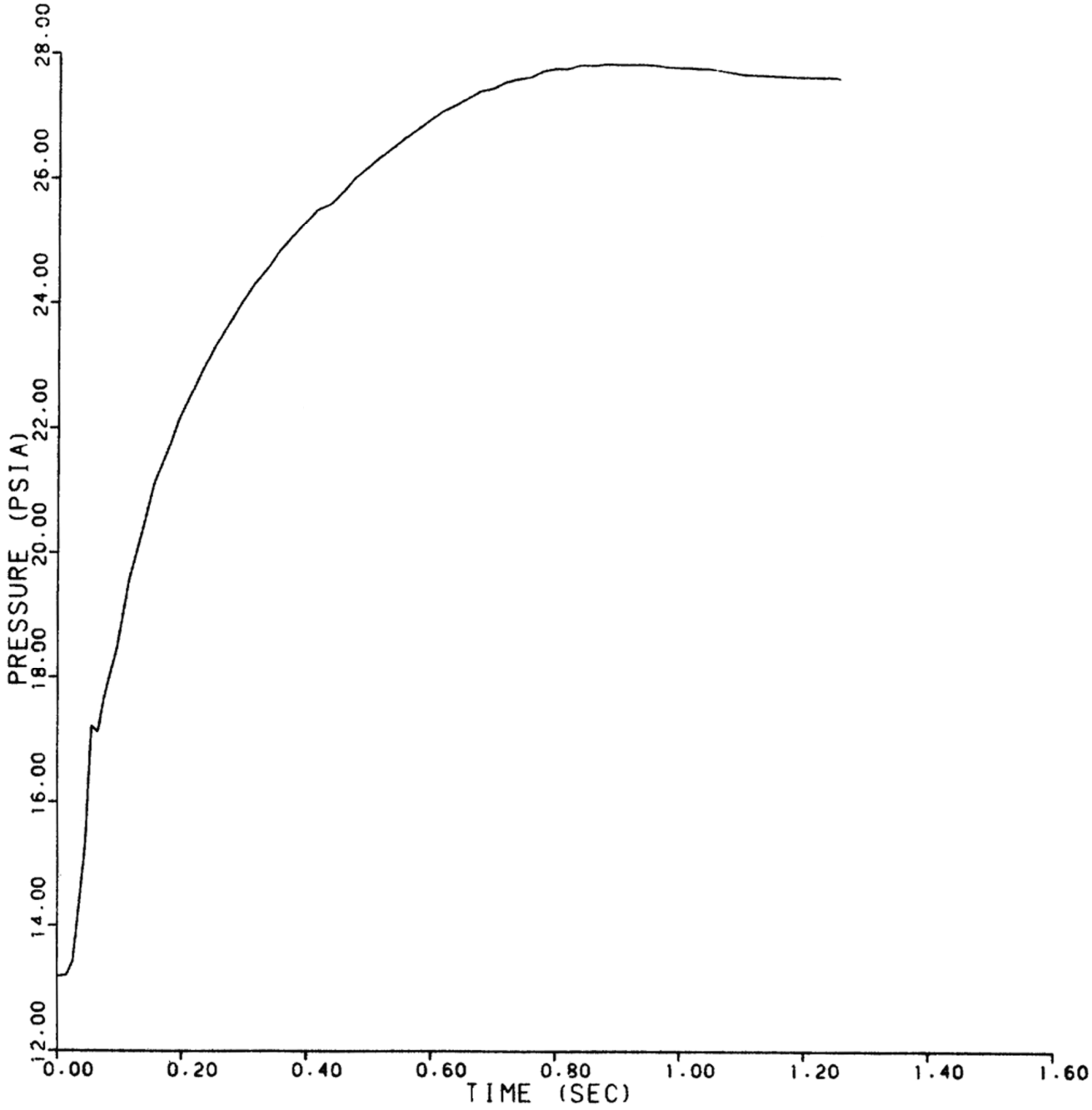
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

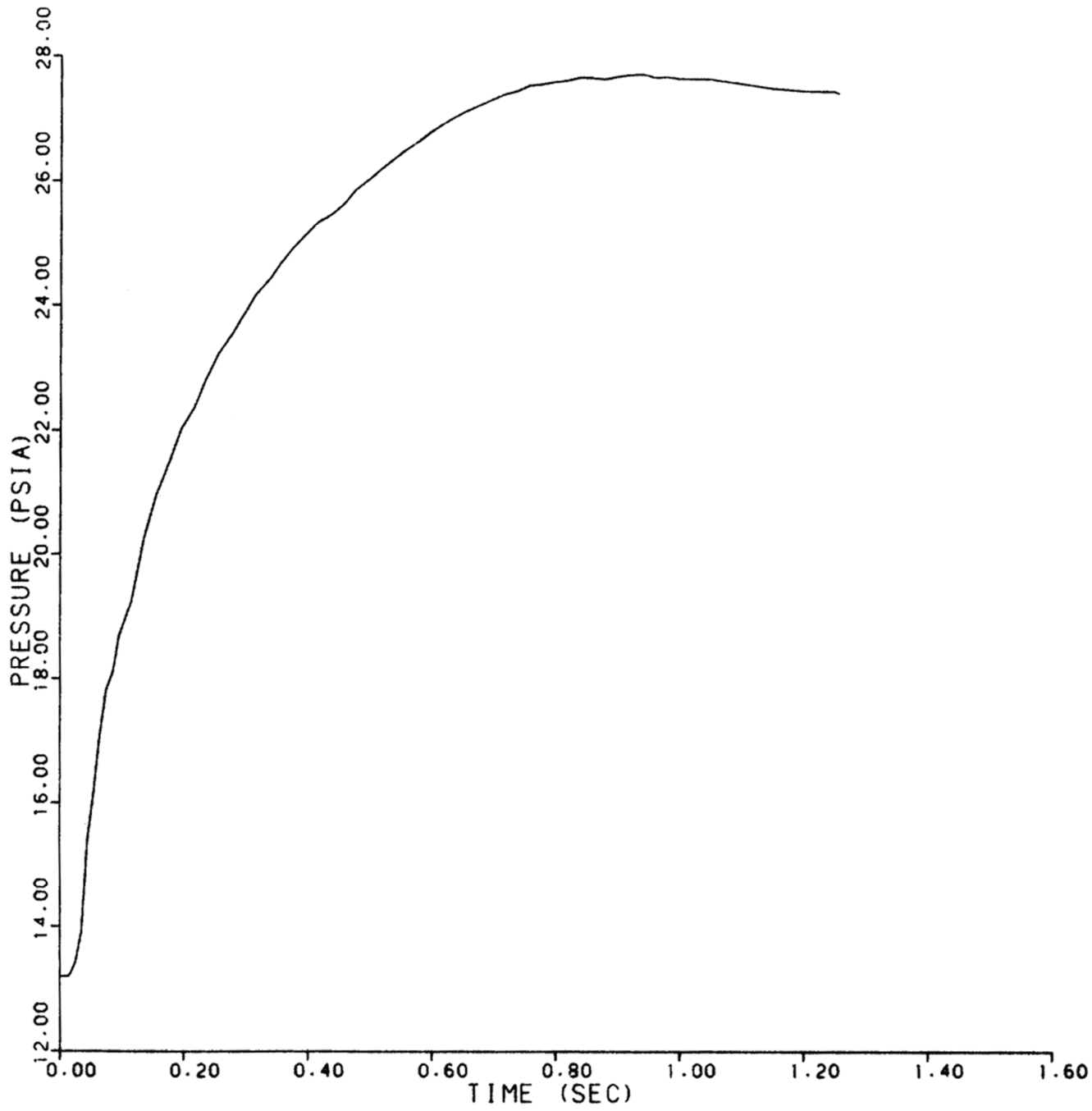
STEAM GENERATOR OUTLET NOZZLE
NODE – E32
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 32 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E33 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 33 OF 74)



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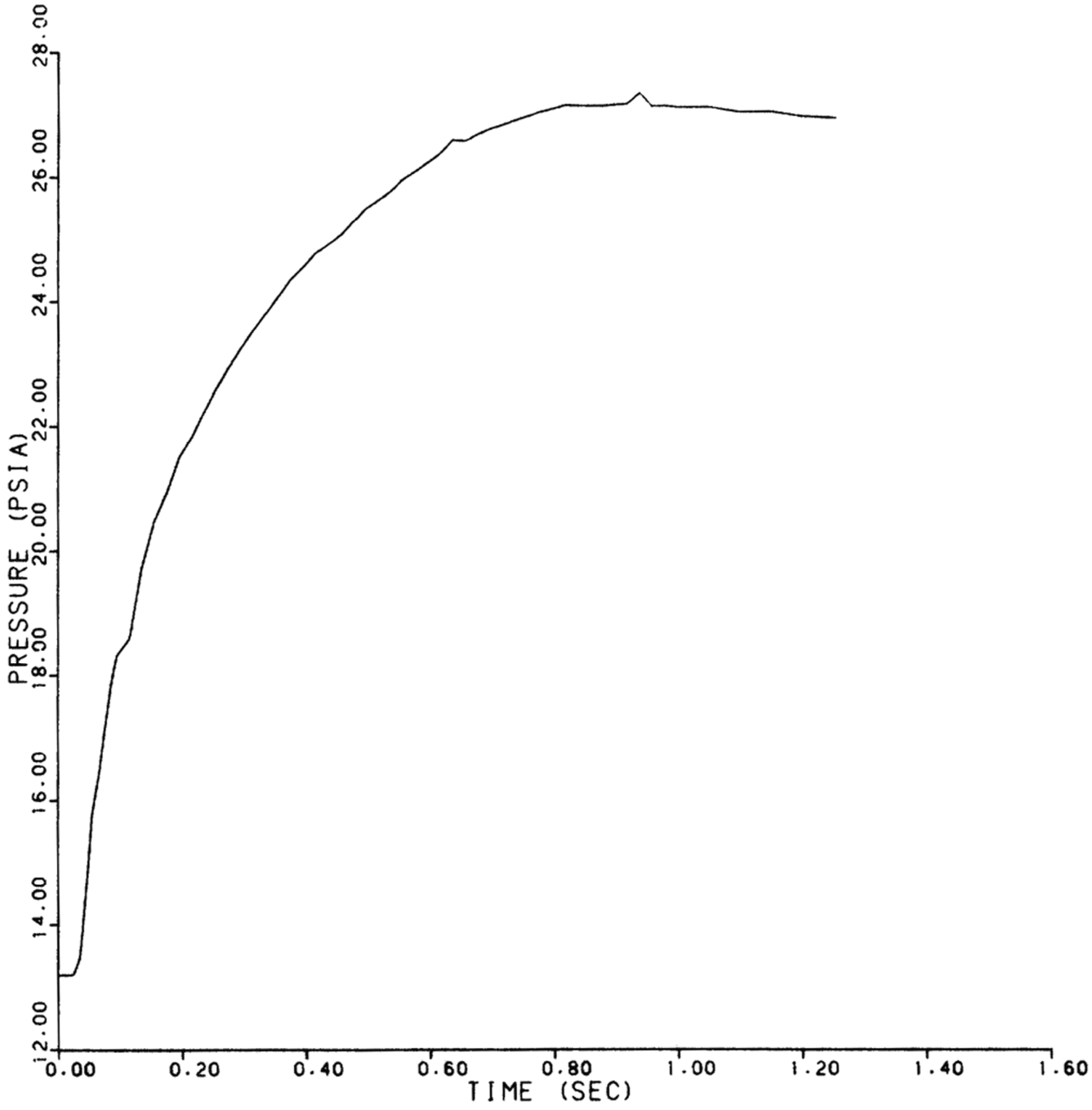
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E34
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 34 OF 74)



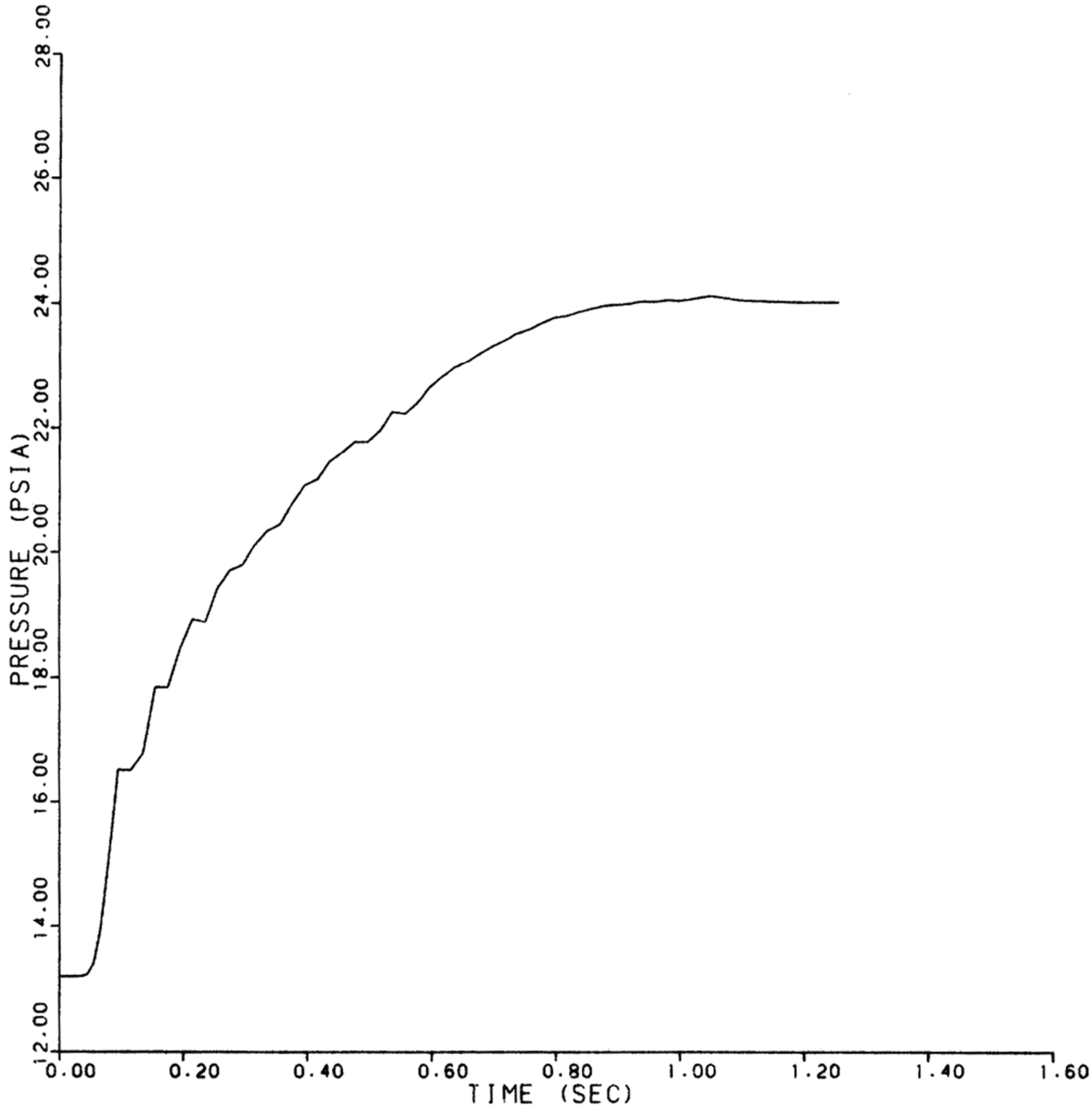
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

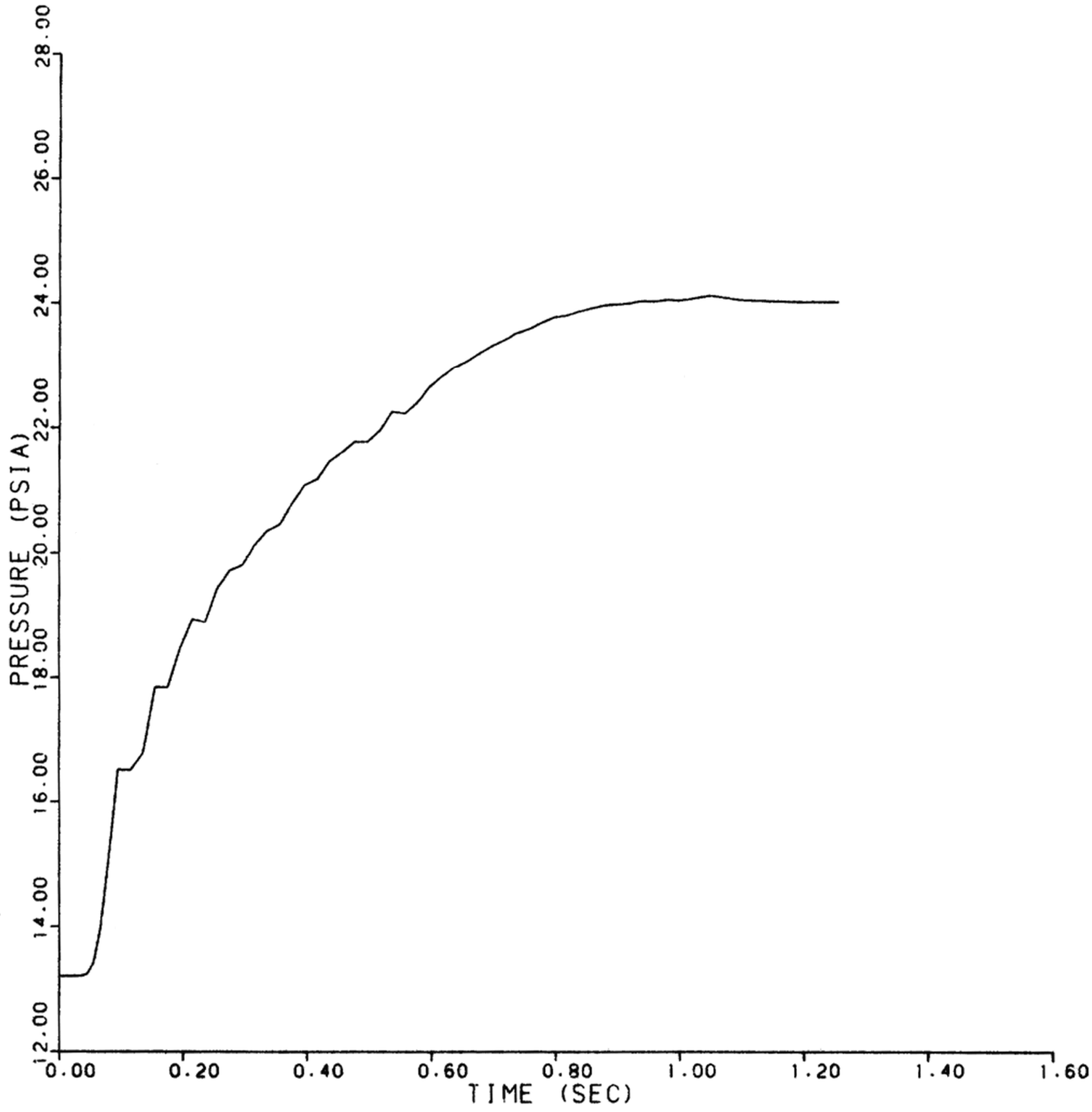
STEAM GENERATOR OUTLET NOZZLE
NODE – E35
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 35 OF 74)



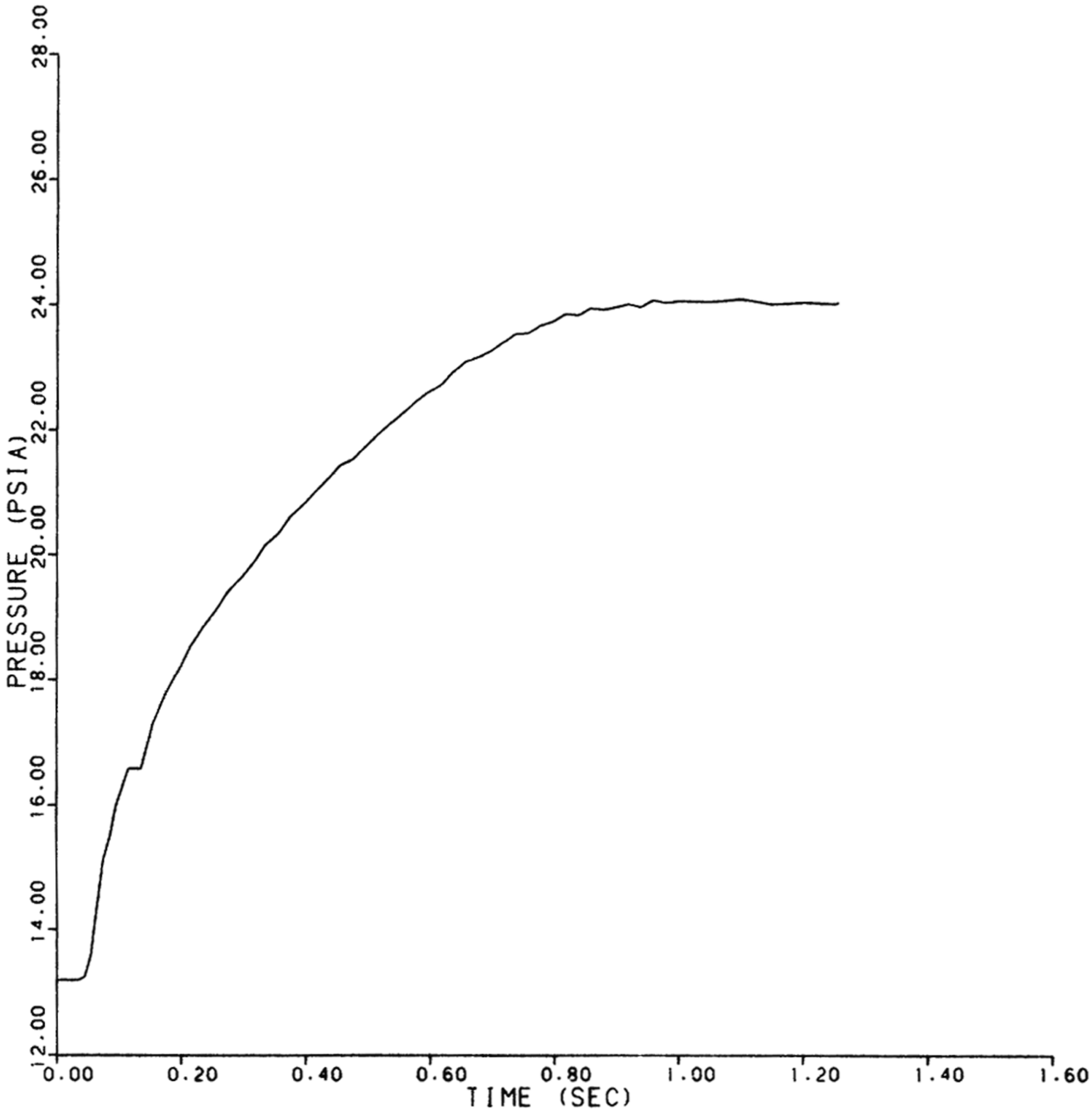
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E36 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 36 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E37 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 37 OF 74)



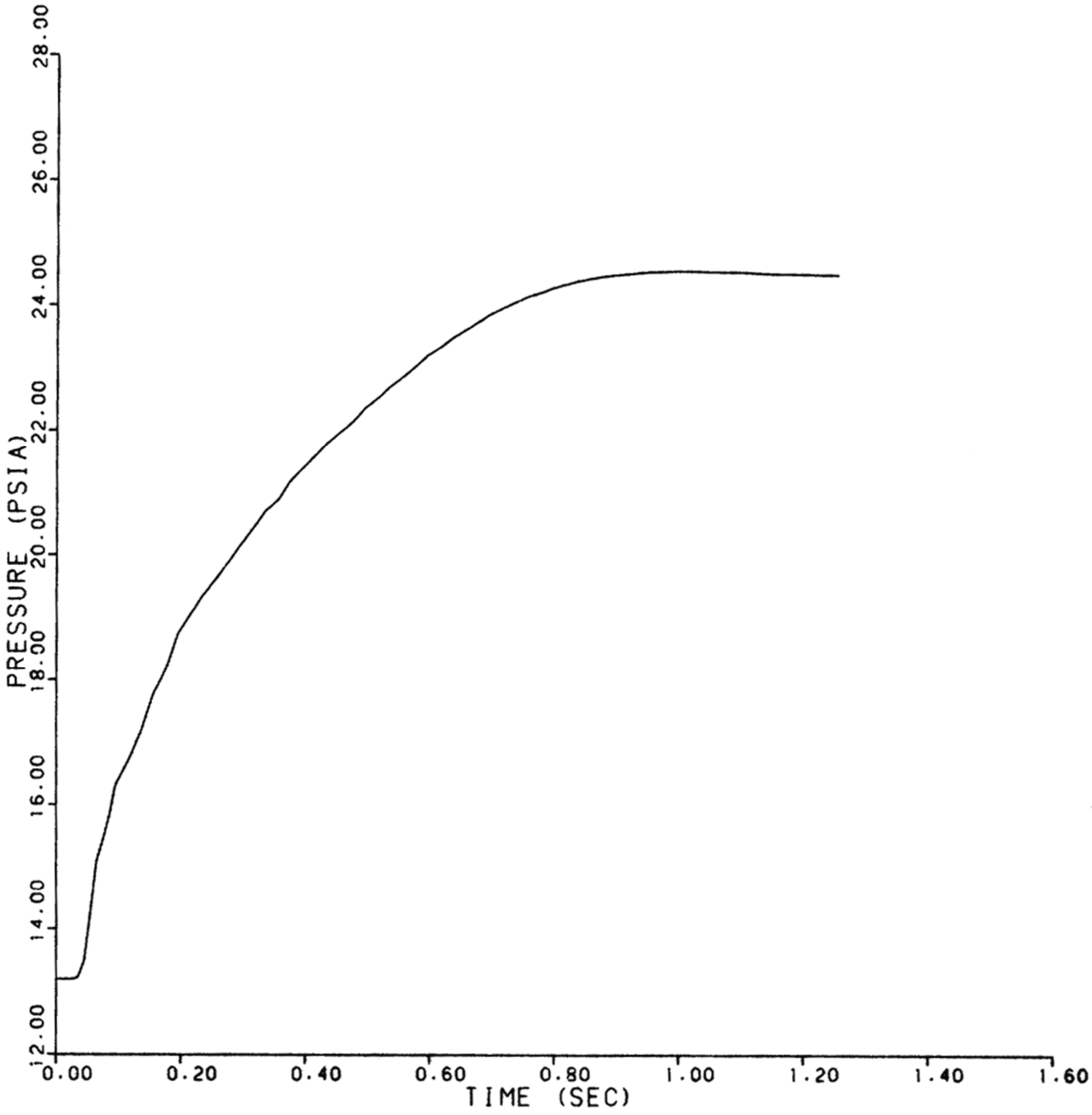
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E38
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 38 OF 74)



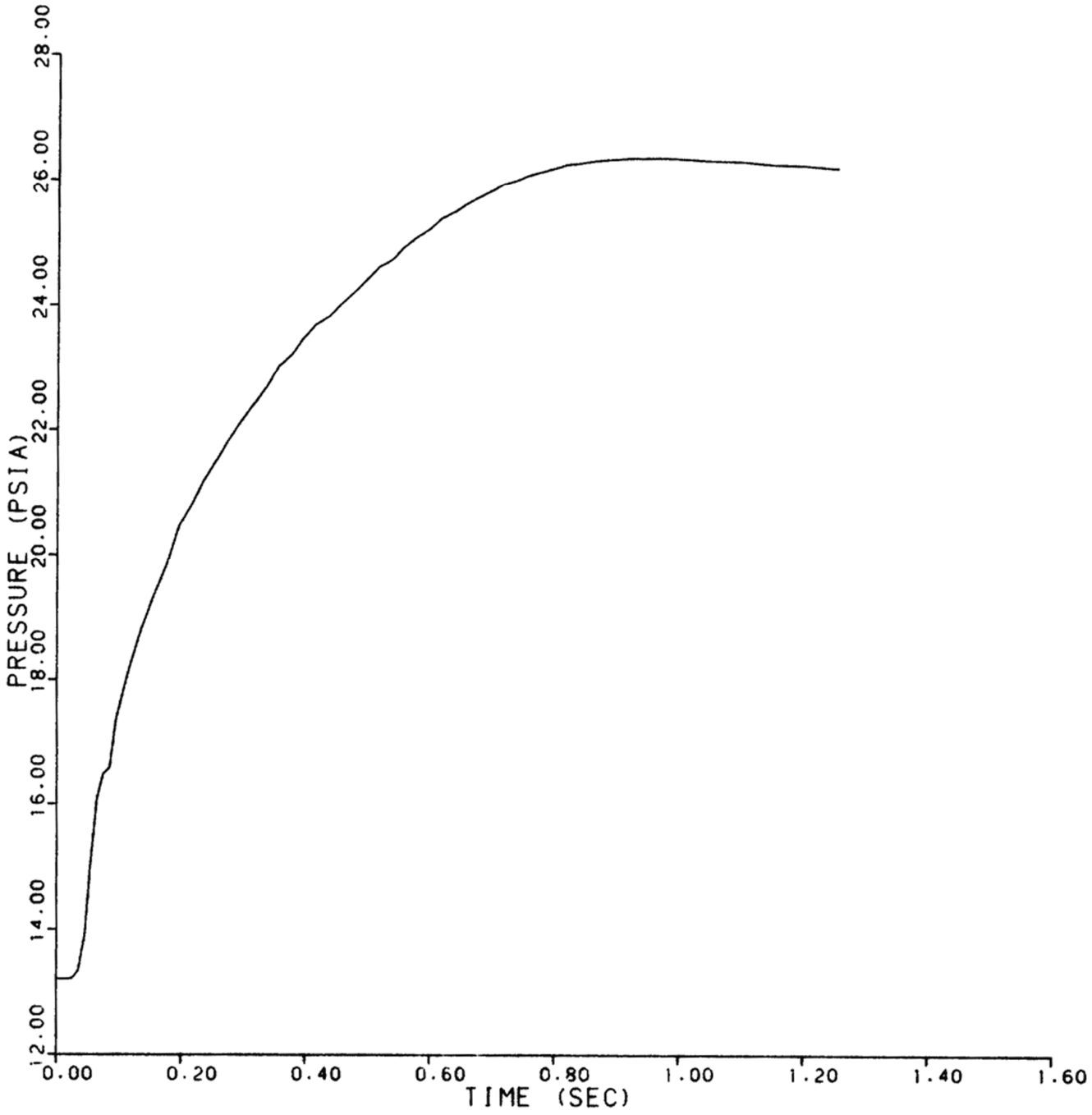
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

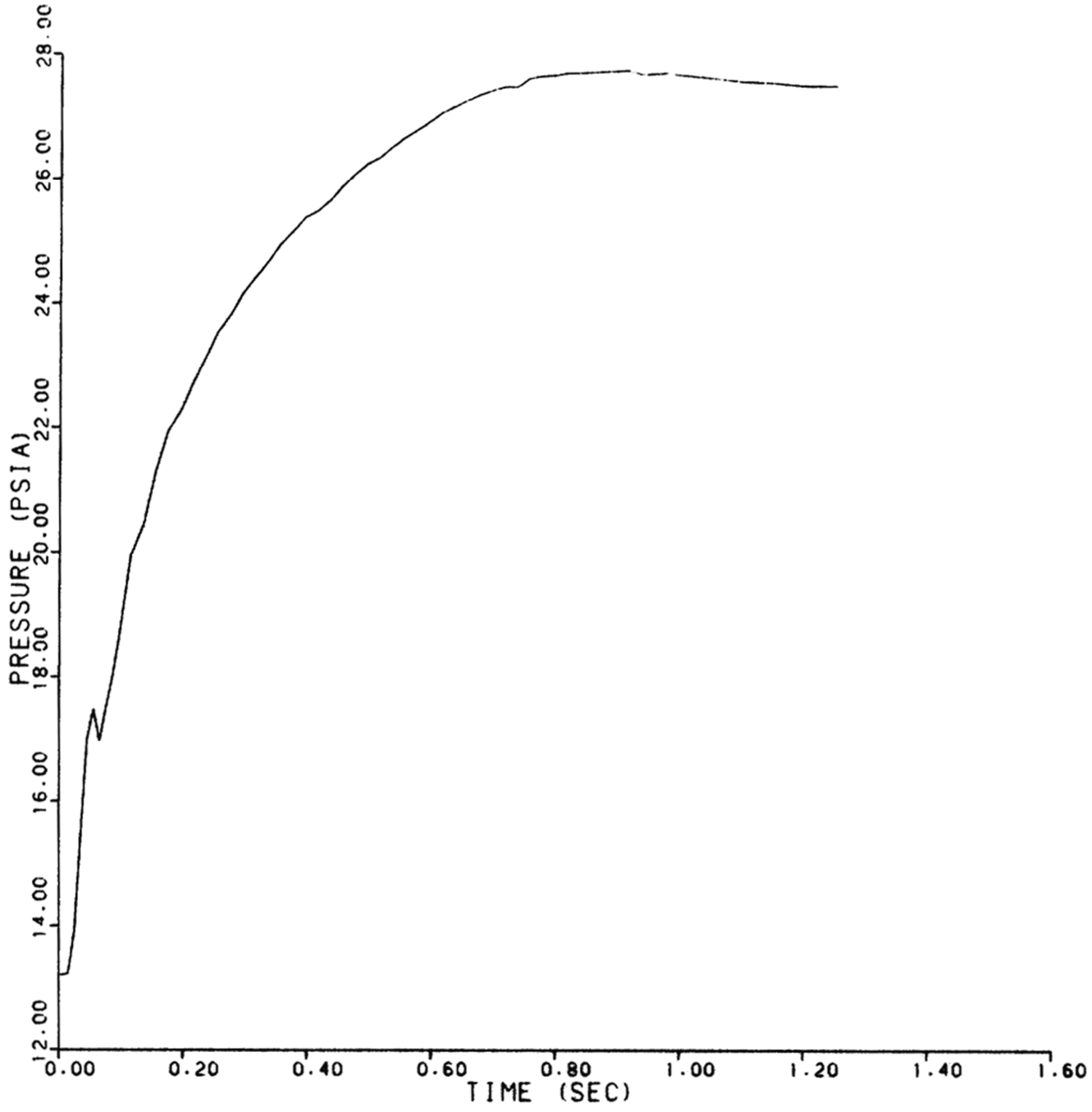
STEAM GENERATOR OUTLET NOZZLE
NODE – E39
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 39 OF 74)



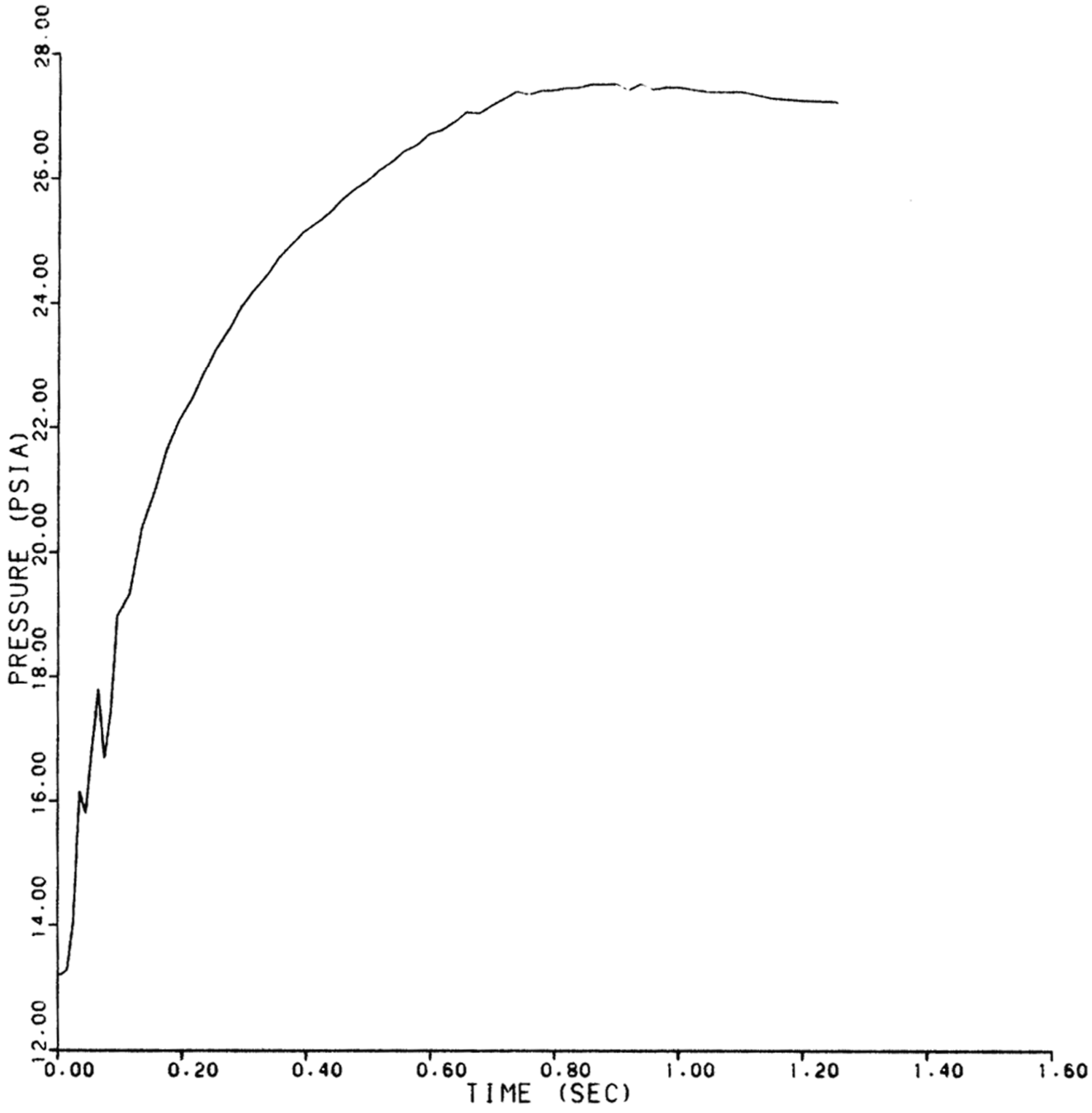
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E40 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 40 OF 74)



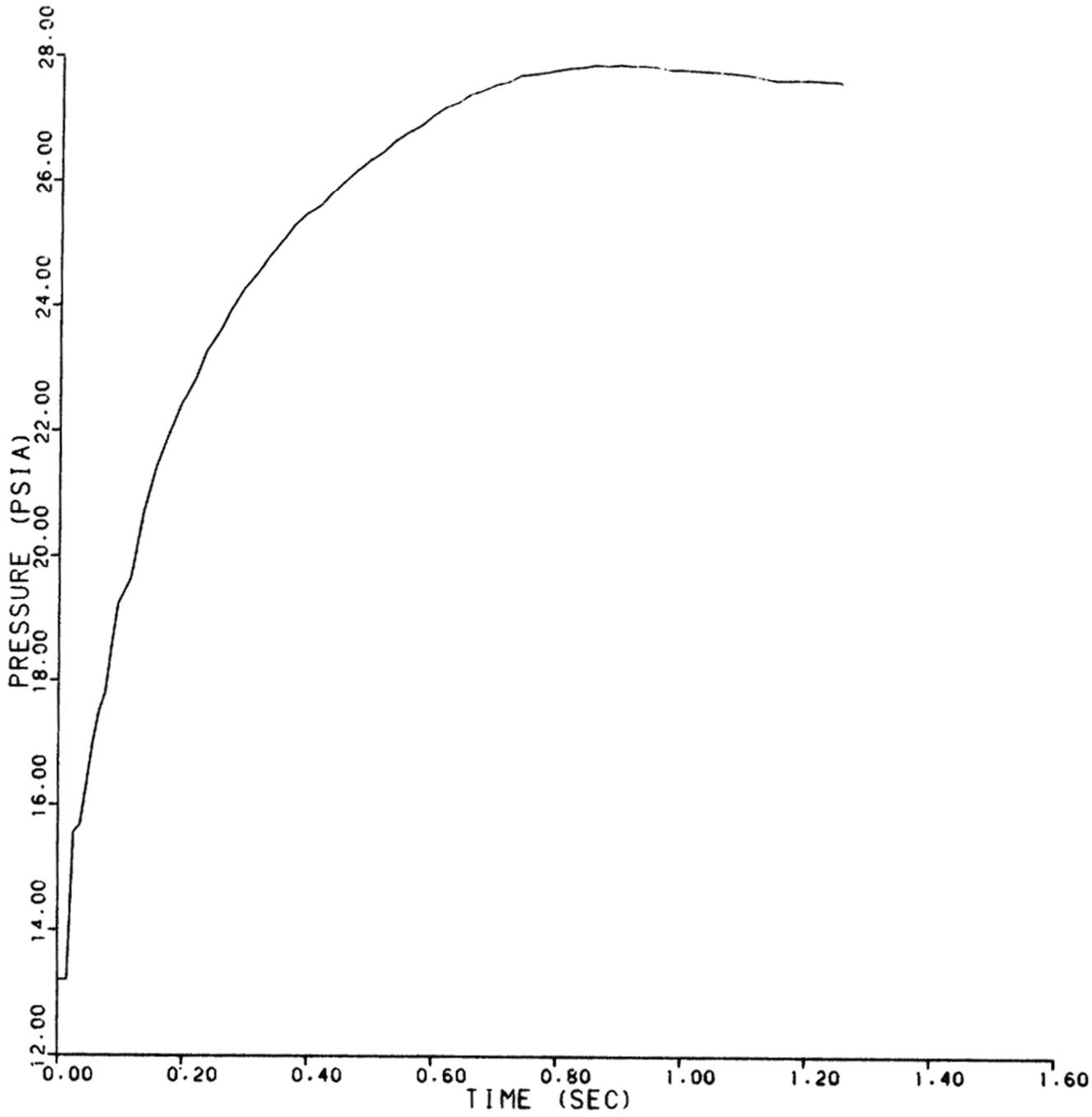
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E41 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 41 OF 74)



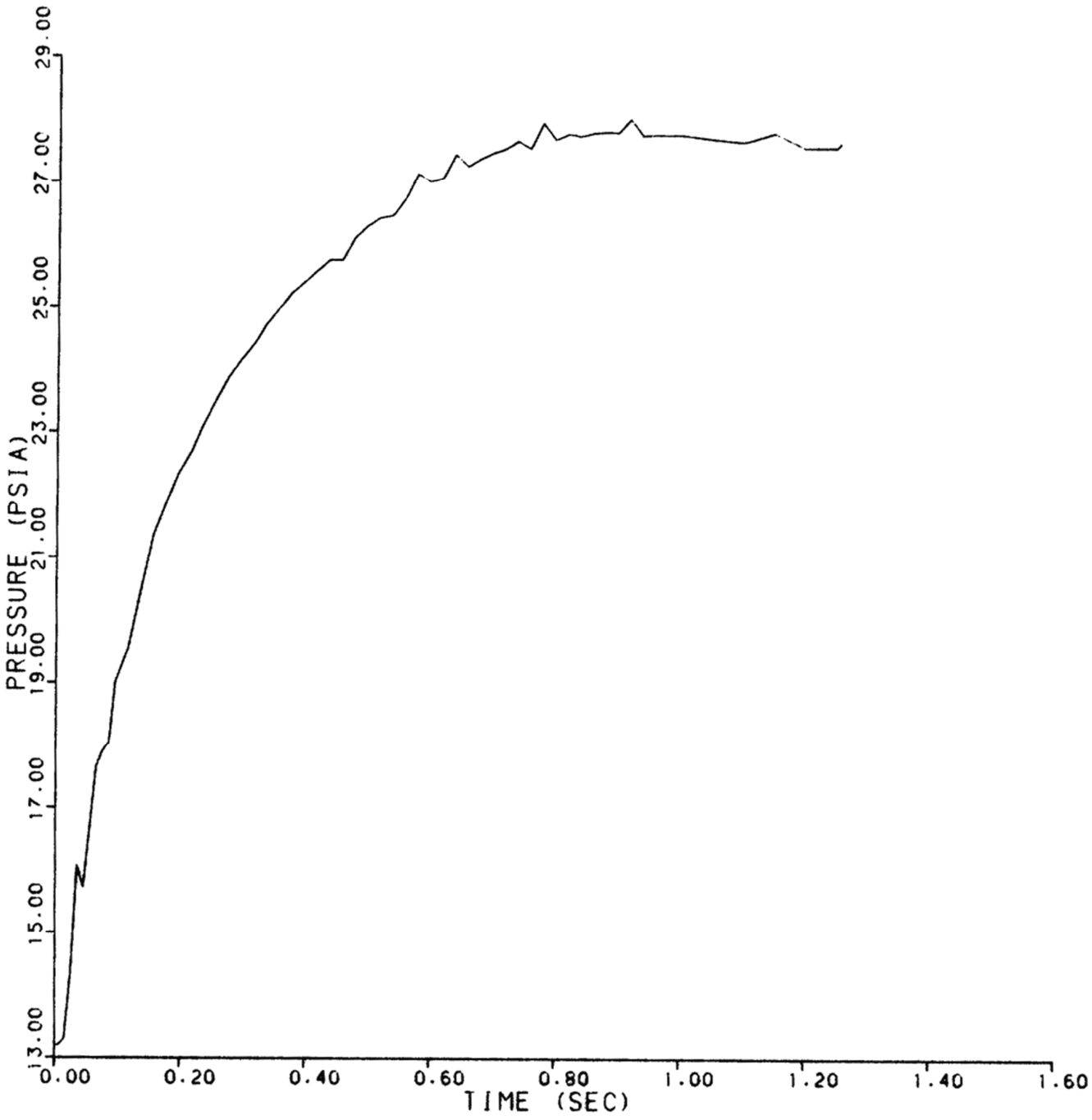
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E42 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 42 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E43 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 43 OF 74)



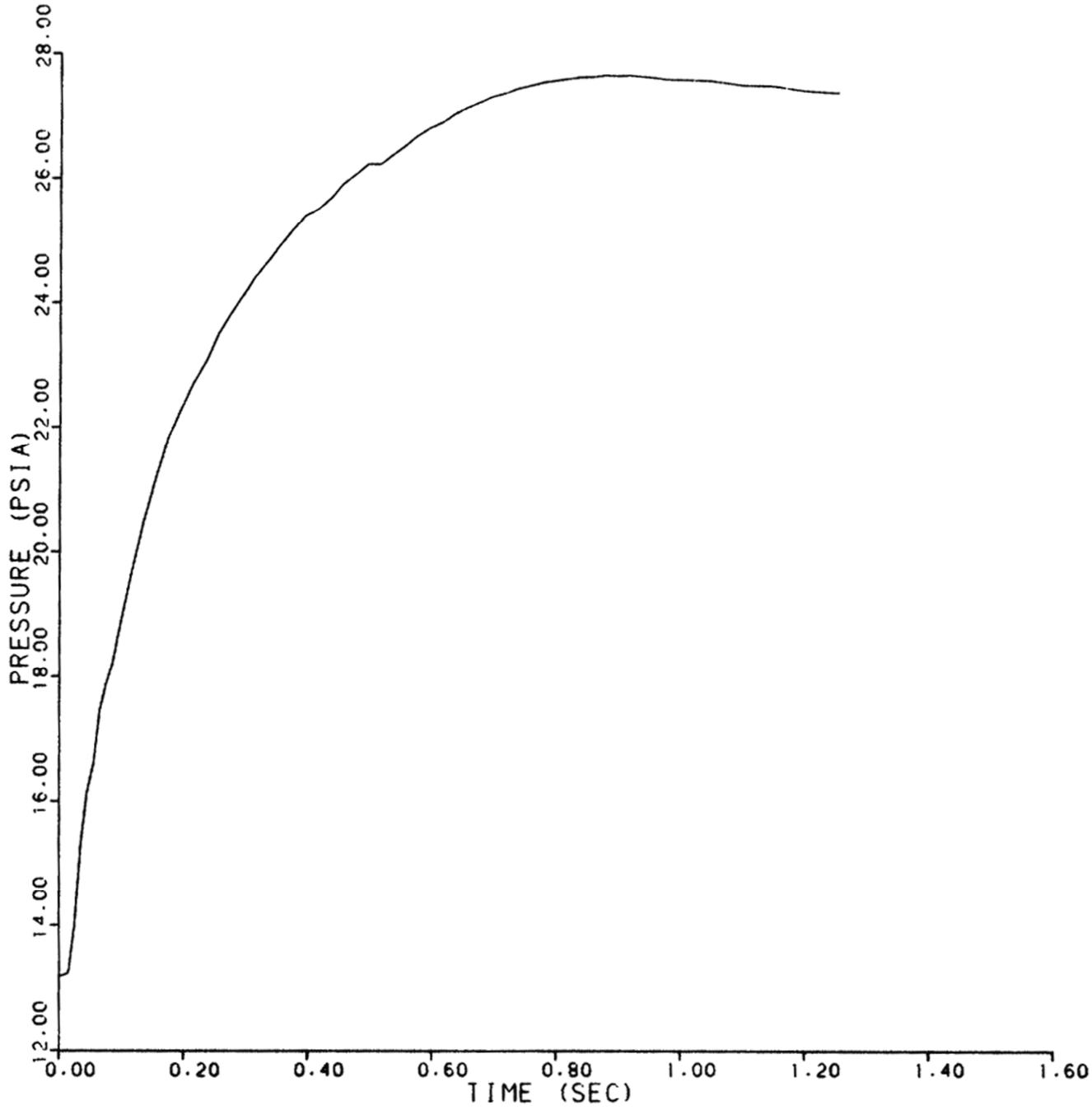
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

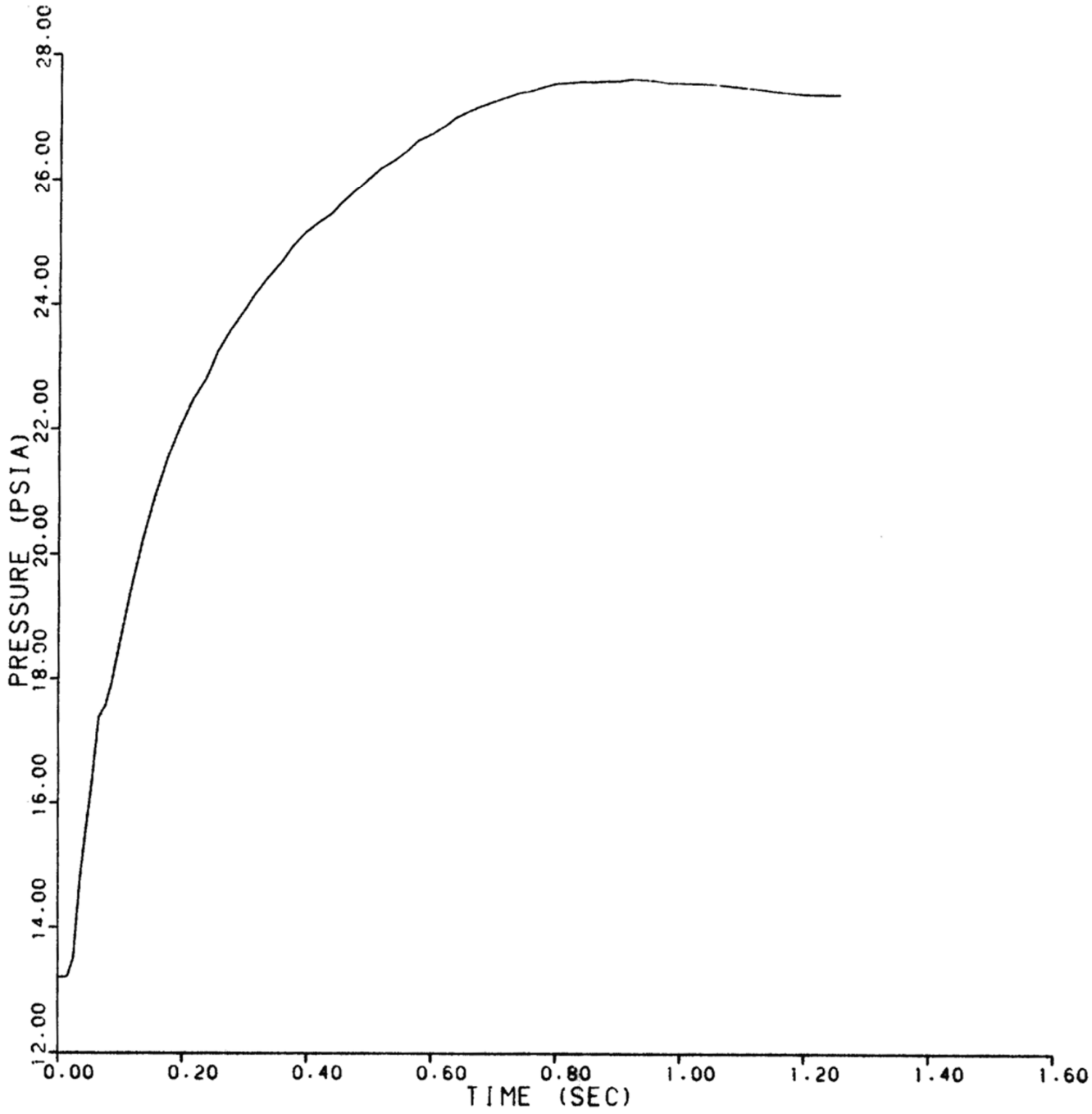
STEAM GENERATOR OUTLET NOZZLE
NODE – E44
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 44 OF 74)



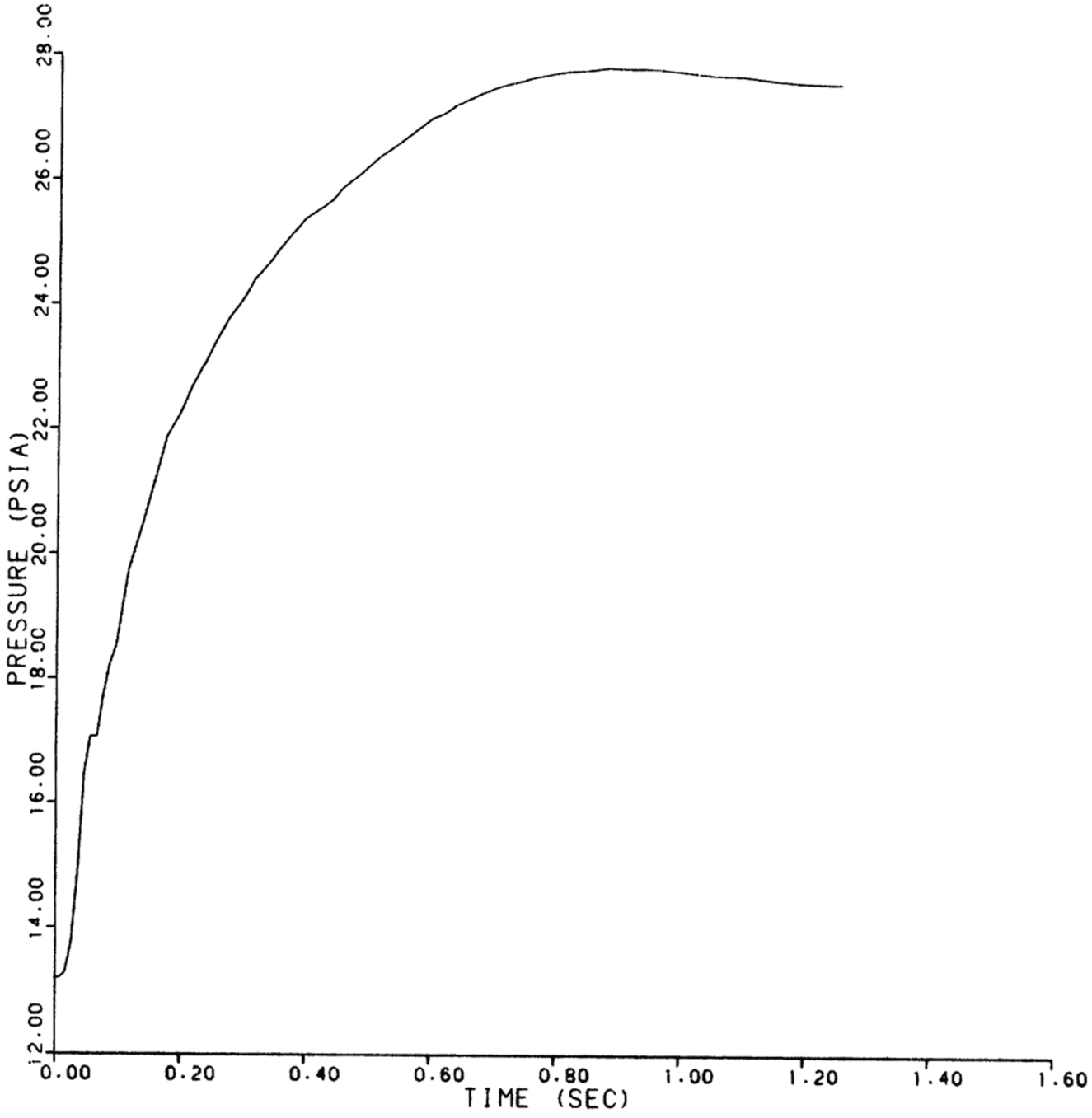
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E45 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 45 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E46 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 46 OF 74)



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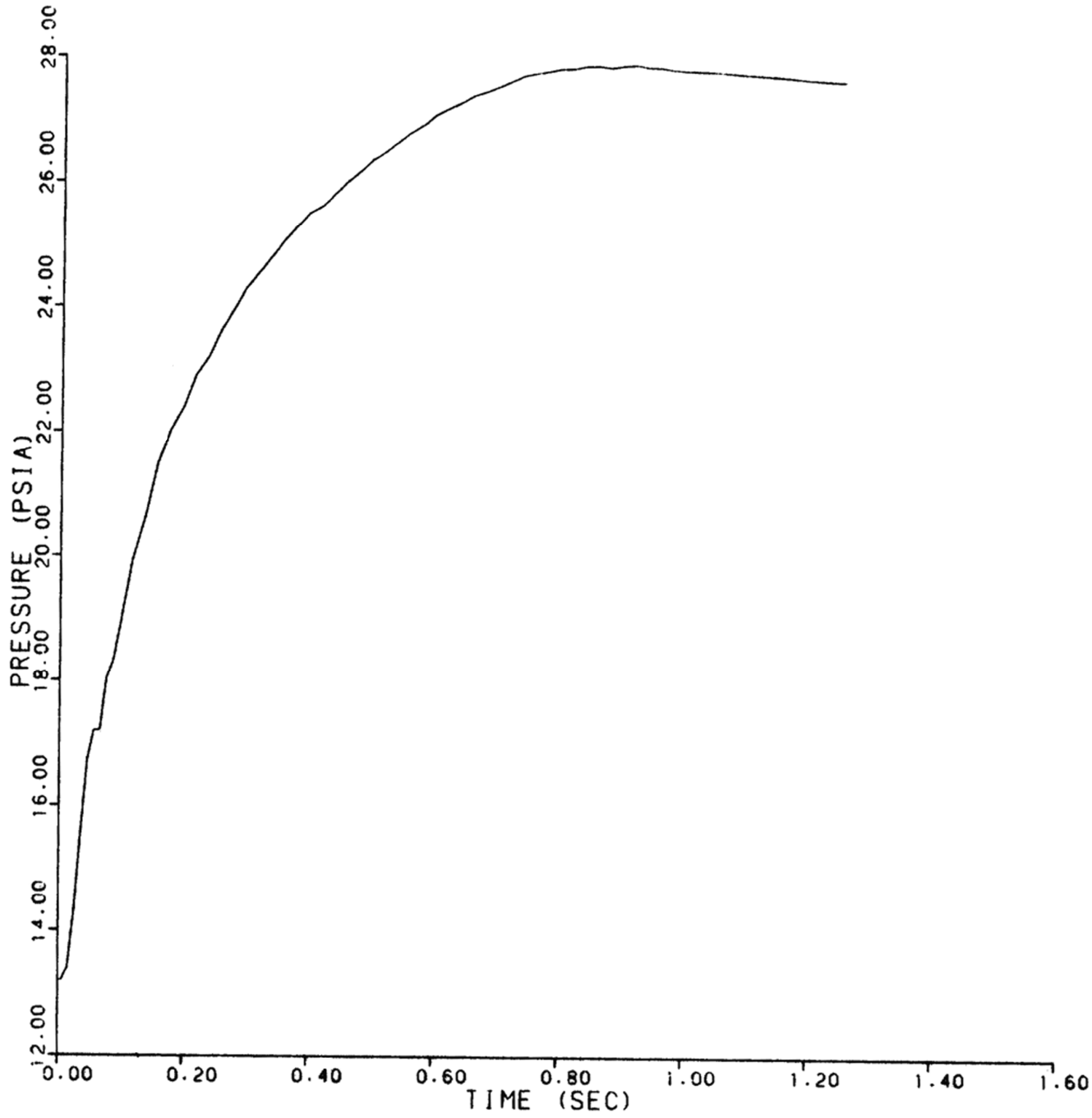
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E47
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 47 OF 74)



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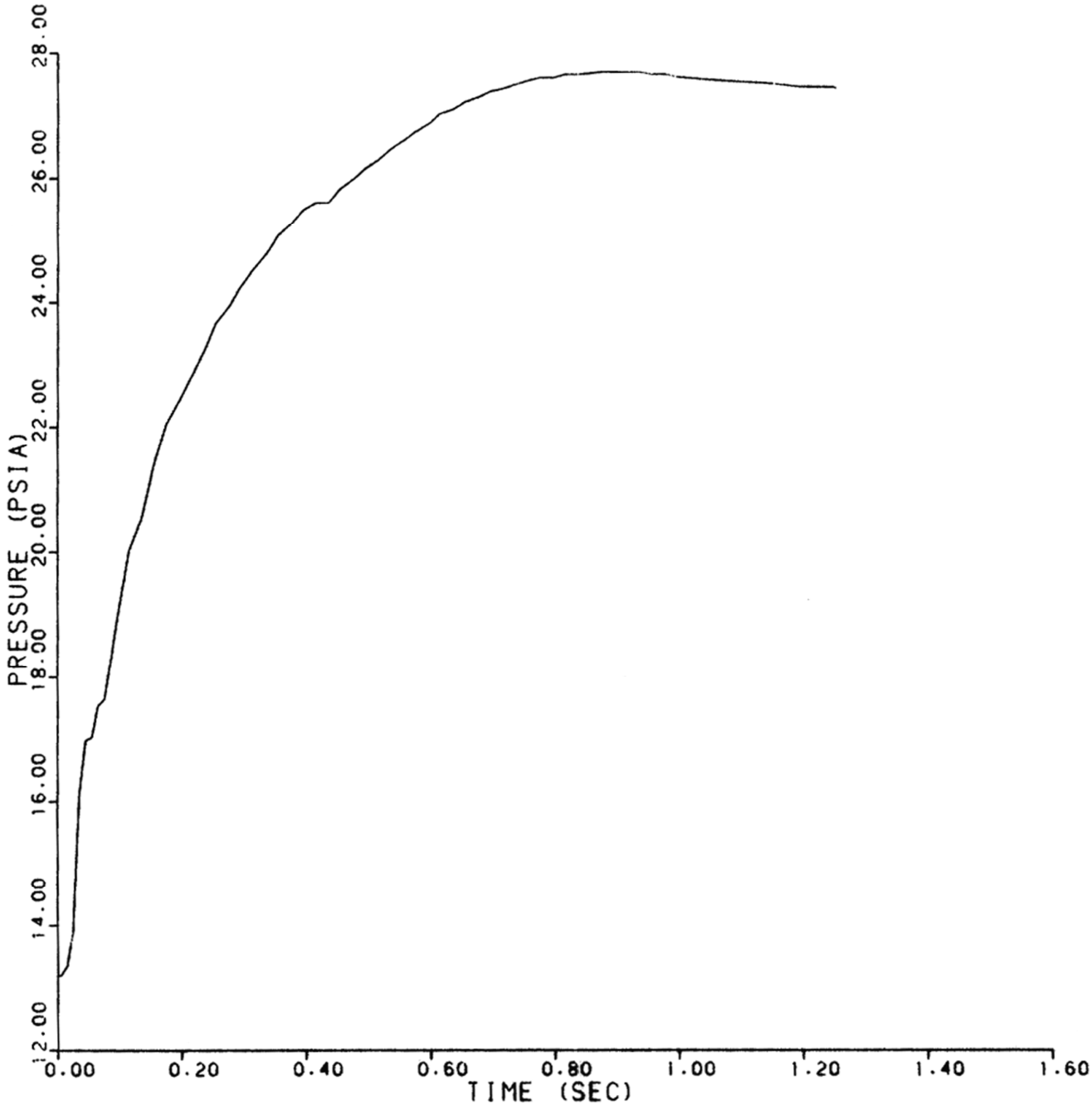
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E48
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 48 OF 74)



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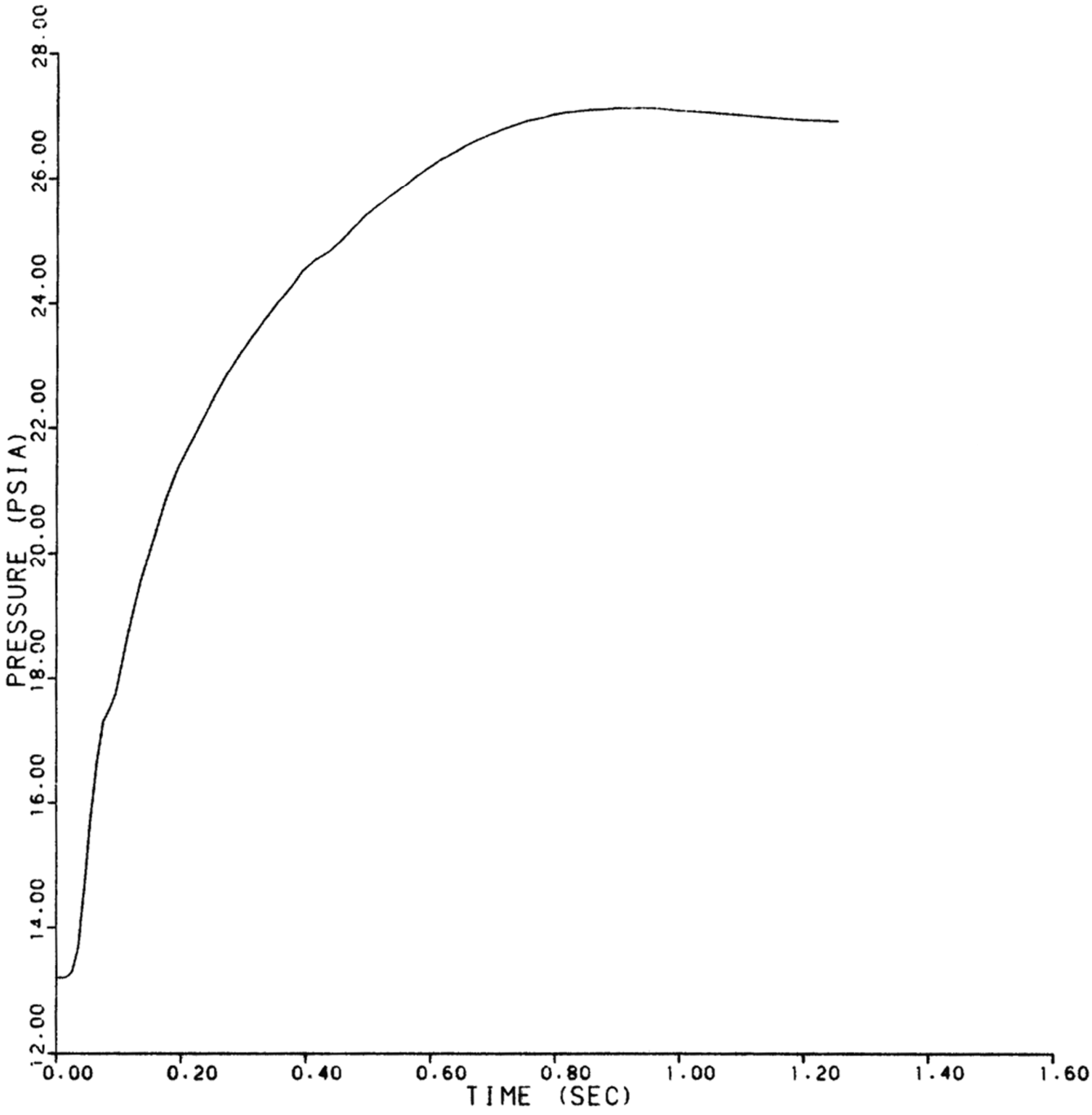
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E49
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 49 OF 74)



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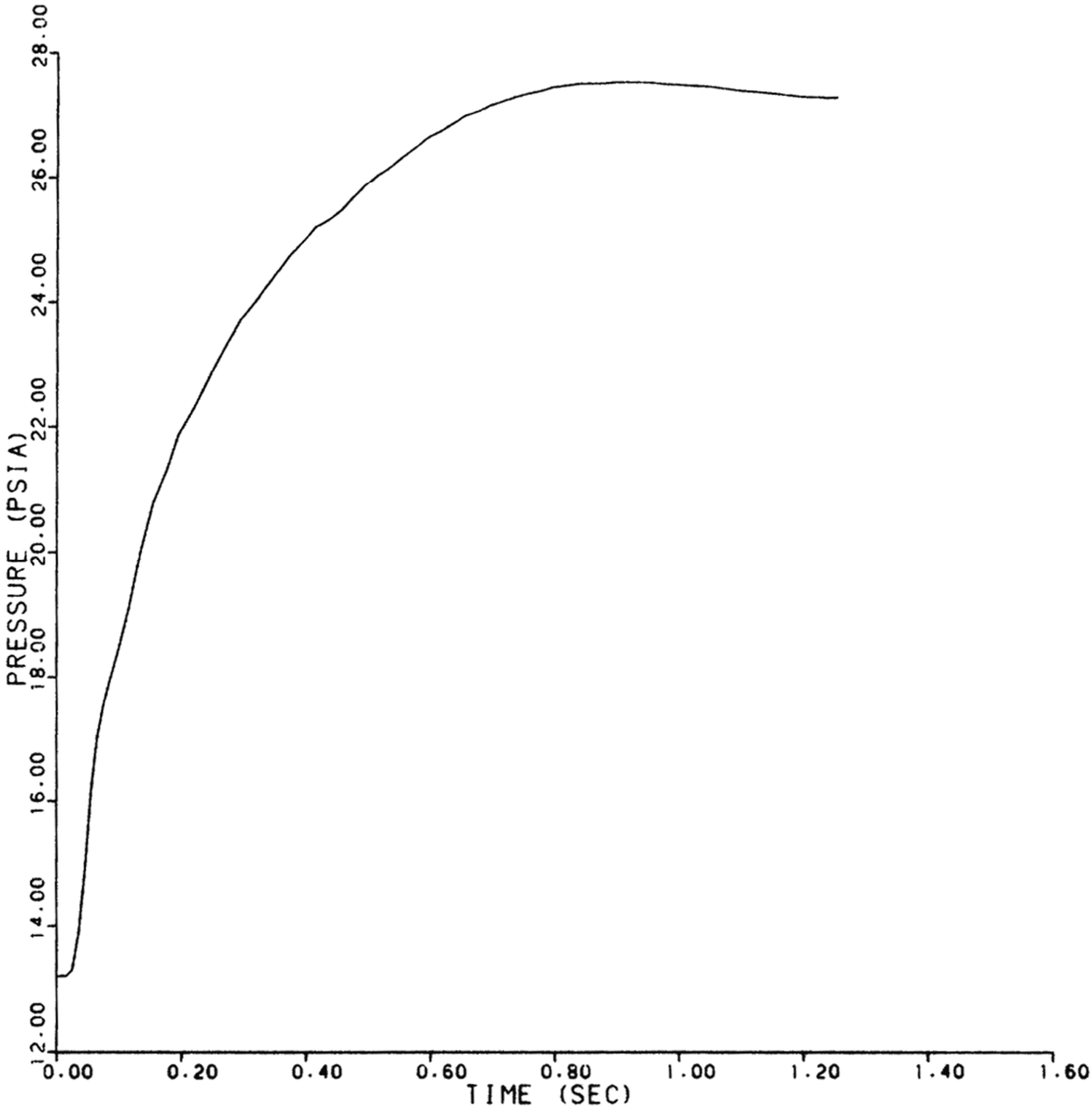
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E50
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 50 OF 74)



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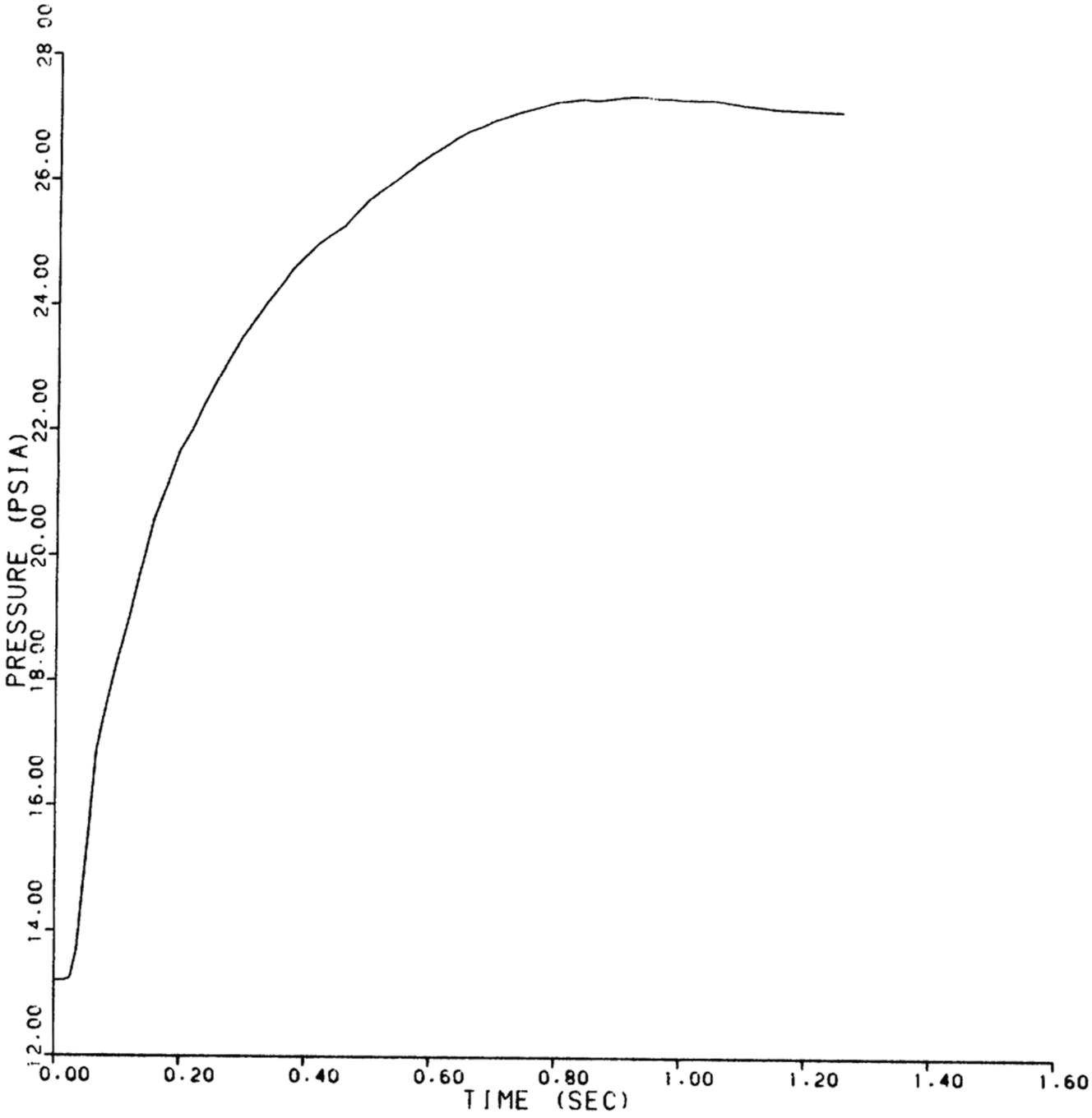
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E51
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 51 OF 74)



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SOUTHERN COMPANY
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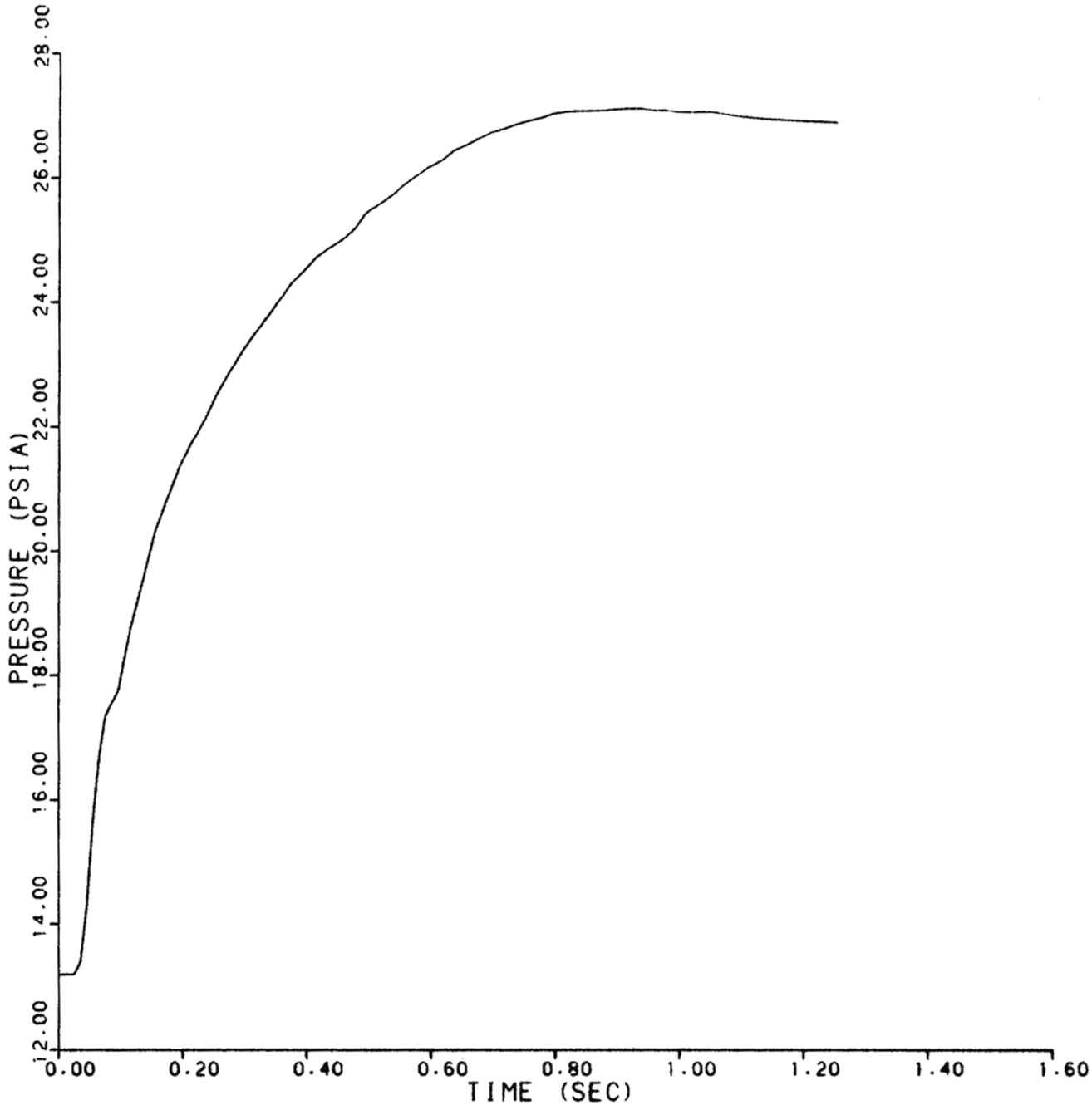
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

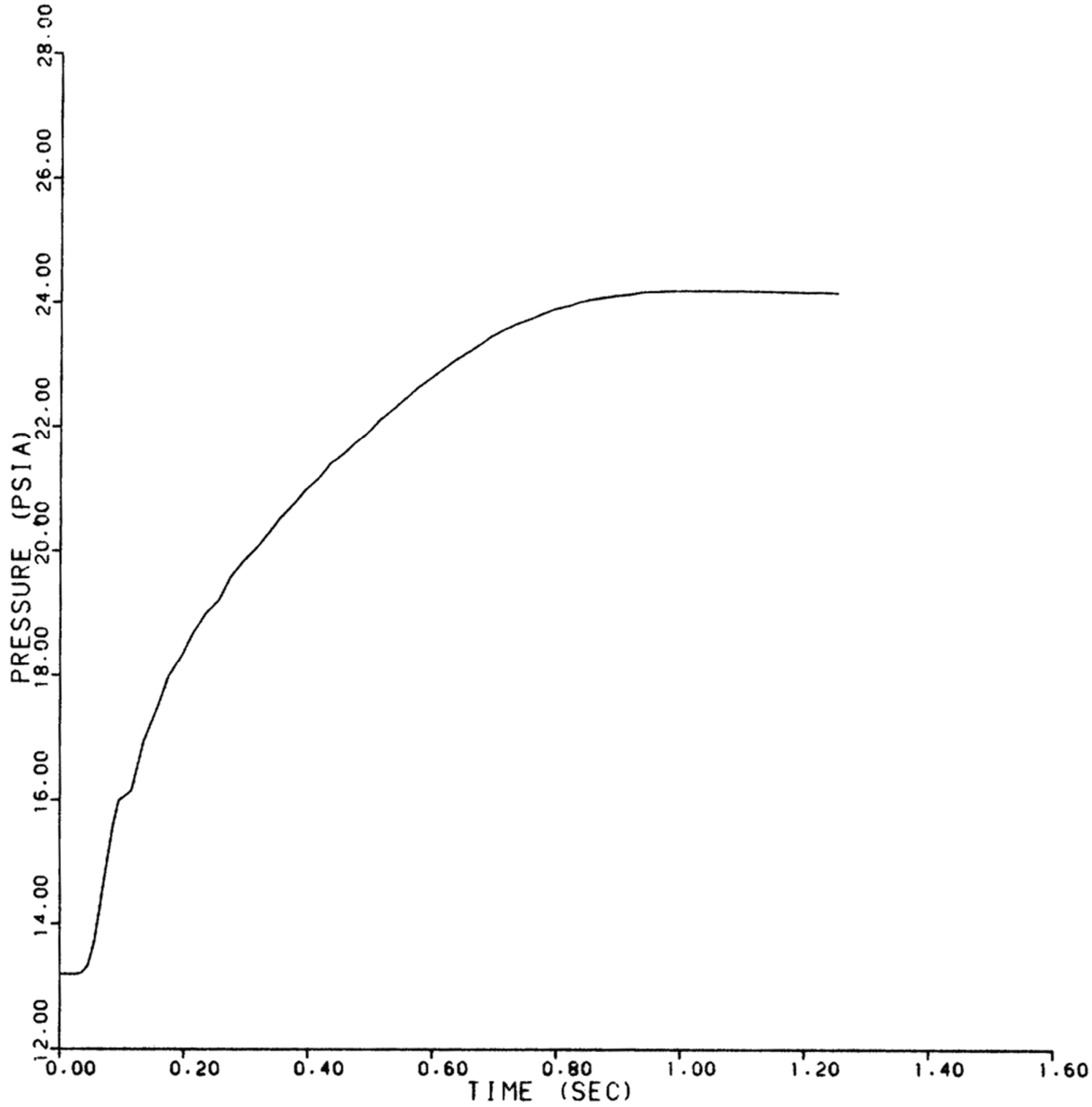
STEAM GENERATOR OUTLET NOZZLE
NODE – E52
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 52 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E53 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 53 OF 74)



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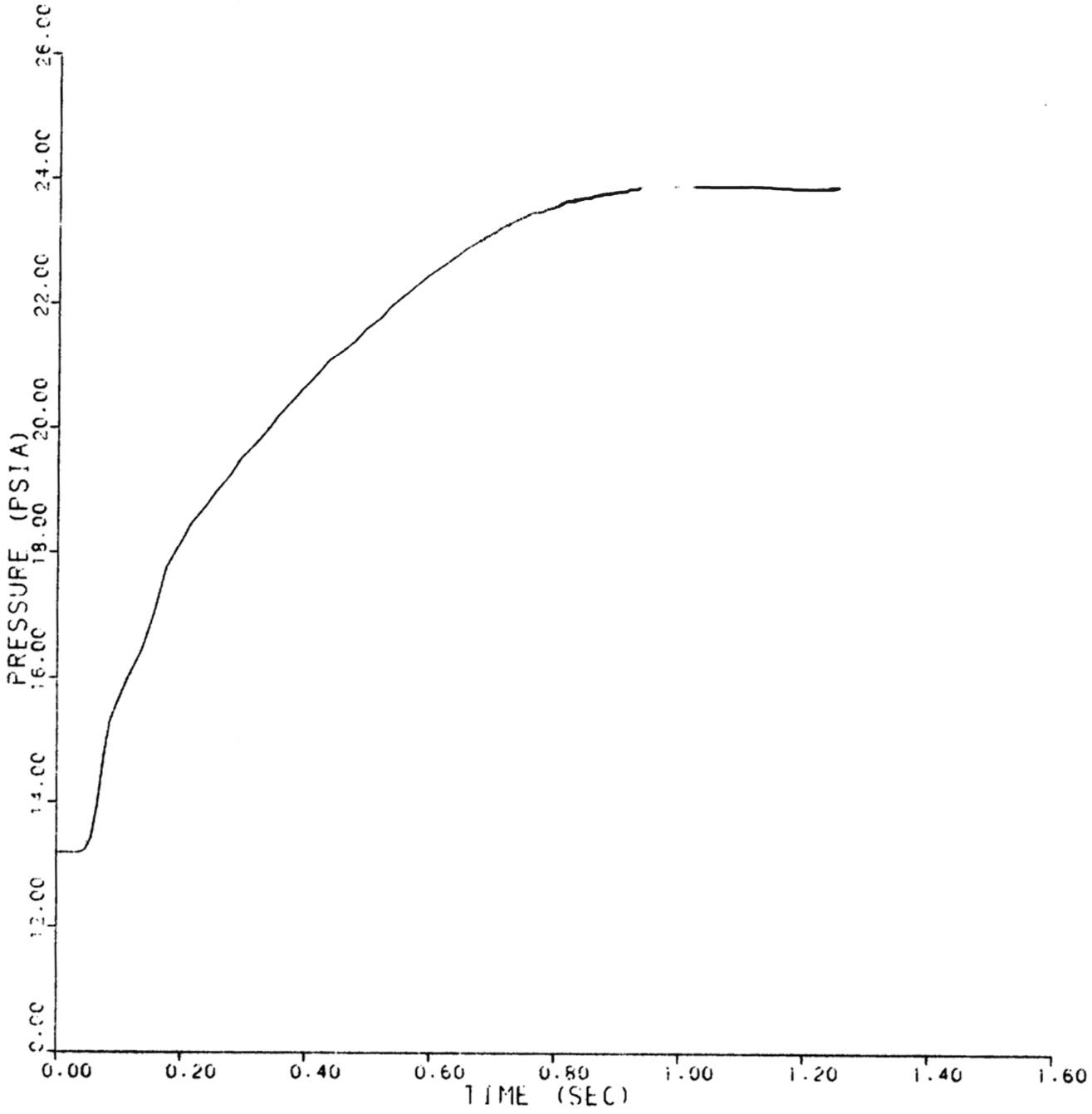
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E54
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 54 OF 74)



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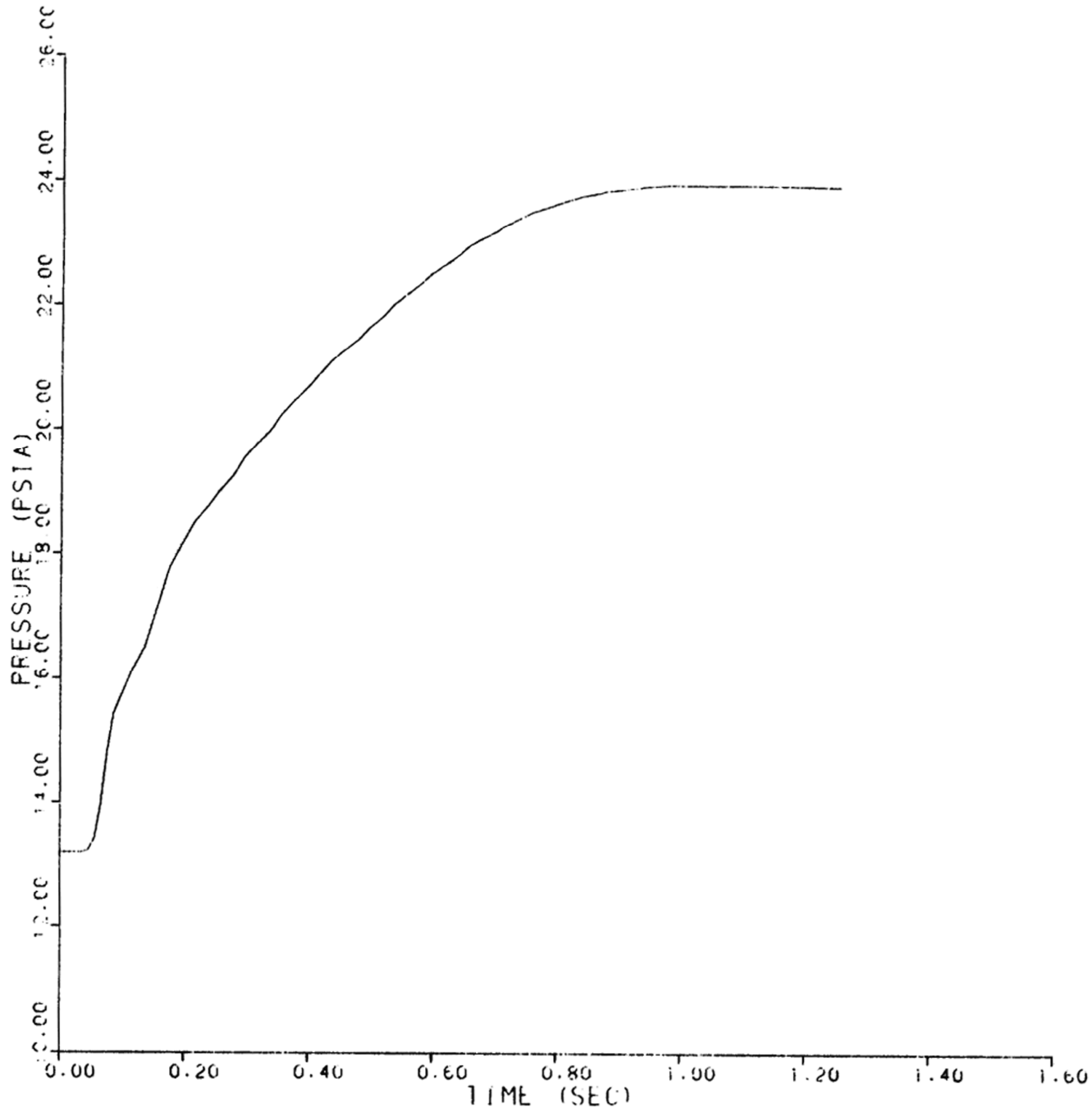
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E55 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 55 OF 74)



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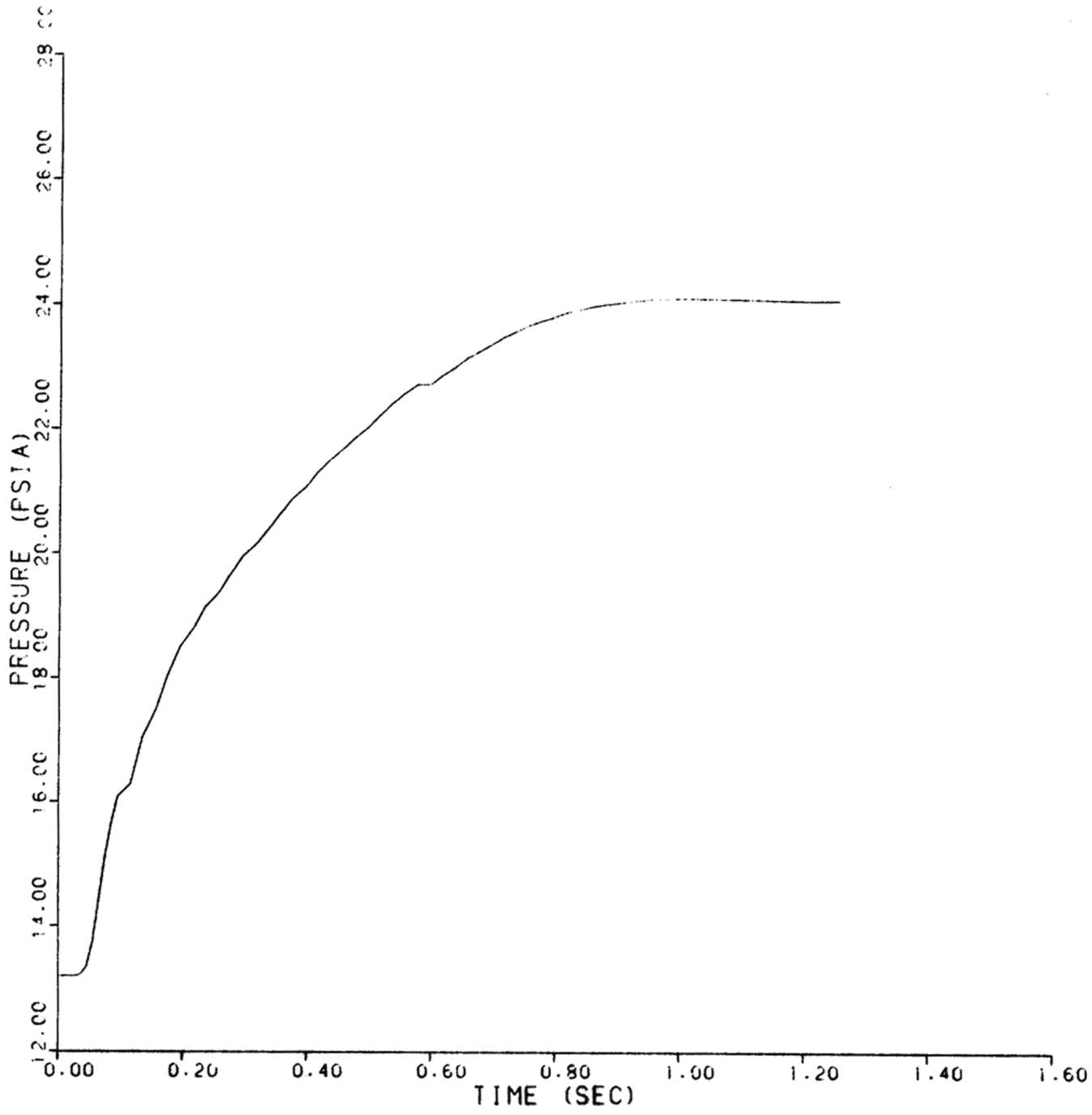
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E56 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 56 OF 74)



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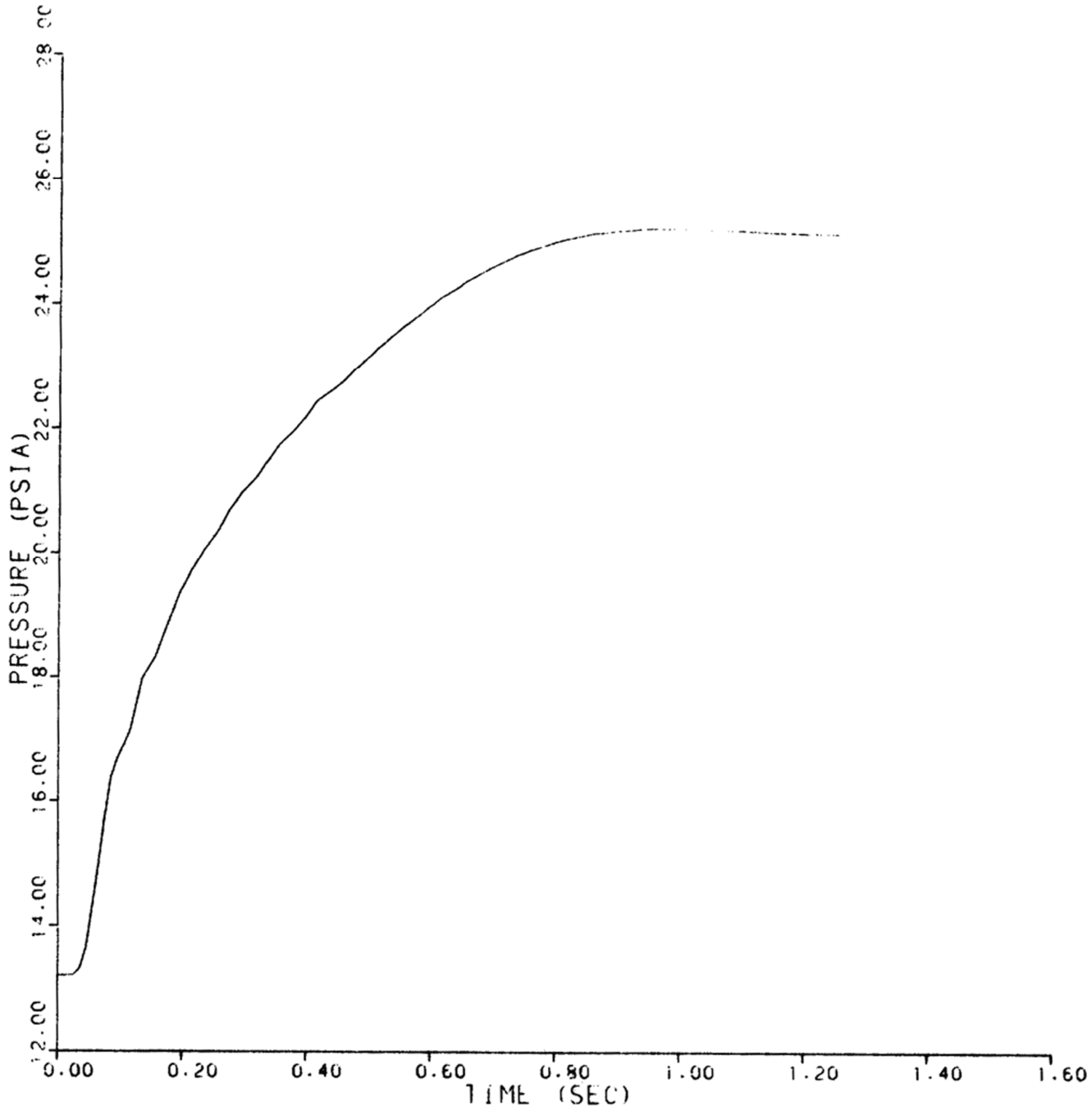
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E57 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 57 OF 74)



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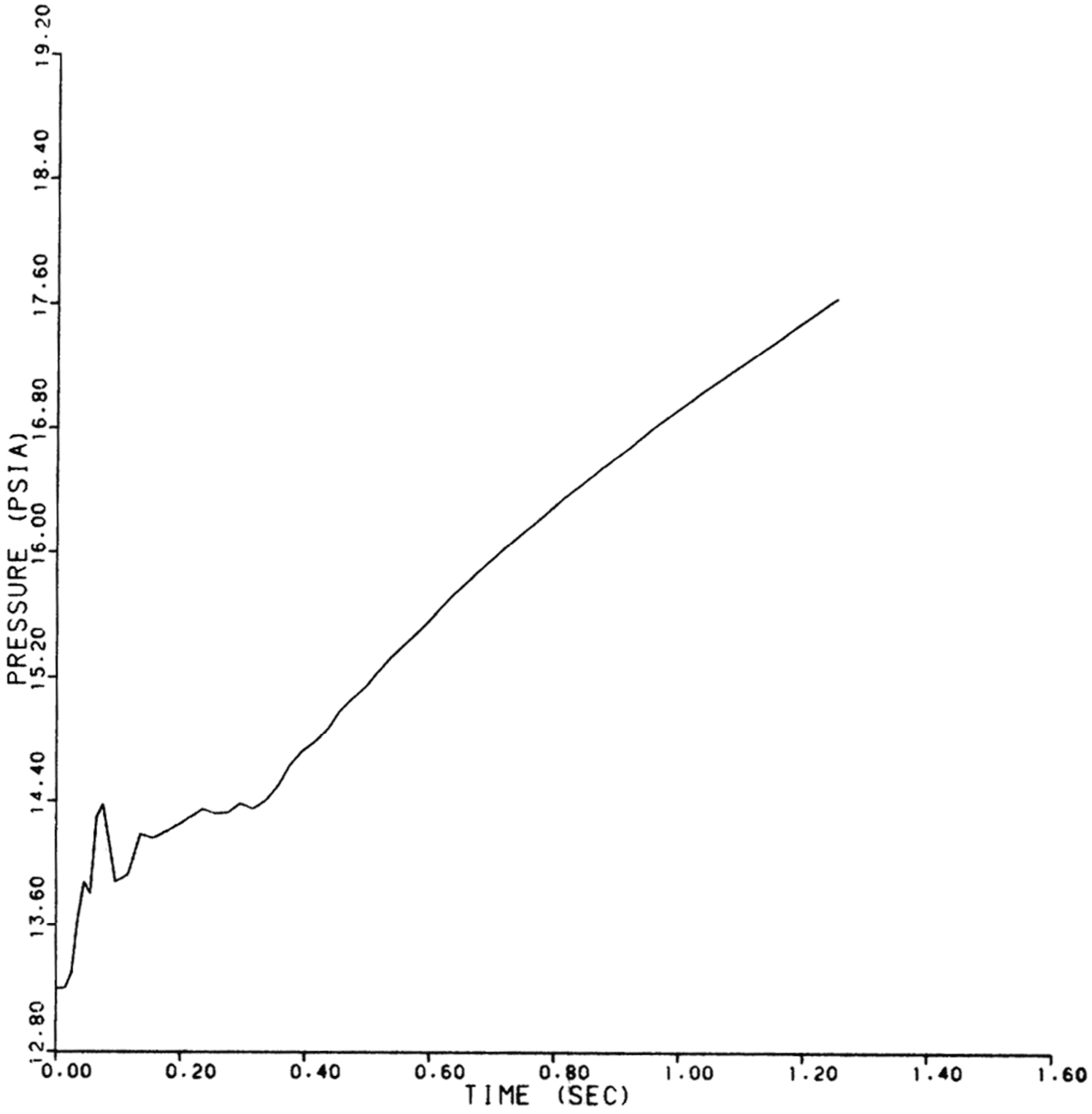
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E58
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 58 OF 74)



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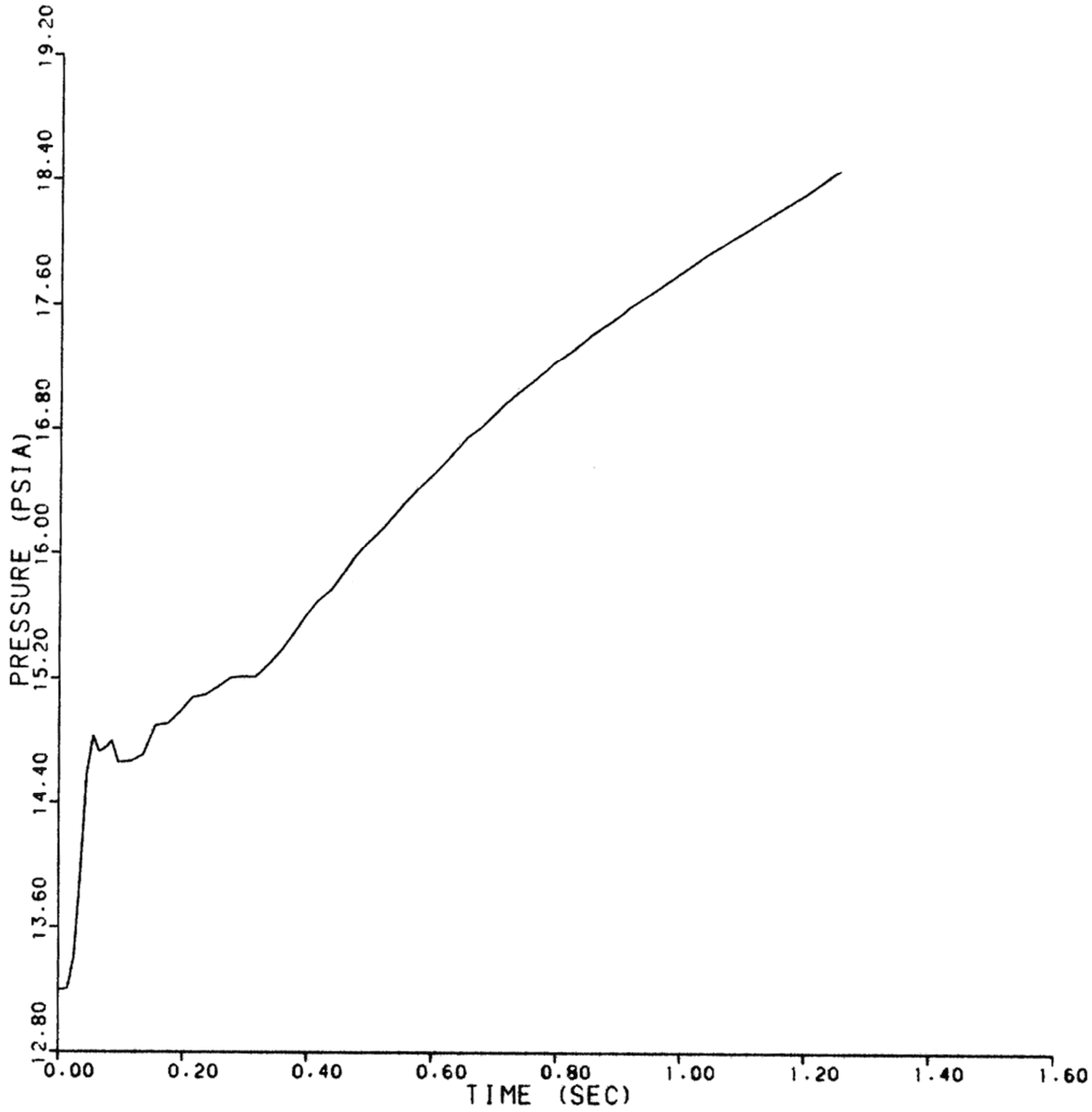
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E59 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 59 OF 74)



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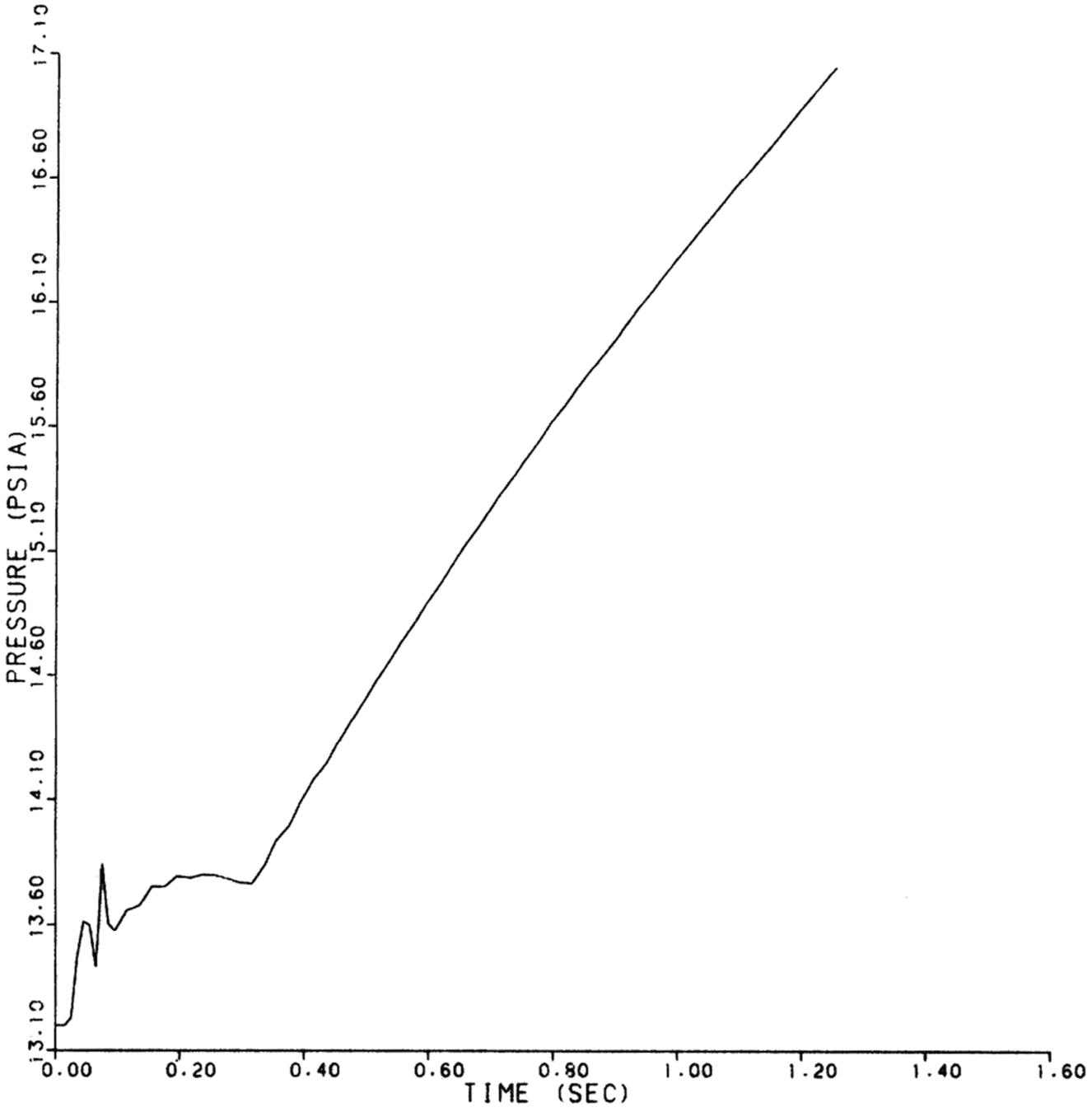
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E60 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 60 OF 74)



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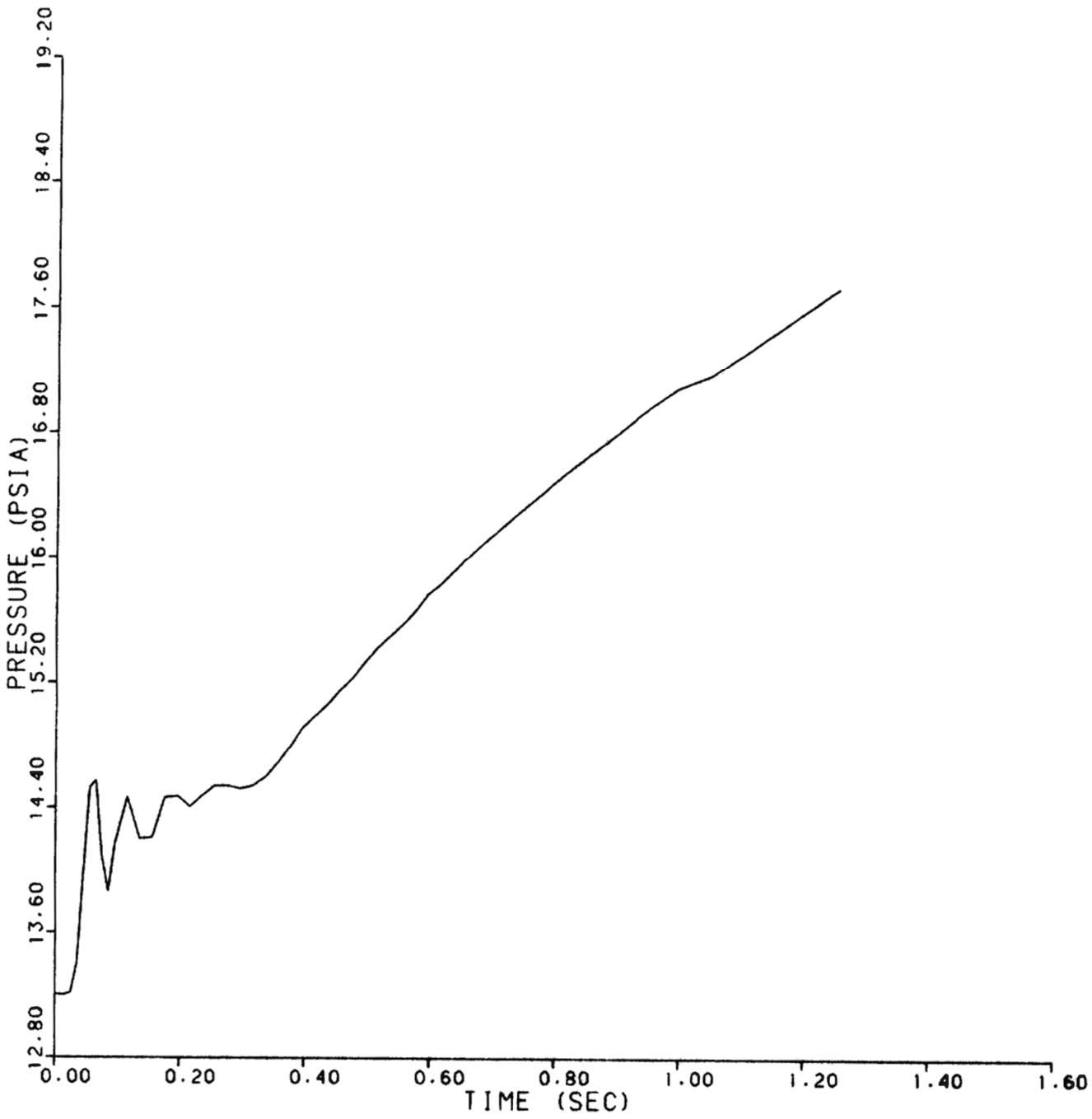
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E61
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 61 OF 74)



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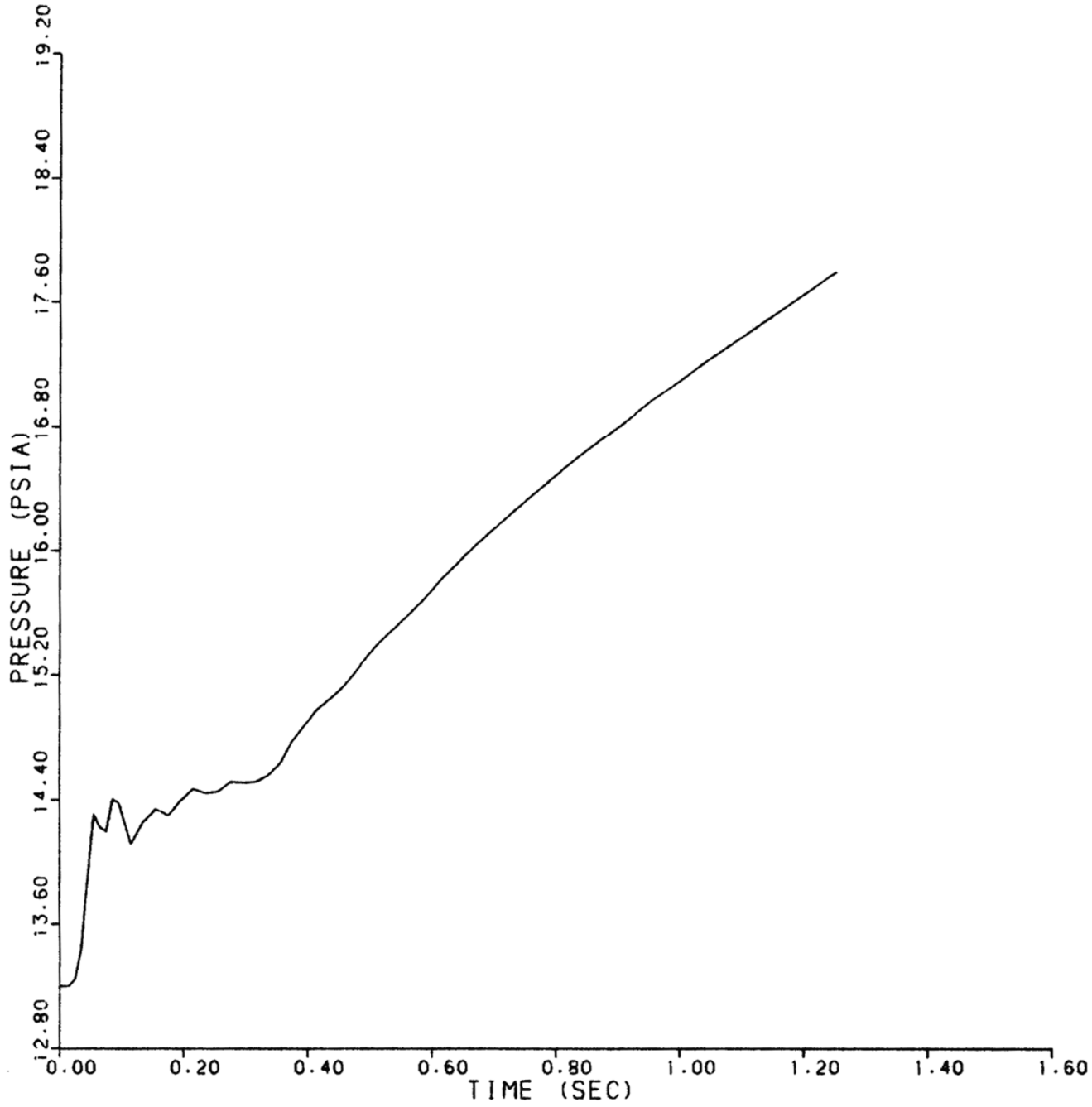
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E62
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 62 OF 74)



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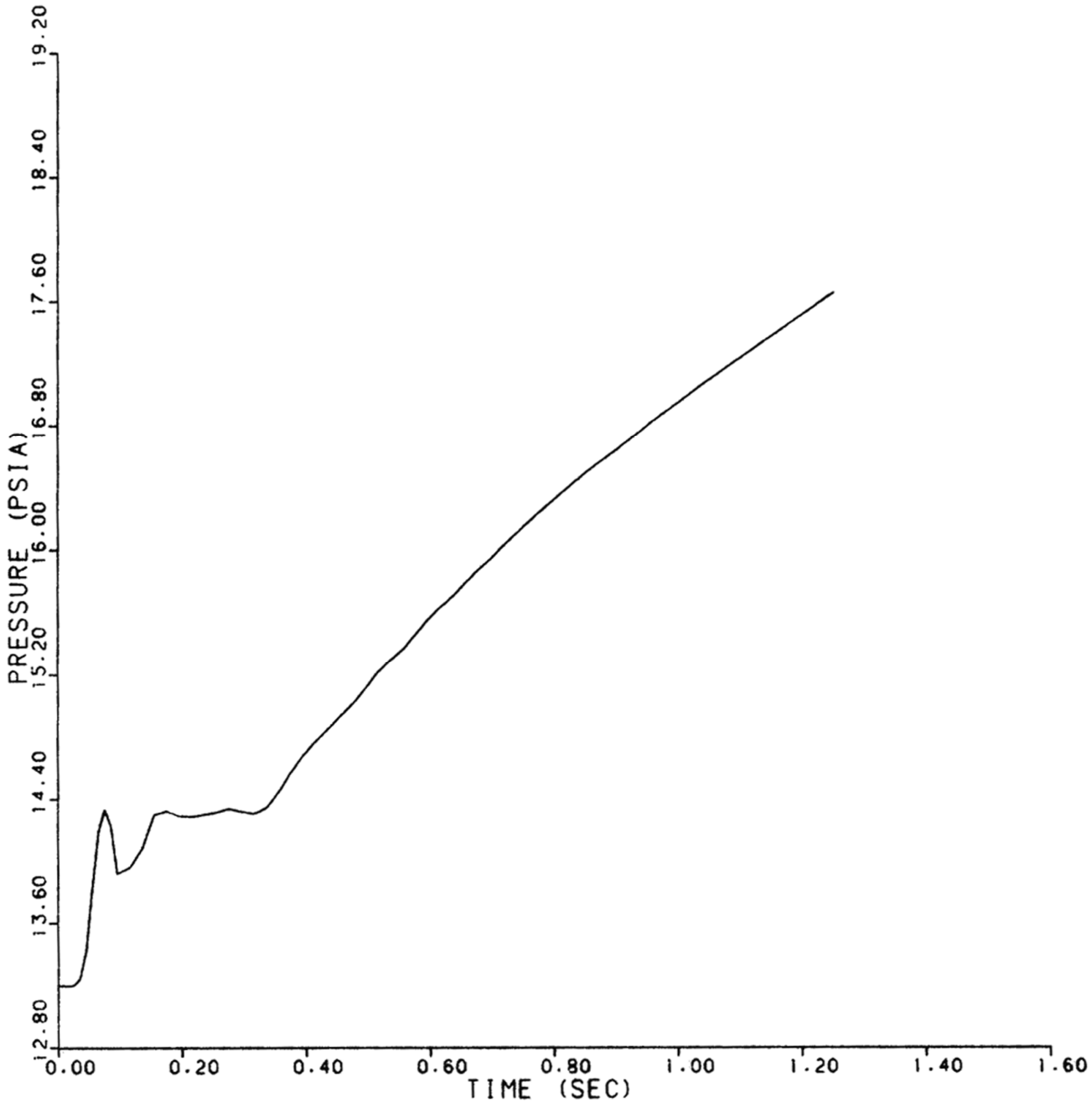
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E63
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 63 OF 74)



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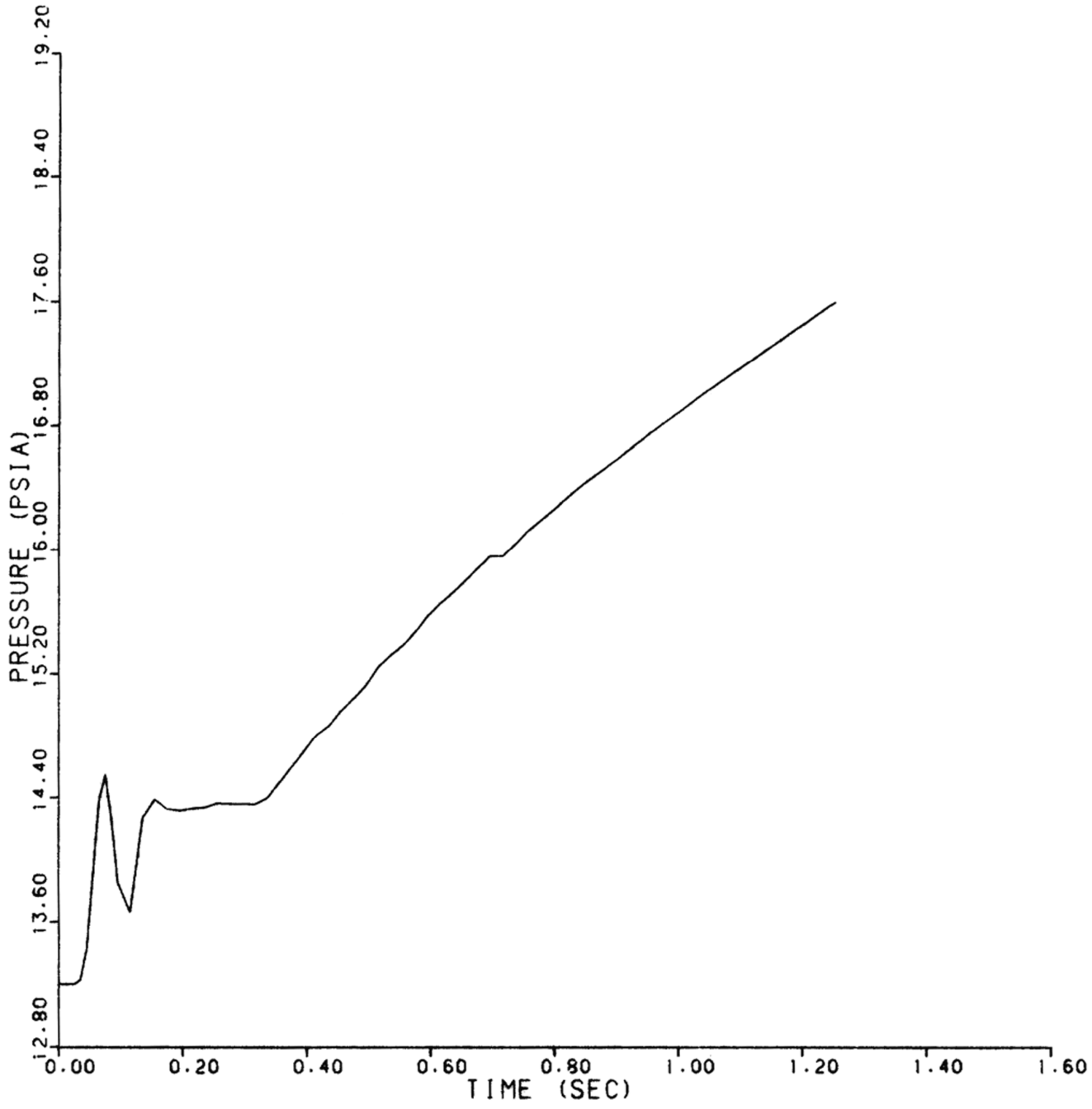
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E64
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 64 OF 74)



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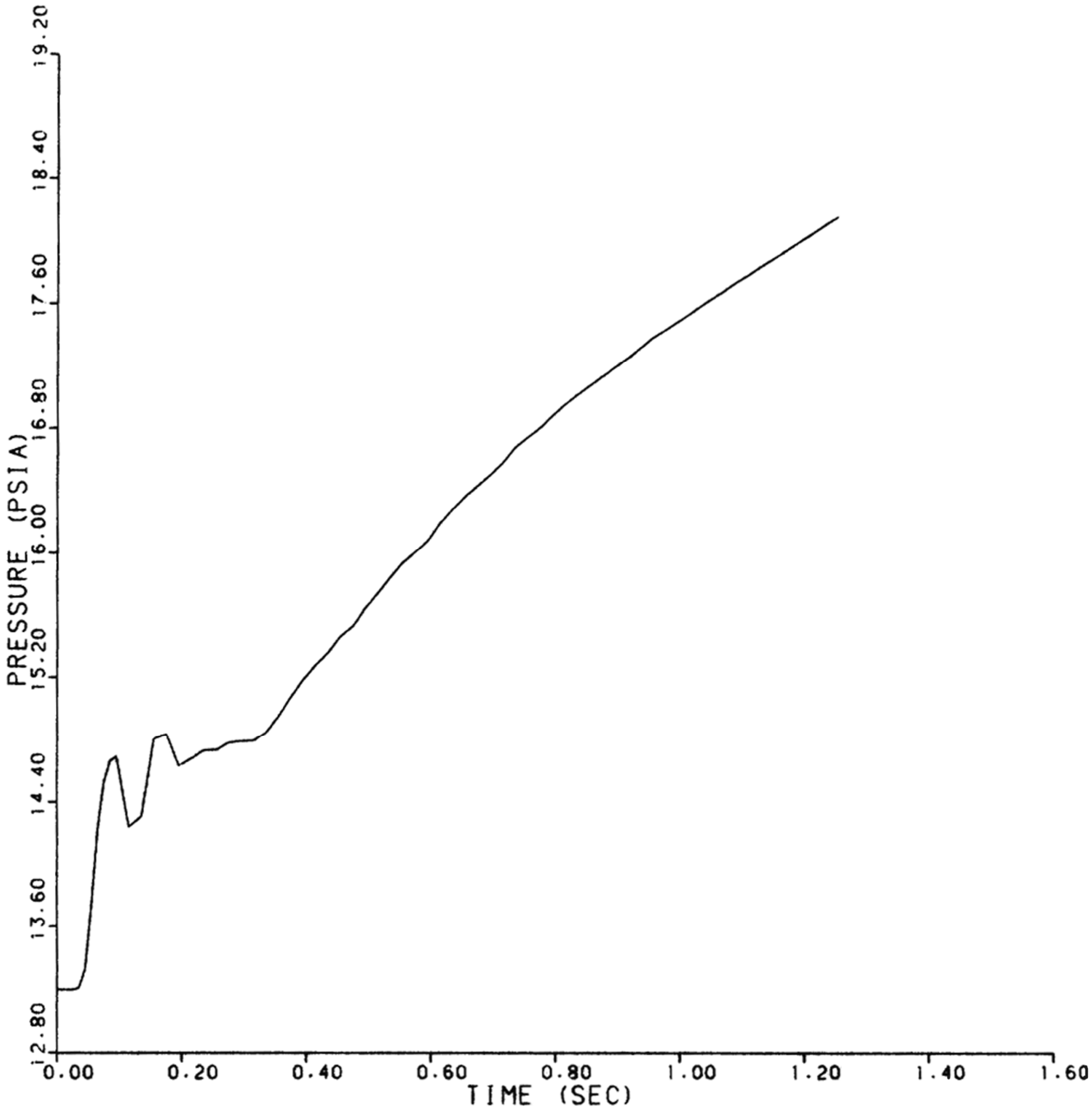
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E65
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 65 OF 74)



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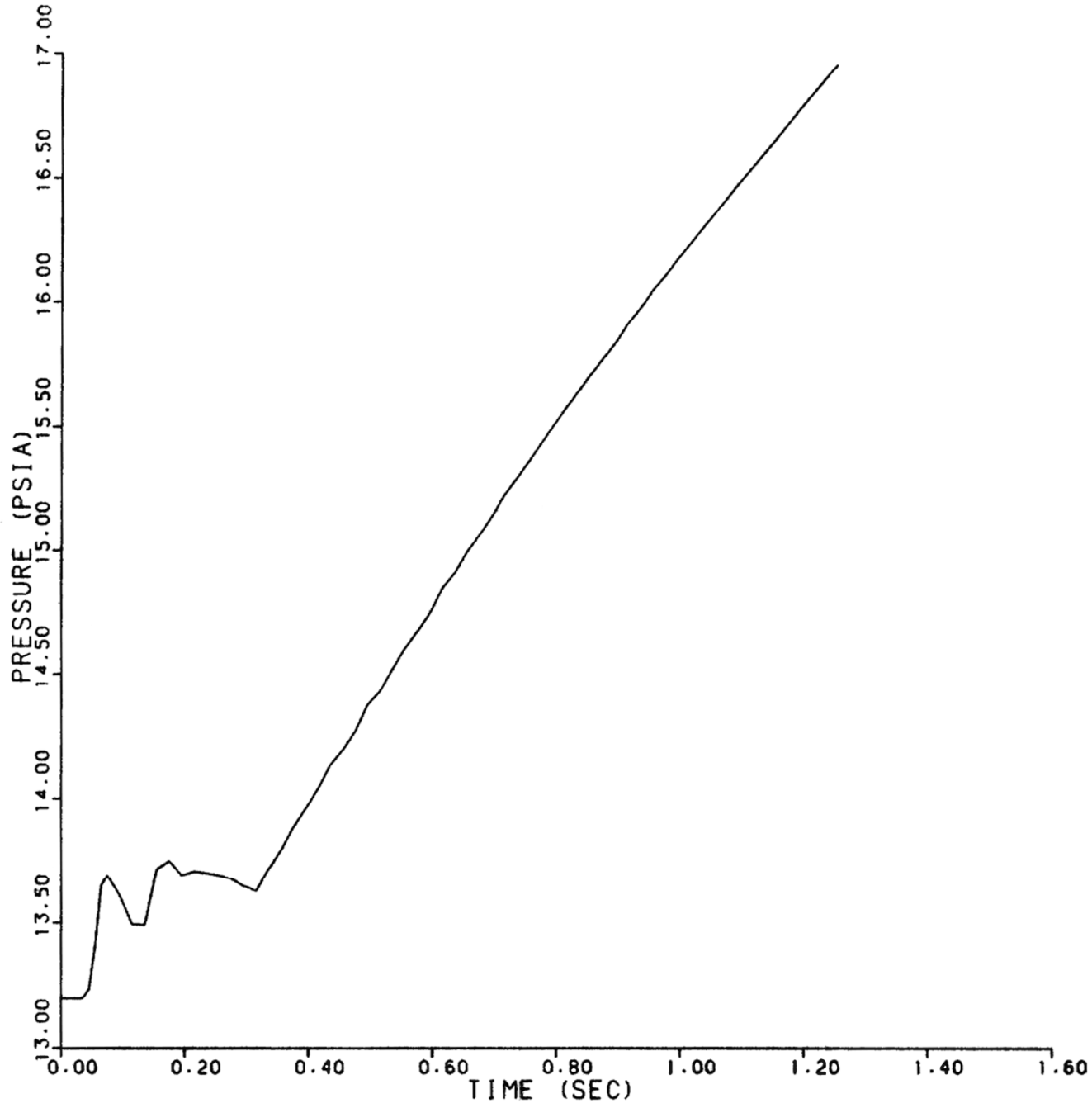
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

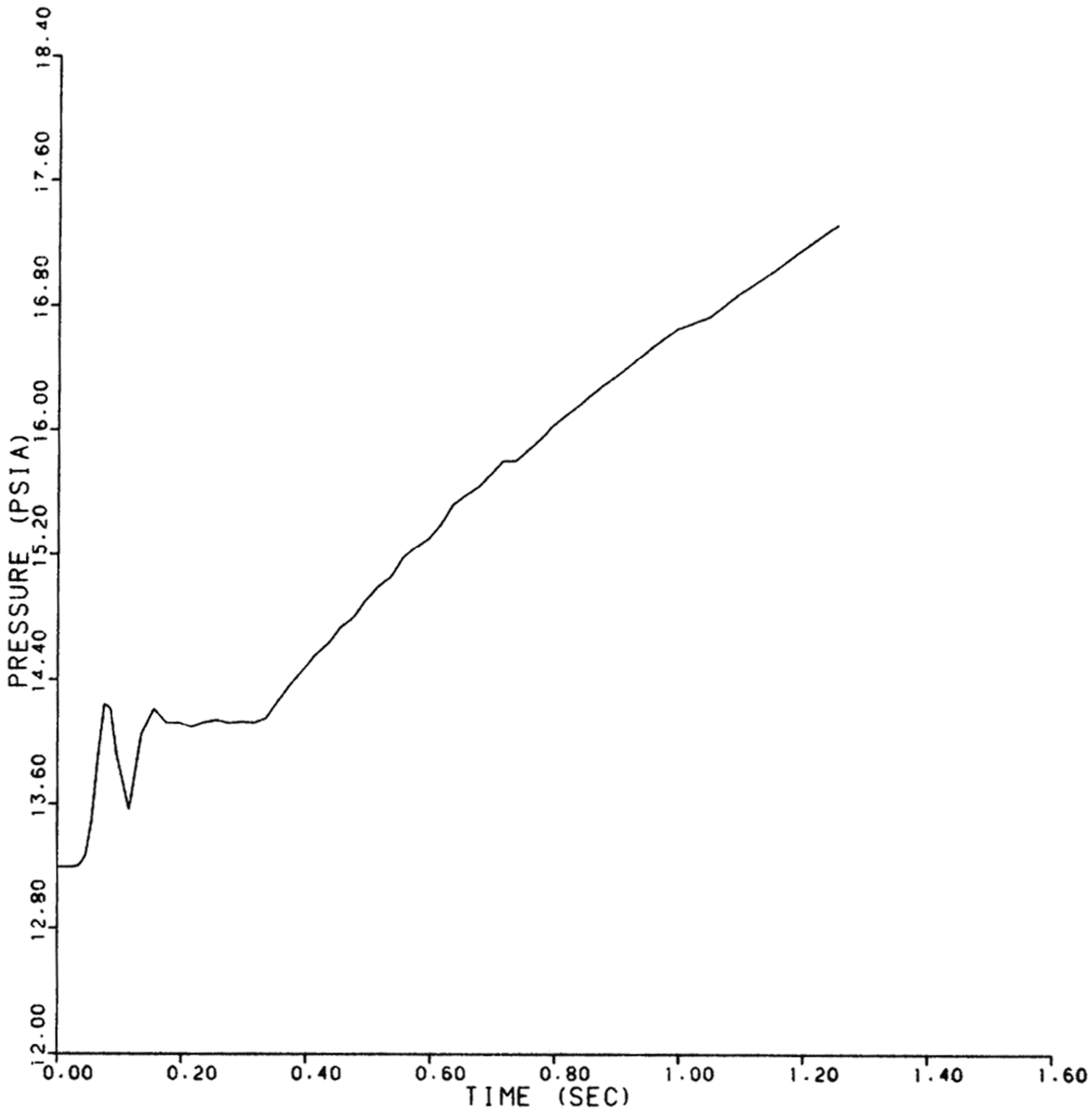
STEAM GENERATOR OUTLET NOZZLE
NODE – E66
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 66 OF 74)



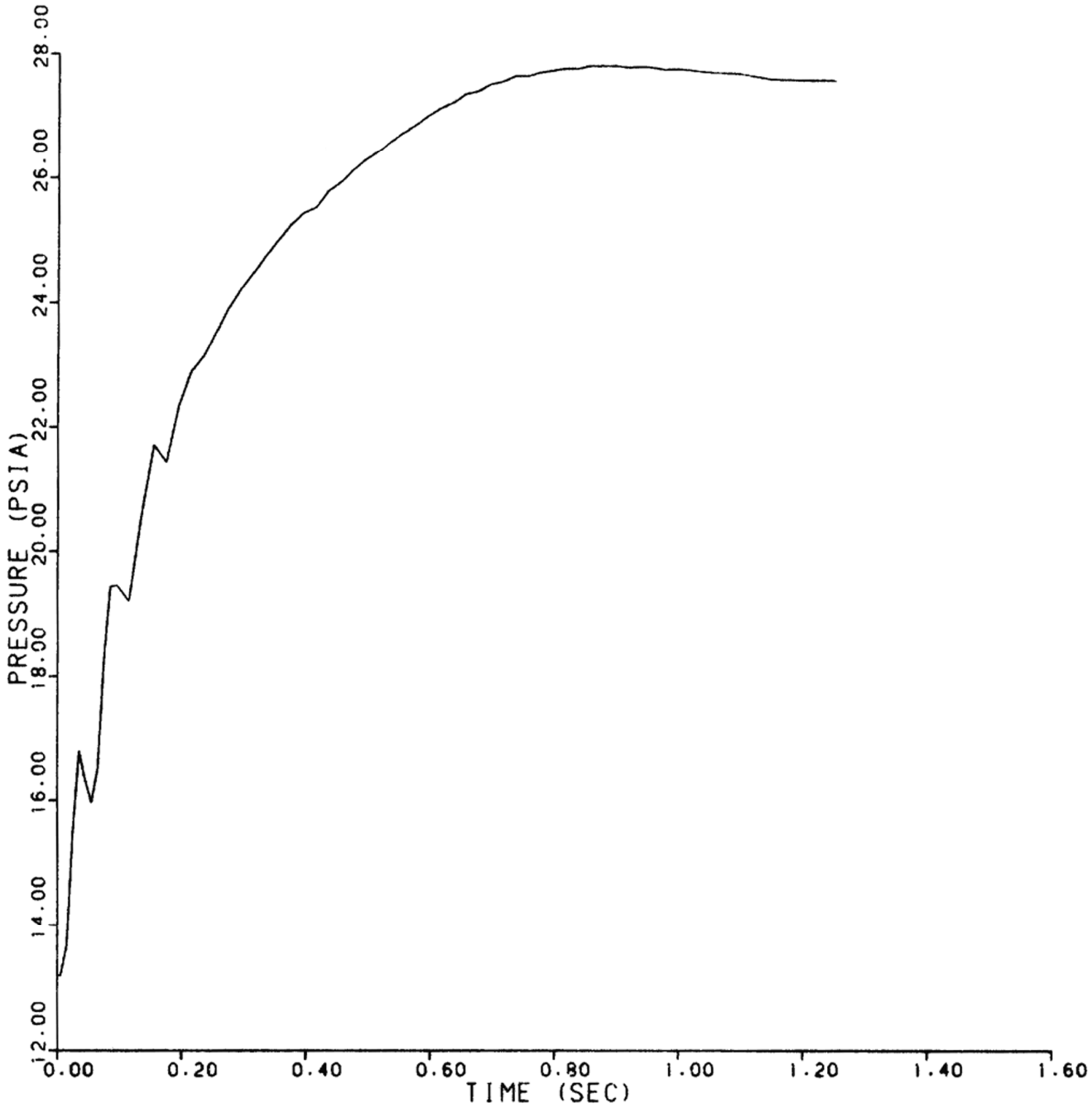
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E67 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 67 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E68 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 68 OF 74)



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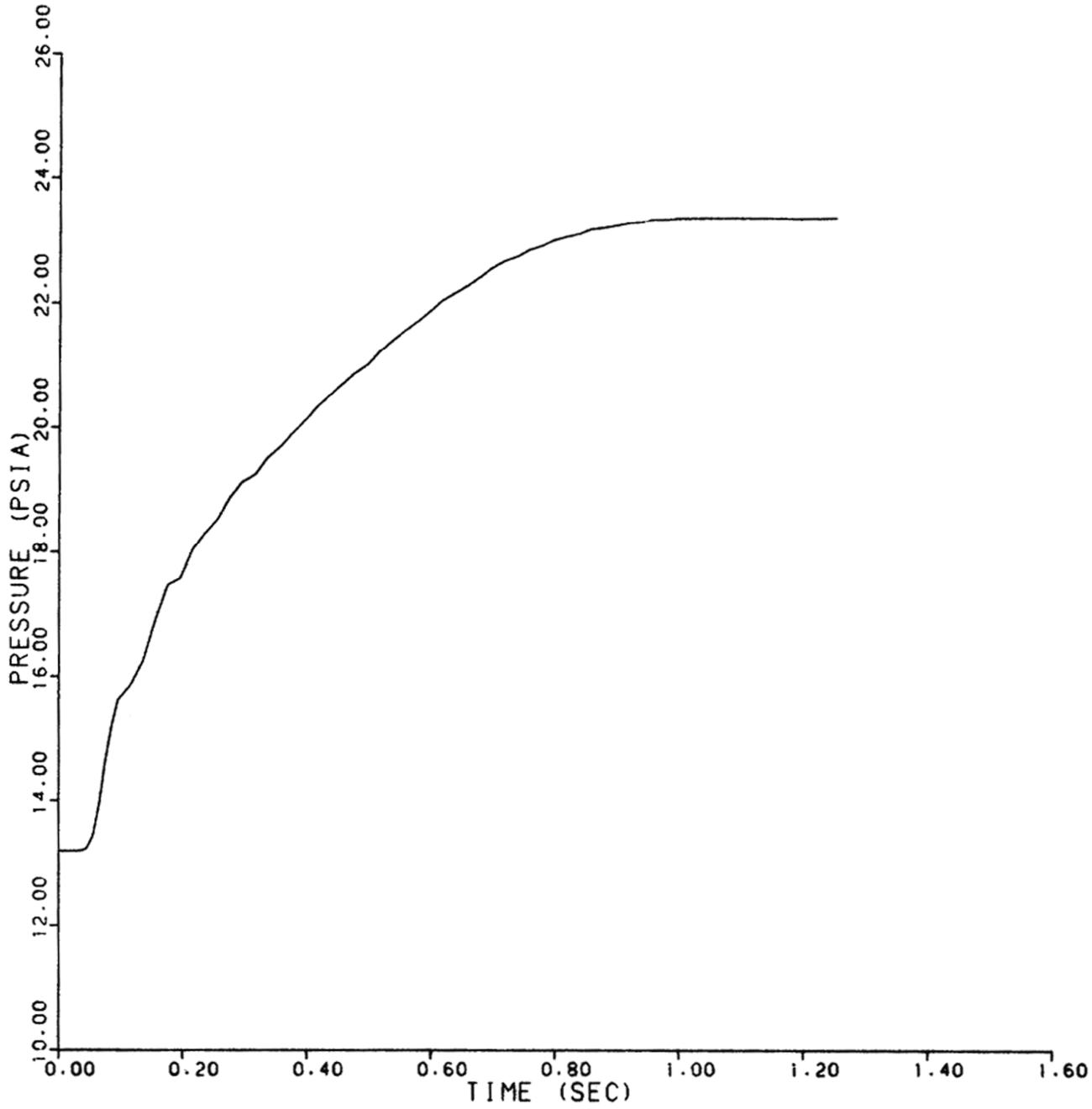
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E69 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 69 OF 74)



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SOUTHERN COMPANY
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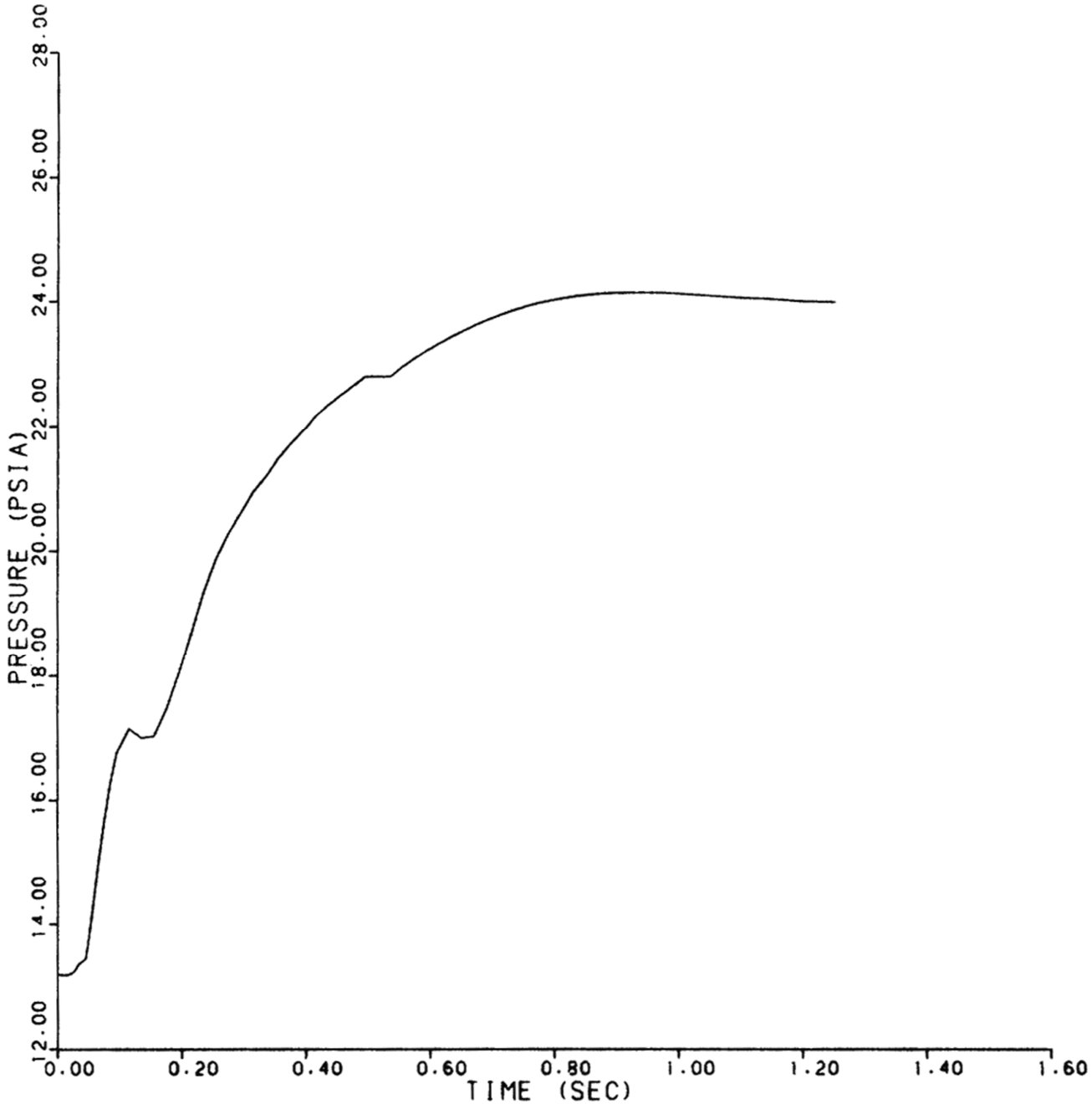
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E70
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 70 OF 74)



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SOUTHERN COMPANY
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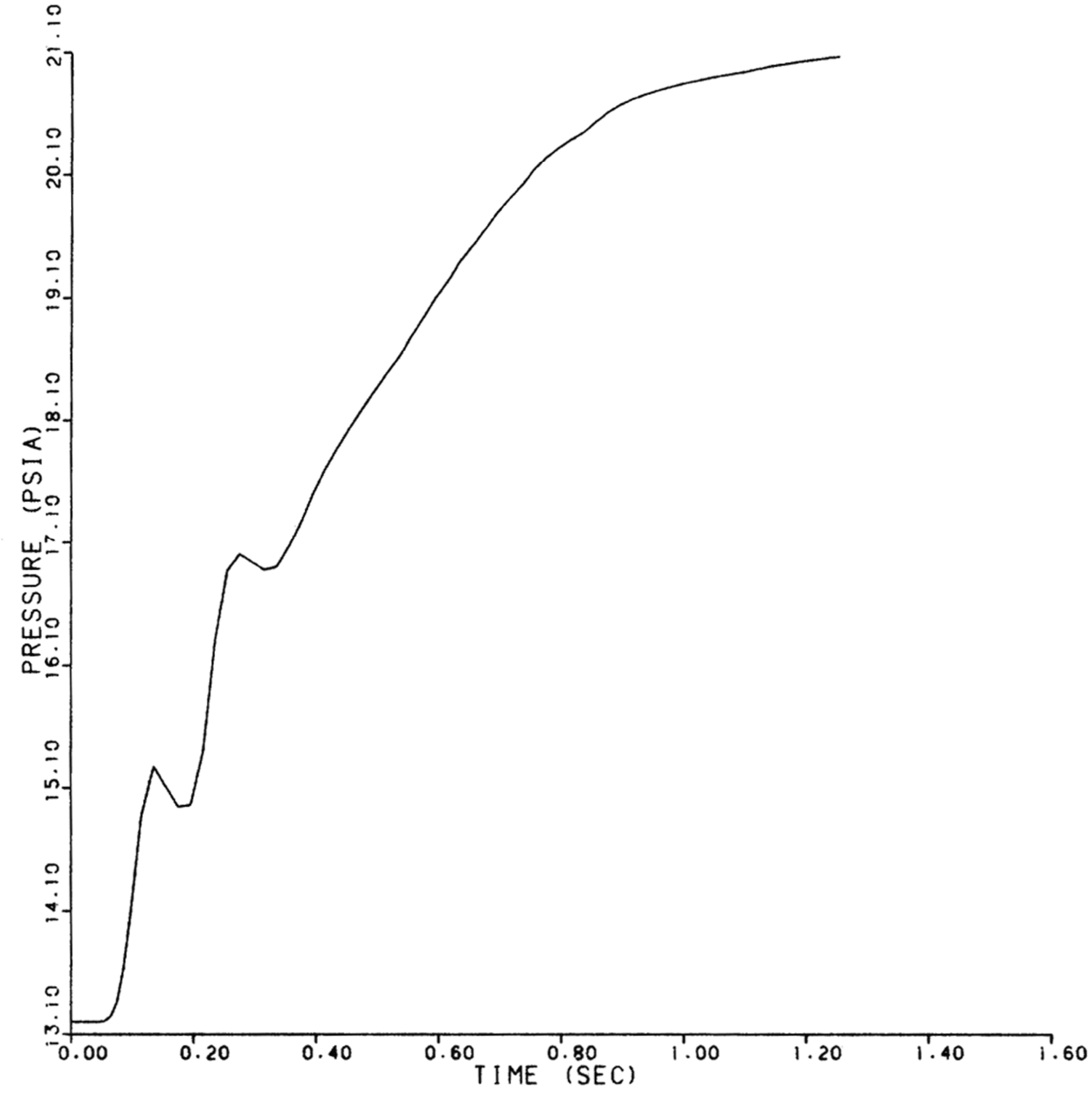
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E71
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 71 OF 74)



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SOUTHERN COMPANY
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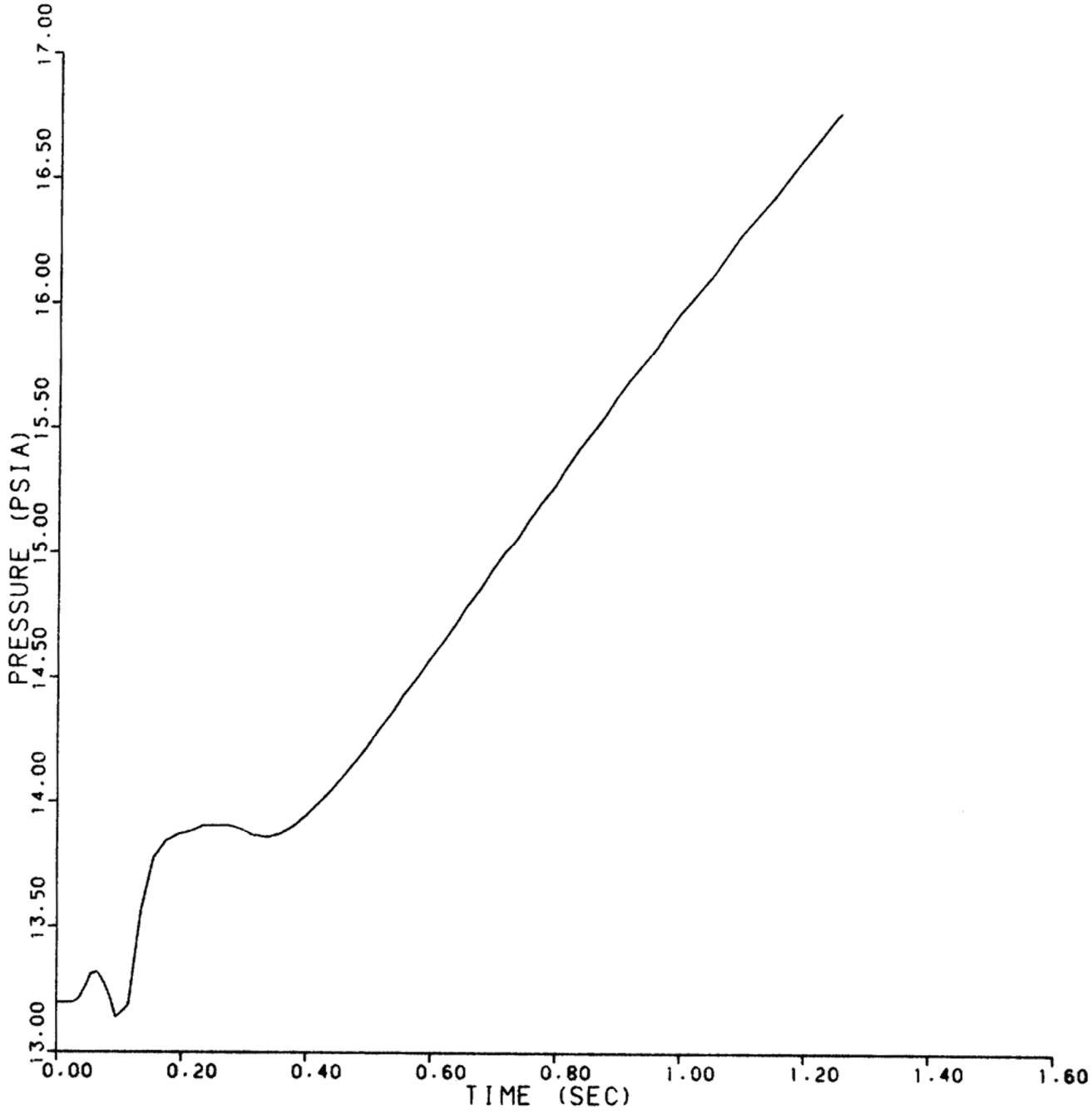
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR OUTLET NOZZLE
NODE – E72
(436–in.^{2} BREAK AREA)

FIGURE 6.2.1–21 (SHEET 72 OF 74)



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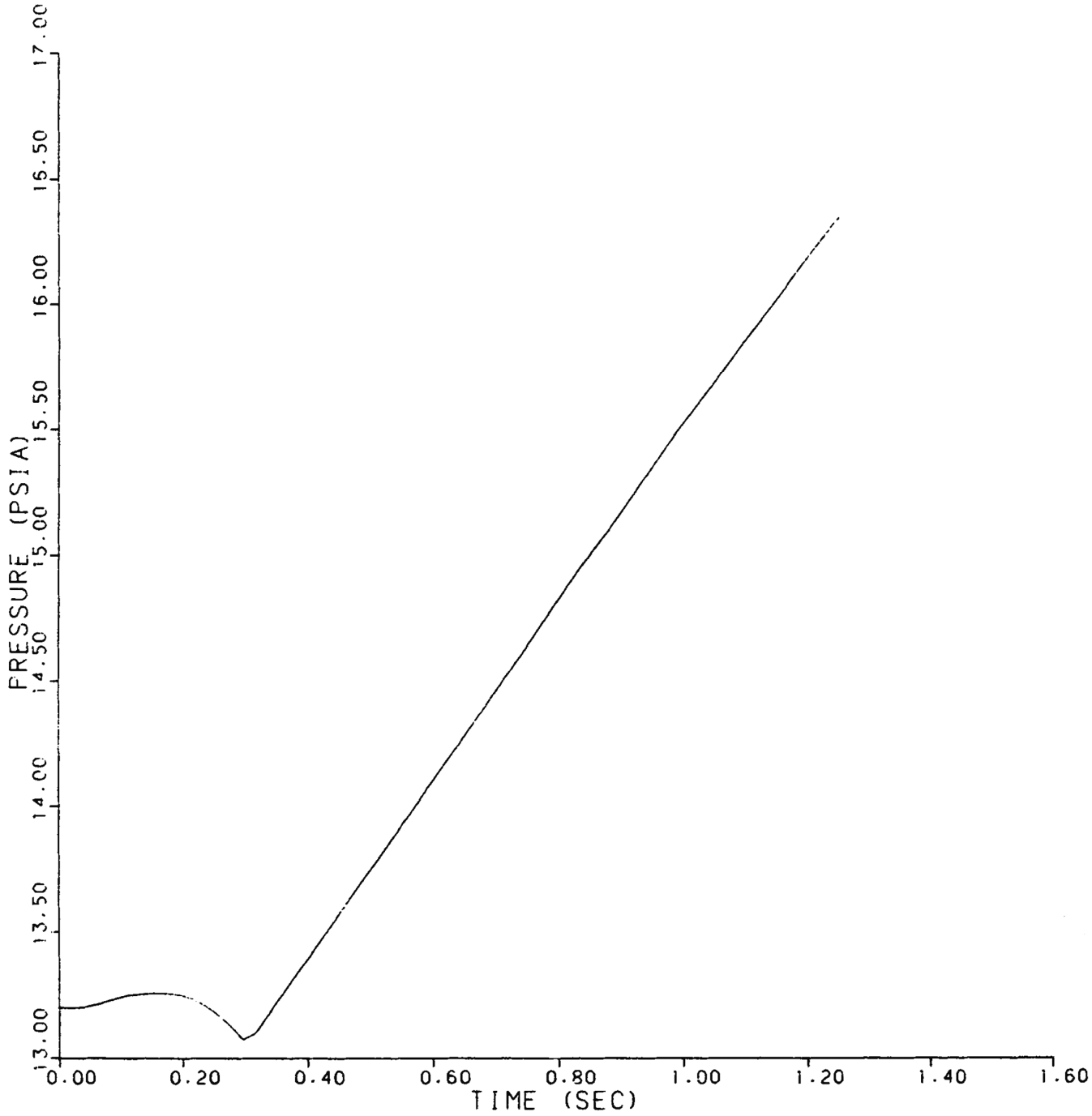
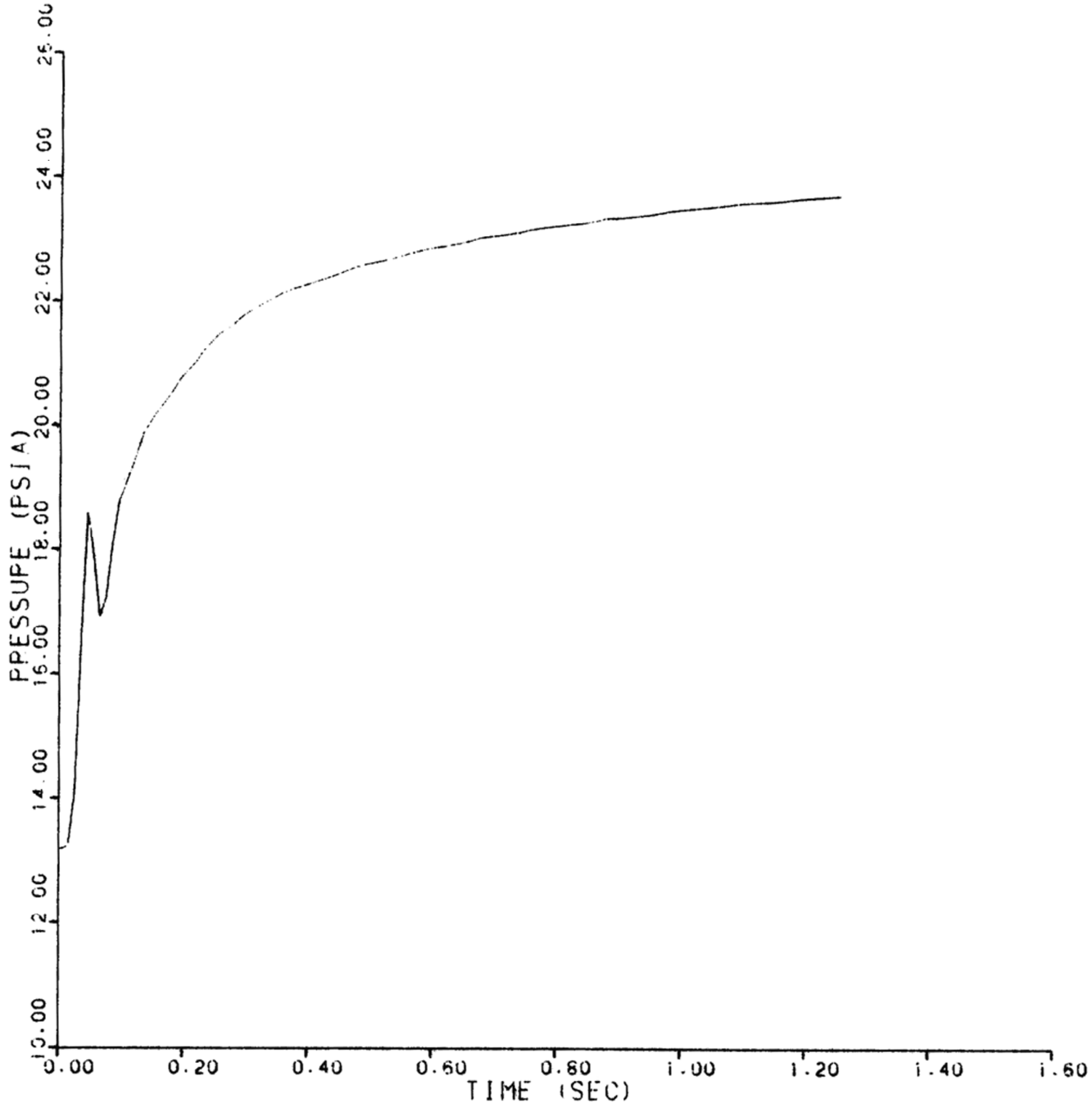
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E73 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 73 OF 74)



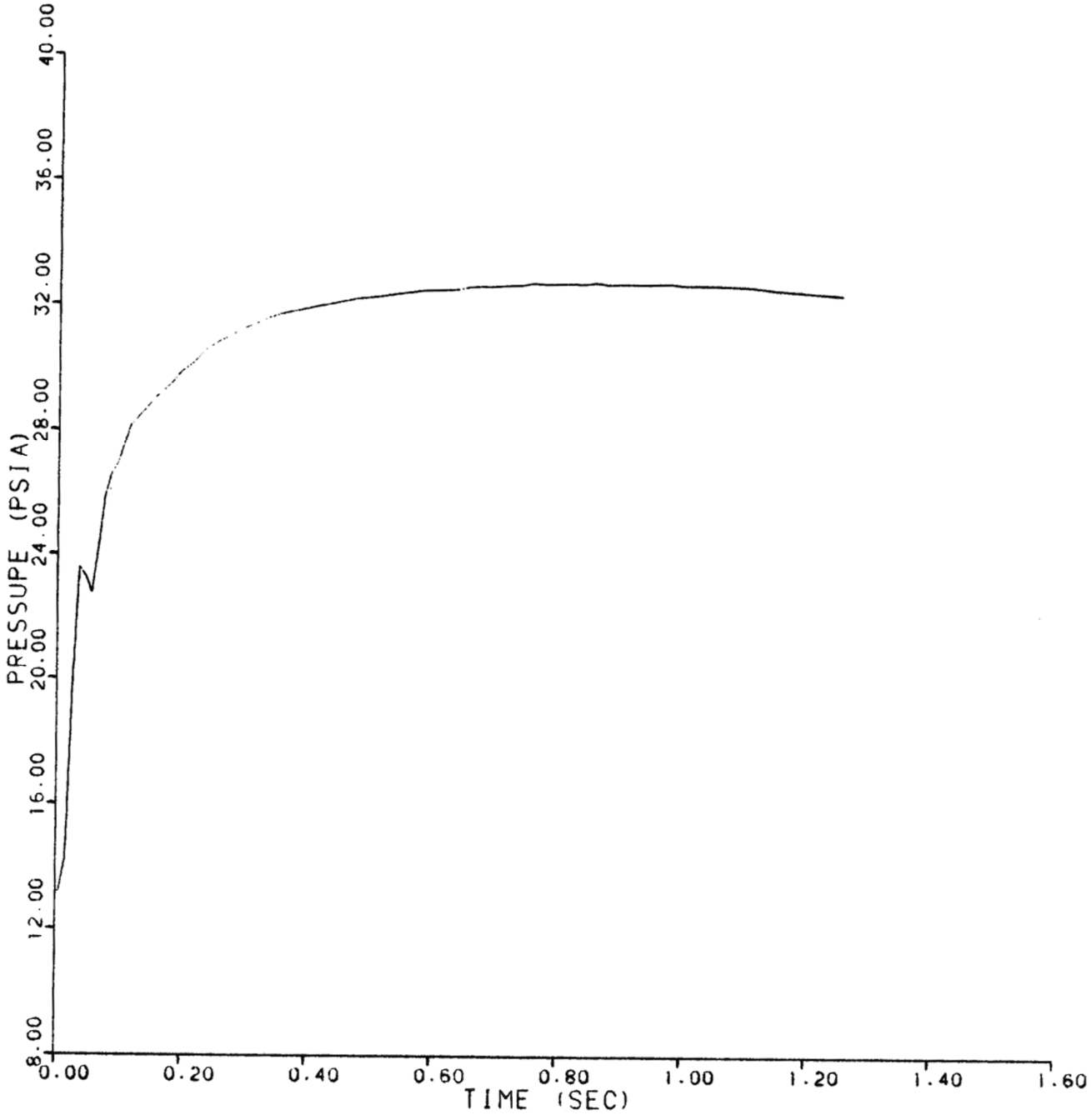
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR OUTLET NOZZLE NODE – E74 (436–in.^{2} BREAK AREA)
	FIGURE 6.2.1–21 (SHEET 74 OF 74)



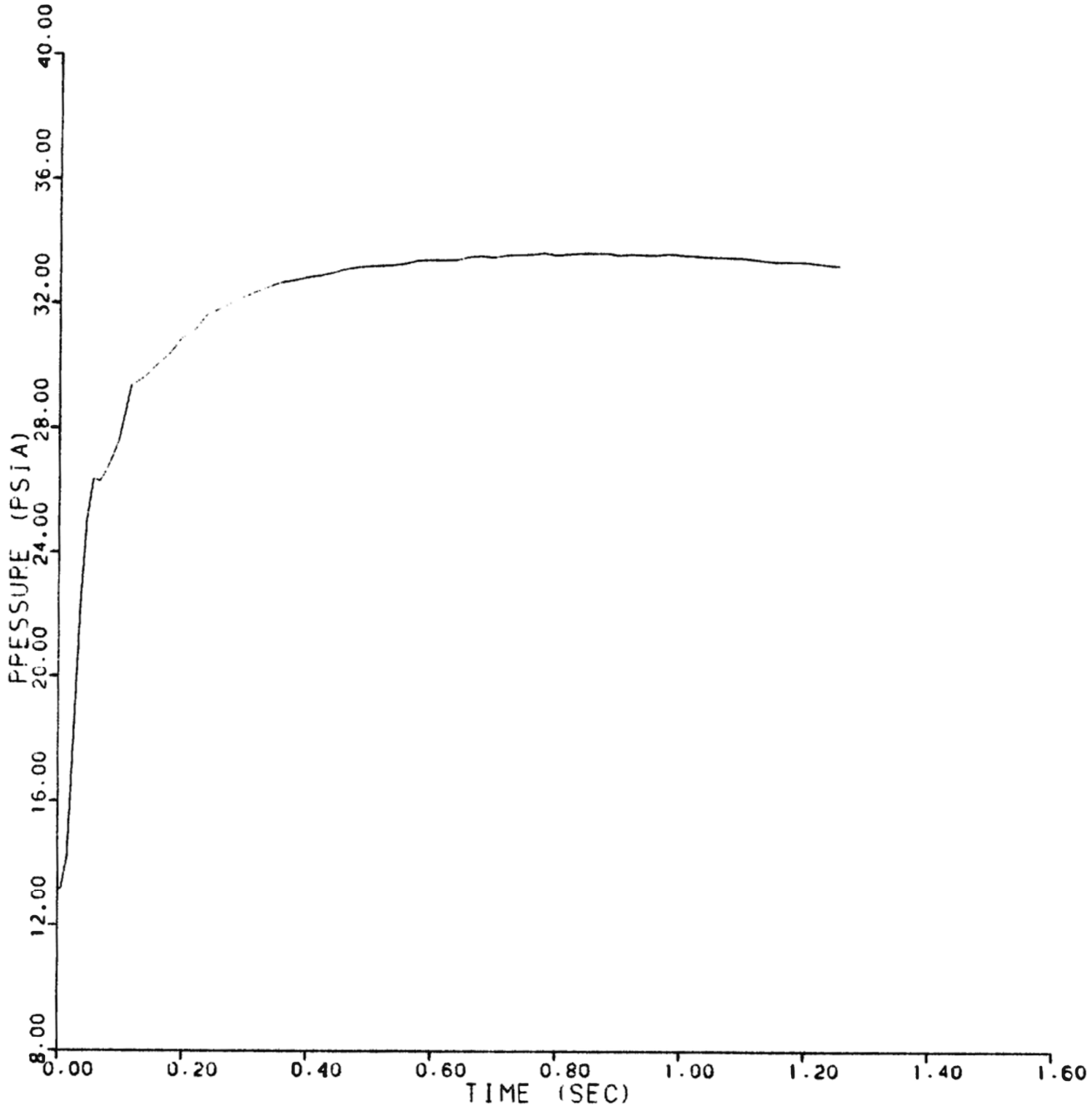
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E1 (763–in.^{2} BRE AK AREA)
	FIGURE 6.2.1–22 (SHEET 1 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E2 (763–in.^{2} BREAK AREA) FIGURE 6.2.1–22 (SHEET 2 OF 74)



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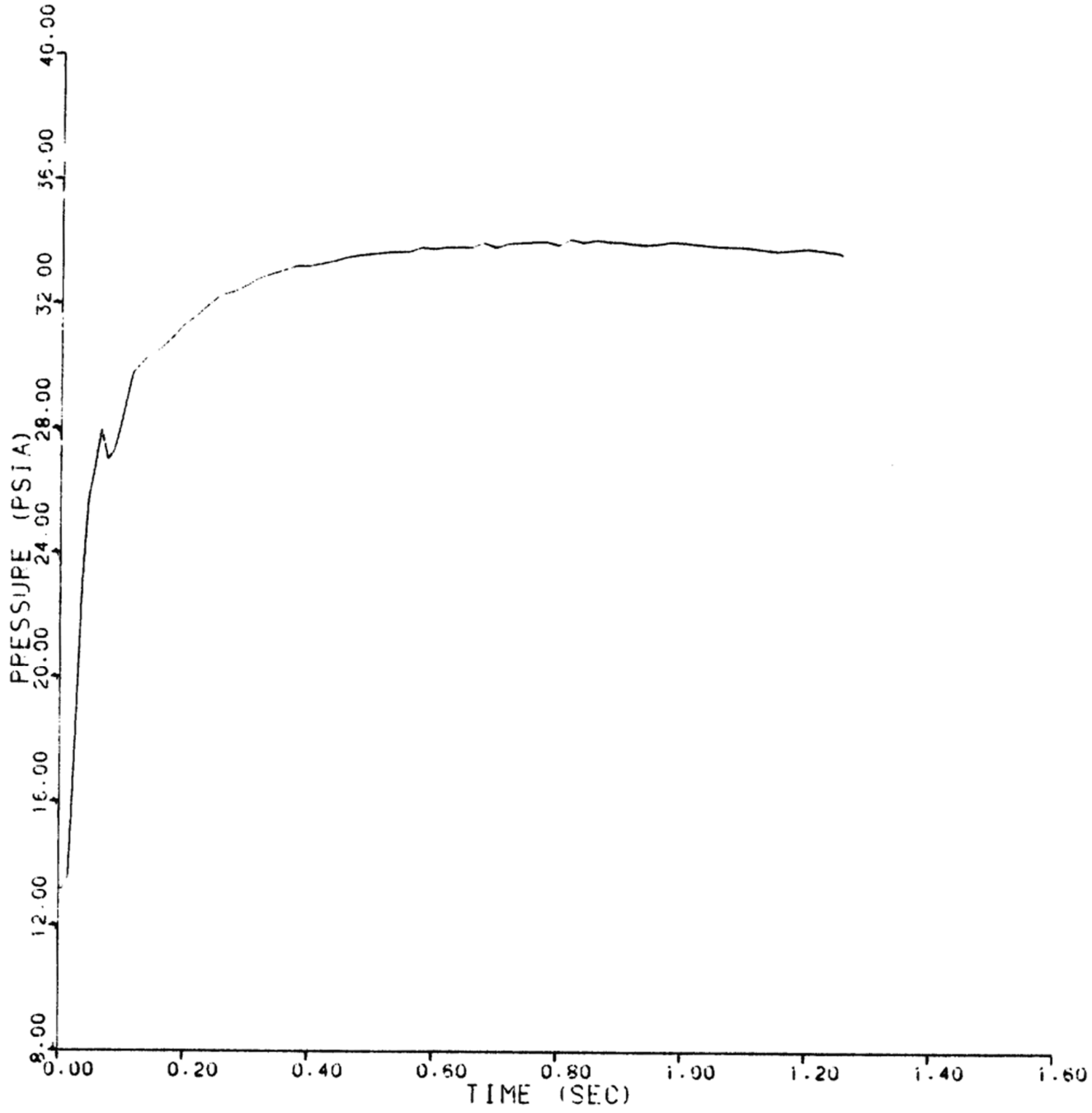
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

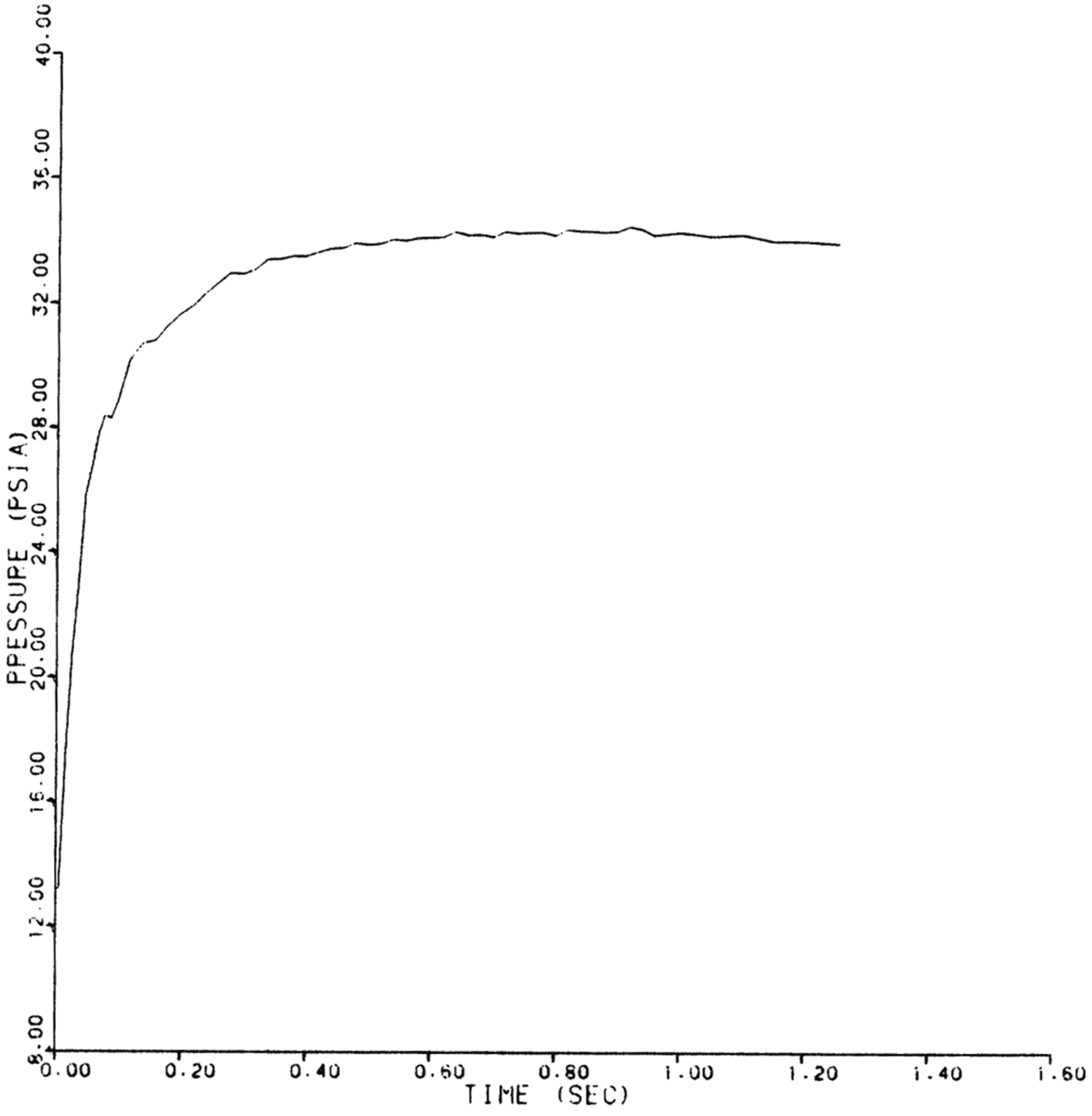
STEAM GENERATOR INLET ELBOW
NODE – E3
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 3 OF 74)



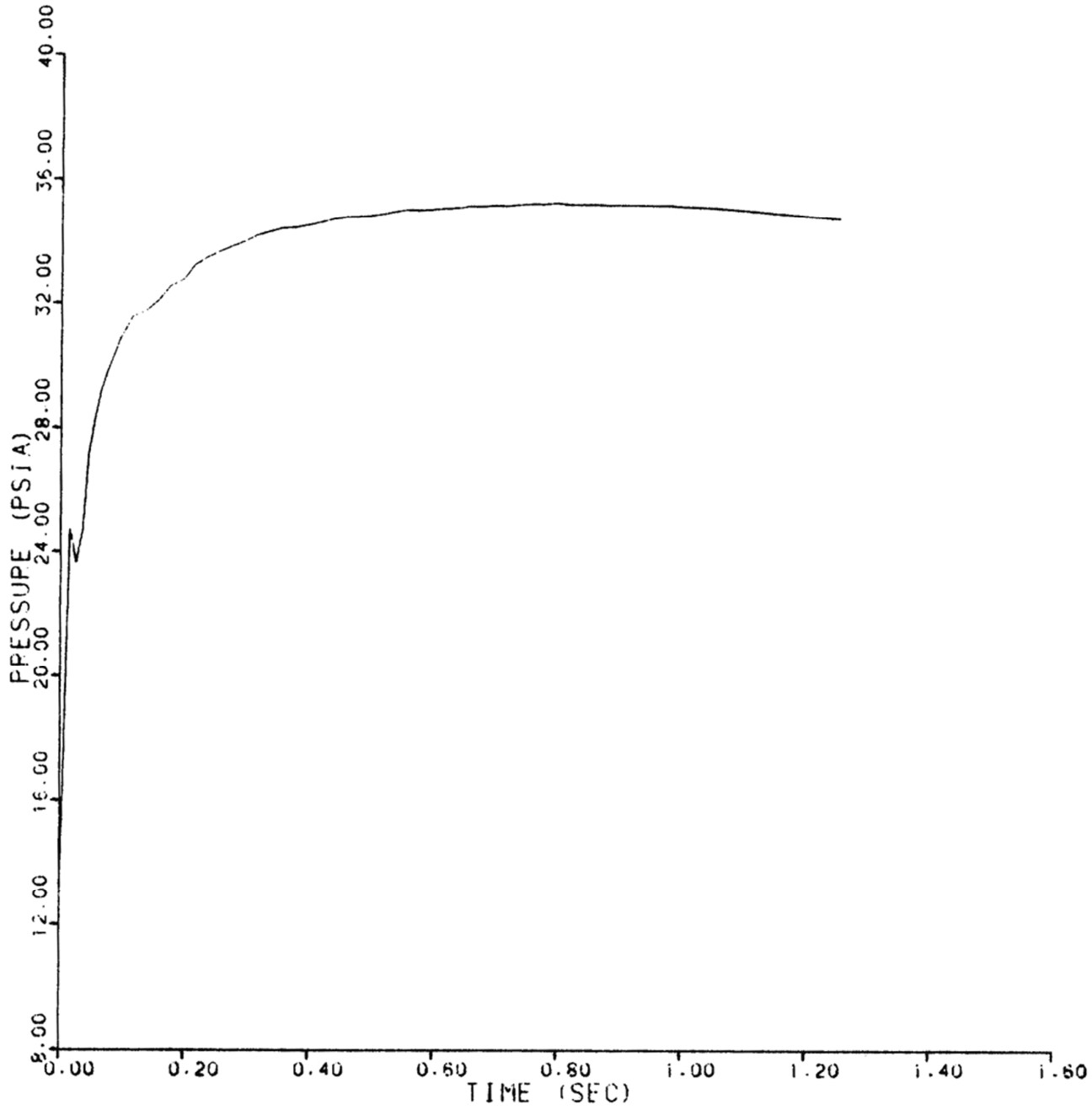
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E4 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 4 OF 74)



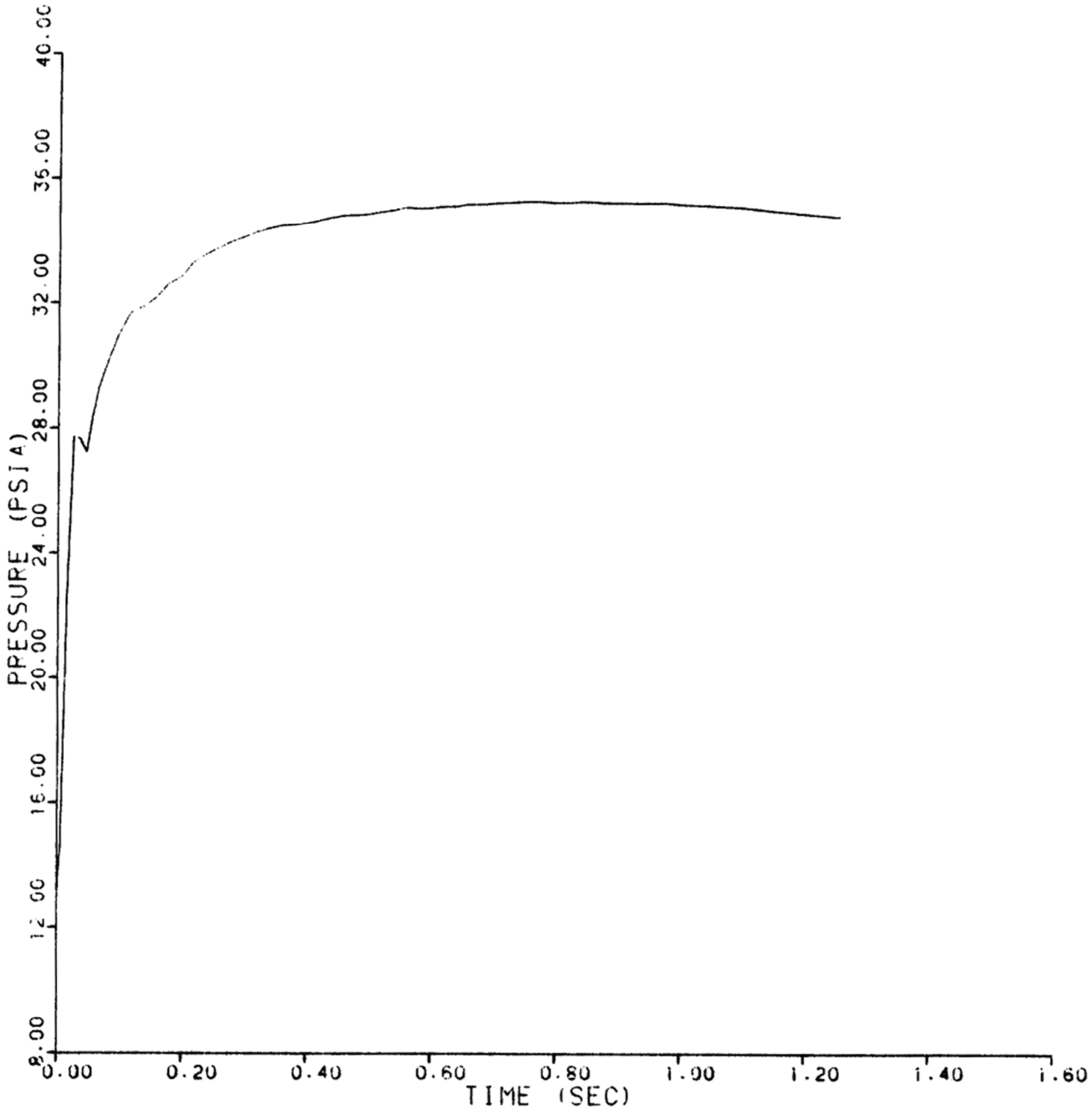
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E5 (763–in.^{2} BREAK AREA) FIGURE 6.2.1–22 (SHEET 5 OF 74)



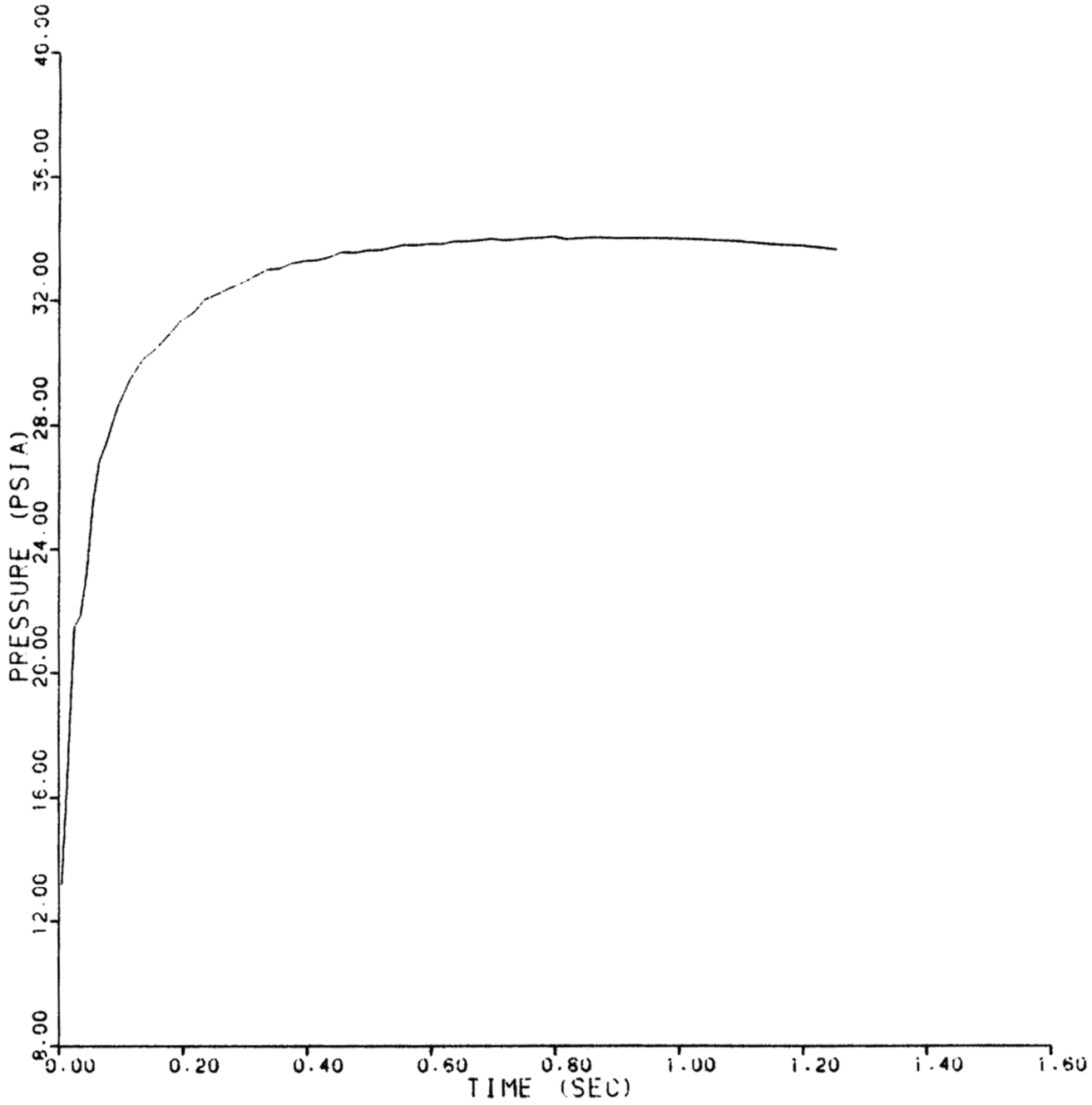
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E6 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 6 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E7 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 7 OF 74)



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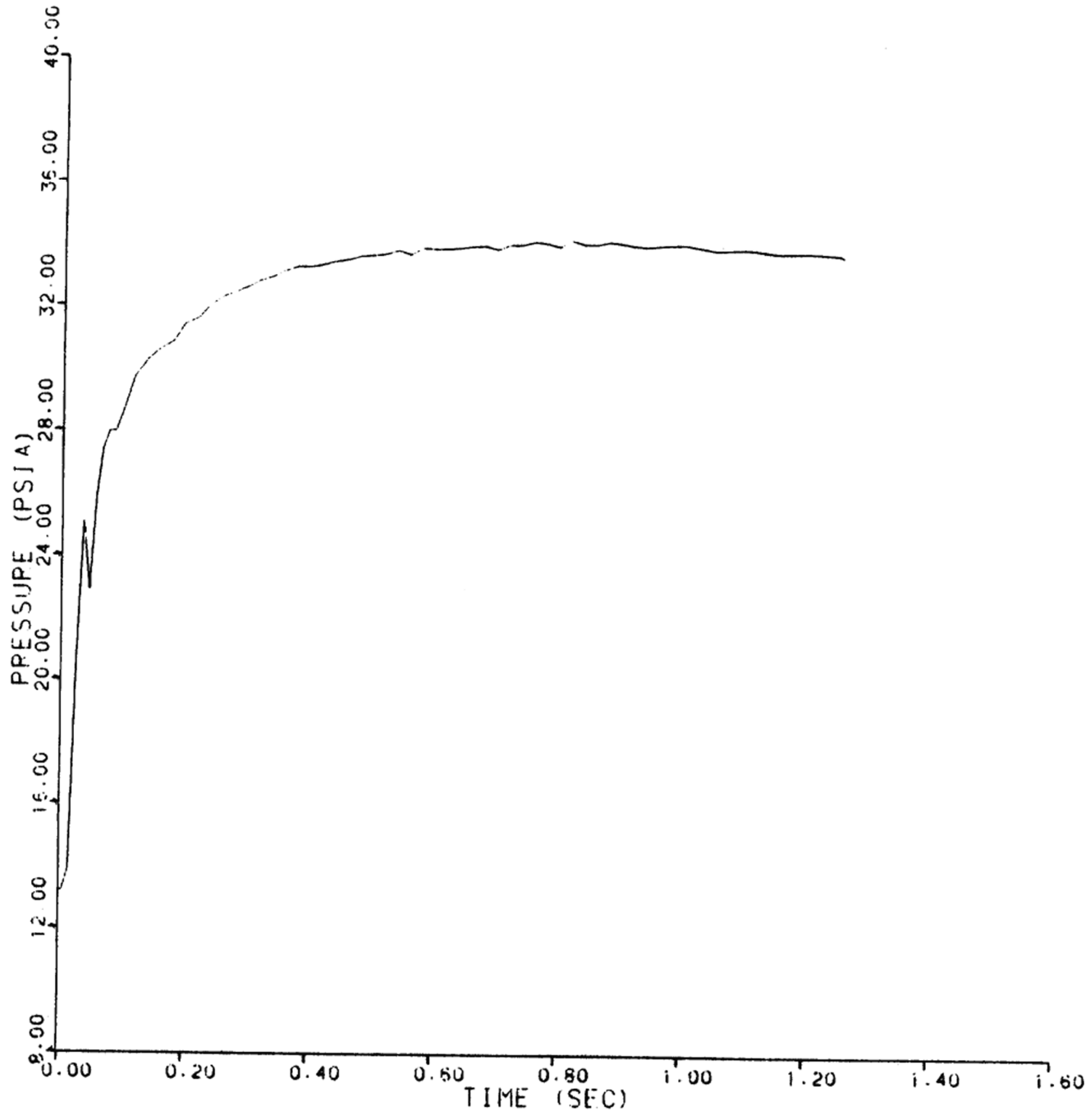
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

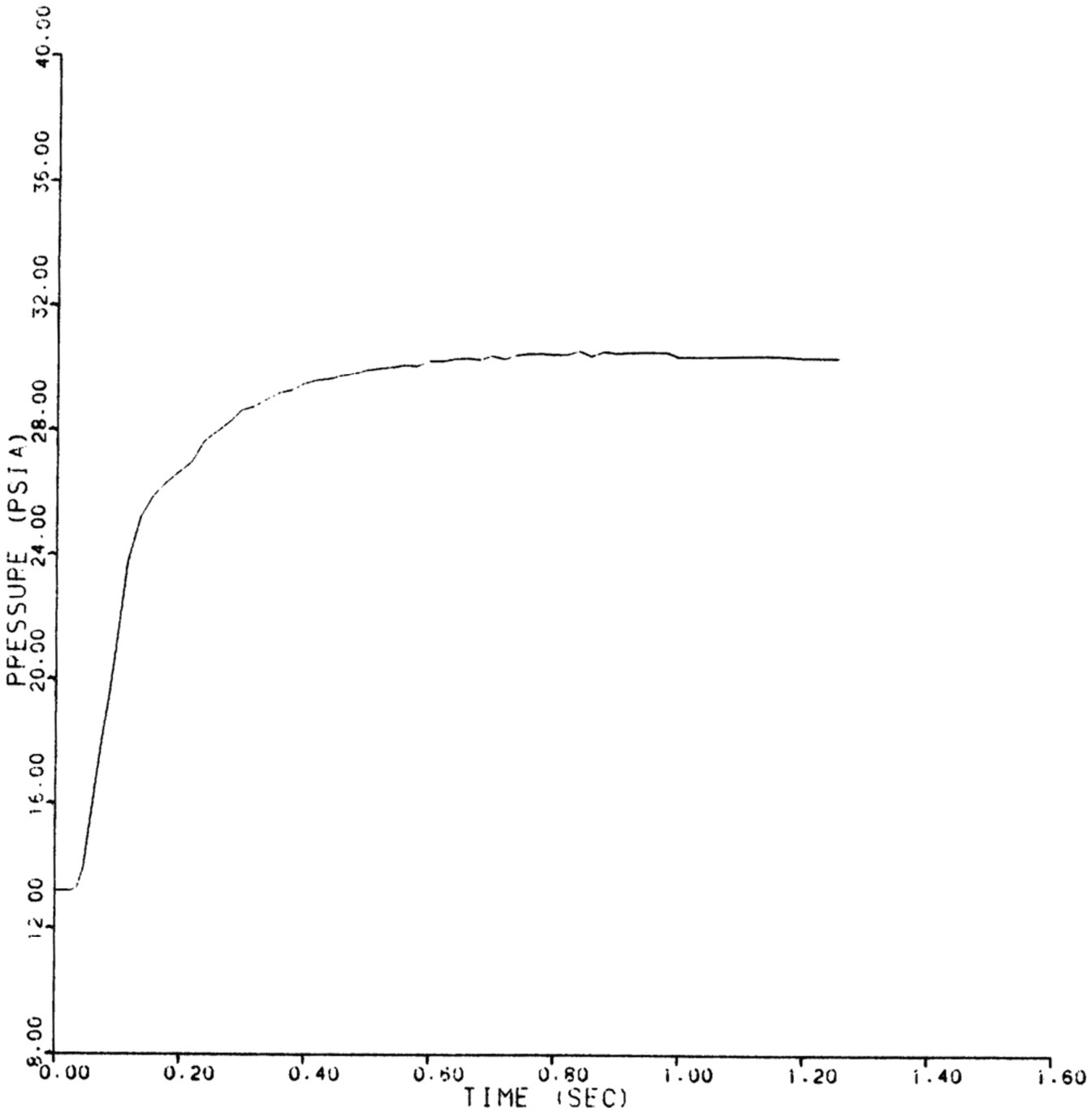
STEAM GENERATOR INLET ELBOW
NODE – E8
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 8 OF 74)



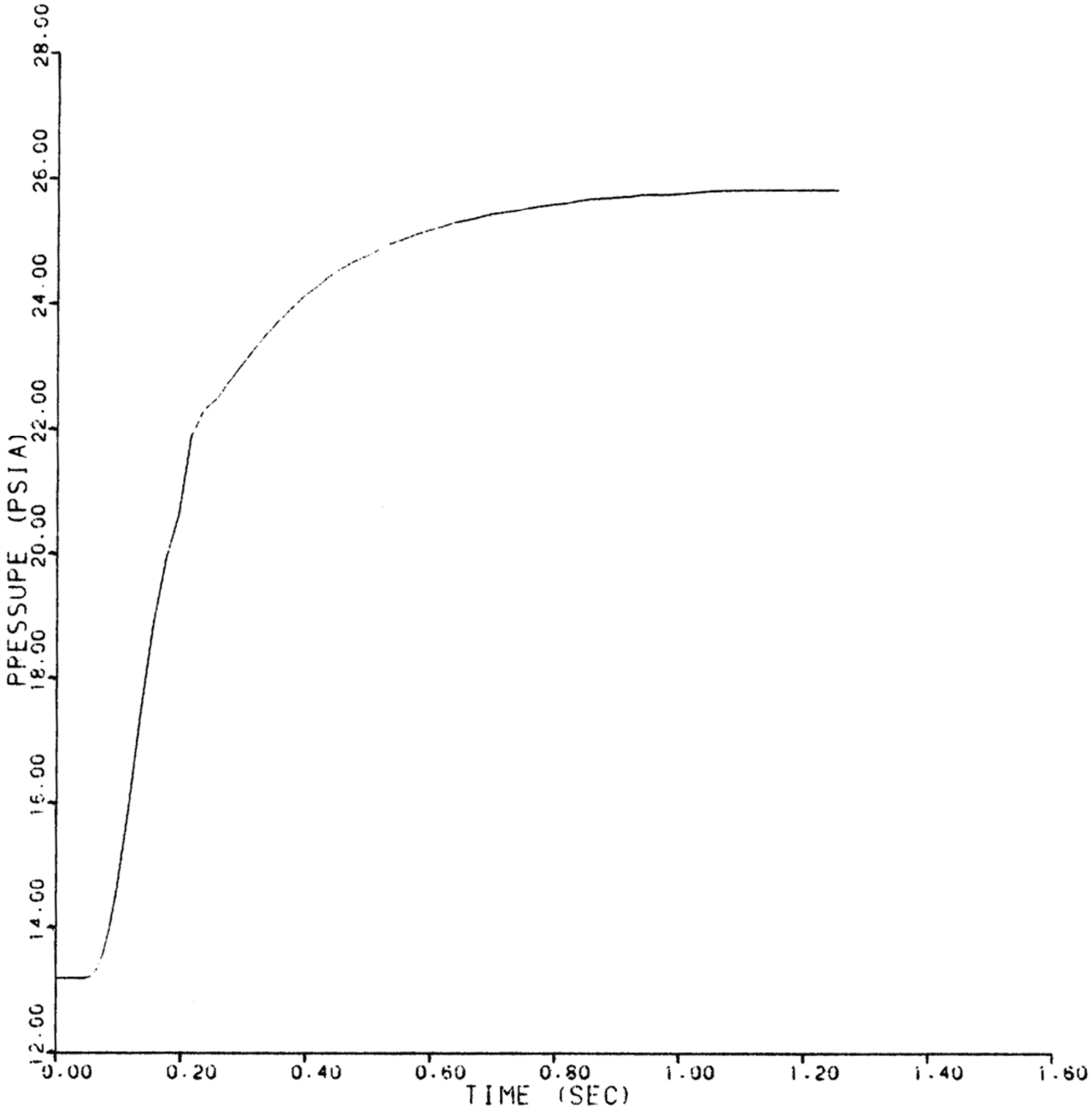
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E9 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 9 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E10 (763–in.^{2} BREAK AREA) FIGURE 6.2.1–22 (SHEET 10 OF 74)



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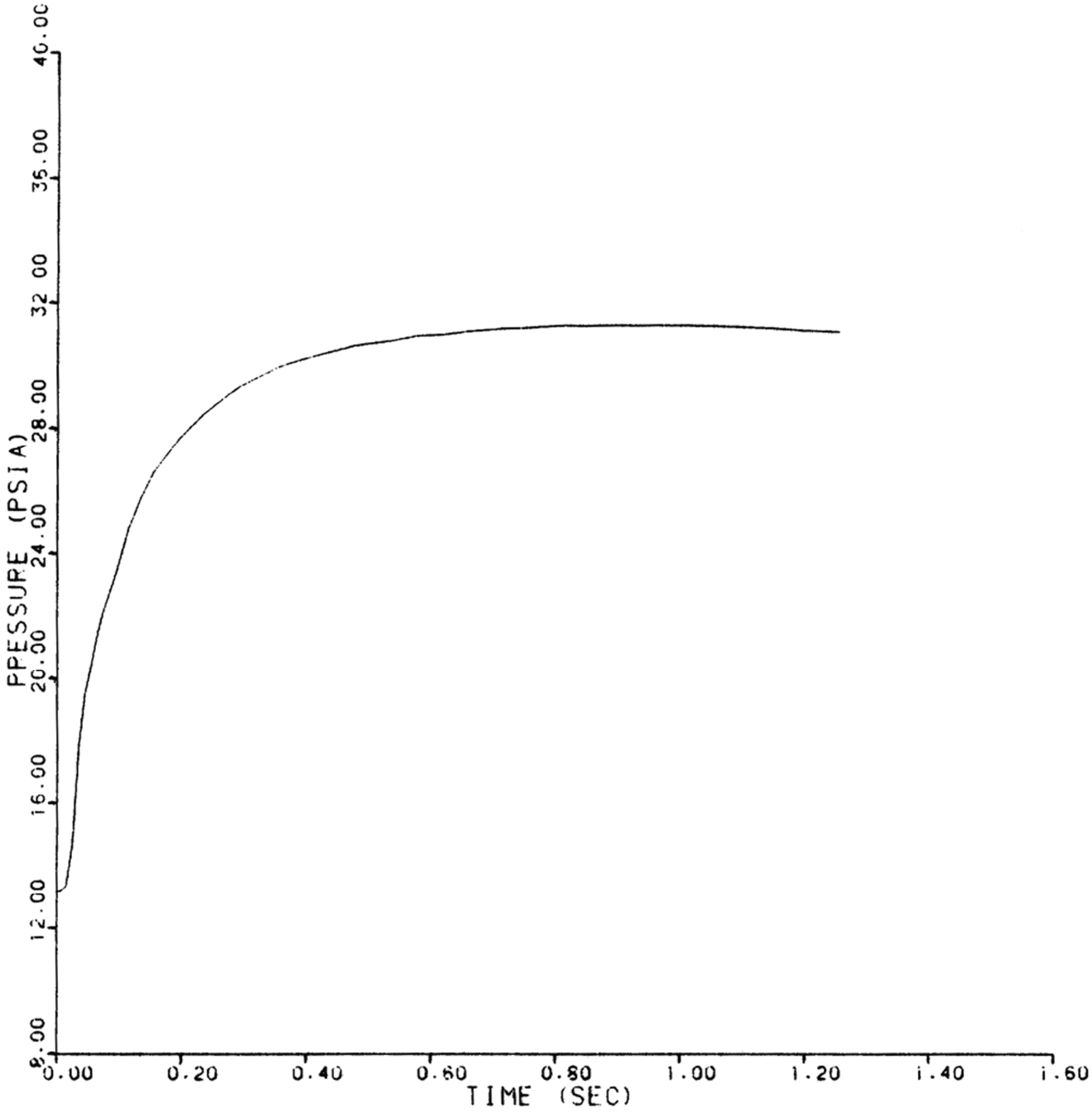
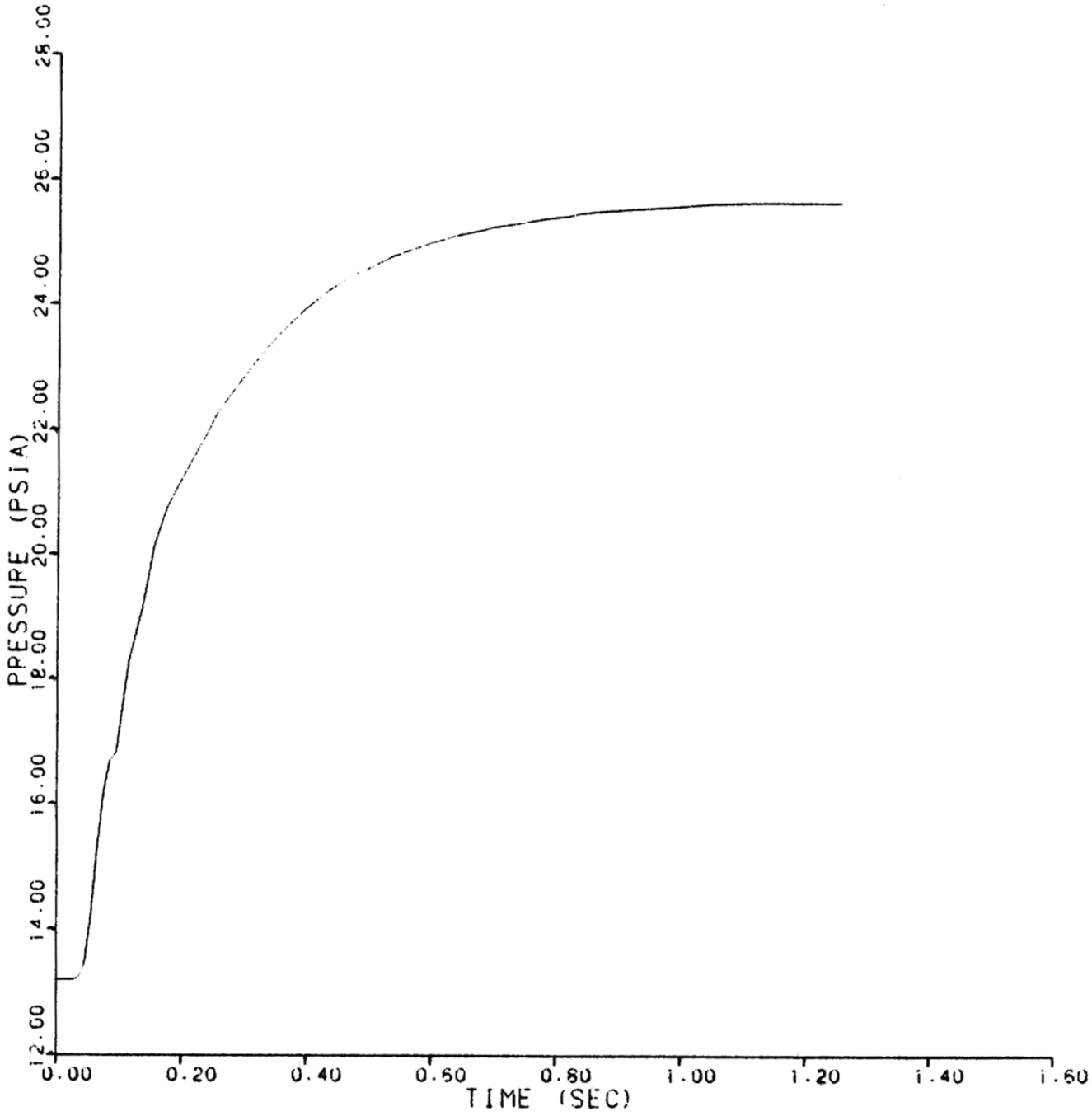
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E11 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 11 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E12 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 12 OF 74)



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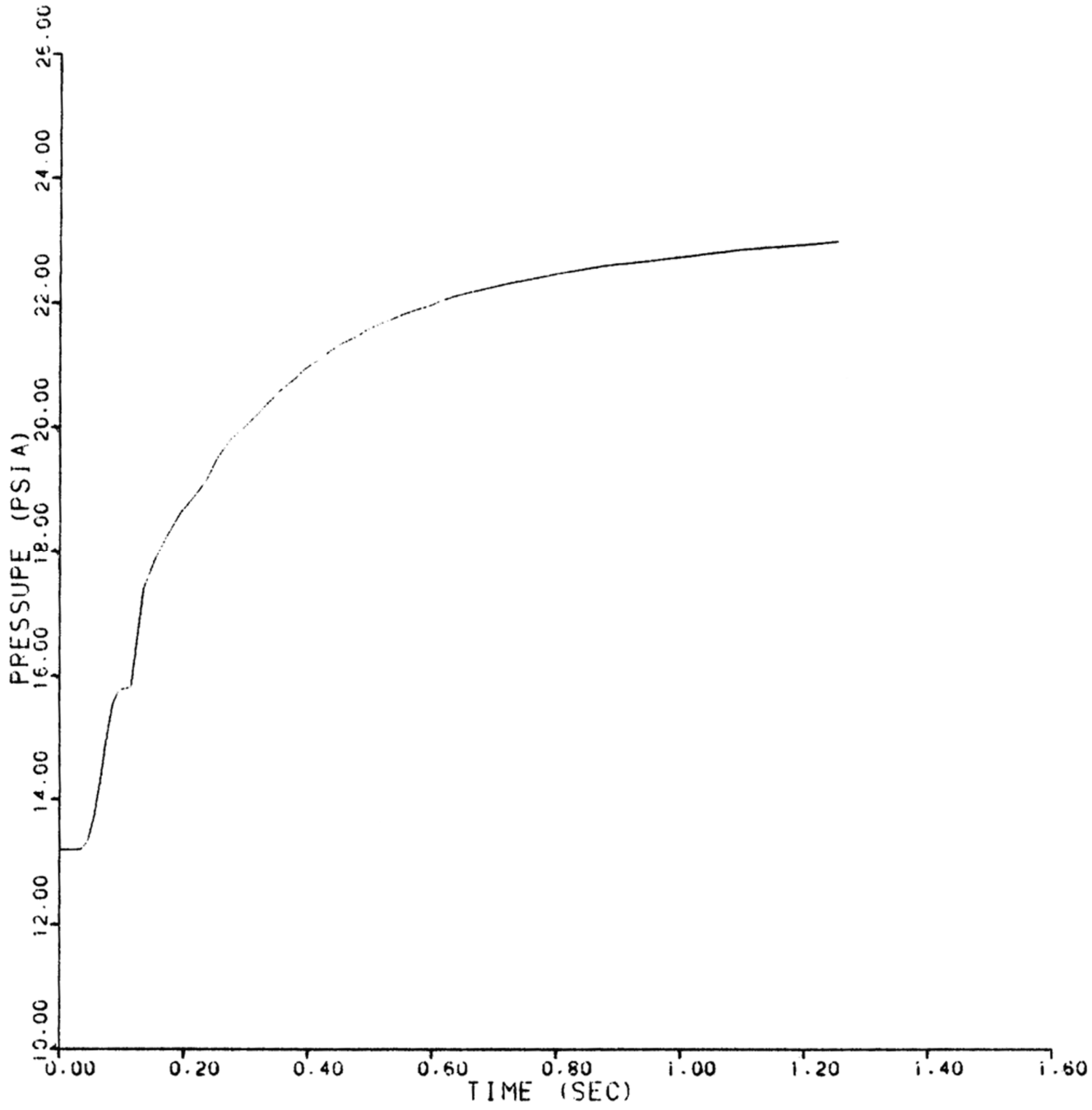
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

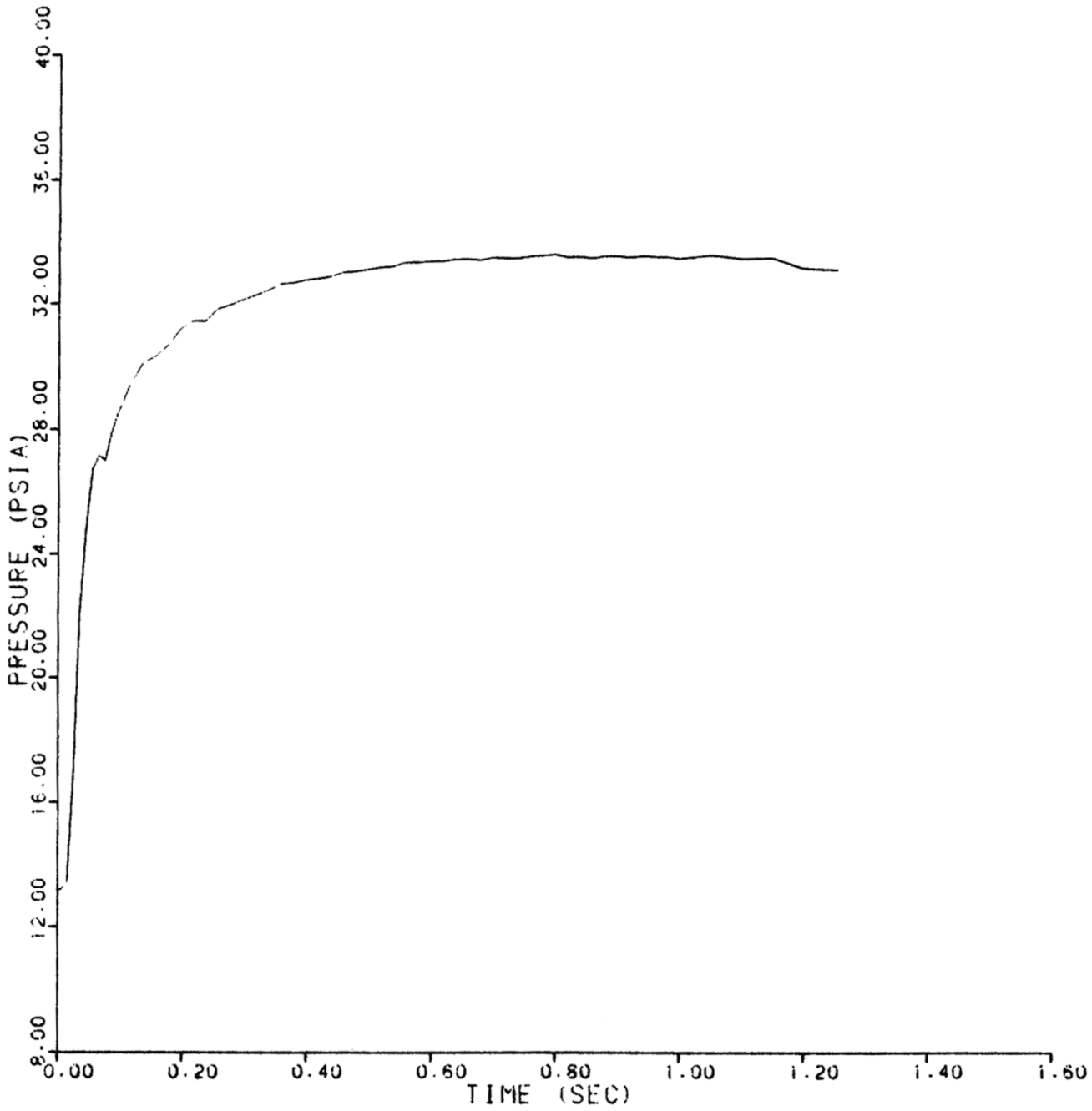
STEAM GENERATOR INLET ELBOW
NODE – E13
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 13 OF 74)



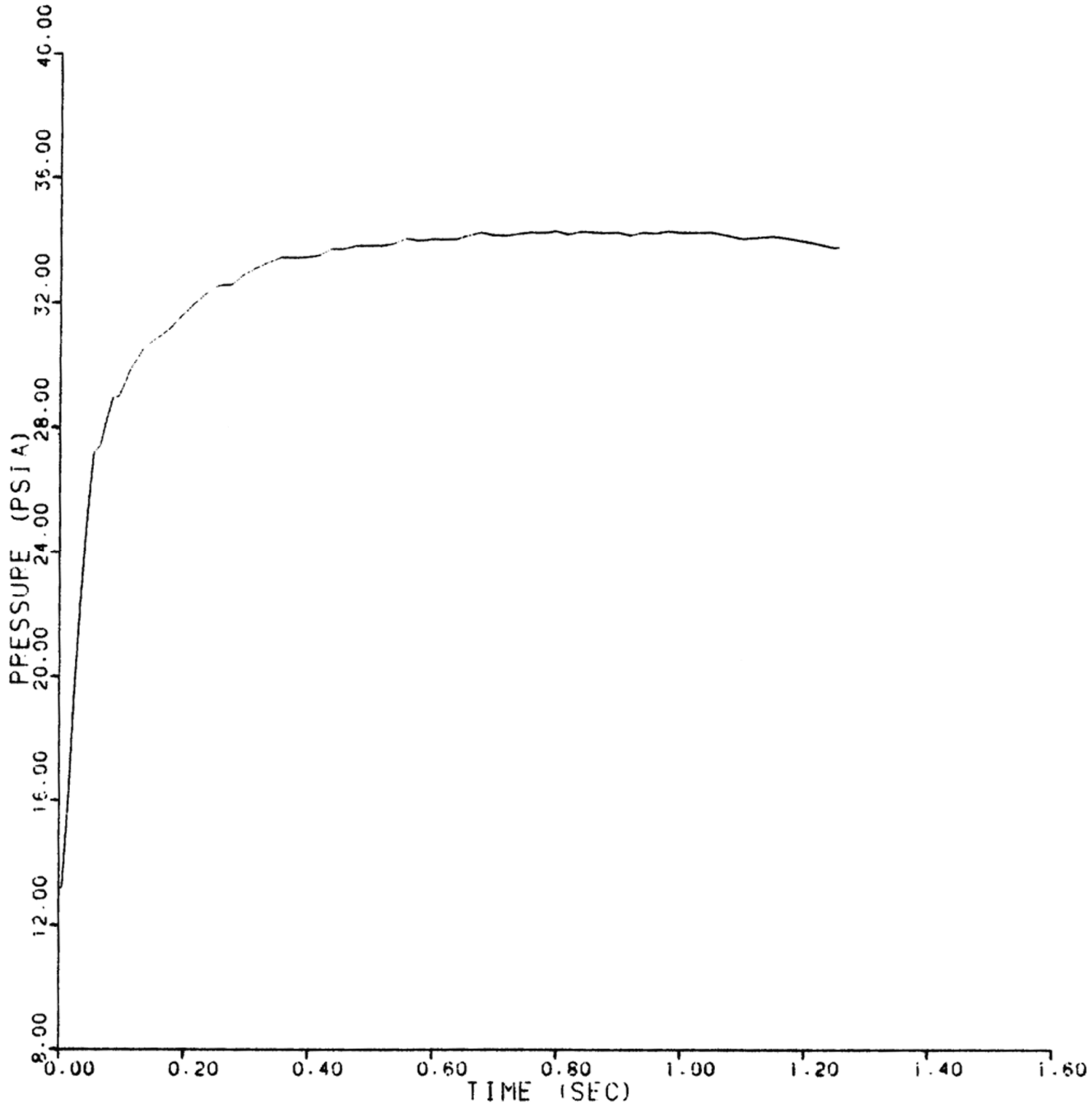
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E14 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 14 OF 74)



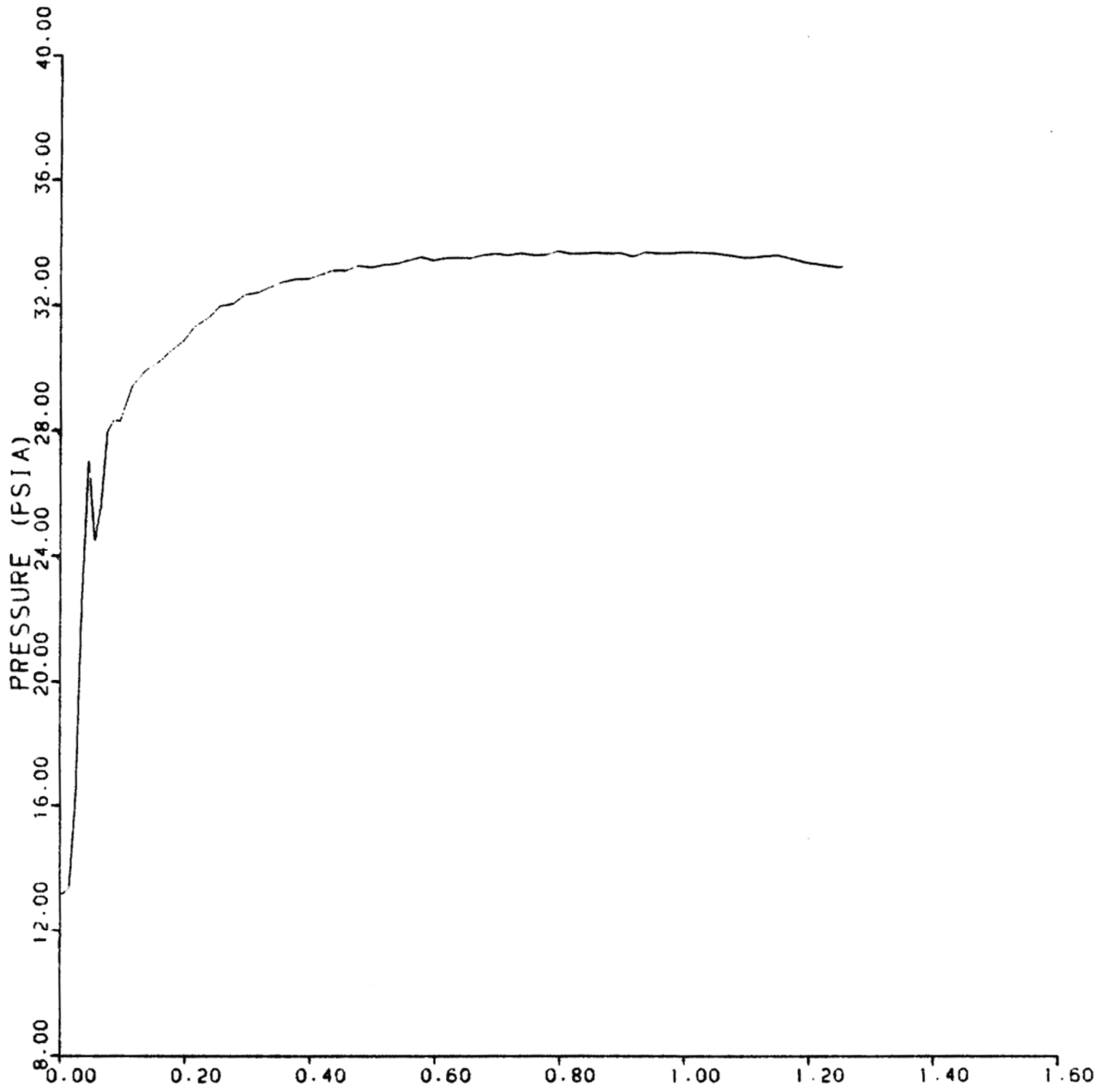
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E15 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 15 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E16 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 16 OF 74)



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SOUTHERN COMPANY
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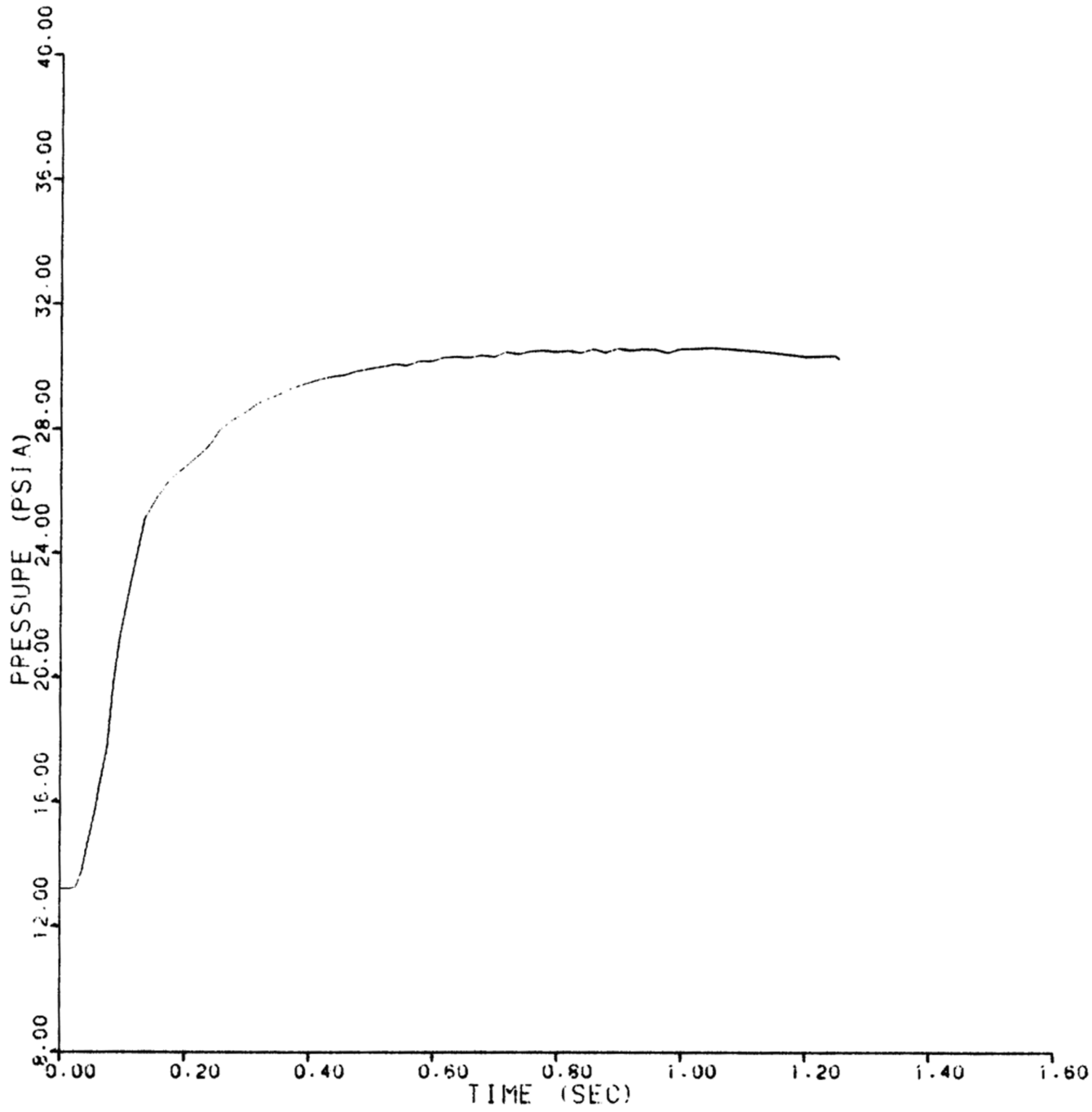
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E17
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 17 OF 74)



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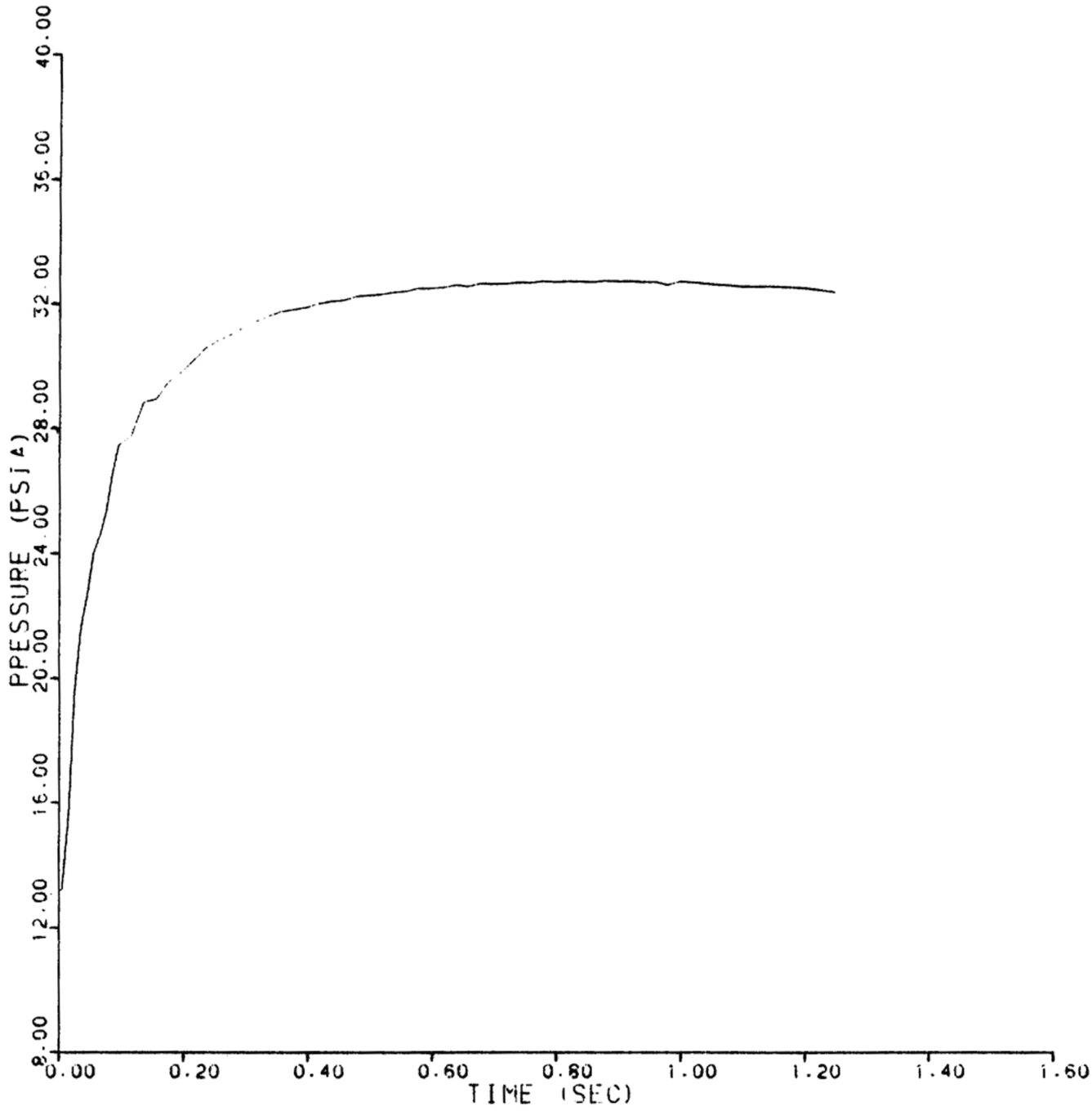
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

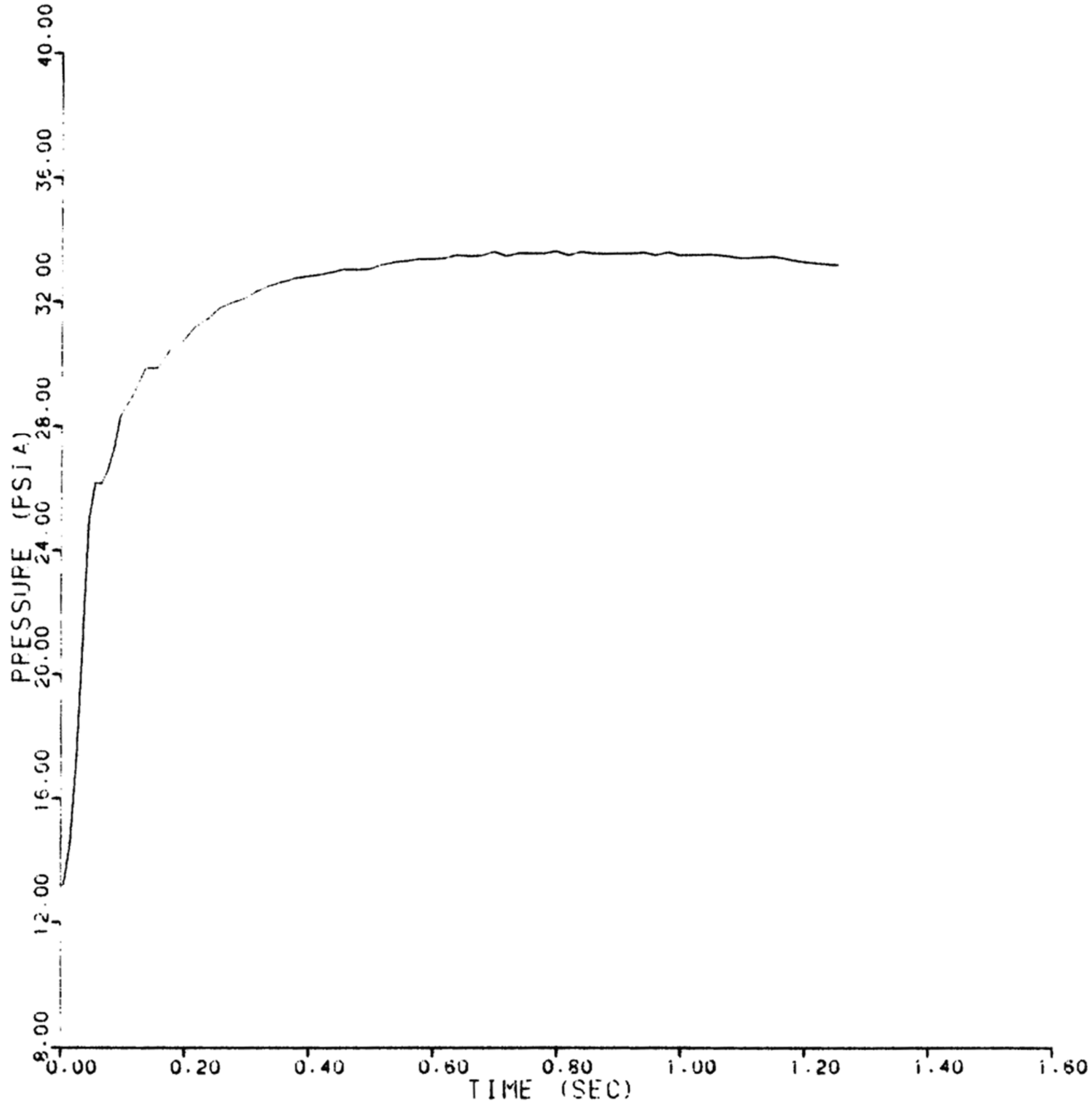
STEAM GENERATOR INLET ELBOW
NODE – E18
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 18 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E19 (763–in.^{2} BREAK AREA) FIGURE 6.2.1–22 (SHEET 19 OF 74)



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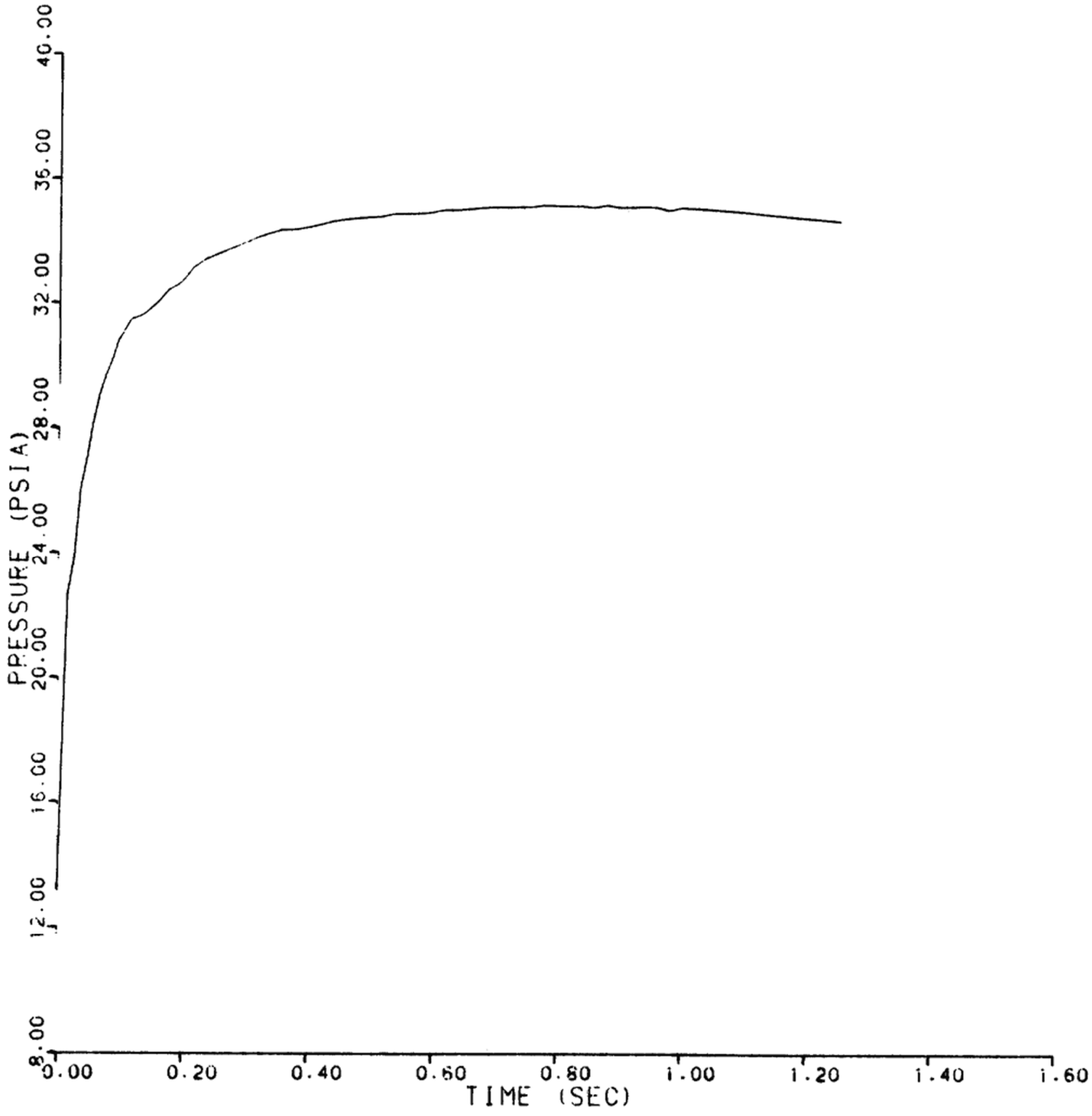
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

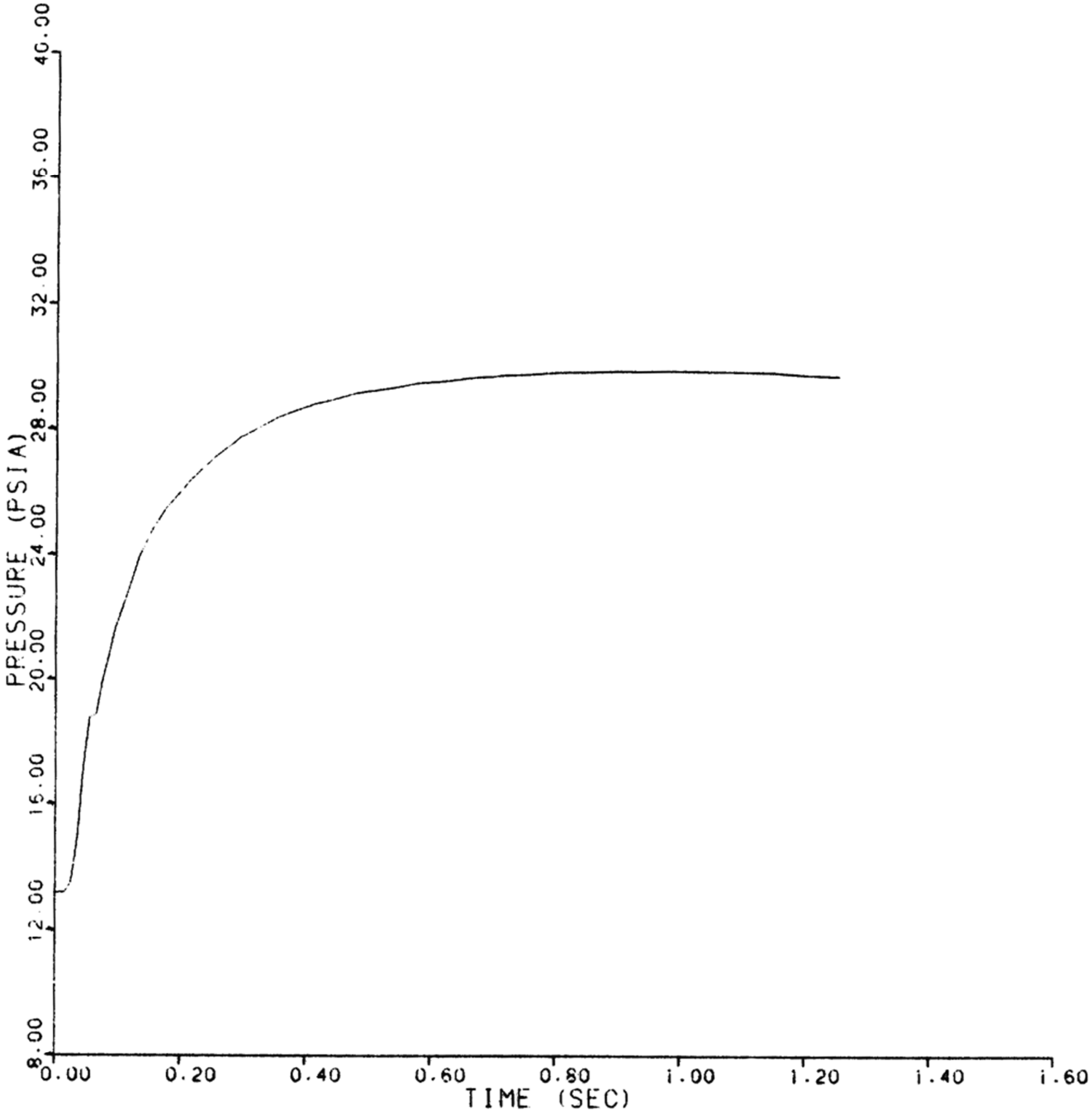
STEAM GENERATOR INLET ELBOW
NODE – E20
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 20 OF 74)



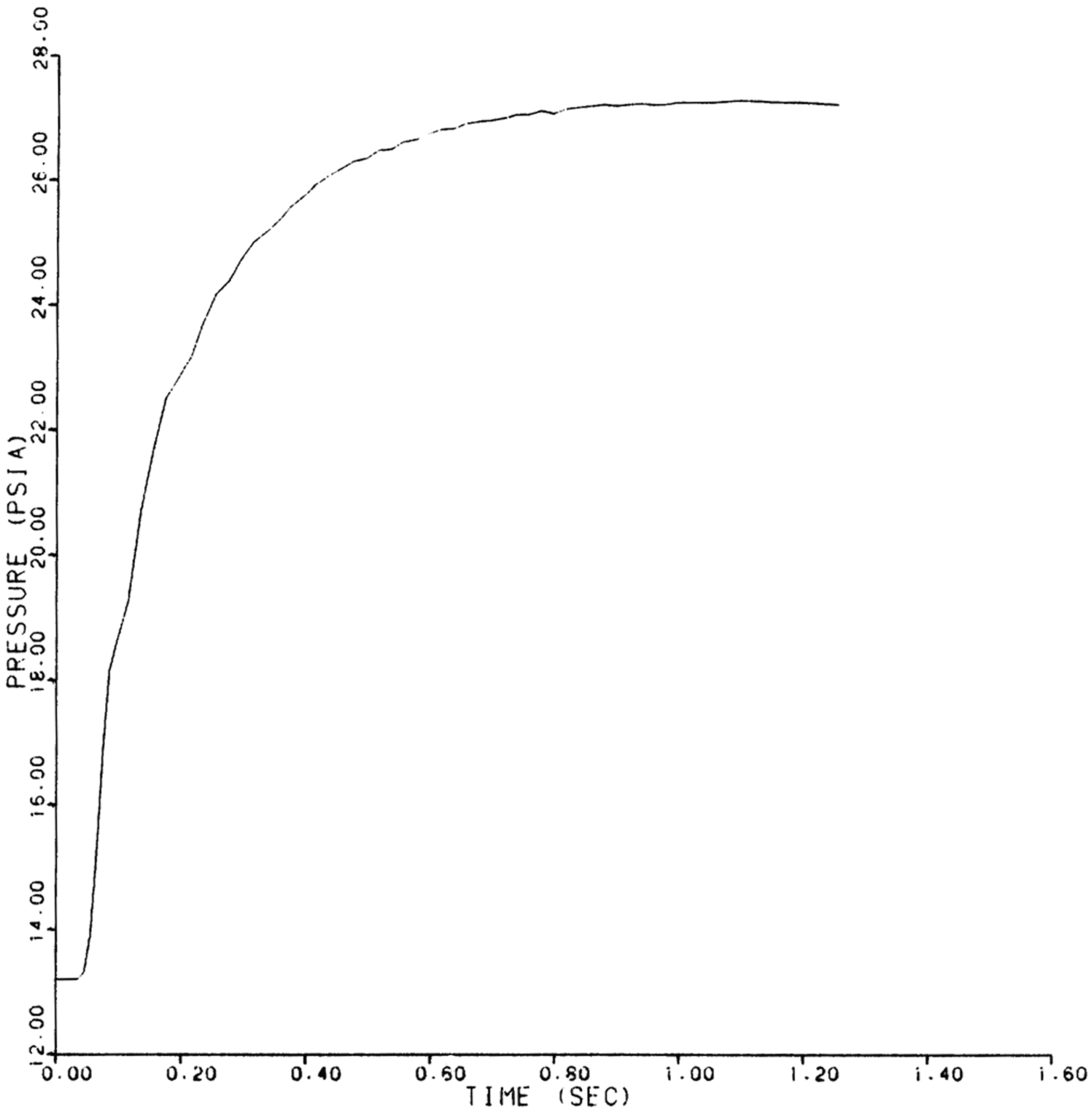
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E21 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 21 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E22 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 22 OF 74)



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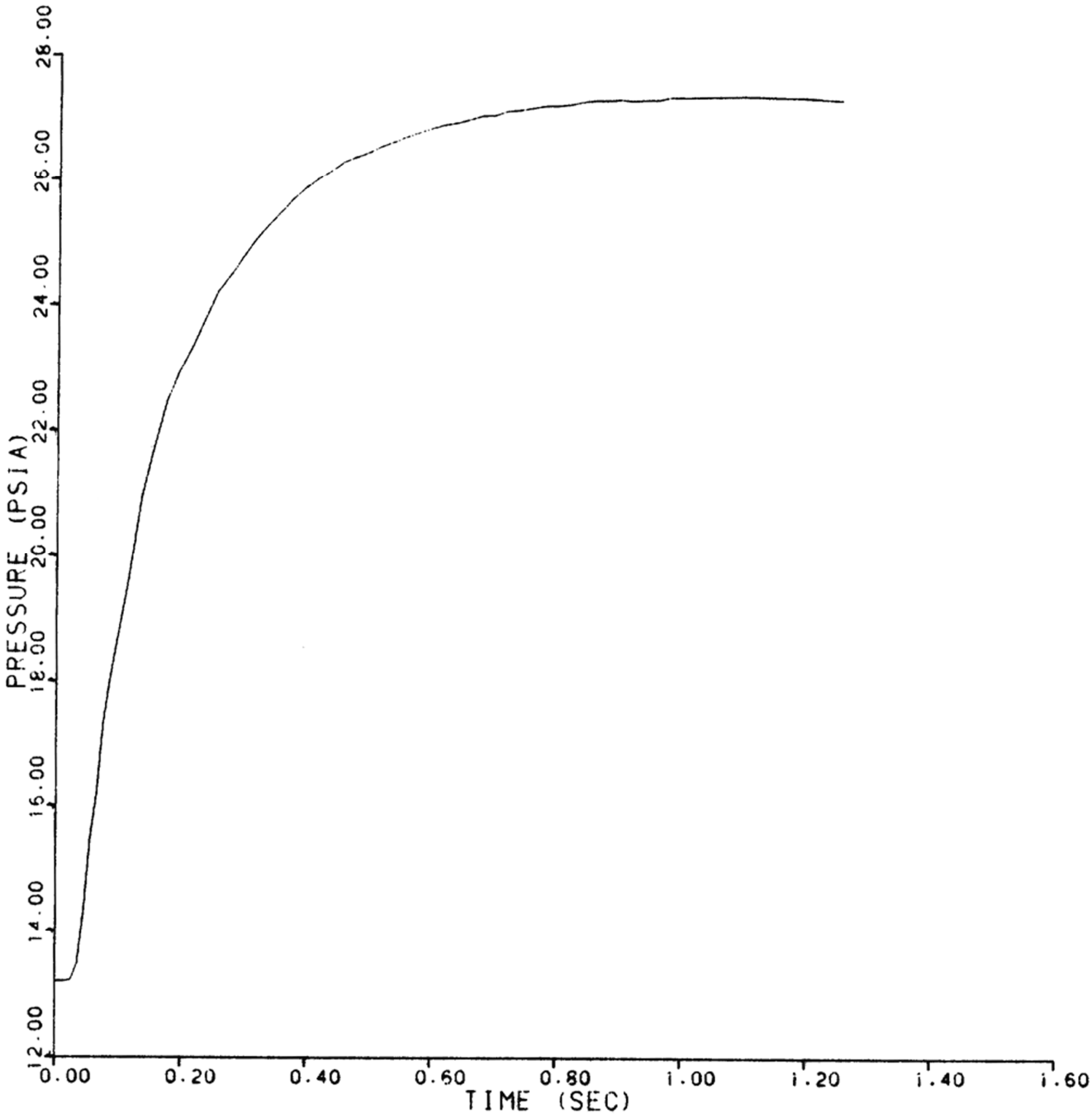
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E23
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 23 OF 74)



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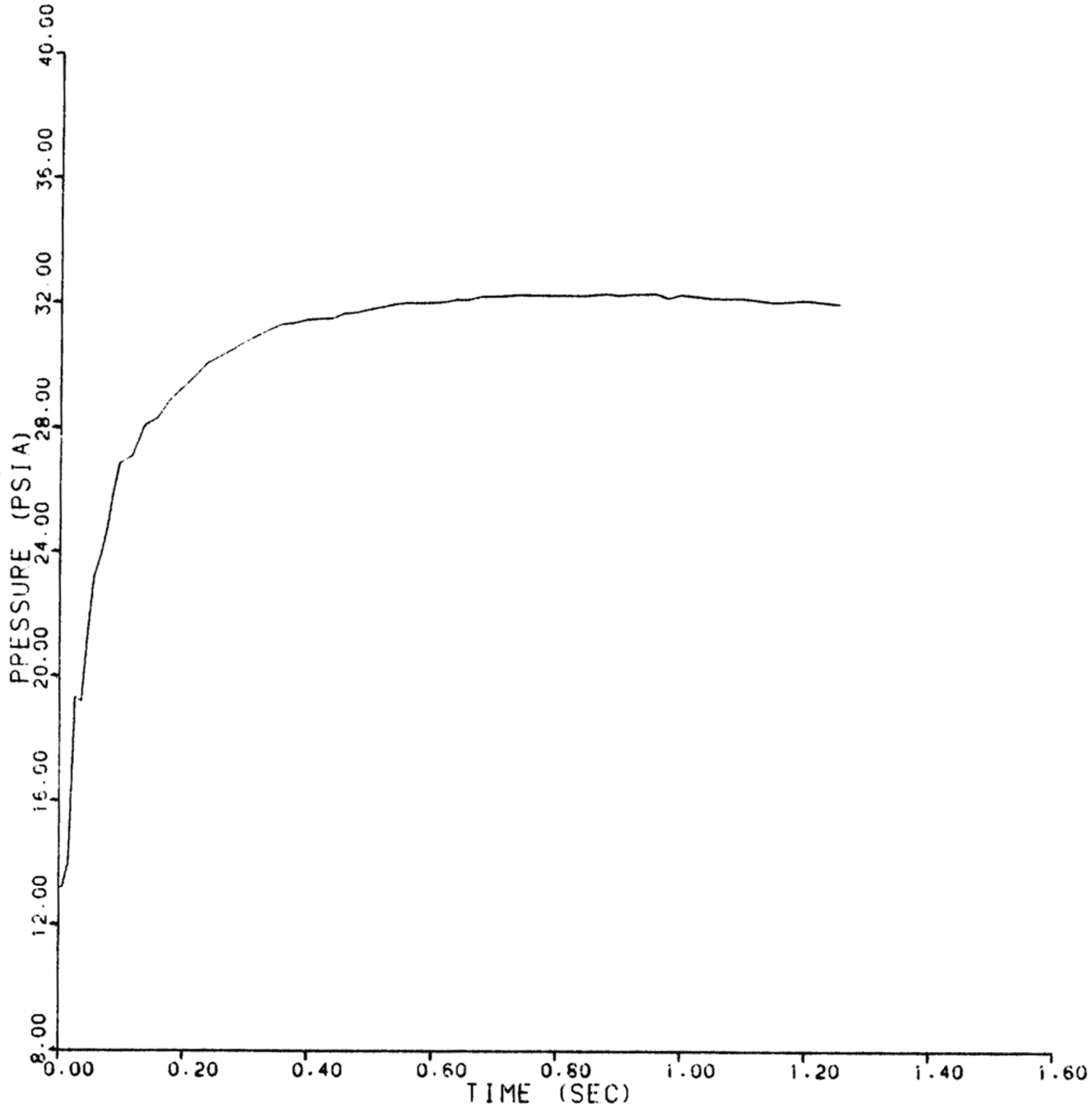
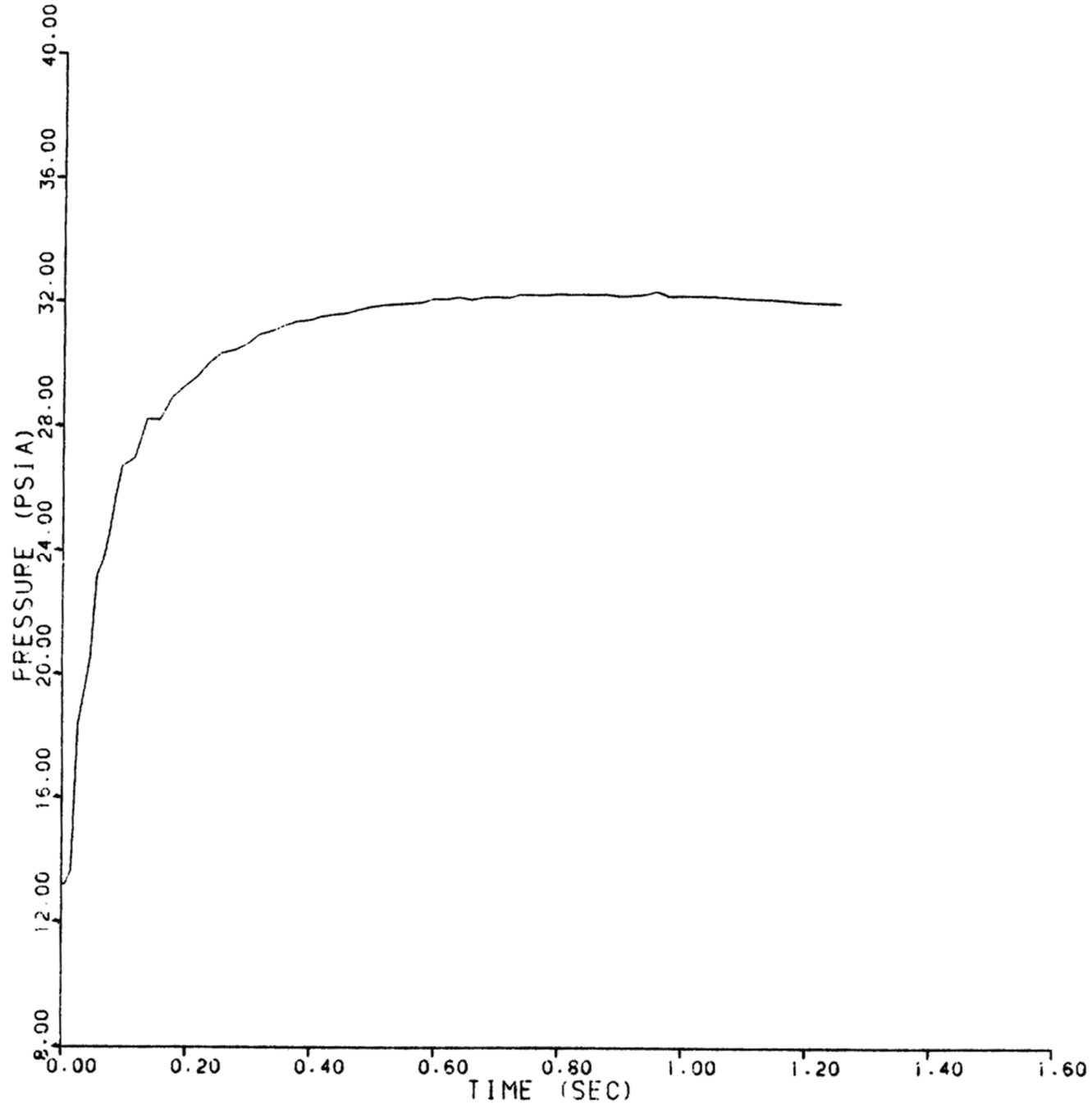
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E24 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 24 OF 74)



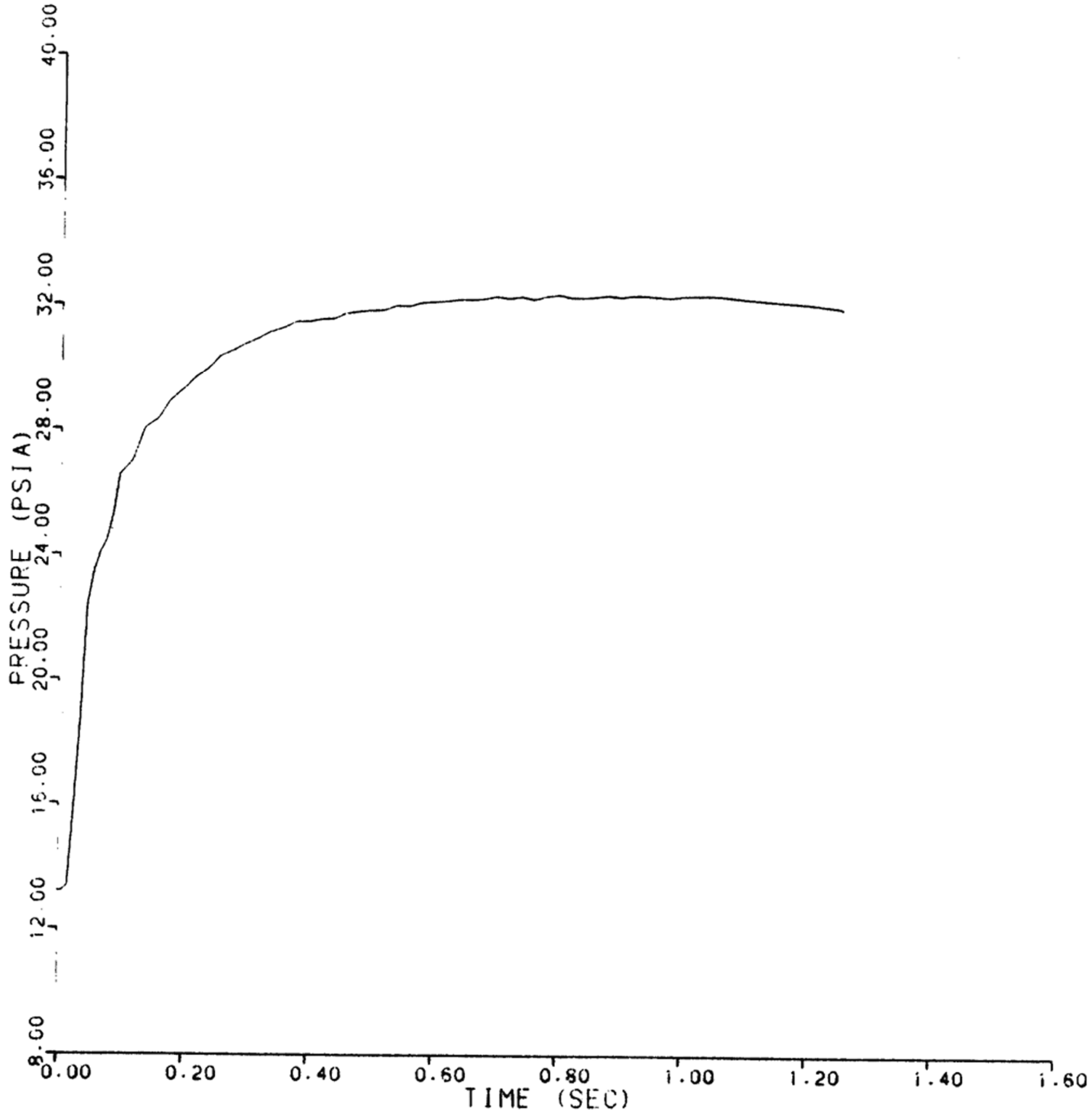
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E25 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 25 OF 74)



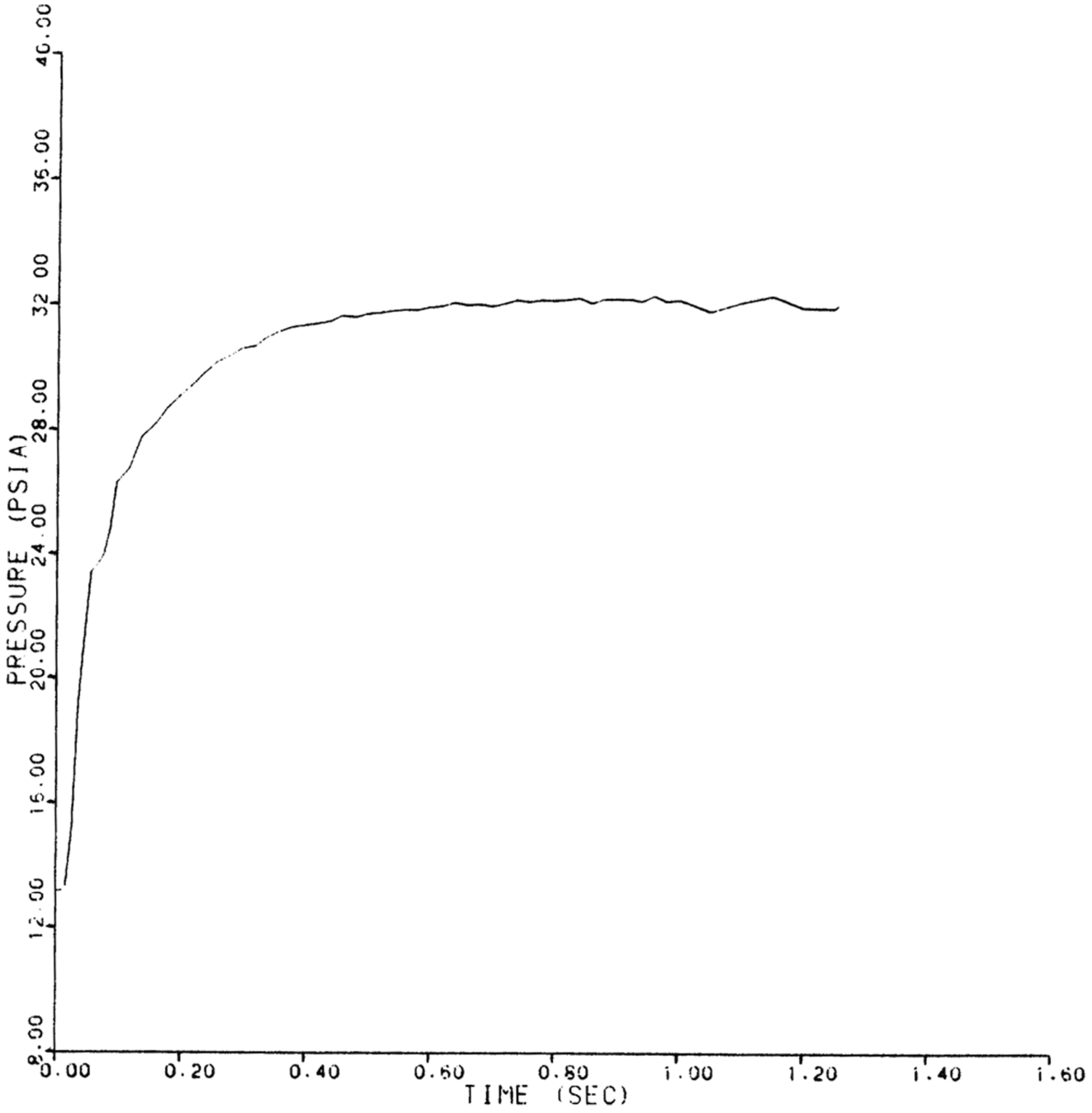
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E26 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 26 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E27 (763–in.^{2} BREAK AREA) FIGURE 6.2.1–22 (SHEET 27 OF 74)



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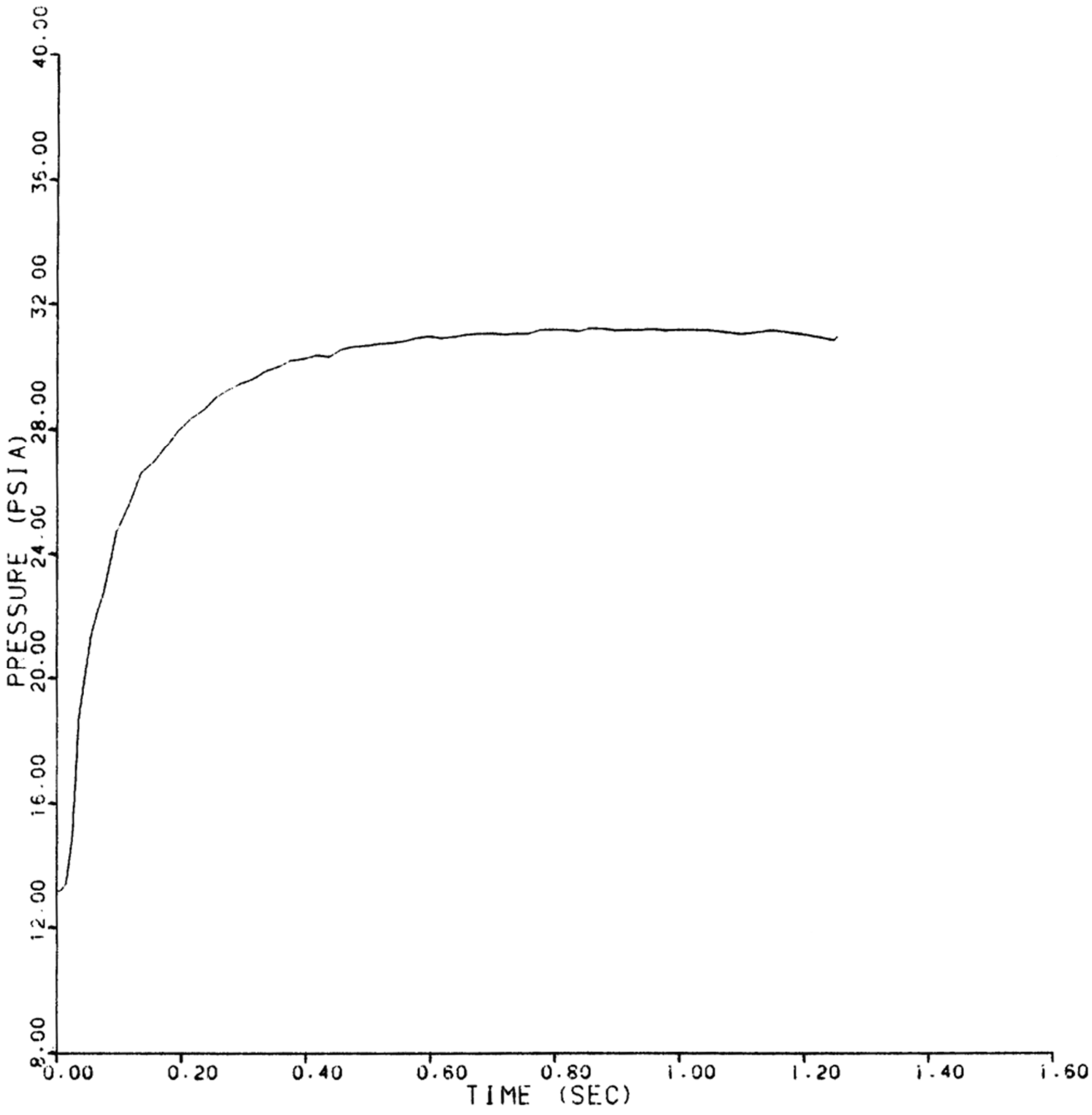
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E28 (763–in.^{2} BREAK AREA) FIGURE 6.2.1–22 (SHEET 28 OF 74)



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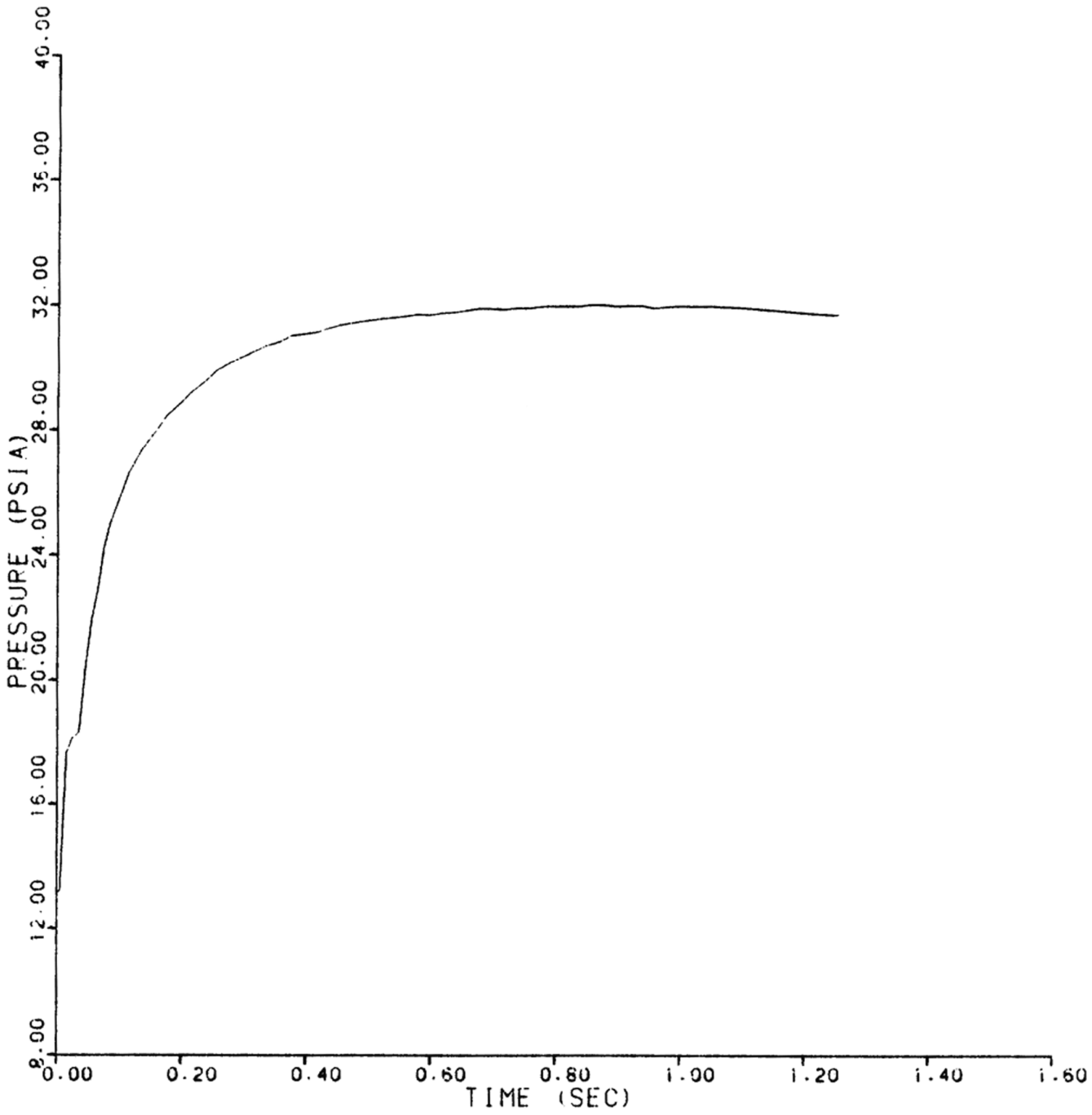
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E29
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 29 OF 74)



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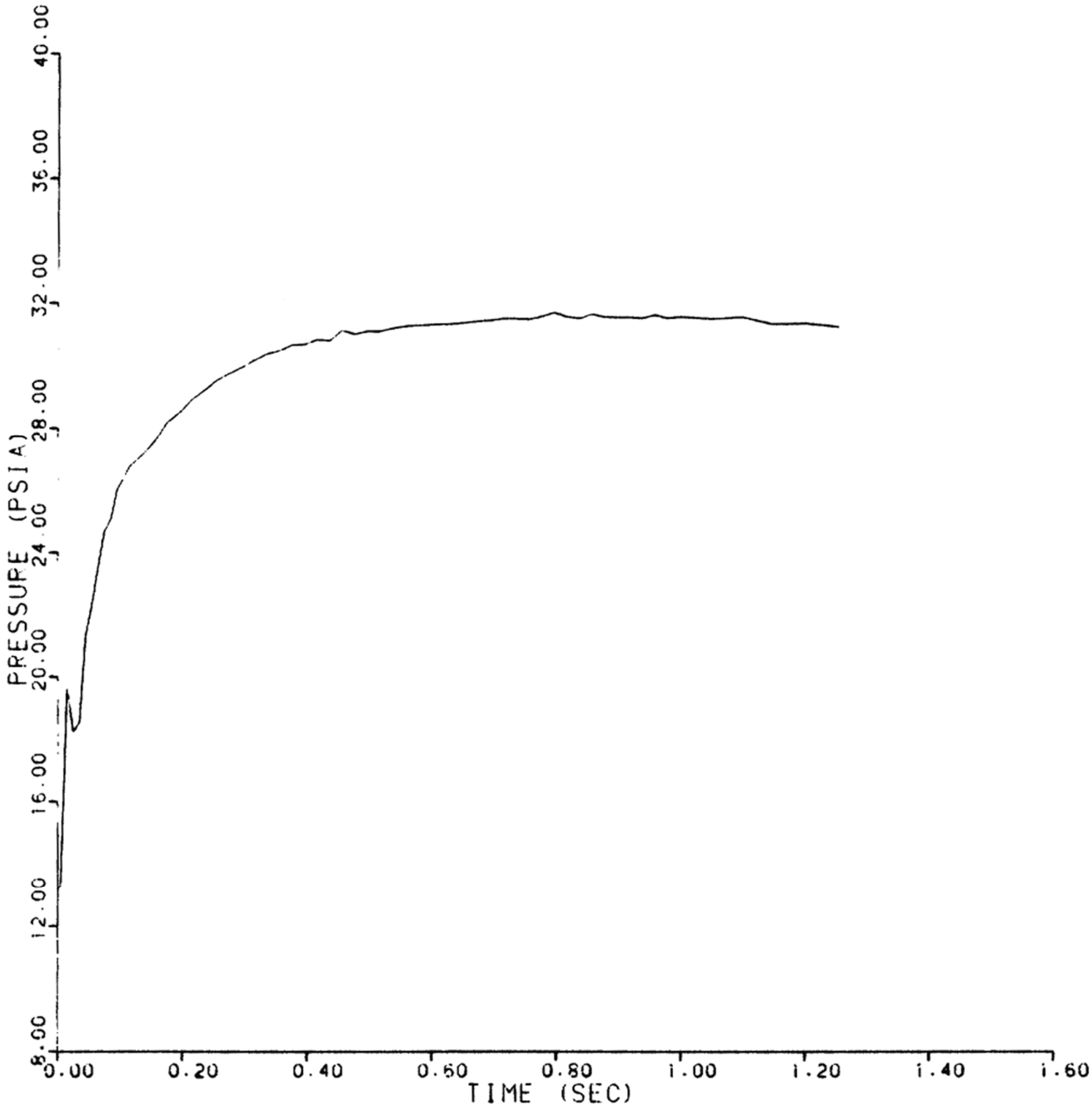
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E30 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 30 OF 74)



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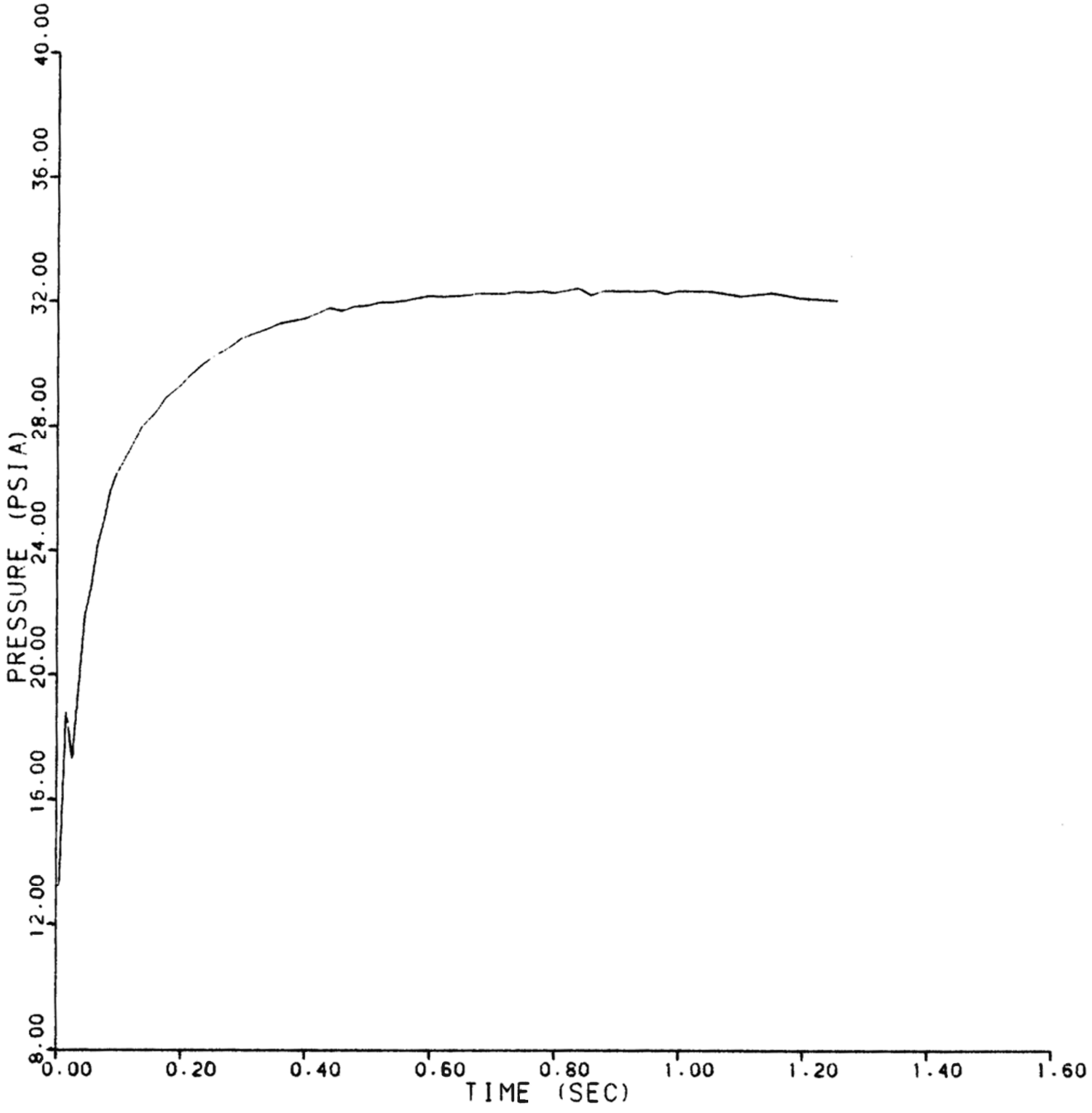
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E31
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 31 OF 74)



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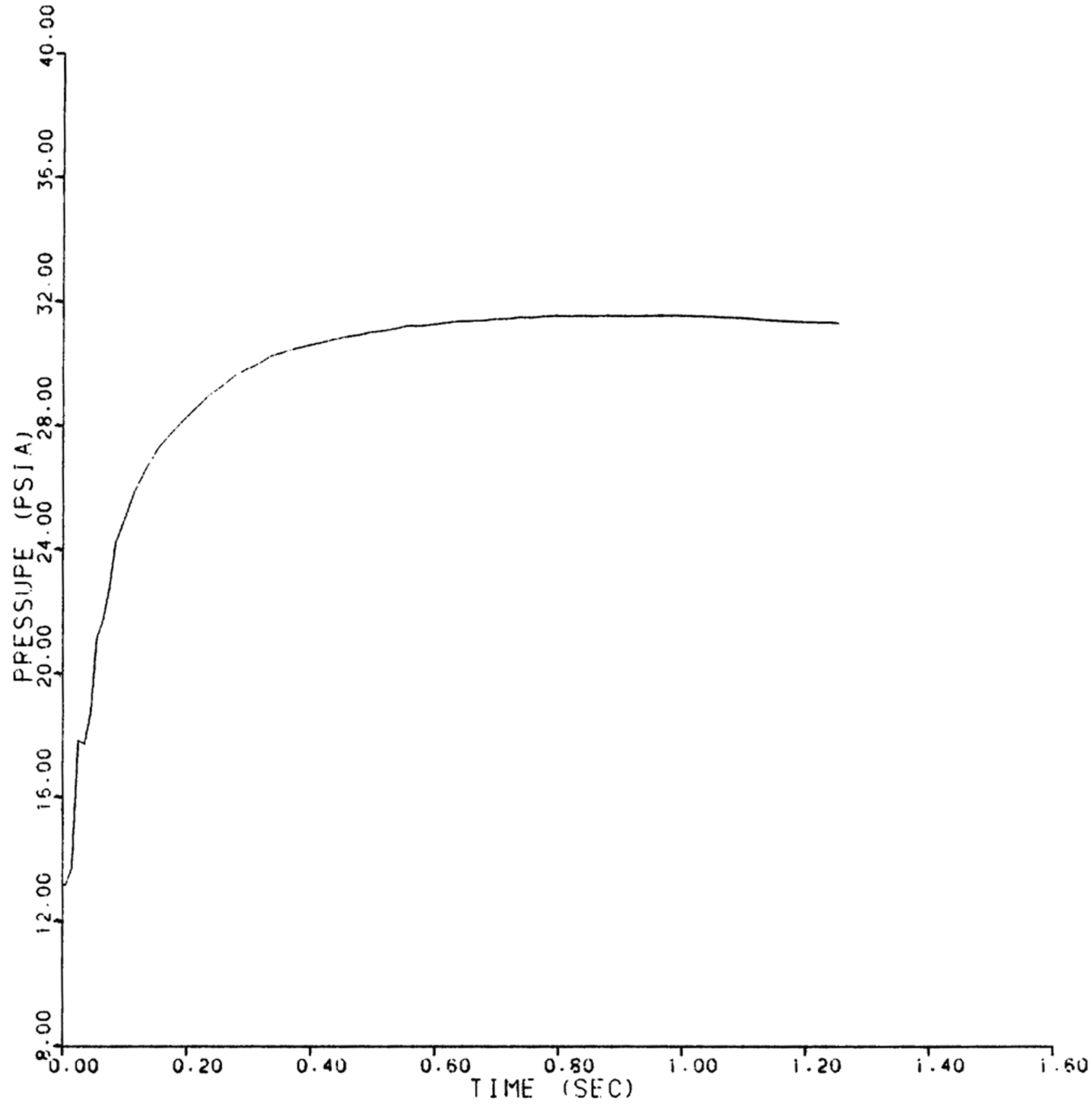
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E32
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 32 OF 74)



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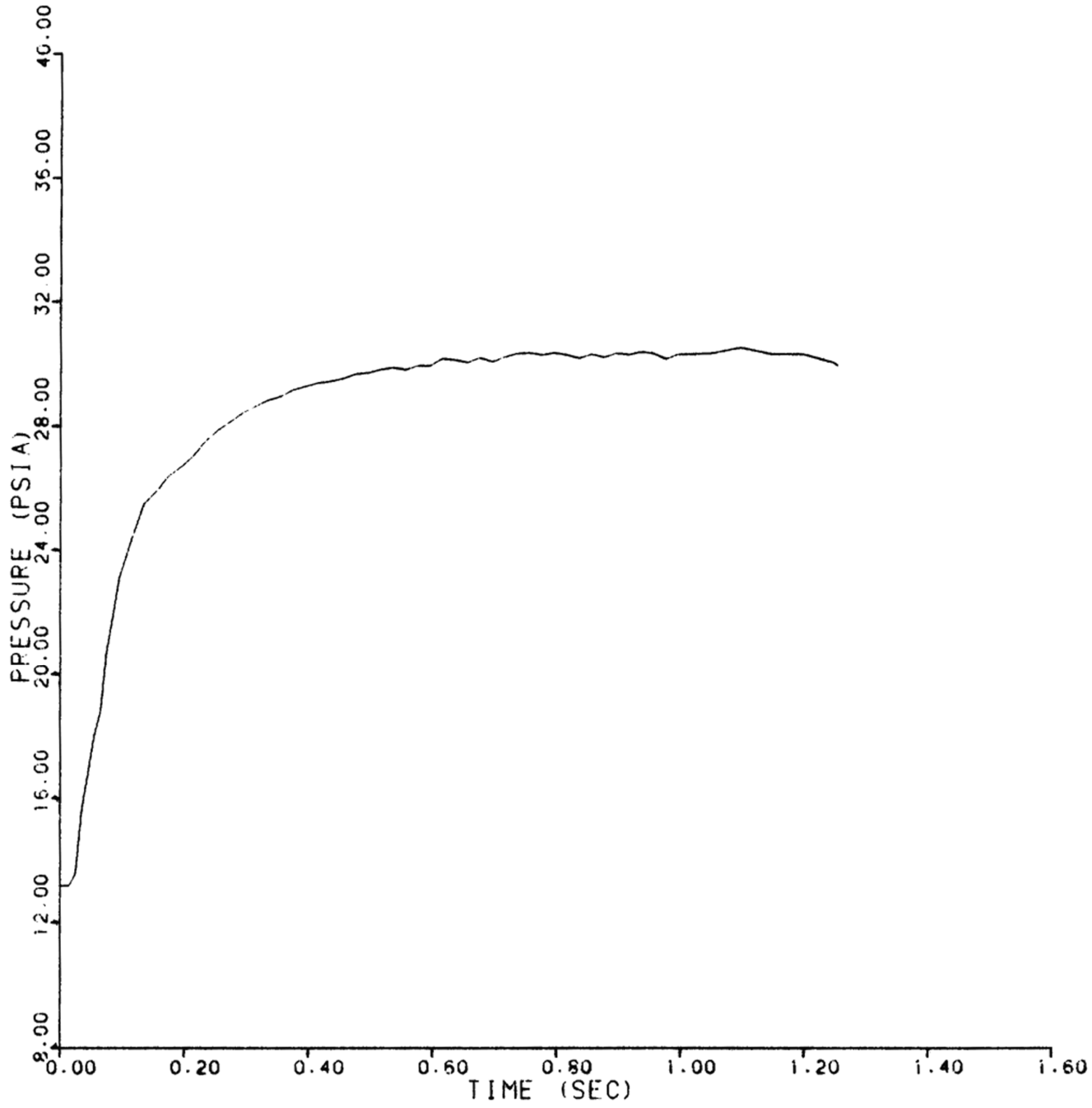
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E33 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 33 OF 74)



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SOUTHERN COMPANY
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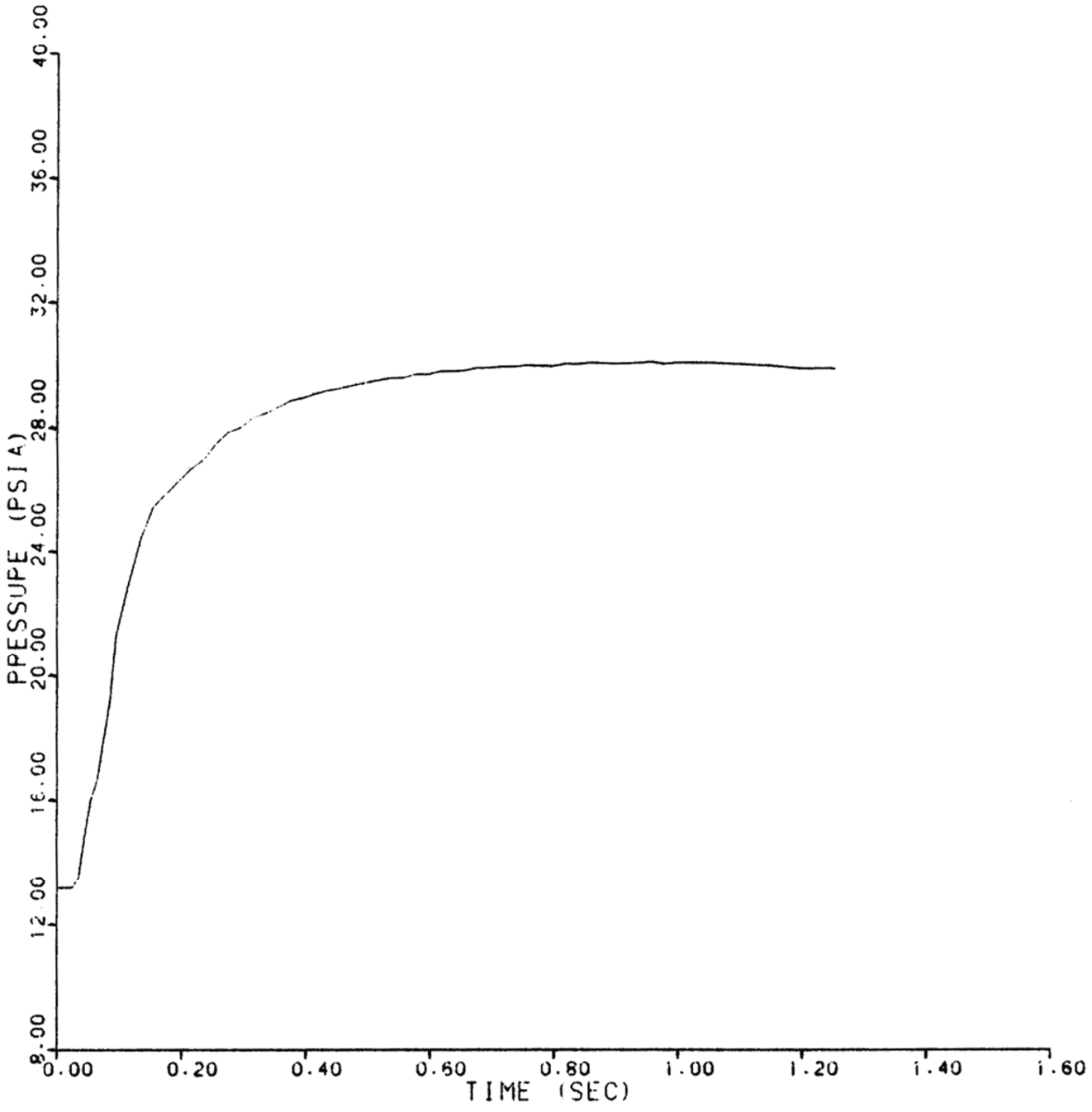
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E34
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 34 OF 74)



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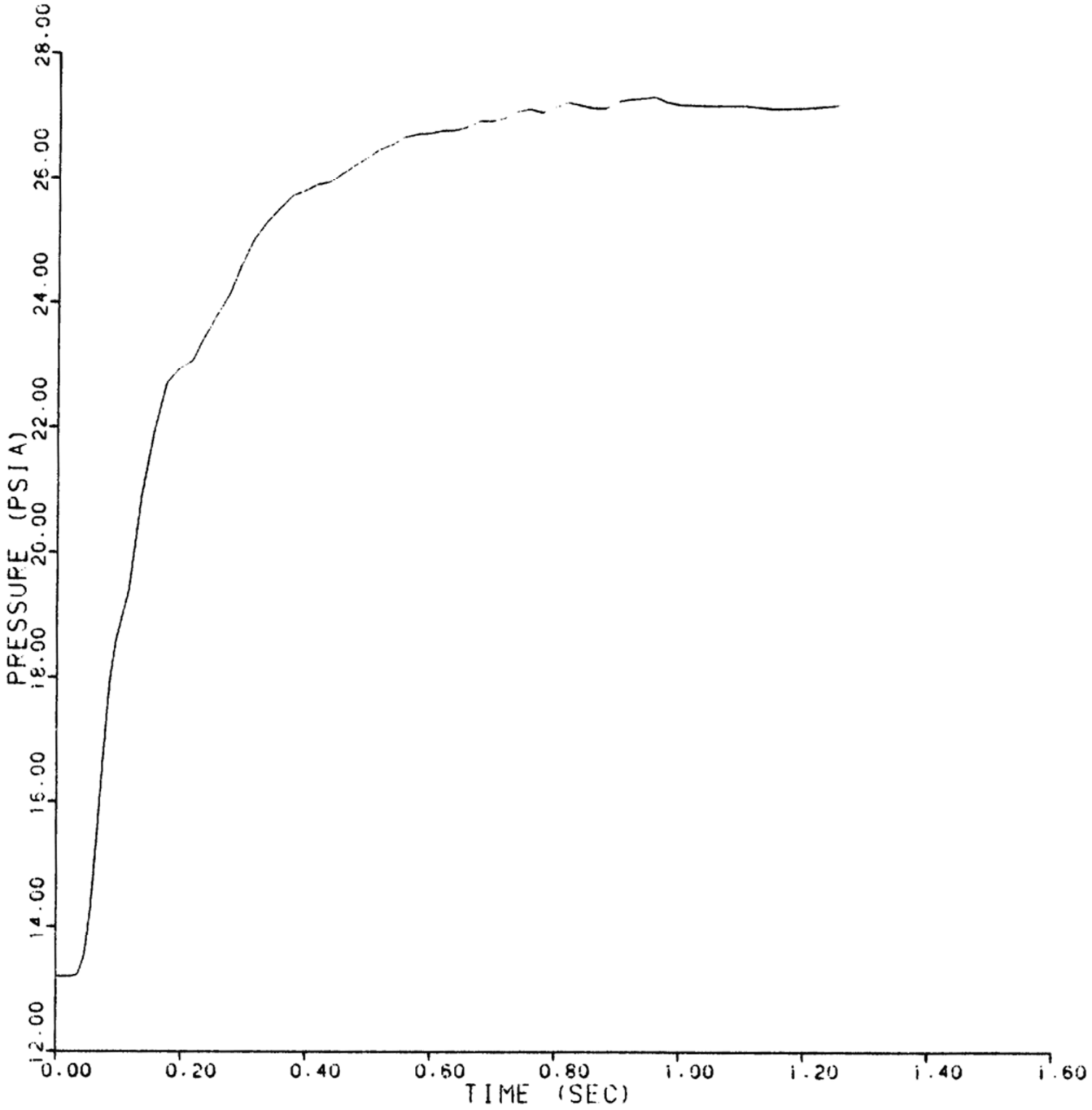
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E35
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 35 OF 74)



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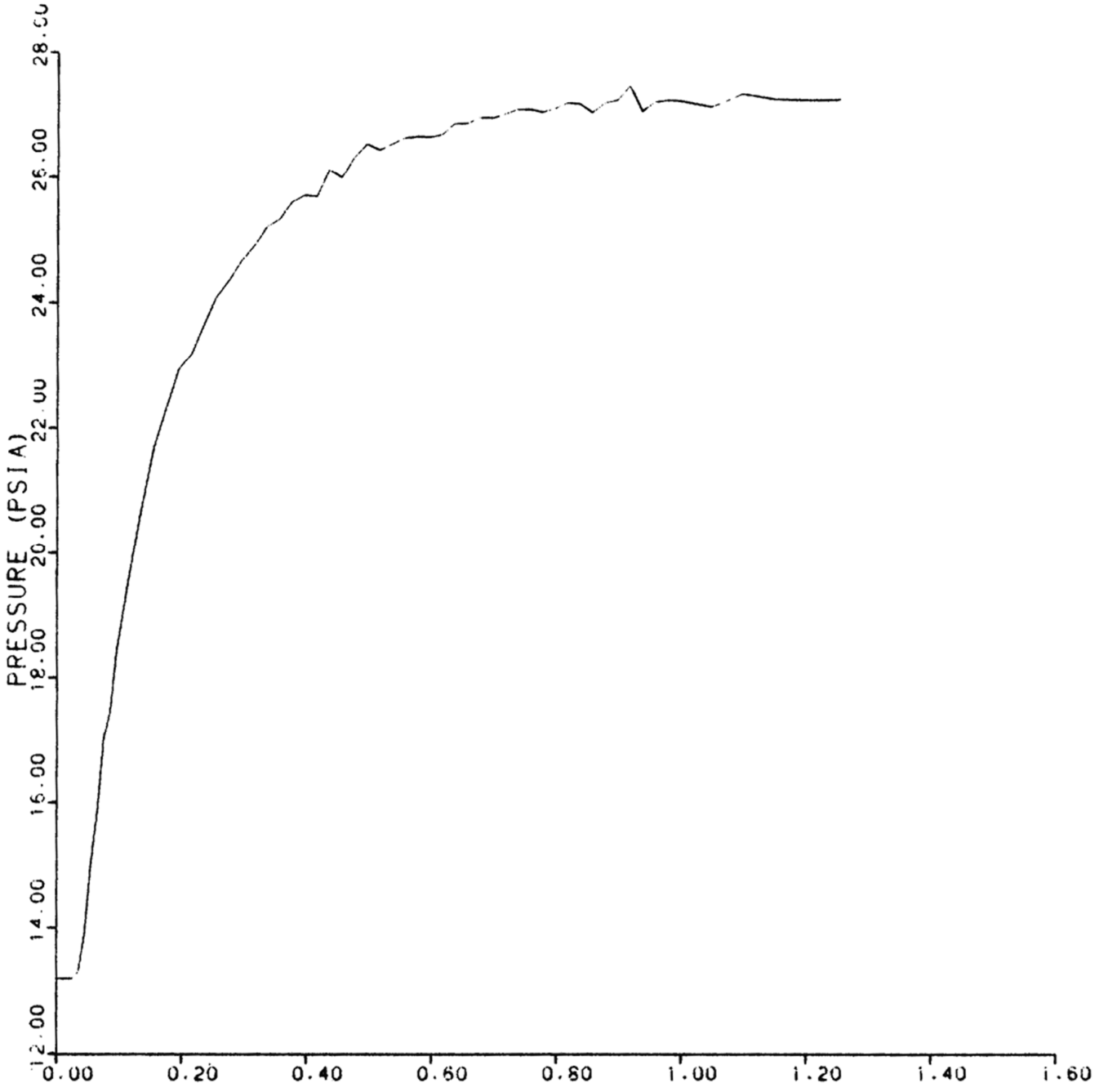
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E36
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 36 OF 74)



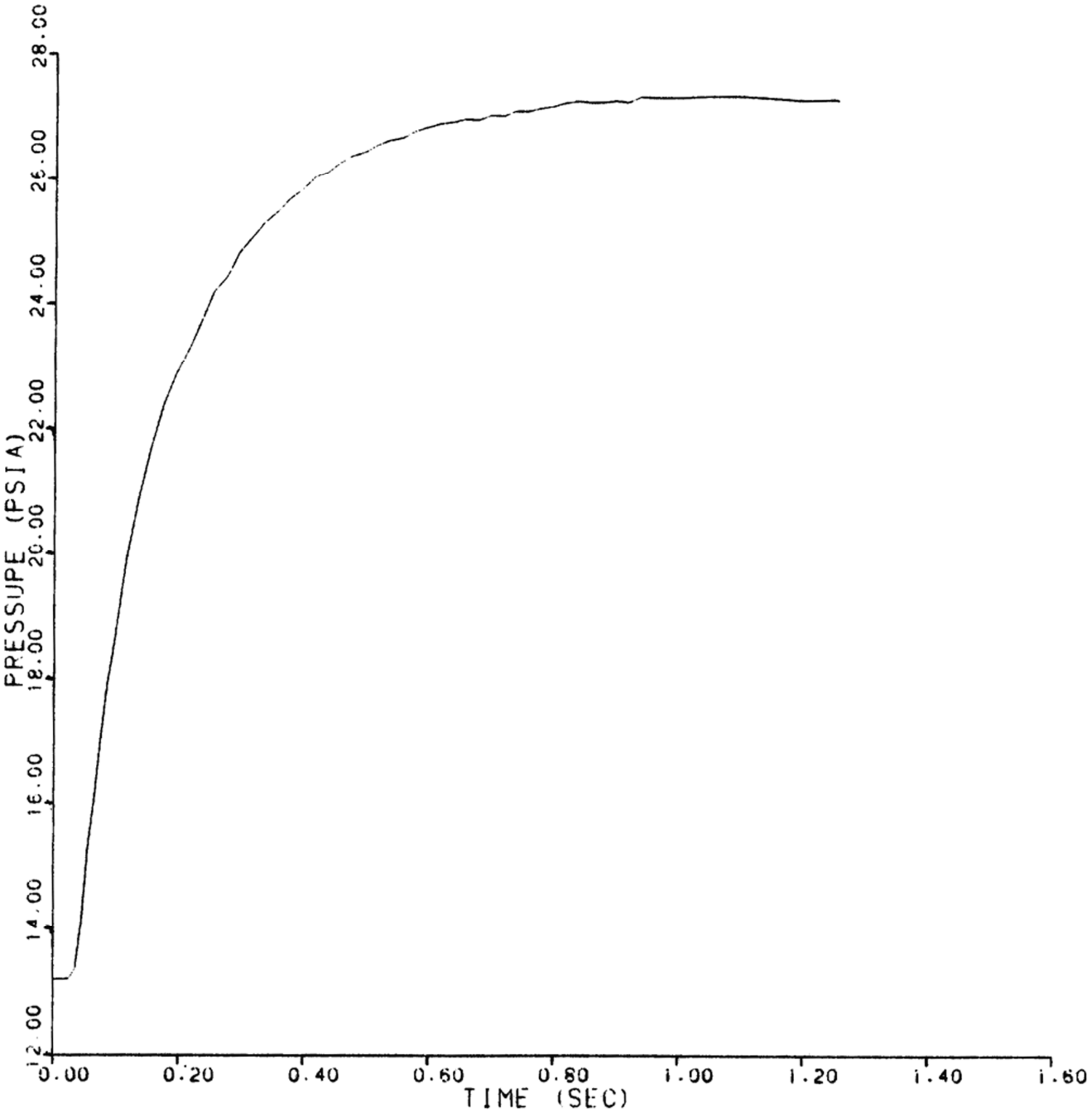
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E37
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 37 OF 74)



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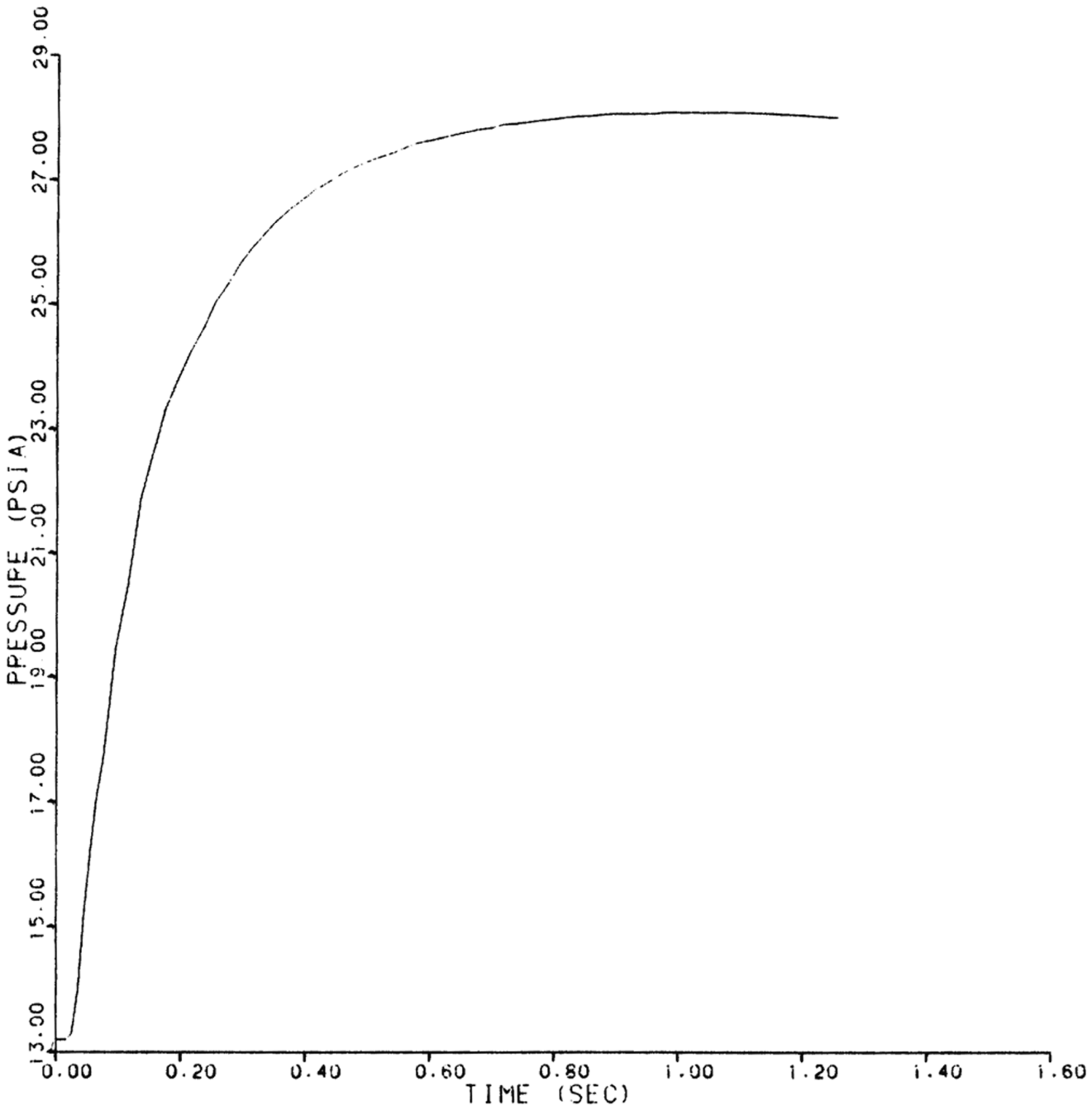
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E38
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 38 OF 74)



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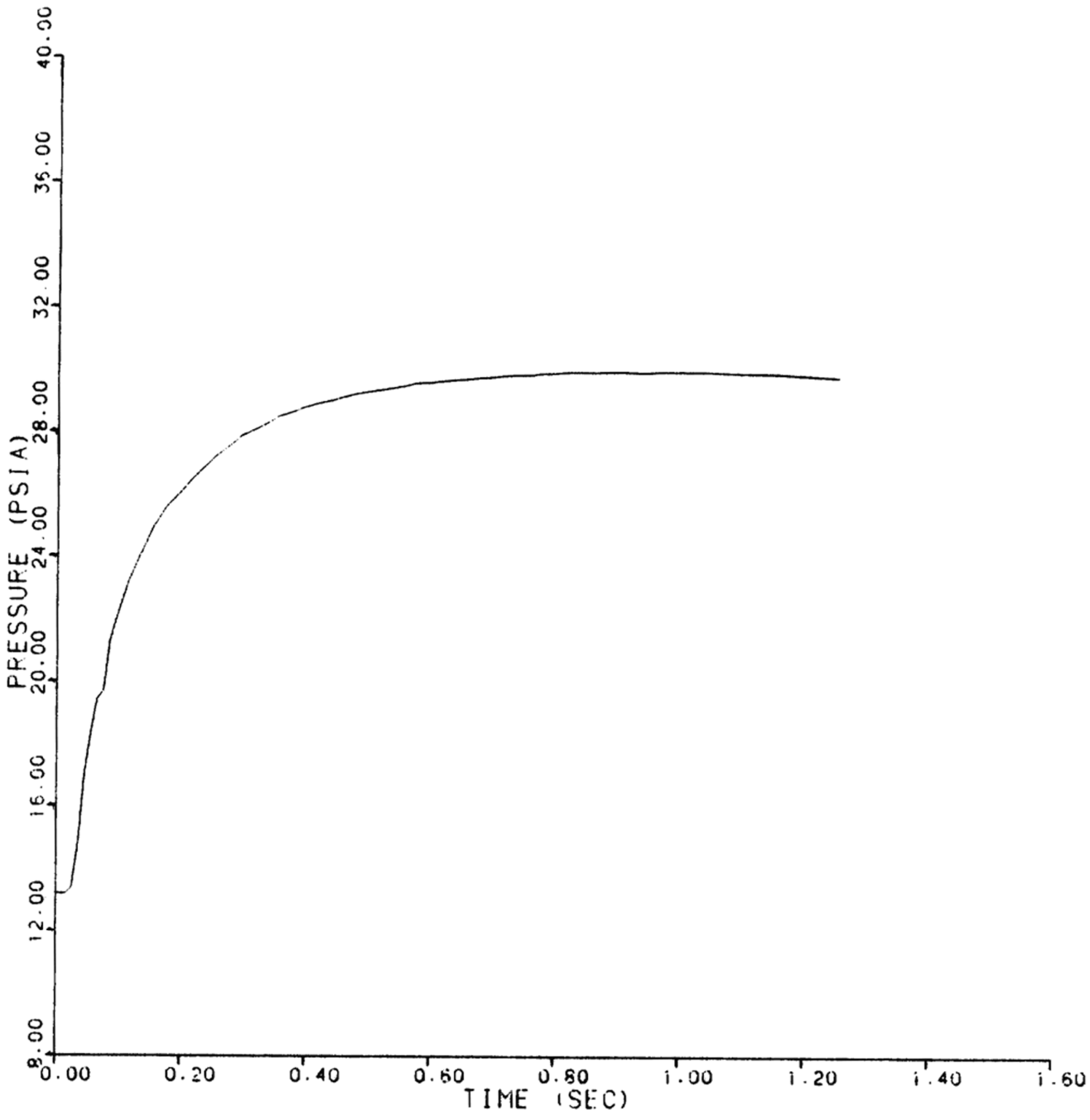
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E39
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 39 OF 74)



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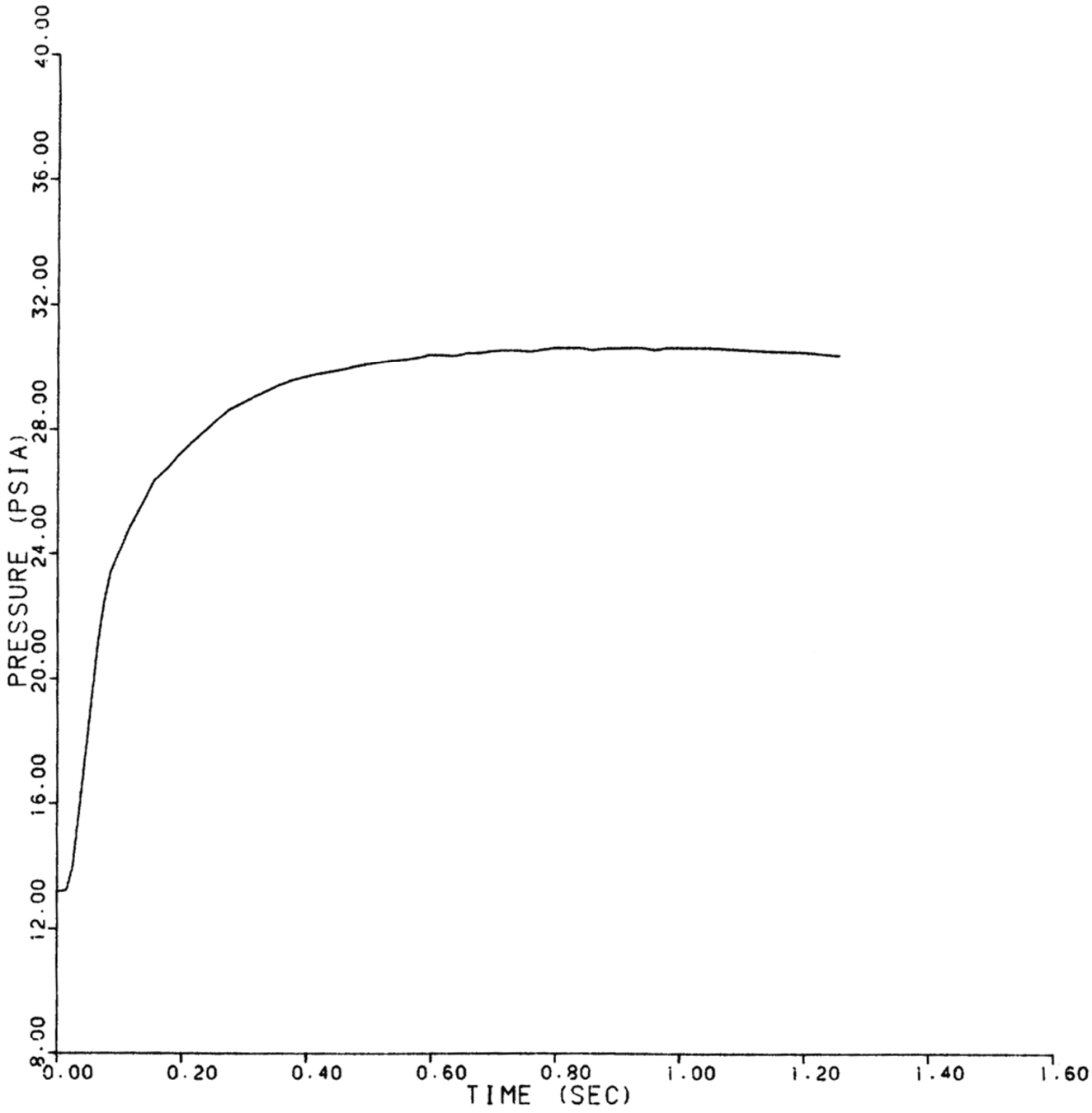
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

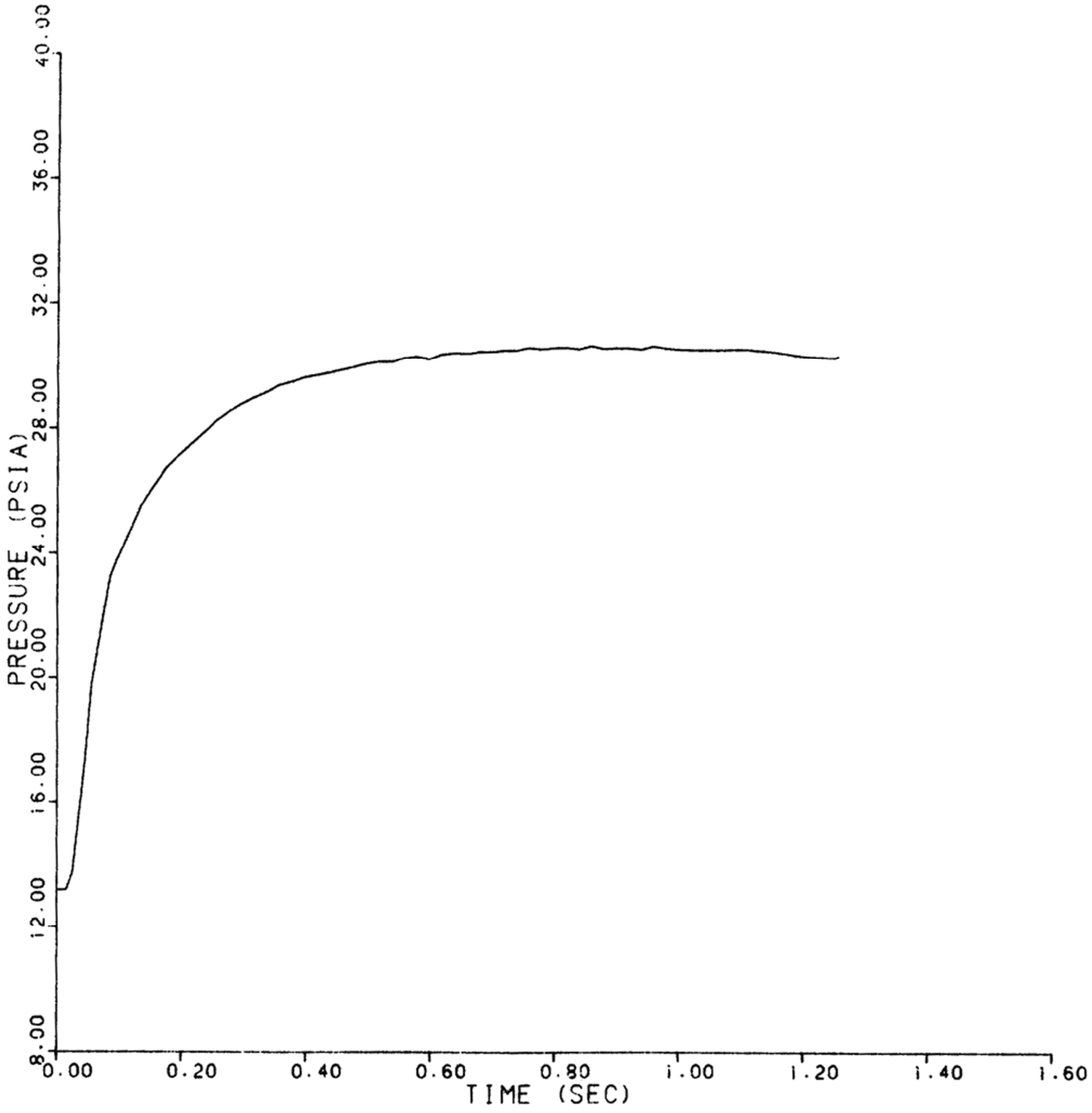
STEAM GENERATOR INLET ELBOW
NODE – E40
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 40 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E41 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 41 OF 74)



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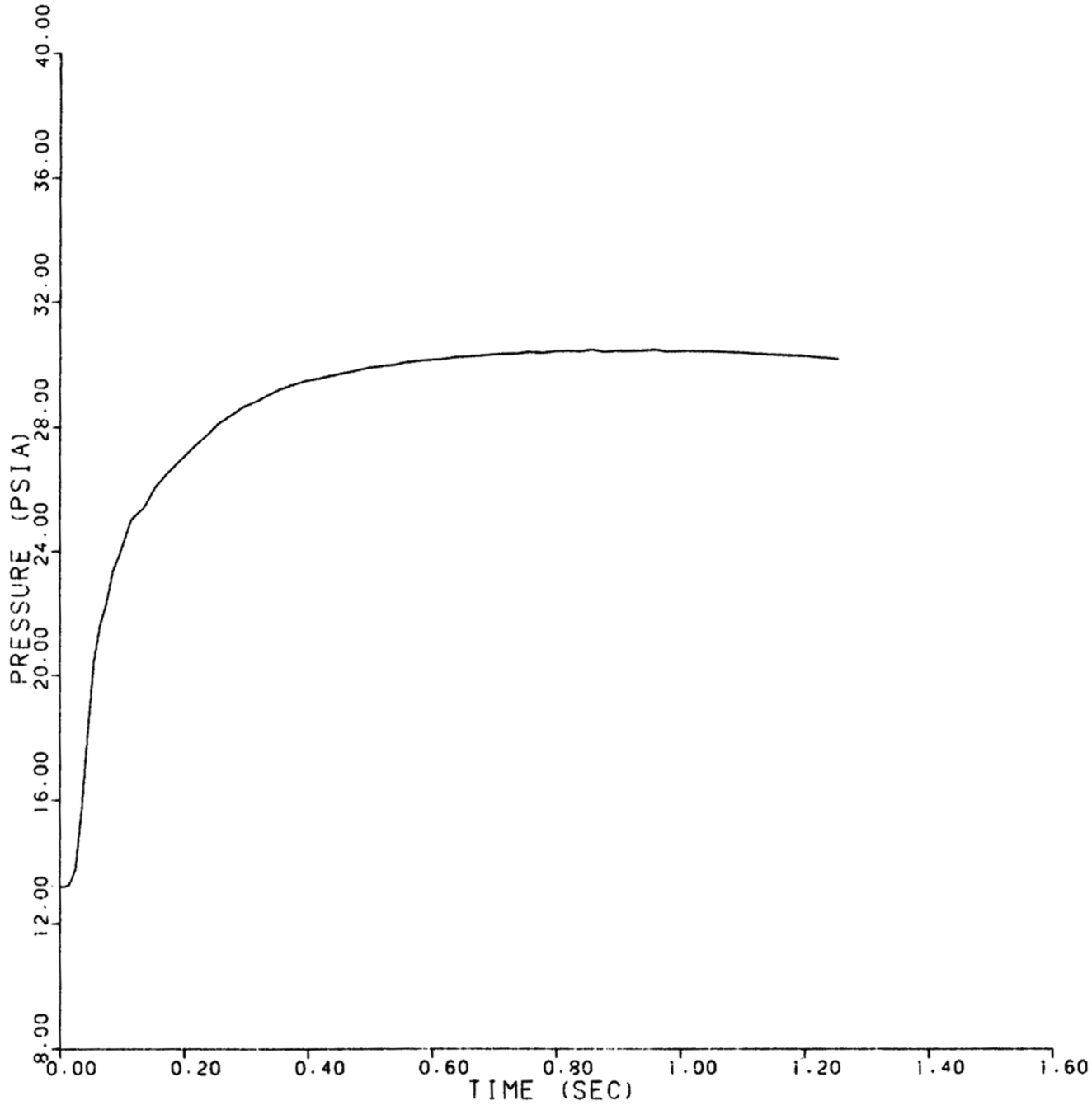
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E42
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 42 OF 74)



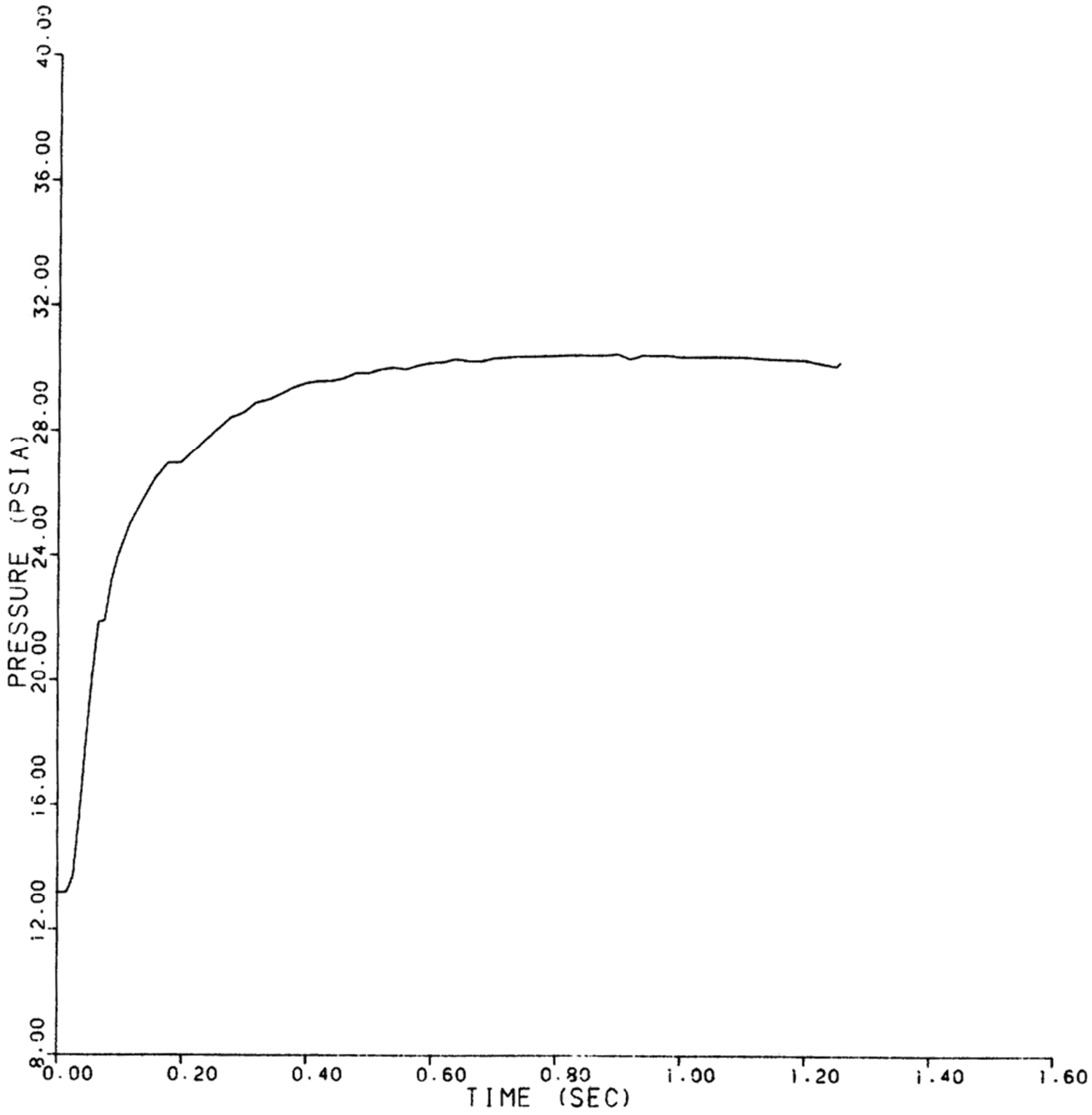
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E43
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 43 OF 74)



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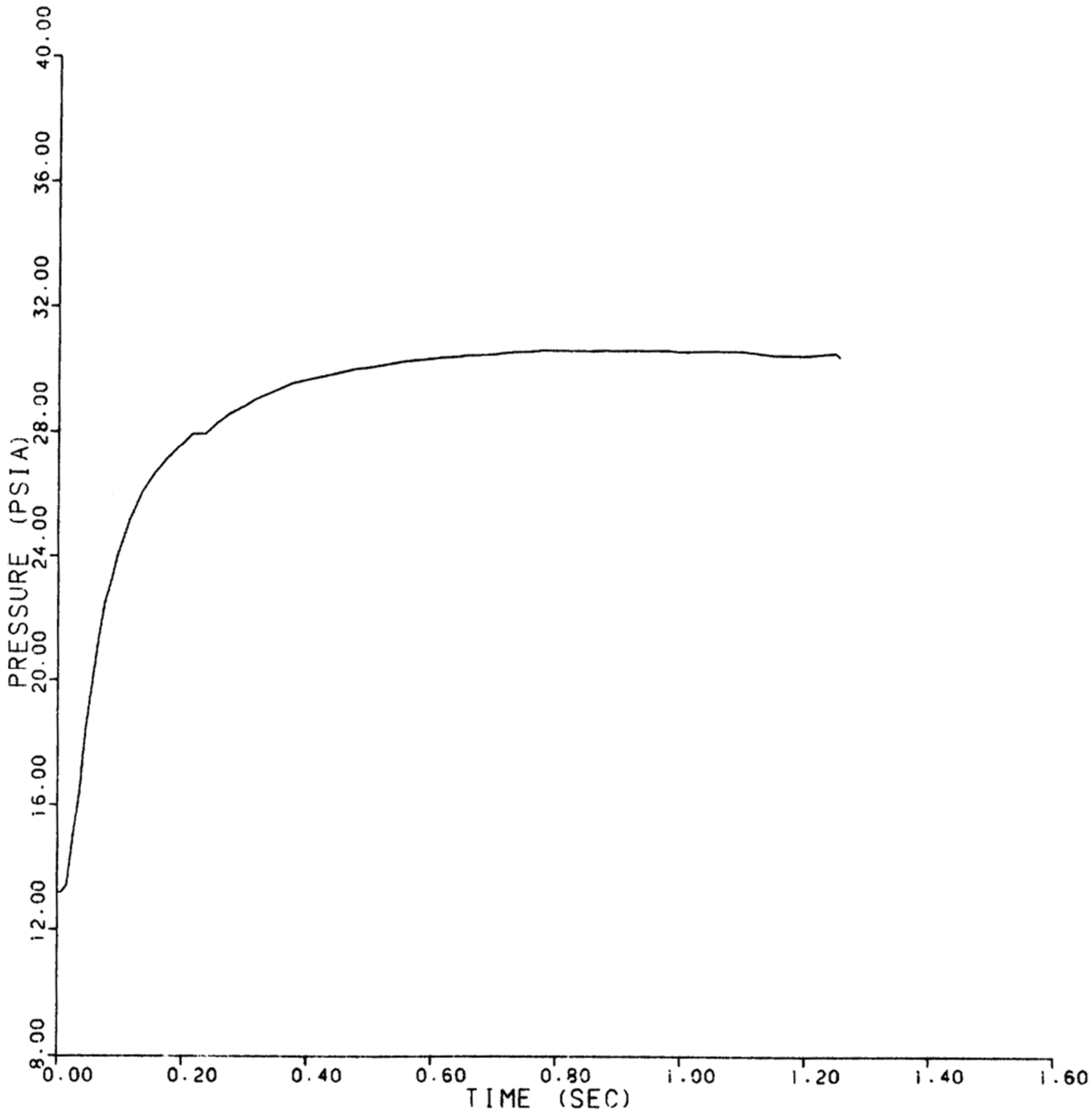
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E44
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 44 OF 74)



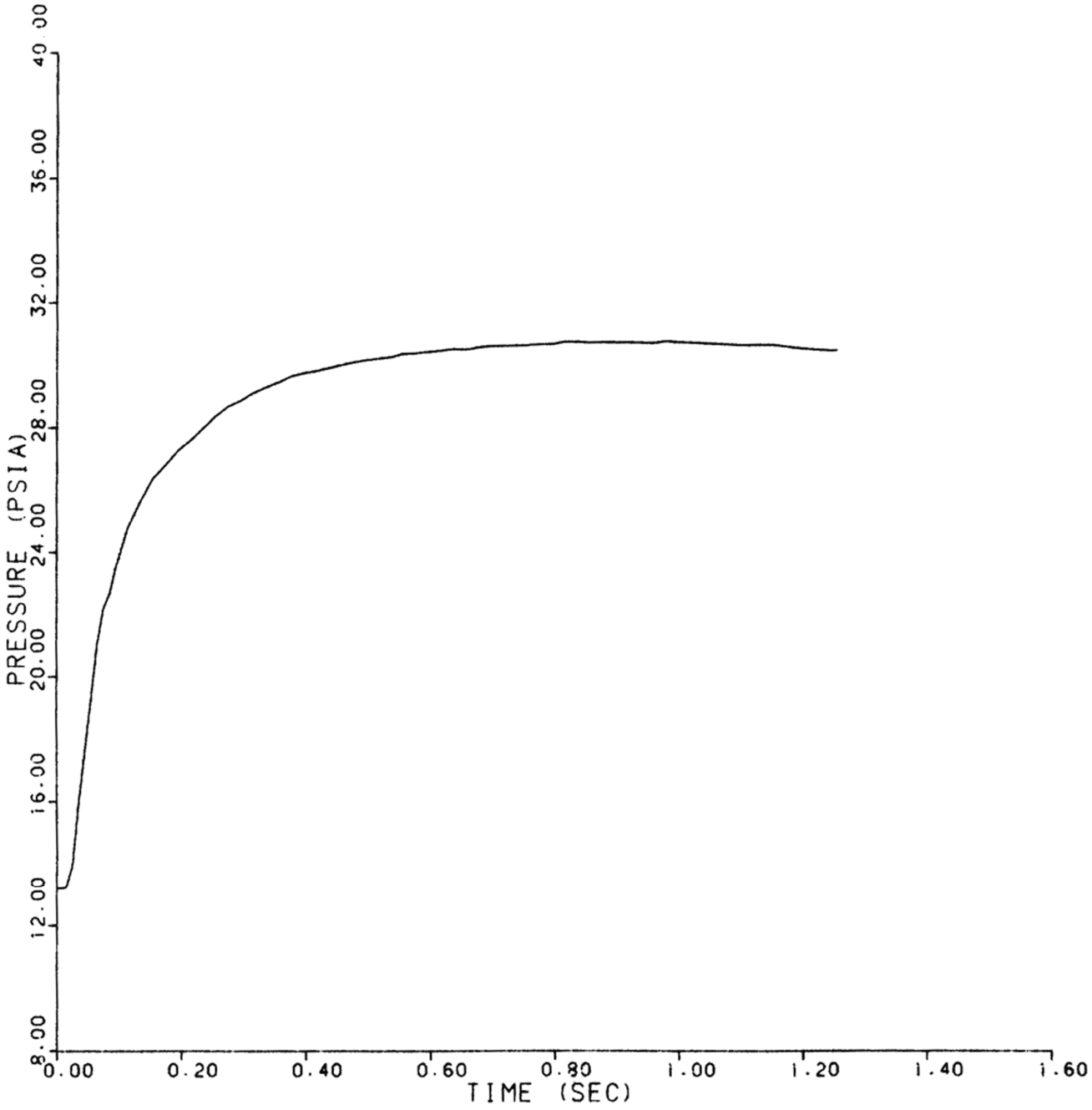
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E45
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 45 OF 74)



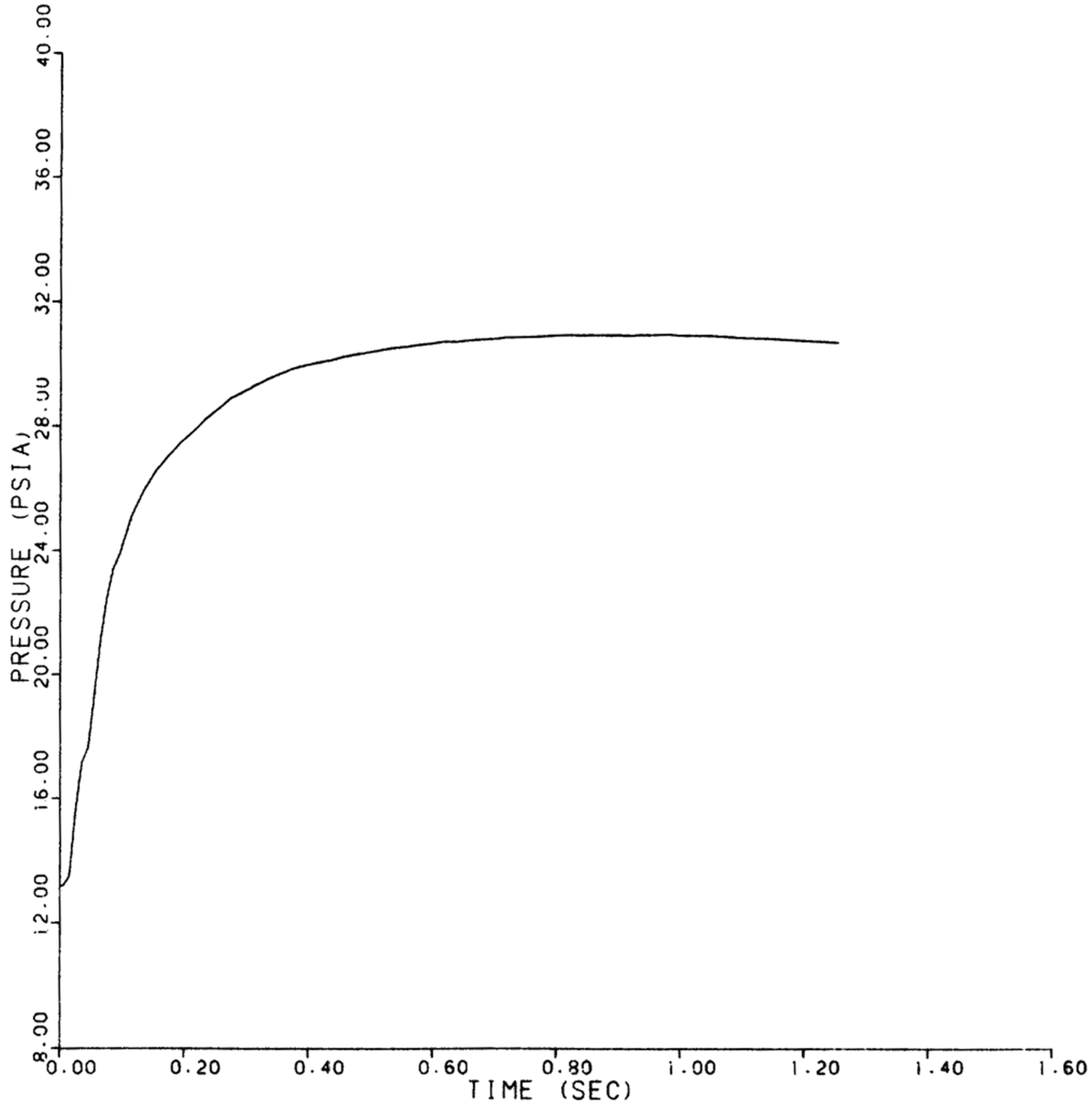
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E46
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 46 OF 74)



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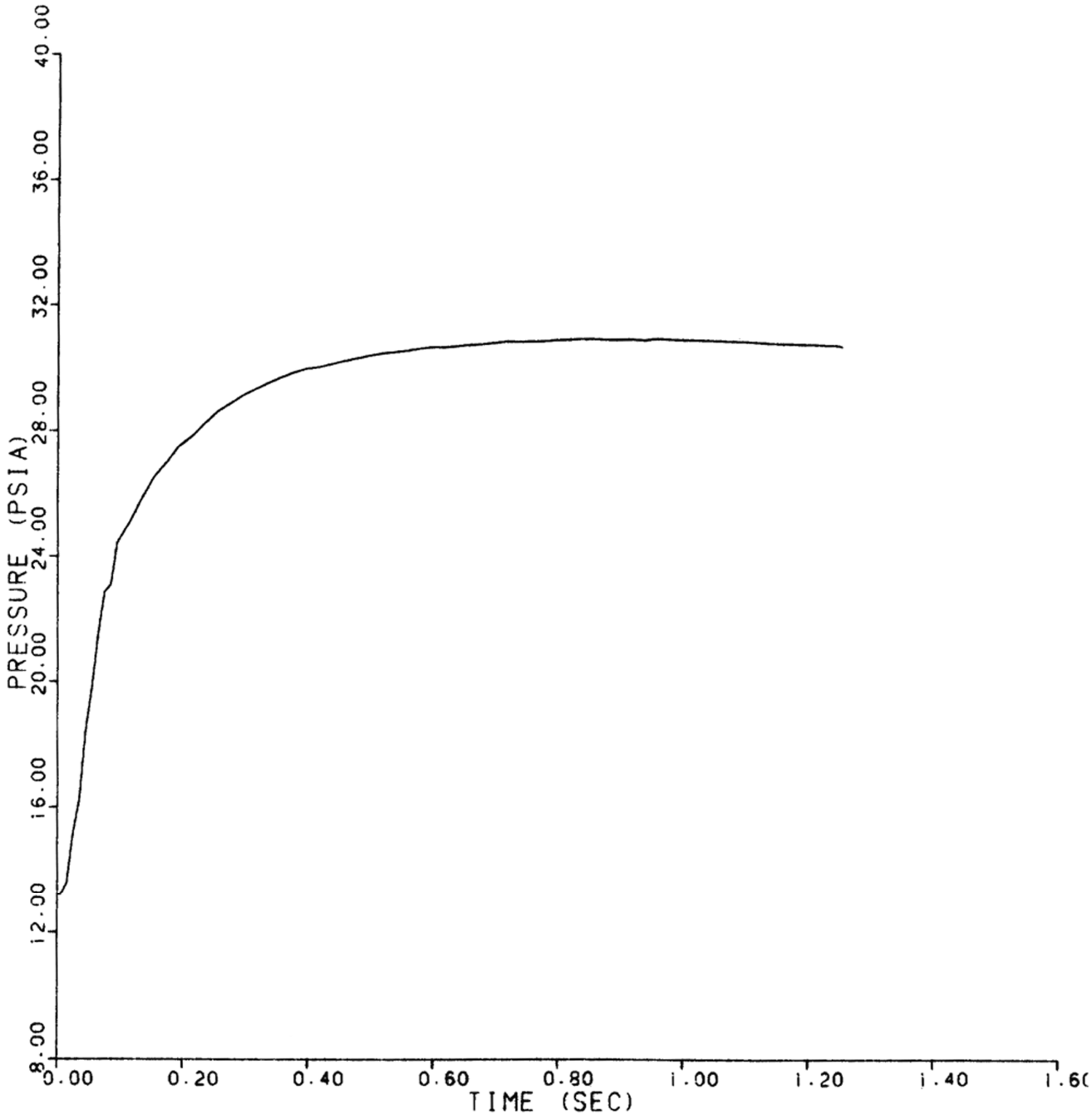
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E47 (763–in.^{2} BREAK AREA) FIGURE 6.2.1–22 (SHEET 47 OF 74)



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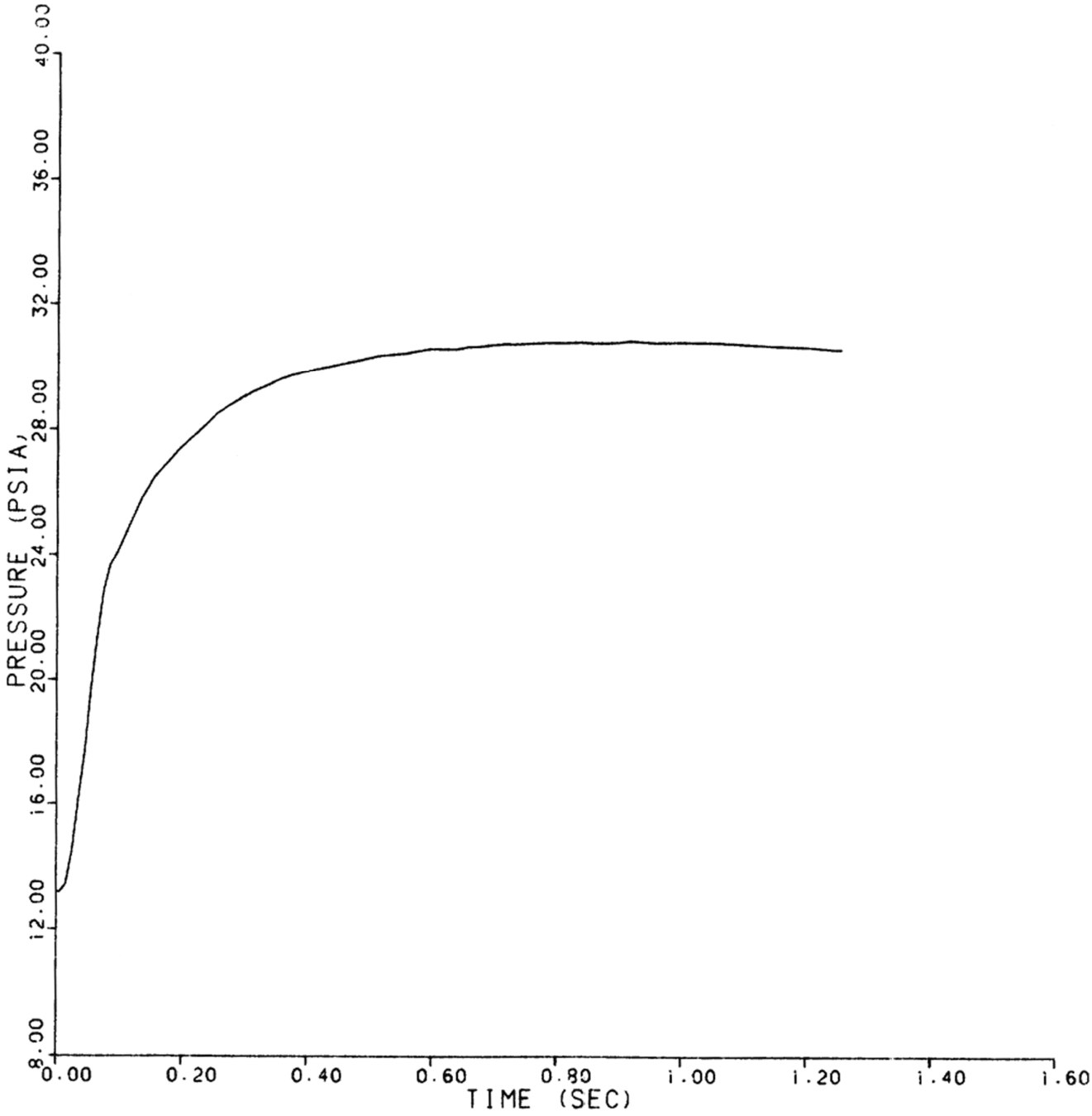
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E48
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 48 OF 74)



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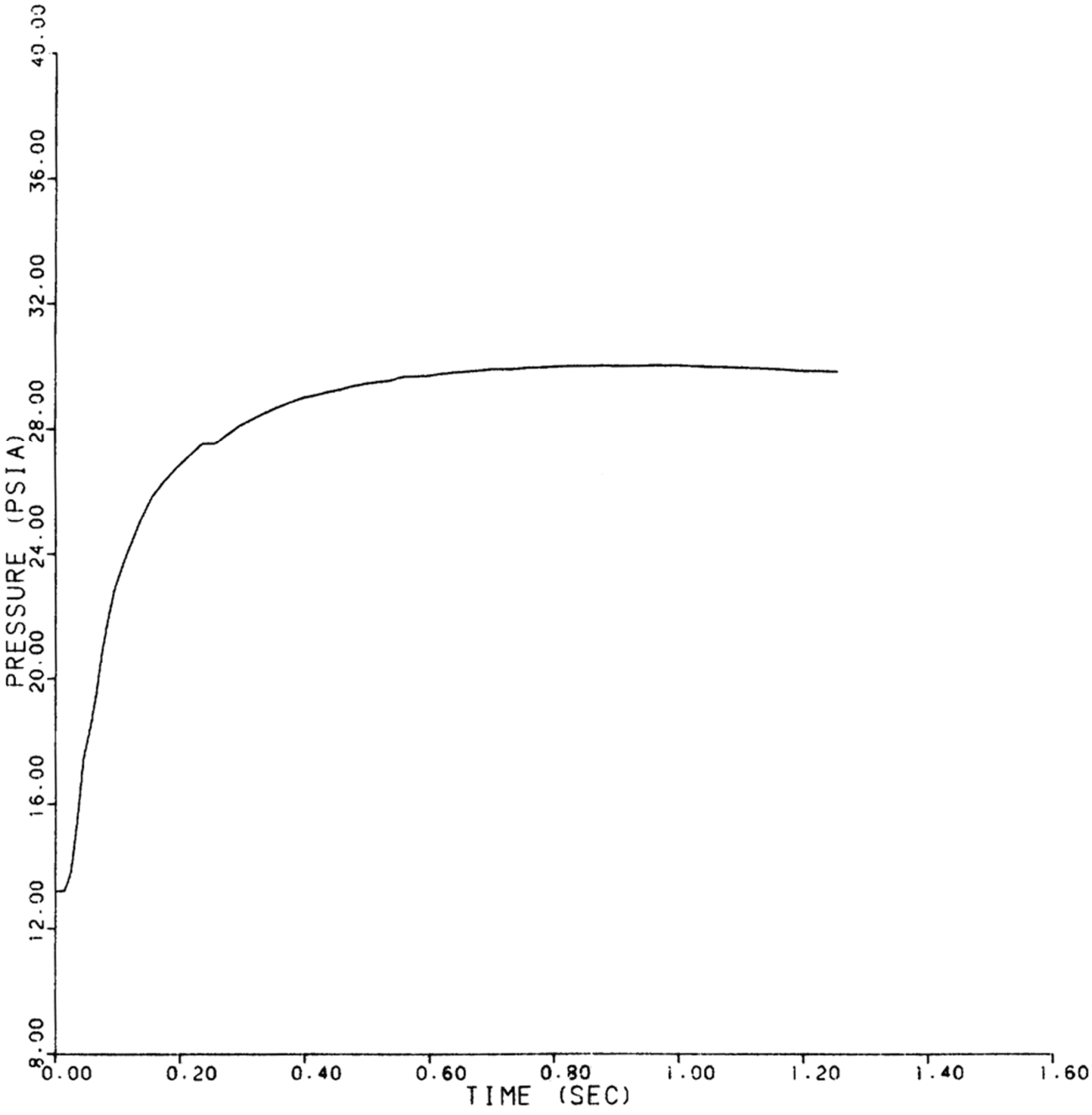
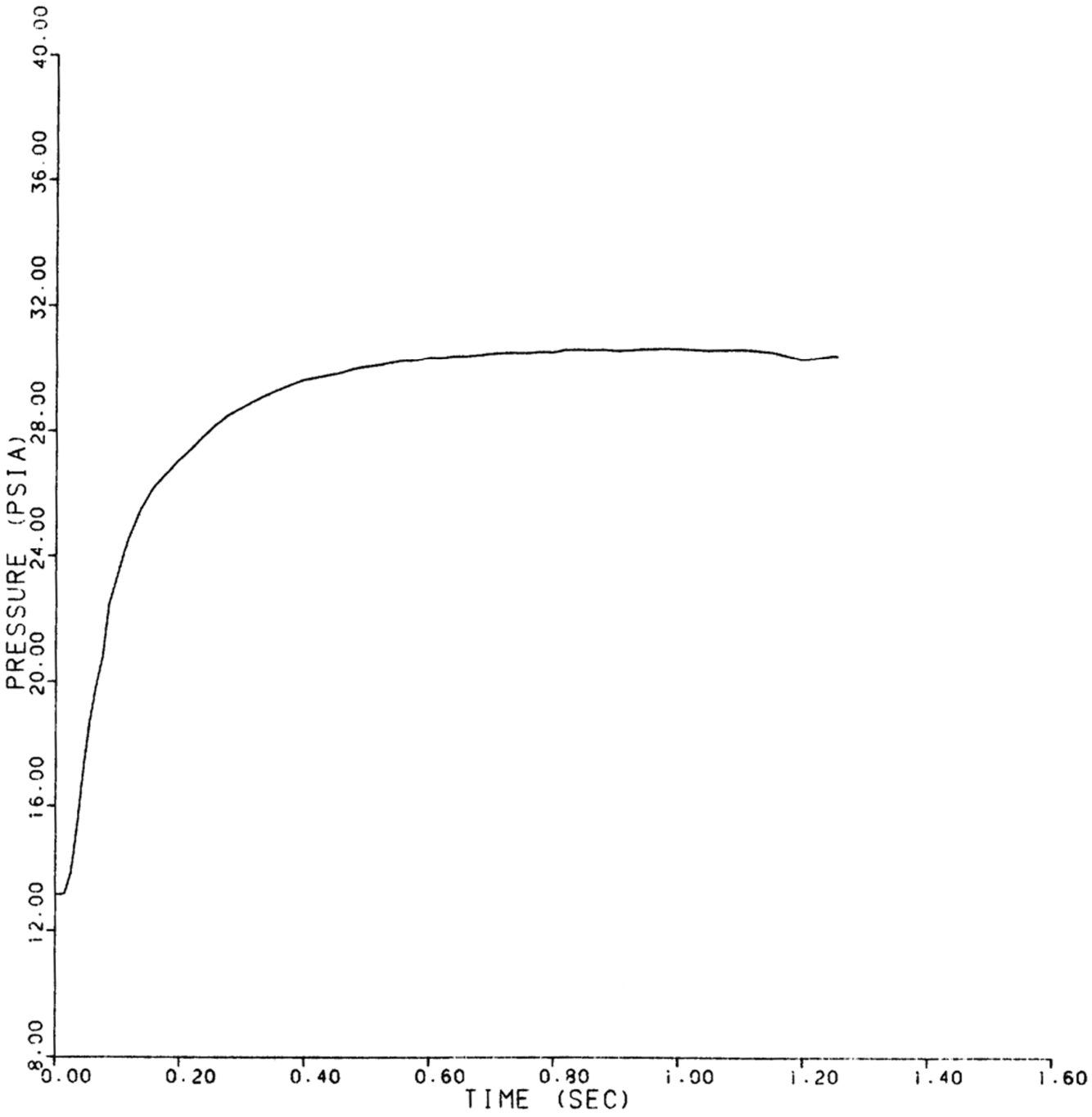
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E49 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 49 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E50 (763–in.^{2} BREAK AREA) FIGURE 6.2.1–22 (SHEET 50 OF 74)



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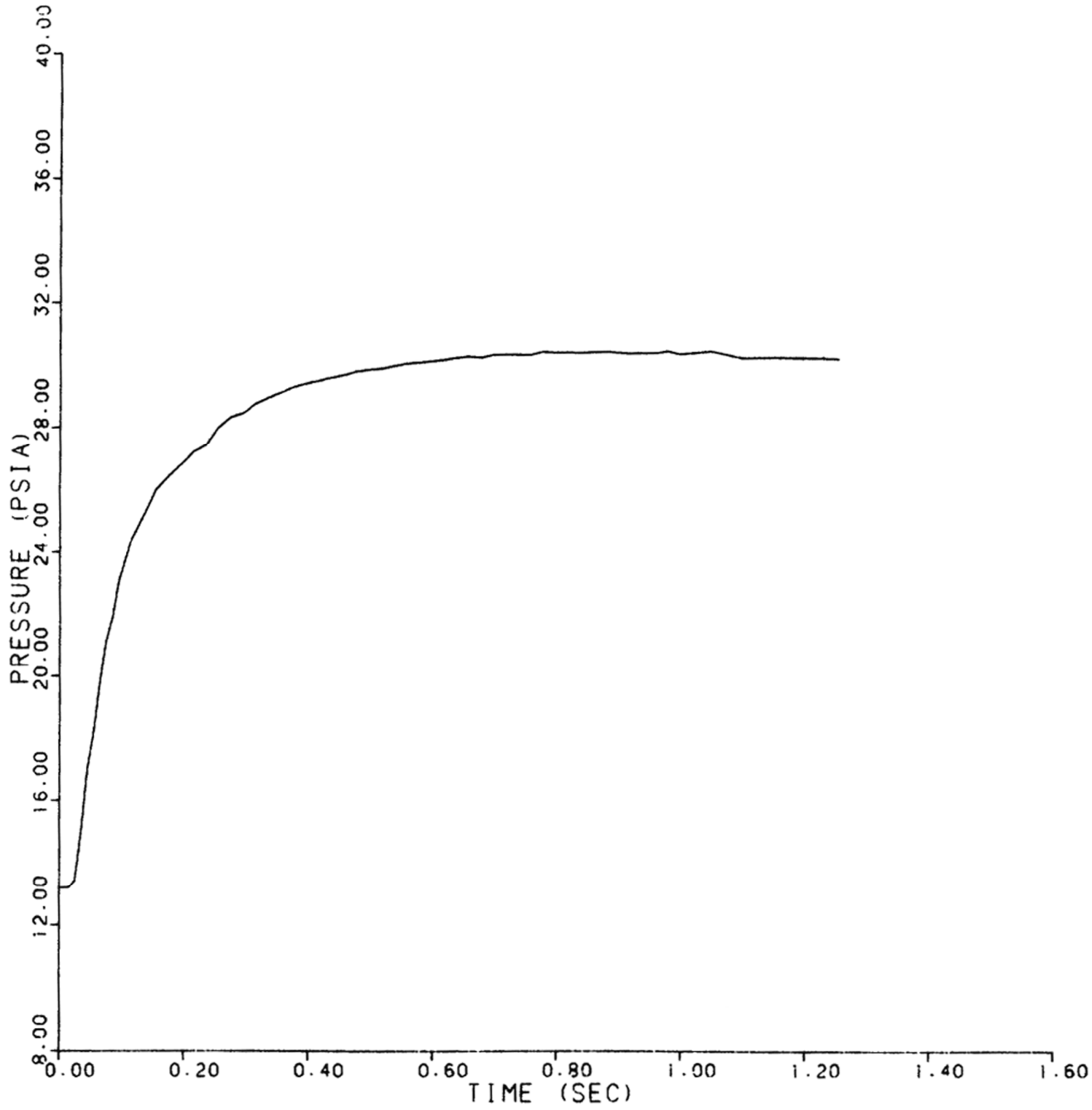
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

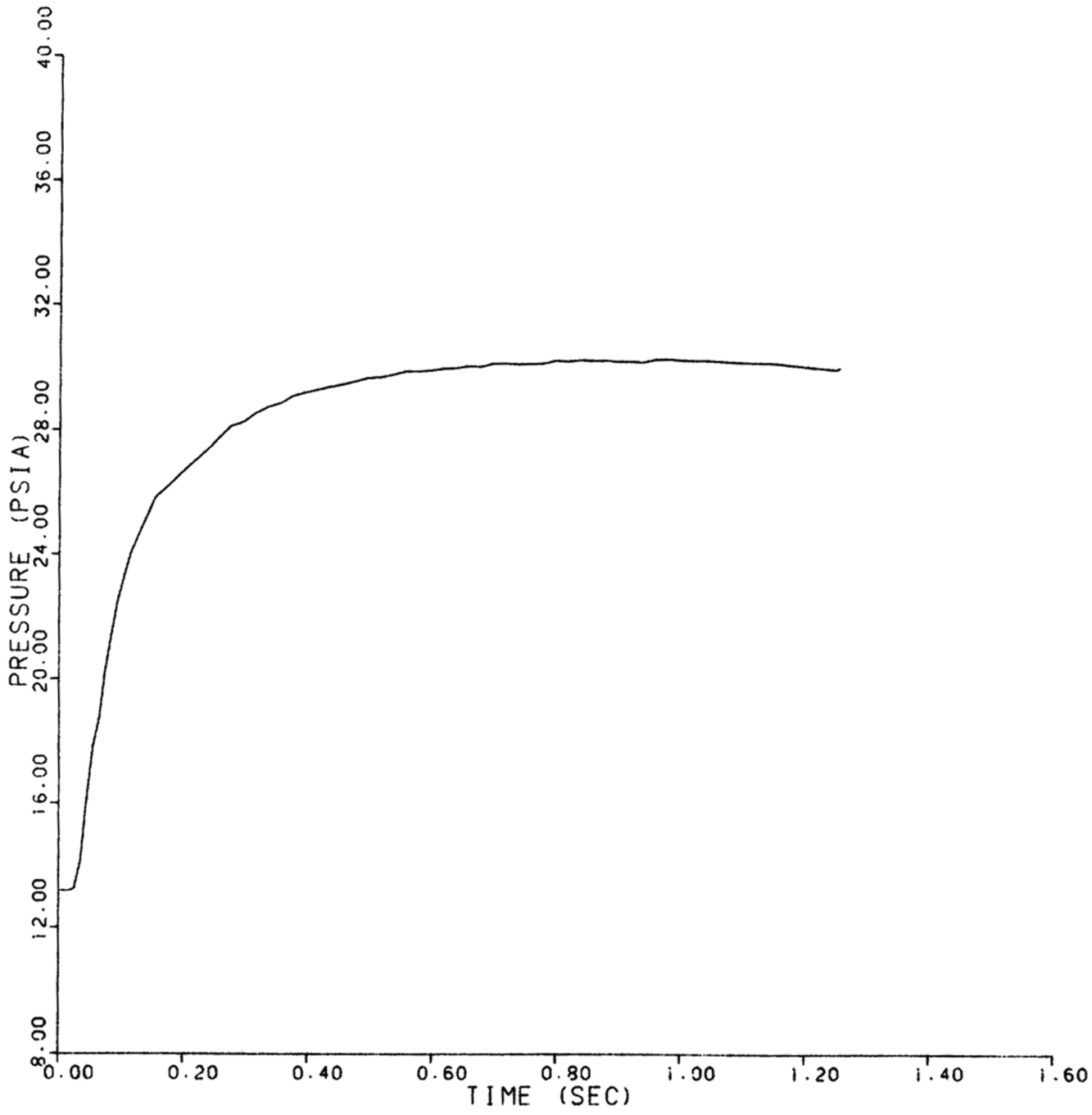
STEAM GENERATOR INLET ELBOW
NODE – E51
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 51 OF 74)



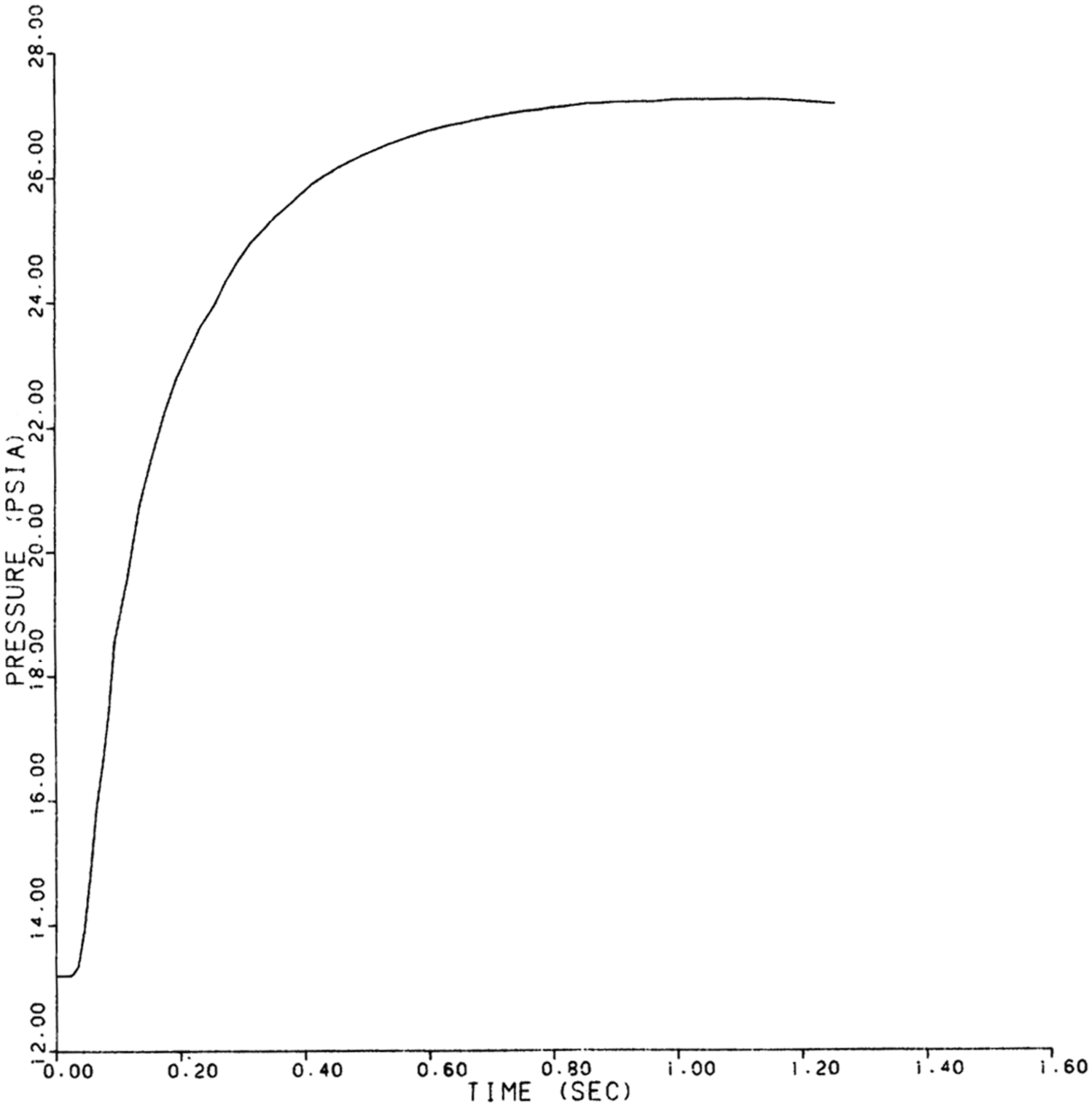
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E52 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 52 OF 74)



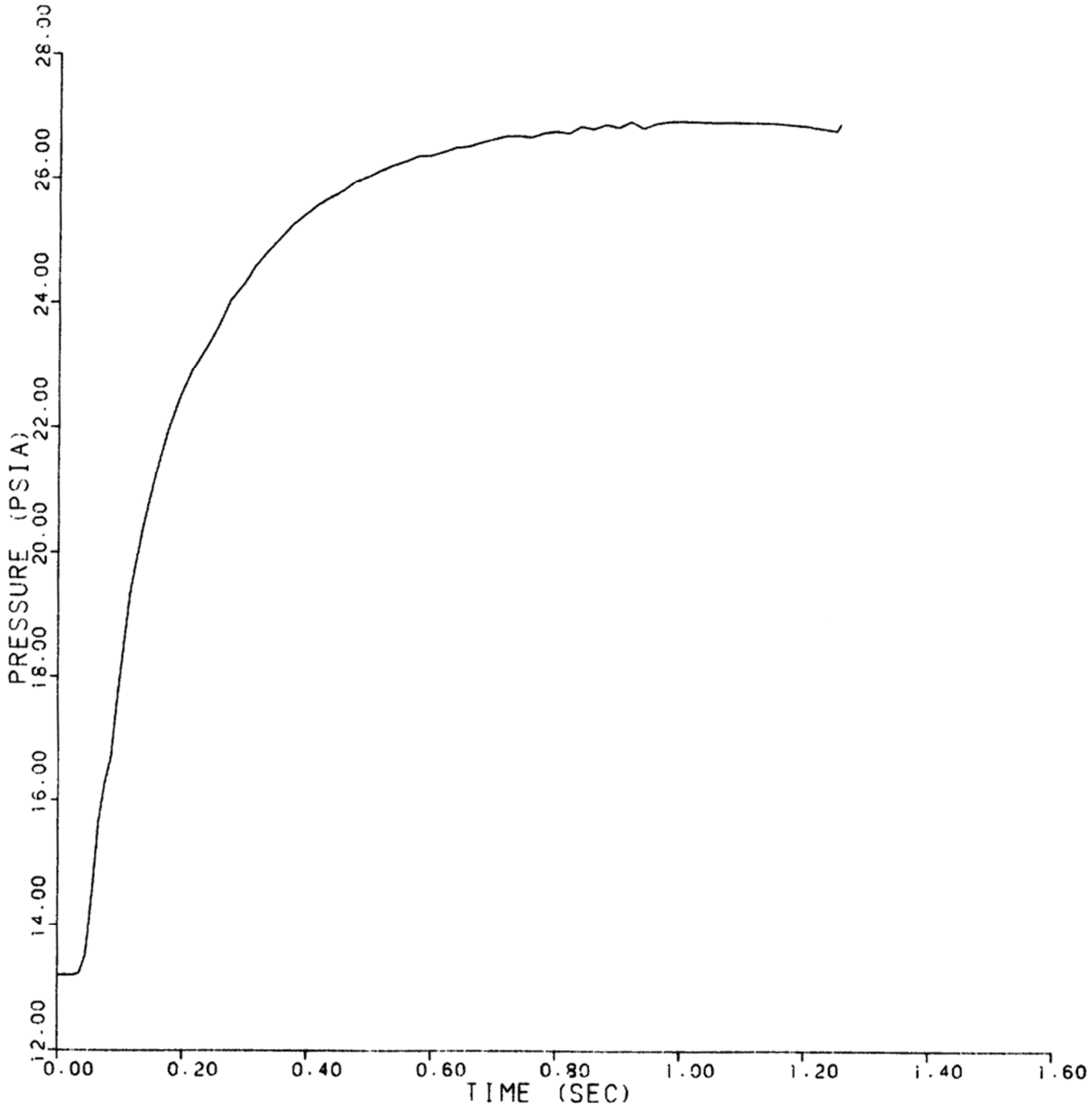
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E53 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 53 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E54 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 54 OF 74)



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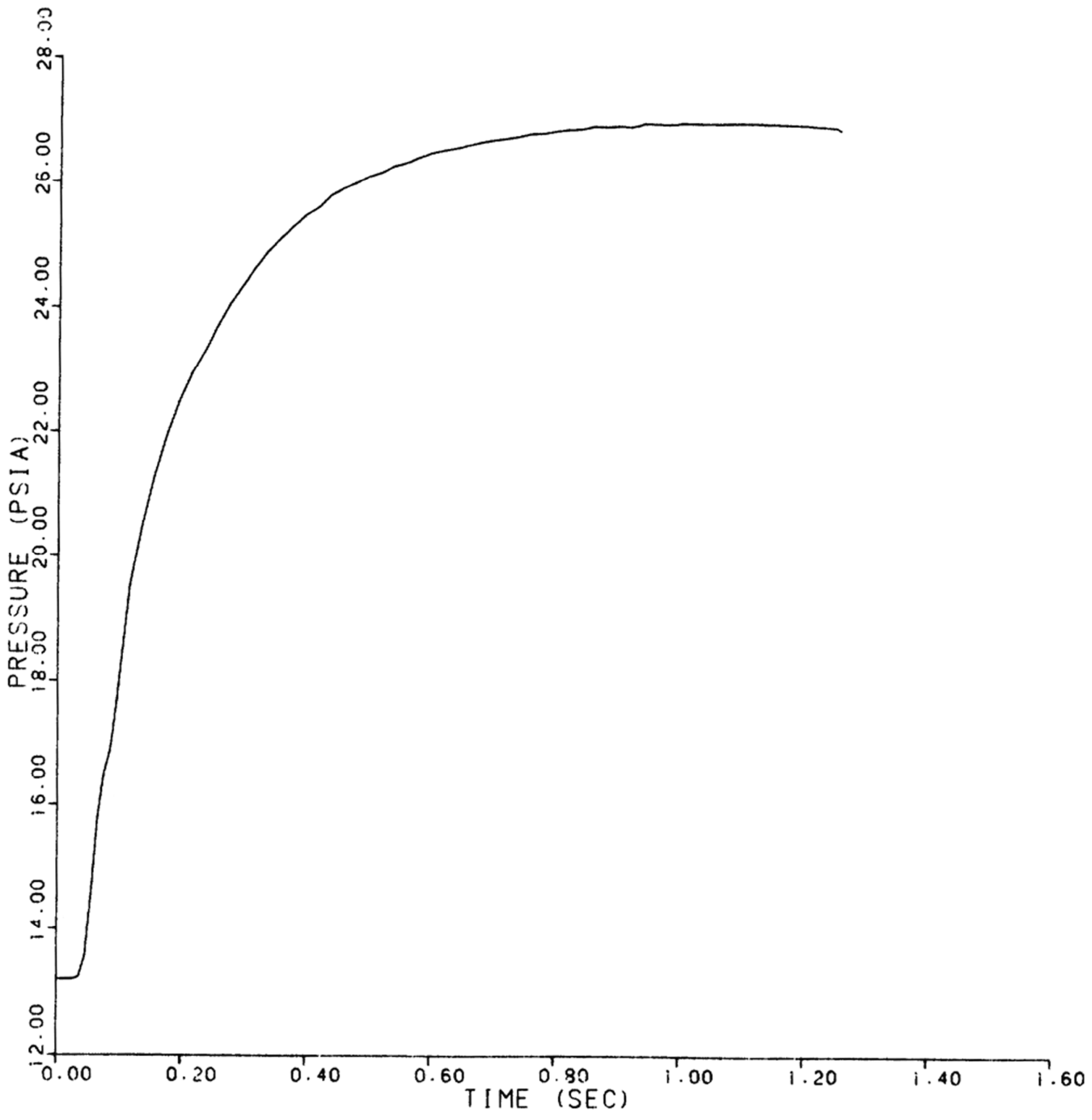
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

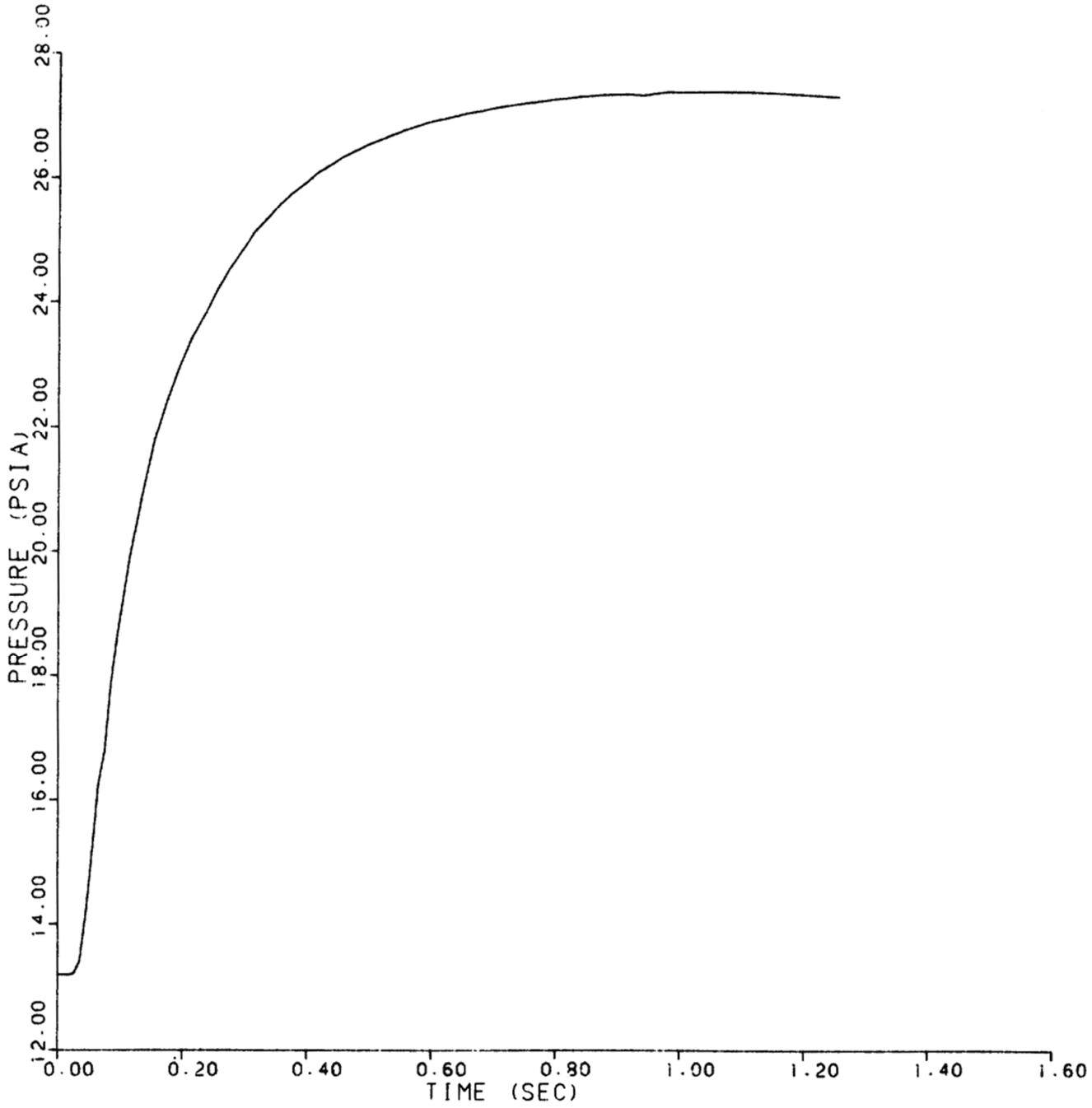
STEAM GENERATOR INLET ELBOW
NODE – E55
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 55 OF 74)



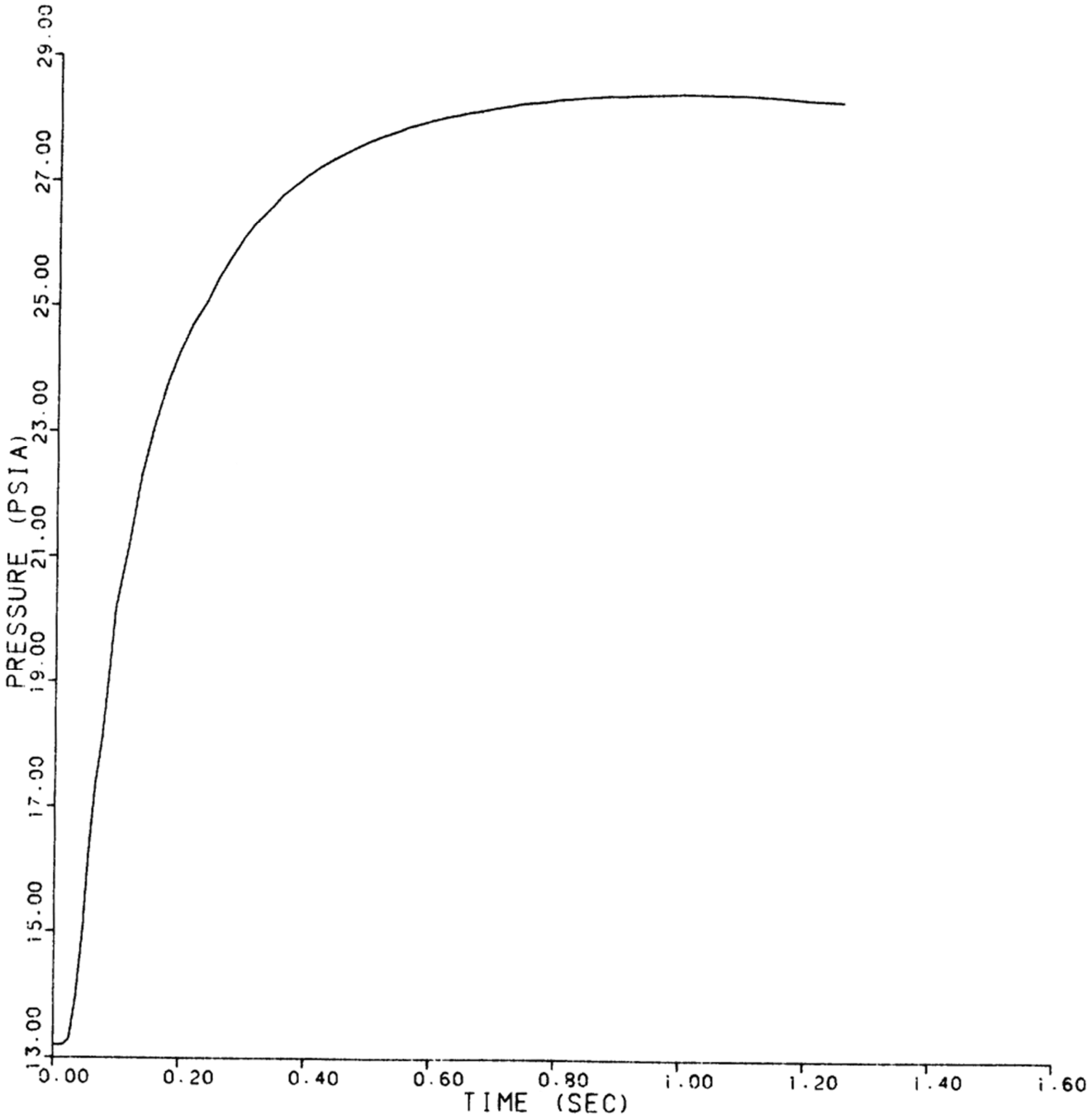
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E56 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 56 OF 74)



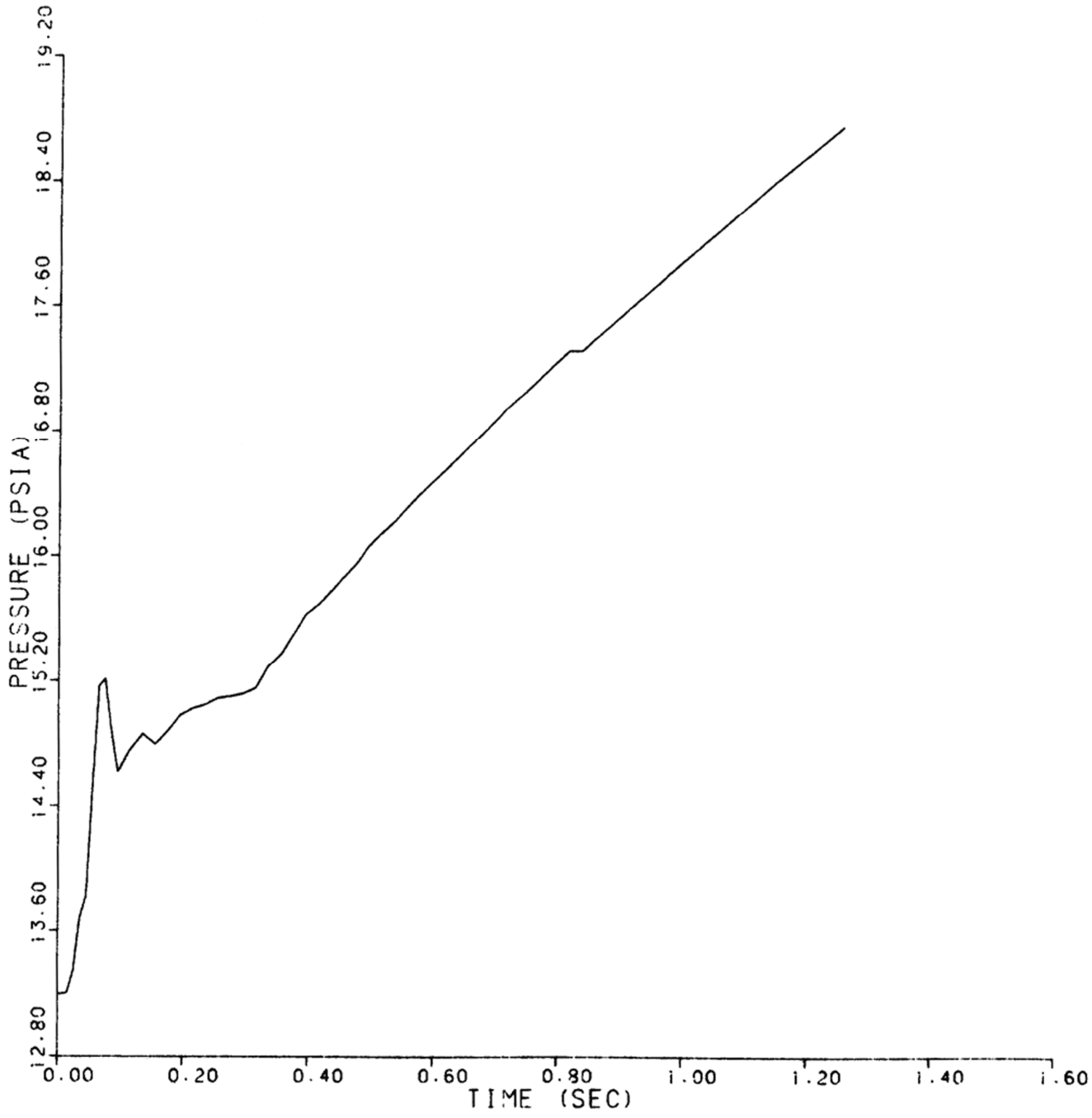
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E57 (763–in.^{2} BREAK AREA) FIGURE 6.2.1–22 (SHEET 57 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E58 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 58 OF 74)



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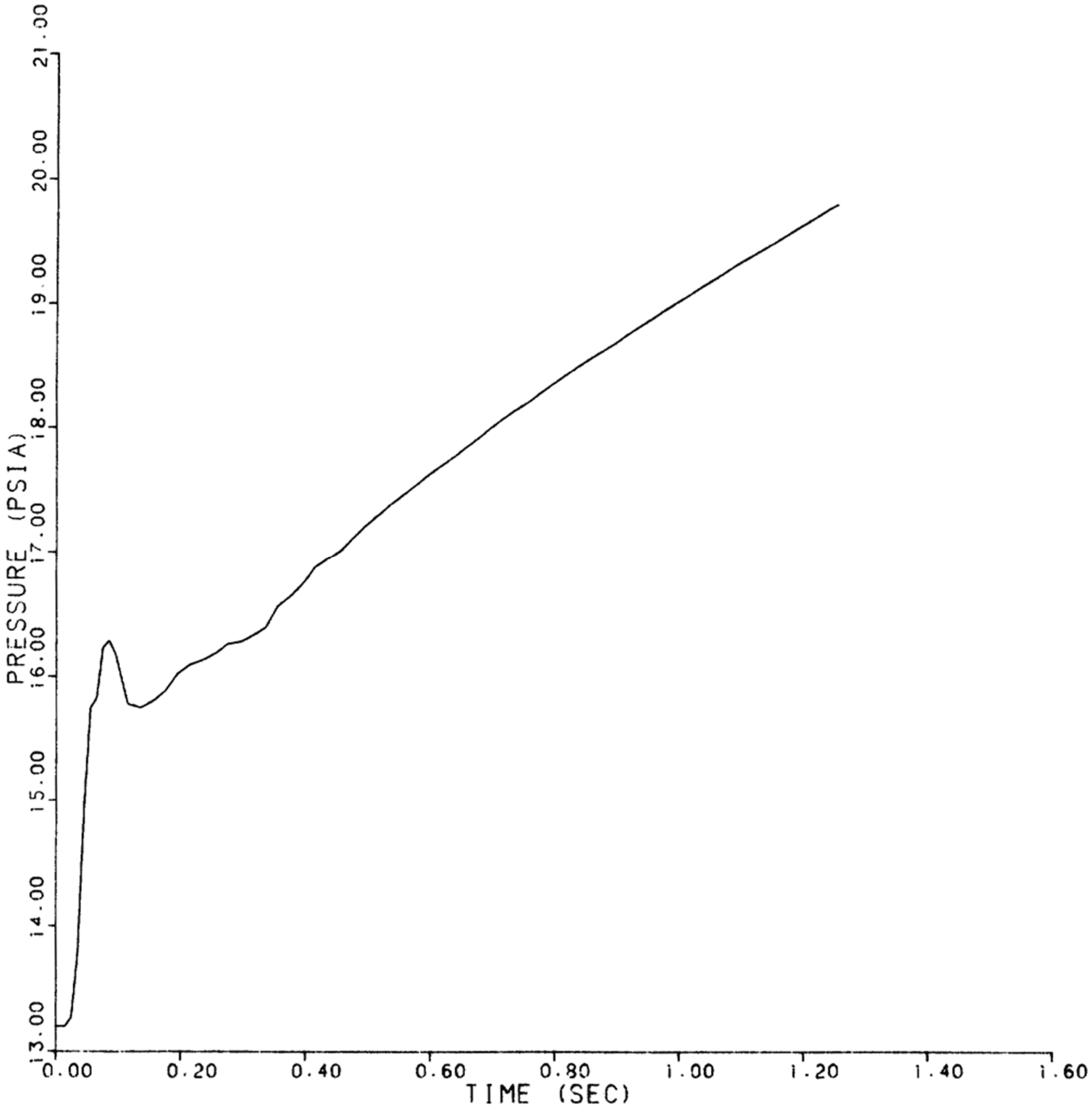
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E59 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 59 OF 74)



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SOUTHERN COMPANY
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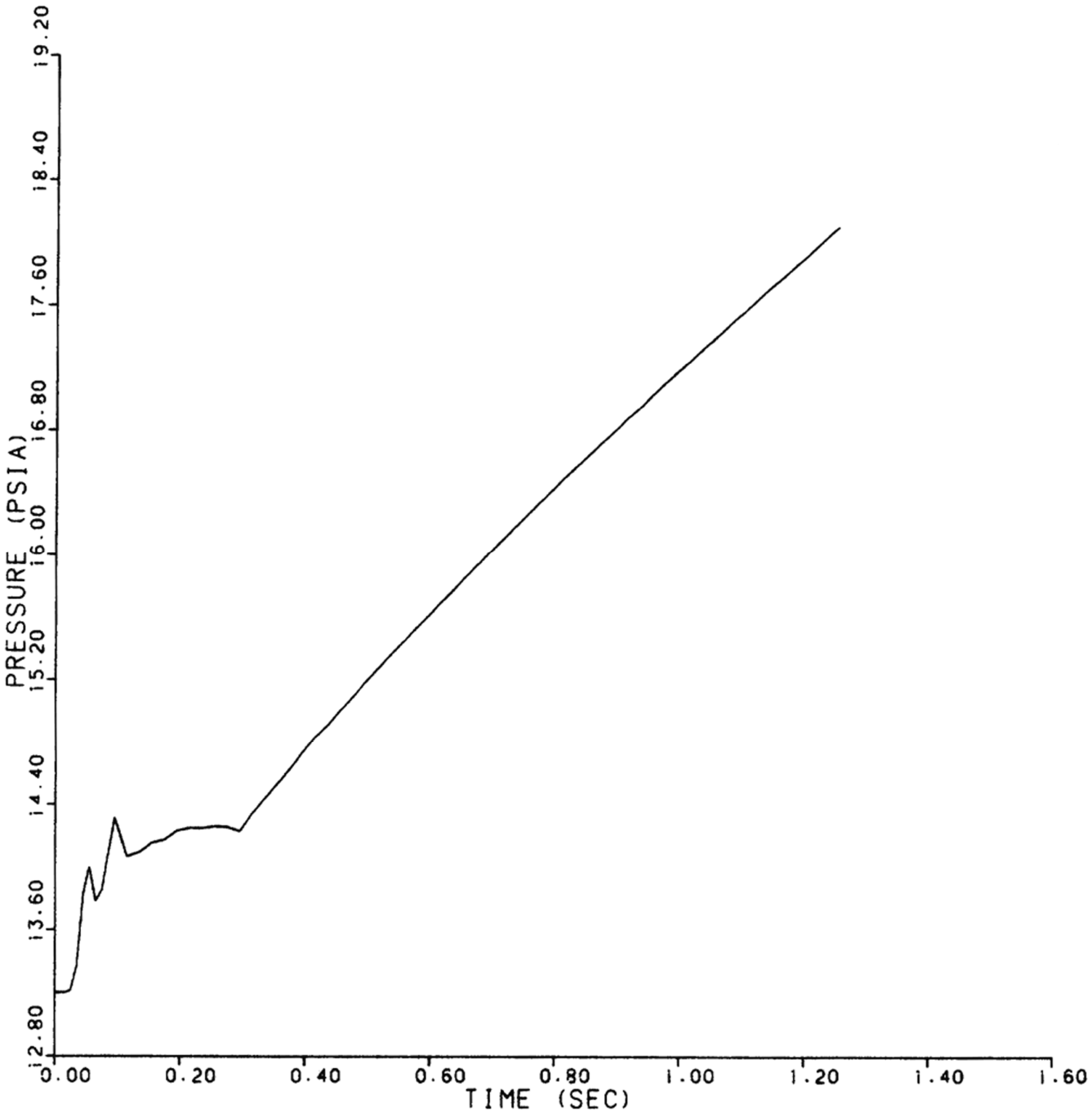
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E60
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 60 OF 74)



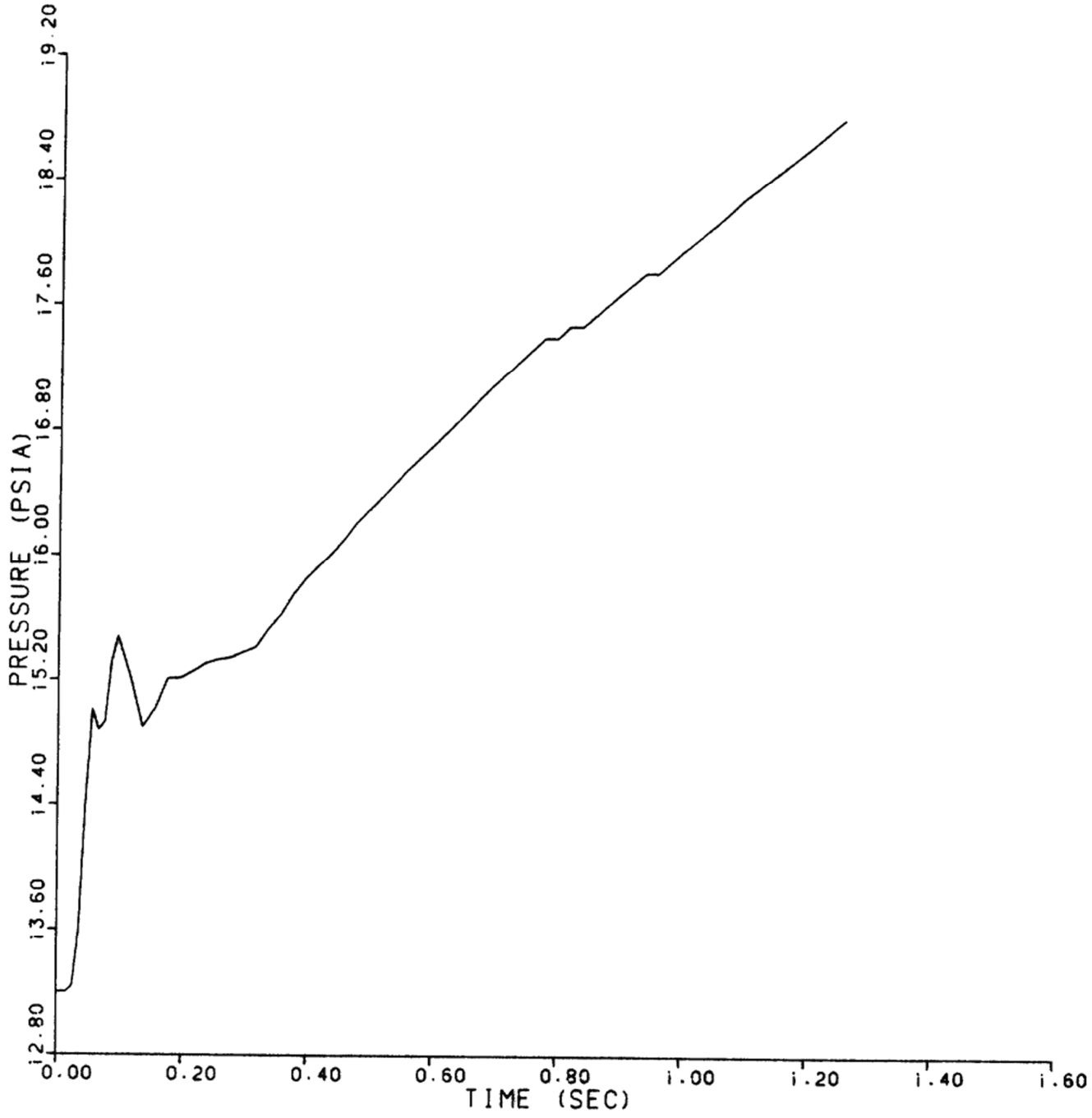
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E61
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 61 OF 74)



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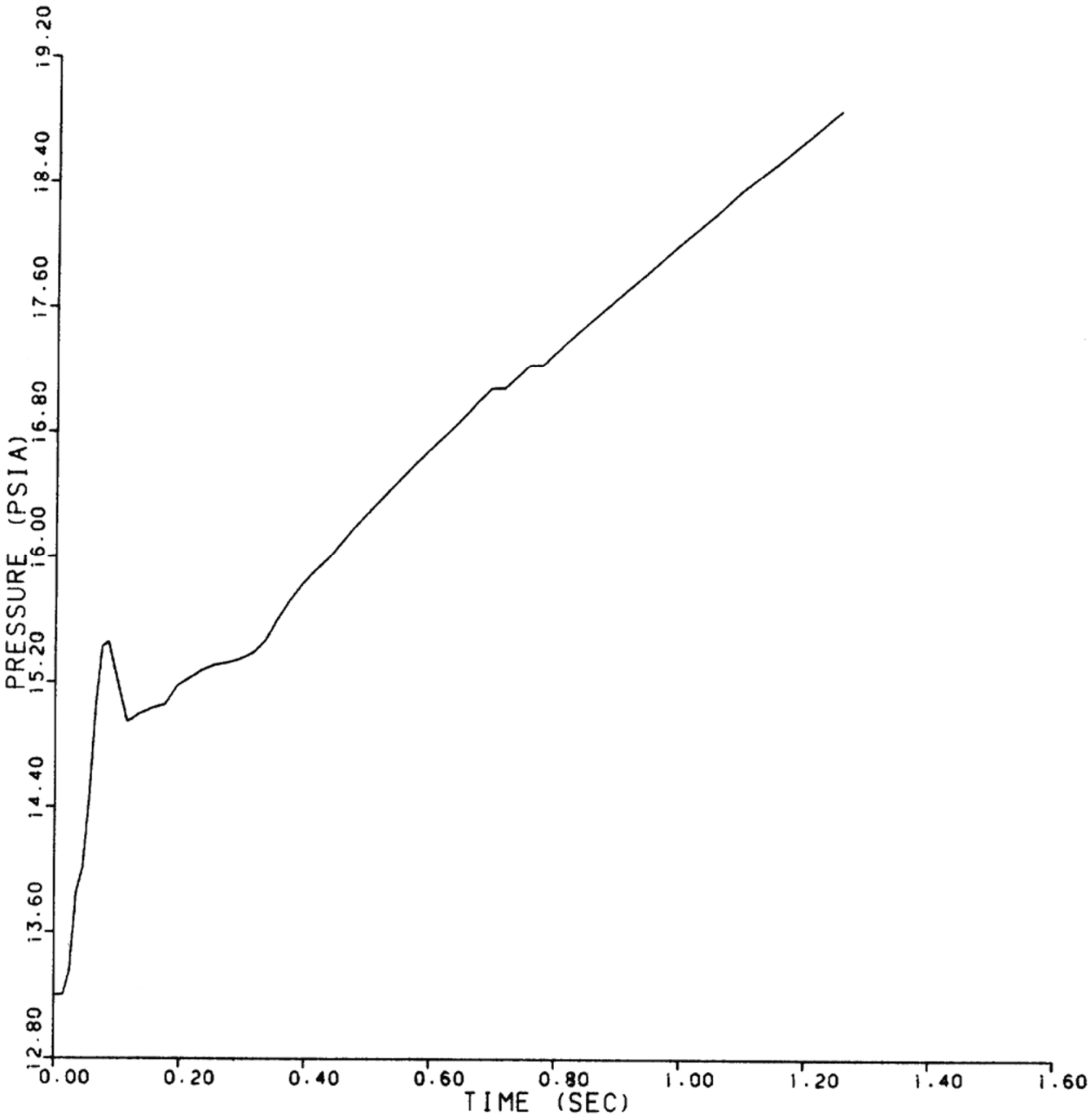
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E62 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 62 OF 74)



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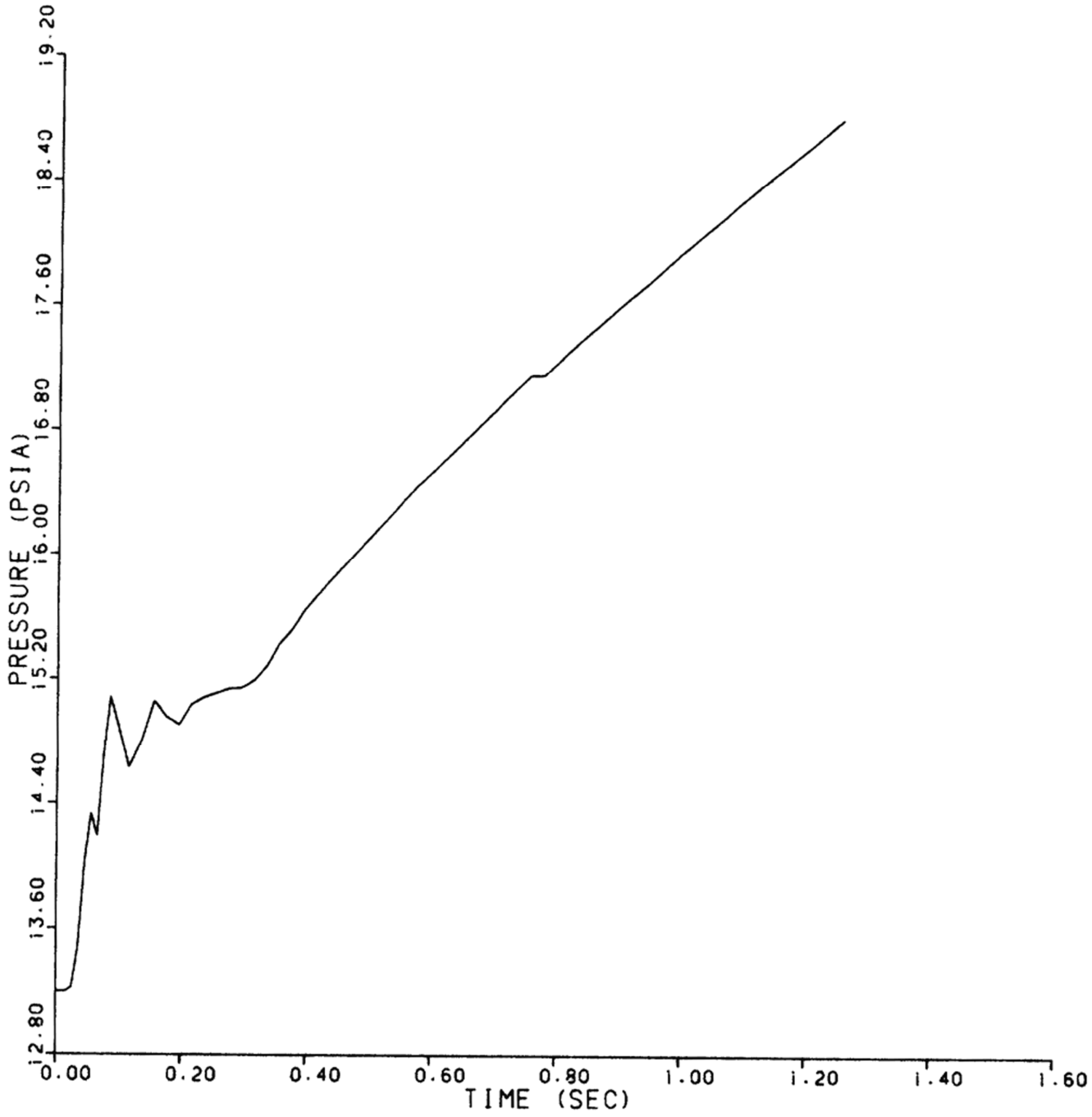
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E63
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 63 OF 74)



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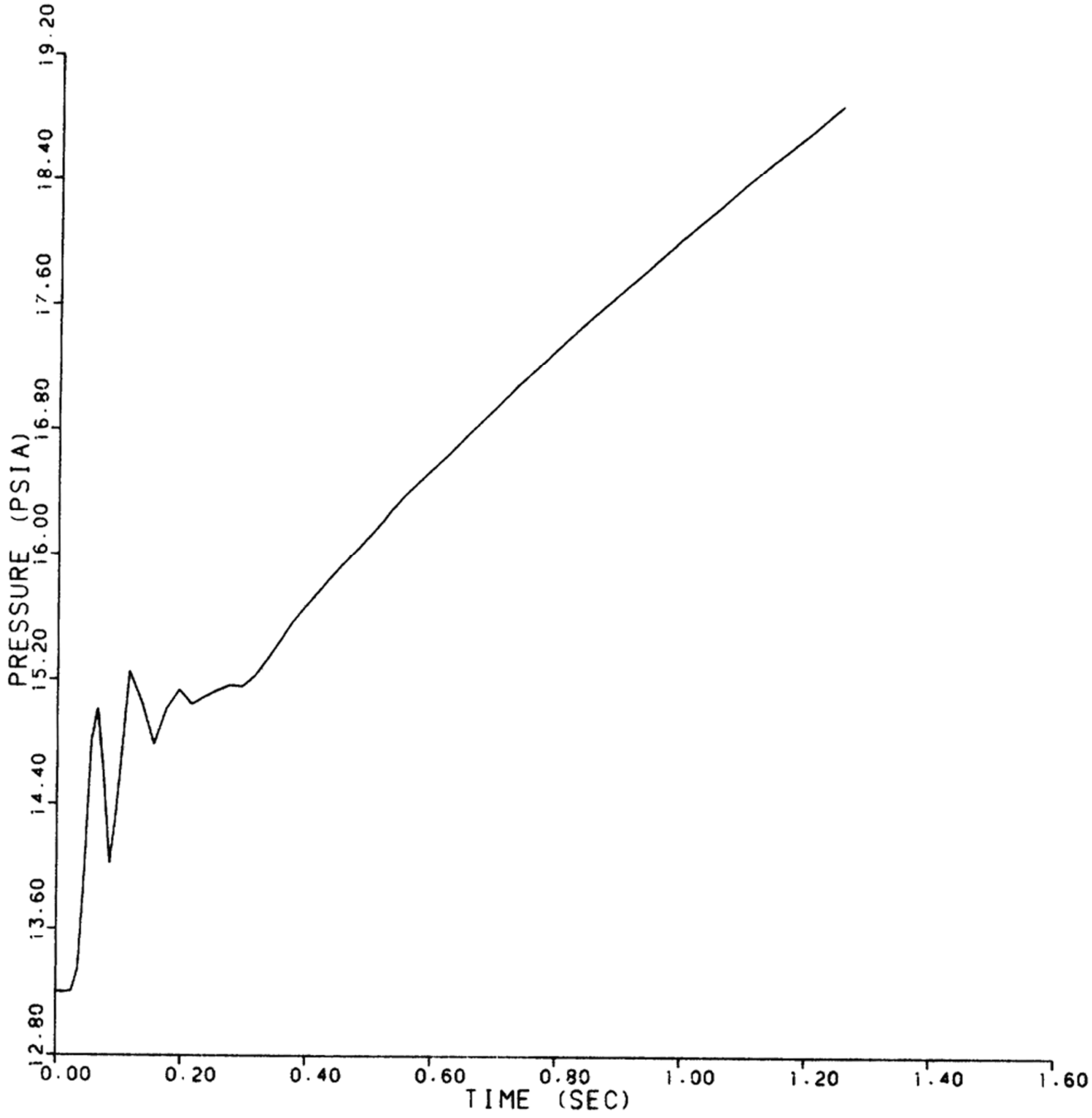
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E64
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 64 OF 74)



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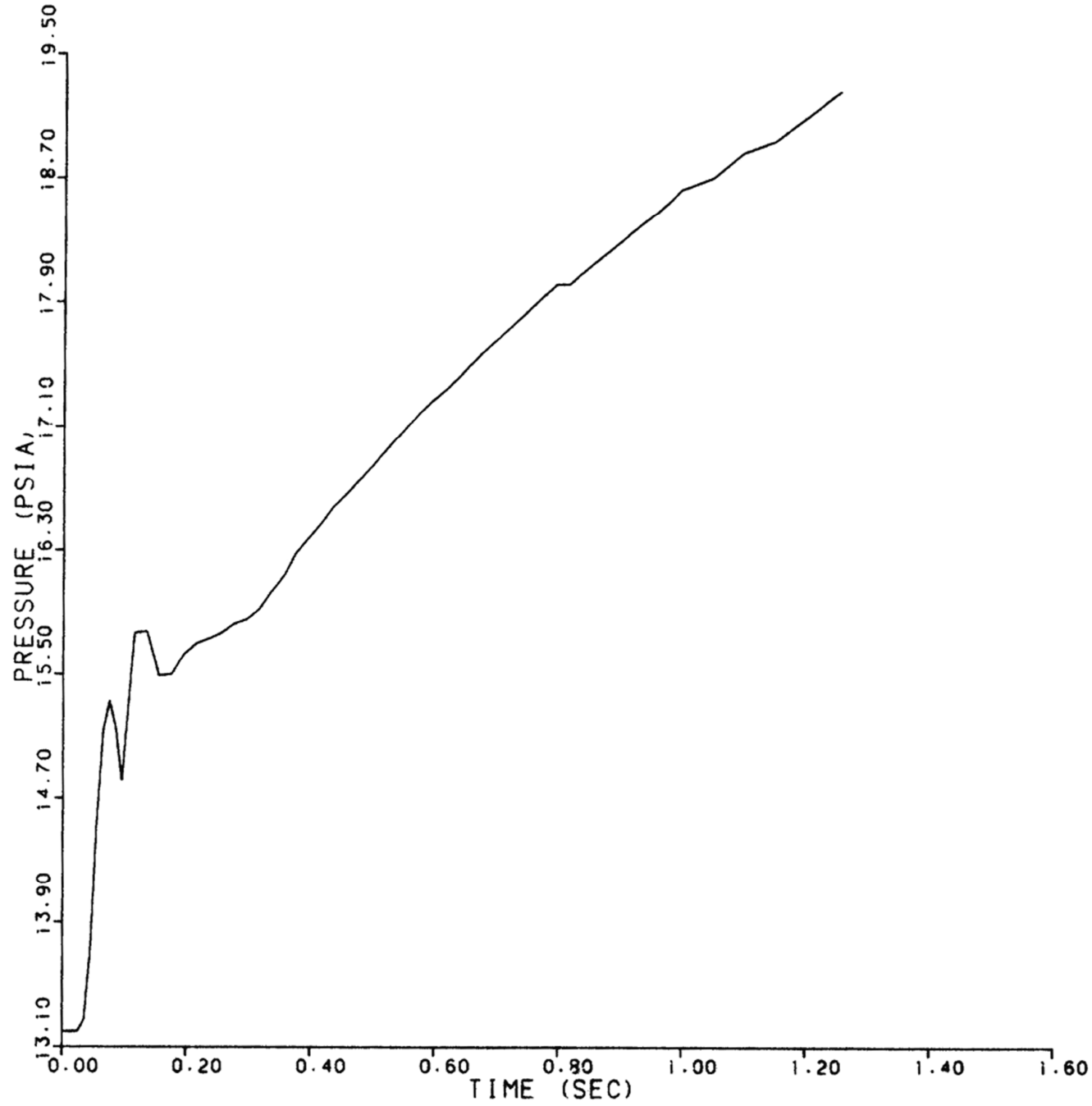
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E65 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 65 OF 74)



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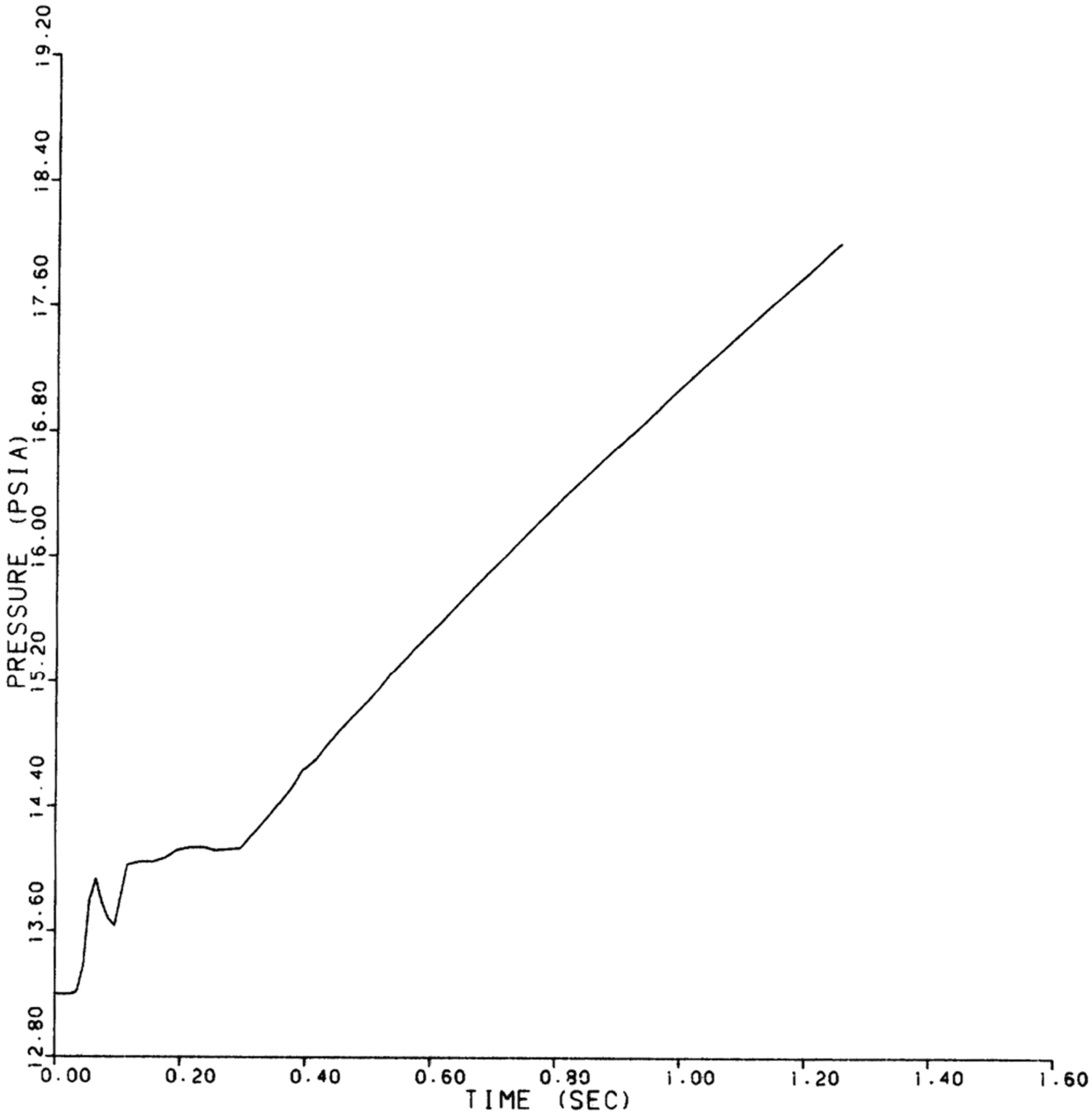
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E66 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 66 OF 74)



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SOUTHERN COMPANY
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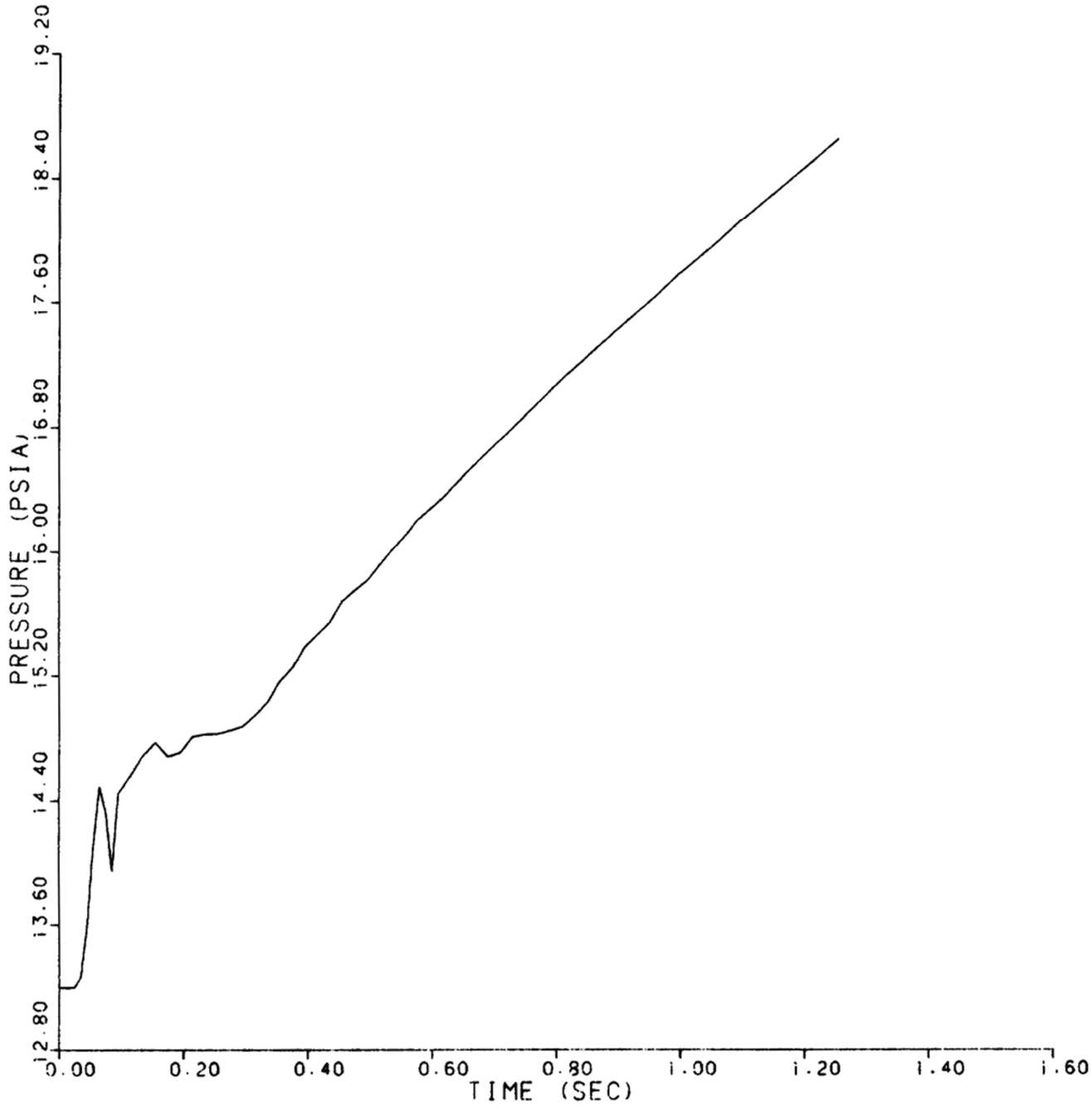
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E67
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 67 OF 74)



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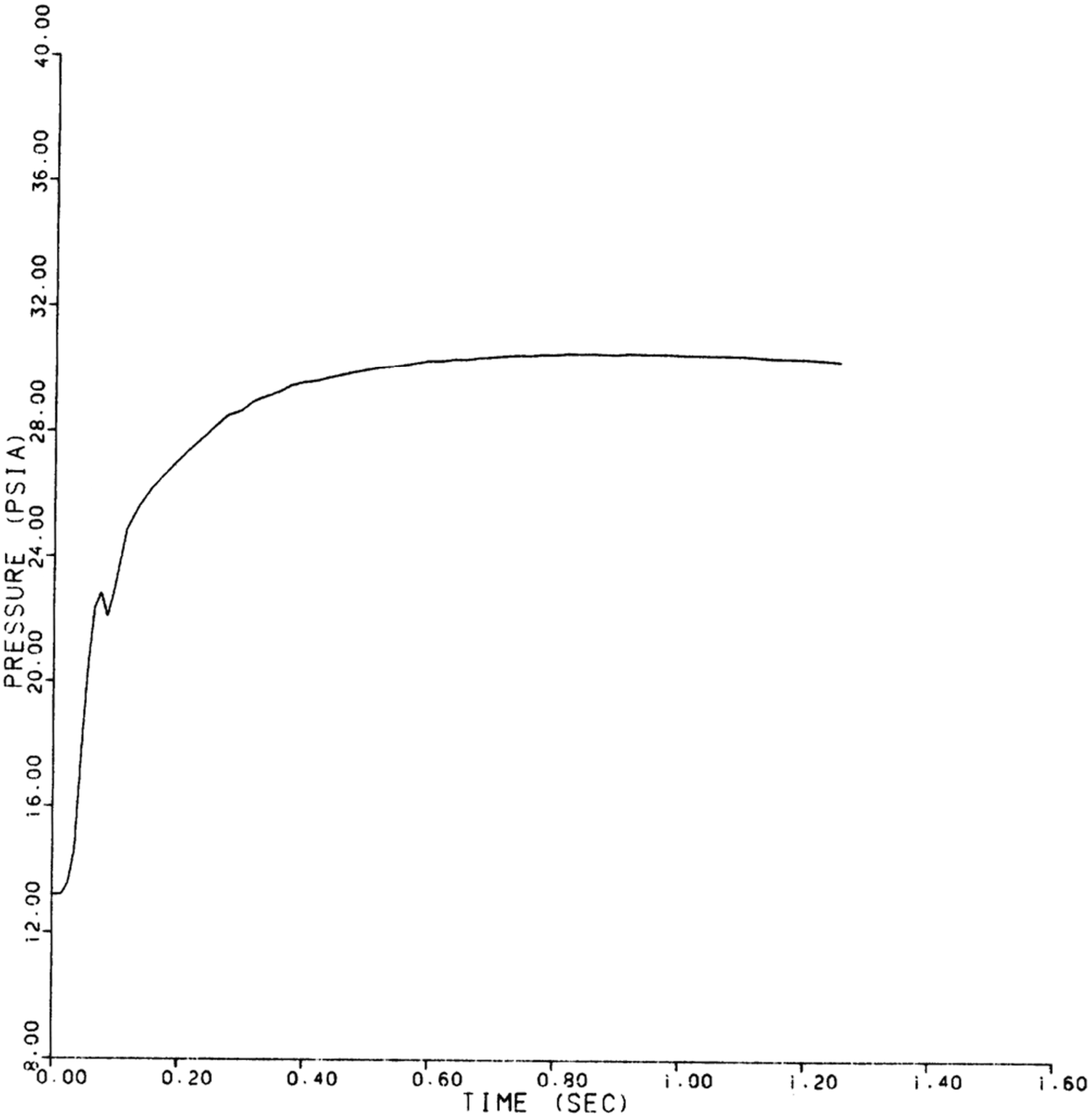
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E68
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 68 OF 74)



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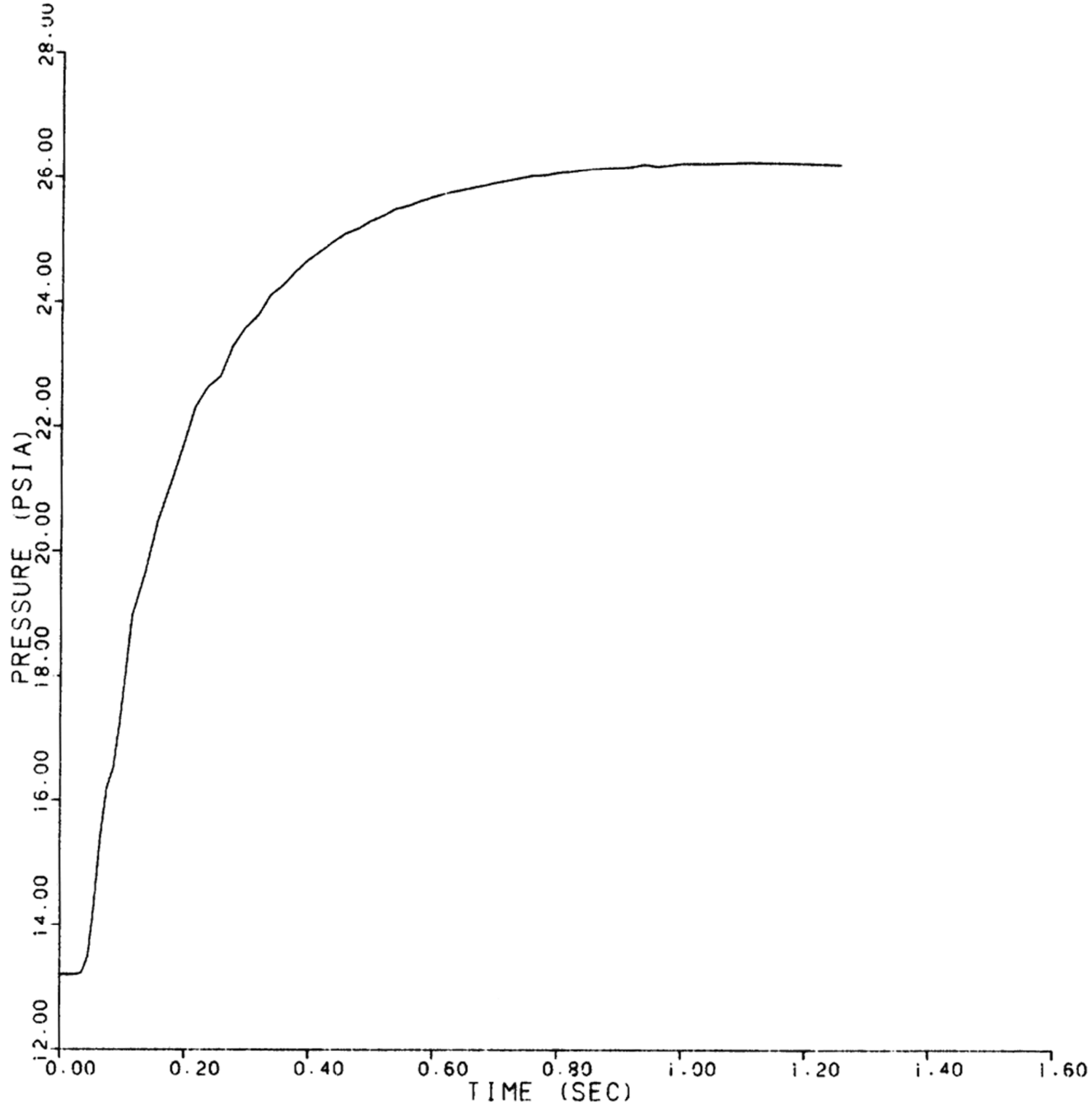
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

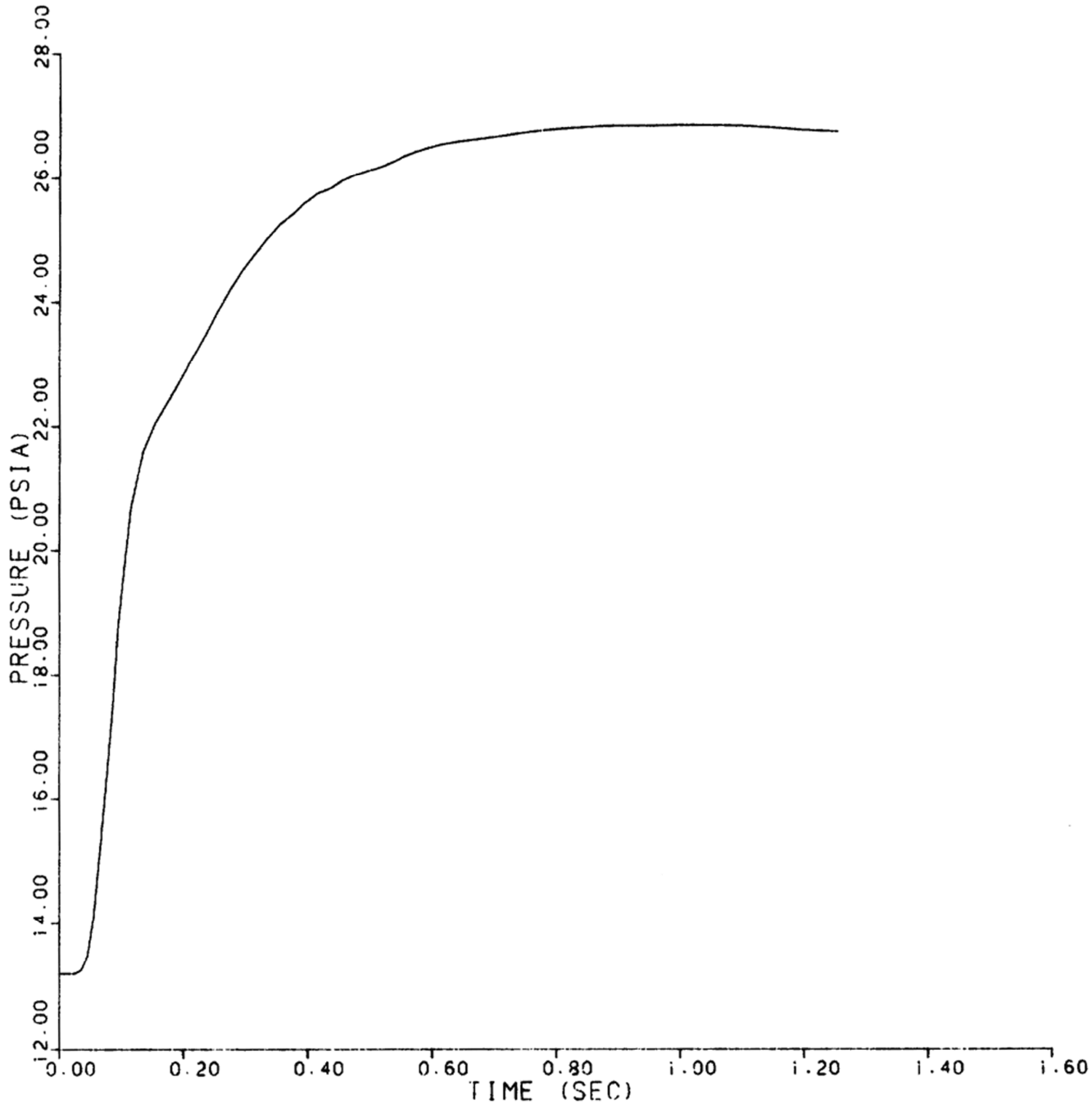
STEAM GENERATOR INLET ELBOW
NODE – E69
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 69 OF 74)



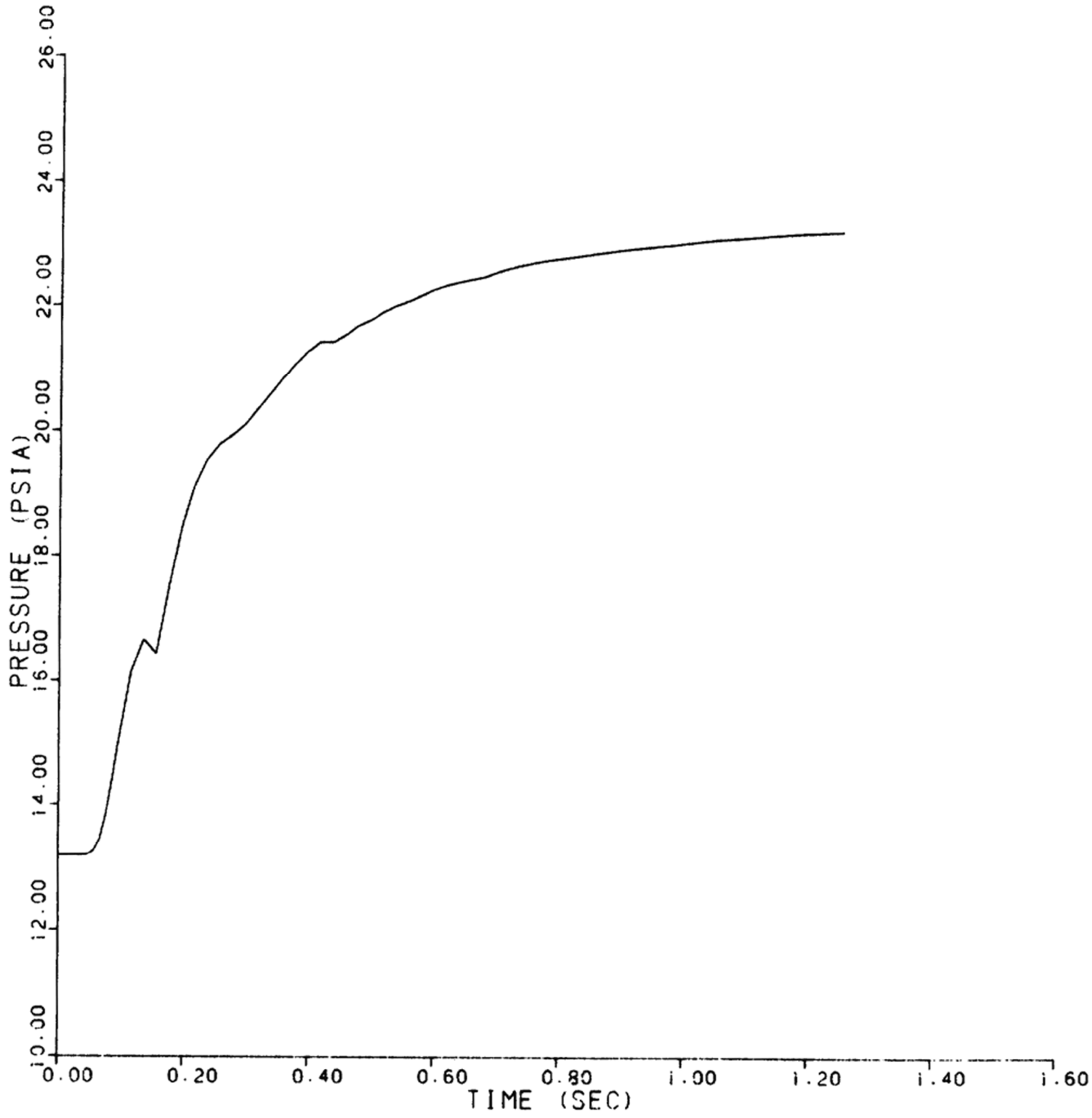
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E70 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 70 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E71 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 71 OF 74)



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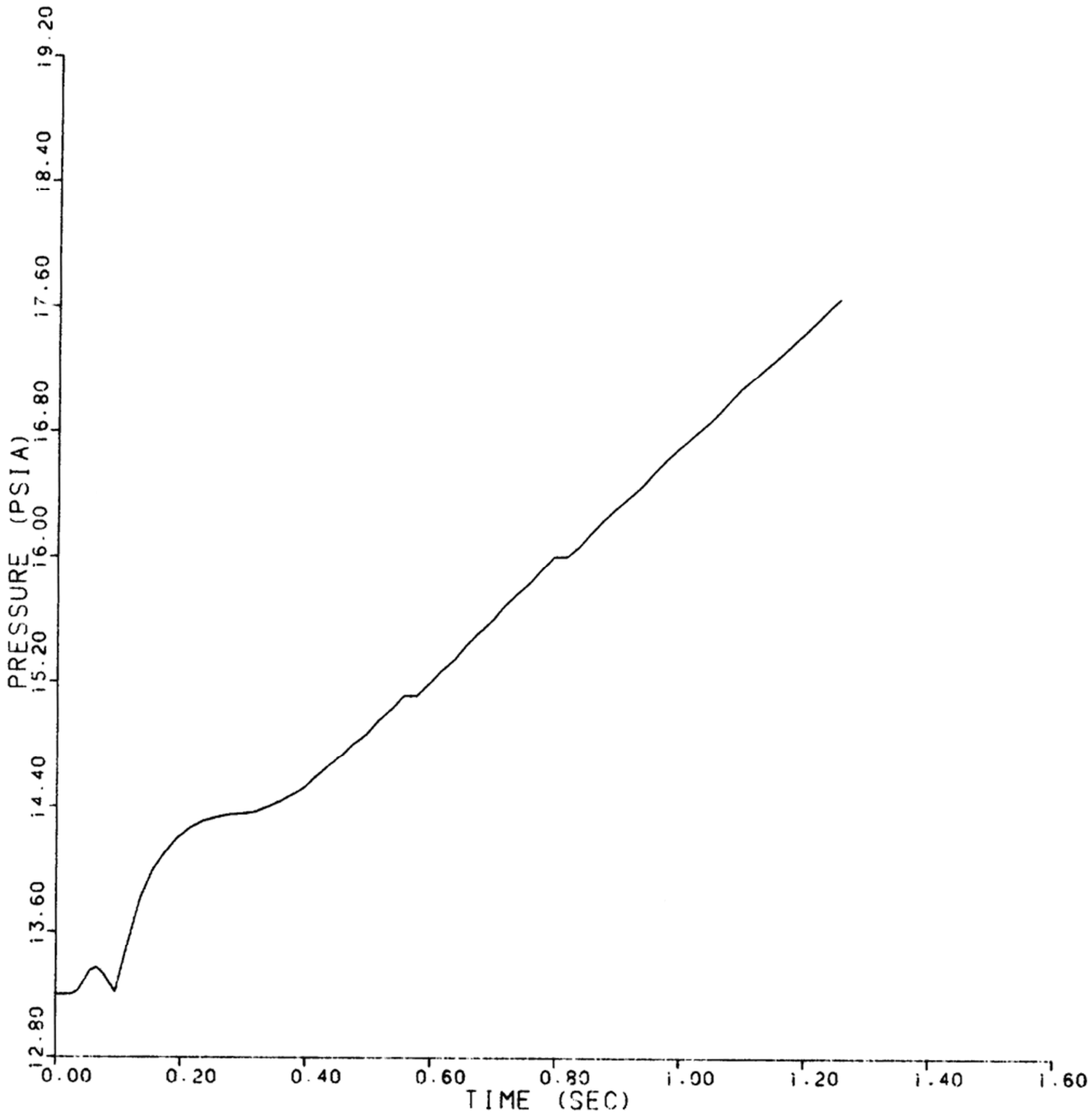
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

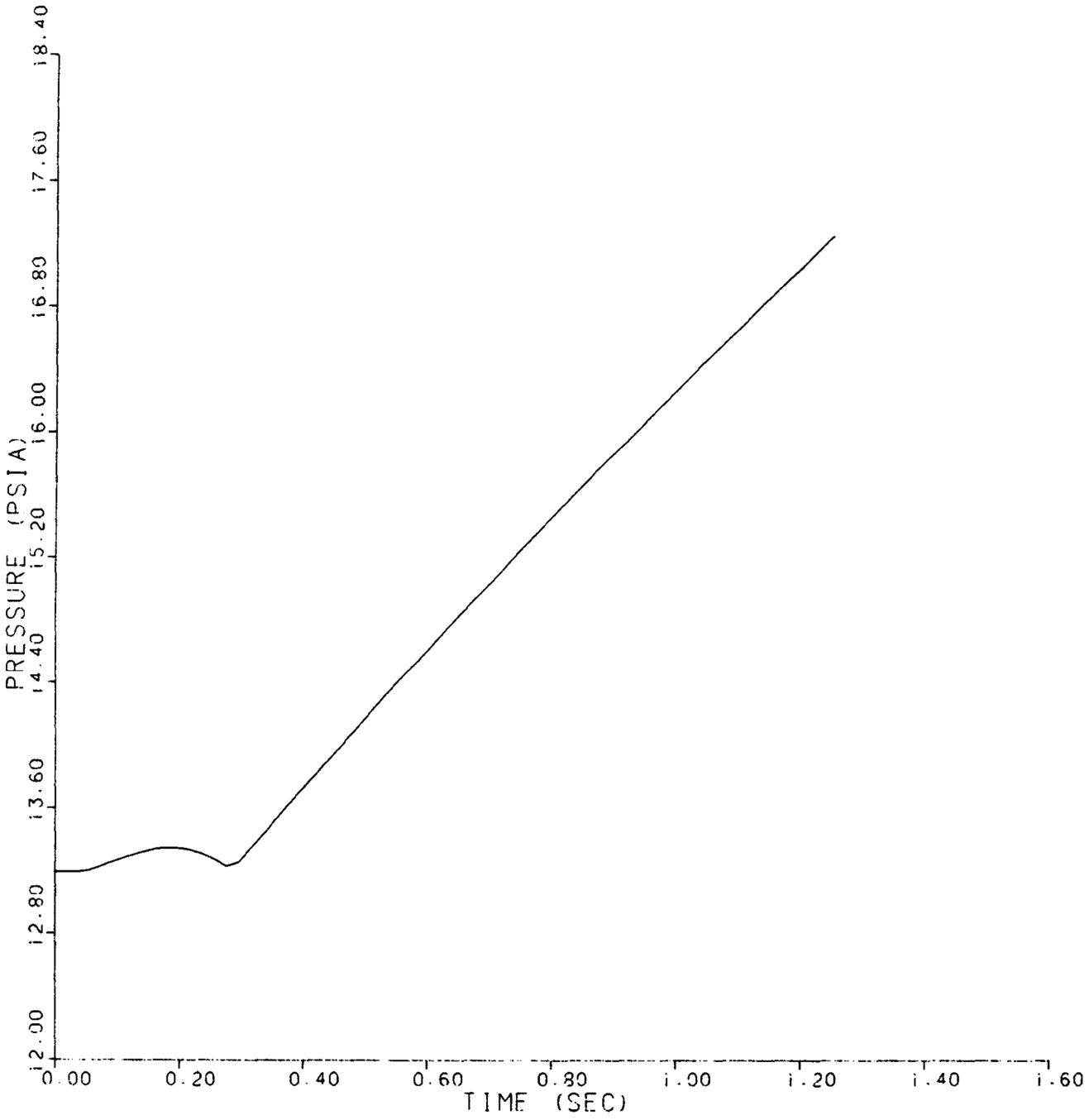
STEAM GENERATOR INLET ELBOW
NODE – E72
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 72 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	STEAM GENERATOR INLET ELBOW NODE – E73 (763–in.^{2} BREAK AREA)
	FIGURE 6.2.1–22 (SHEET 73 OF 74)



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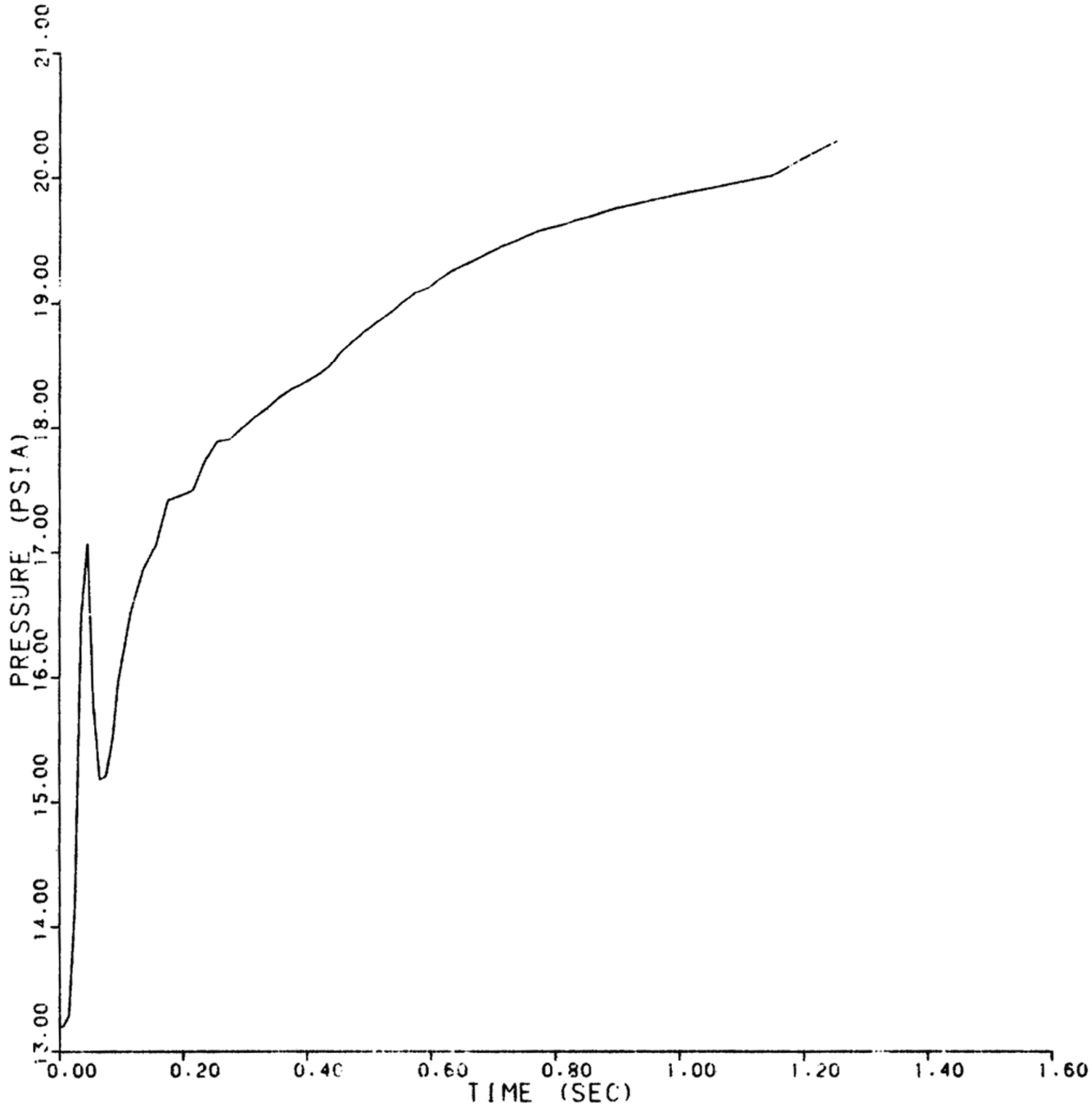
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

STEAM GENERATOR INLET ELBOW
NODE – E74
(763–in.^{2} BREAK AREA)

FIGURE 6.2.1–22 (SHEET 74 OF 74)



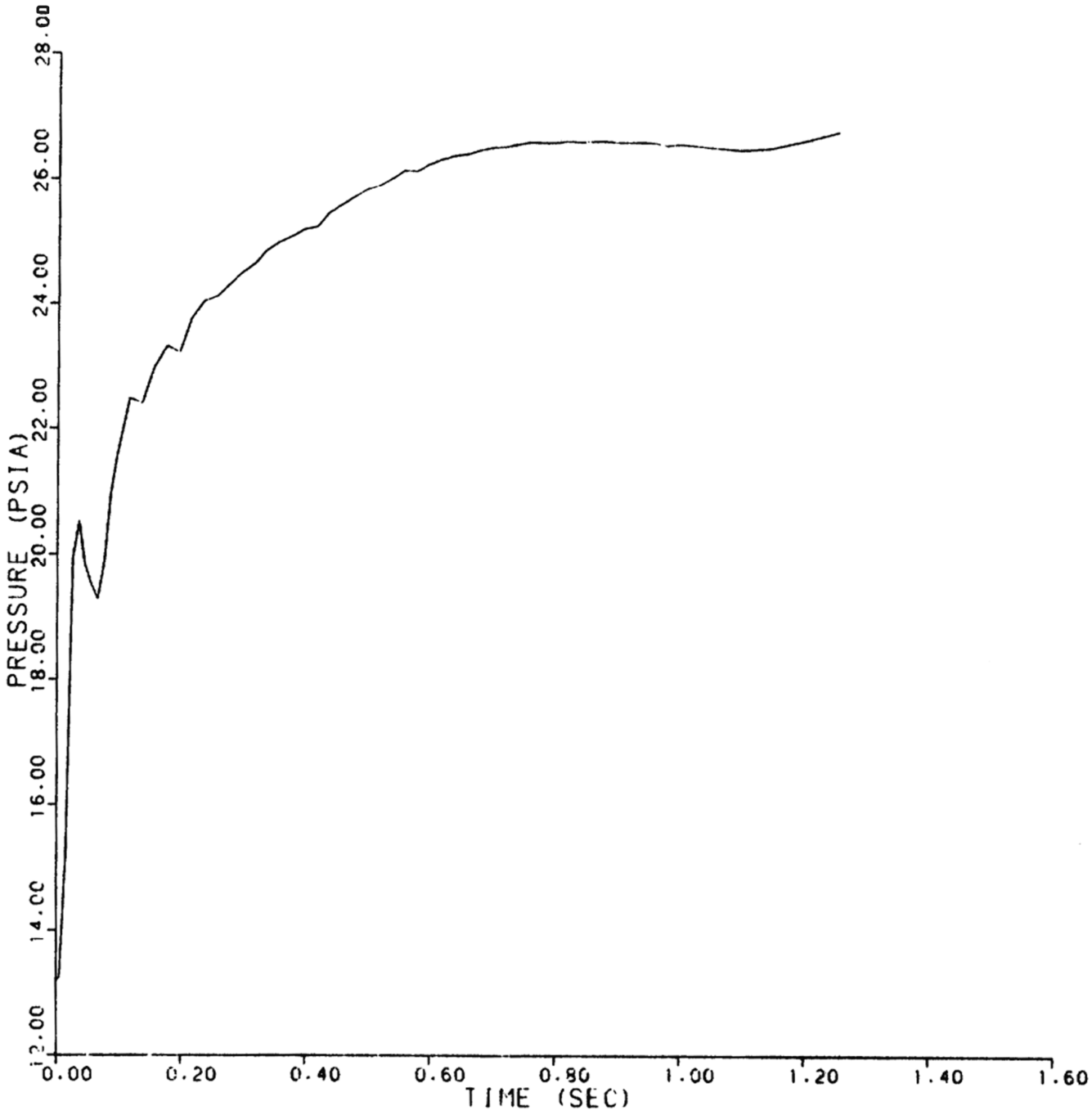
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

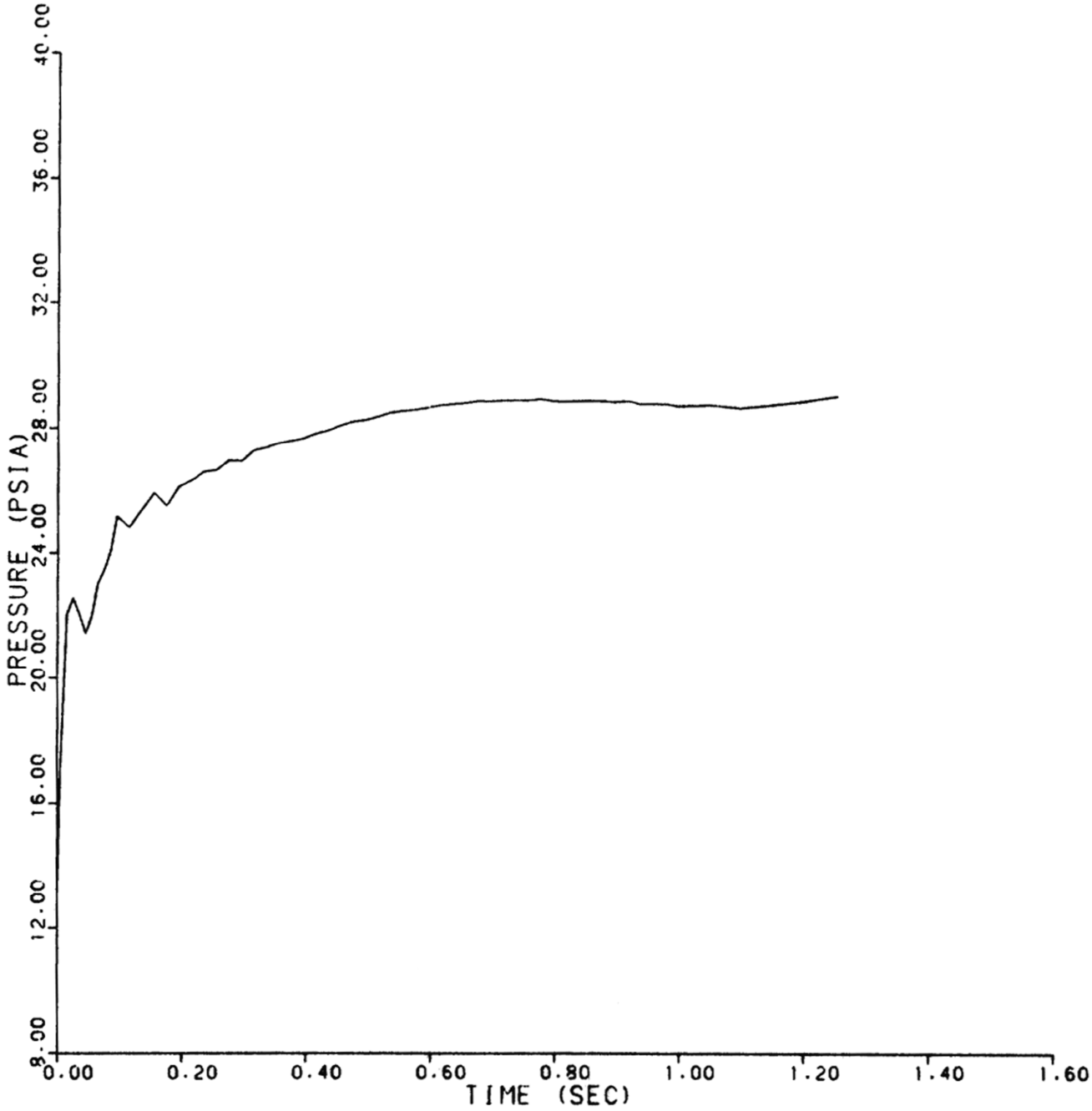
LOOP CLOSURE WELD
NODE E1
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 1 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E2 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 2 OF 74)



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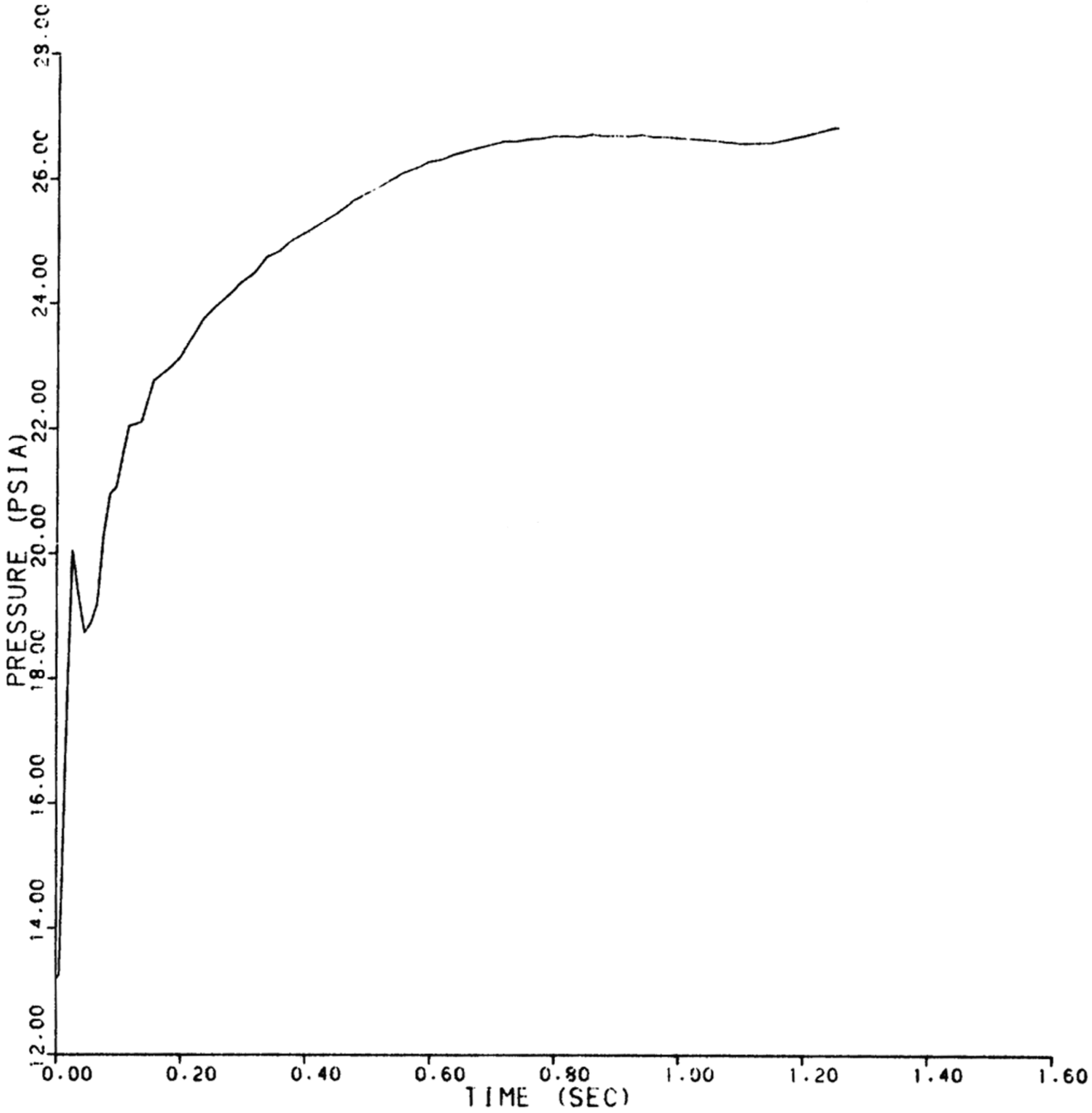
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E3
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 3 OF 74)



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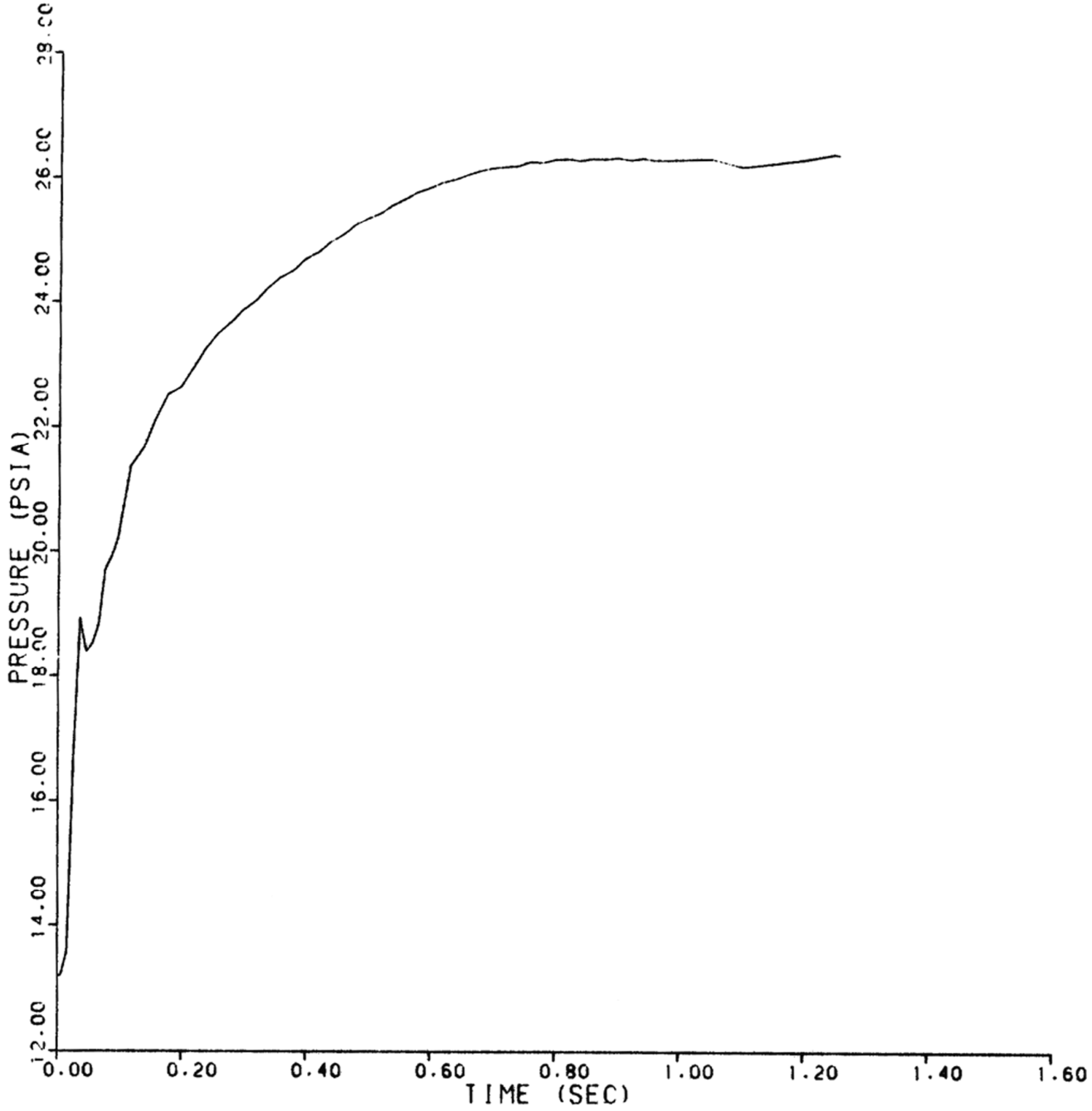
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E4 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 4 OF 74)



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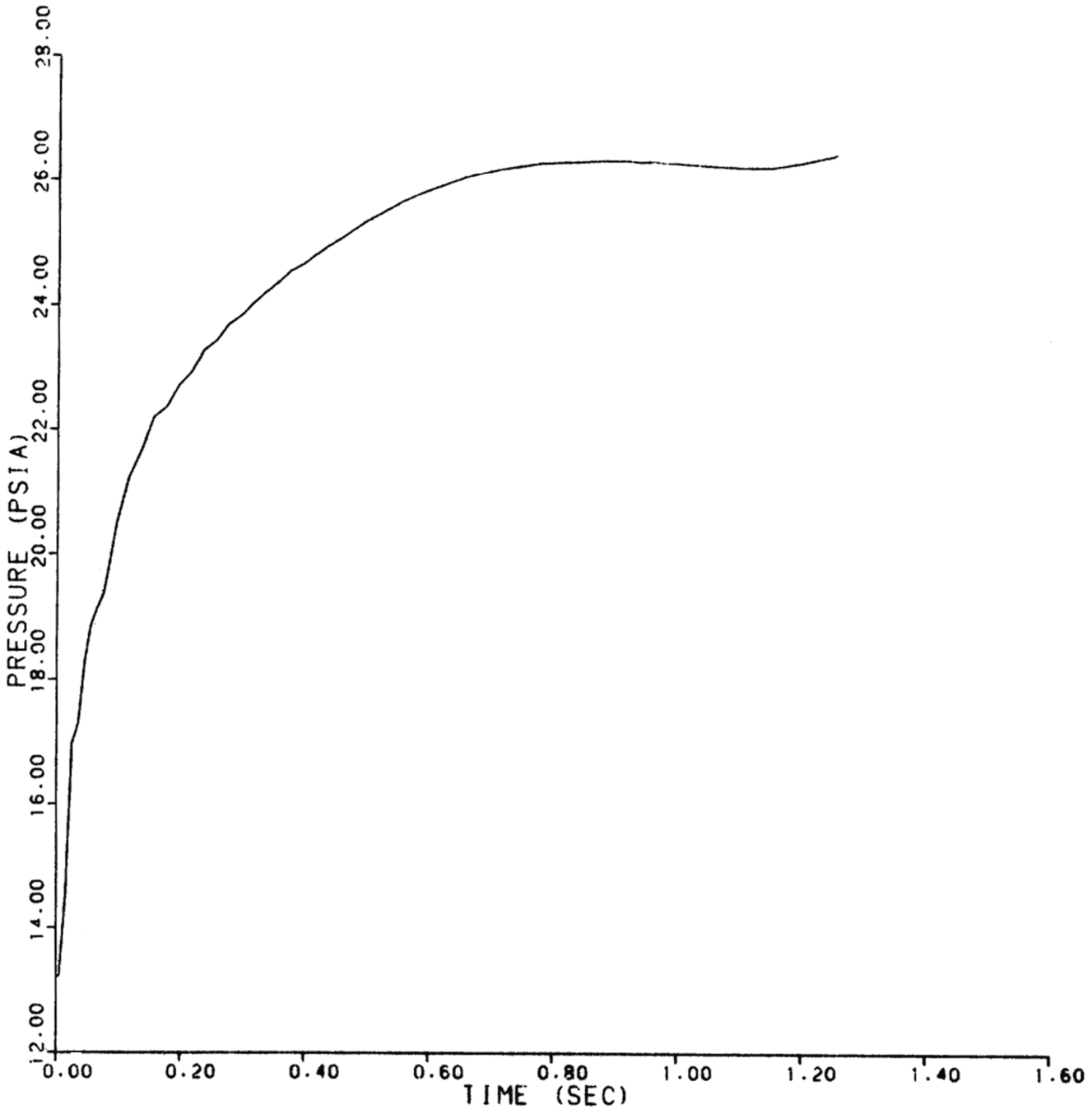
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E5
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 5 OF 74)



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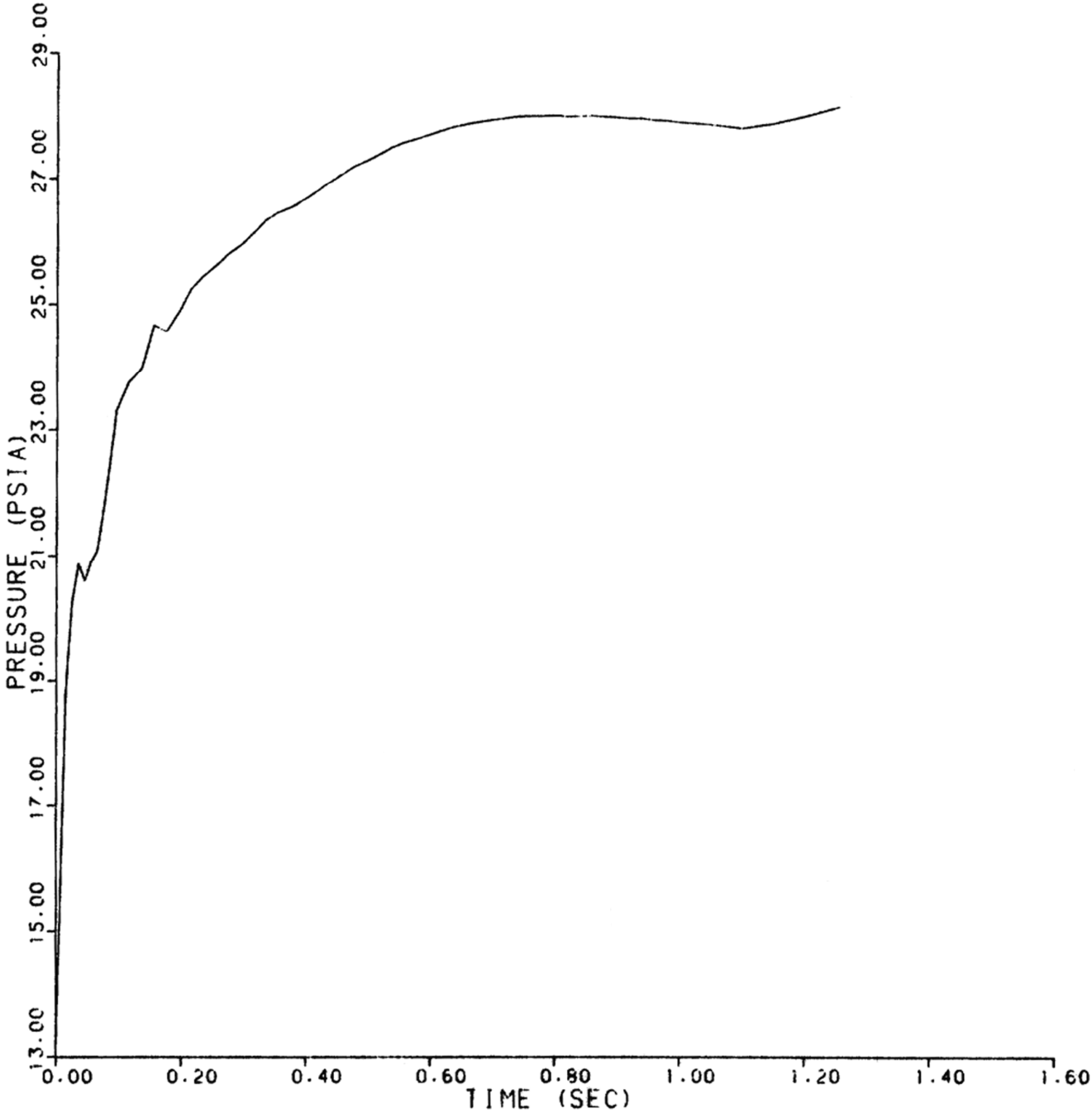
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E6
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 6 OF 74)



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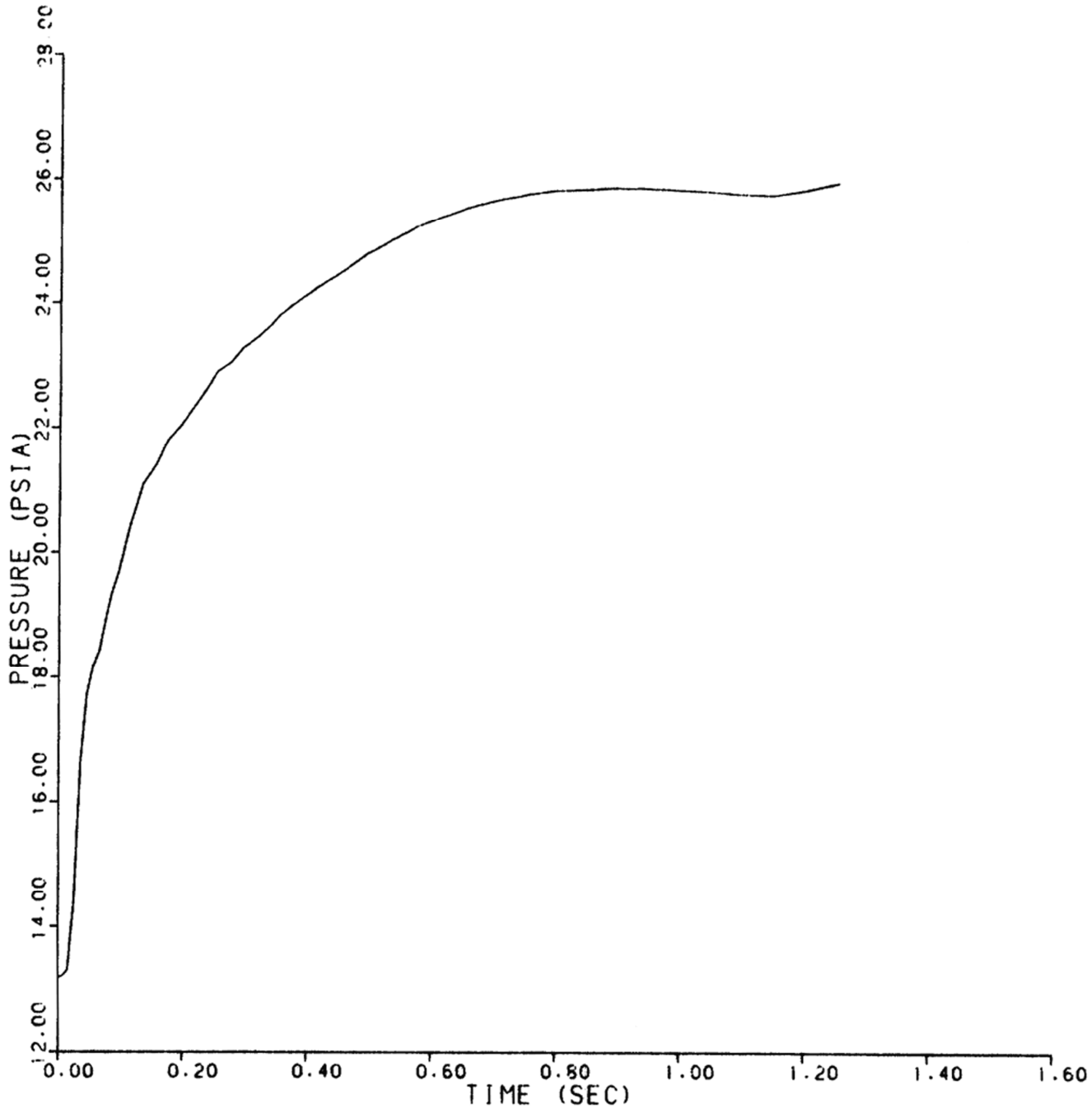
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

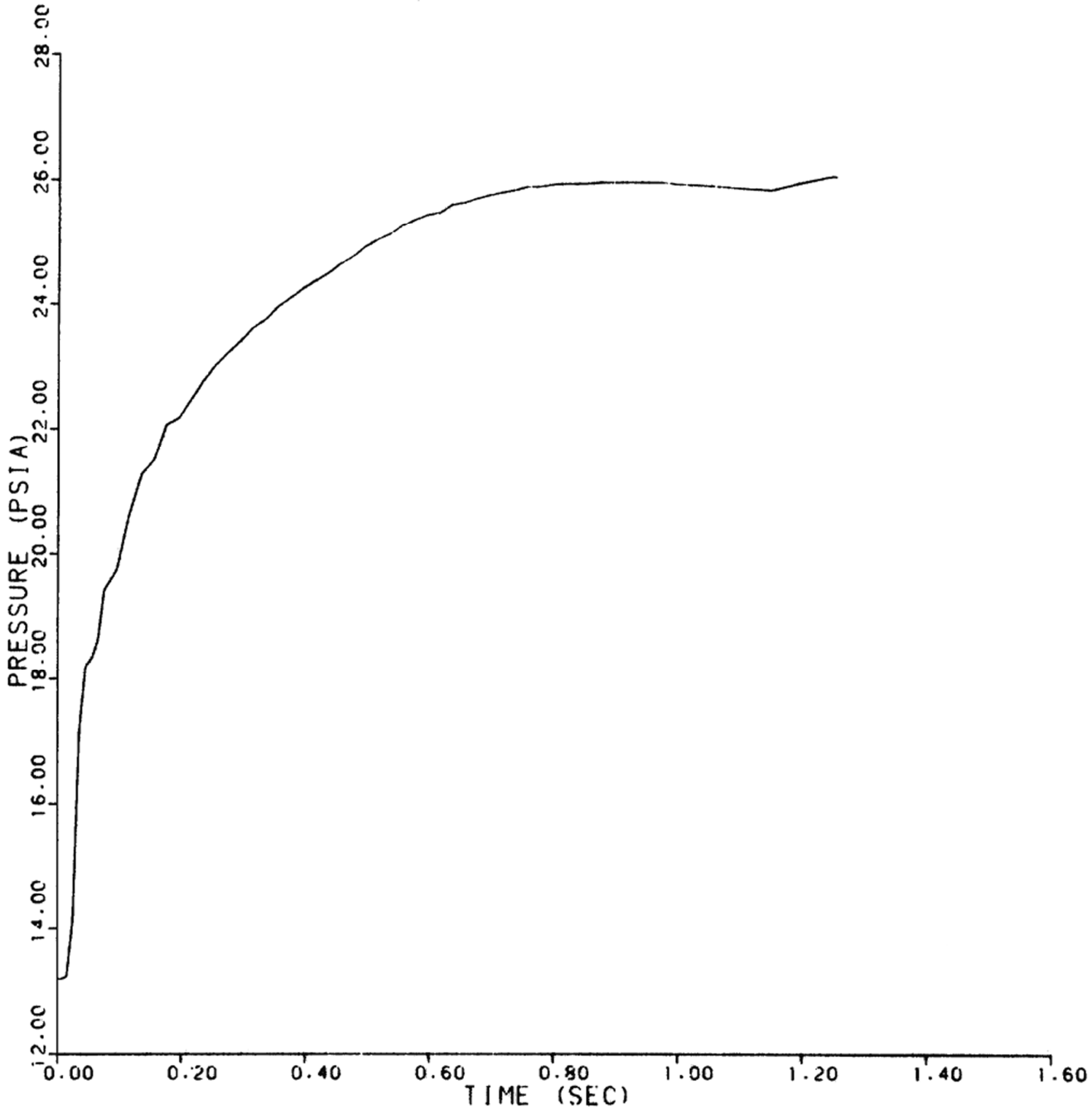
LOOP CLOSURE WELD
NODE E7
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 7 OF 74)



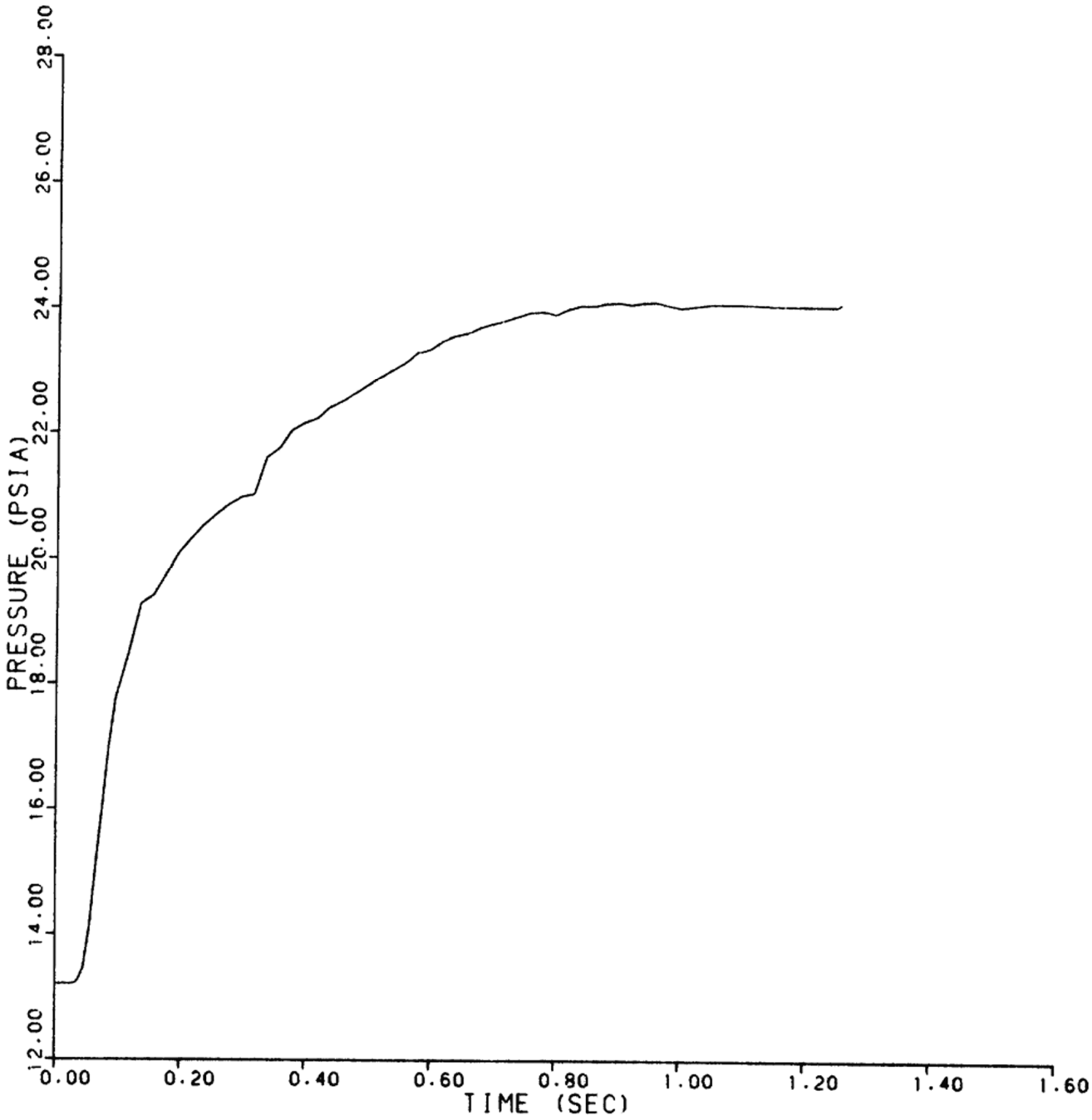
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E8 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 8 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E9 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 9 OF 74)



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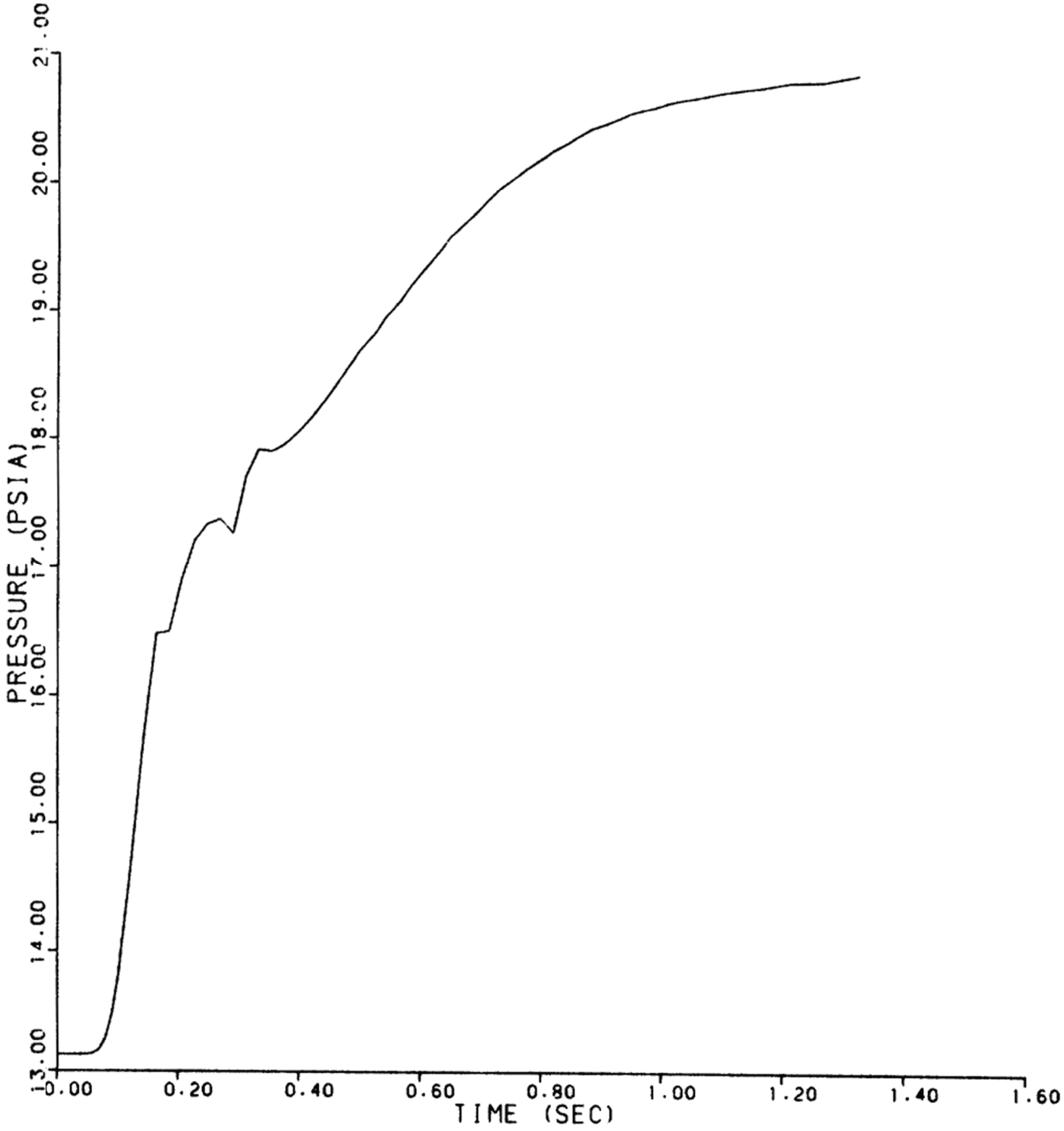
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

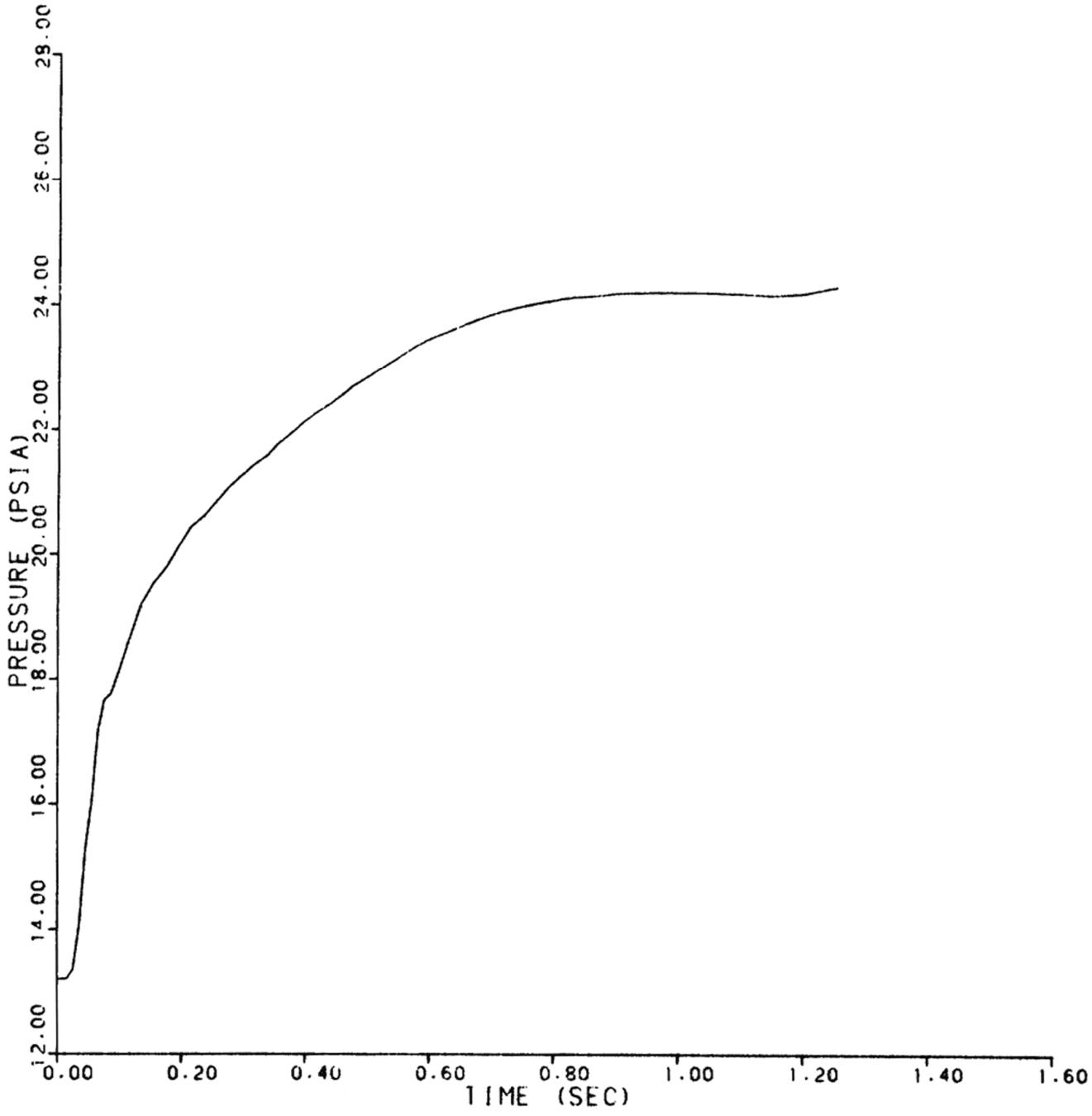
LOOP CLOSURE WELD
NODE E10
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 10 OF 74)



REV 13 4/06

SOUTHERN COMPANY Energy to Serve Your World® VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	LOOP CLOSURE WELD NODE E11 (336–in.^{2} BREAK AREA) FIGURE 6.2.1–23 (SHEET 11 OF 74)



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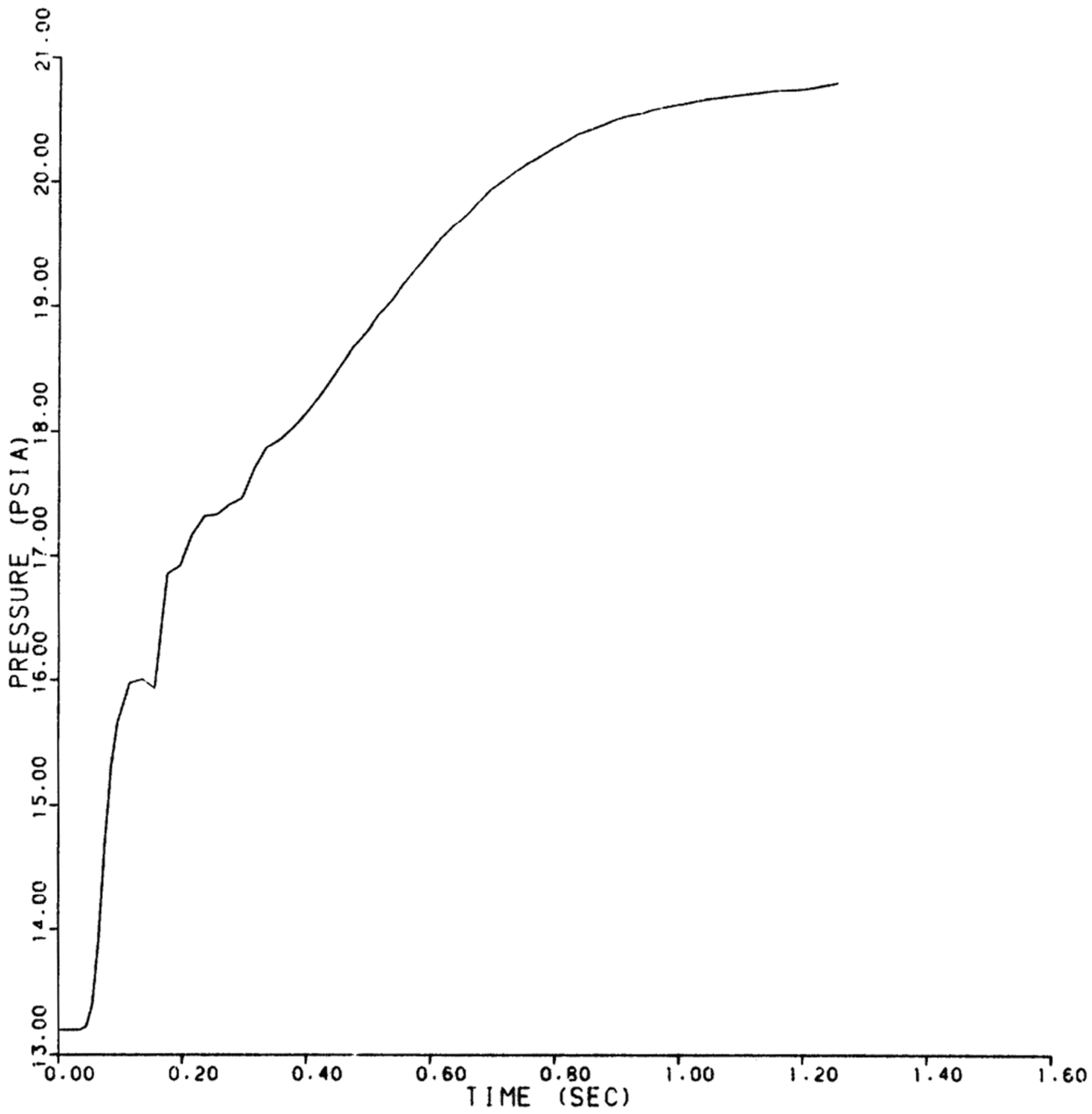
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E12
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 12 OF 74)



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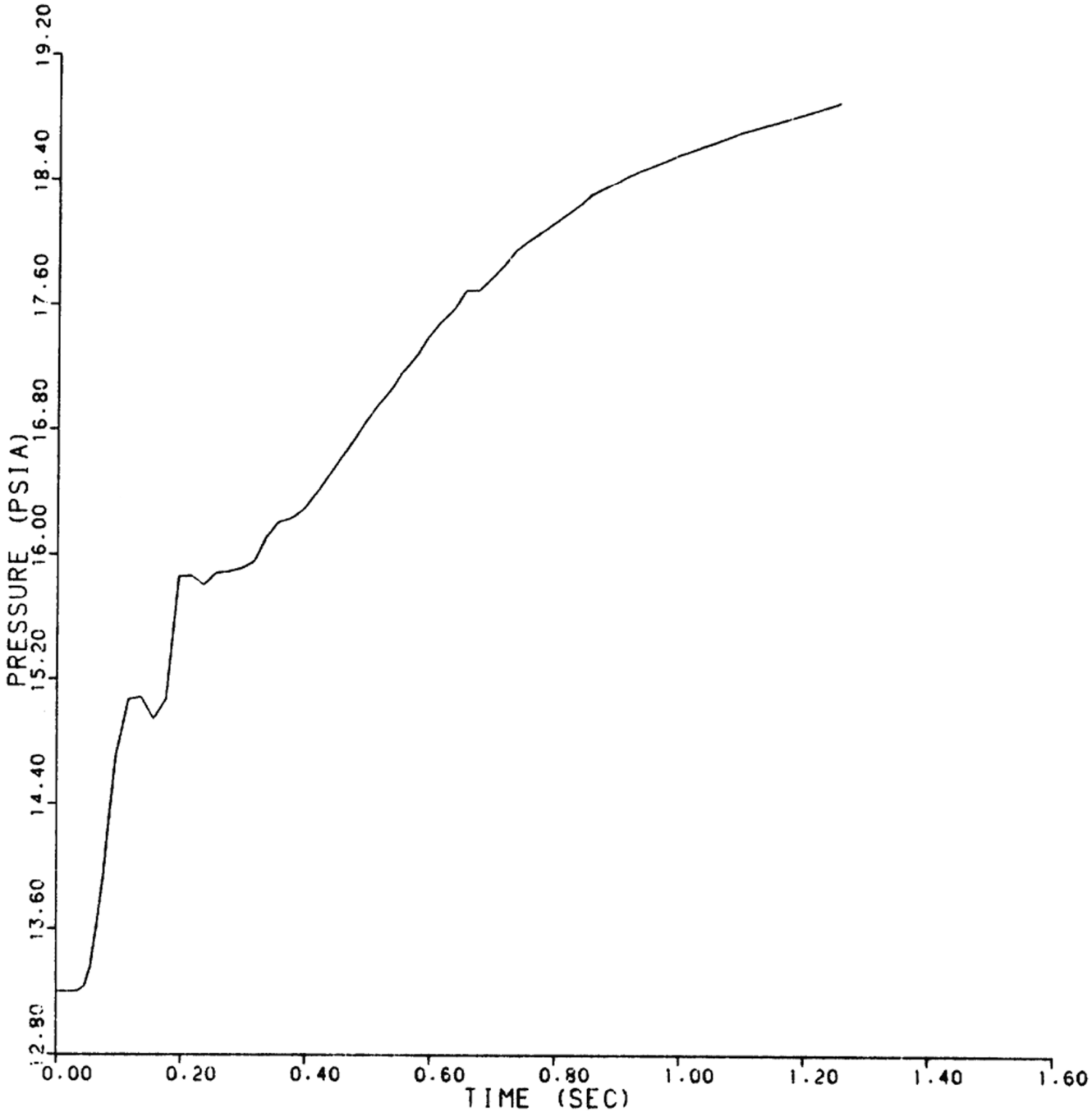
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E13 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 13 OF 74)



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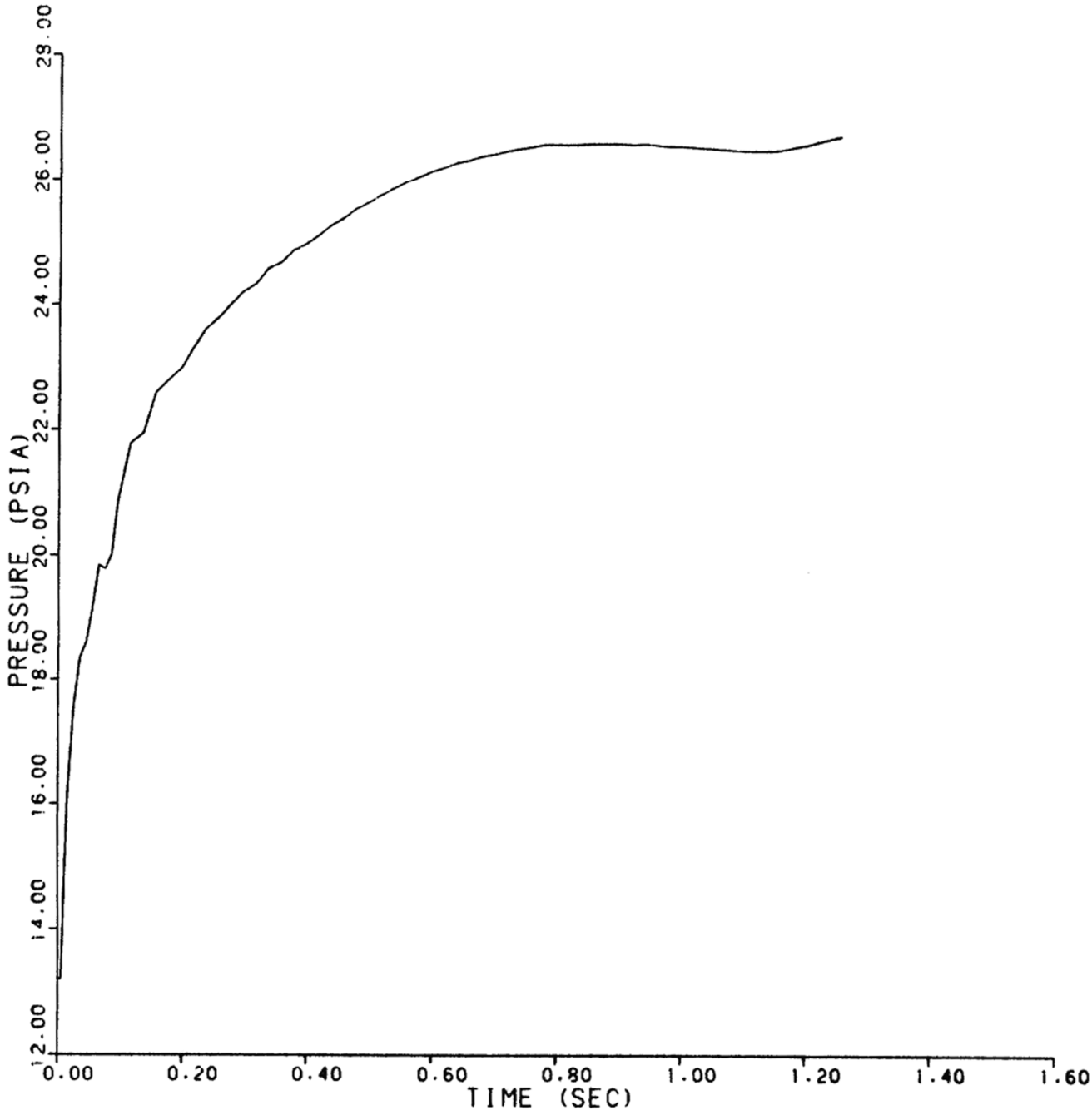
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E14 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 14 OF 74)



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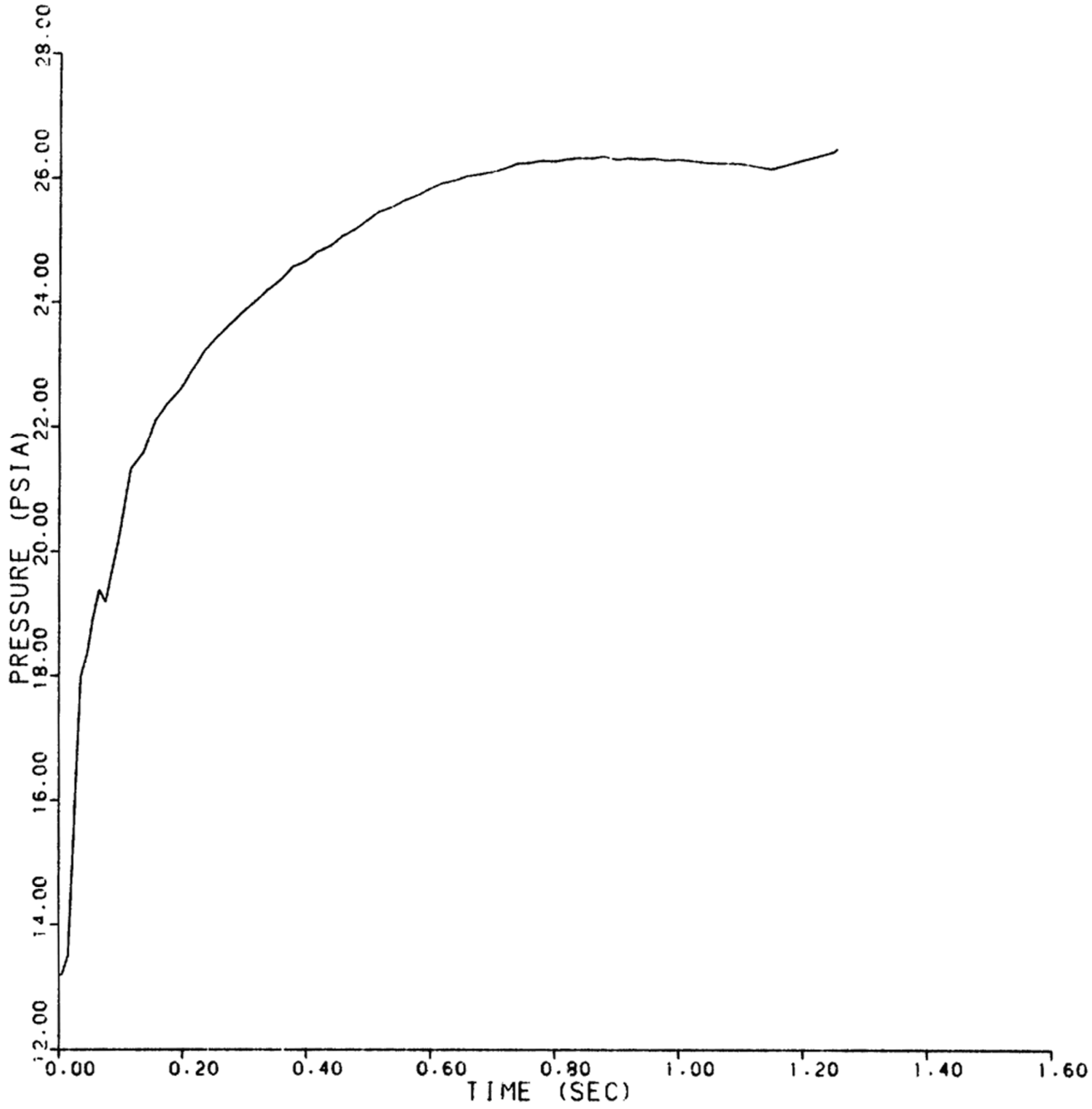
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E15
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 15 OF 74)



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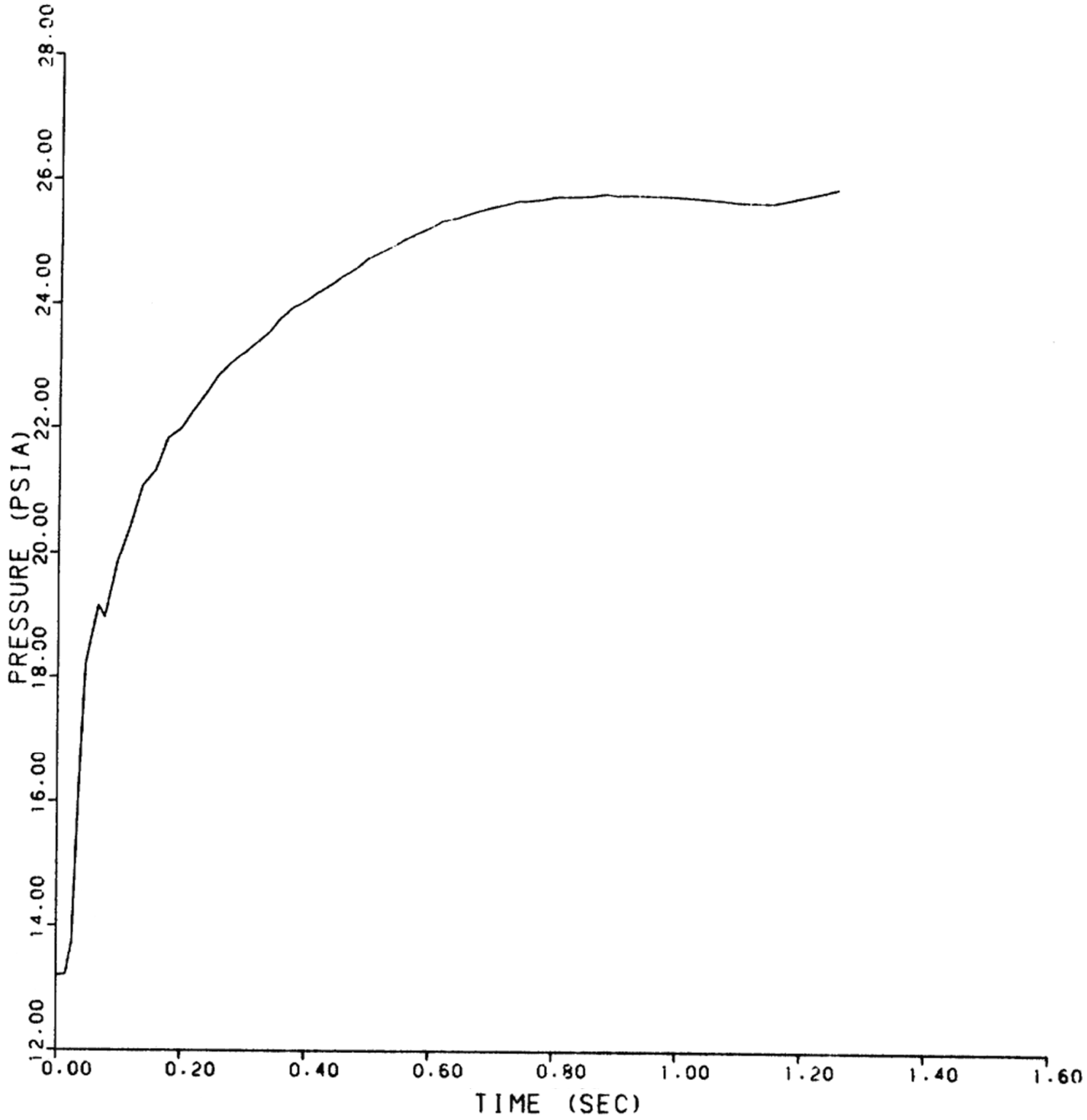
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E16
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 16 OF 74)



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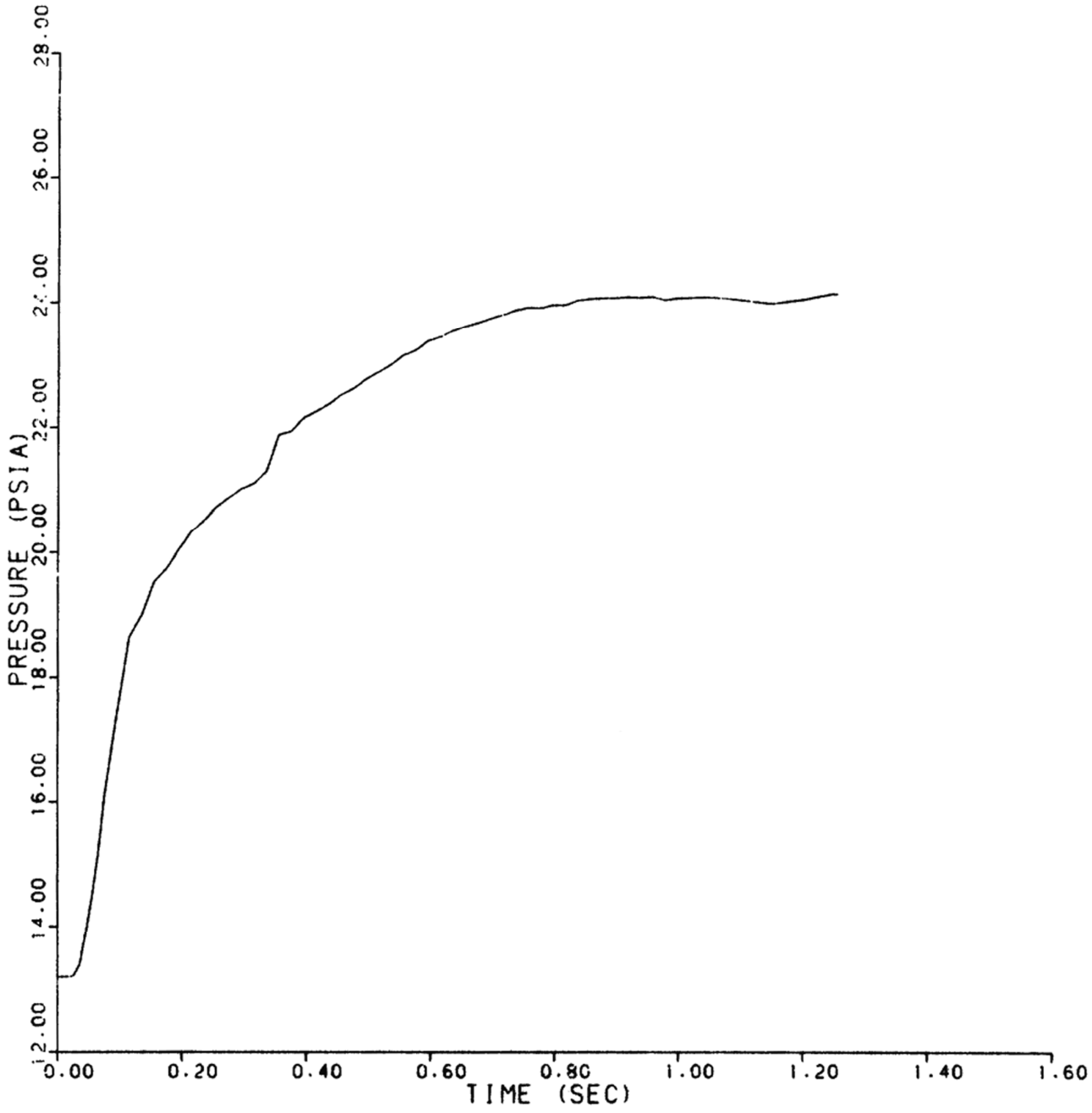
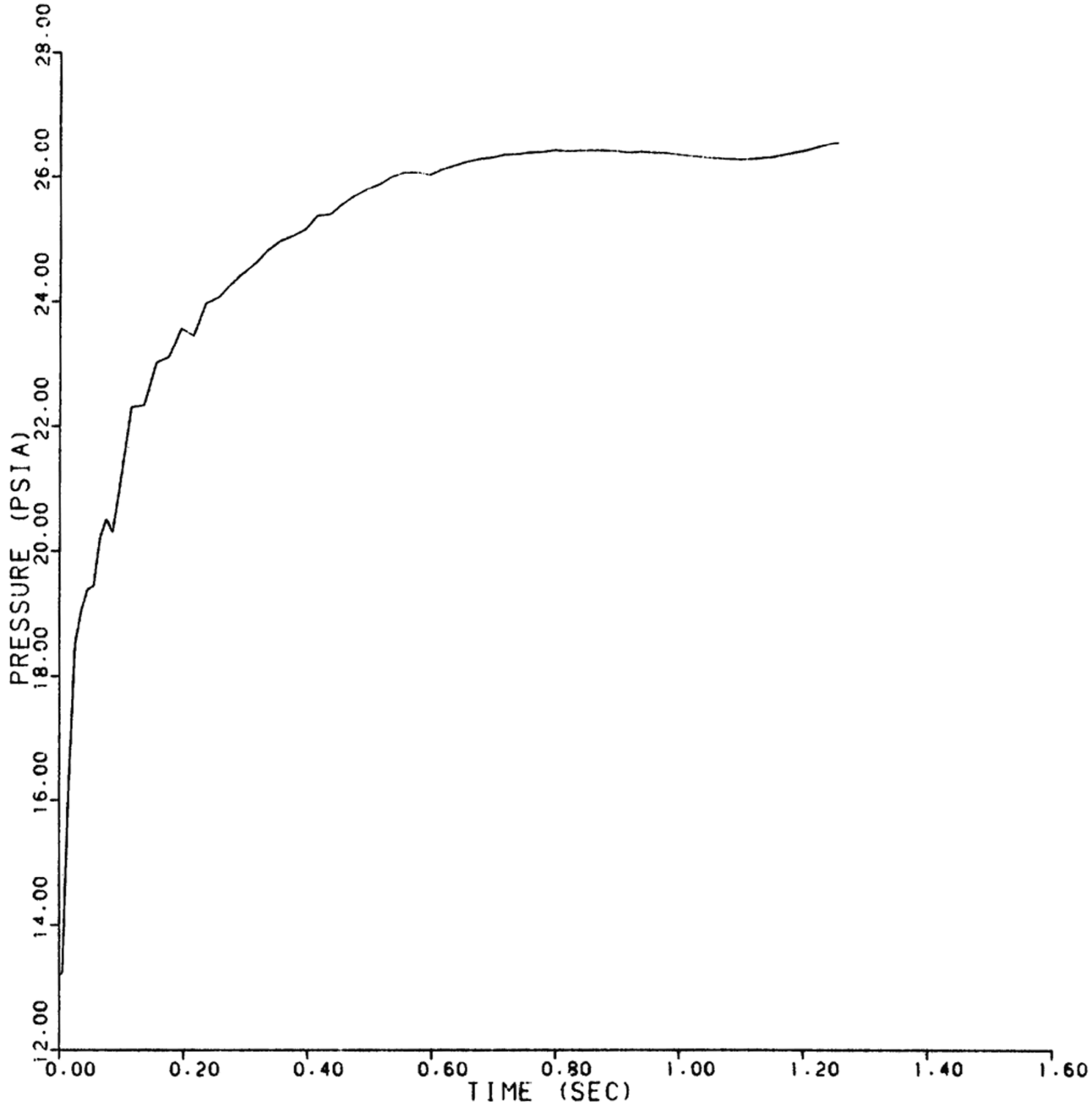
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E17 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 17 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E18 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 18 OF 74)



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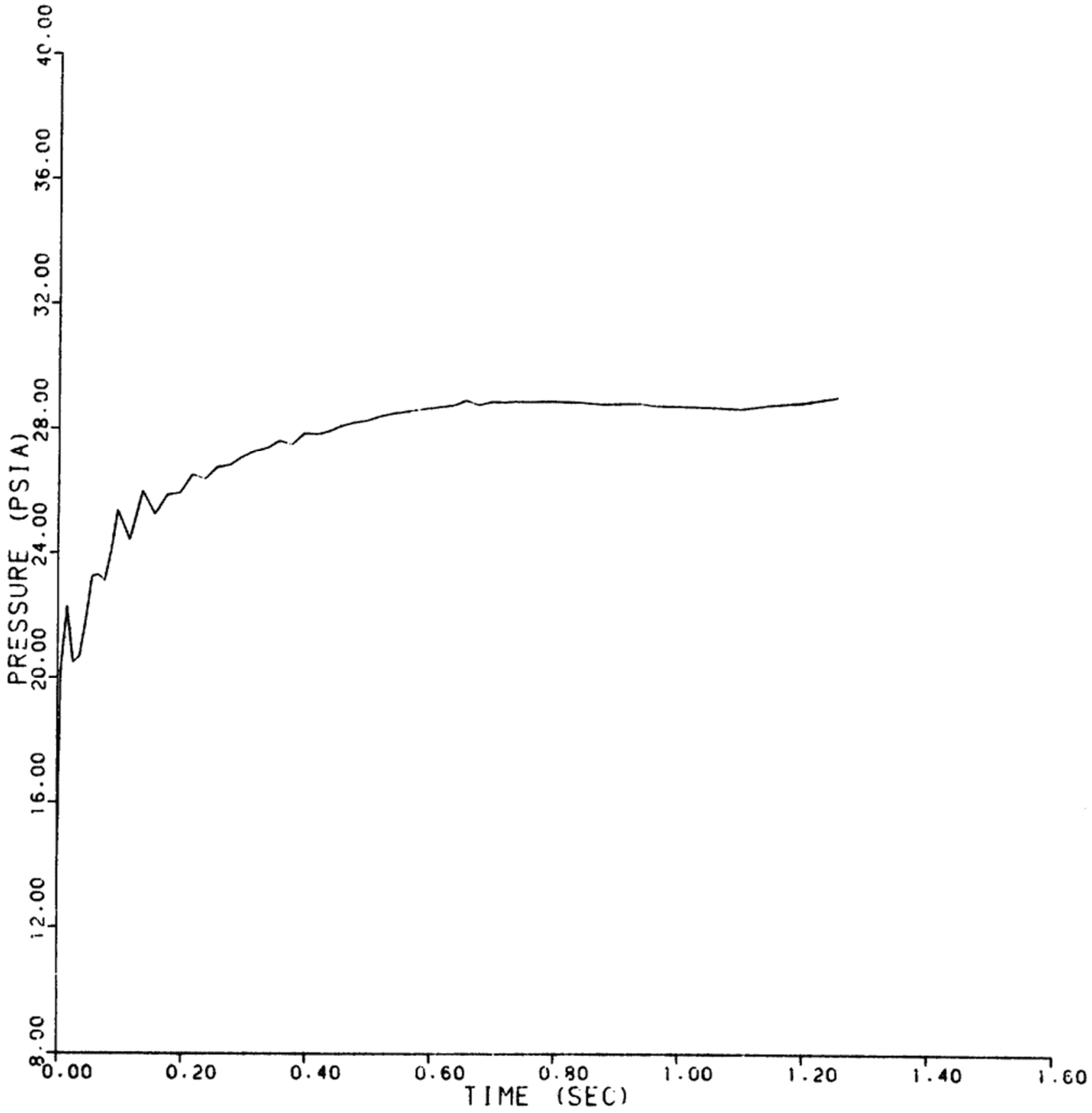
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E19
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 19 OF 74)



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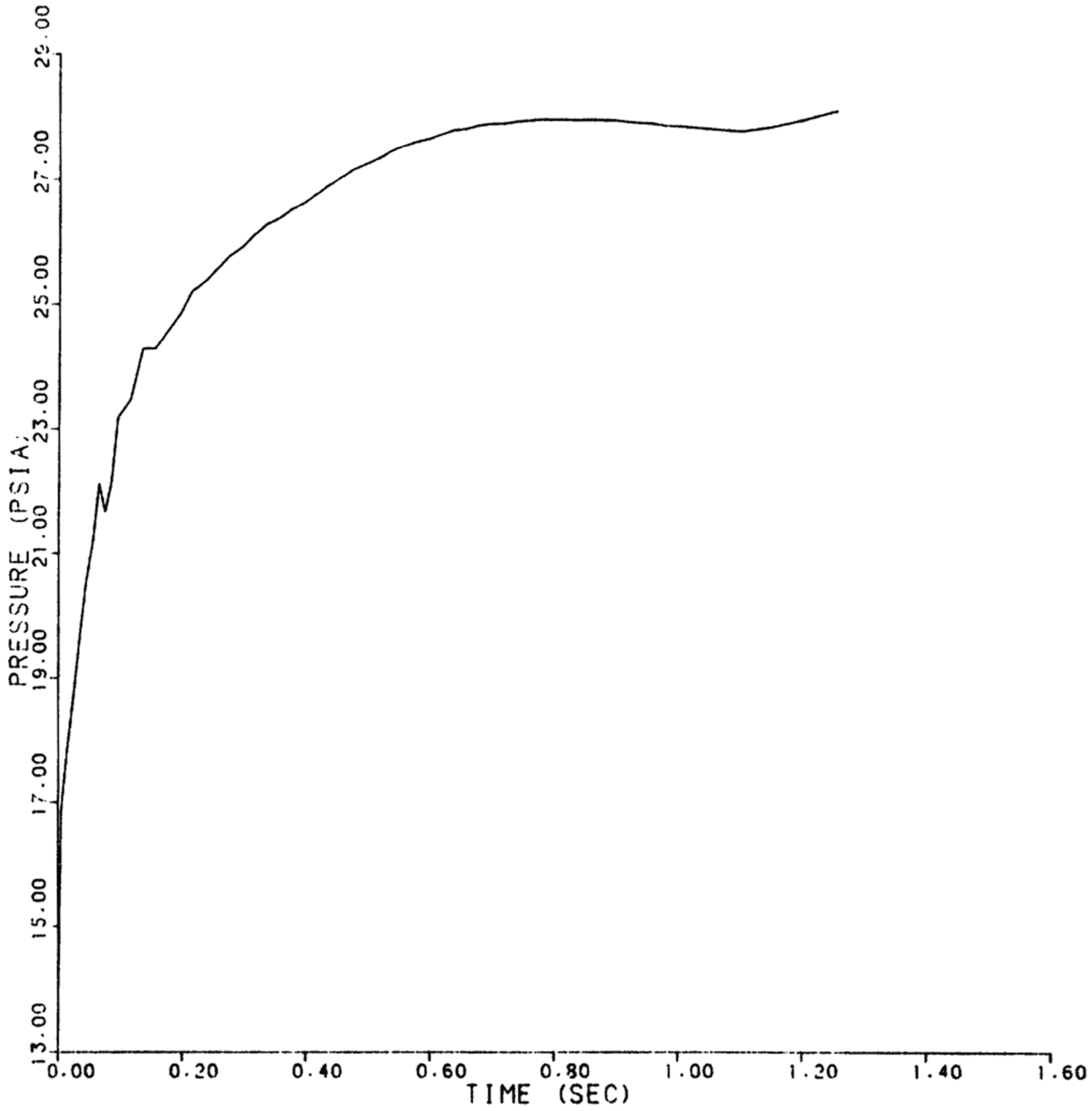
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E20 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 20 OF 74)



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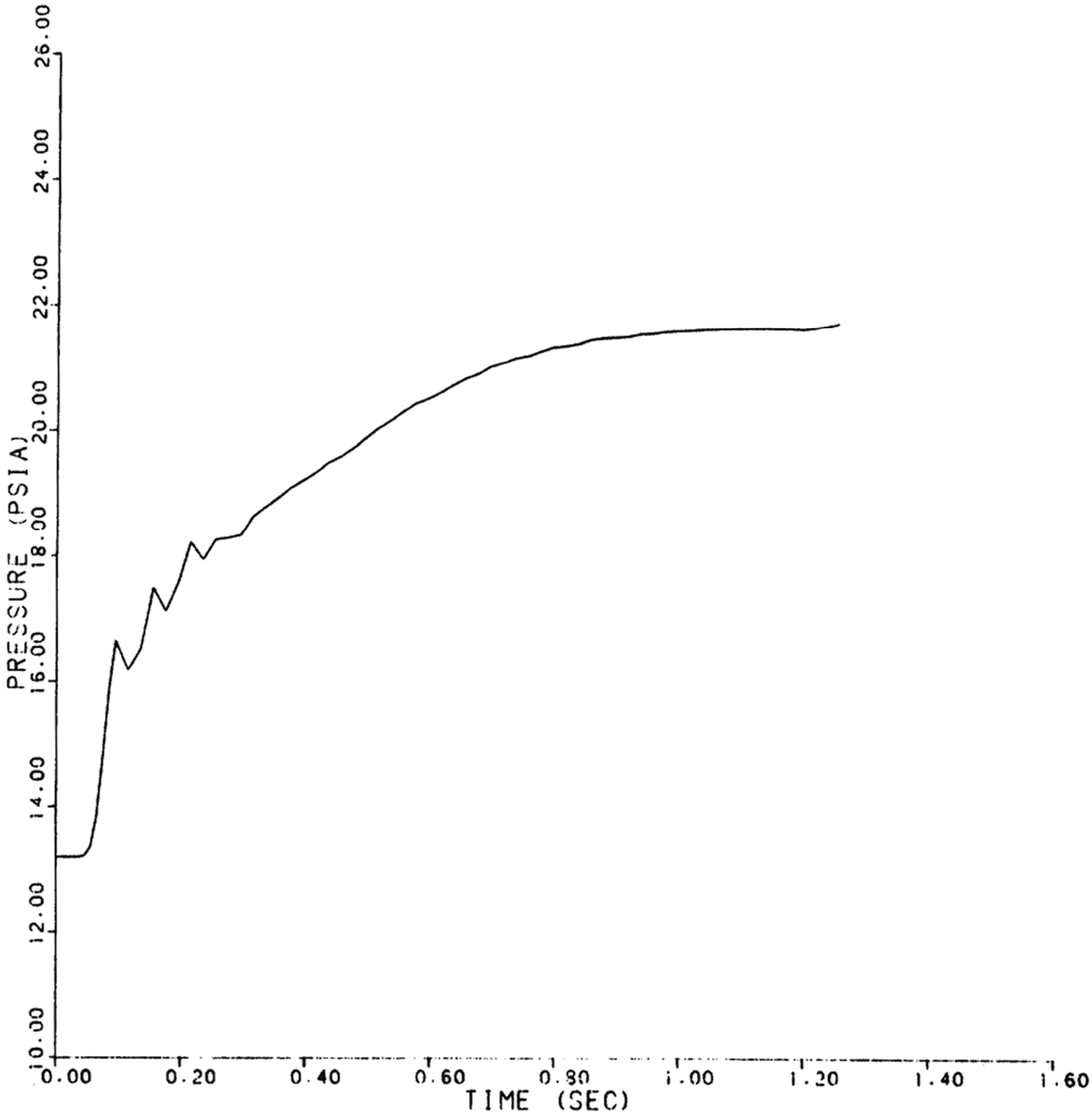
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E21
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 21 OF 74)



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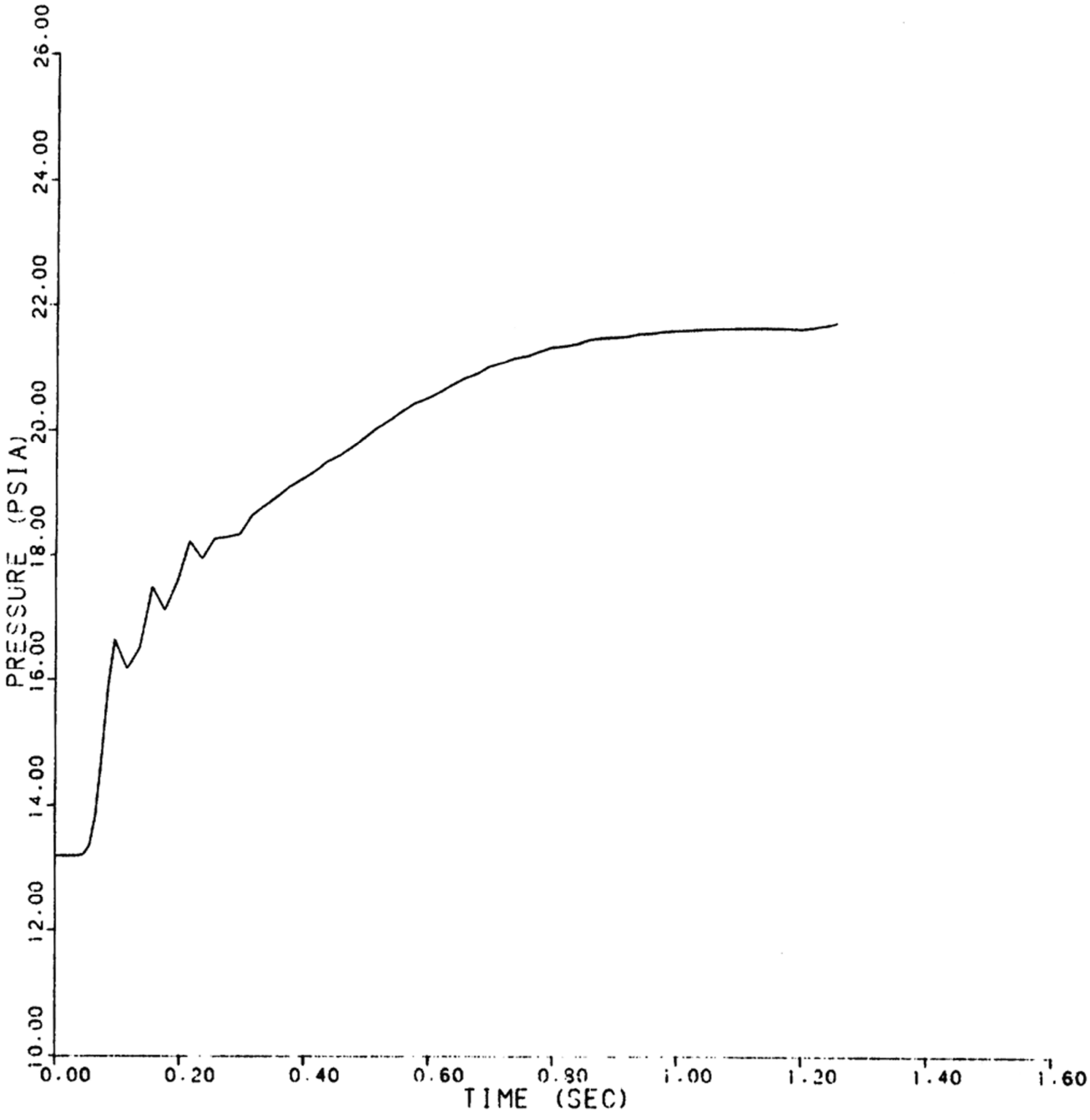
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E22 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 22 OF 74)



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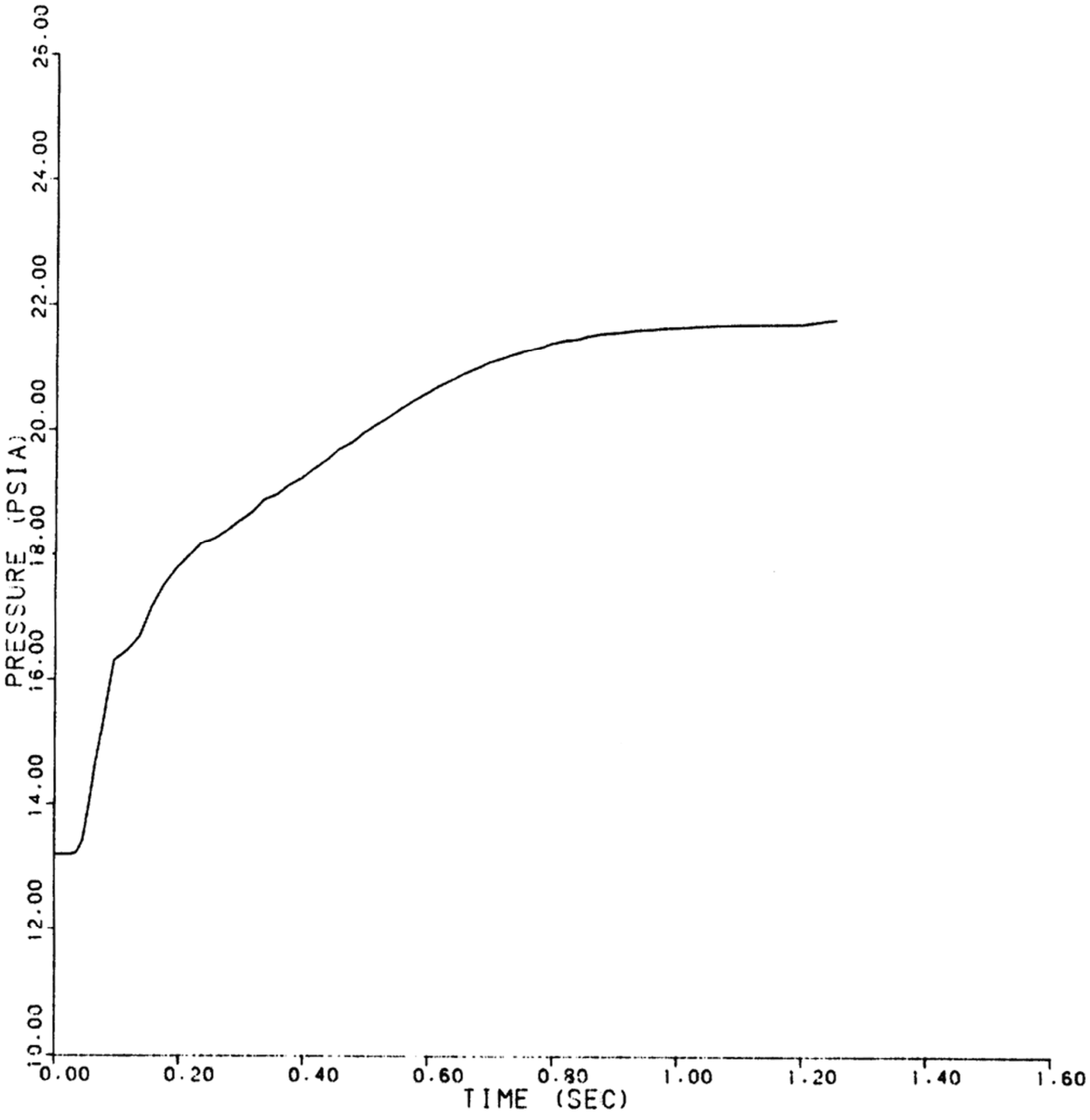
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E23
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 23 OF 74)



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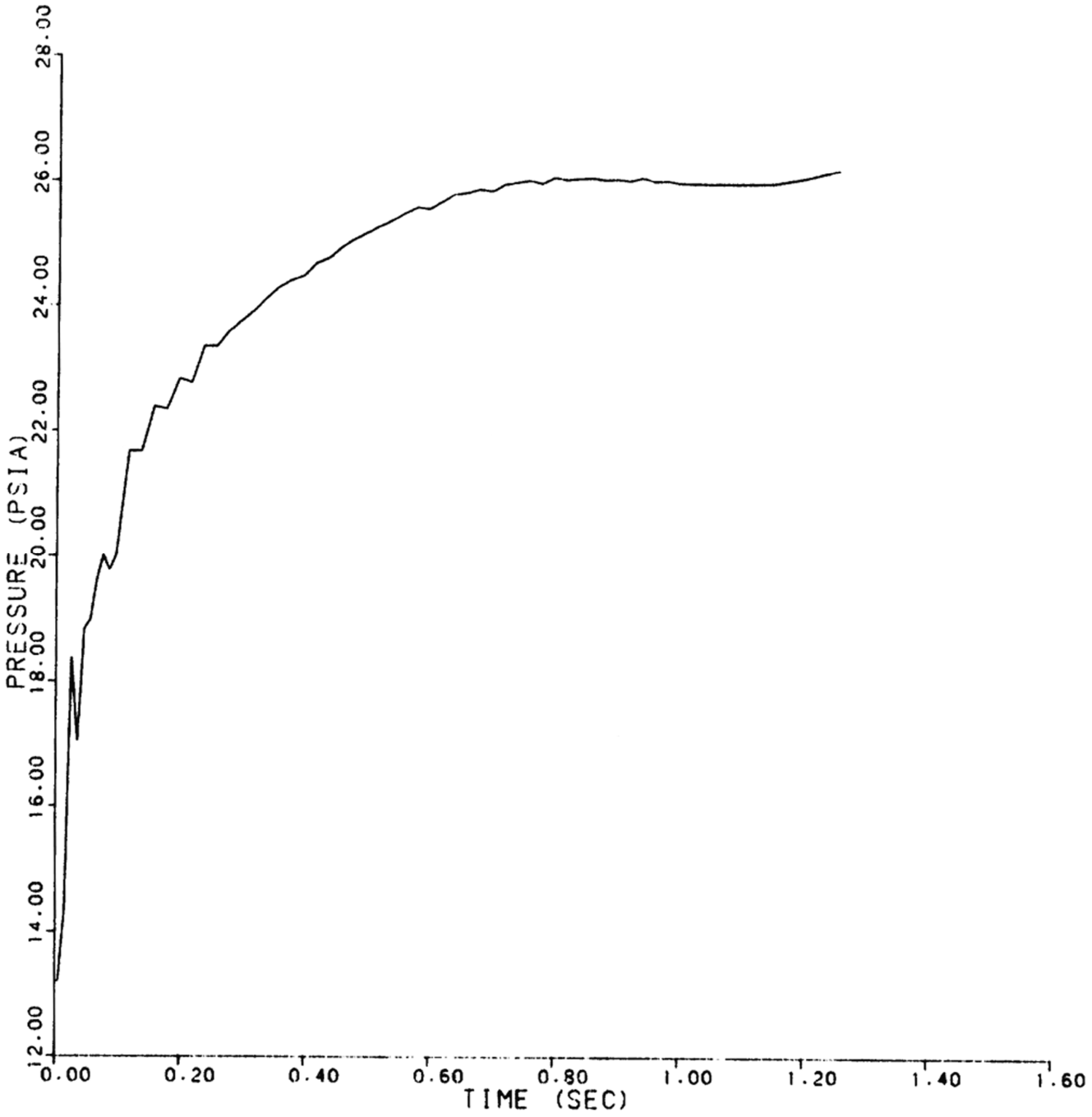
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E24
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 24 OF 74)



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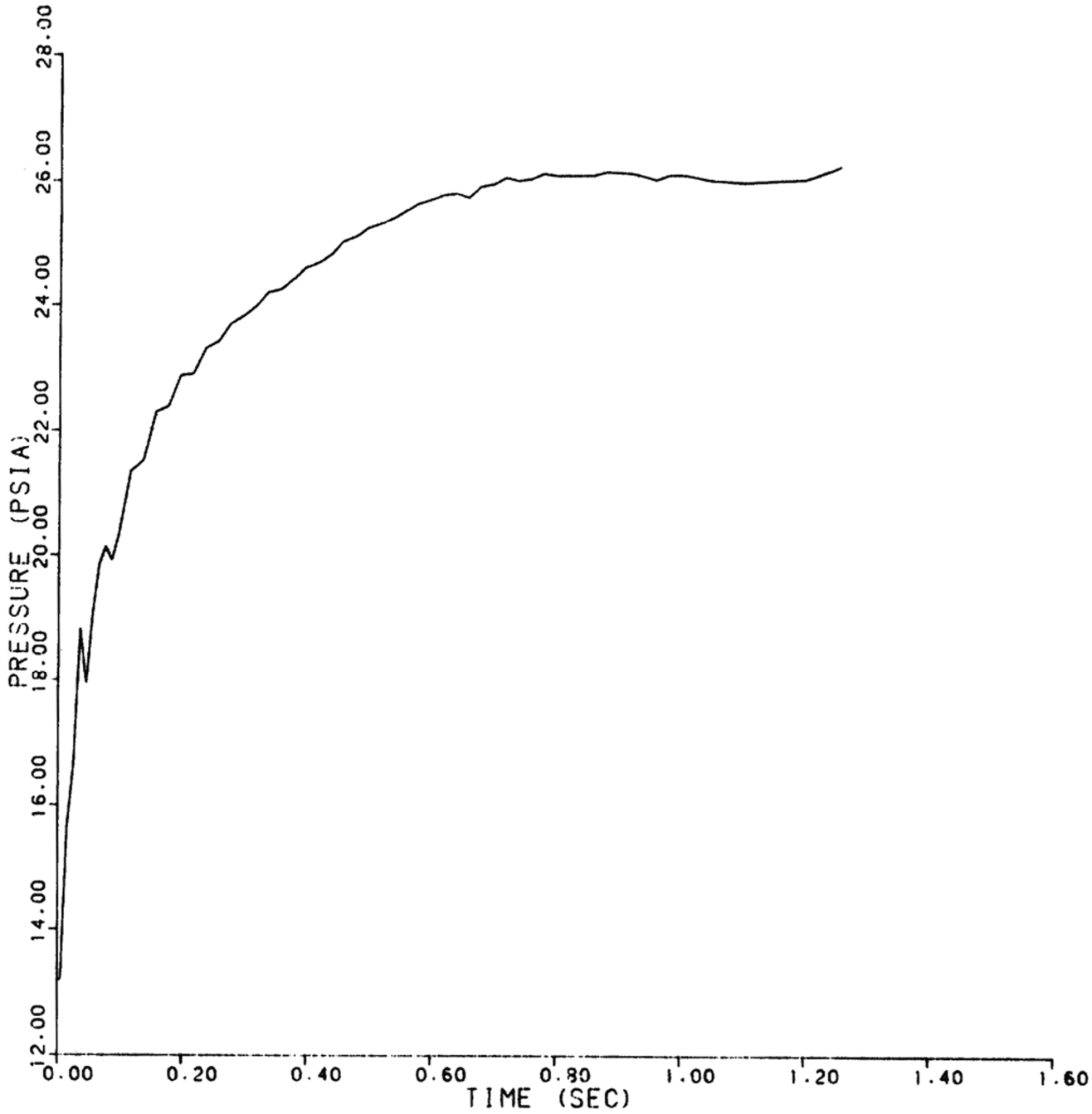
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

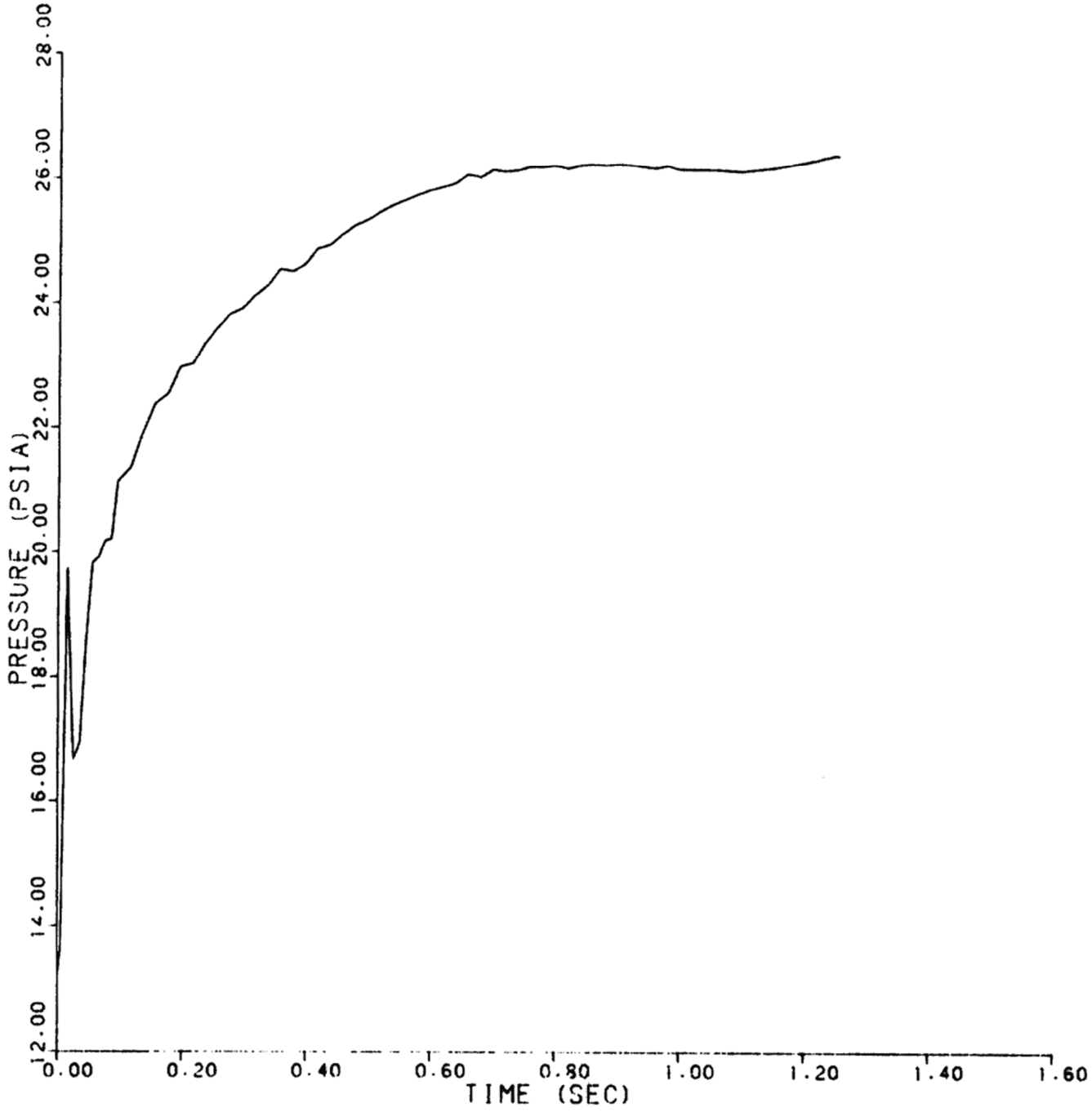
LOOP CLOSURE WELD
NODE E25
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 25 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E26 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 26 OF 74)



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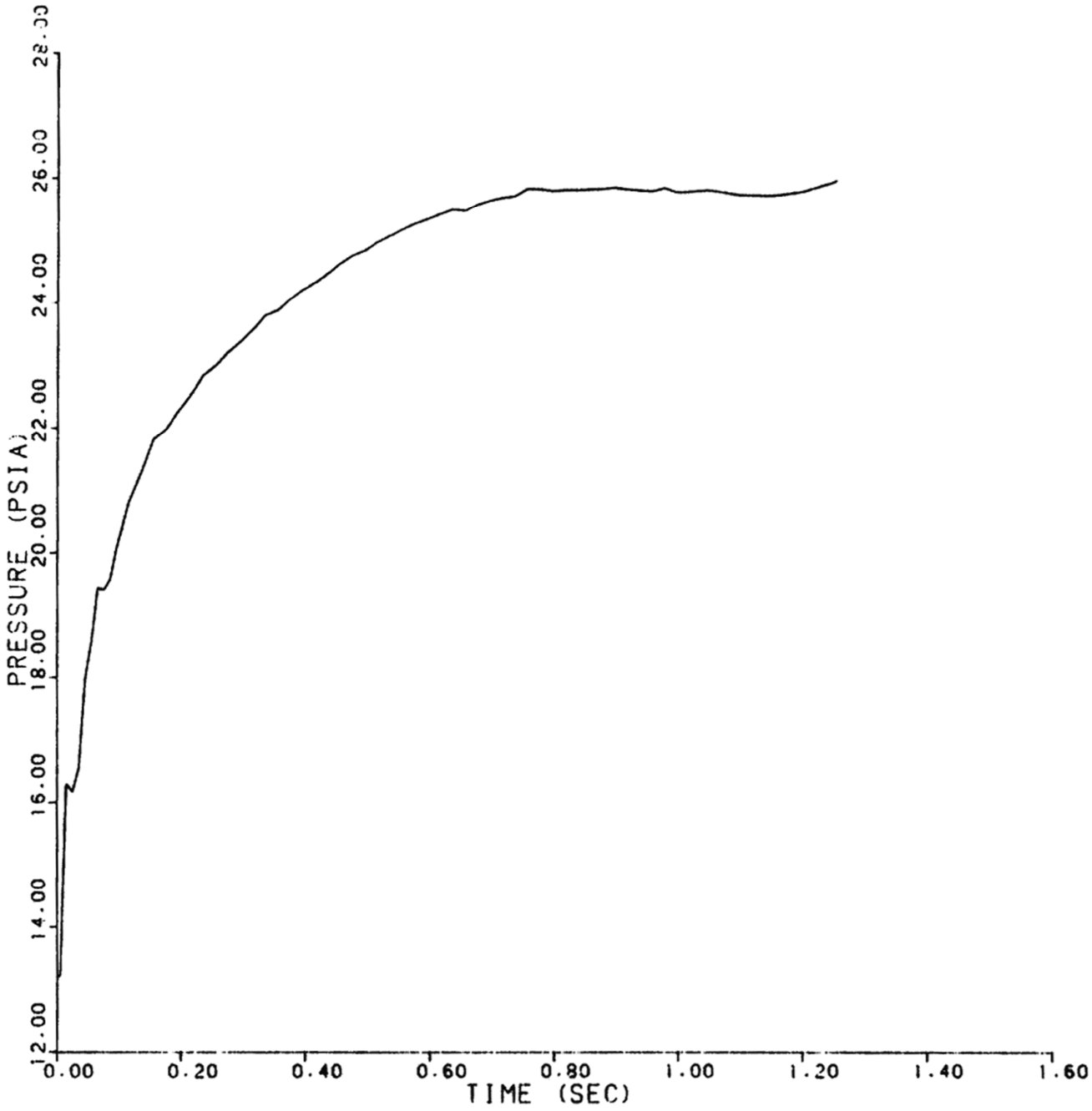
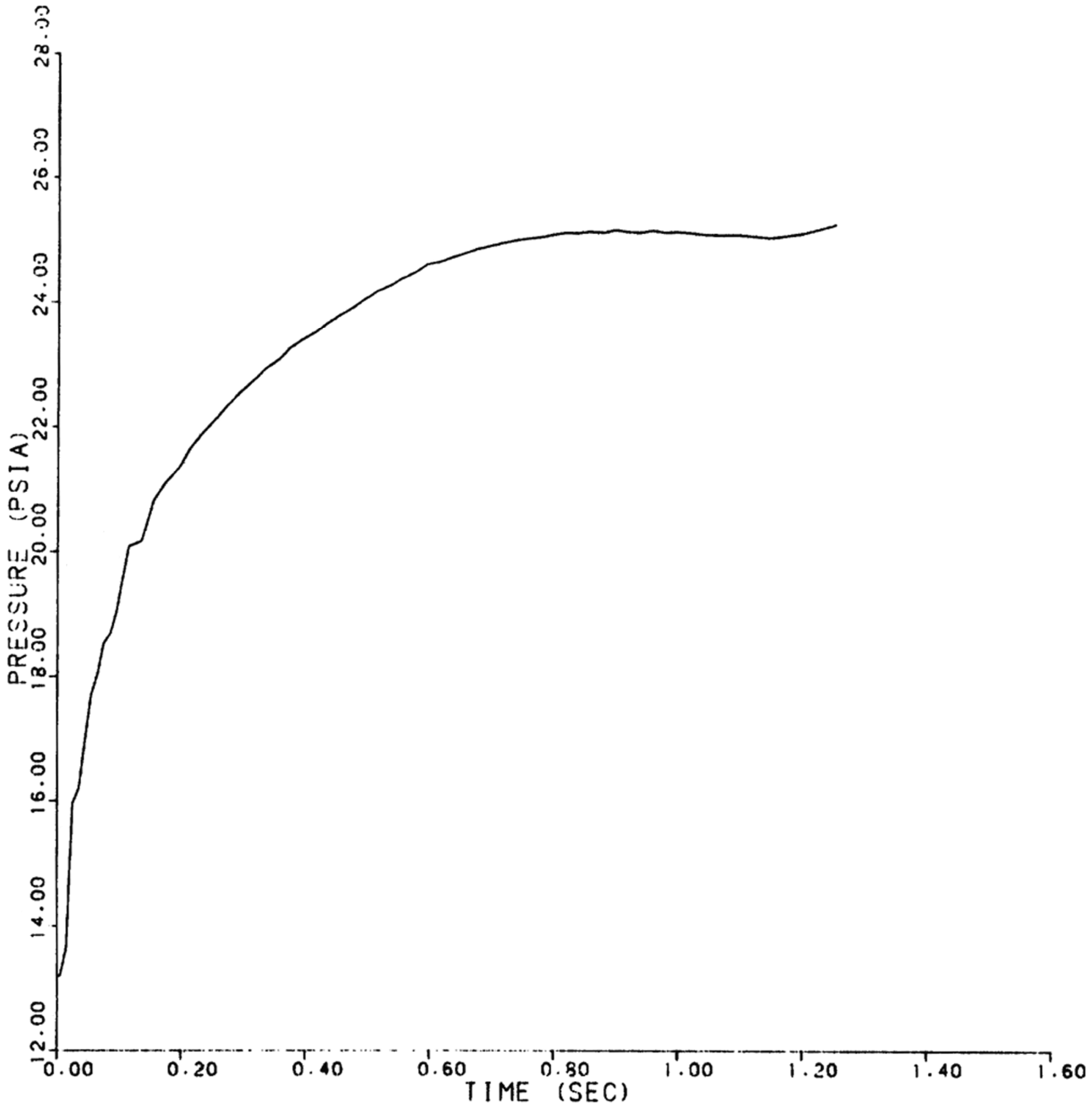
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E27 (336–in.^{2} BREAK AREA) FIGURE 6.2.1–23 (SHEET 27 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E28 (336–in.^{2} BREAK AREA) FIGURE 6.2.1–23 (SHEET 28 OF 74)



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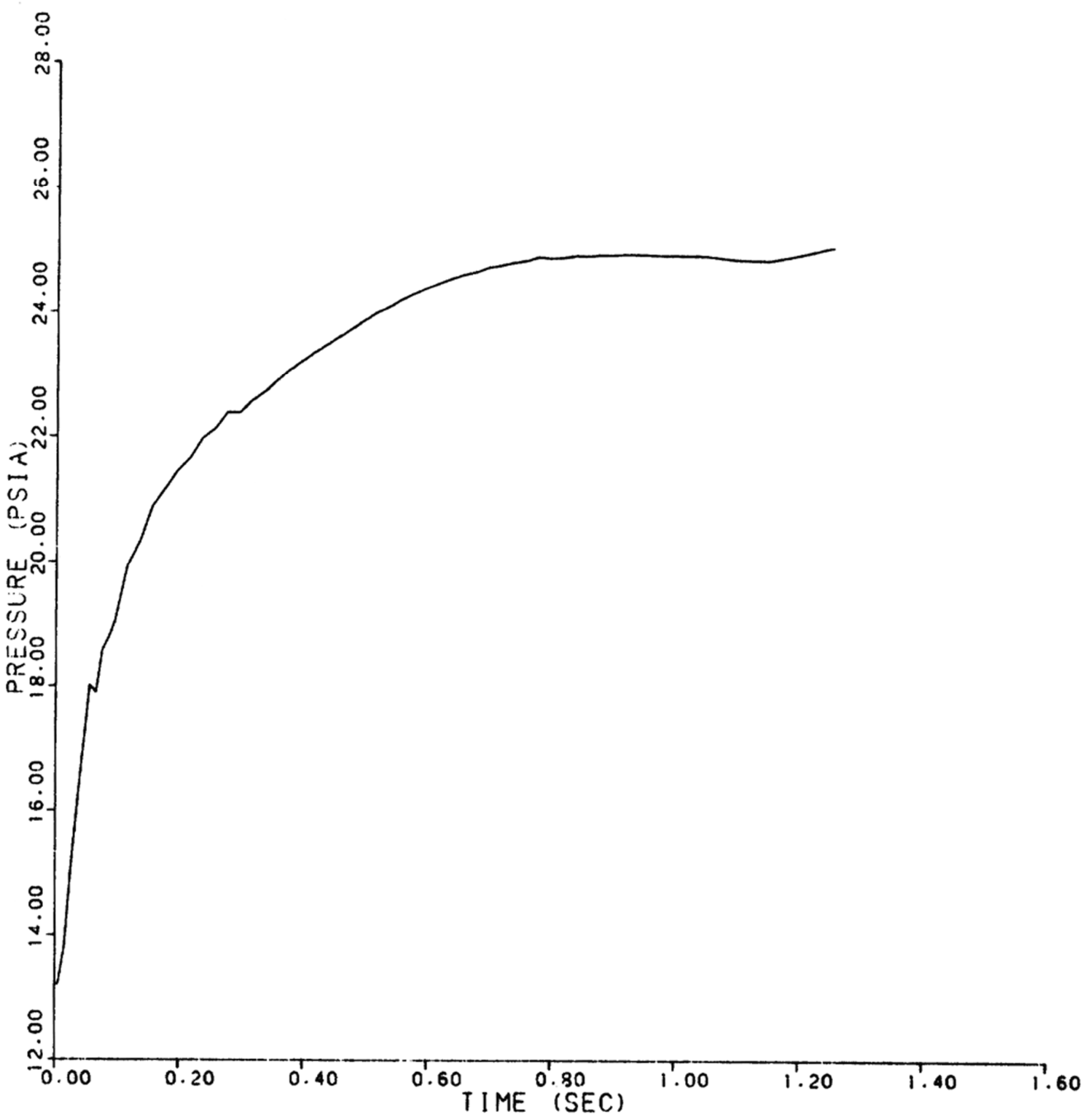
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

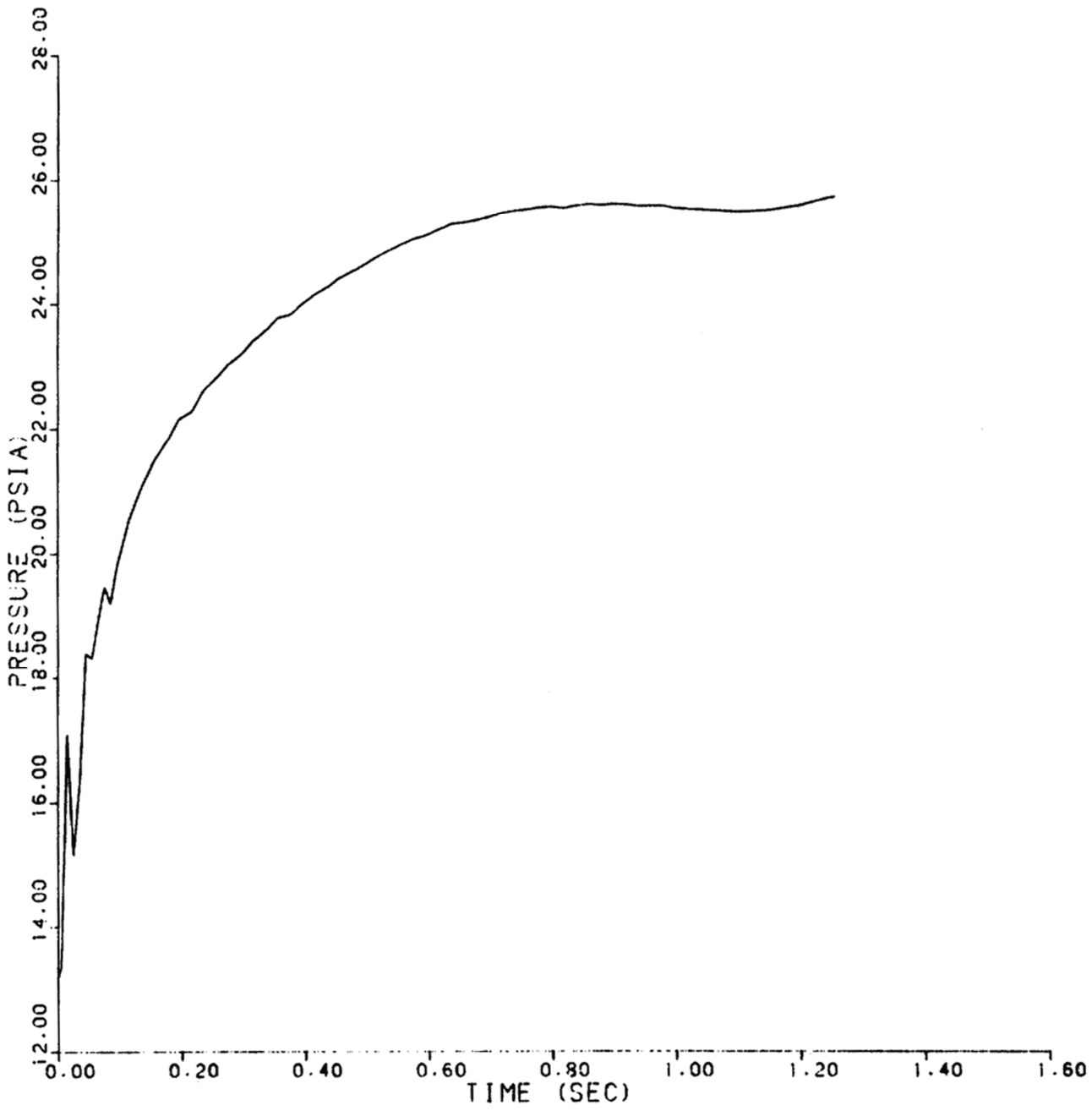
LOOP CLOSURE WELD
NODE E29
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 29 OF 74)



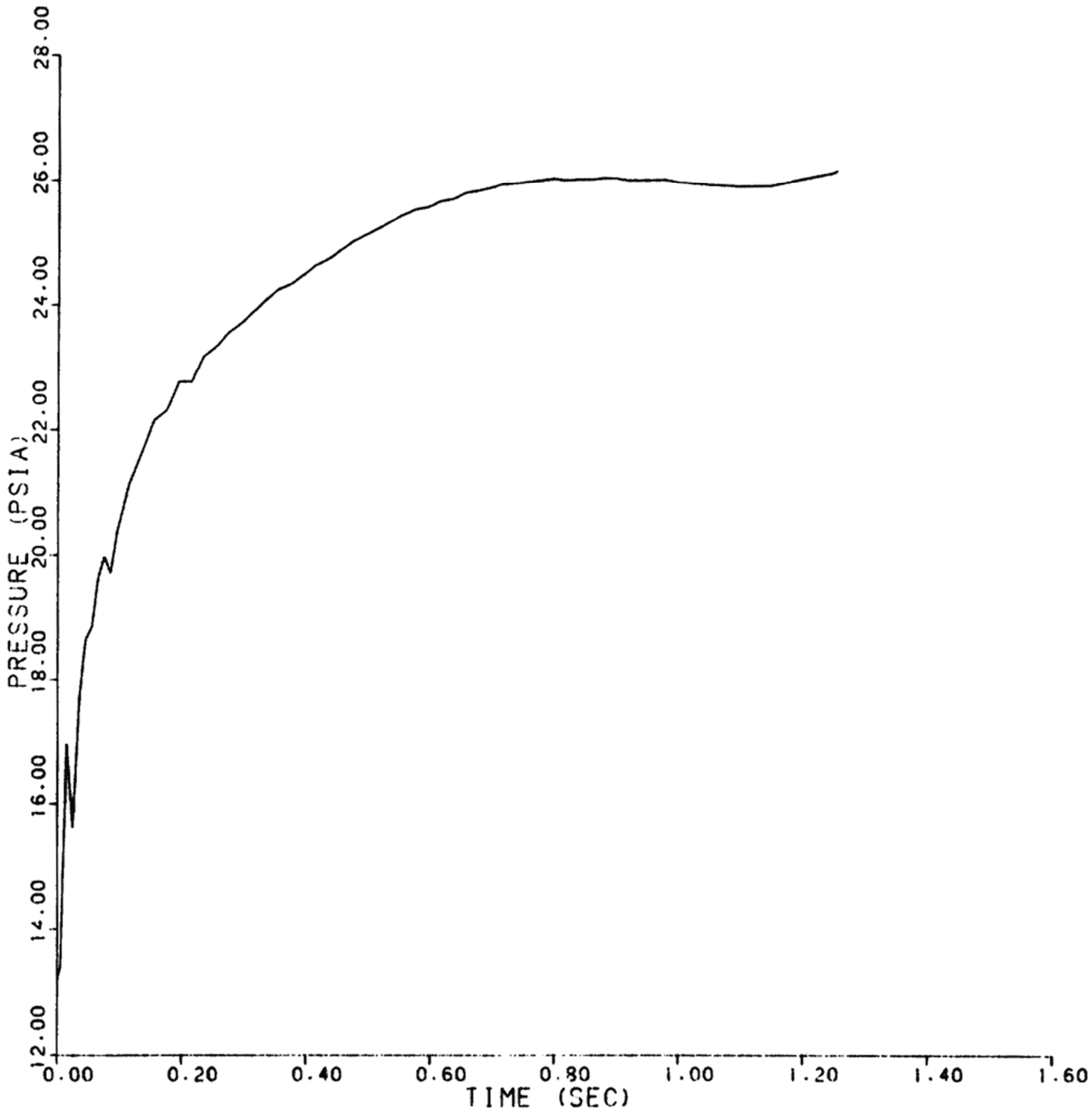
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E30 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 30 OF 74)



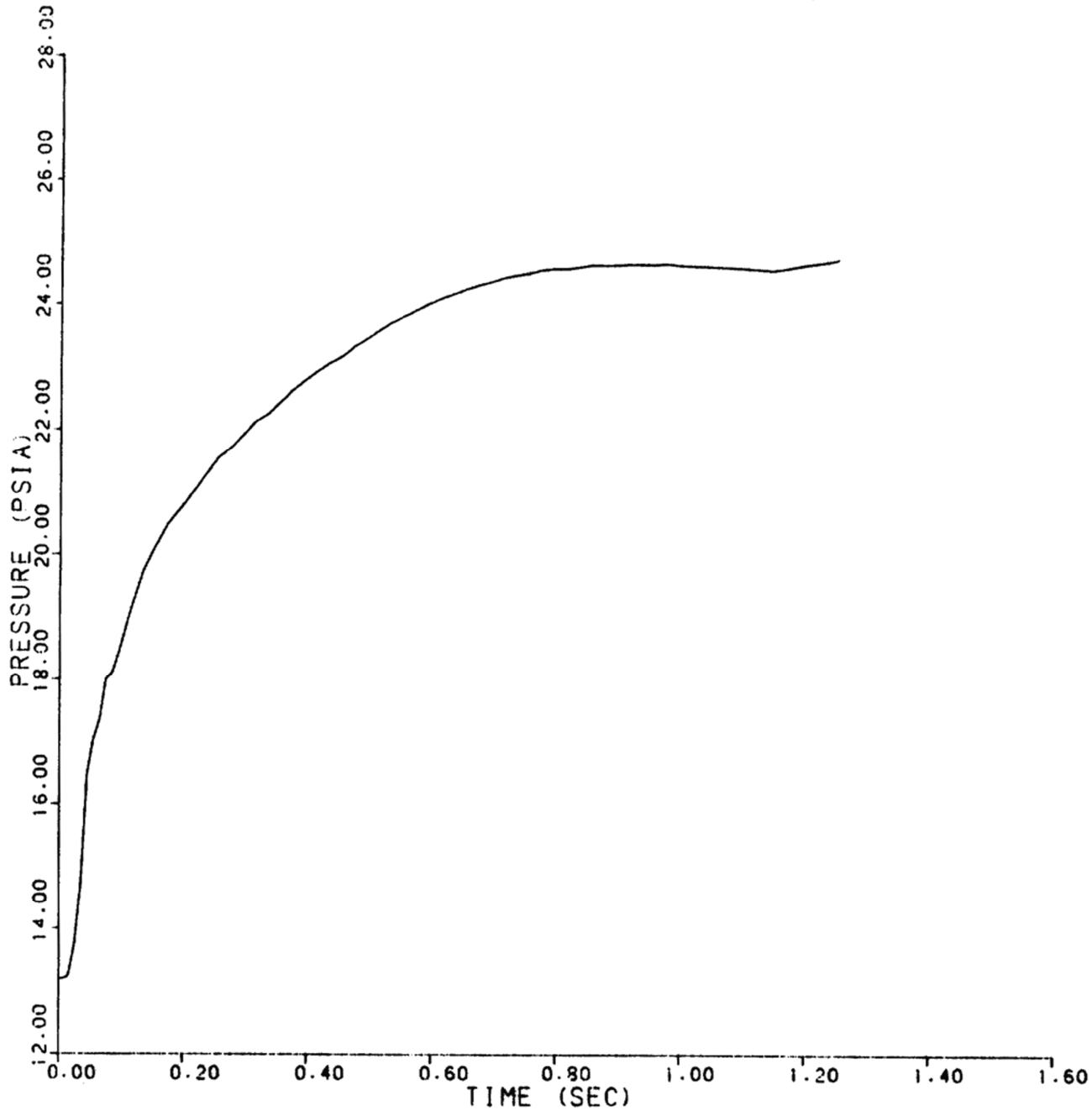
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E31 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 31 OF 74)



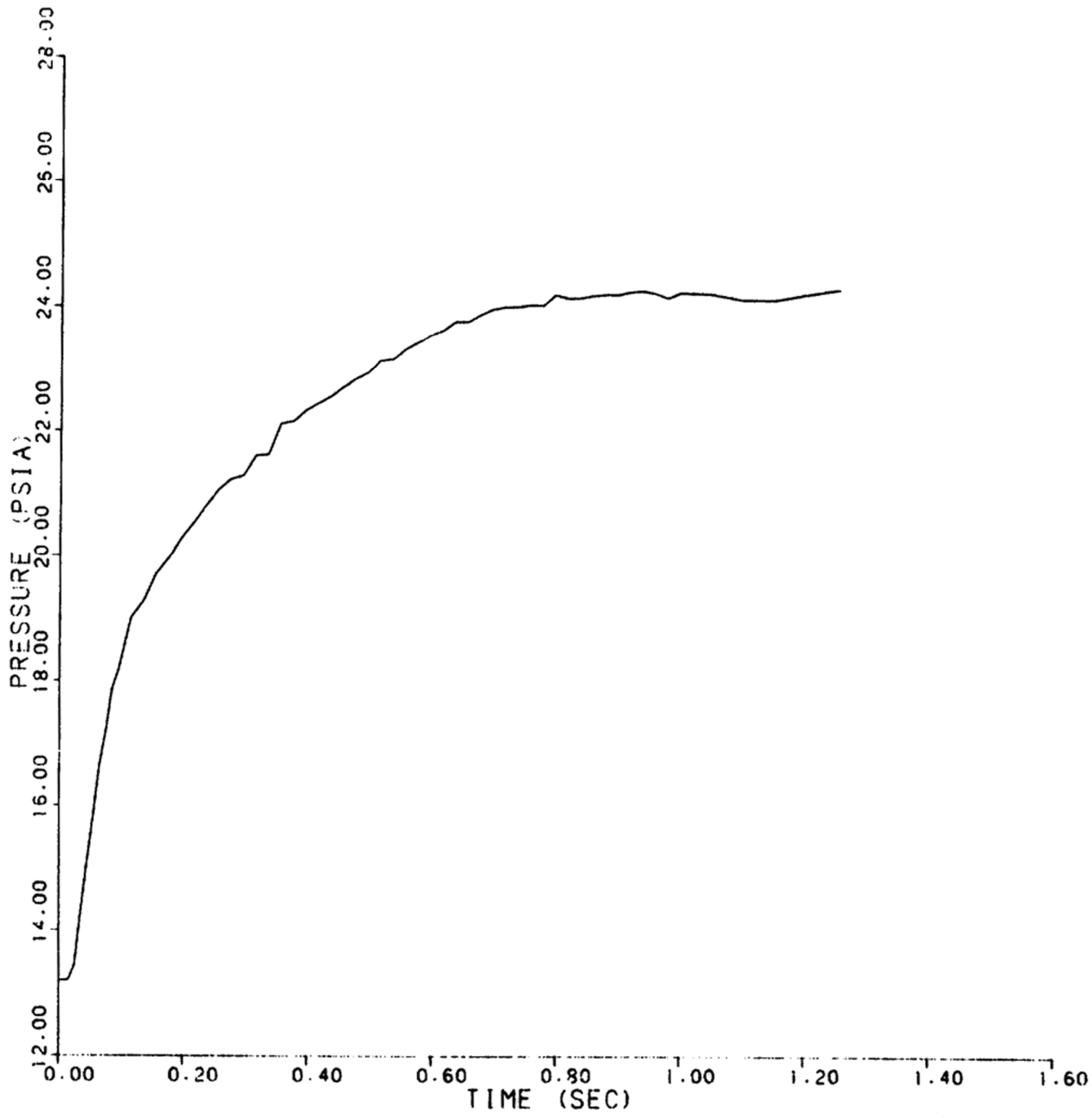
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E32 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 32 OF 74)



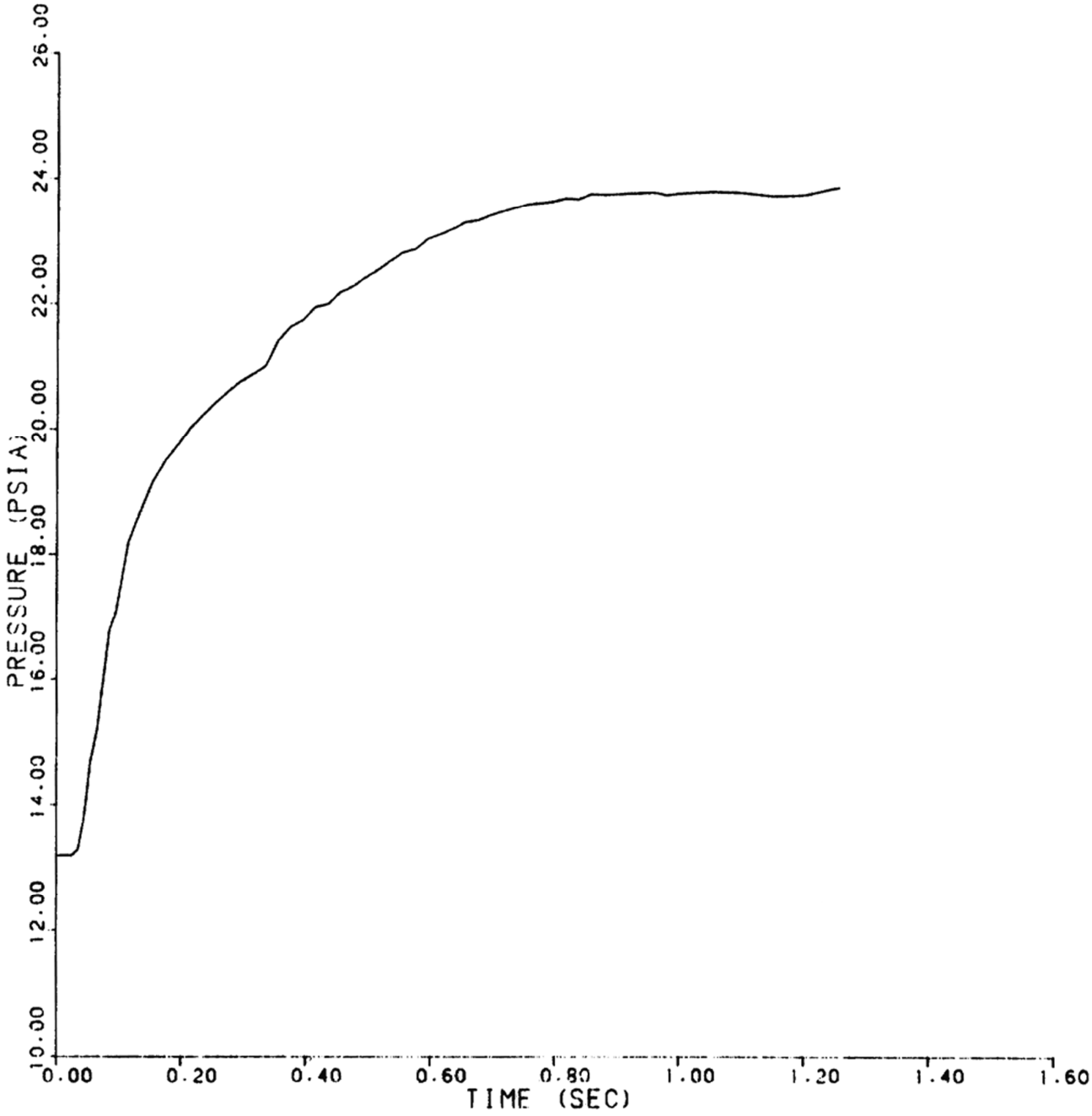
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E33 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 33 OF 74)



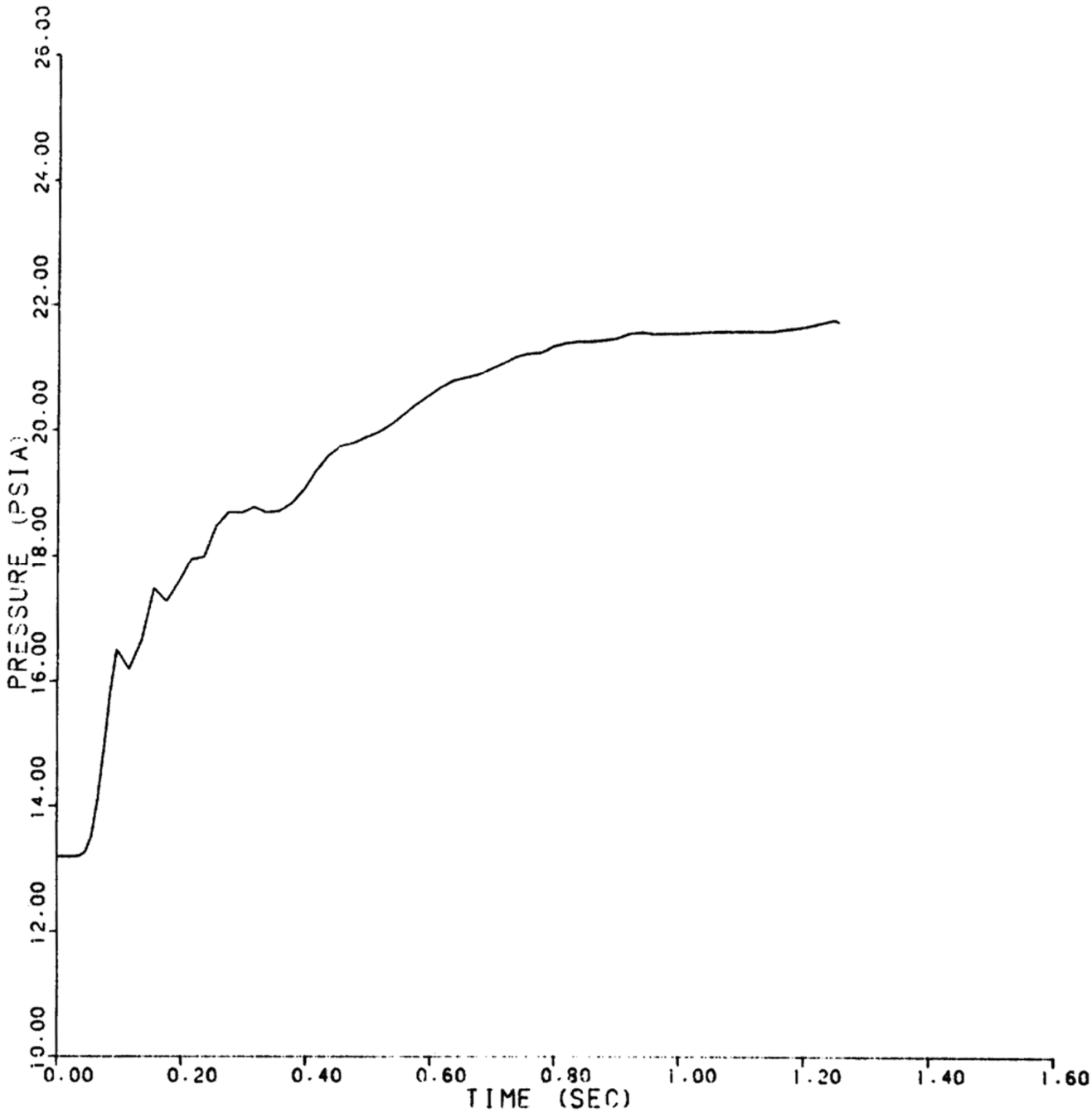
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E34 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 34 OF 74)



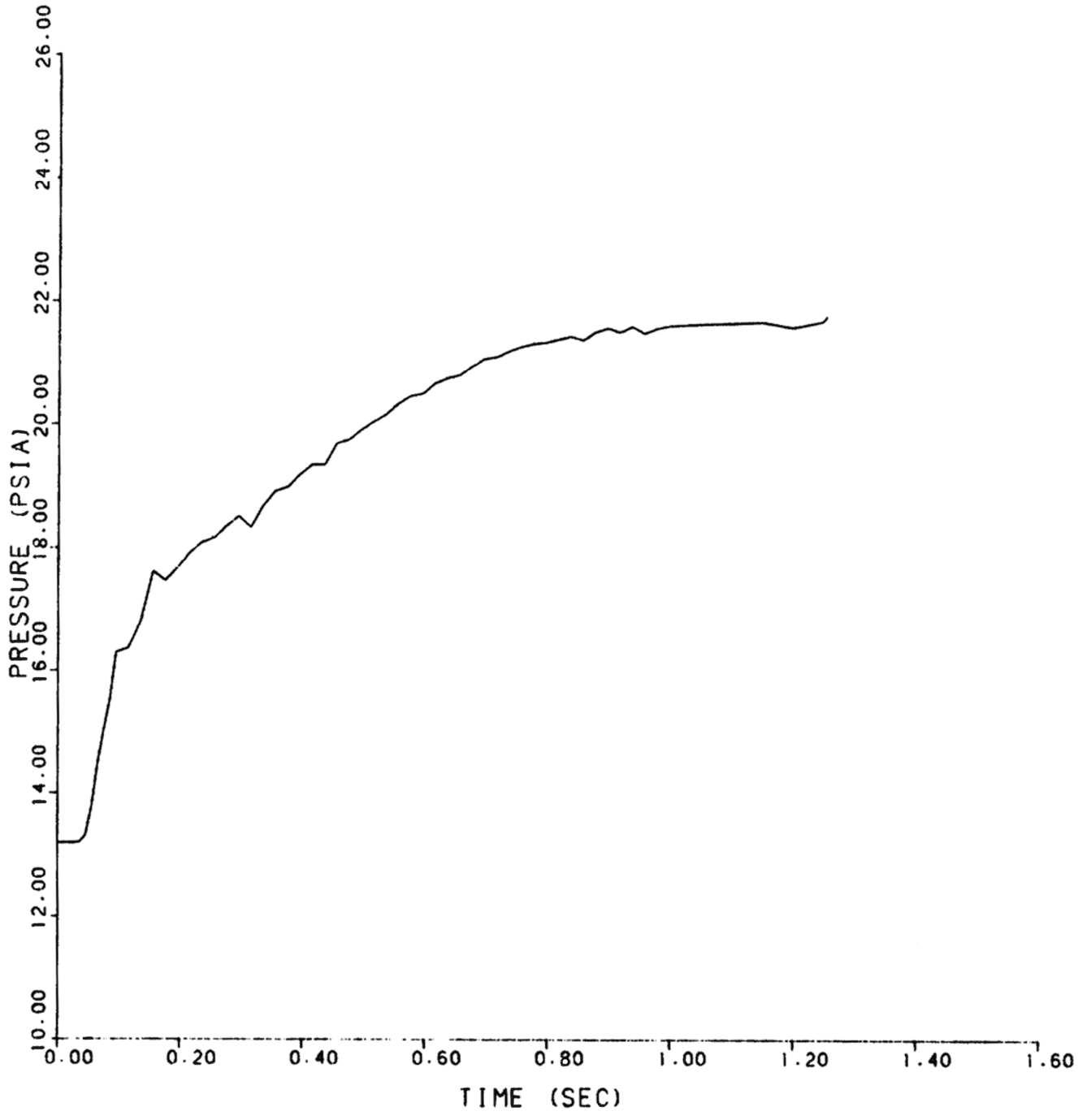
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E35 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 35 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E36 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 36 OF 74)



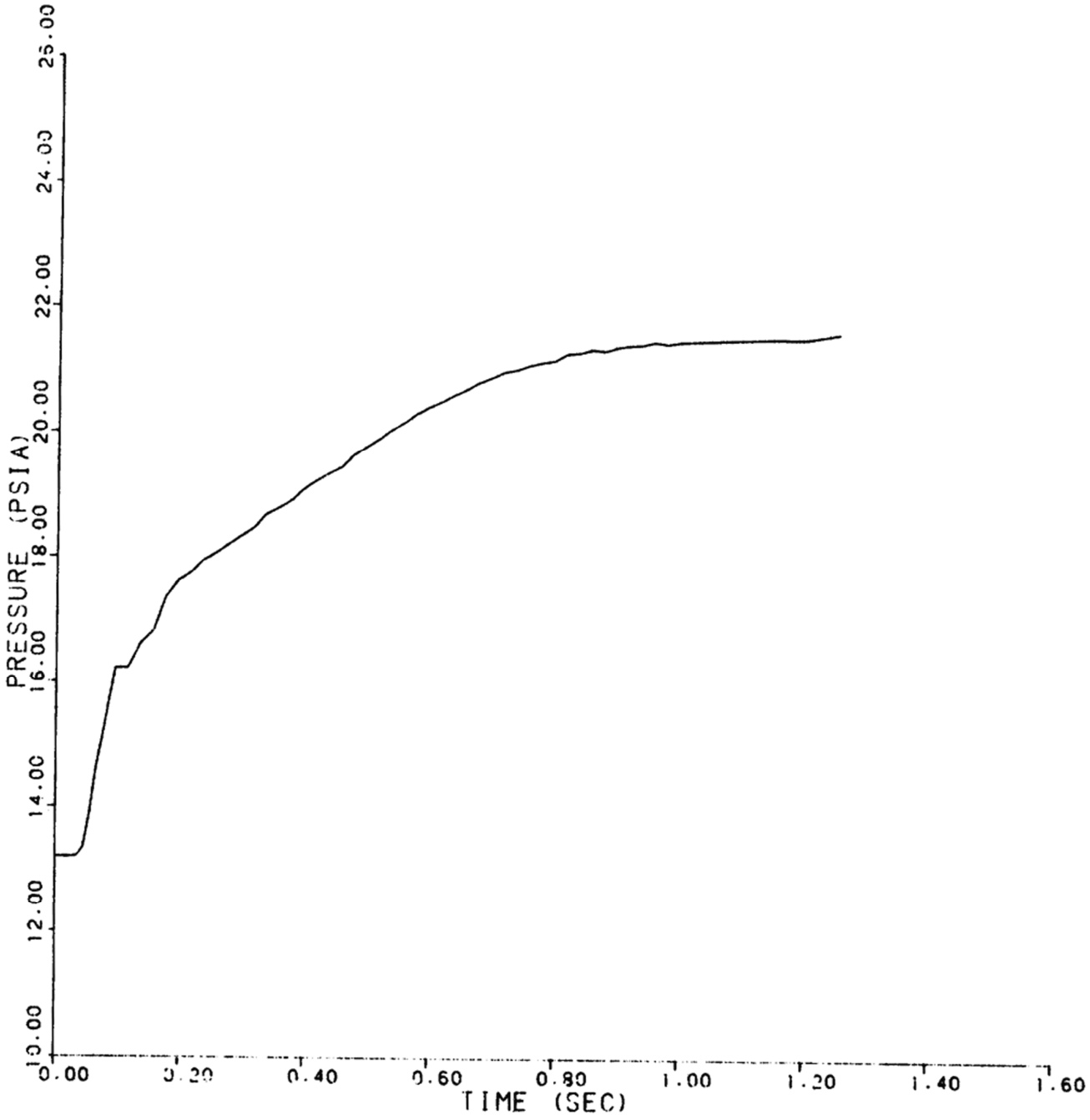
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

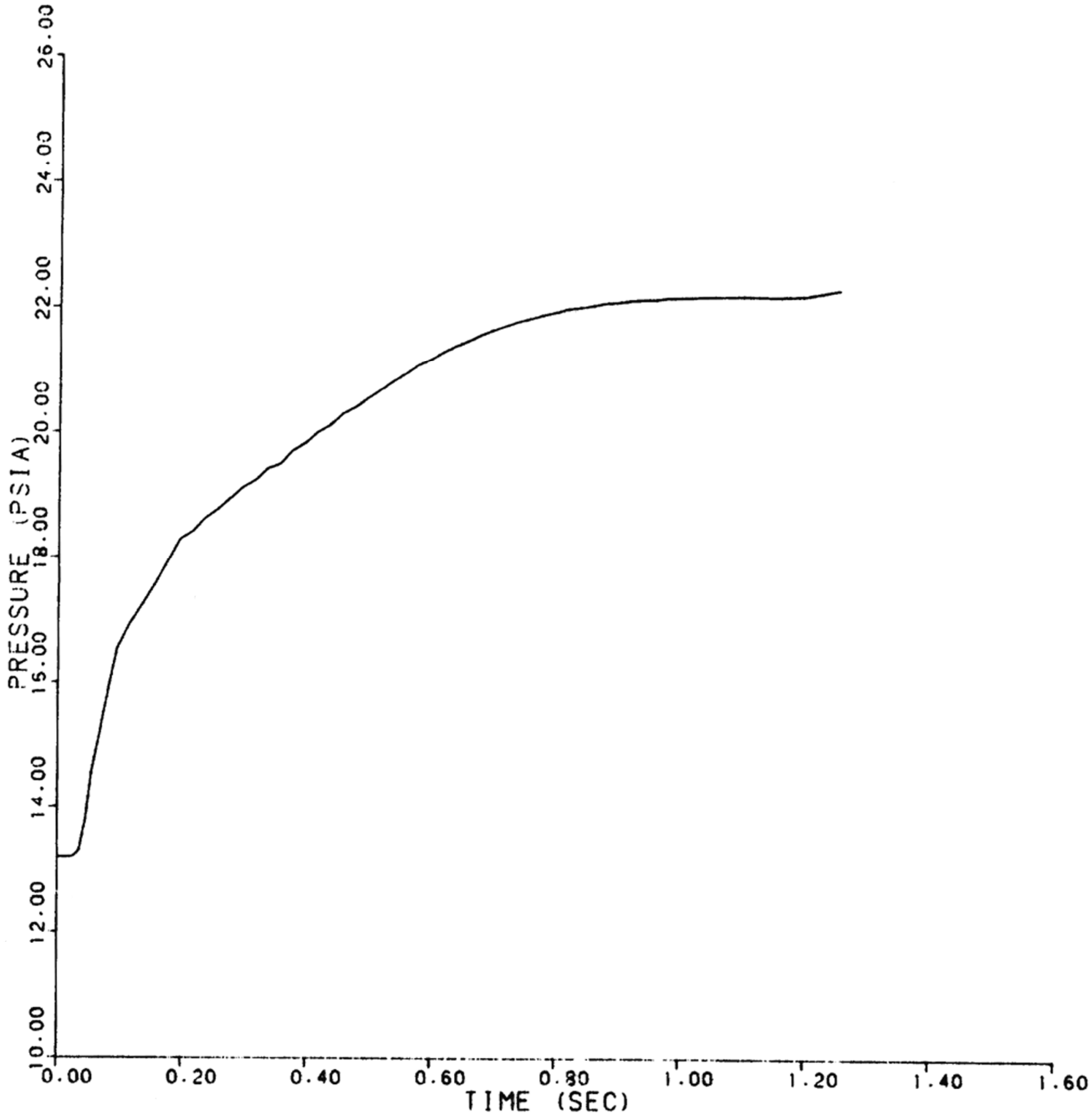
LOOP CLOSURE WELD
NODE E37
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 37 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E38 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 38 OF 74)



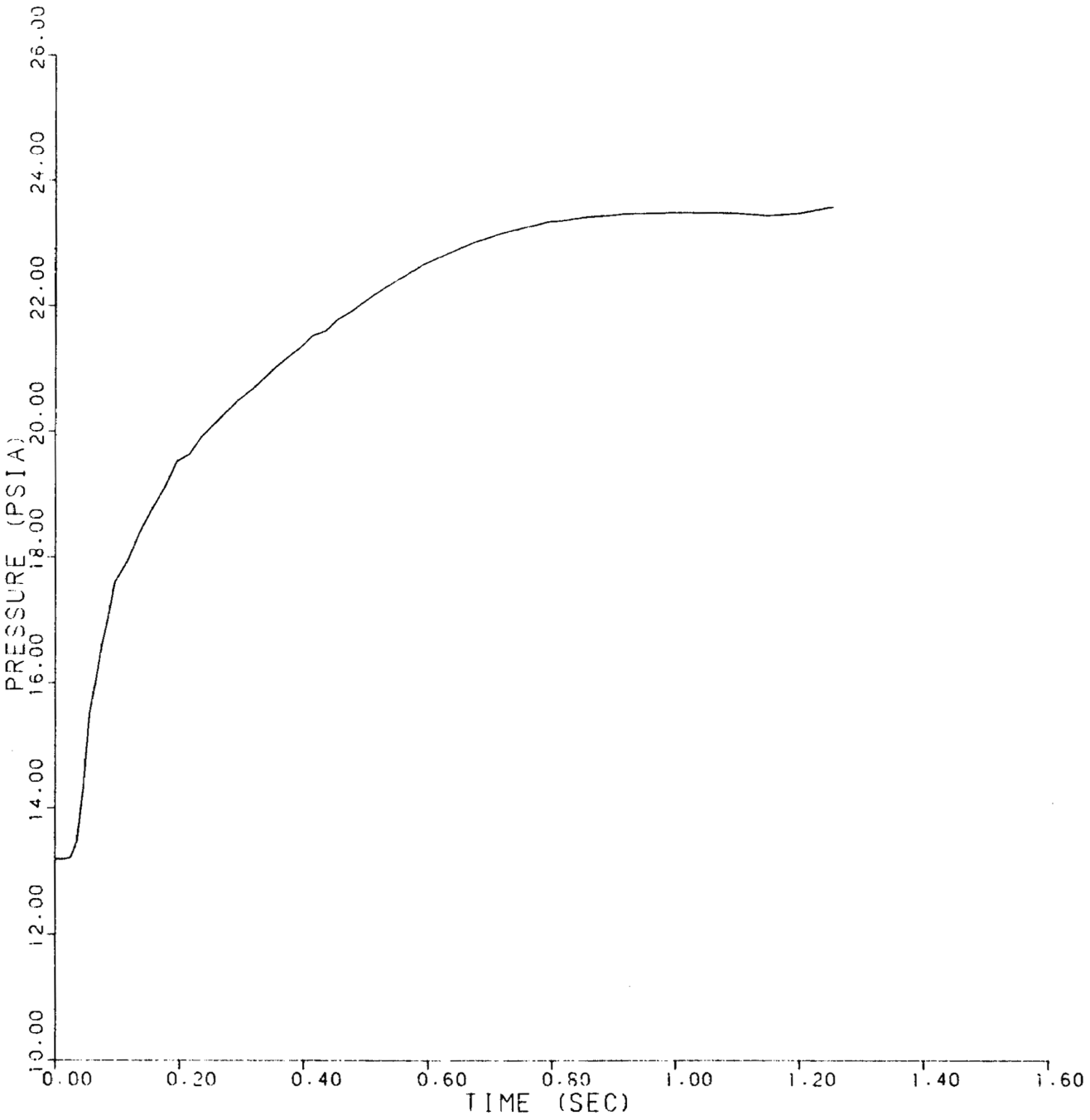
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

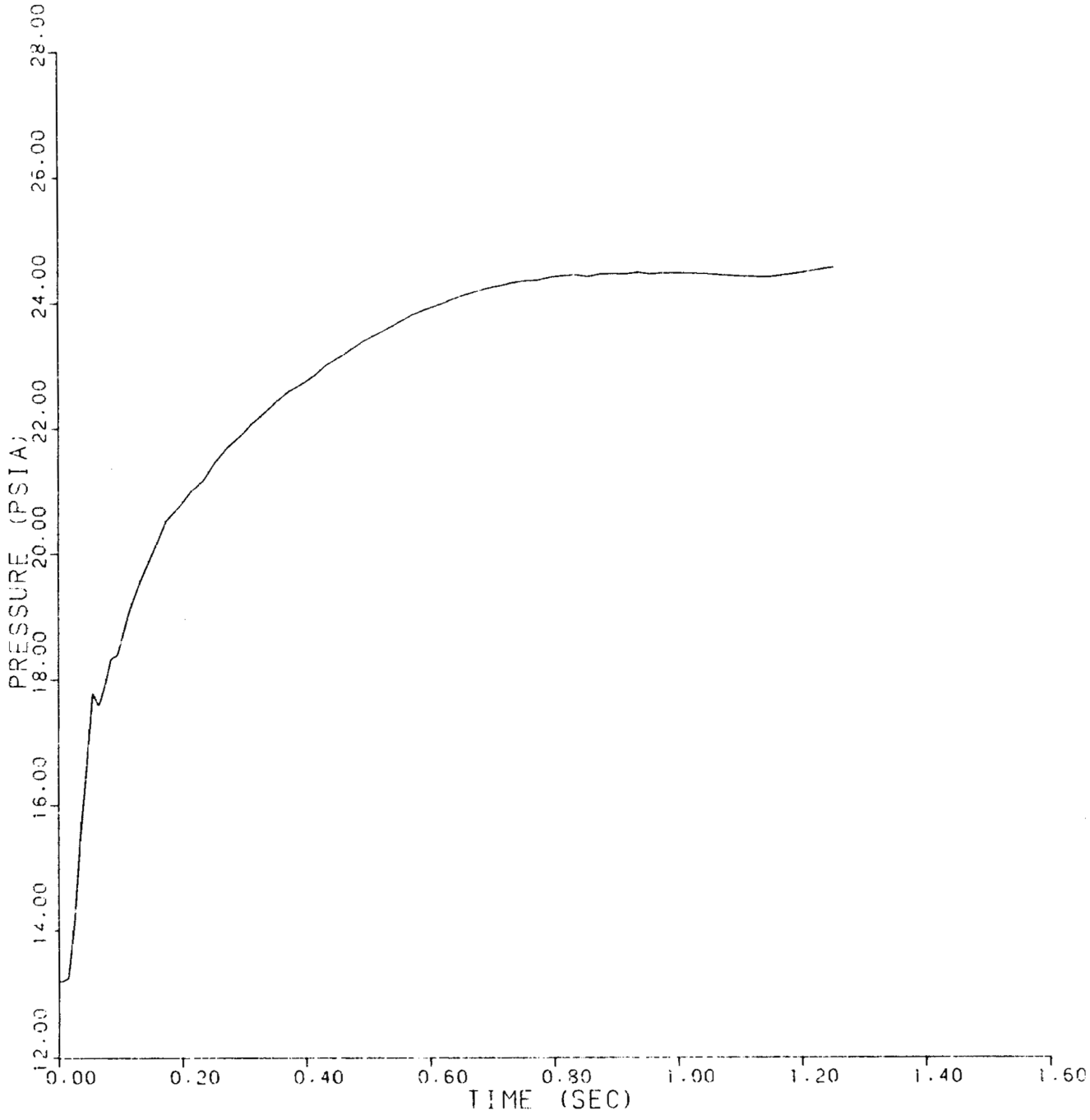
LOOP CLOSURE WELD
NODE E39
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 39 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E40 (336–in.^{2} BREAK AREA) FIGURE 6.2.1–23 (SHEET 40 OF 74)



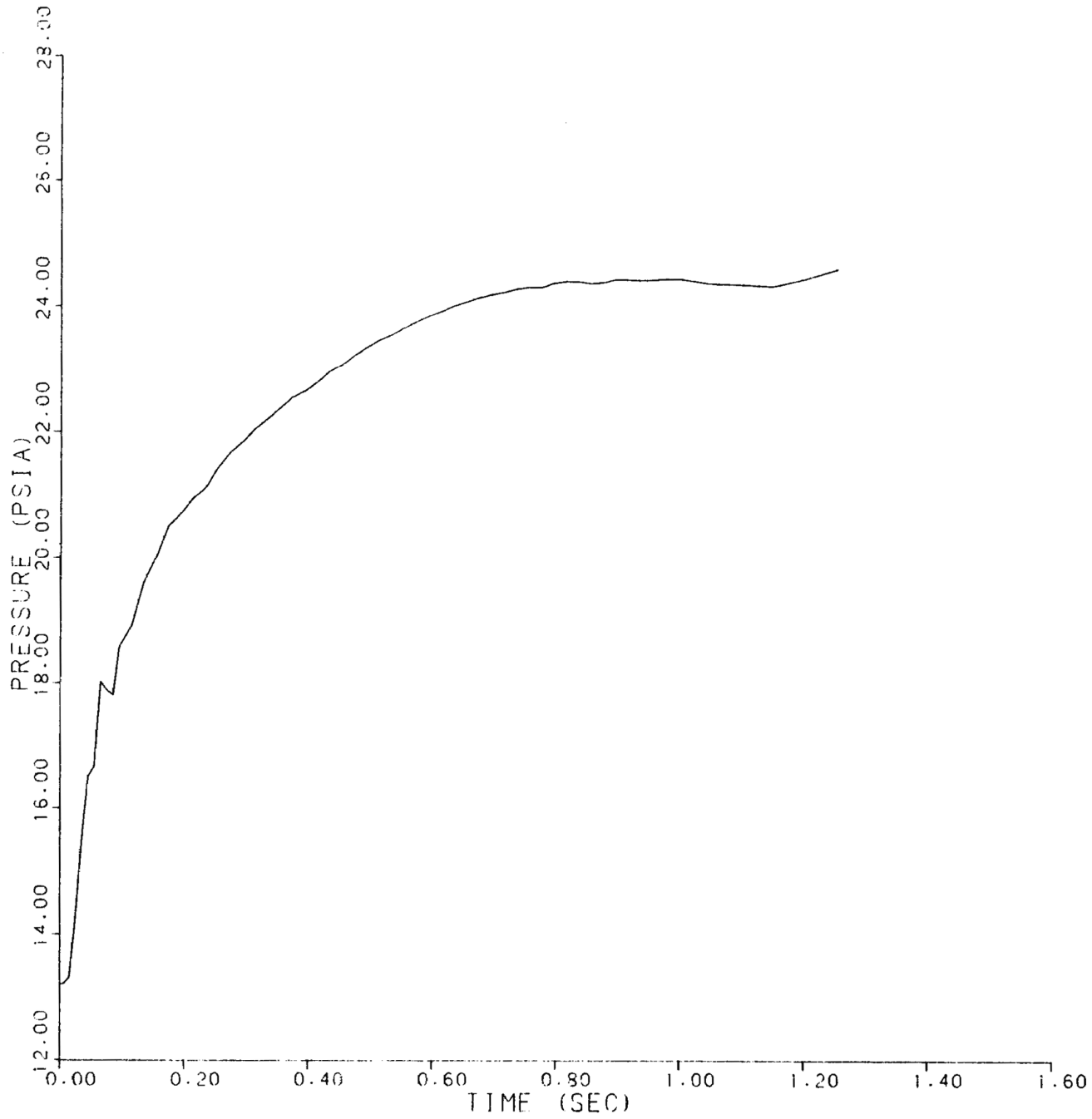
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

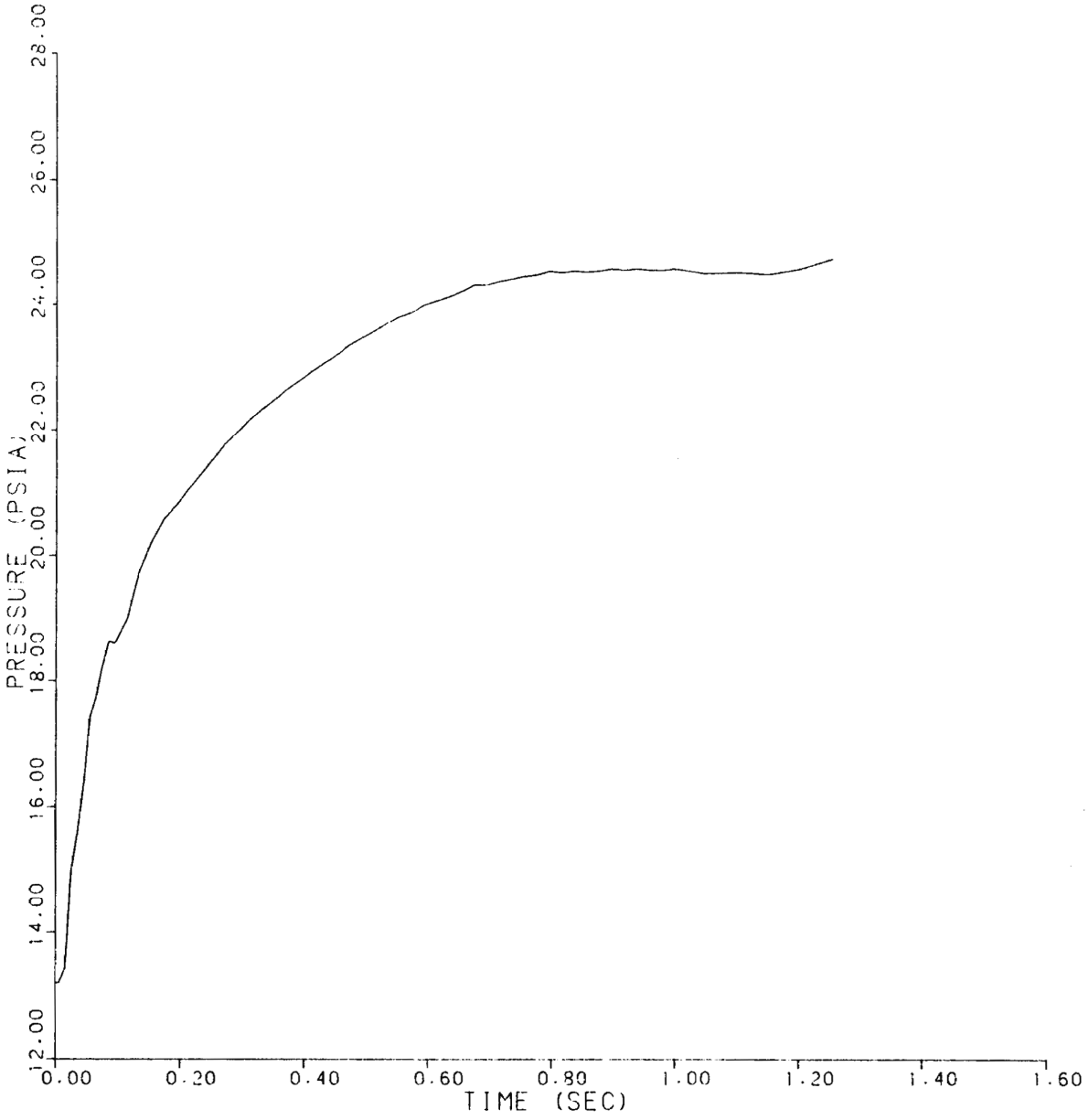
LOOP CLOSURE WELD
NODE E41
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 41 OF 74)



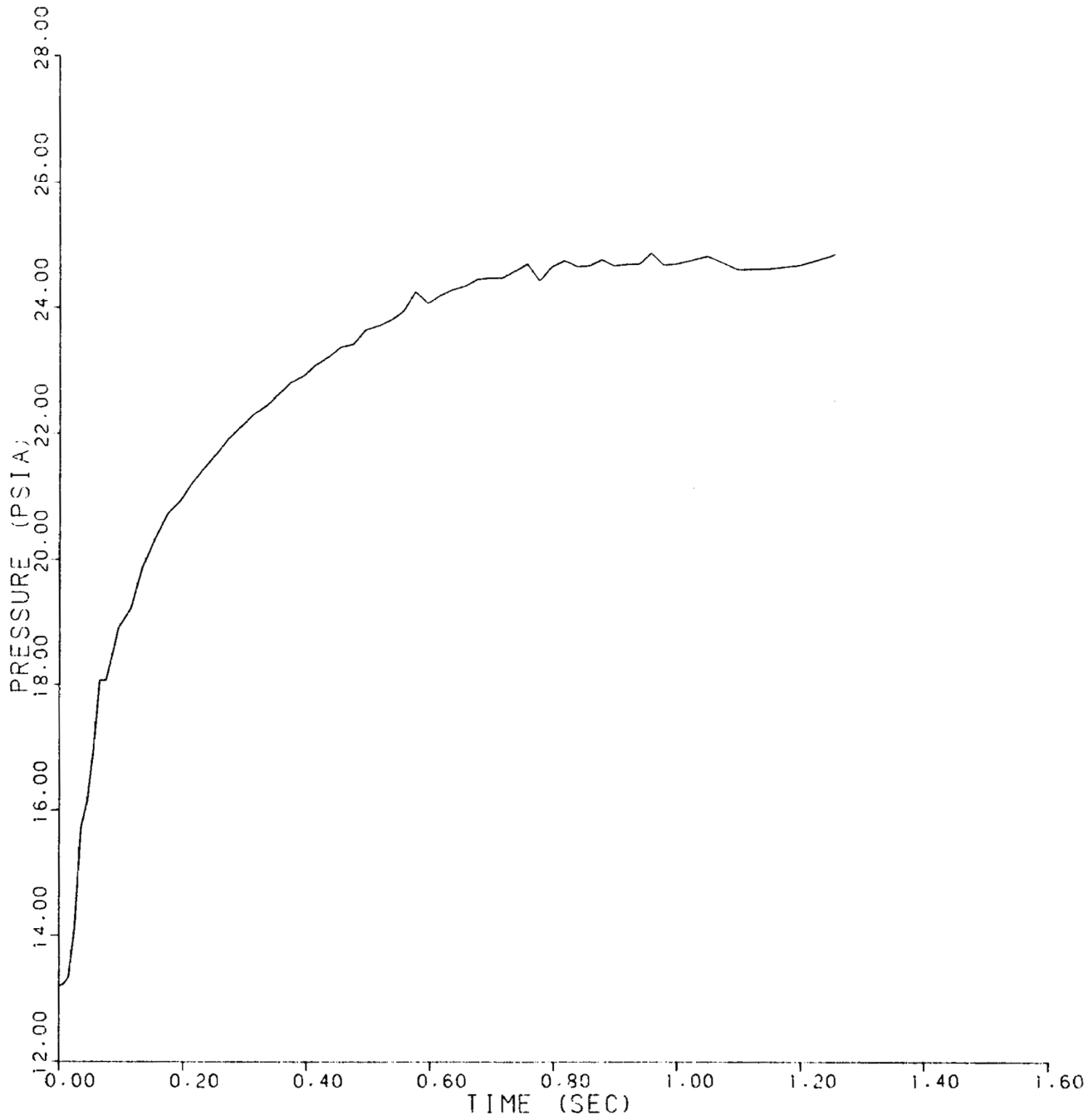
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E42 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 42 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E43 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 43 OF 74)



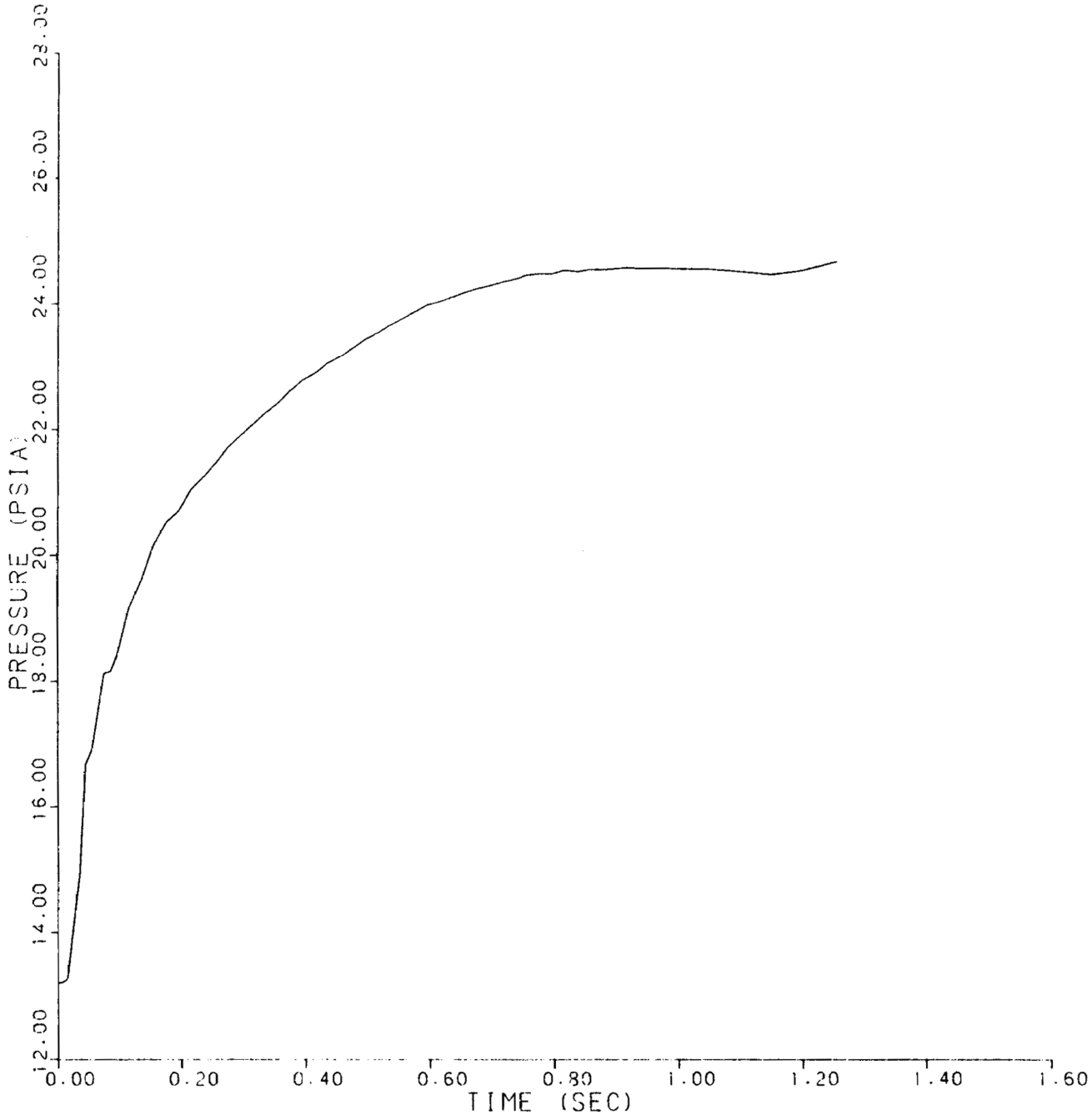
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E44
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 44 OF 74)



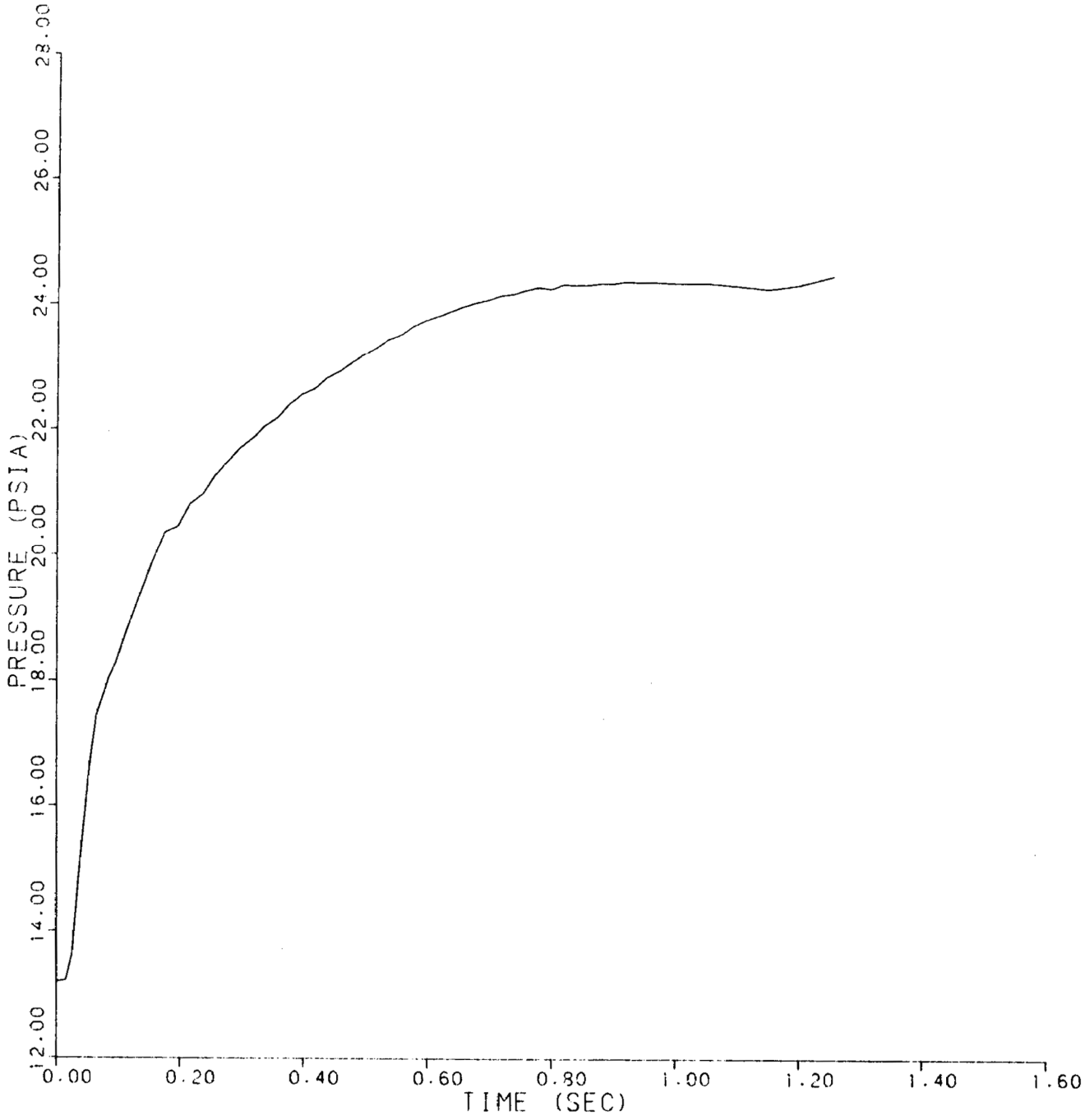
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E45
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 45 OF 74)



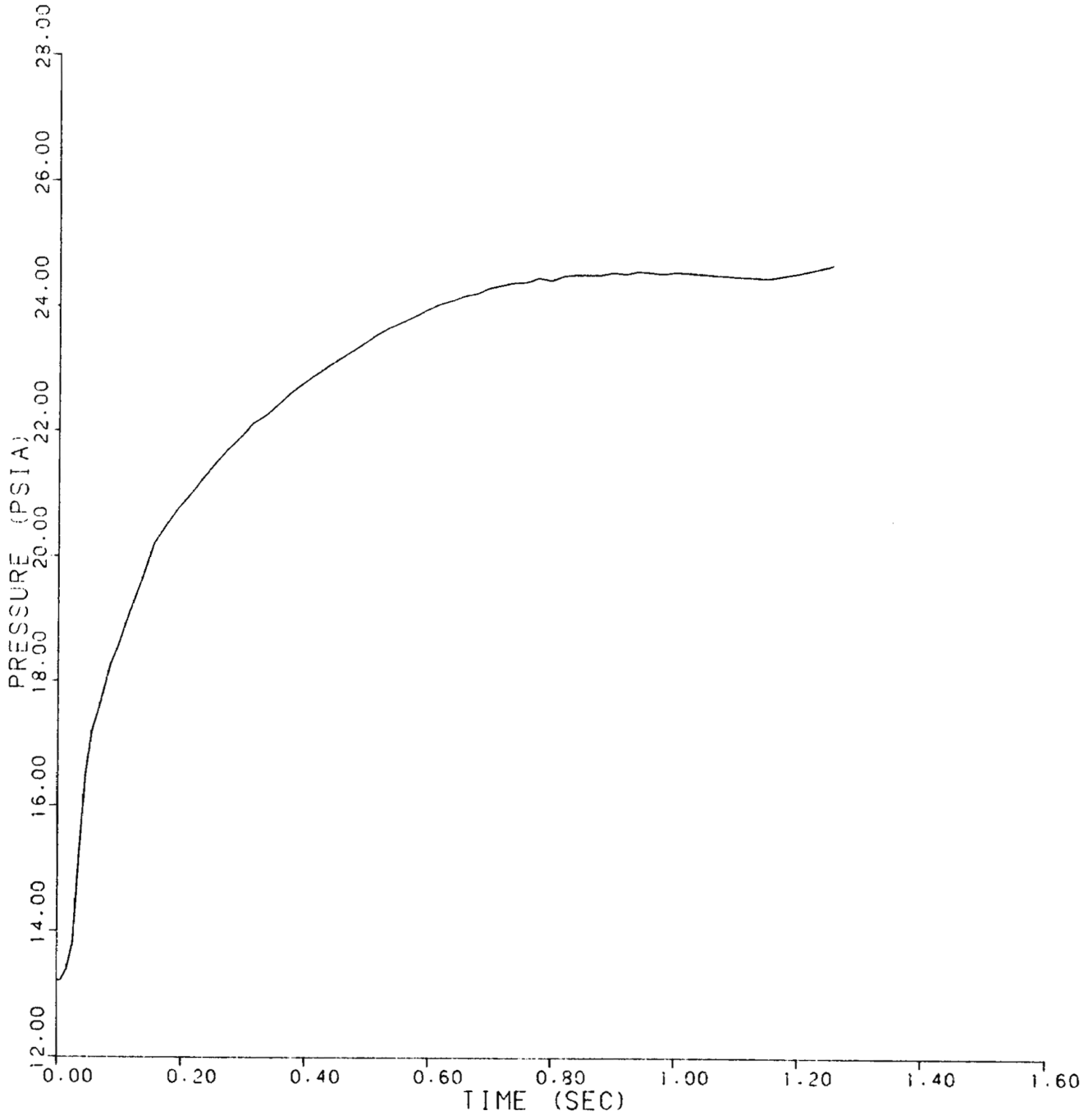
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

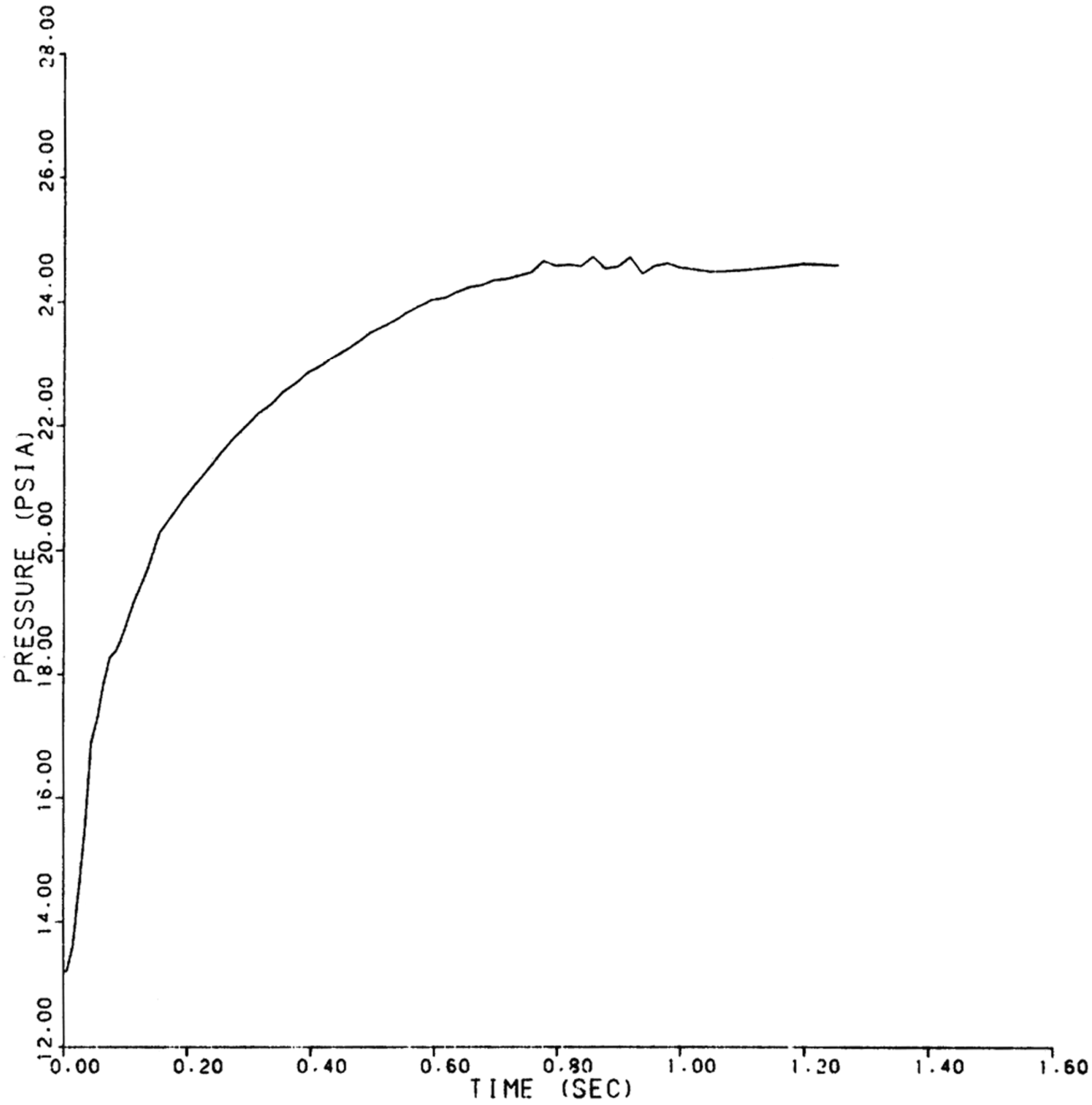
LOOP CLOSURE WELD
NODE E46
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 46 OF 74)



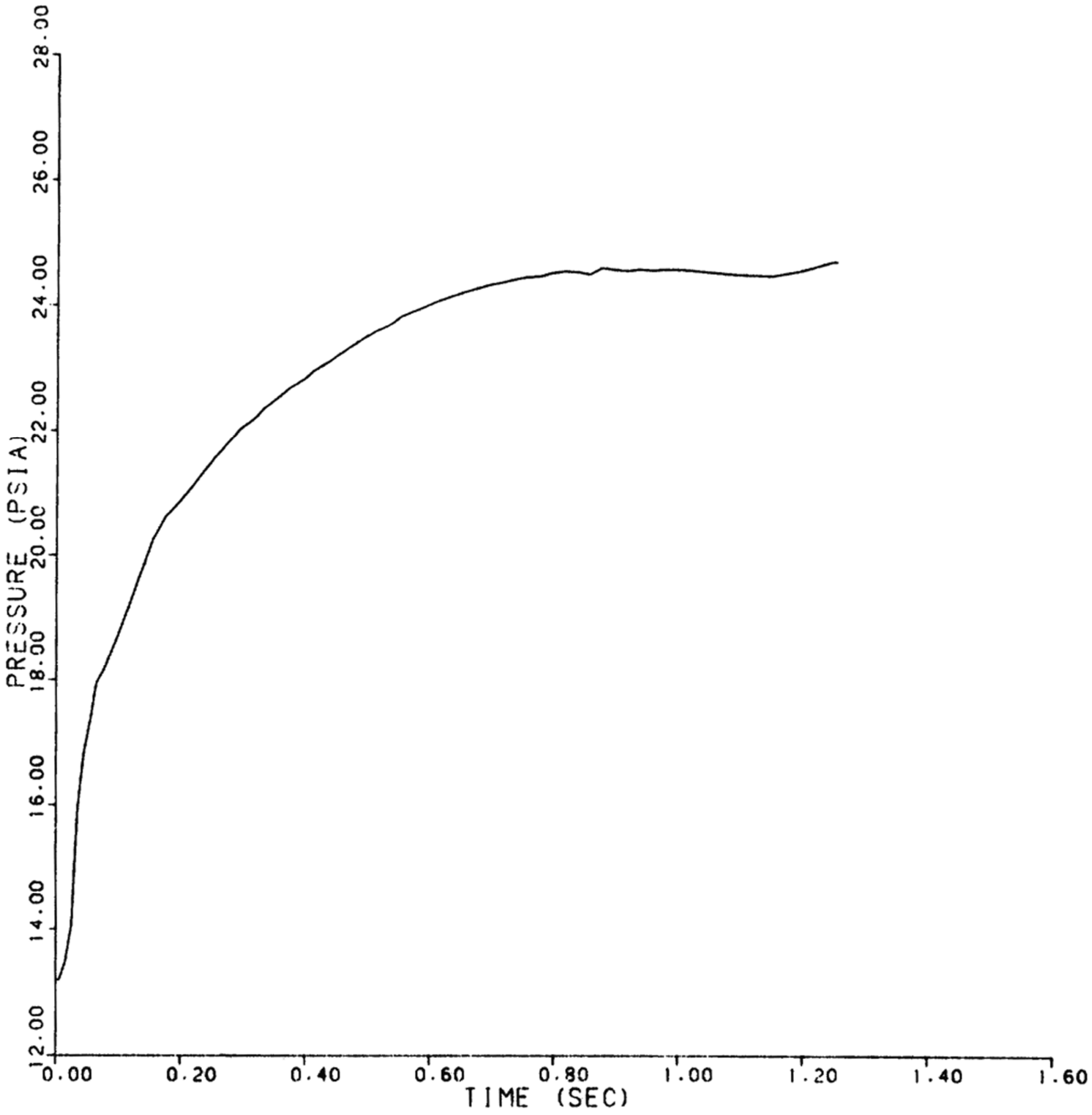
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E47 (336–in.^{2} BREAK AREA) FIGURE 6.2.1–23 (SHEET 47 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E48 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 48 OF 74)



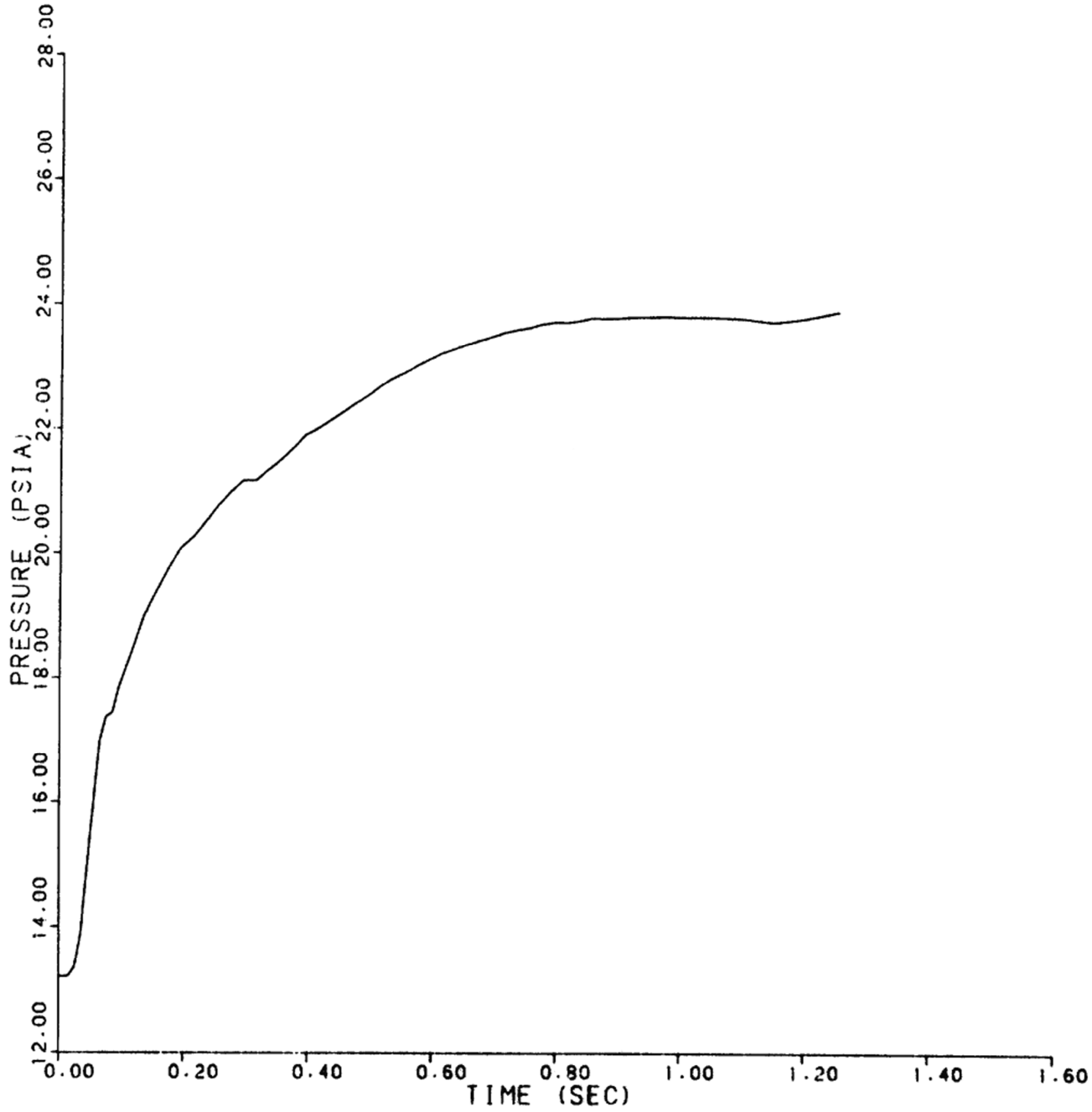
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

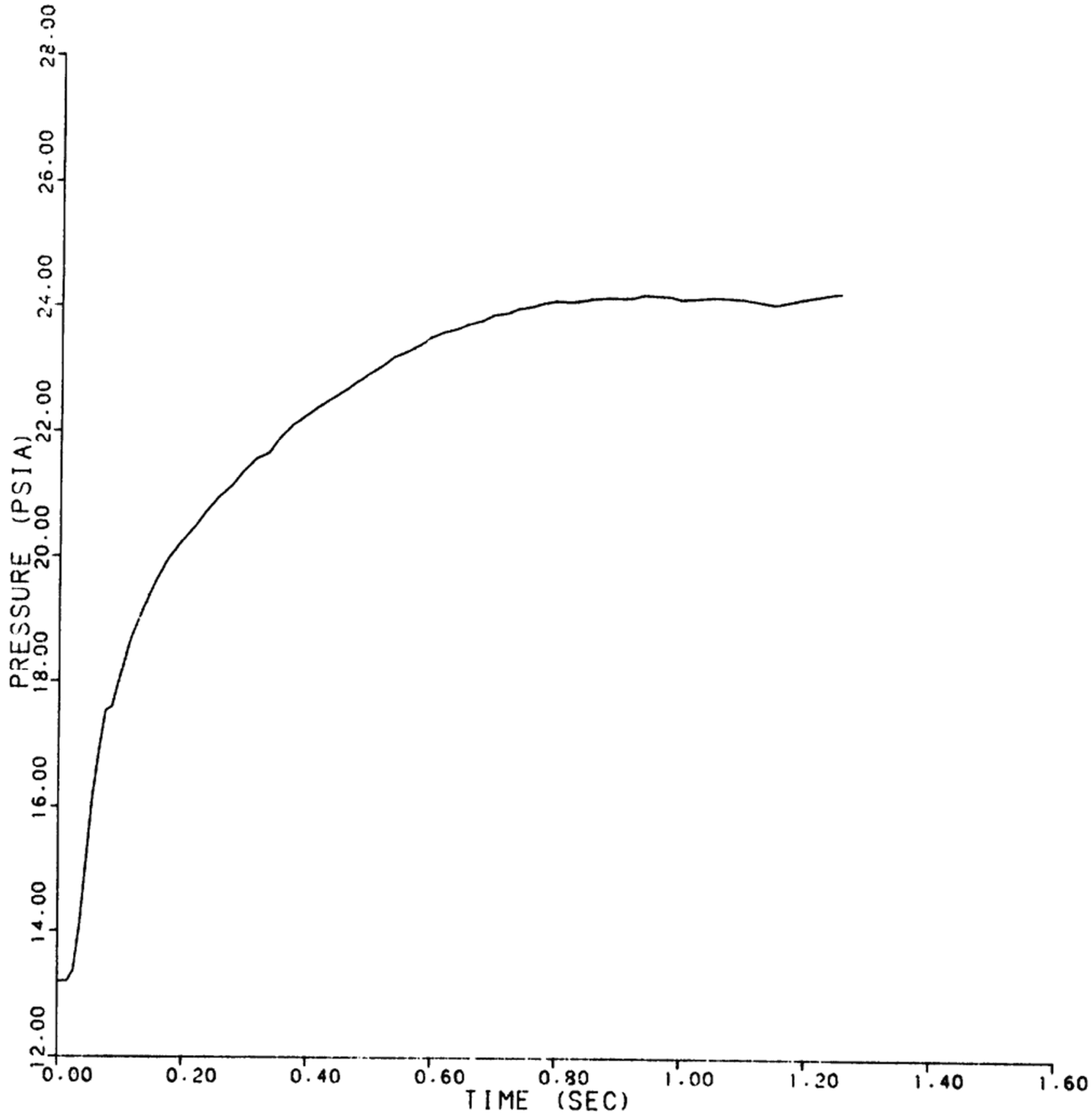
LOOP CLOSURE WELD
NODE E49
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 49 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E50 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 50 OF 74)



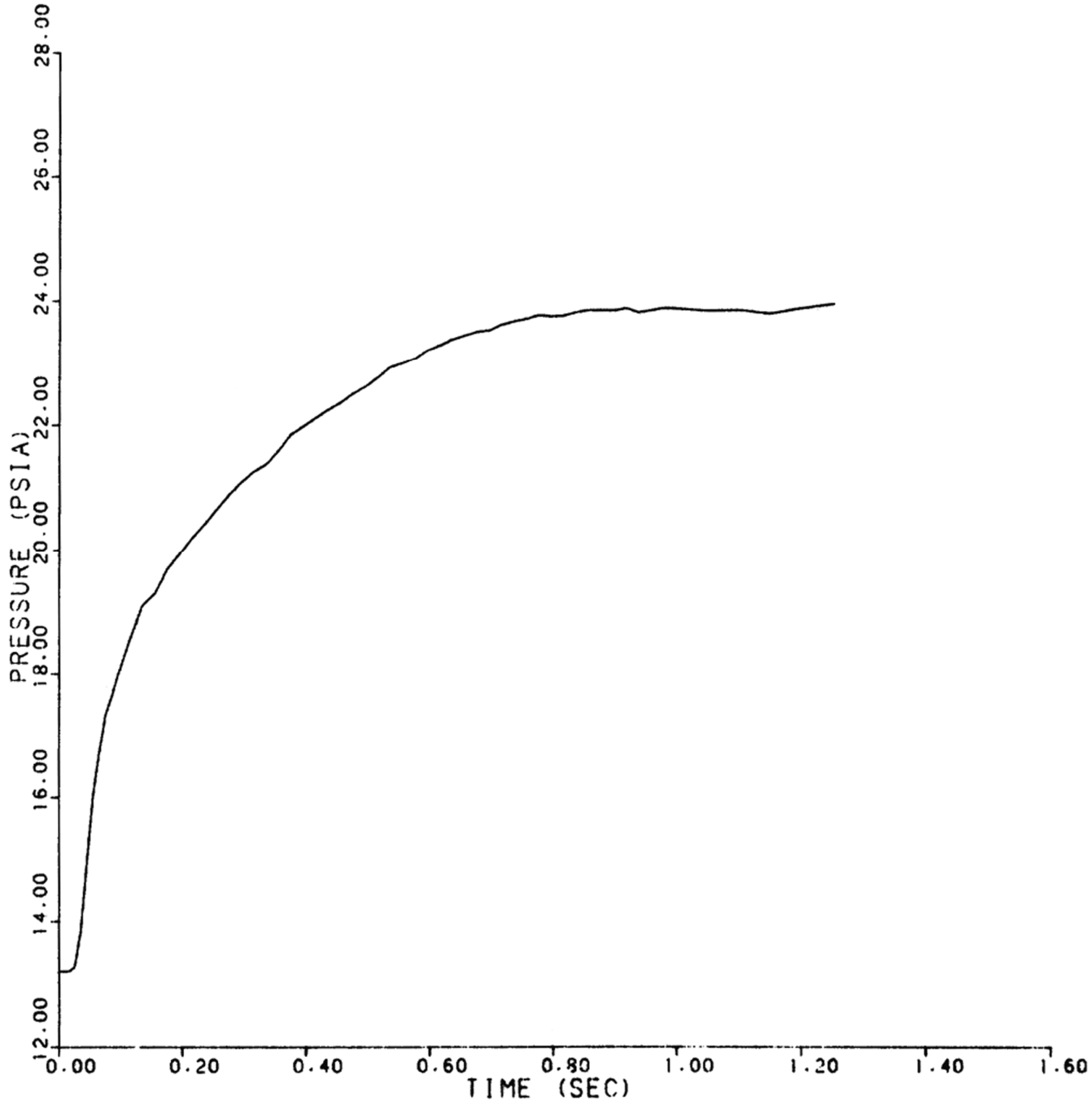
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

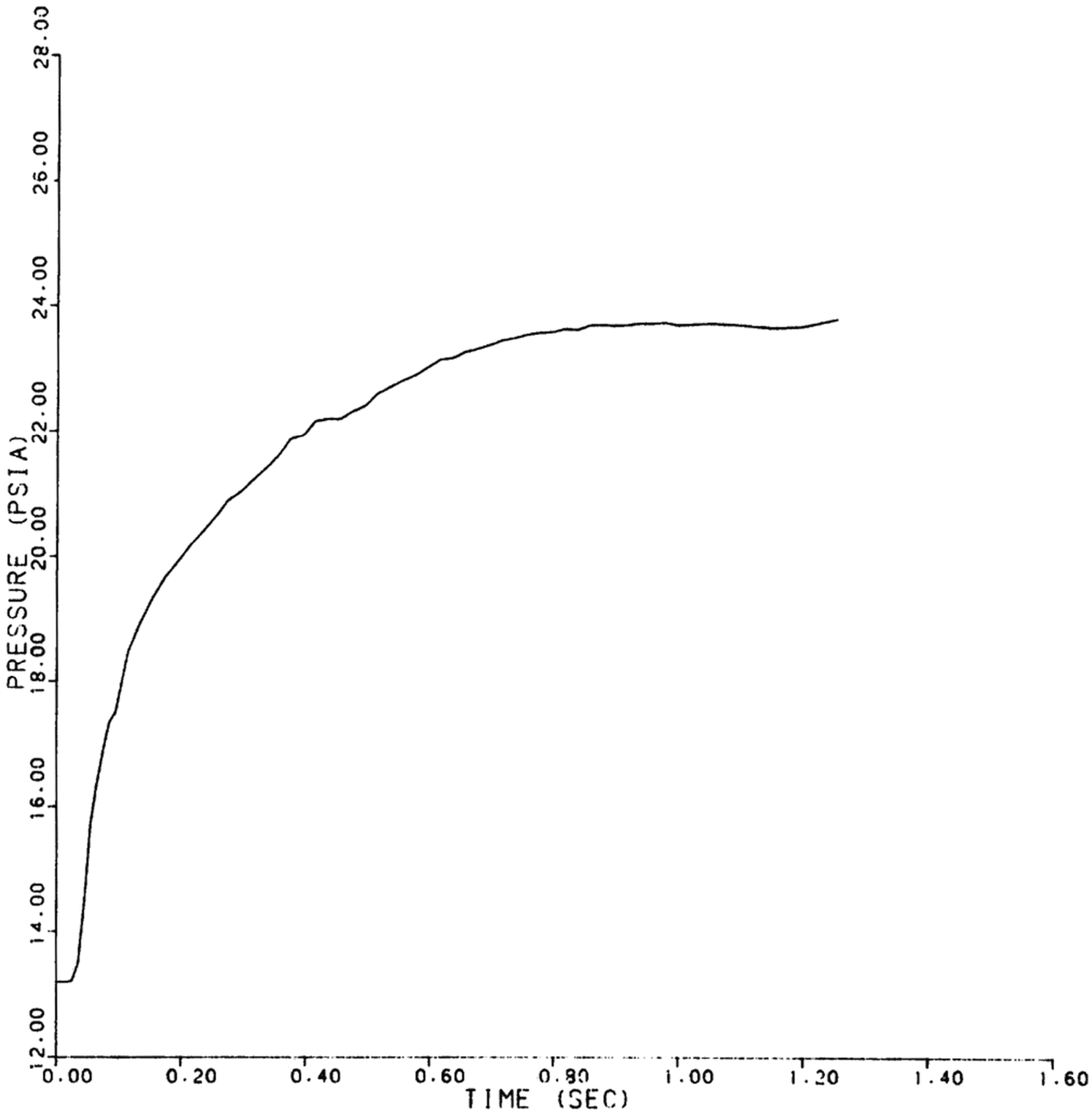
LOOP CLOSURE WELD
NODE E51
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 51 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E52 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 52 OF 74)



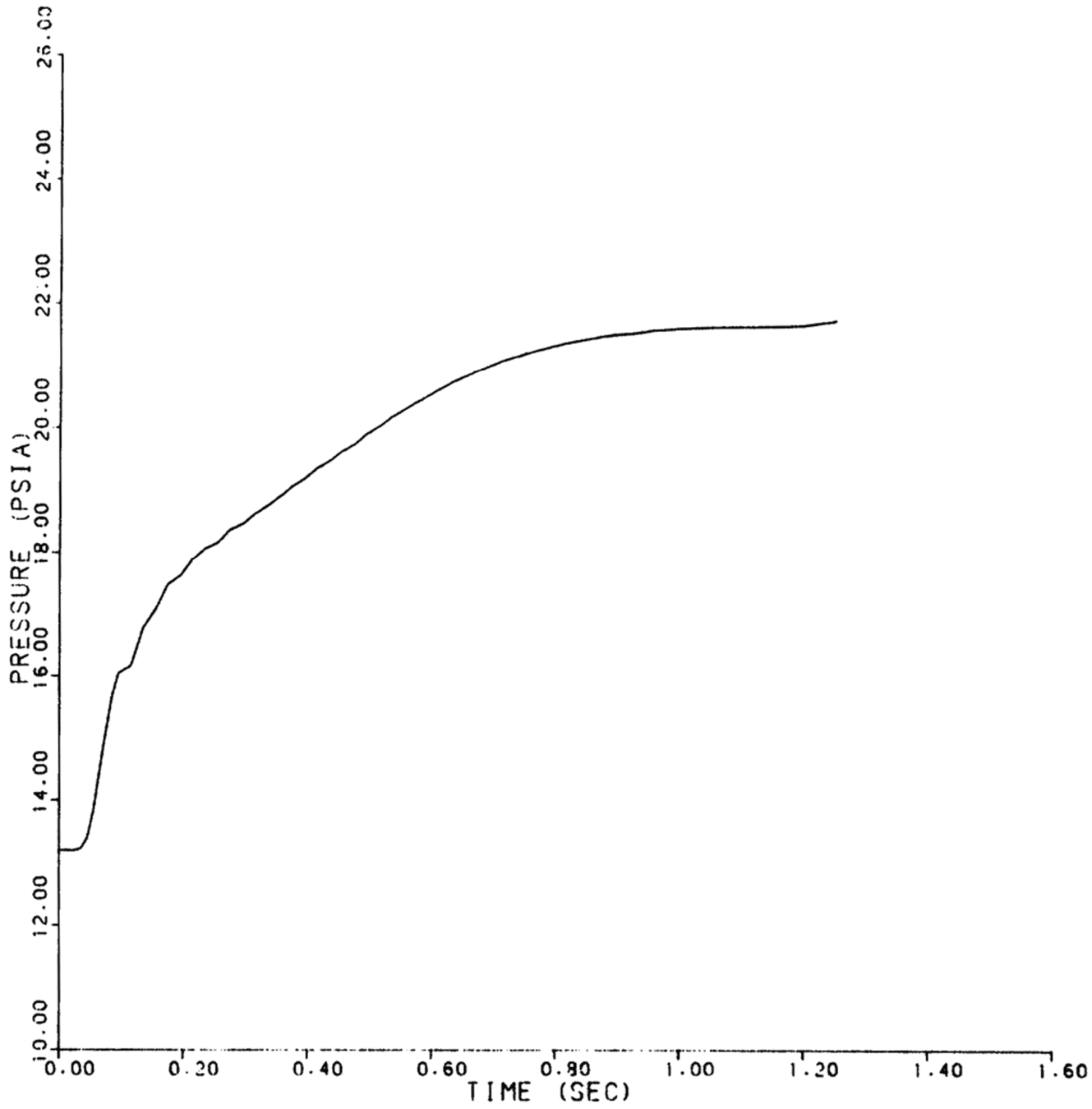
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

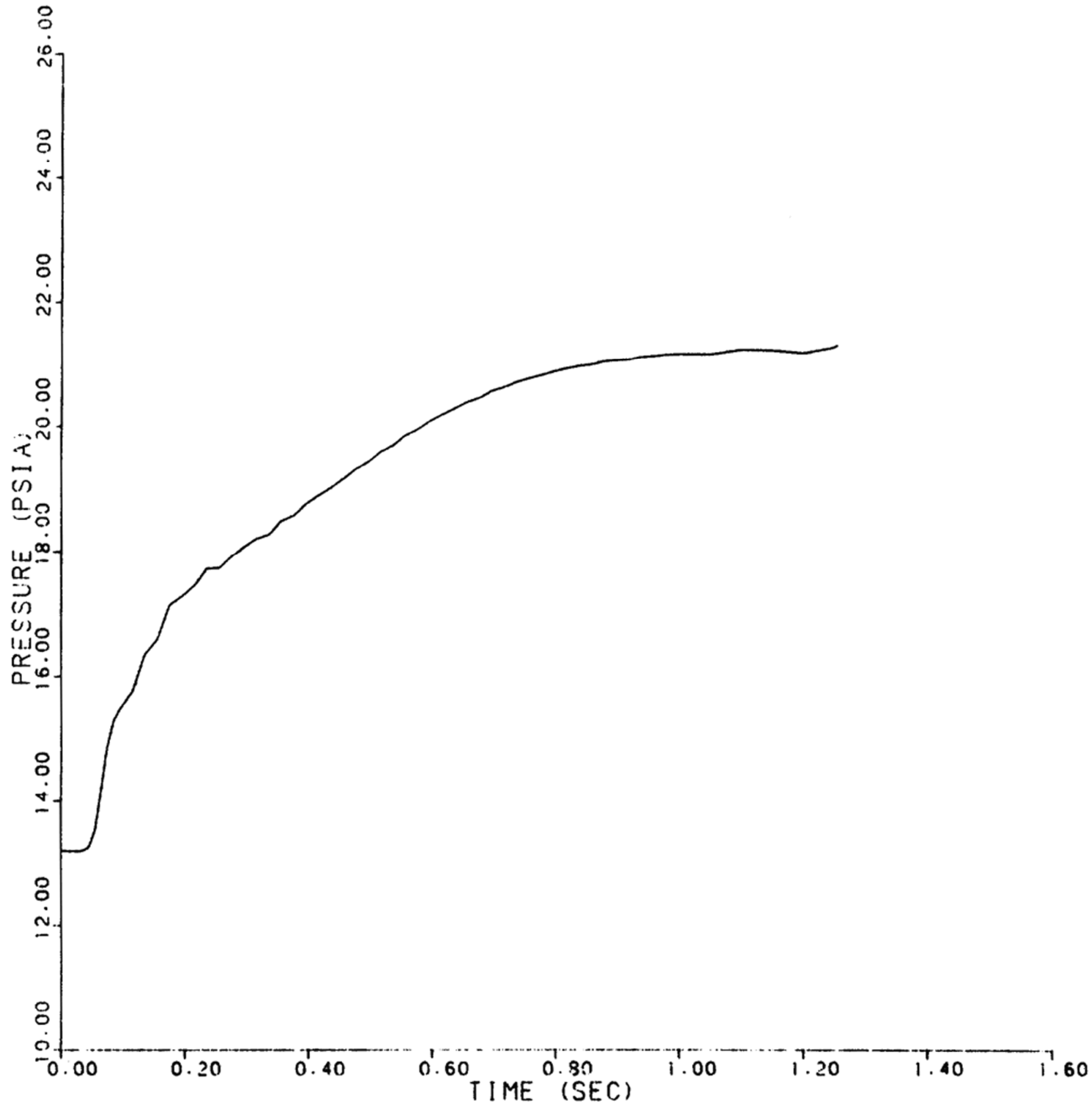
LOOP CLOSURE WELD
NODE E53
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 53 OF 74)



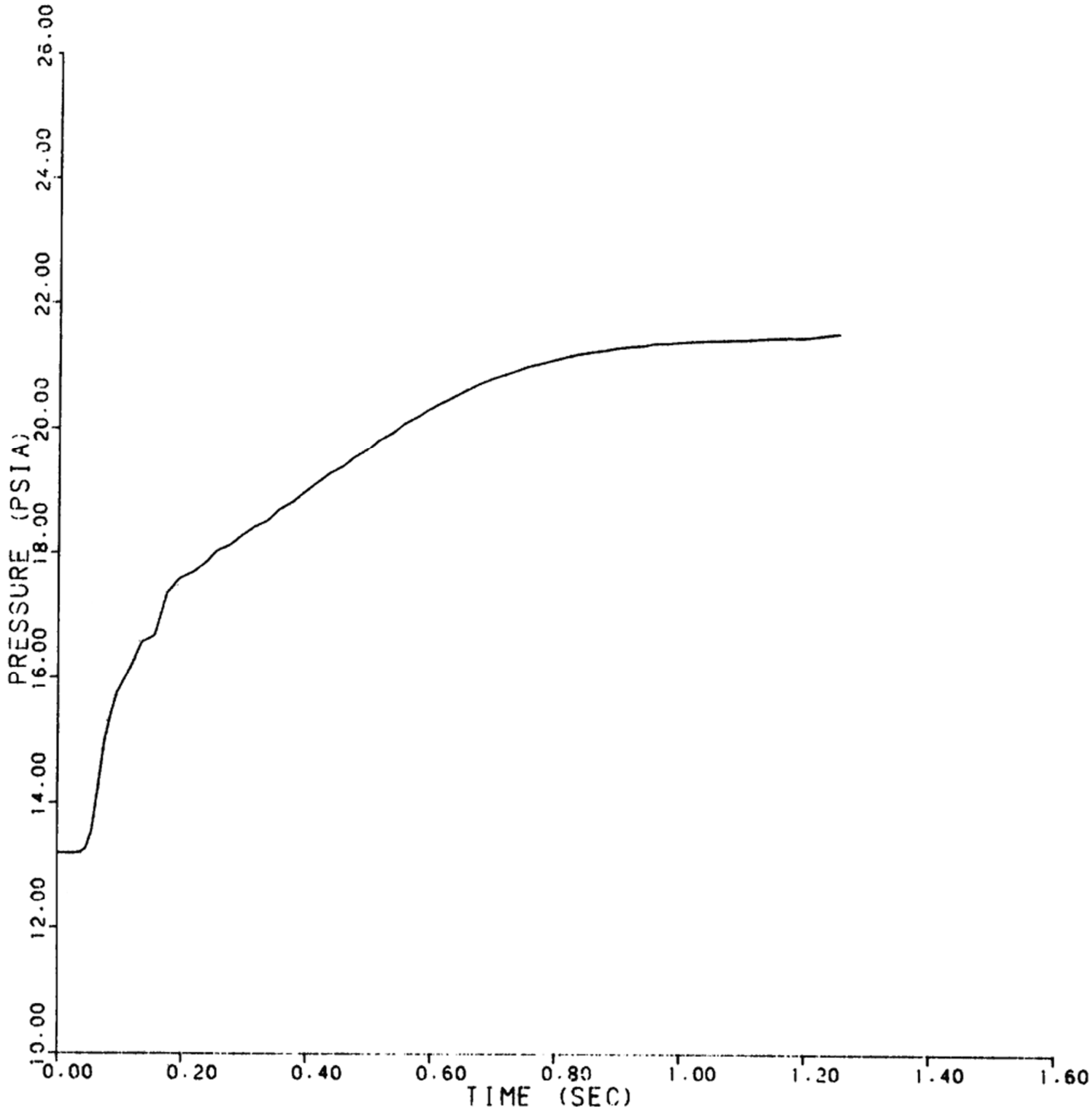
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E54 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 54 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E55 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 55 OF 74)



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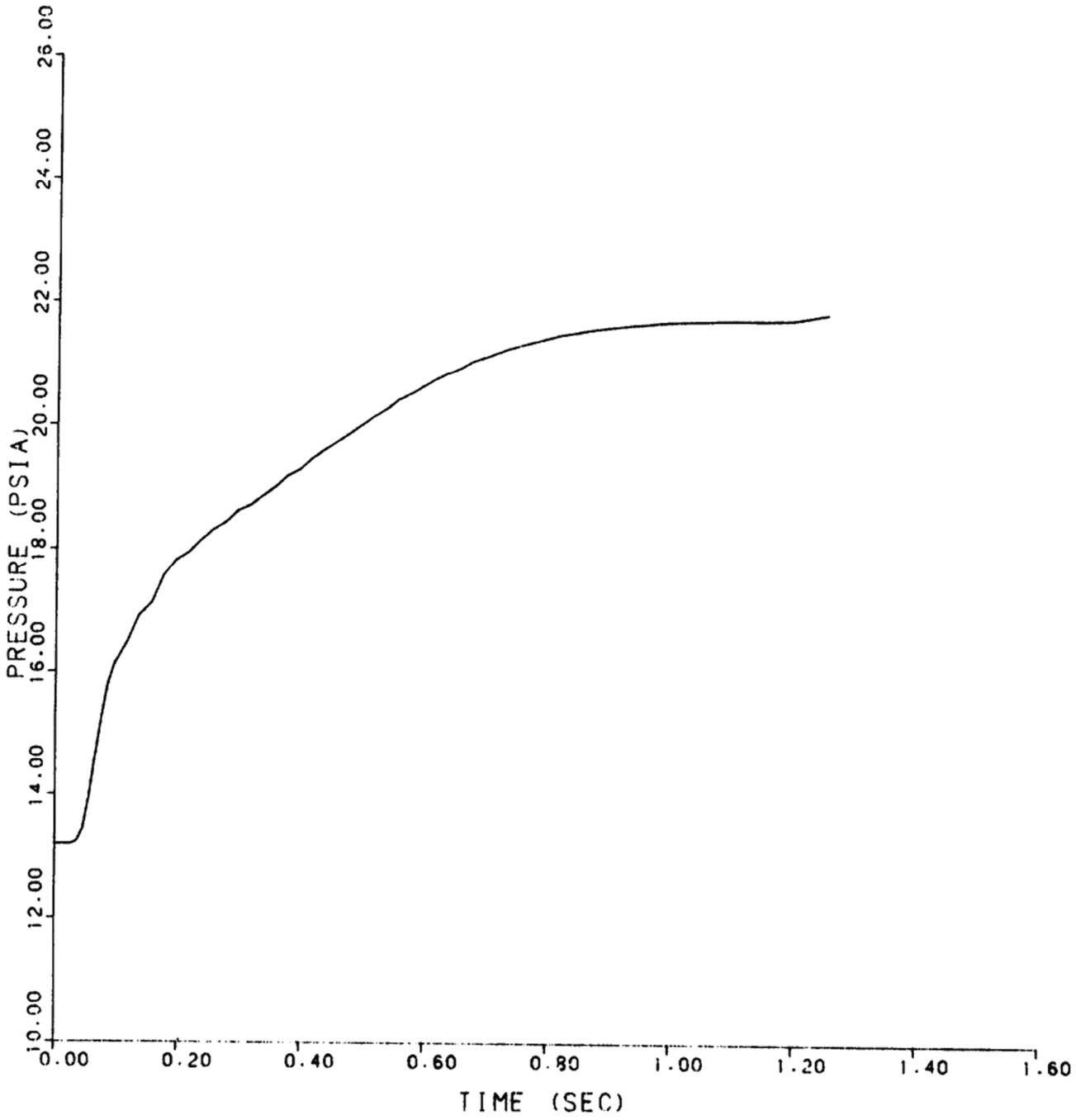
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

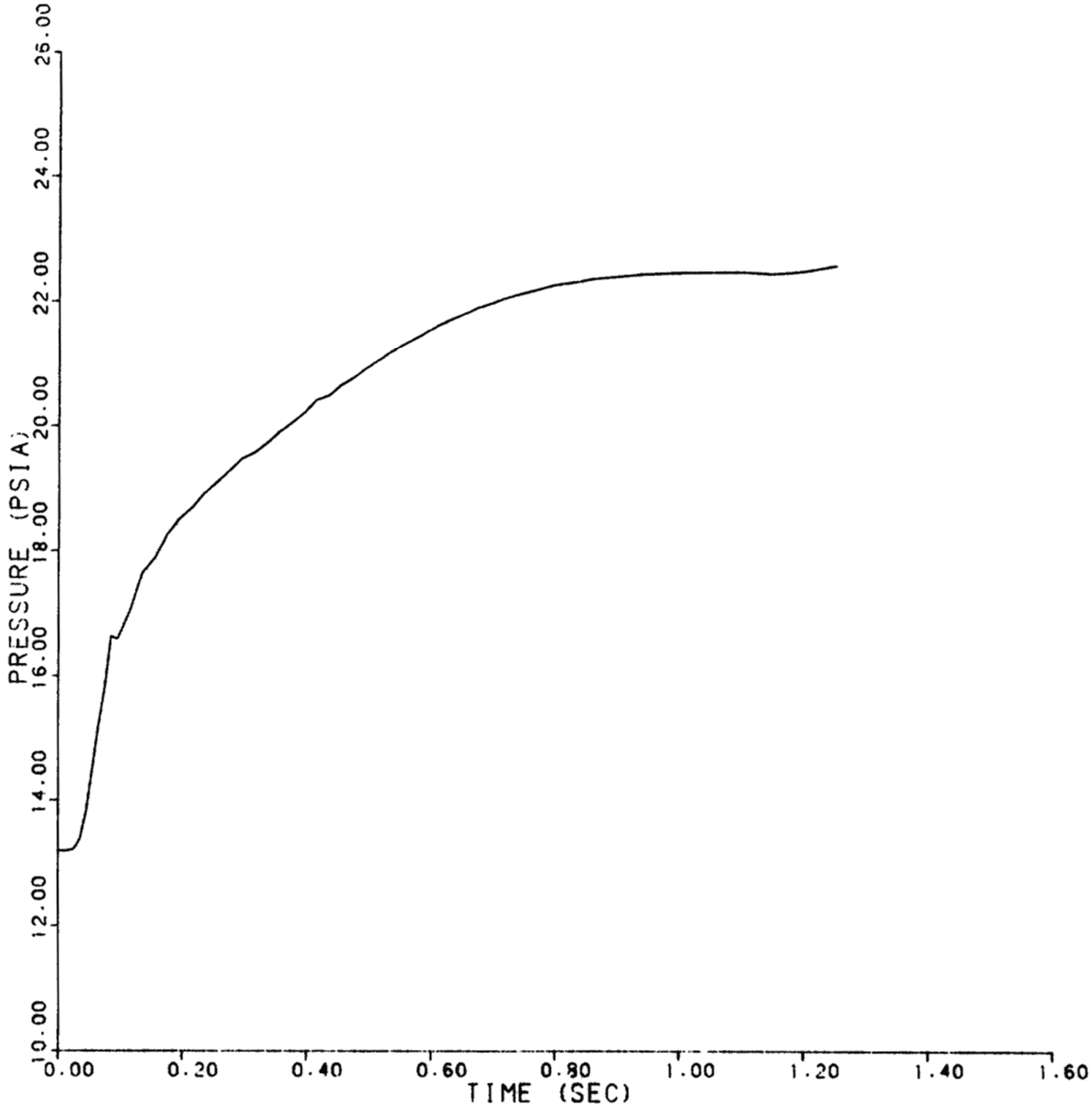
LOOP CLOSURE WELD
NODE E56
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 56 OF 74)



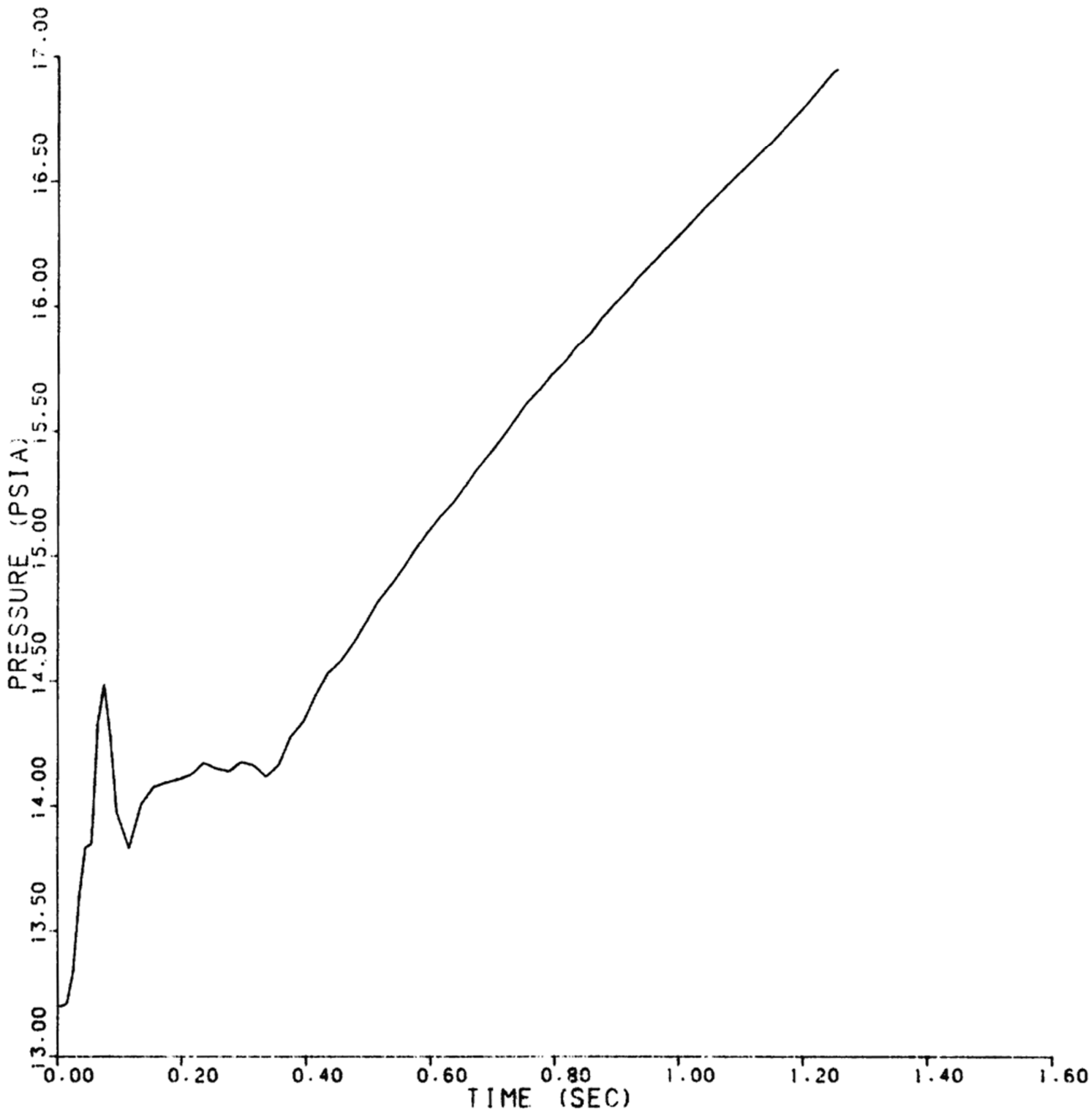
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E57 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 57 OF 74)



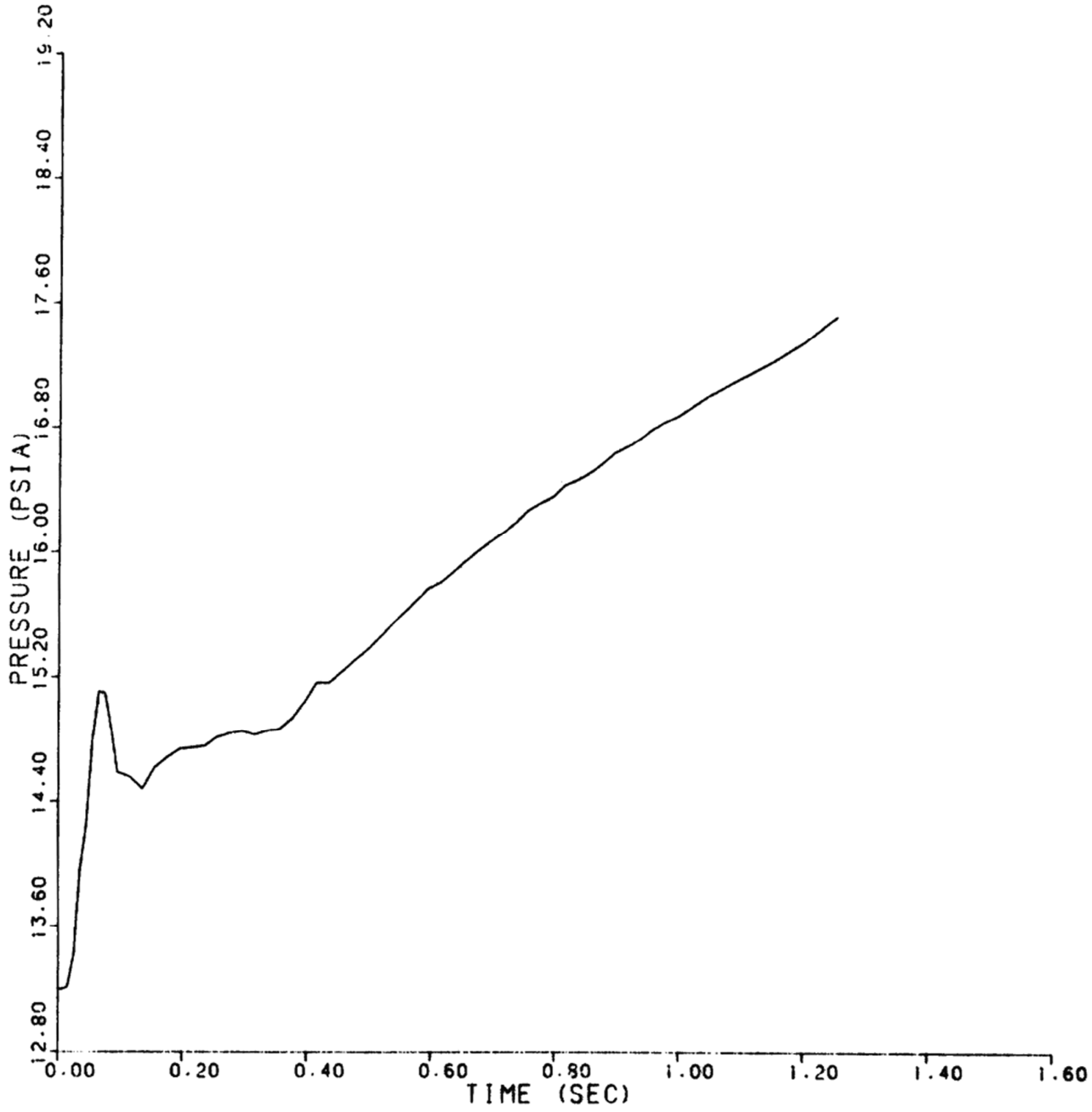
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E58 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 58 OF 74)



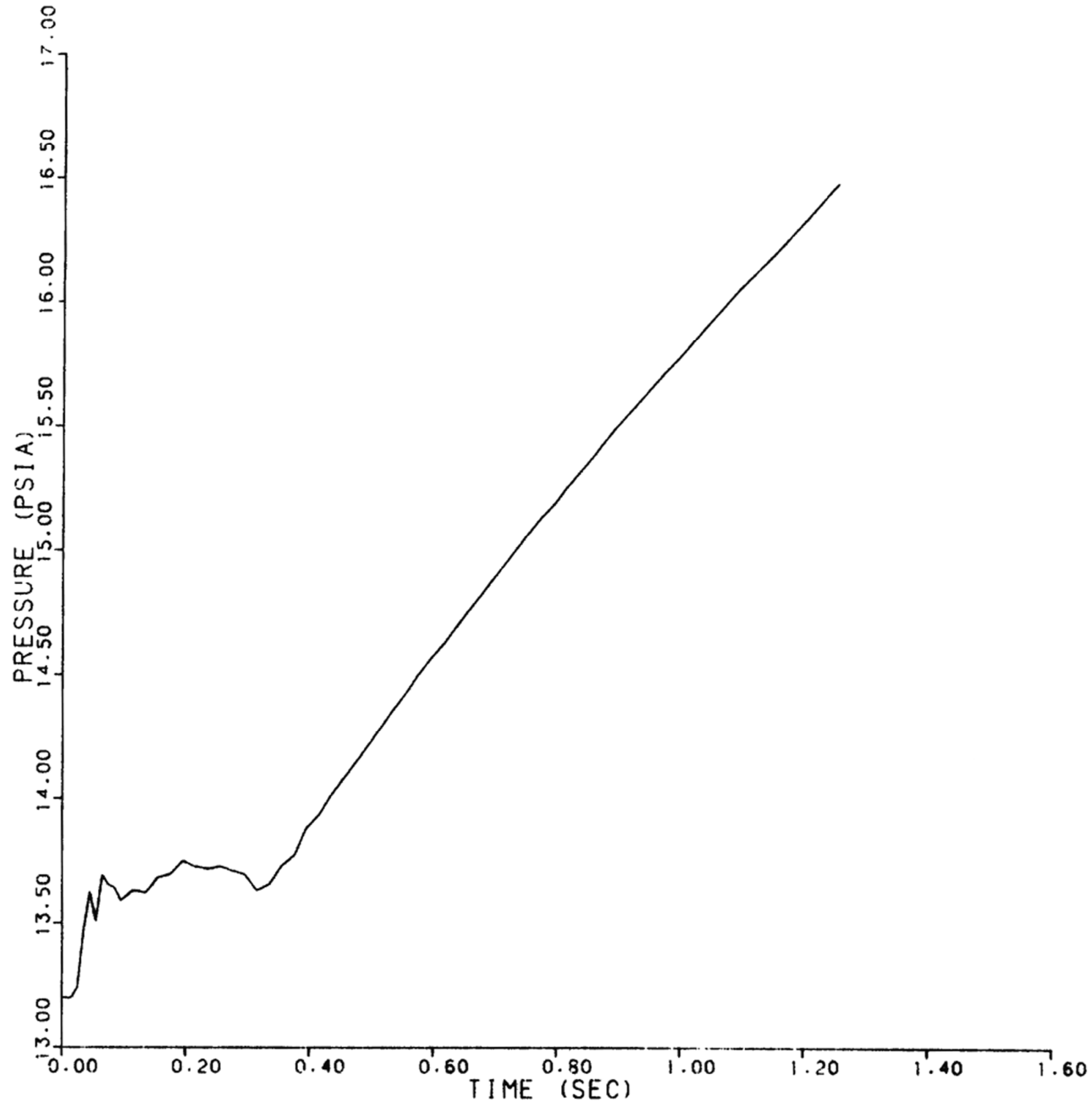
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E59 (336–in.^{2} BREAK AREA) FIGURE 6.2.1–23 (SHEET 59 OF 74)



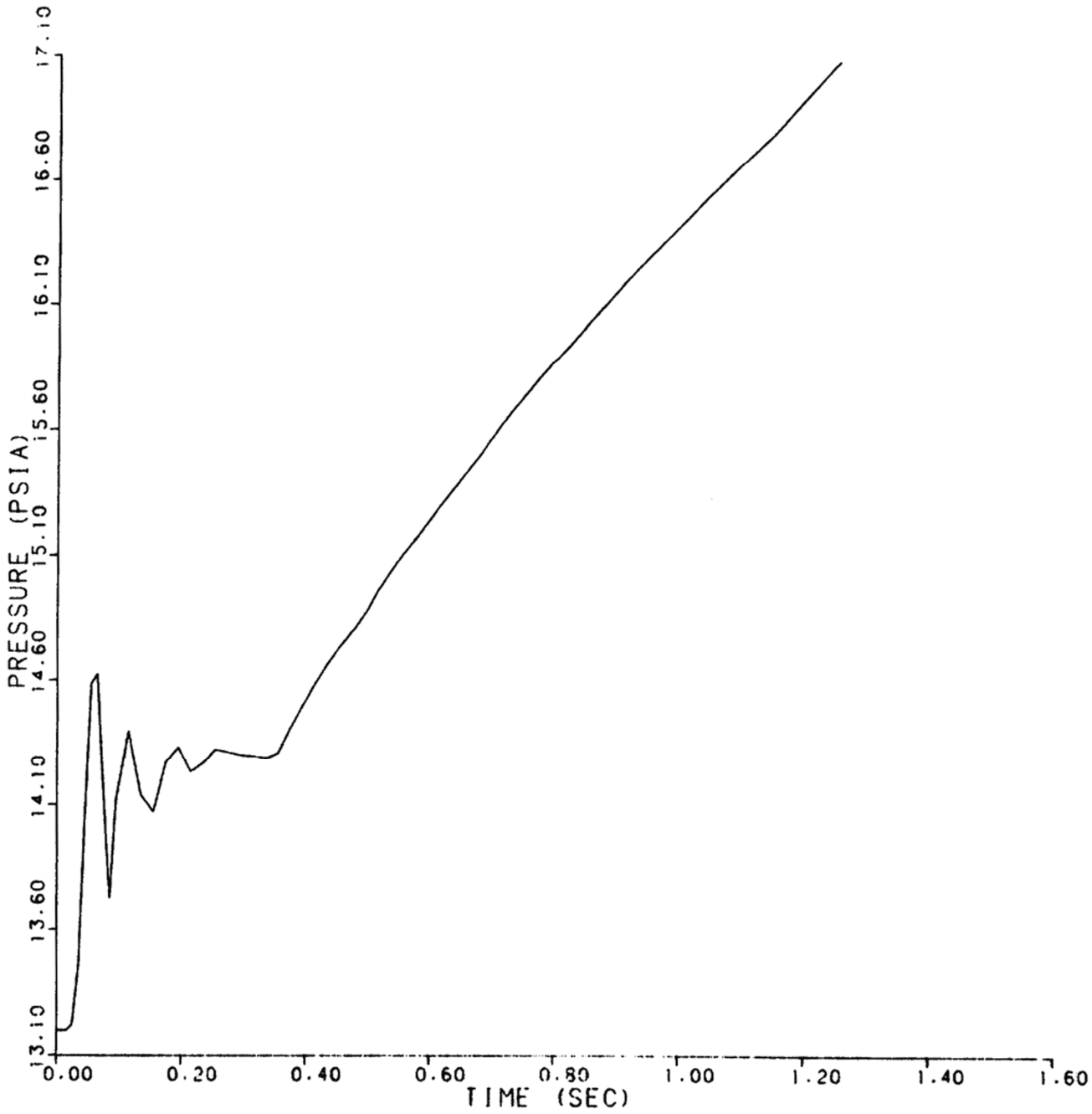
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E60 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 60 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E61 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 61 OF 74)



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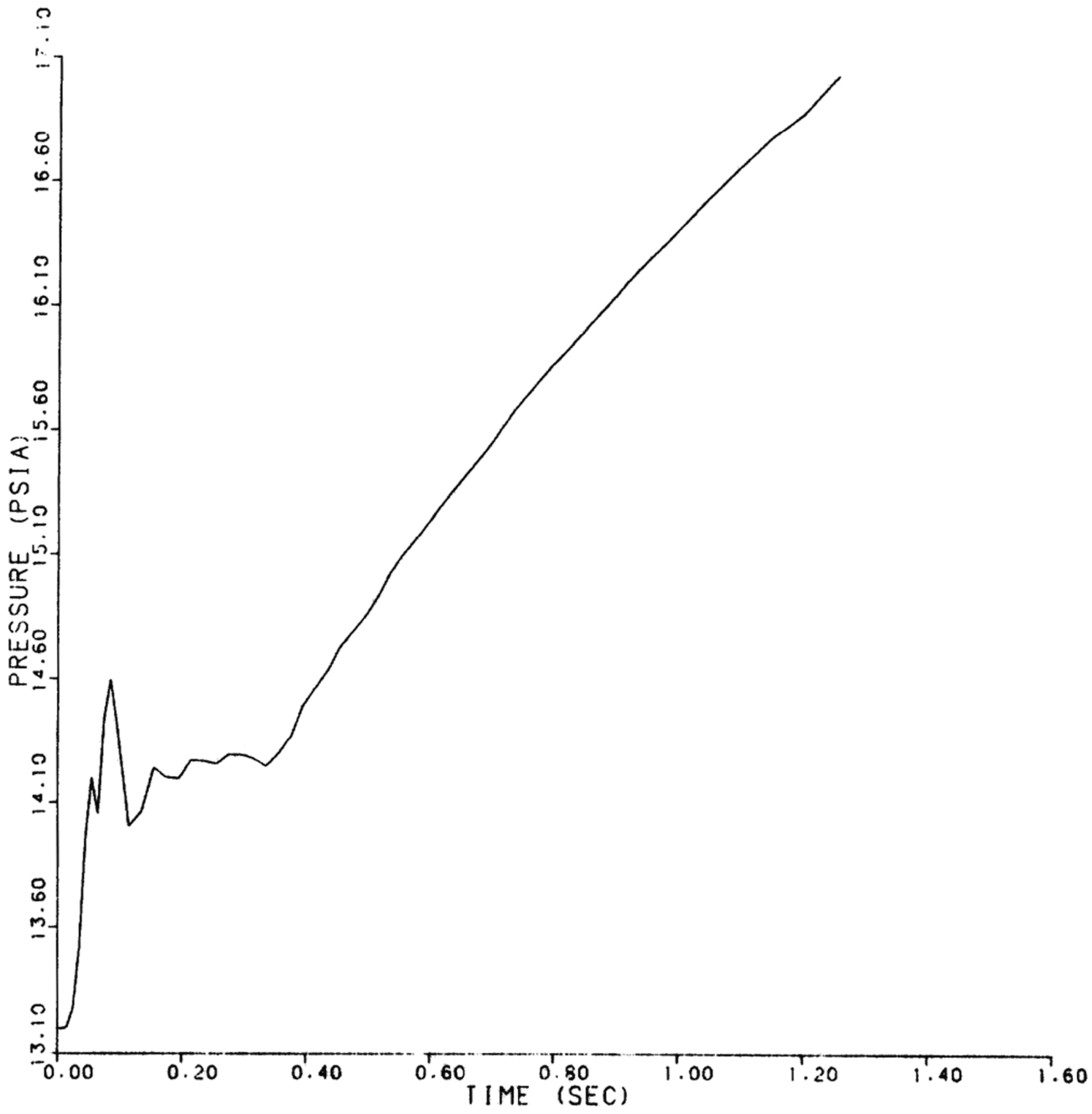
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

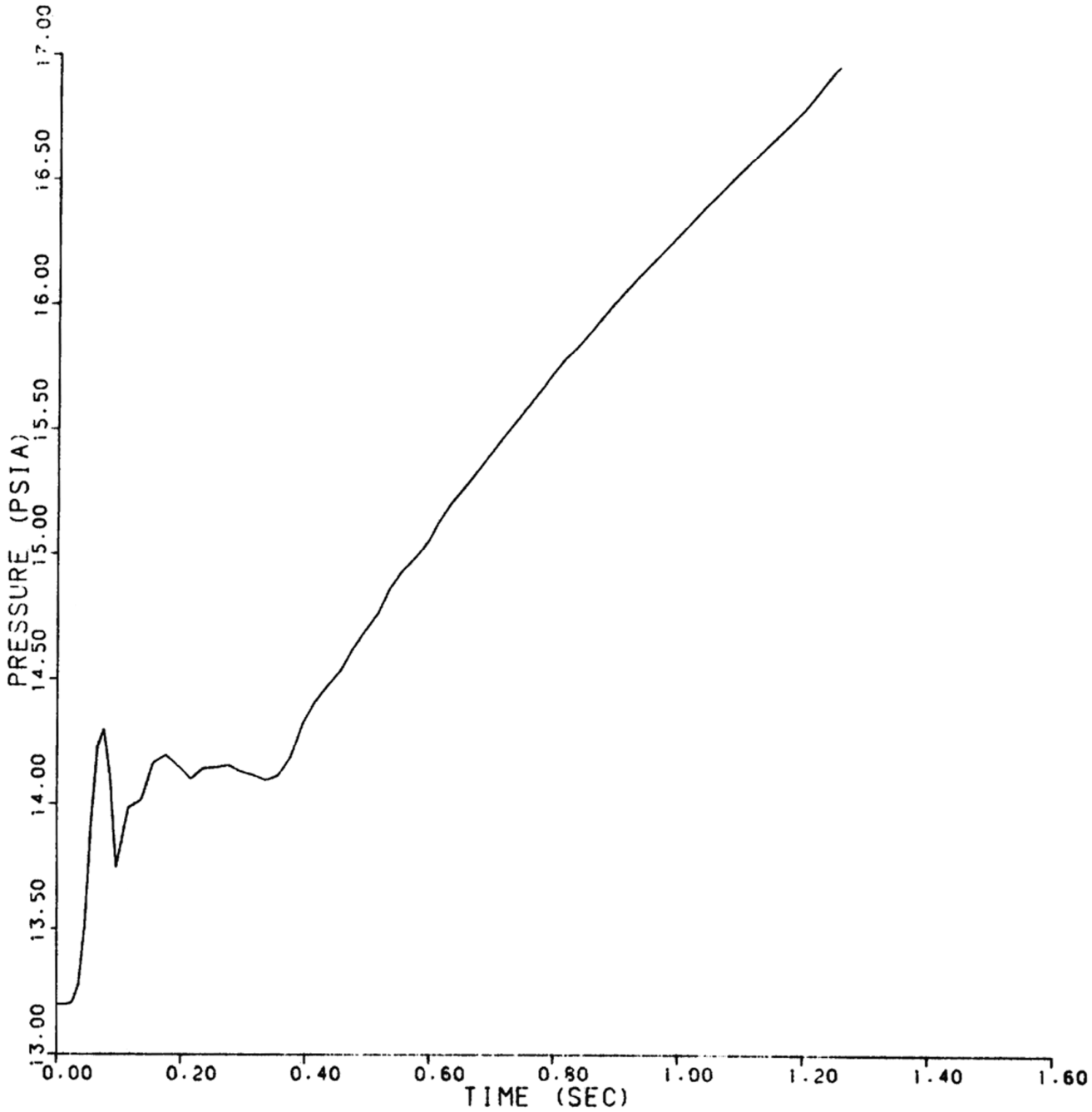
LOOP CLOSURE WELD
NODE E62
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 62 OF 74)



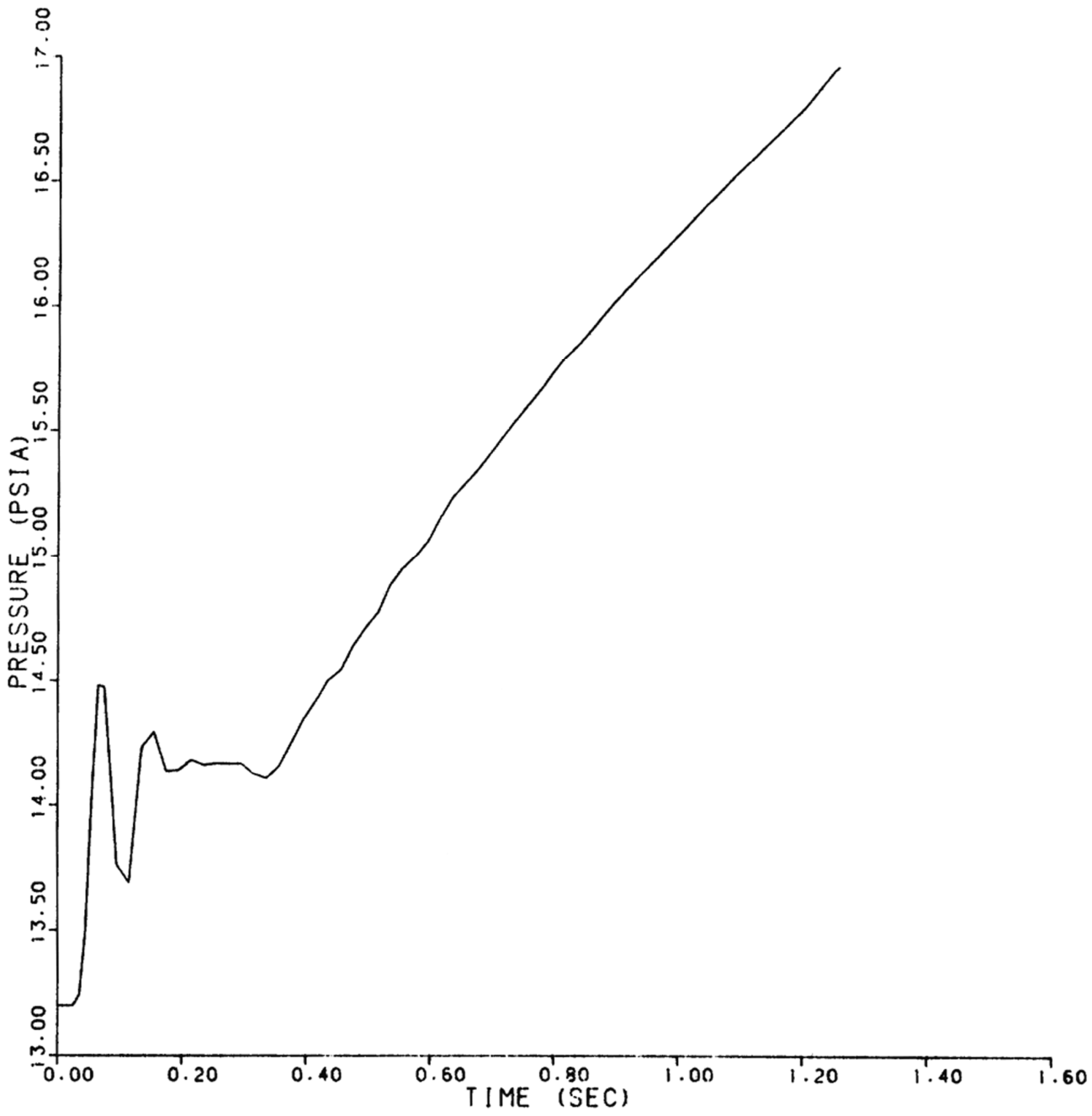
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E63 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 63 OF 74)



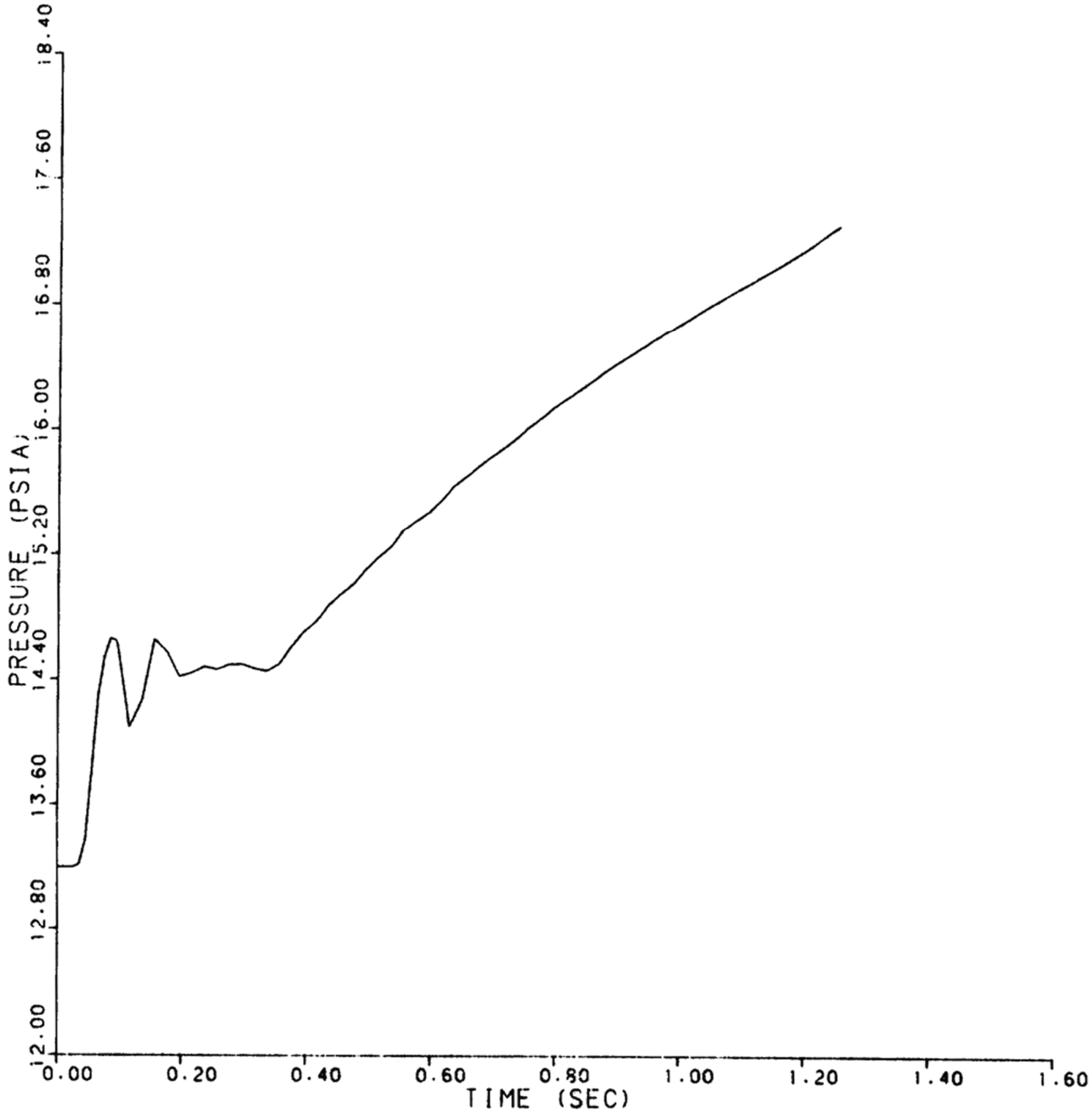
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E64 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 64 OF 74)



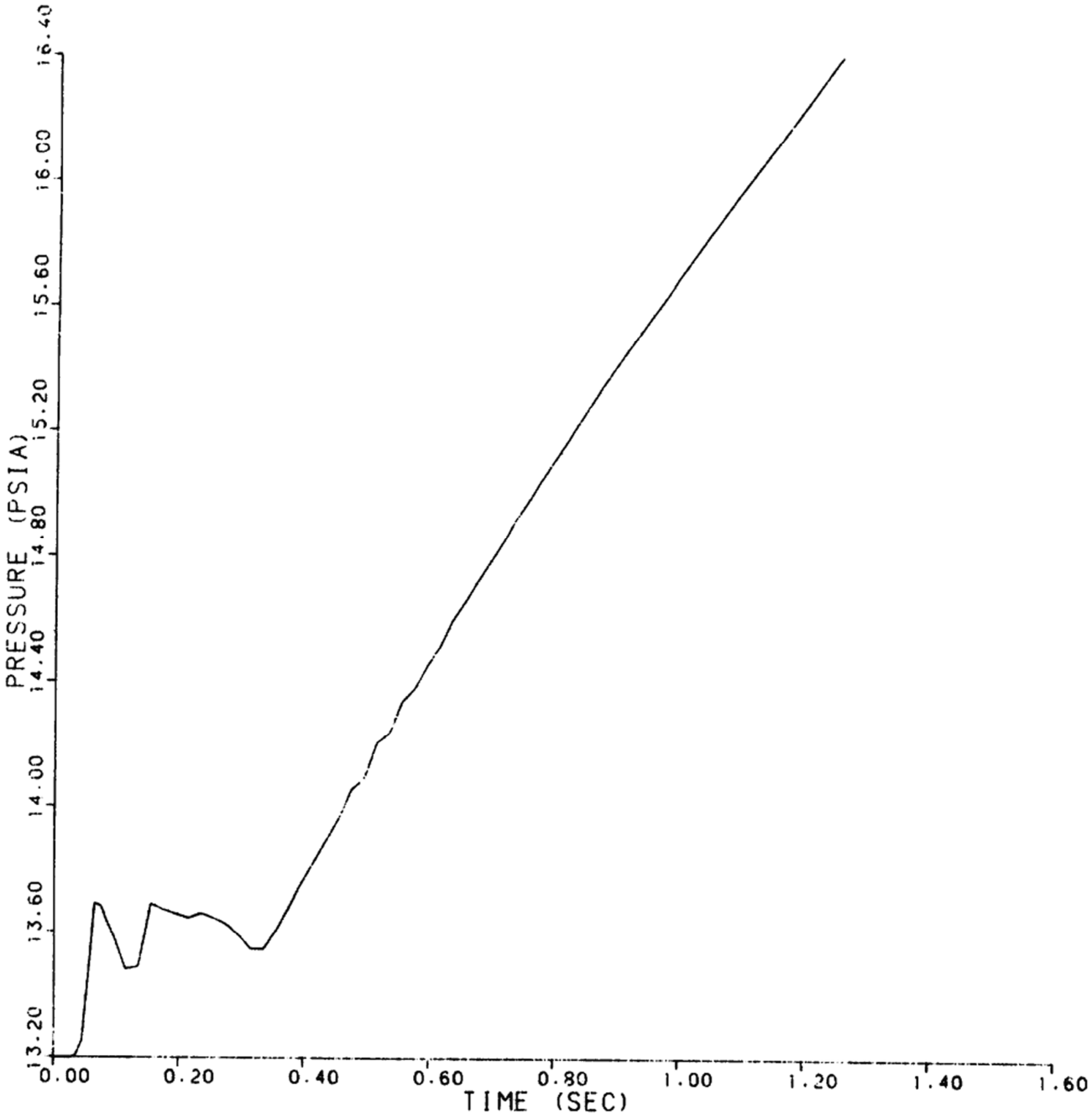
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E65 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 65 OF 74)



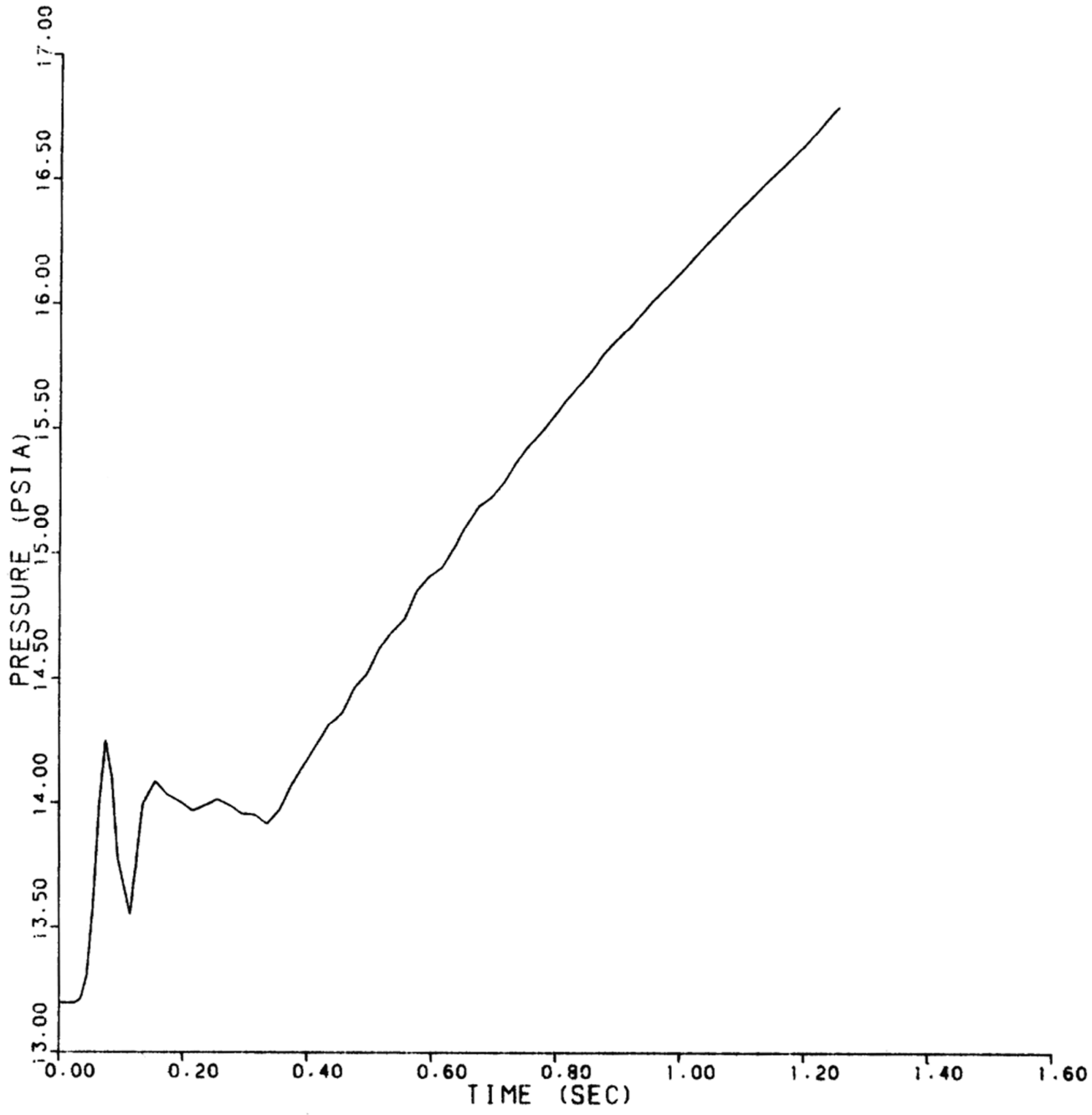
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E66 (336–in.^{2} BREAK AREA) FIGURE 6.2.1–23 (SHEET 66 OF 74)



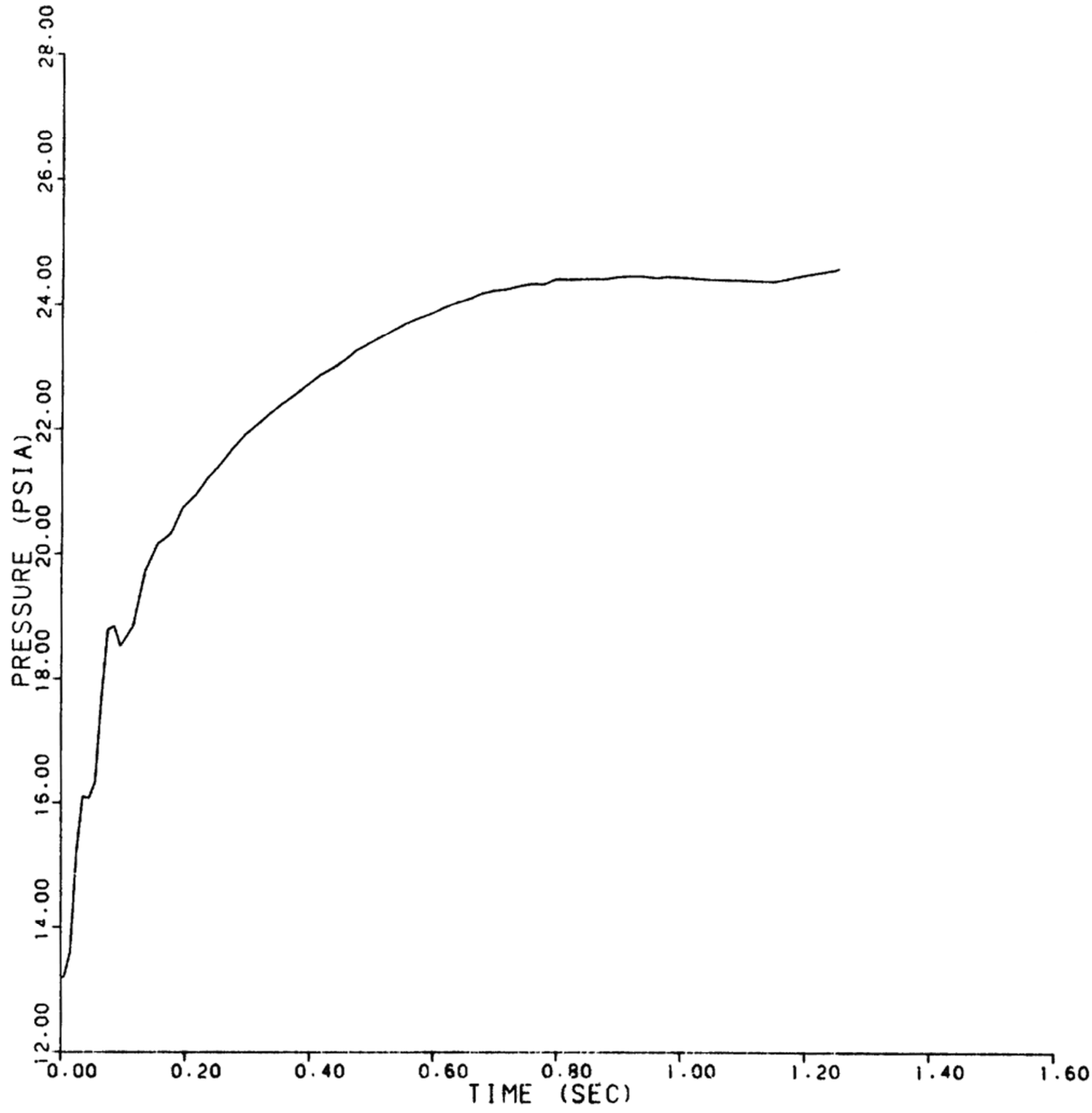
REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E67 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 67 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E68 (336–in.^{2} BREAK AREA) FIGURE 6.2.1–23 (SHEET 68 OF 74)



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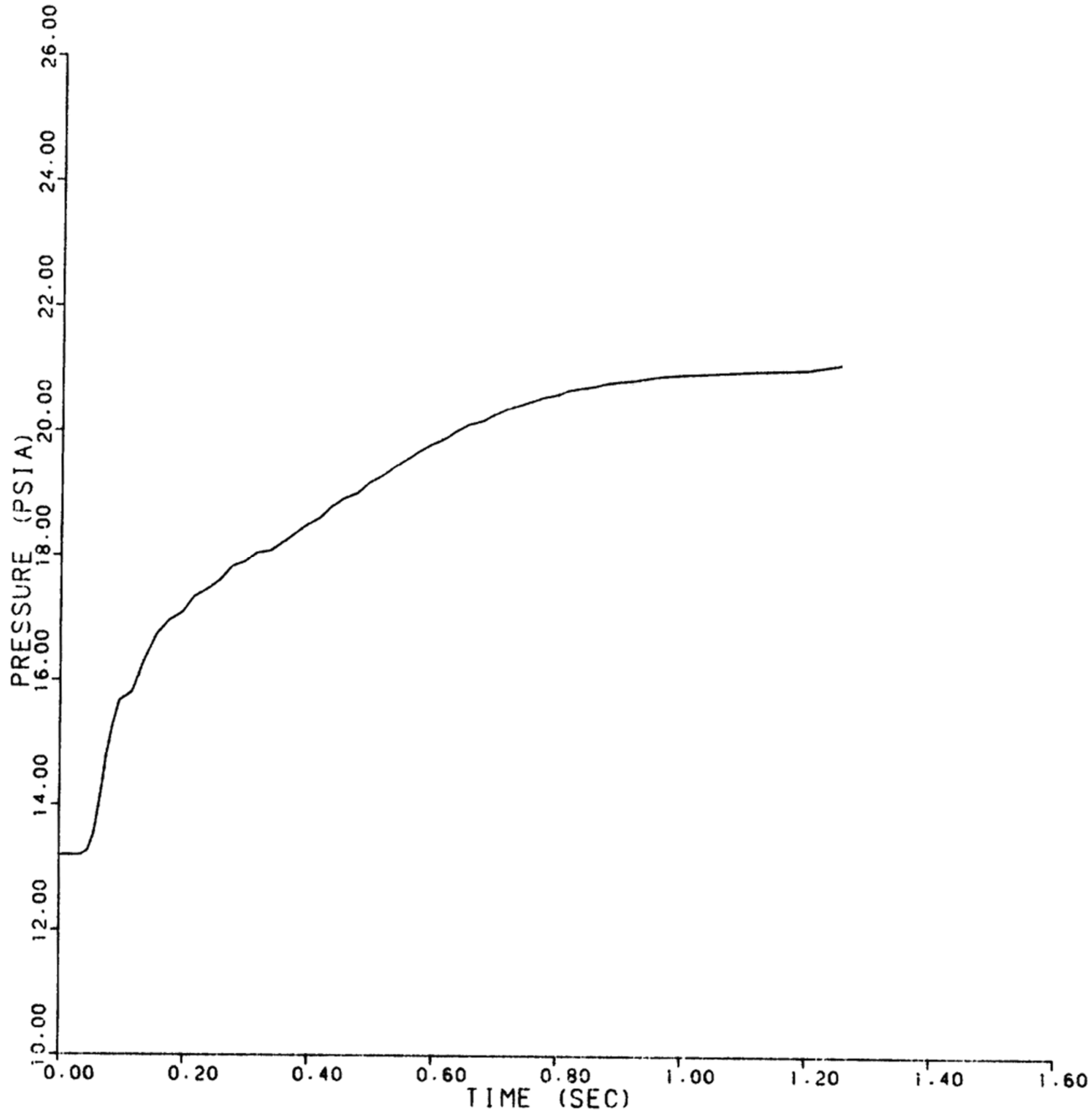
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

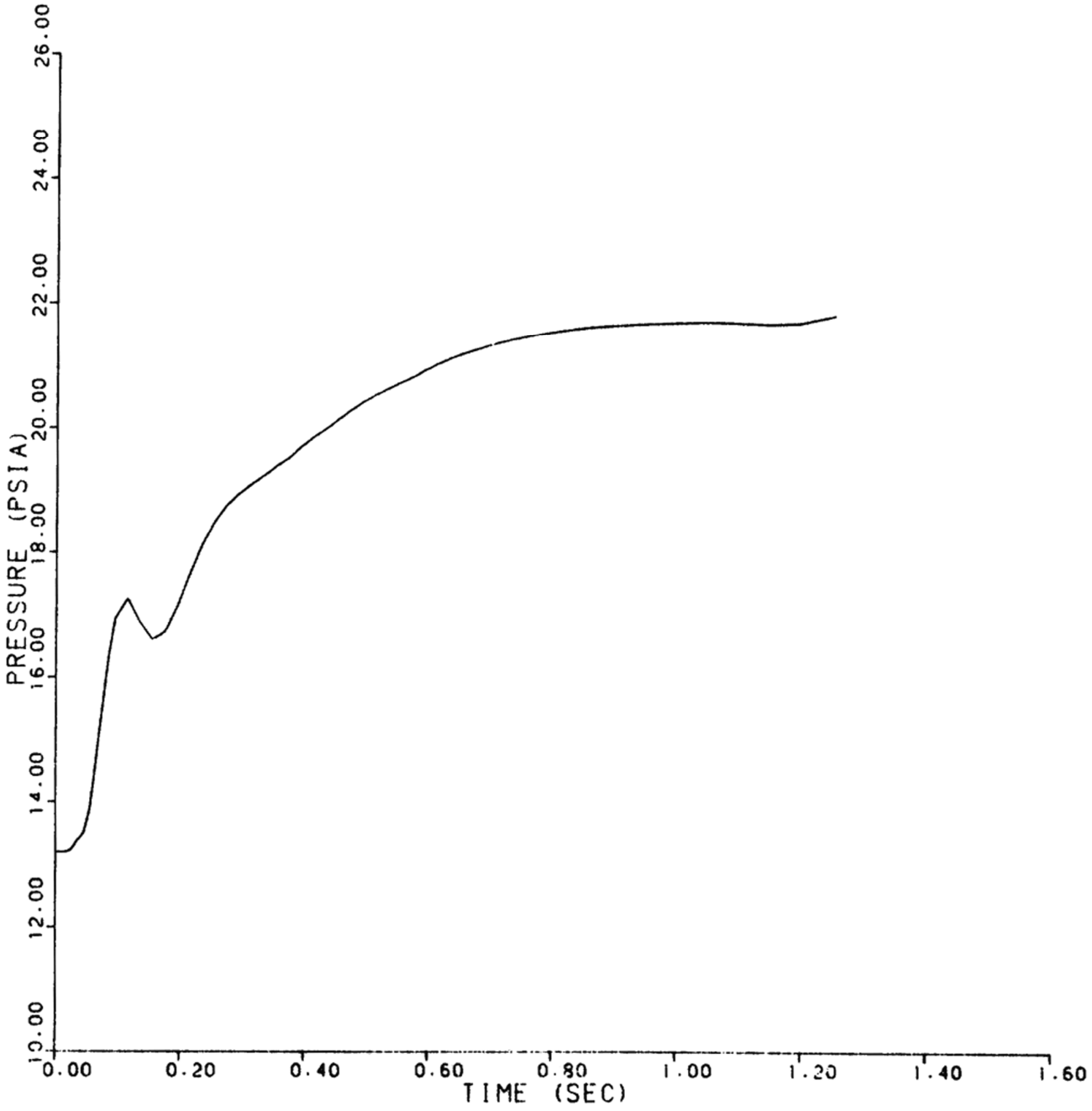
LOOP CLOSURE WELD
NODE E69
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 69 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E70 (336–in.^{2} BREAK AREA)
	FIGURE 6.2.1–23 (SHEET 70 OF 74)



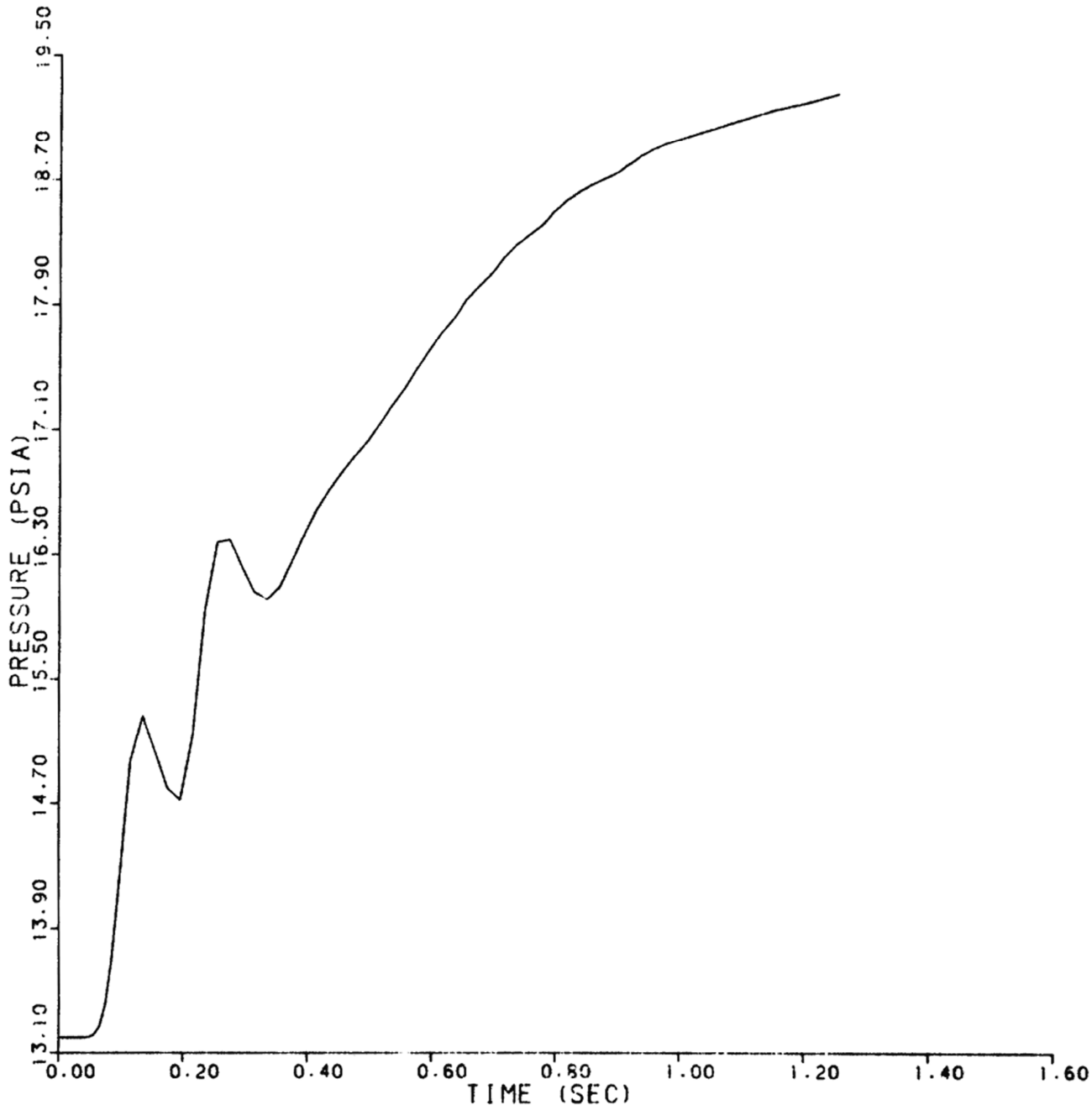
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E71
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 71 OF 74)



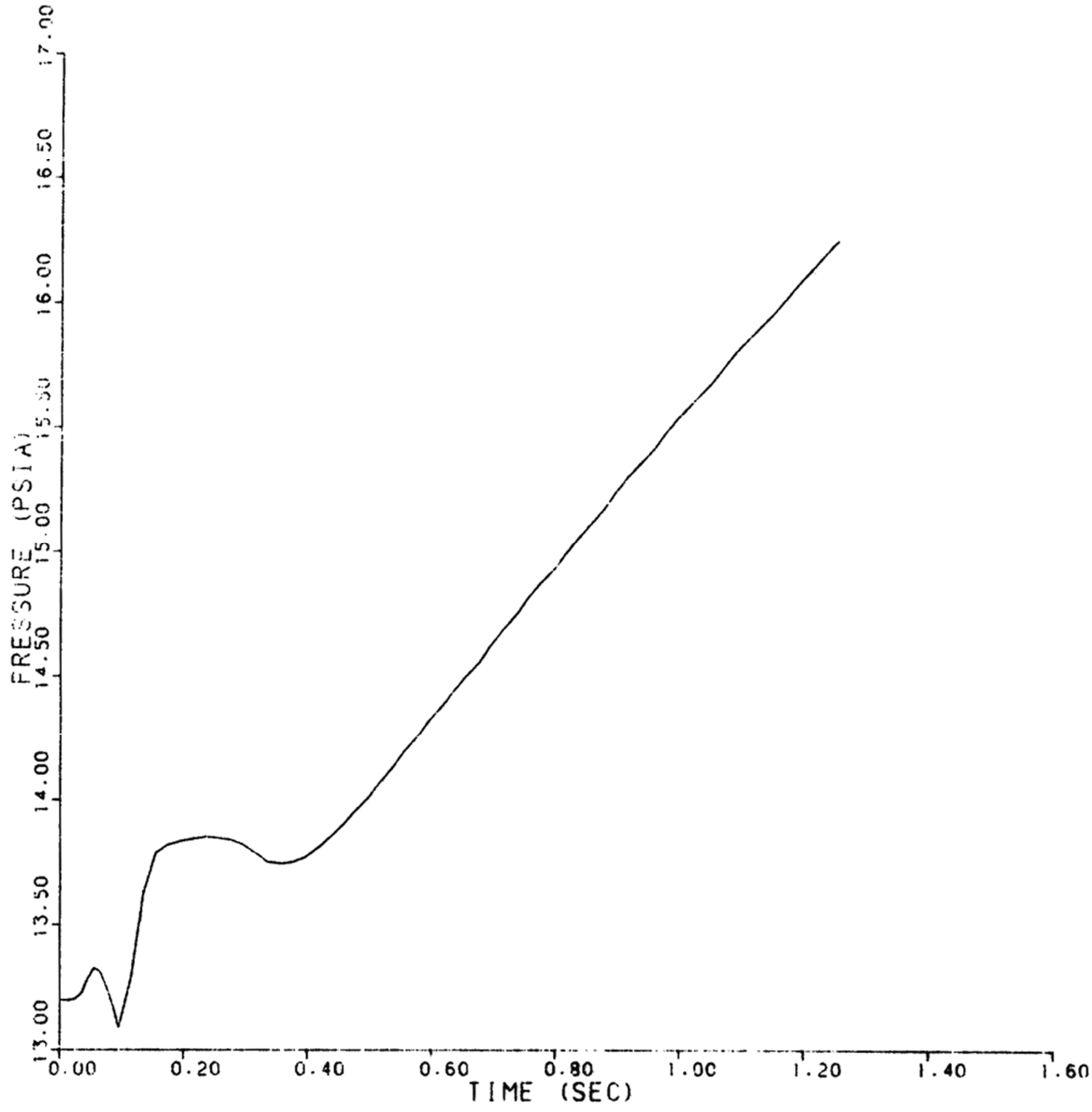
REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

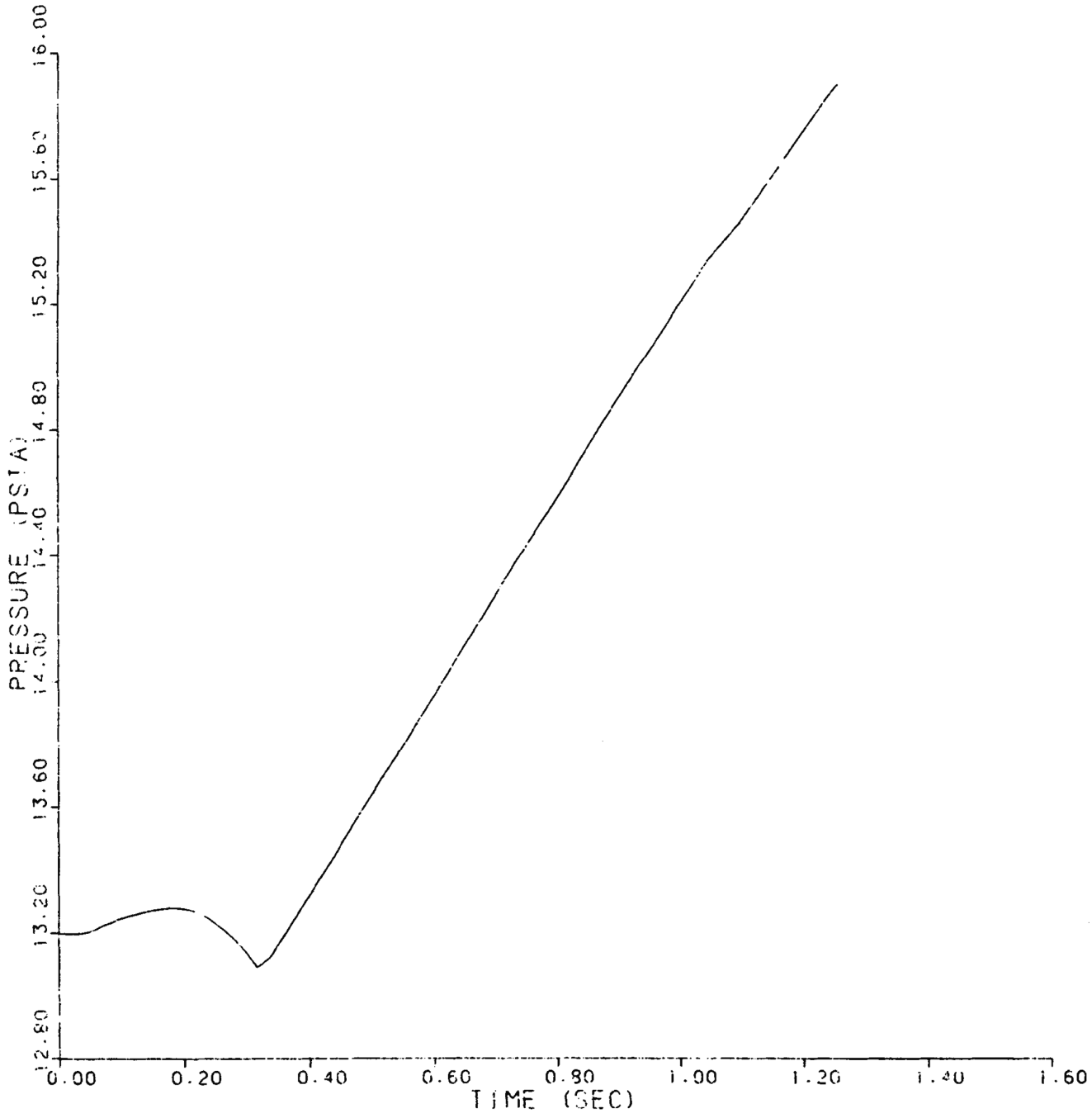
LOOP CLOSURE WELD
NODE E72
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 72 OF 74)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	LOOP CLOSURE WELD NODE E73 (336–in.^{2} BREAK AREA) FIGURE 6.2.1–23 (SHEET 73 OF 74)



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SOUTHERN COMPANY
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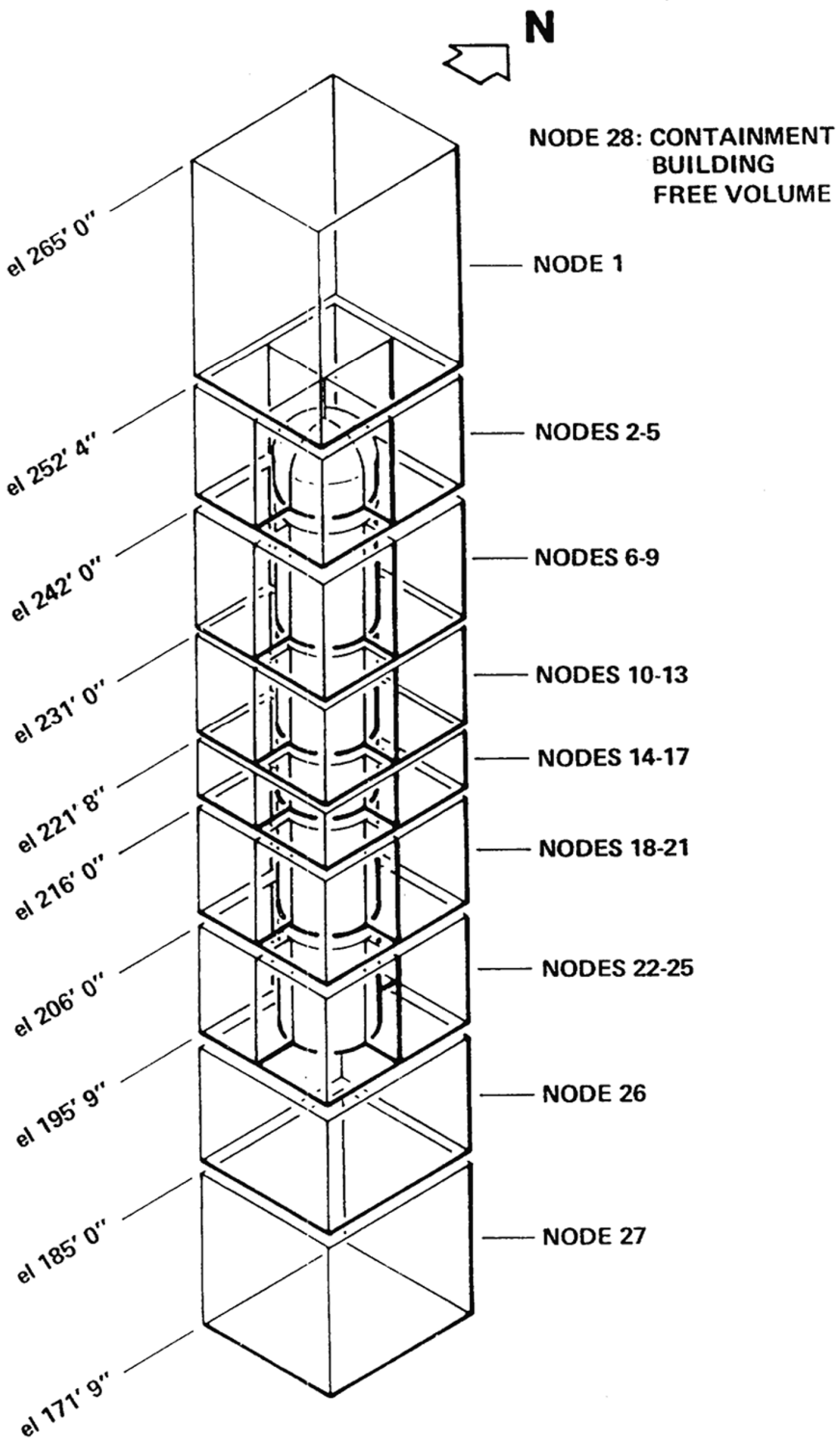
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

LOOP CLOSURE WELD
NODE E74
(336–in.^{2} BREAK AREA)

FIGURE 6.2.1–23 (SHEET 74 OF 74)



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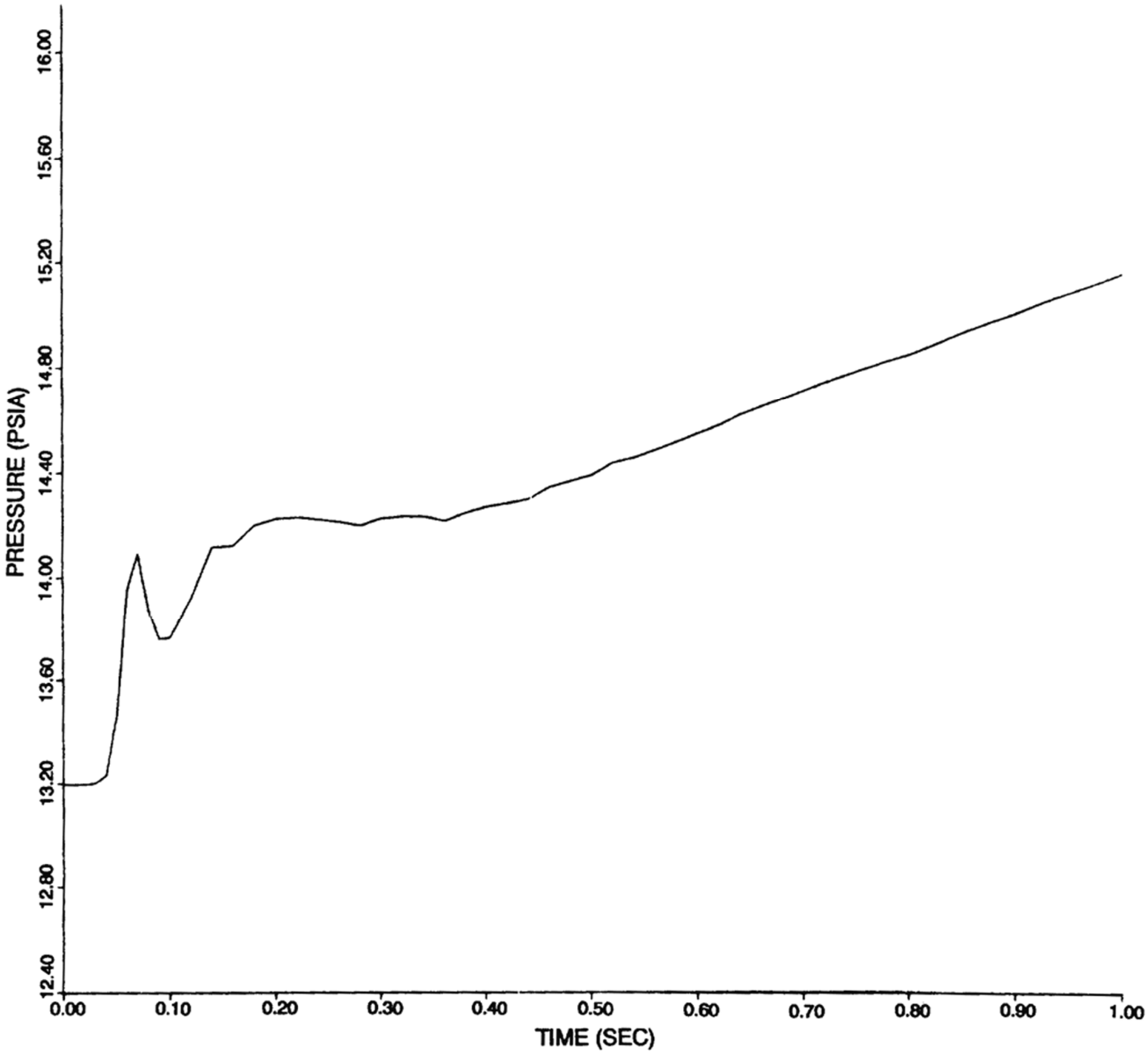
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	PRESSURIZER MODEL
	FIGURE 6.2.1–24



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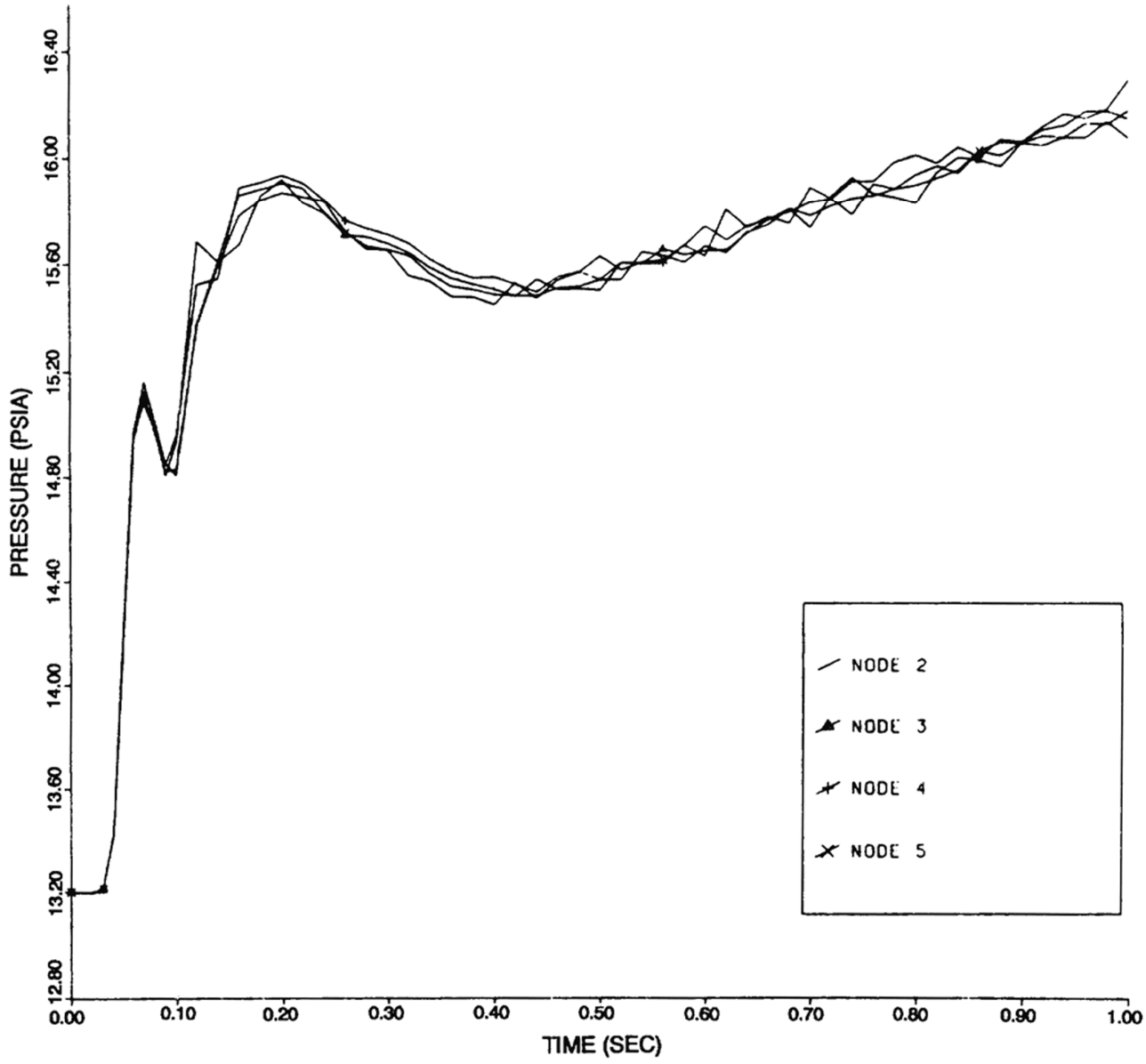
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 1
SURGE LINE BREAK
(308–in^2 BREAK AREA)

FIGURE 6.2.1–25 (SHEET 1 OF 8)



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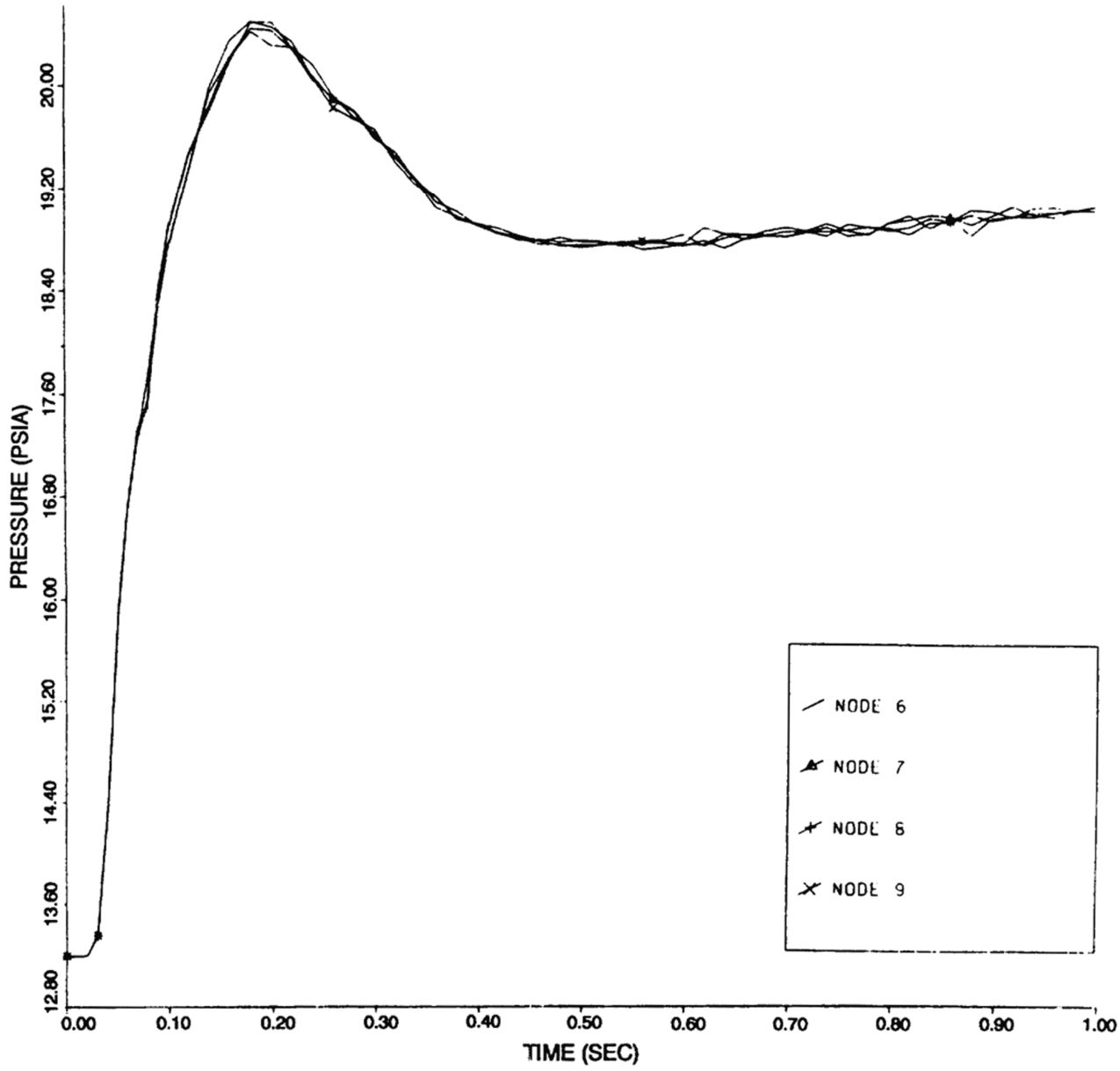
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	PRESSURIZER COMPARTMENT PRESSURE RESPONSE – NODE 2, 3, 4, 5 SURGE LINE BREAK (308–in^2 BREAK AREA)
	FIGURE 6.2.1–25 (SHEET 2 OF 8)



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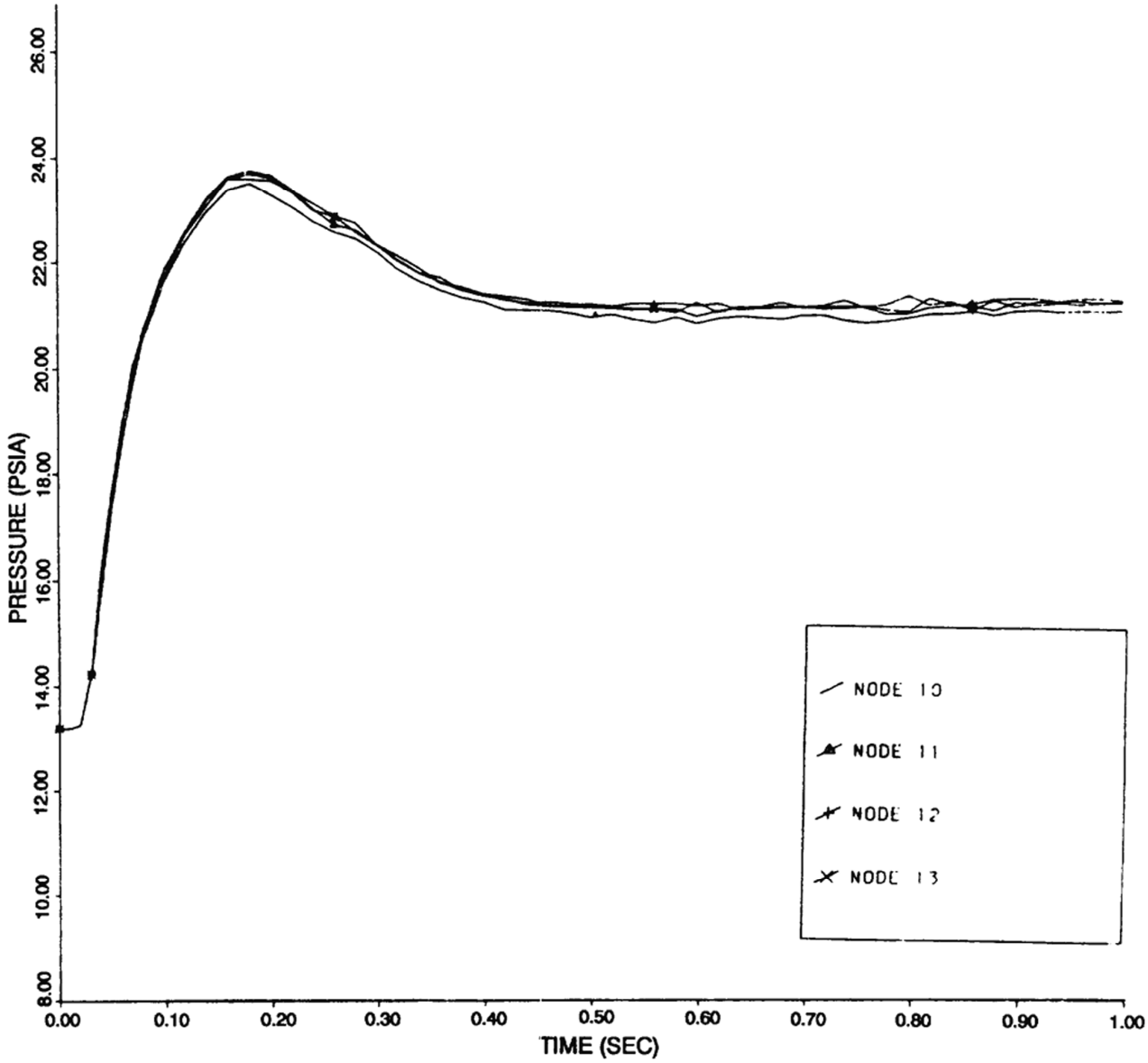
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 6, 7, 8, 9
SURGE LINE BREAK
(308–in^2 BREAK AREA)

FIGURE 6.2.1–25 (SHEET 3 OF 8)



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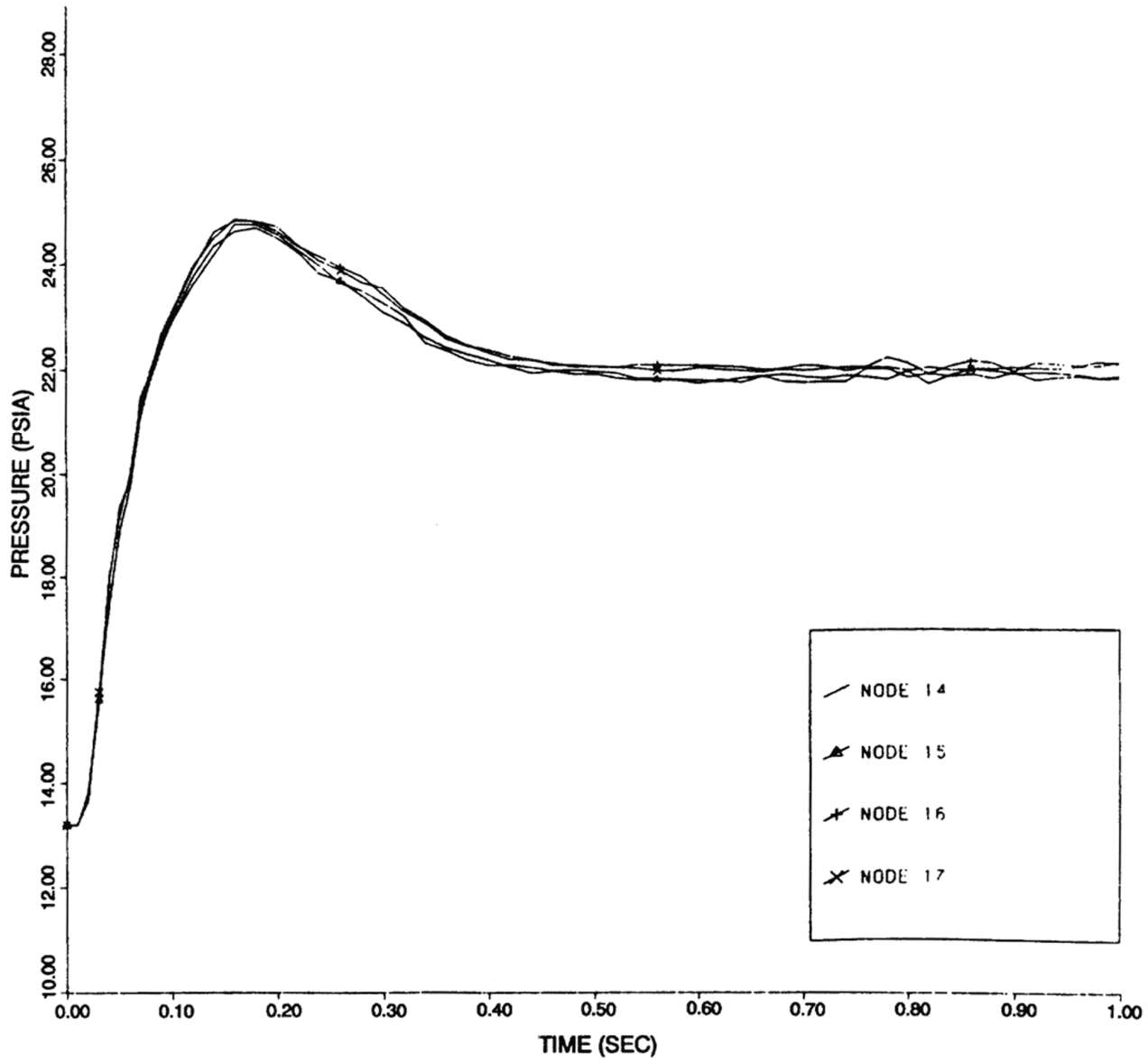
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 10, 11, 12, 13
SURGE LINE BREAK
(308–in^2 BREAK AREA)

FIGURE 6.2.1–25 (SHEET 4 OF 8)



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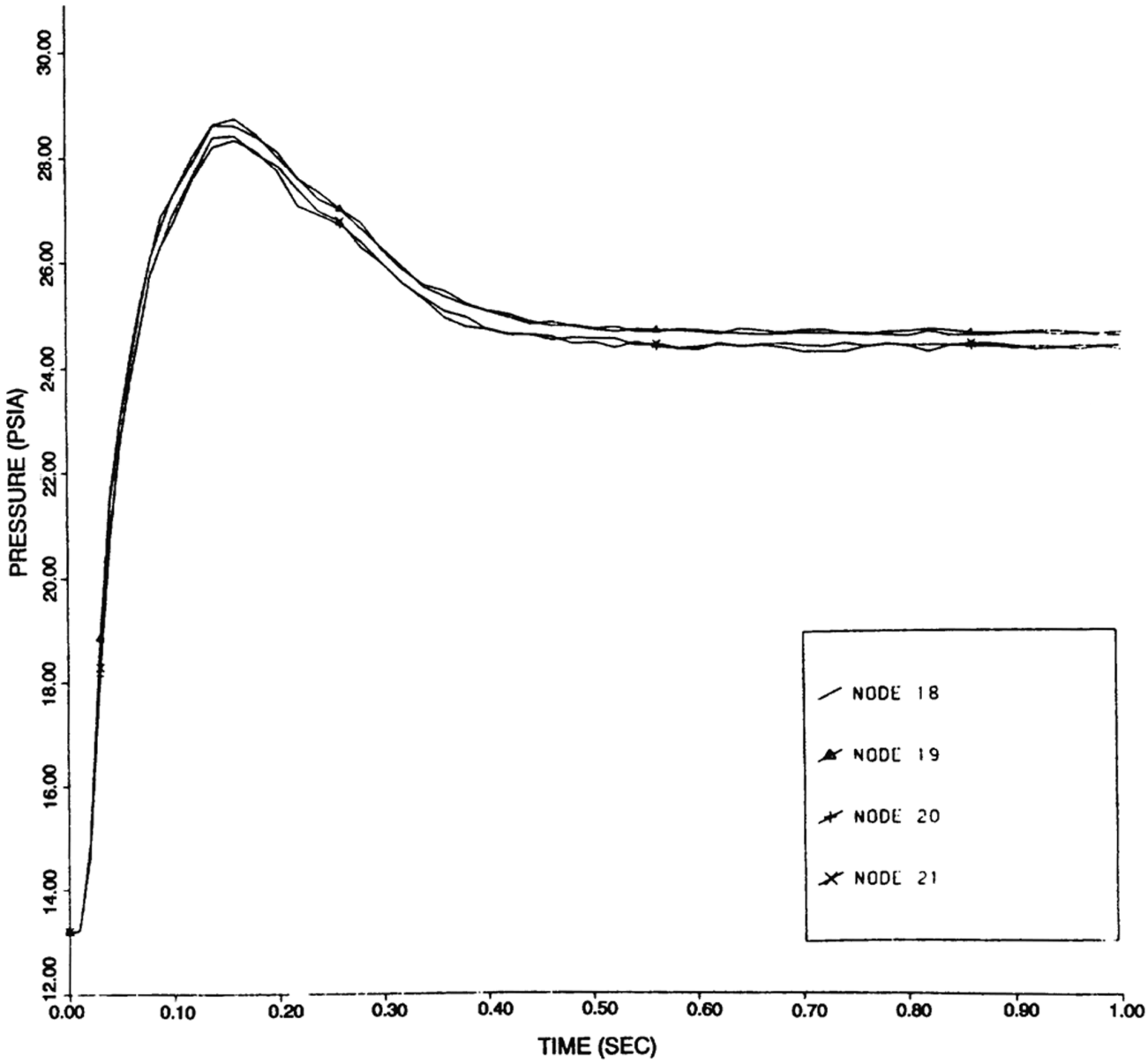
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 14, 15, 16, 17
SURGE LINE BREAK
(308–in^2 BREAK AREA)

FIGURE 6.2.1–25 (SHEET 5 OF 8)



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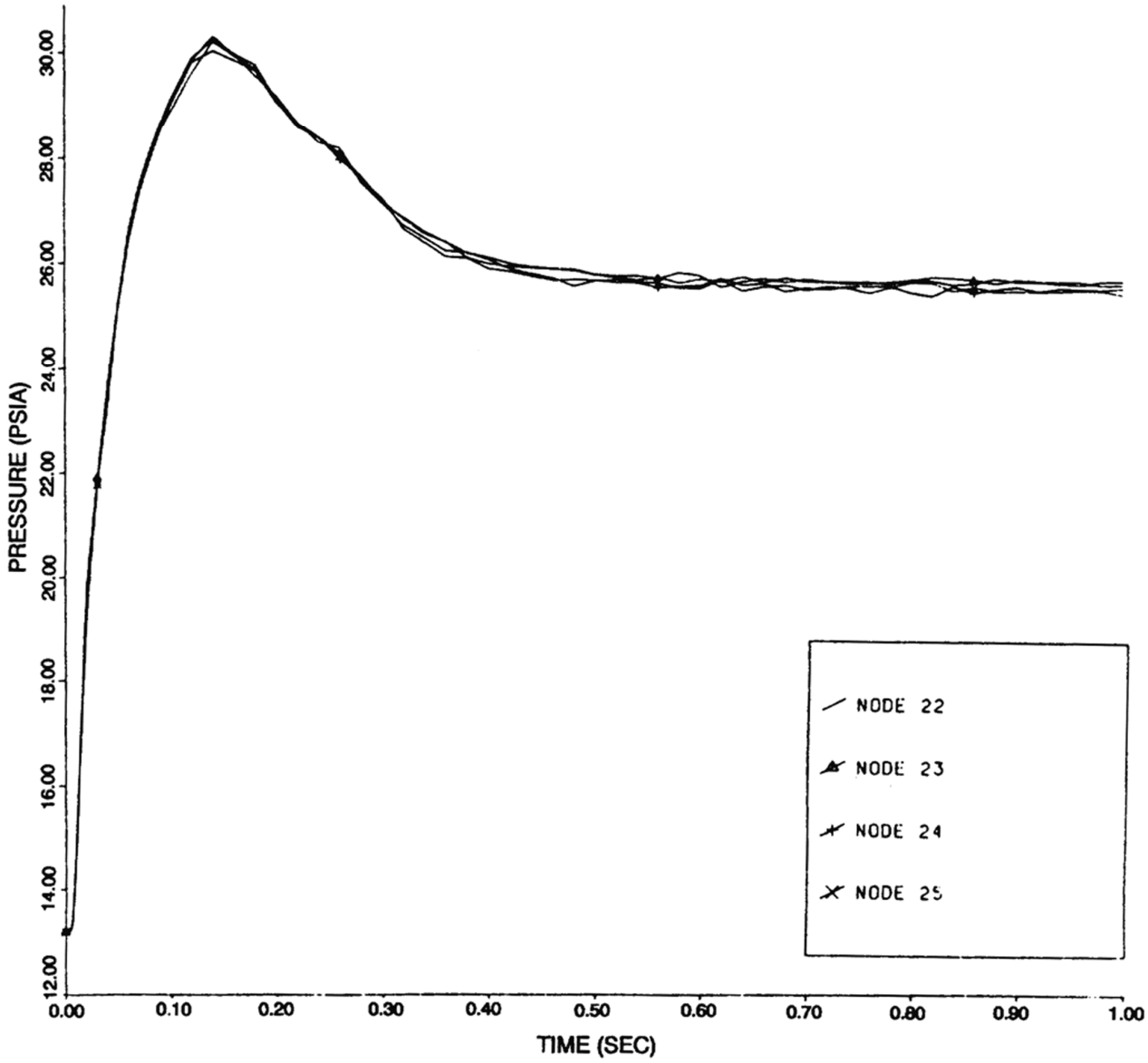
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 18, 19, 20, 21
SURGE LINE BREAK
(308–in^2 BREAK AREA)

FIGURE 6.2.1–25 (SHEET 6 OF 8)



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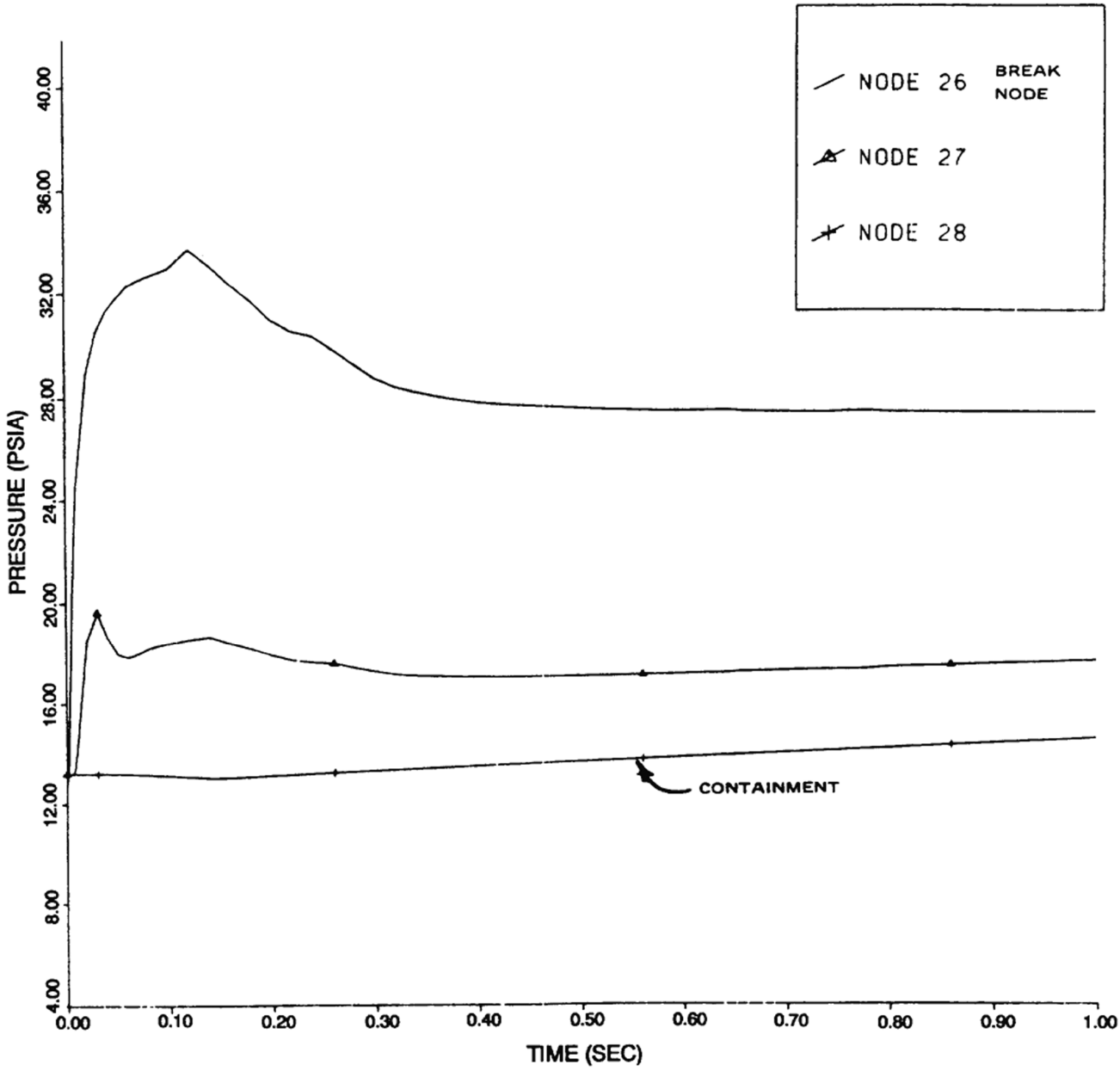
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 22, 23, 24, 25
SURGE LINE BREAK
(308–in^2 BREAK AREA)

FIGURE 6.2.1–25 (SHEET 7 OF 8)



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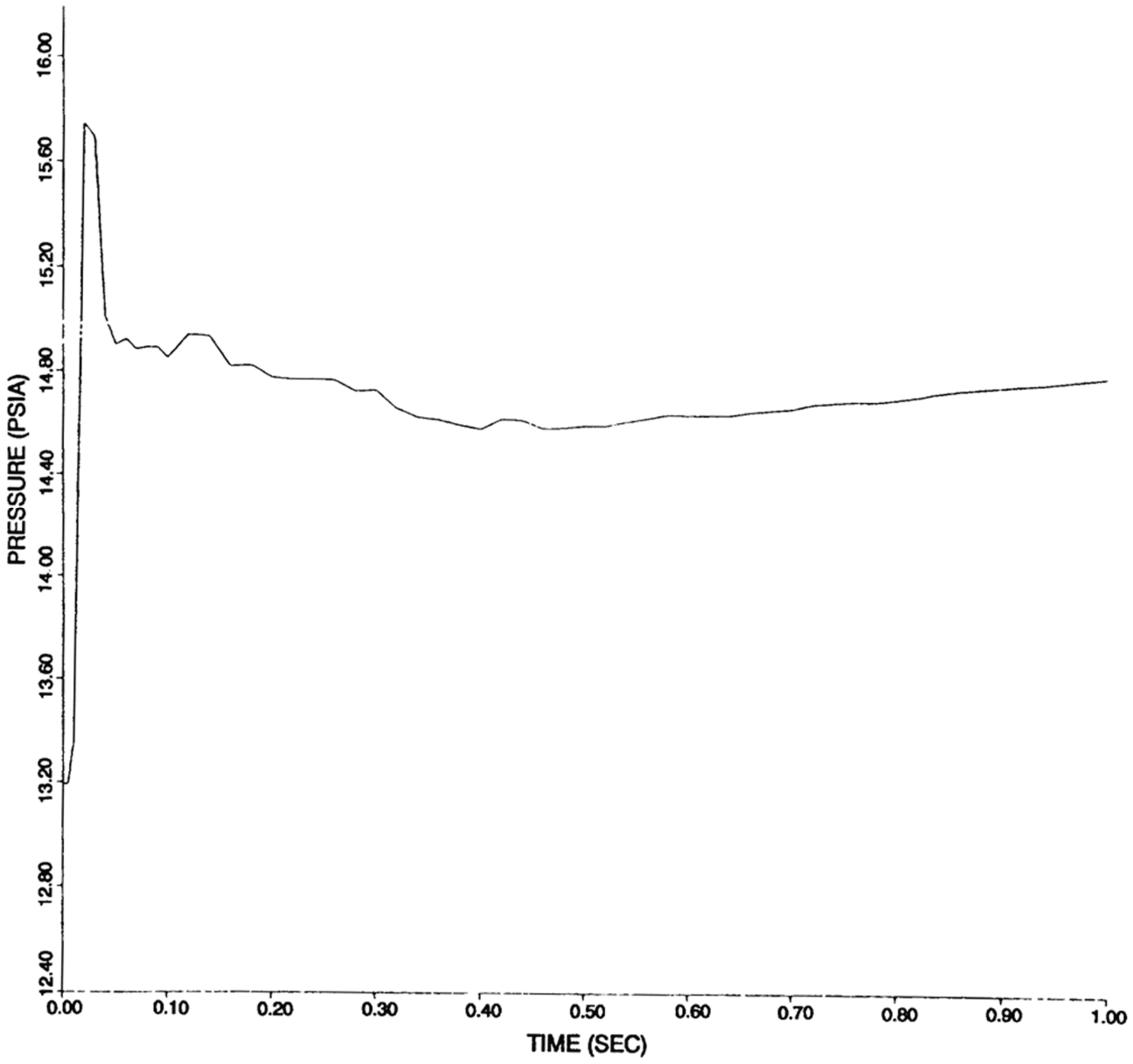
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 26, 27, 28
SURGE LINE BREAK
(308–in^2 BREAK AREA)

FIGURE 6.2.1–25 (SHEET 8 OF 8)



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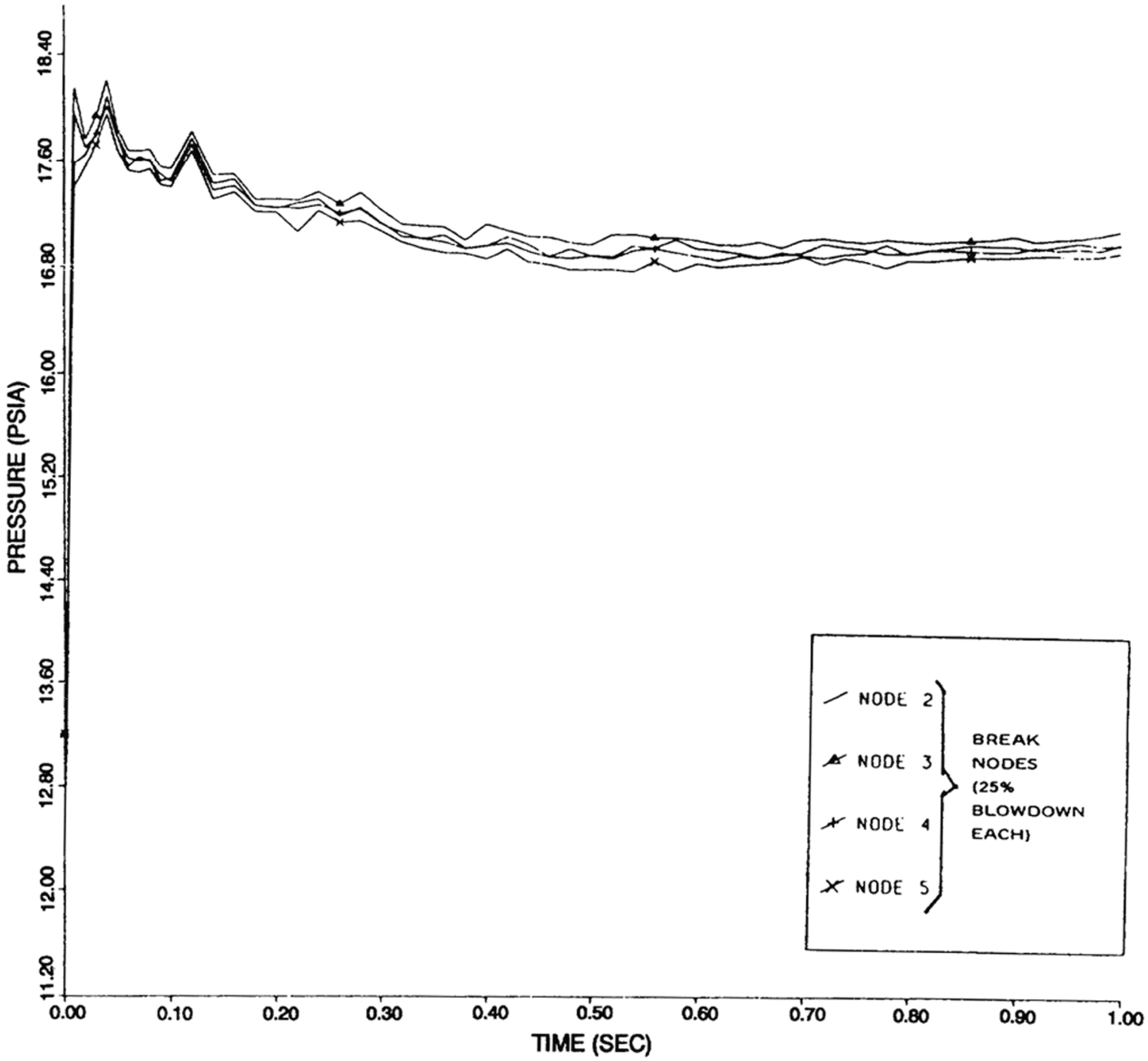
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	PRESSURIZER COMPARTMENT PRESSURE RESPONSE – NODE 1 SPRAY LINE BREAK AT TOP OF PRESSURIZER
	FIGURE 6.2.1–25a (SHEET 1 OF 8)



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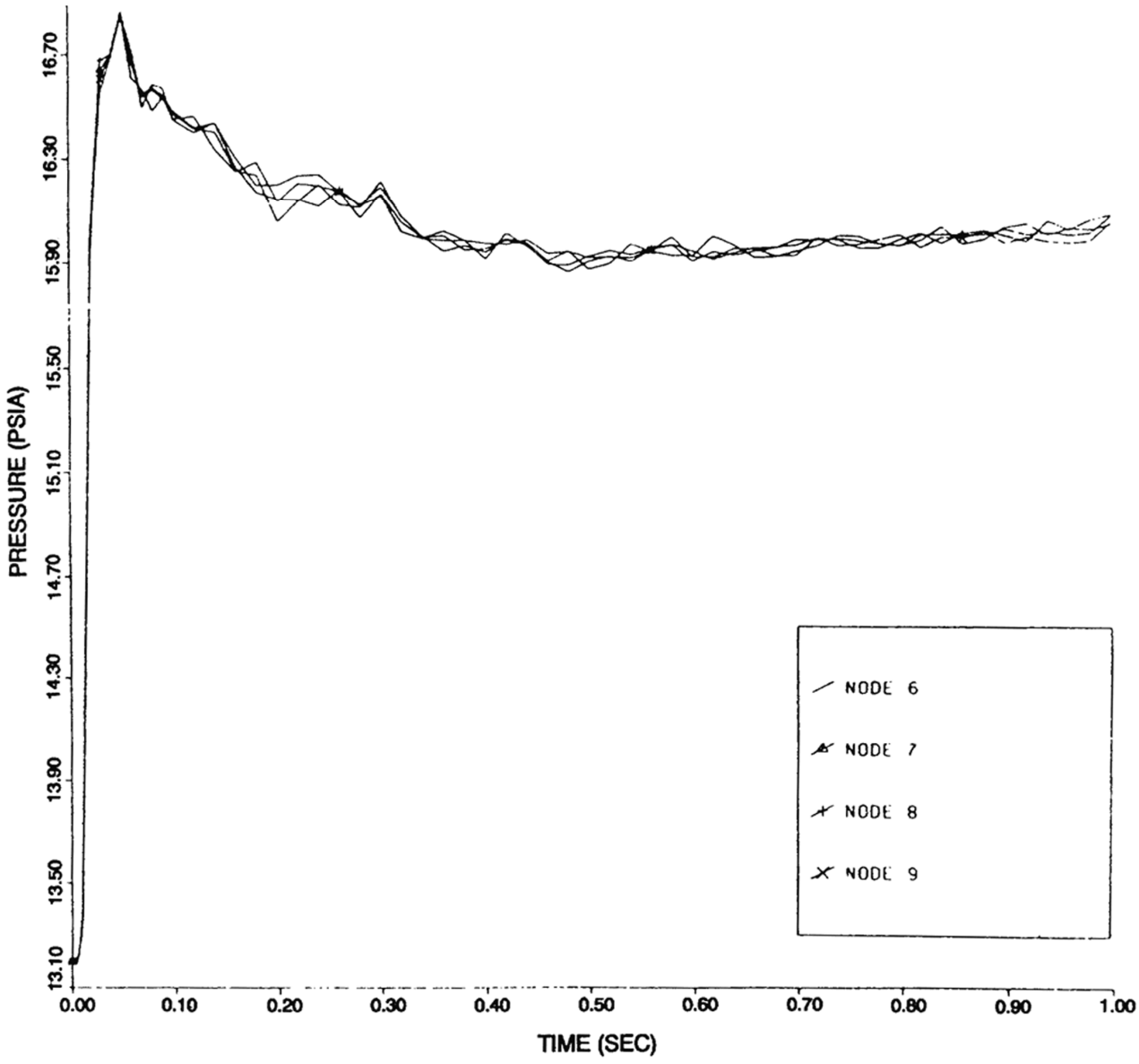
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 2, 3, 4, 5
SPRAY LINE BREAK AT TOP OF PRESSURIZER

FIGURE 6.2.1–25a (SHEET 2 OF 8)



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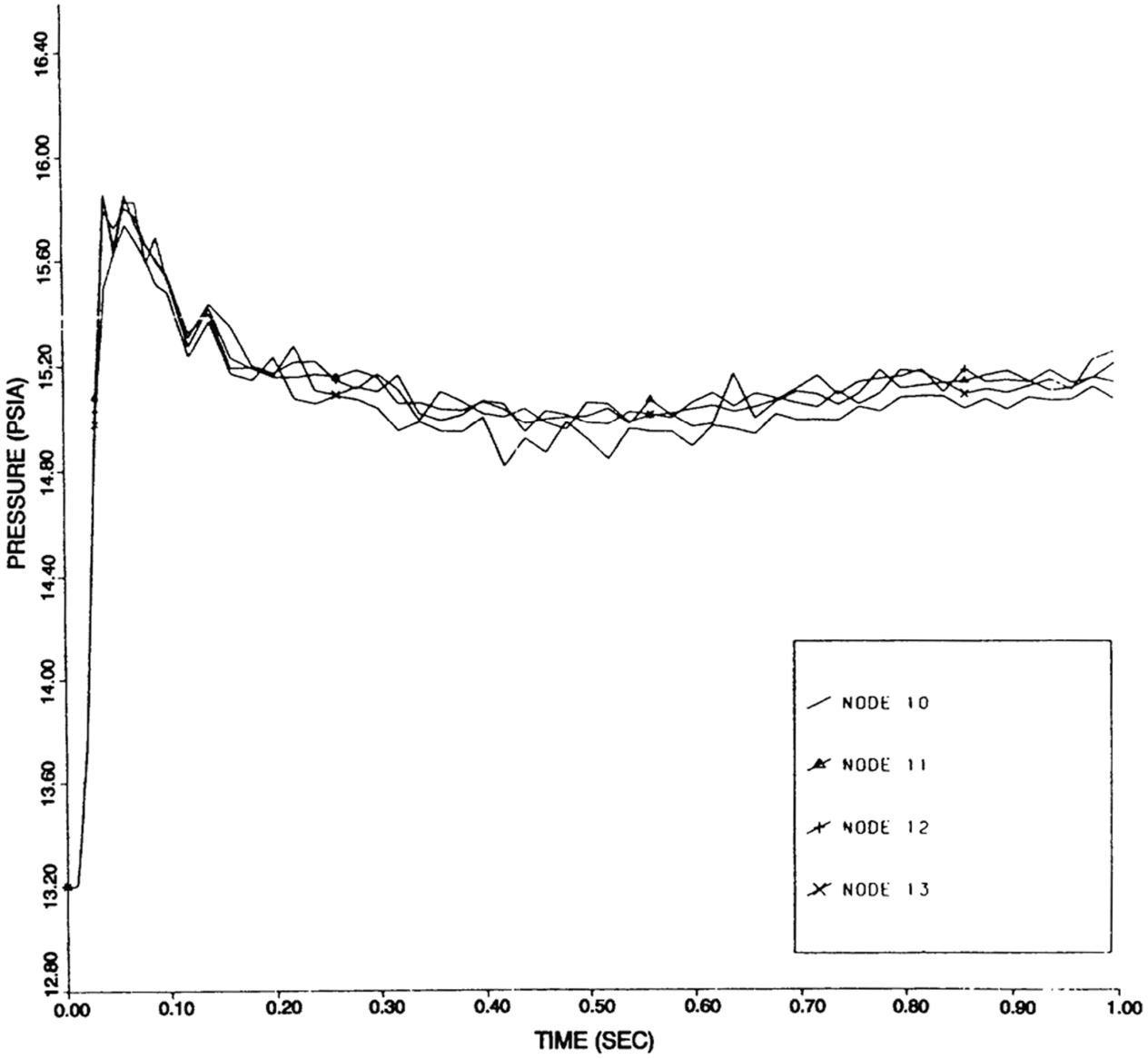
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 6, 7, 8, 9 SPRAY
LINE BREAK AT TOP OF PRESSURIZER

FIGURE 6.2.1–25a (SHEET 3 OF 8)



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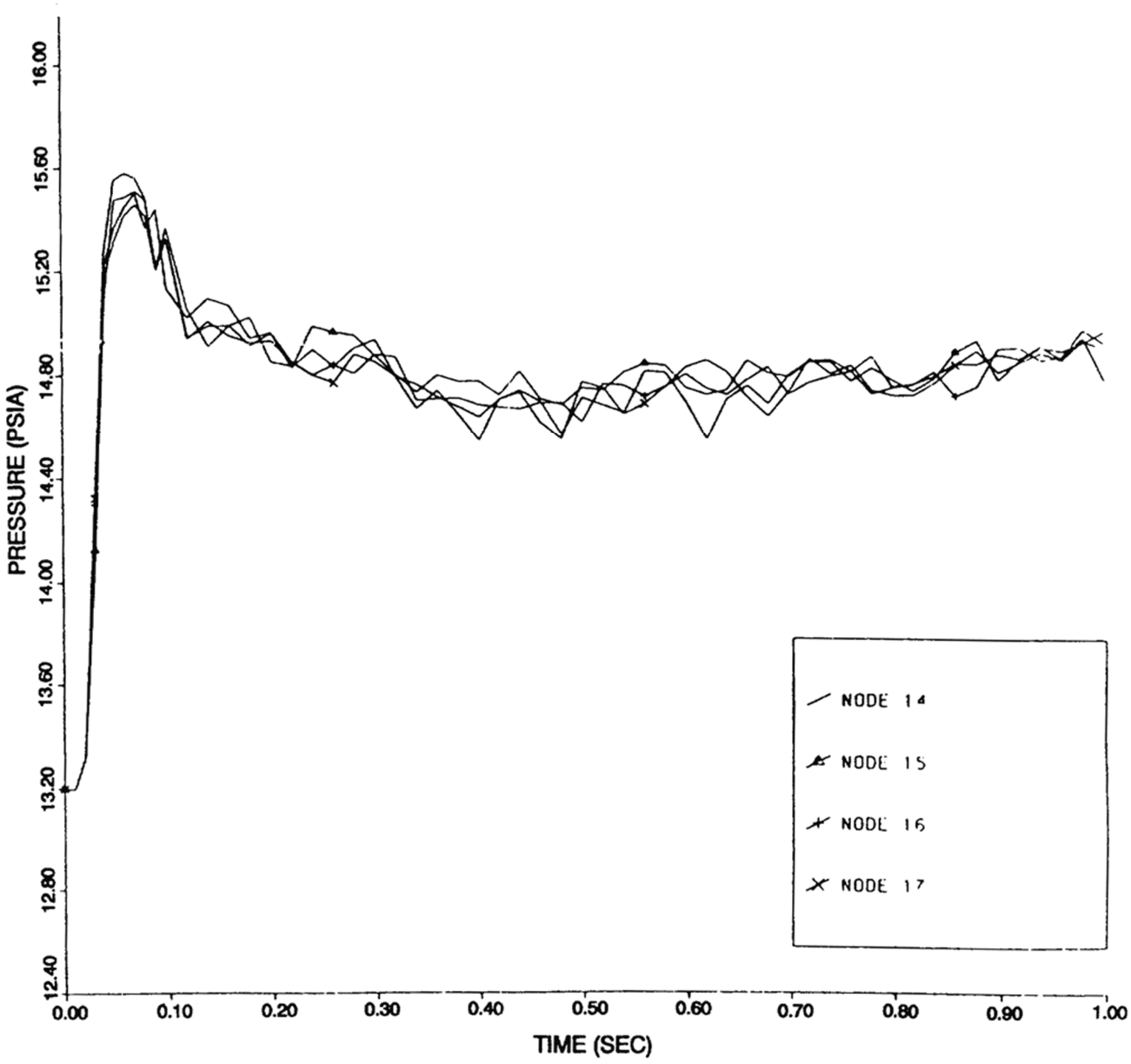
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 10, 11, 12, 13
SPRAY LINE BREAK AT TOP OF PRESSURIZER

FIGURE 6.2.1–25a (SHEET 4 OF 8)



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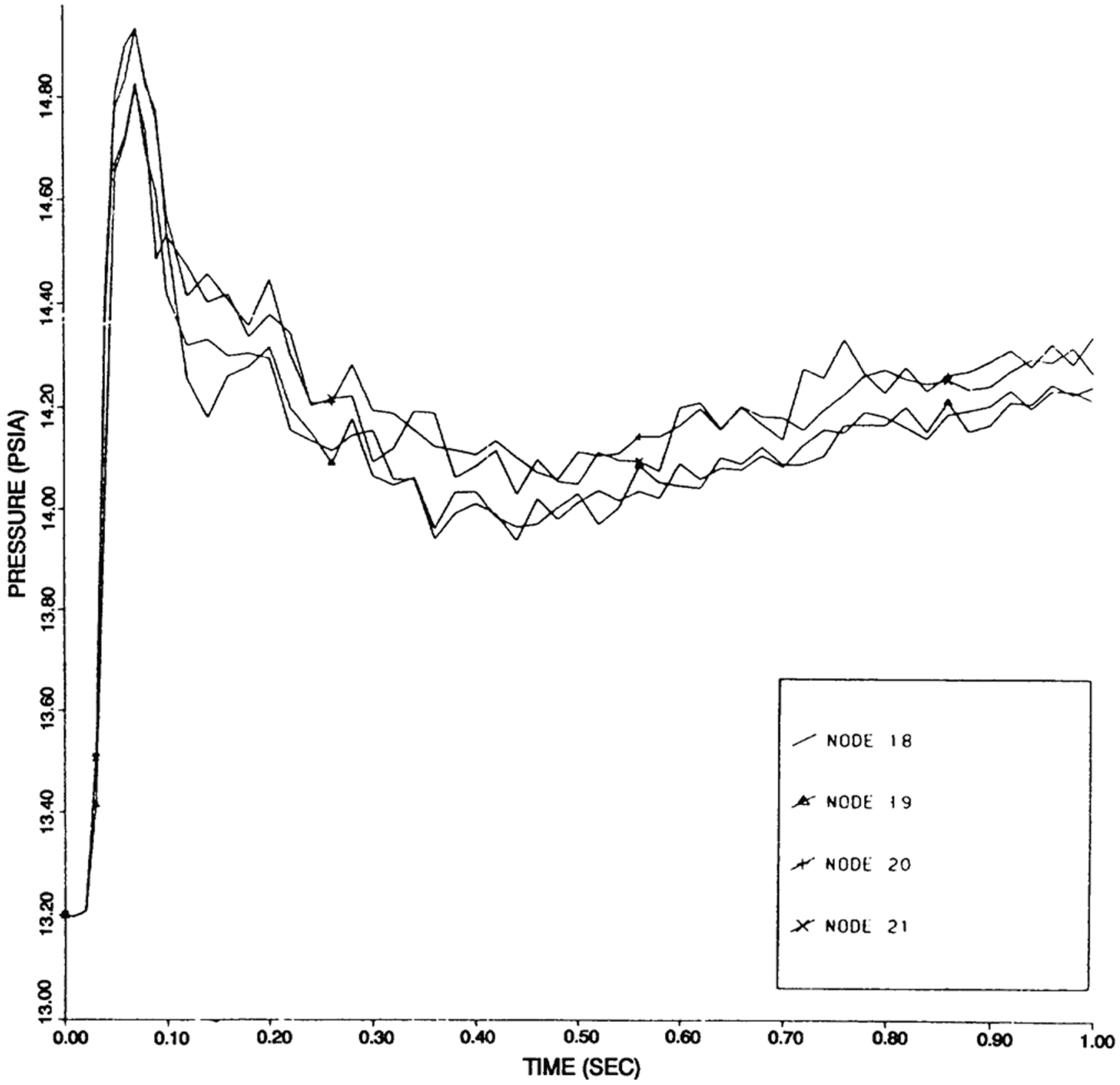
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 14, 15, 16, 17
SPRAY LINE BREAK AT TOP OF PRESSURIZER

FIGURE 6.2.1–25a (SHEET 5 OF 8)



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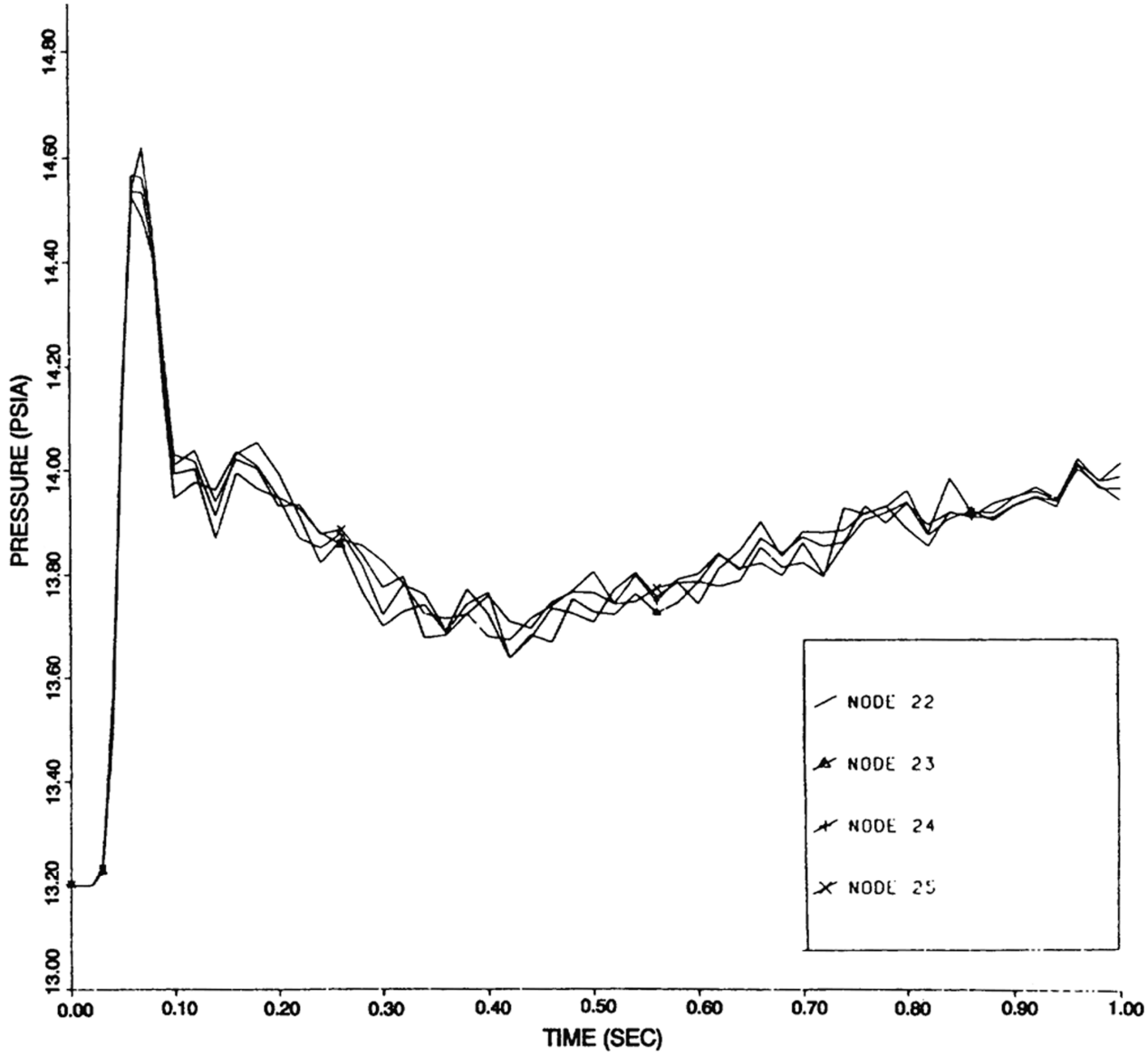
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 18, 19, 20, 21
SPRAY LINE BREAK AT TOP OF PRESSURIZER

FIGURE 6.2.1–25a (SHEET 6 OF 8)



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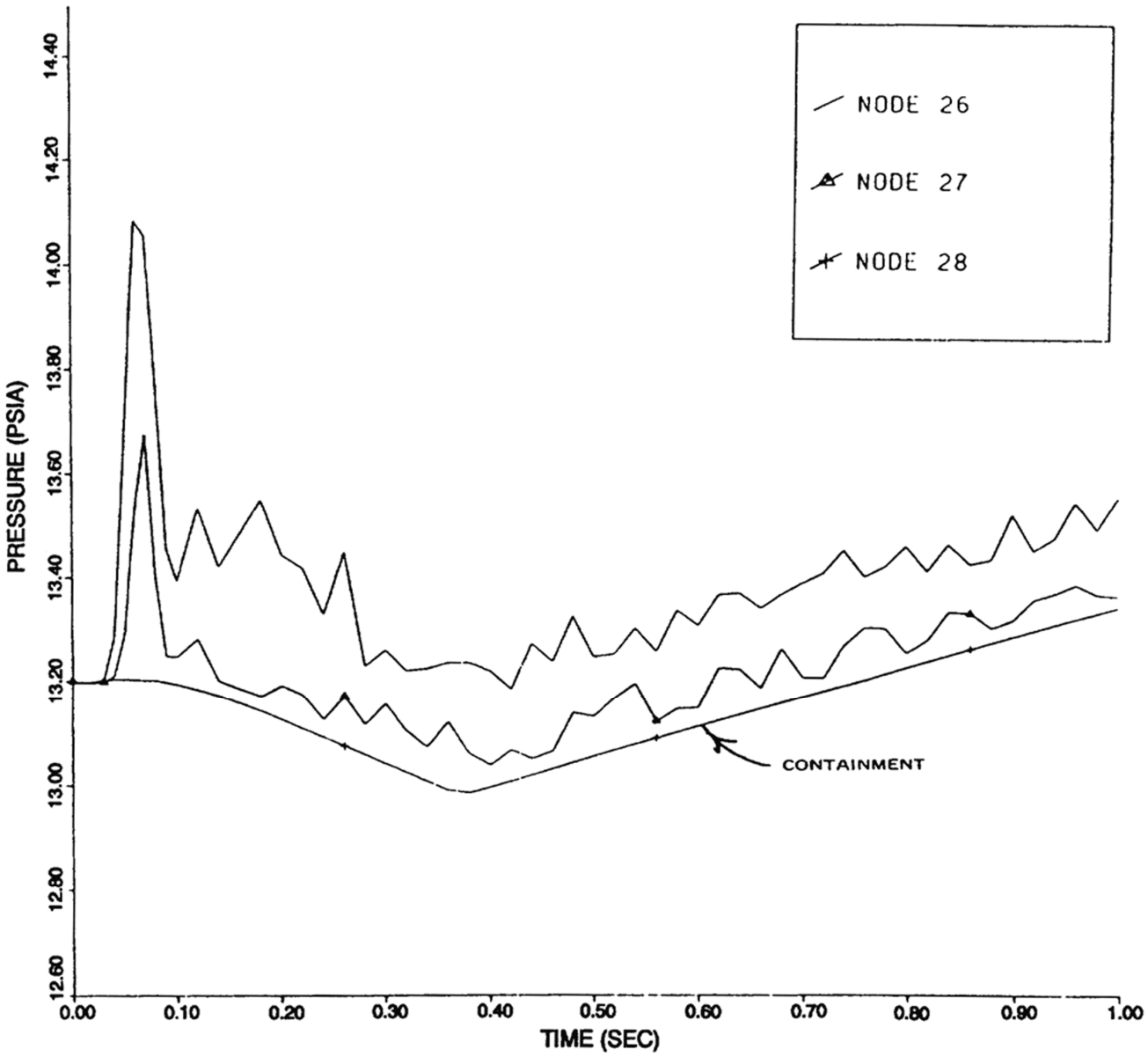
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 22, 23, 24, 25
SPRAY LINE BREAK AT TOP OF PRESSURIZER

FIGURE 6.2.1–25a (SHEET 7 OF 8)



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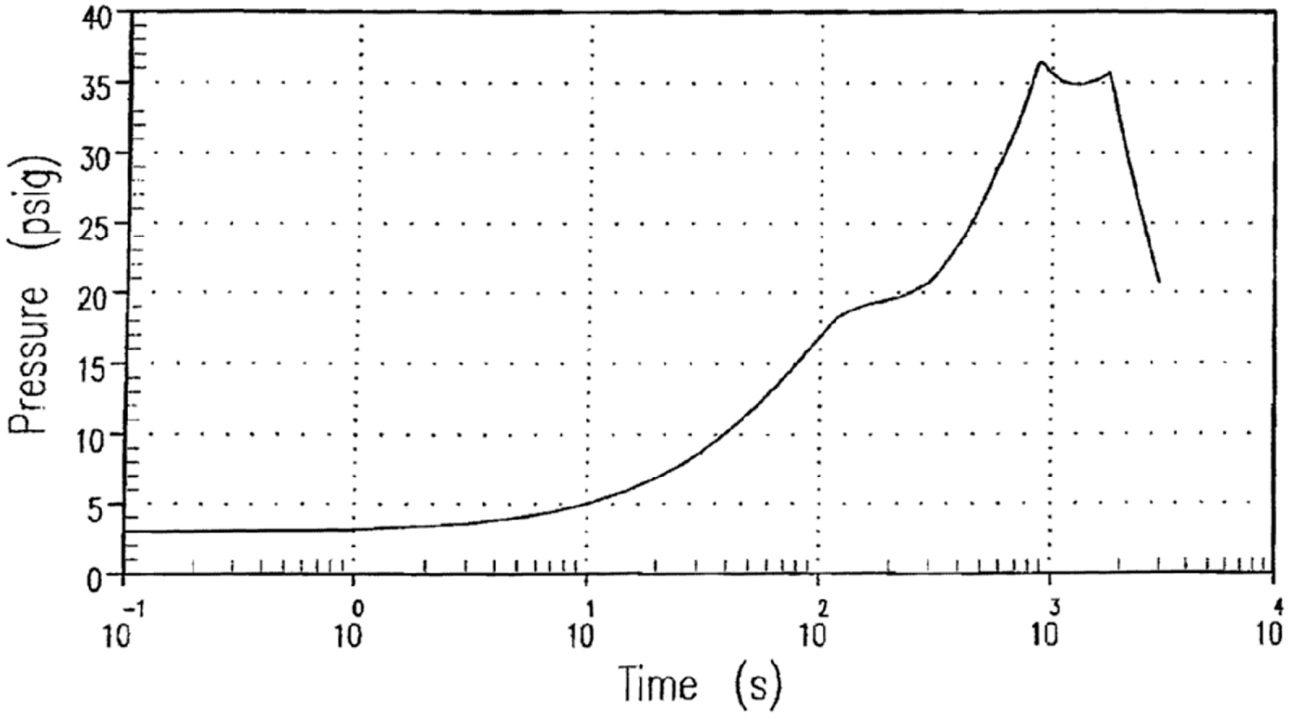
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PRESSURIZER COMPARTMENT
PRESSURE RESPONSE – NODE 26, 27, 28
SPRAY LINE BREAK AT TOP OF PRESSURIZER

FIGURE 6.2.1–25a (SHEET 8 OF 8)



REV 15 4/09

SOUTHERN COMPANY
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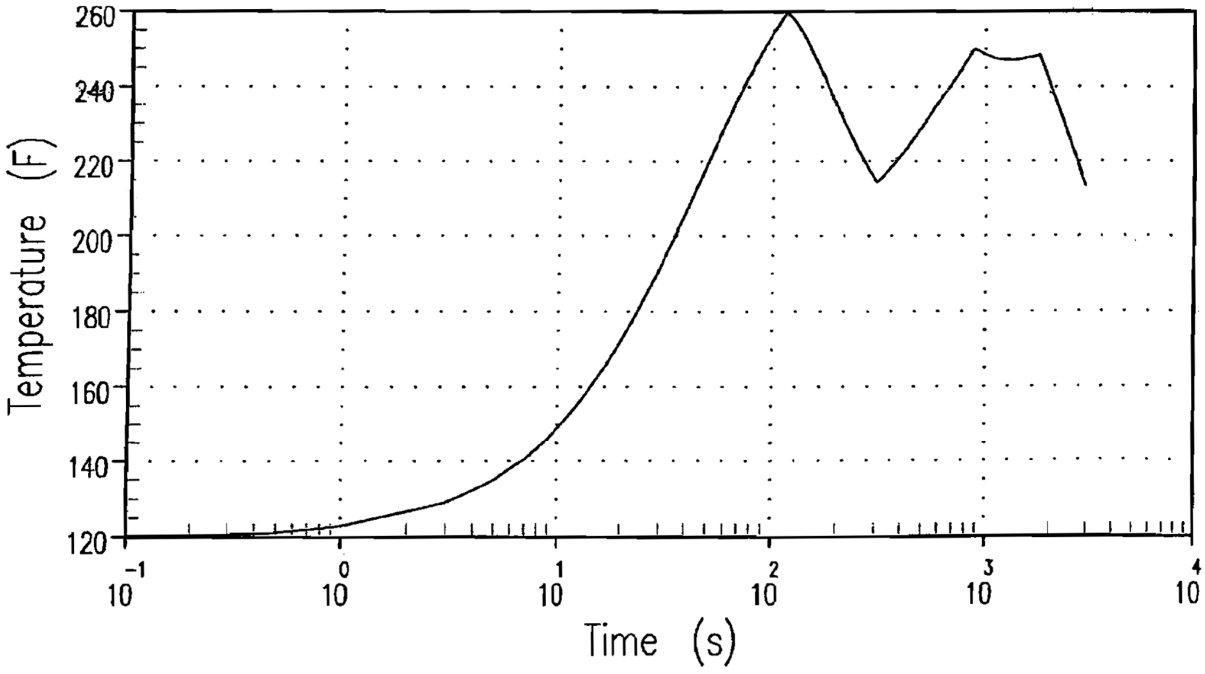
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTAINMENT PRESSURE TRANSIENT
MSLB – 0.4 ft^2 SPLIT RUPTURE– 0% POWER
CASE 16

FIGURE 6.2.1–26



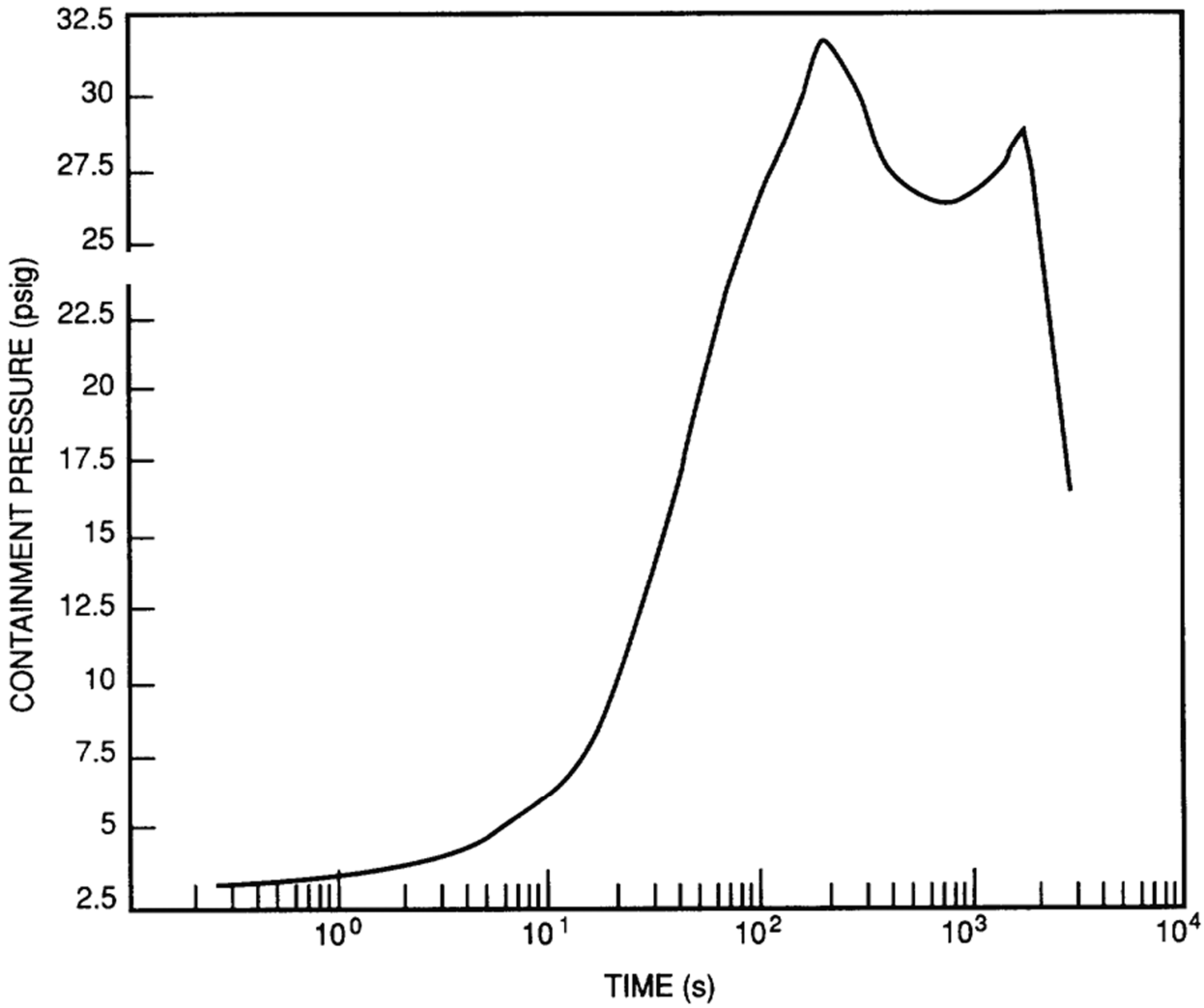
REV 15 4/09

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTAINMENT TEMPERATURE TRANSIENT
MSLB – 0.4 ft^2 SPLIT RUPTURE– 0% POWER
CASE 16

FIGURE 6.2.1–27



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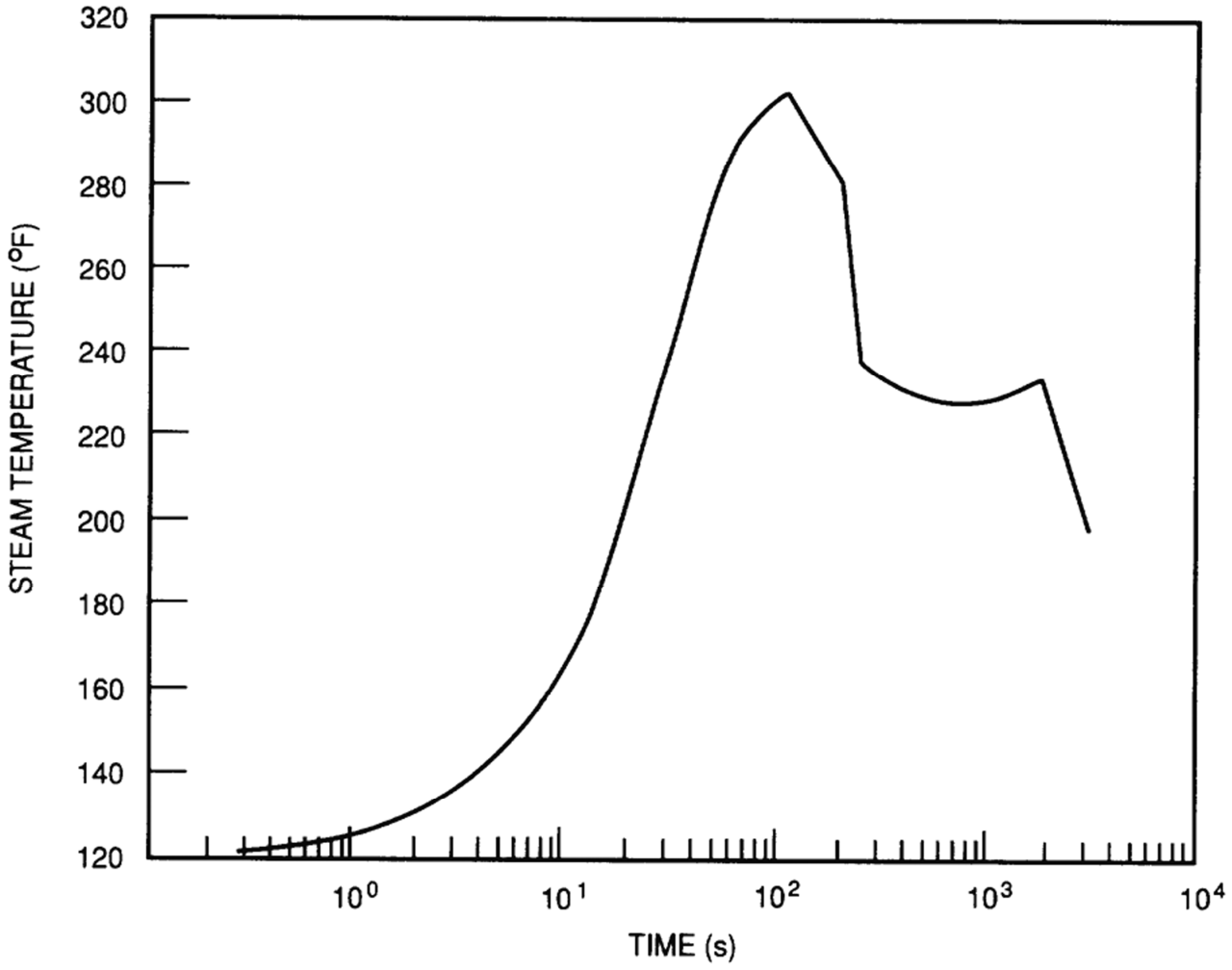
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	CONTAINMENT PRESSURE TRANSIENT MSLB – 0.86 ft^2 SPLIT RUPTURE – 102% POWER CASE 13
	FIGURE 6.2.1–28



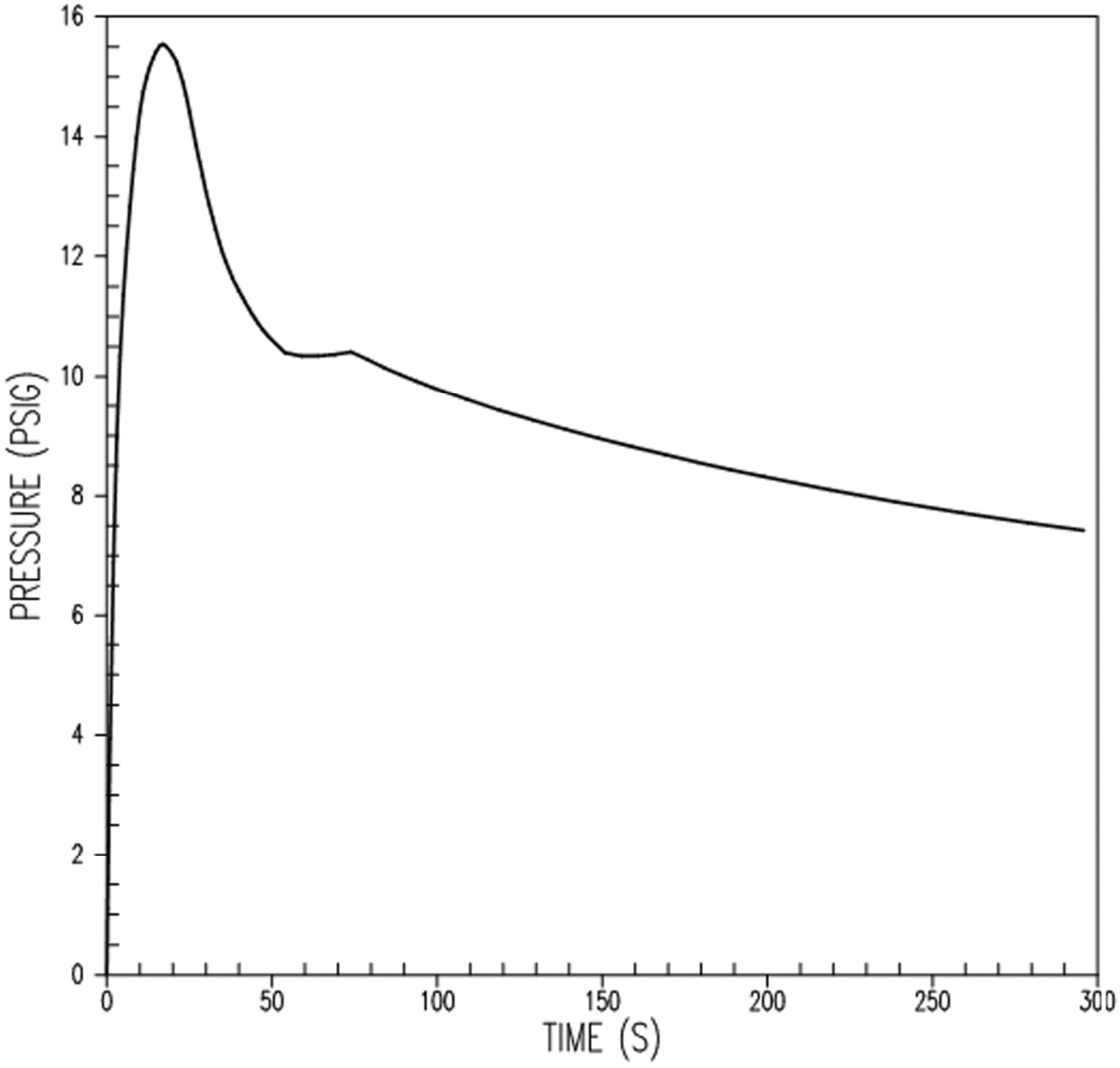
REV 18 9/13

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CONTAINMENT TEMPERATURE TRANSIENT
MSLB – 0.86 ft^2 SPLIT RUPTURE – 102% POWER
CASE 13

FIGURE 6.2.1–29



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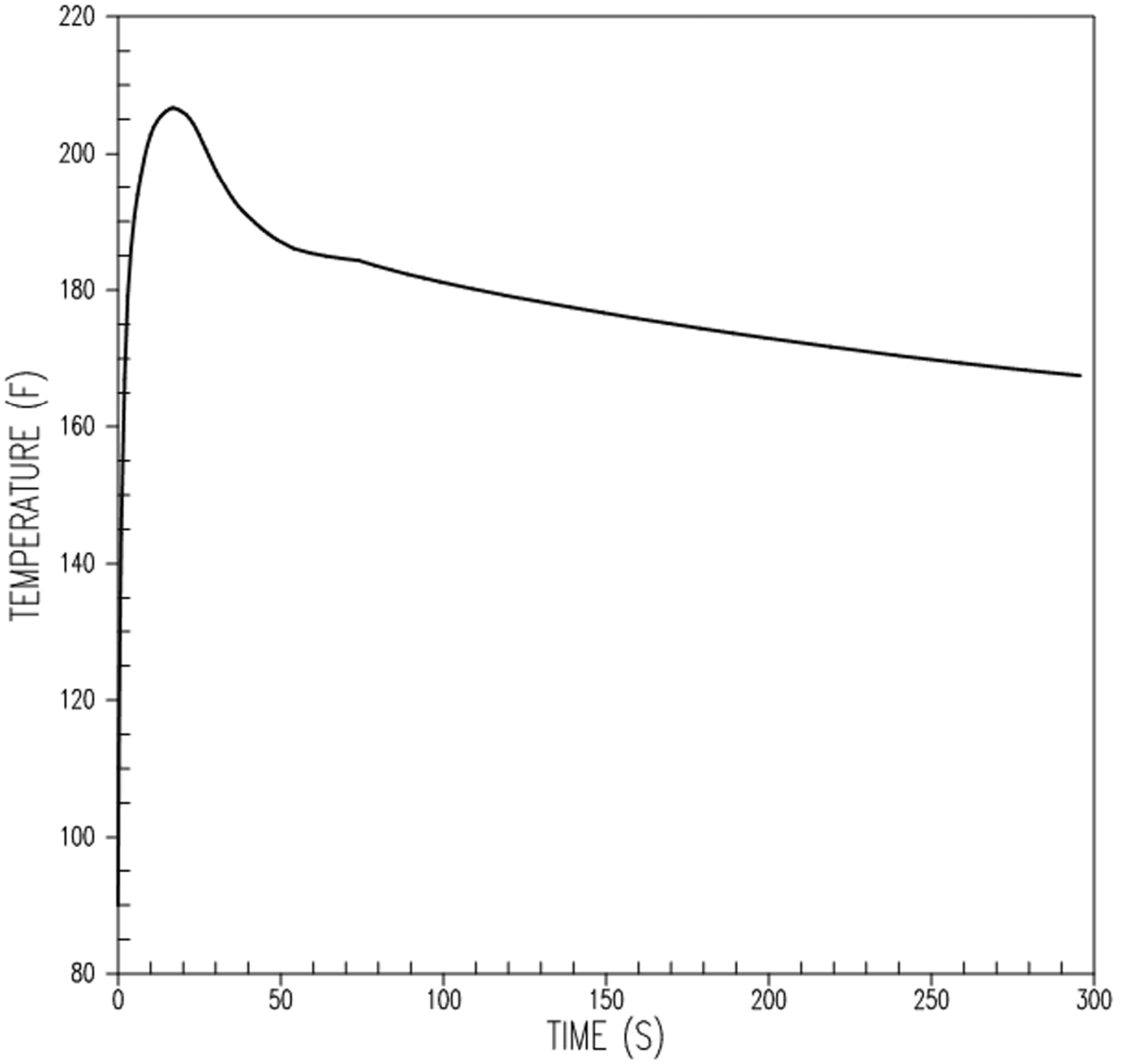
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	CONTAINMENT PRESSURE DECLG (C_D = 0.6, LOW T_{AVG}, MIN SI, COSINE) FIGURE 6.2.1–30



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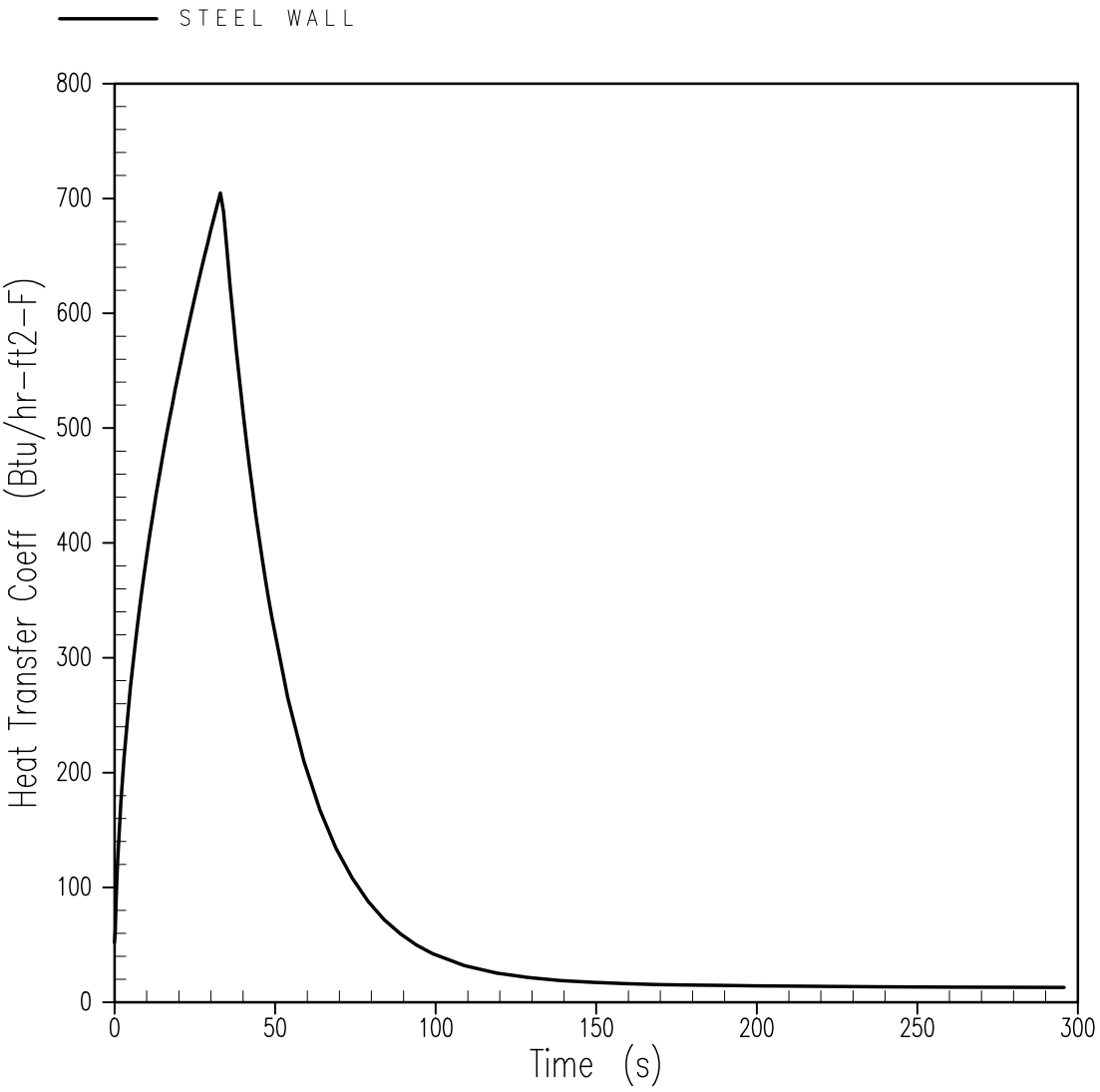
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	CONTAINMENT TEMPERATURE DECLG (C_D = 0.6, LOW T_{AVG}, MIN SI, COSINE) FIGURE 6.2.1–31



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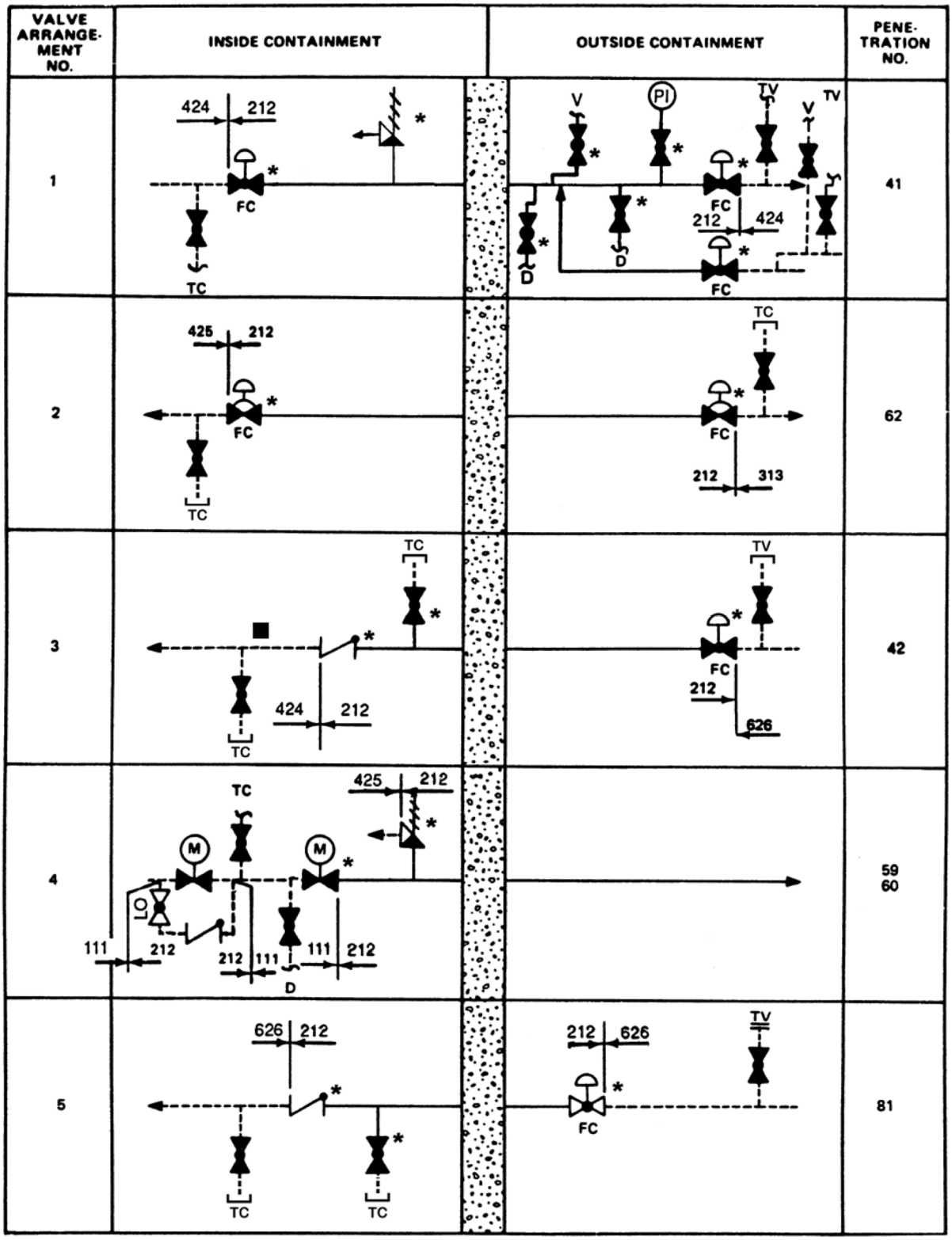
SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	CONTAINMENT WALL CONDENSATION HEAT TRANSFER COEFFICIENT DECLG (C_D = 0.6, LOW T_{AVG}, MIN SI, COSINE) FIGURE 6.2.1–32



* Valve listed on table 6.2.4-1.

** Bypass line (includes globe & check valve) Unit 1, Loops 1 & 4; Unit 2, Loop 4 only.

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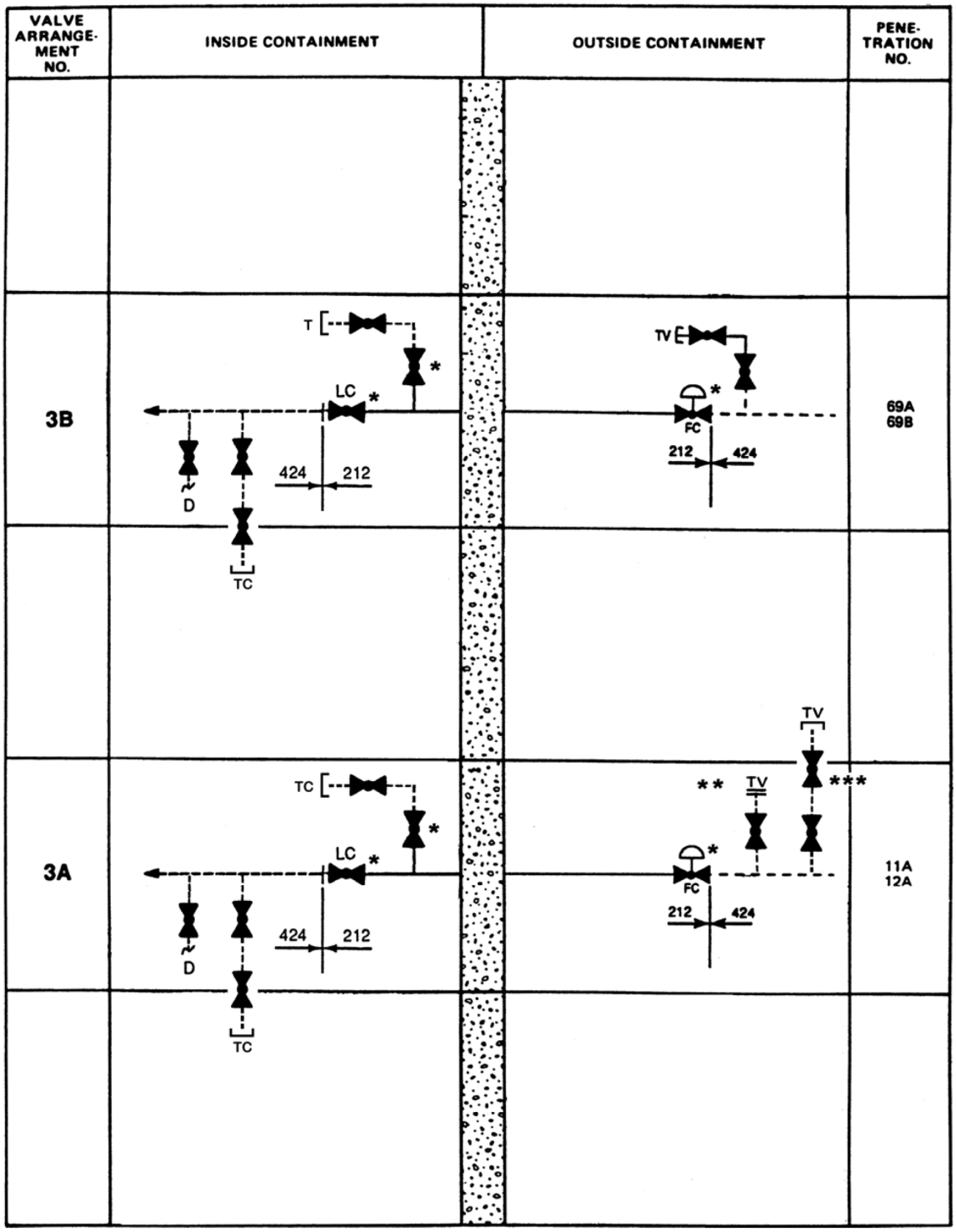
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 1 OF 13)



* Valve listed on table 6.2.4-1

** Applicable Unit 1 only

*** Applicable Unit 2 only

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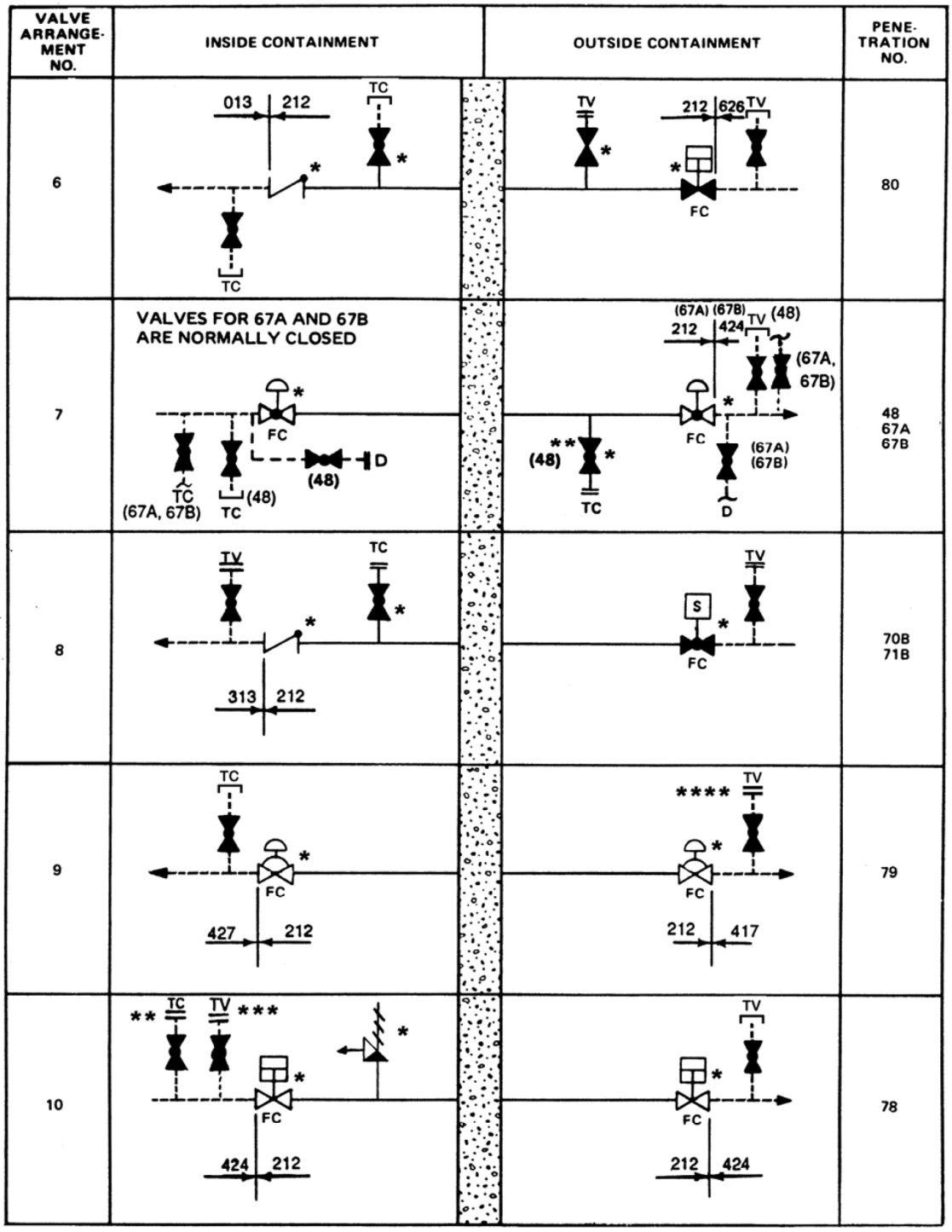

* Valve listed on table 6.2.4-1
** Applicable Unit 1 only
*** Applicable Unit 2 only
**** Blind flange applies to Unit 2 only

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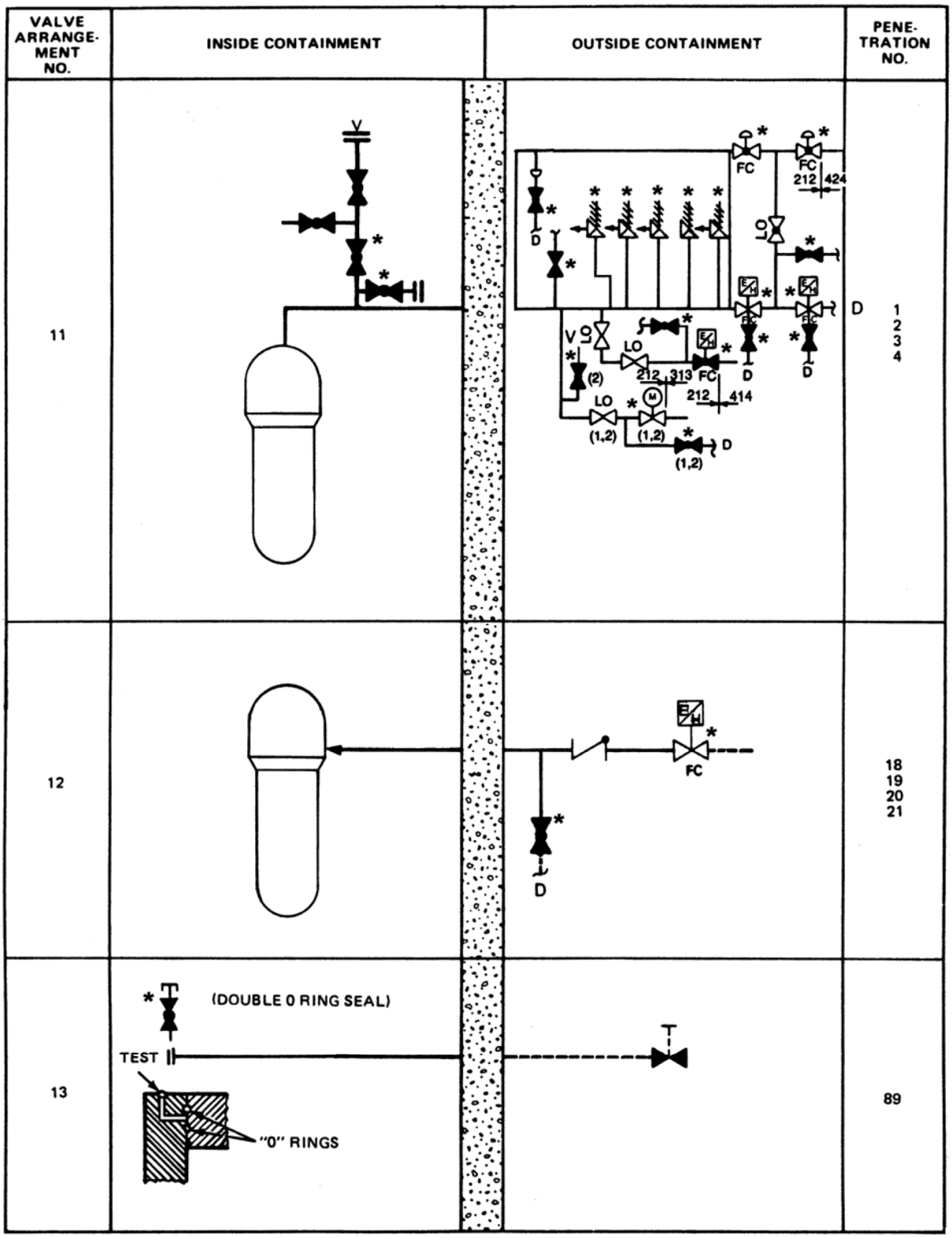
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 3 OF 13)



* Valve listed on table 6.2.4-1

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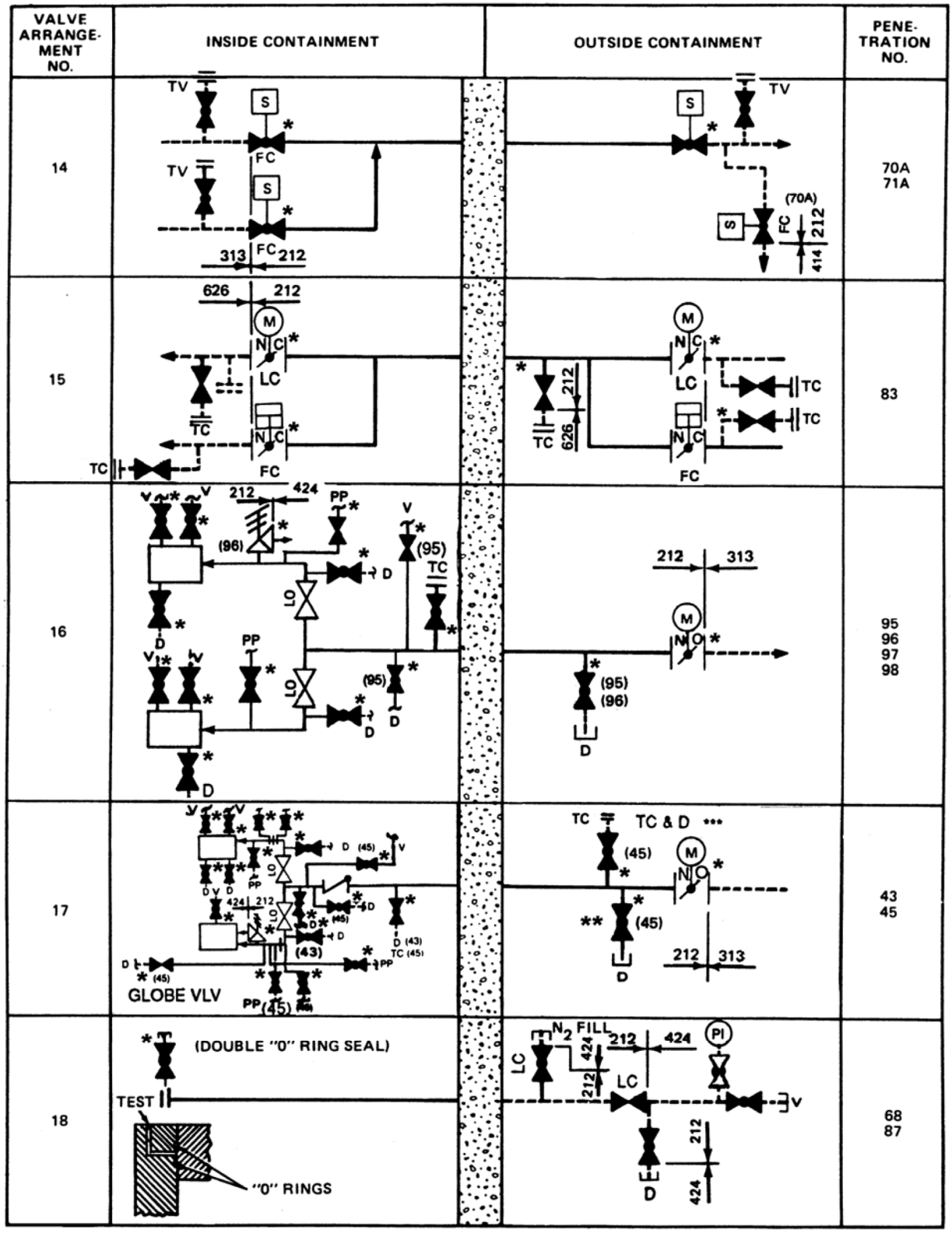
SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 4 OF 13)



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SOUTHERN COMPANY
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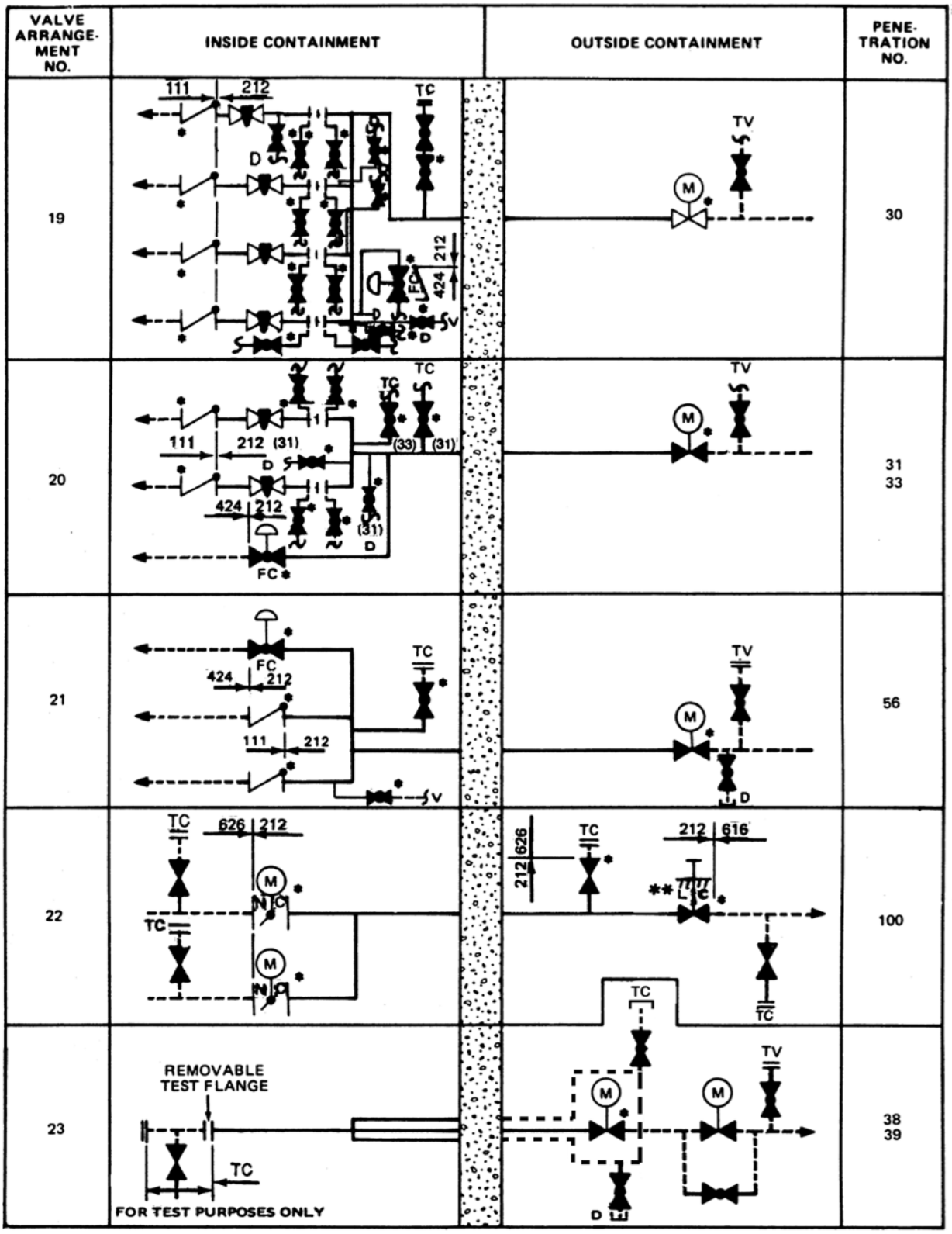
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 5 OF 13)



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SOUTHERN COMPANY
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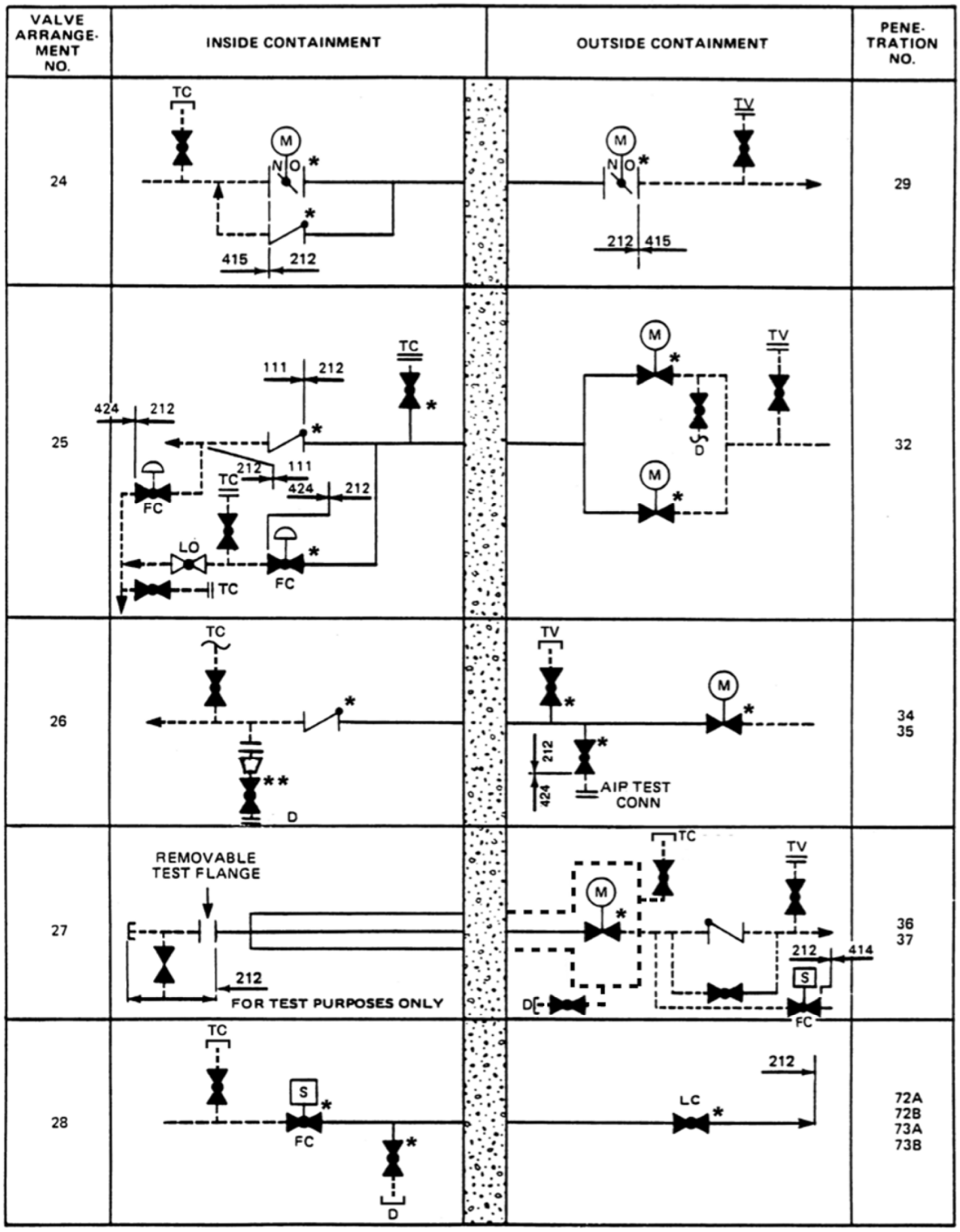
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 6 OF 13)



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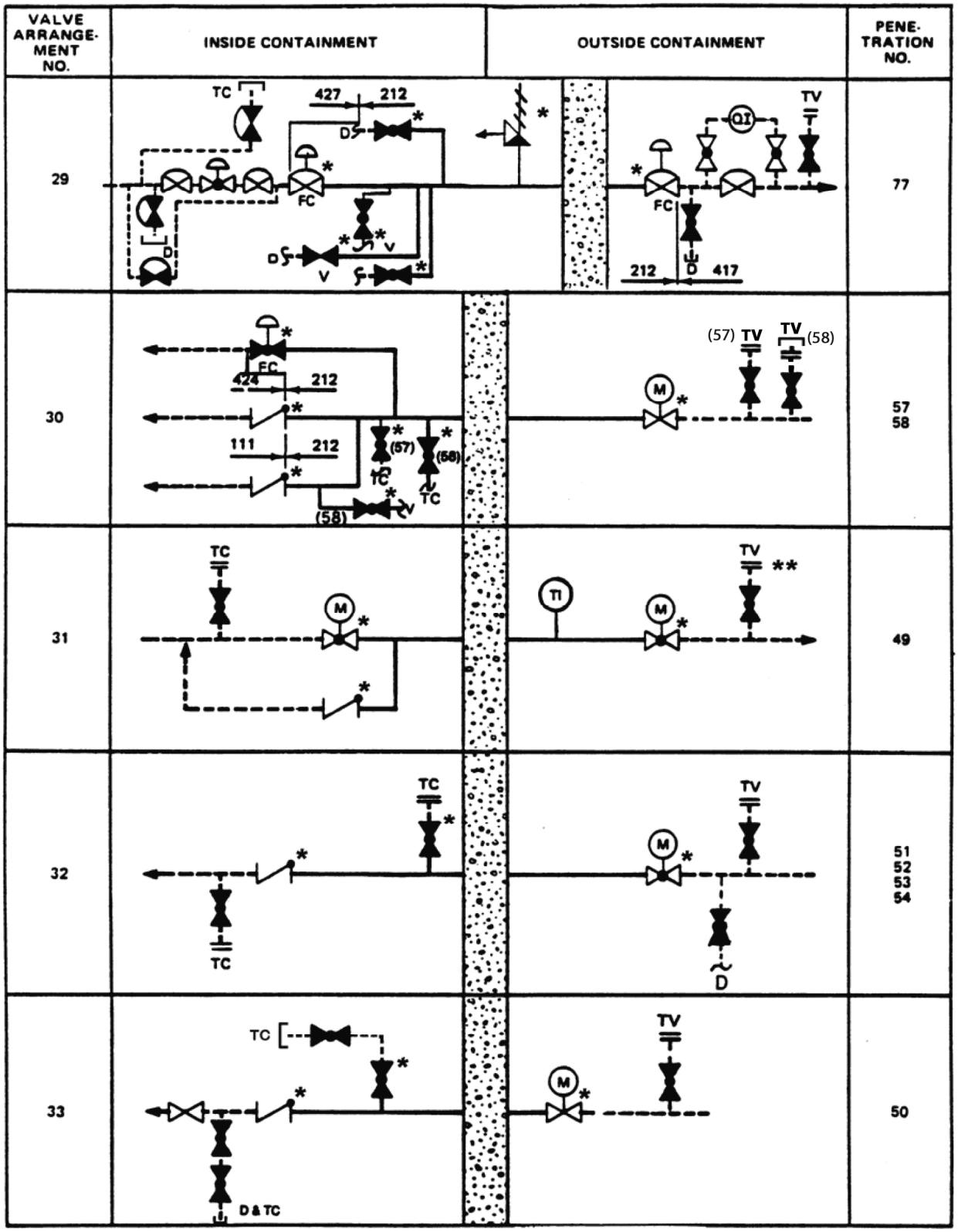
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 7 OF 13)



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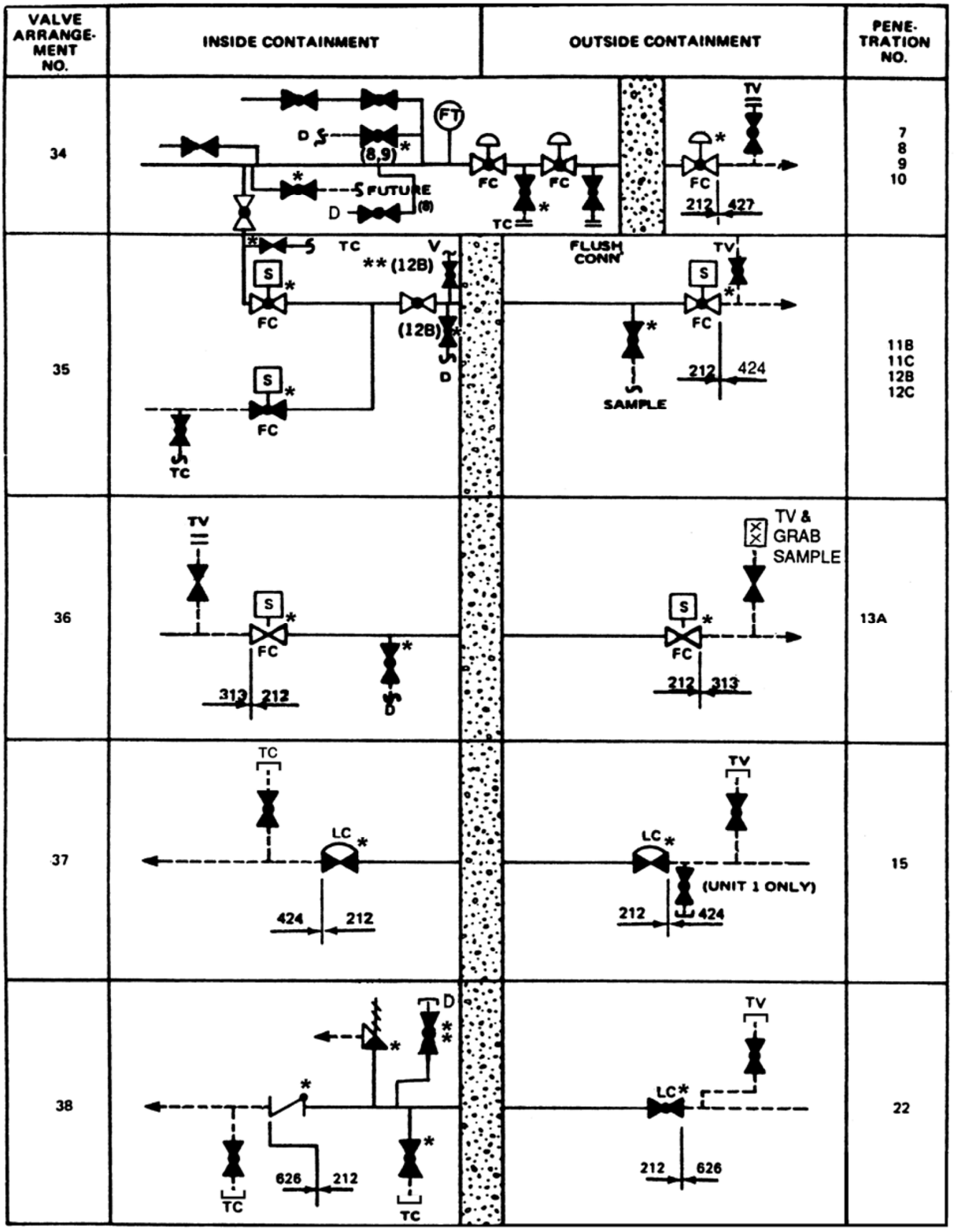
VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 8 OF 13)



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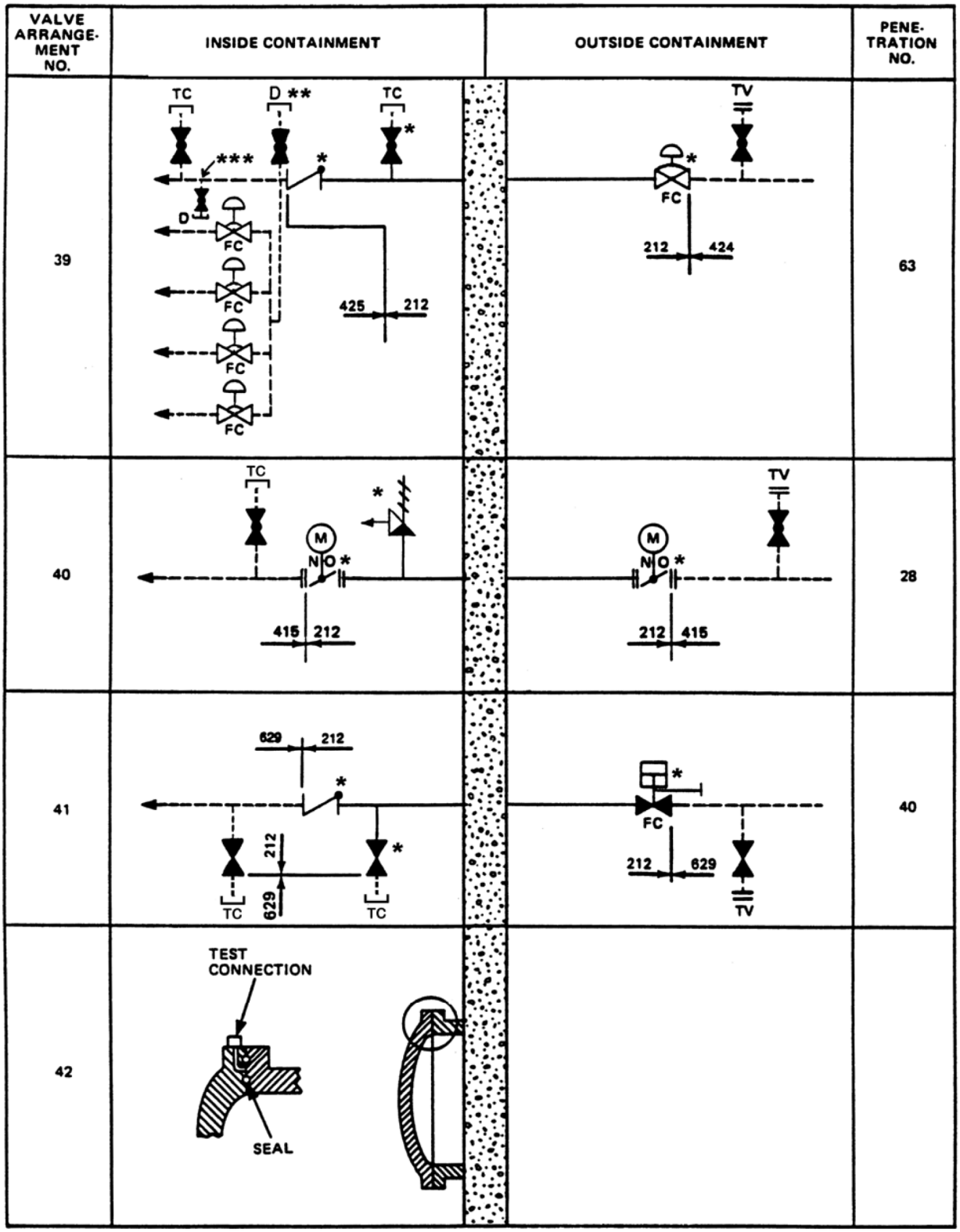
SOUTHERN COMPANY *Energy to Serve Your World®*	VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	VALVE ARRANGEMENT
		FIGURE 6.2.4-1 (SHEET 9 OF 13)



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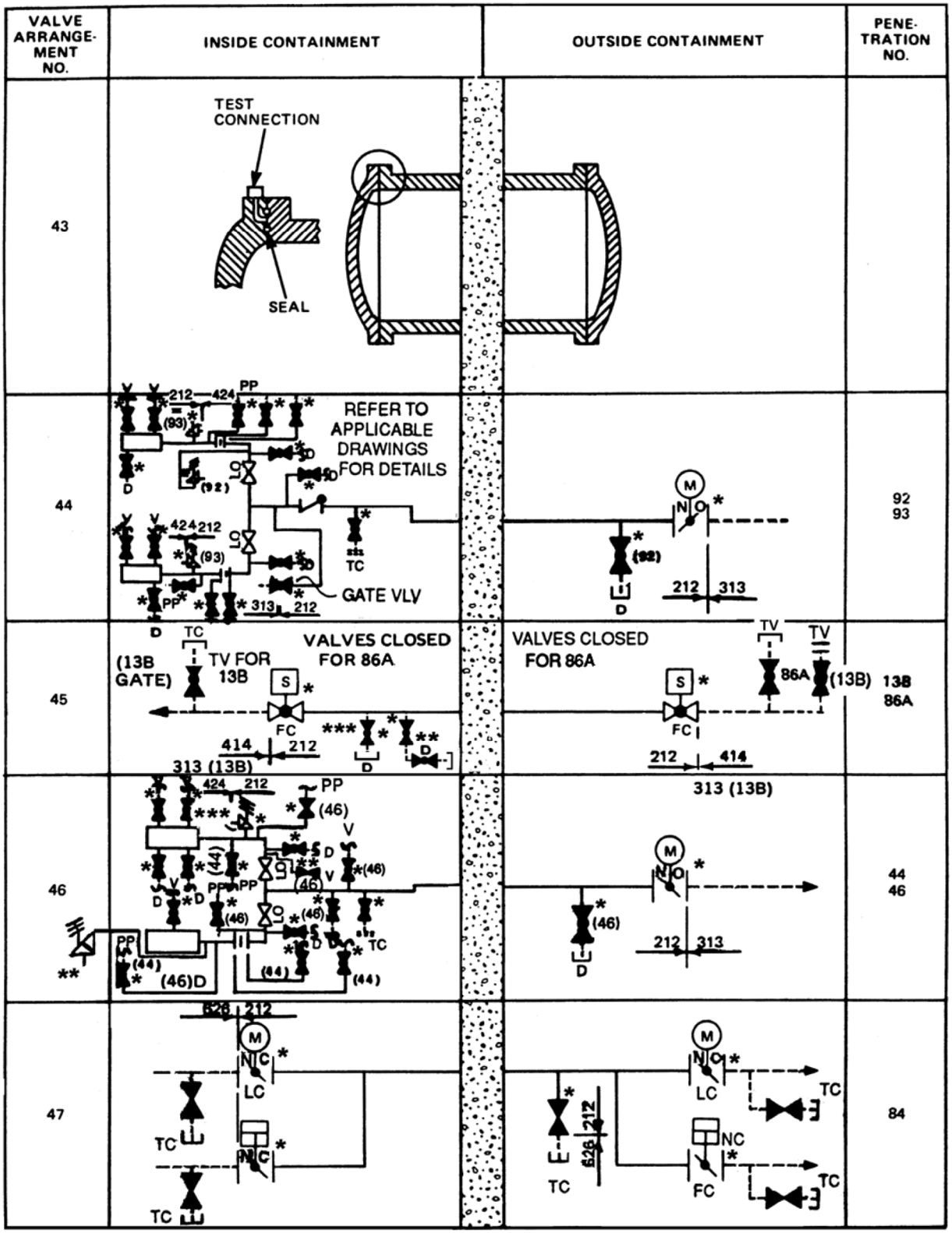
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 10 OF 13)



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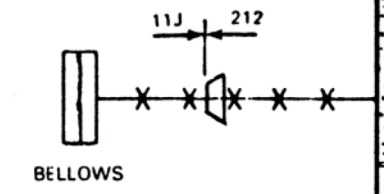
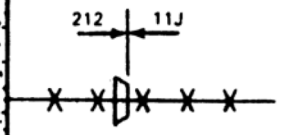
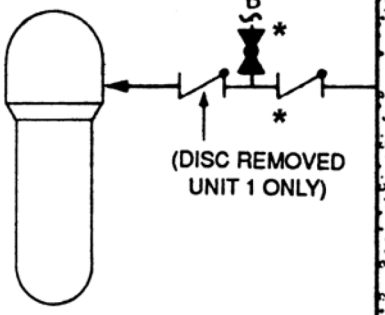
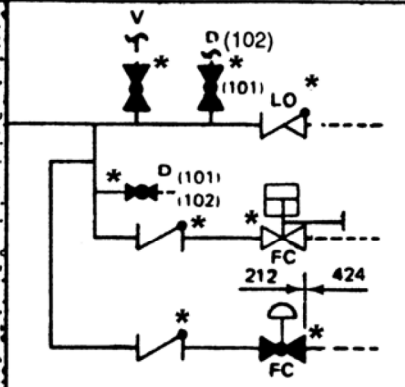
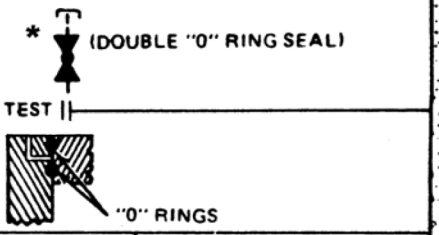
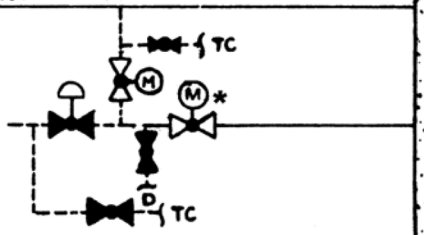
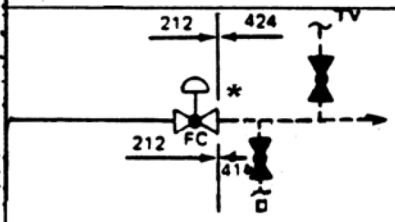
VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 11 OF 13)

VALVE ARRANGEMENTS NO.	INSIDE CONTAINMENT	OUTSIDE CONTAINMENT	PENETRATION NO.
48			13C 69C 70C 71C 85C 67C
49			101 102 103 104
51			64A 64B
52			24

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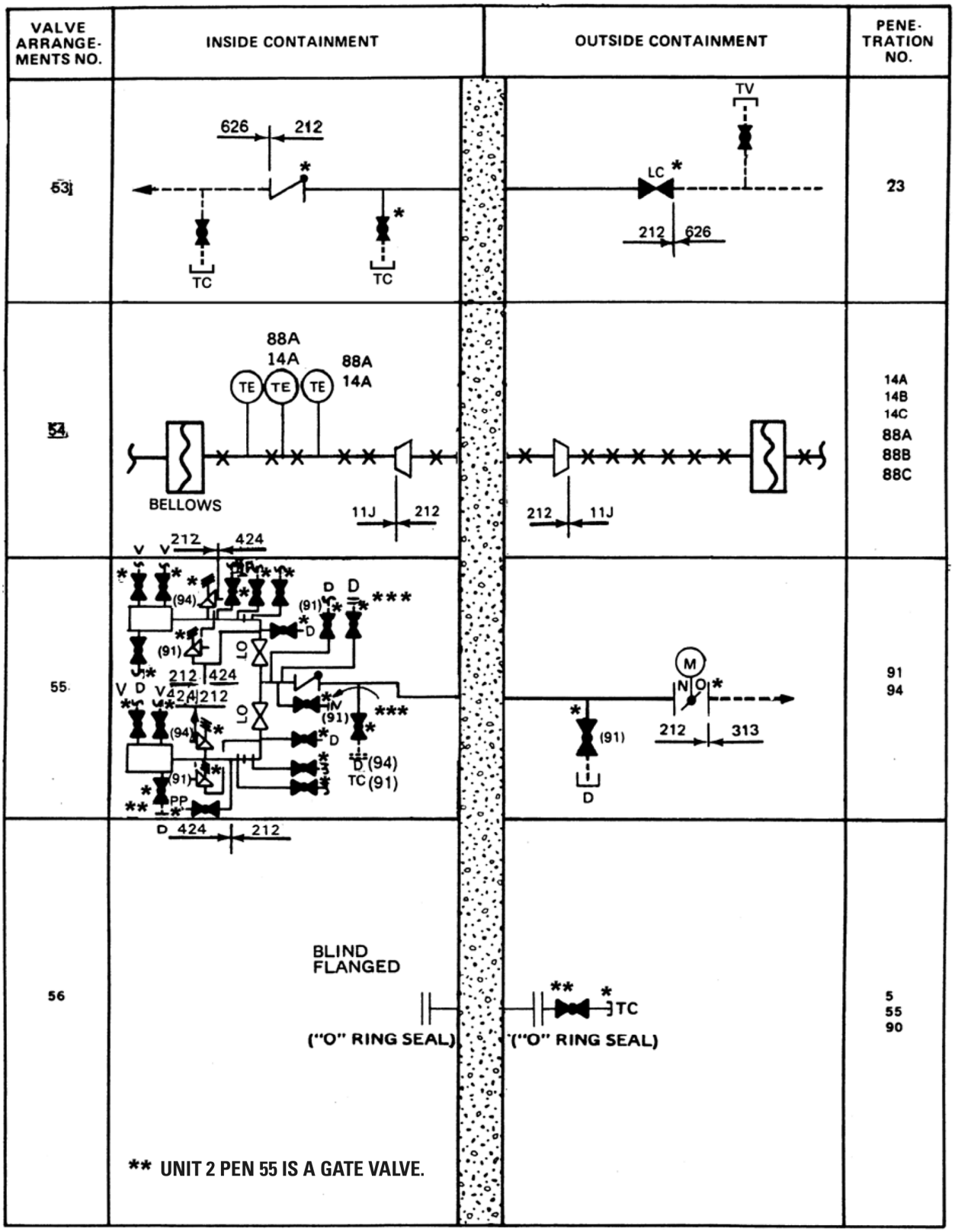
UNIT 1 AND UNIT 2

VALVE ARRANGEMENT

FIGURE 6.2.4-1 (SHEET 12 OF 13)



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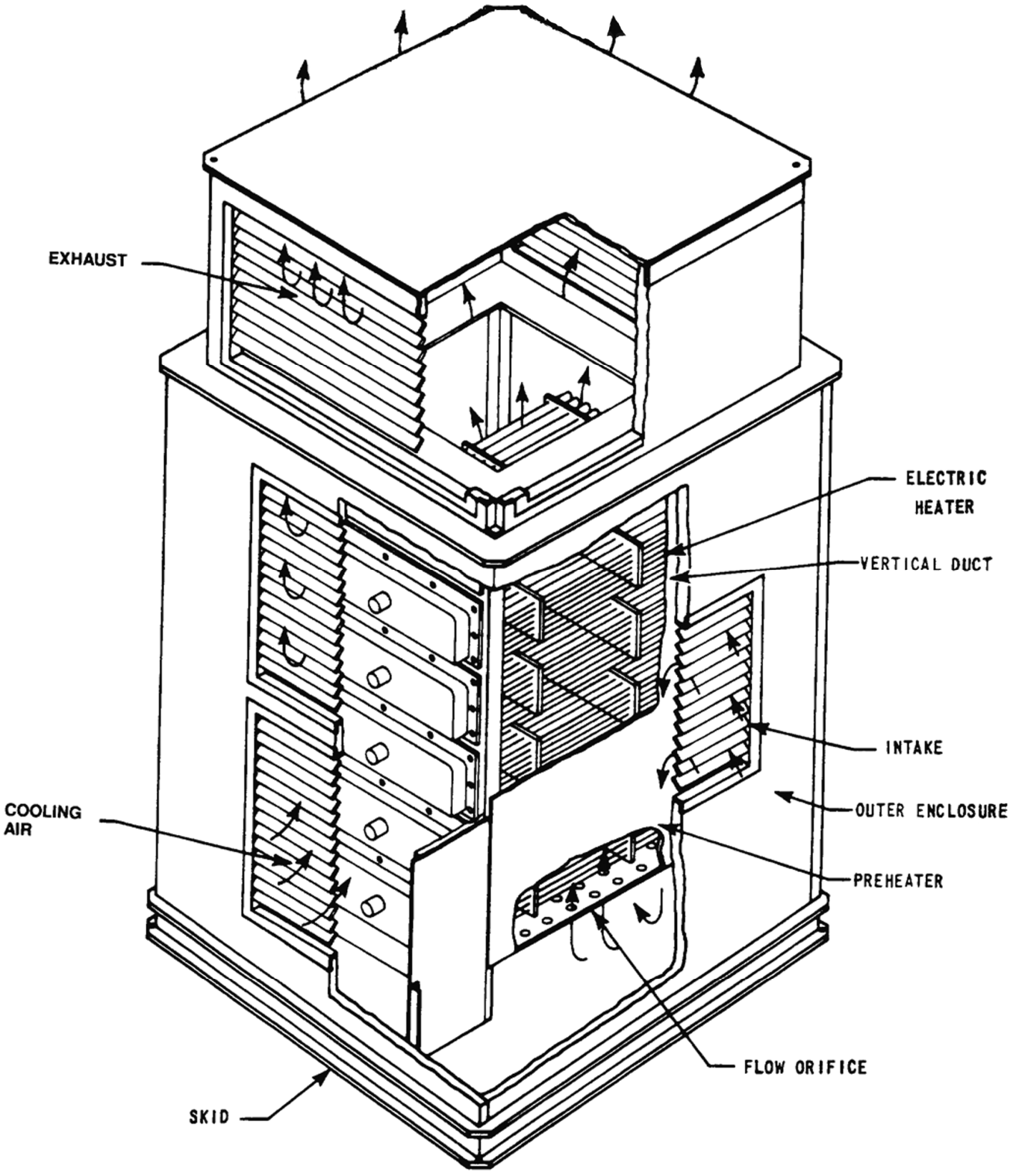
SOUTHERN COMPANY *Energy to Serve Your World®*	VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	VALVE ARRANGEMENT
		FIGURE 6.2.4-1 (SHEET 13 OF 13)



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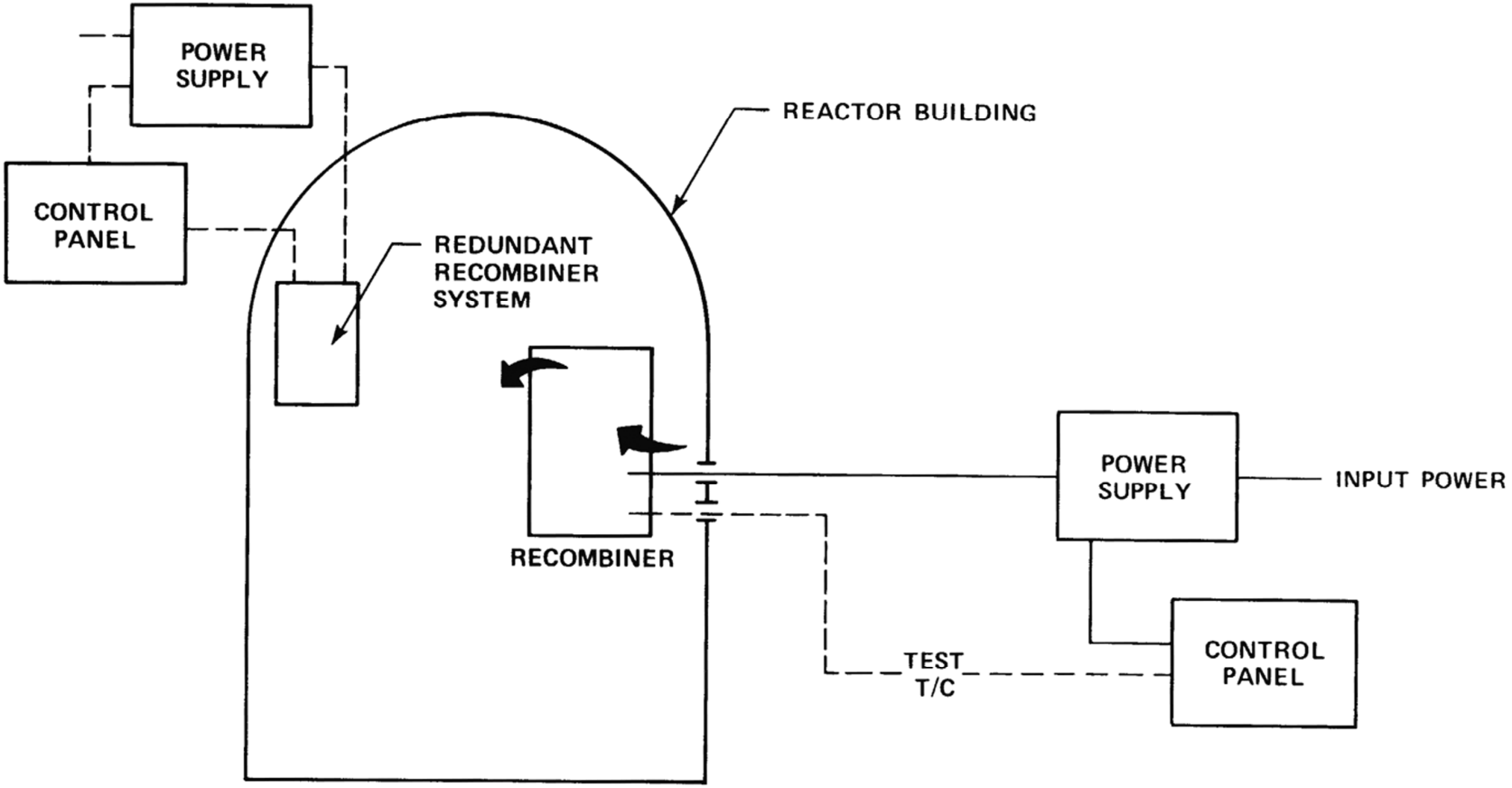
UNIT 1 AND UNIT 2

ELECTRIC HRYDROGEN RECOMBINER (TYPICAL)

FIGURE 6.2.5–1



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ELECTRIC HYDROGEN RECOMBINER SYSTEM SCHEMATIC

FIGURE 6.2.5–2

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COMPARISON OF ANS 5.1 DECAY ENERGY CURVE AT 650 DAYS IRRADIATION + 20% TO DECAY ENERGY VALUES USED FOR H_2 PRODUCTION CALCULATION

FIGURE 6.2.5–3

DELETED

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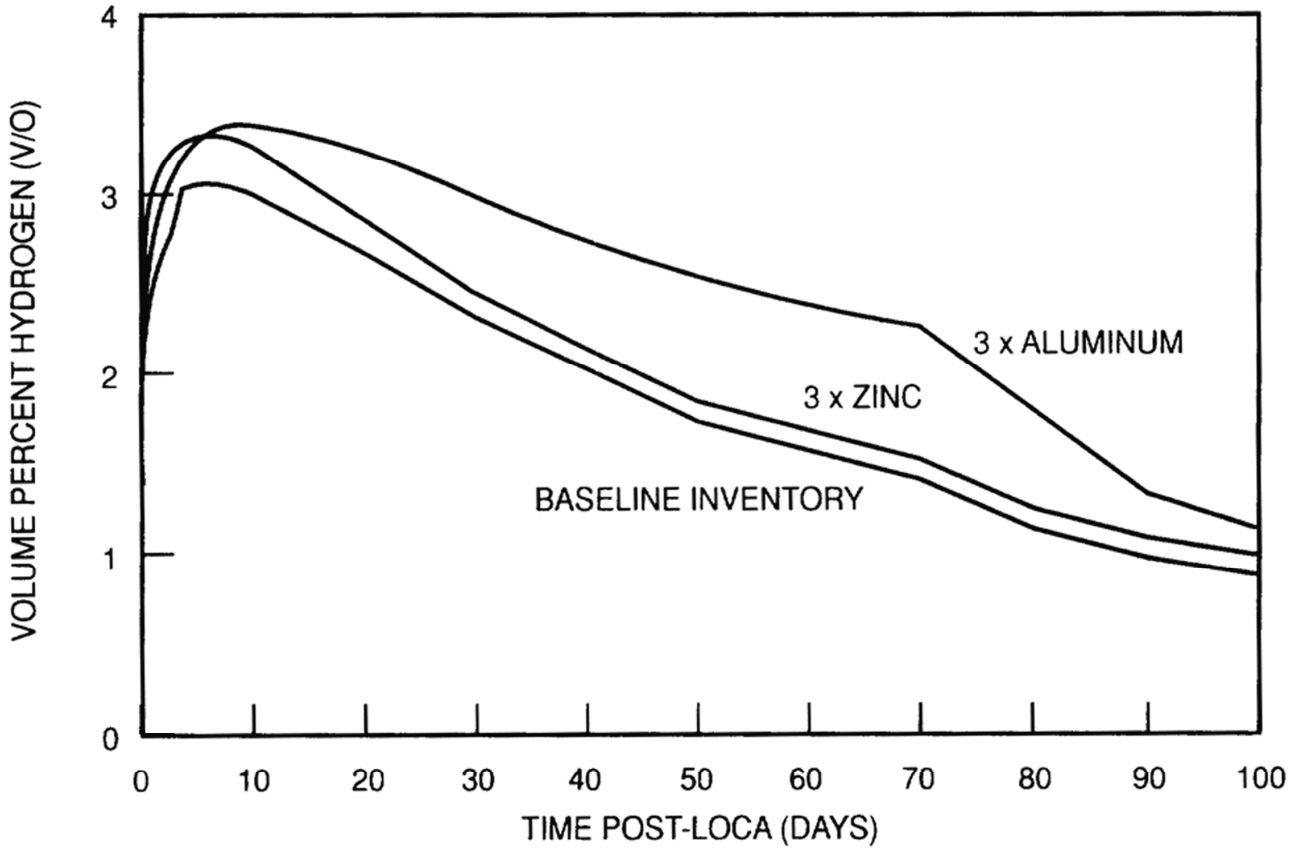
SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	ALUMINUM AND ZINC CORROSION RATE DESIGN CURVES
		FIGURE 6.2.5–4



THIS FIGURE REPRESENTS THE CONCENTRATION OF HYDROGEN CONTRIBUTED FROM ALL SOURCES.

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CONTAINMENT HYDROGEN CONCENTRATION
(ONE RECOMBINER ON AT 3.0 V/O)

FIGURE 6.2.5–5

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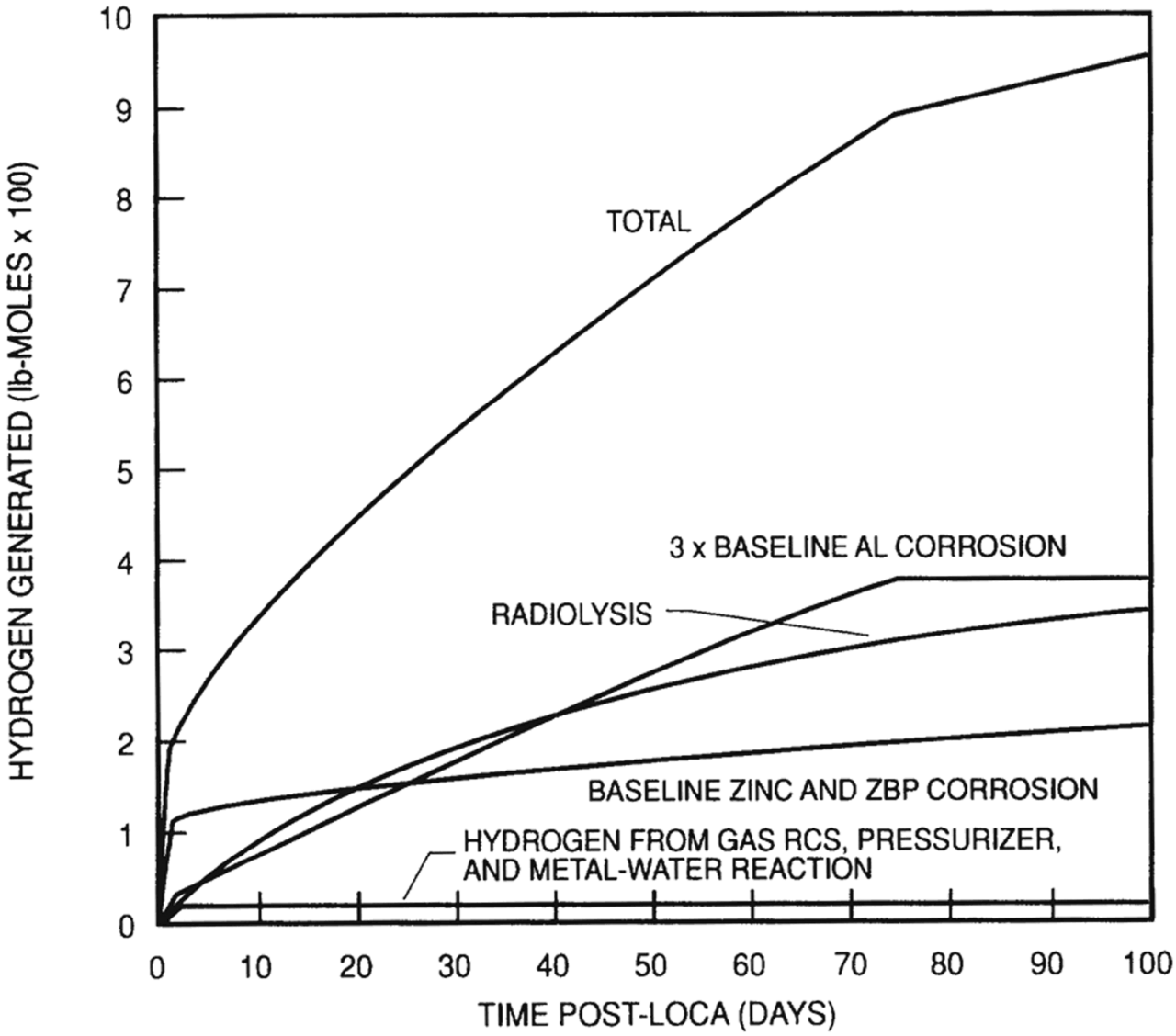
UNIT 1 AND UNIT 2

HYDROGEN PRODUCTION FROM ALL SOURCES

FIGURE 6.2.5–6



This curve bounds 3x baseline zinc case or an equivalent combination of zinc and aluminum.

Note: Total hydrogen generation curve remains bounding for MUR power uprate conditions. Individual curves showing contributions of the four individual hydrogen generation sources were not updated as part of the MUR power uprate.

REV 15 4/09

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	HYDROGEN ACCUMULATION FROM ALL SOURCES (FOR 3 X BASELINE ALUMINUM SURFACE AREA)
	FIGURE 6.2.5–7

6.3 EMERGENCY CORE COOLING SYSTEM

6.3.1 DESIGN BASES

The emergency core cooling system (ECCS) is a Seismic Category 1 safety-related system. It consists of the centrifugal charging pumps, safety injection pumps, residual heat removal pumps, accumulators, boron injection tank (Unit 1 only), residual heat removal heat exchangers, refueling water storage tank (RWST), and the associated piping, valves, instrumentation, and other related equipment. Nuclear plants employing the same ECCS design as VEGP are given in section 1.3.

The primary function of the ECCS following an accident is to remove the stored and fission product decay heat from the reactor core such that fuel rod damage, to the extent that it would impair effective cooling of the core, is prevented.

The ECCS is designed to cool the reactor core and to provide additional shutdown capability following initiation of the following accident conditions:

A. Loss-of-coolant accident (LOCA) including a pipe break or a spurious relief or safety valve opening in the reactor coolant system (RCS) which would result in a discharge larger than that which could be made up by the normal makeup system.

B. Loss-of-secondary-coolant accident including a pipe break or a spurious relief or safety valve opening in the secondary steam system which would result in an uncontrolled steam release or a pipe break in the secondary feedwater system.

C. A steam generator tube rupture accident.

The acceptance criteria for the consequences of each of these accidents are described in chapter 15 in the respective accident analyses sections.

The bases used in design and for selection of ECCS functional requirements are derived from 10 CFR 50, Appendix K, limits for fuel cladding temperature, etc., following any of the above accidents as delineated in 10 CFR 50.46. The subsystem functional parameters are selected so that, when integrated, the Appendix K requirements are met over the range of anticipated accidents and single failure assumptions.

Portions of the ECCS also operate in conjunction with the other systems of the cold shutdown design. The primary function of the ECCS during a safety grade cold shutdown is to ensure a means for injecting and throttling boration and makeup flow. Details of the cold shutdown design bases are discussed in subsection 5.4.7.

The primary function of the safety injection system (SIS) is to provide emergency core cooling in the event of a LOCA resulting from a break in the primary RCS or to provide emergency boration in the event of a steam line or feed line break accident resulting from a break in the secondary system.

Emergency core cooling following a LOCA is divided into three phases:

A. Short-Term Core Cooling/Cold Leg Injection Phase

The cold leg injection phase is defined as that period during which borated water is delivered from the RWST and accumulators to the RCS cold legs.

B. Long-Term Core Cooling/Cold Leg Recirculation

The cold leg recirculation phase is that period during which borated water is recirculated from the containment emergency sump to the RCS cold legs.

C. Long-Term Core Cooling/Hot Leg Recirculation Phase

The hot leg recirculation phase is that period during which borated water is recirculated from the containment emergency sump to both the RCS hot legs and RCS cold legs.

The emergency boration following a steam break accident would occur only during the injection phase. The function of the SIS during this phase would be to inject borated water into the RCS with sufficient shutdown reactivity to compensate for the change in RCS volume and to counteract any reactivity increase caused by the resulting cooldown. The SIS would continue to inject borated water from the RWST until the RCS conditions have stabilized, the accident has been identified as a steam break, and the criteria for safety injection termination are satisfied.

The reliability of the ECCS has been considered in selection of the functional requirements, selection of the particular components and location of components, and connected piping. Redundant components are provided where the loss of one component would impair reliability. Valves are provided in series where isolation is desired and in parallel when flow paths are to be established for ECCS performance. Redundant sources of the safety injection actuation signal are available so that the proper and timely operation of the ECCS is ensured. Sufficient instrumentation is available so that failure of an instrument will not impair readiness of the system. The active components of the ECCS are normally powered from separate buses which are energized from offsite power supplies. In addition, redundant sources of emergency onsite power are available through the use of the emergency diesel generators to ensure adequate power for all ECCS requirements. Each diesel generator is capable of driving all pumps, valves, and necessary instruments associated with one train of the ECCS.

All valves required to be actuated during ECCS operation are located to prevent vulnerability to flooding. Repositioning of valves due to spurious actuation coincident with an accident has been analyzed and is not considered credible as a design basis.

To address the Three Mile Island requirements of II.K.1.5, operations surveillance procedures (OSP) have controls which address the manipulation of valves during testing. The procedures are written to ensure positive control over all valve movements and to independently verify that each valve is returned to its original position, i.e., safeguards position.

After extended outages systems are aligned by their individual procedures which include safety-related systems. Safety-related systems are aligned again using the flow path verification surveillance procedure. This in effect provides an independent verification of flow paths for safety-related system valves.

These surveillance procedures are then performed on a periodic basis during plant operation to ensure proper alignment. Also, as mentioned earlier, if a valve position is changed during a surveillance test, it is procedurally repositioned to original position and independently verified in that position.

Using the methods described above, OSPs maintain positive control over safety system valves.

The environmental qualification of active ECCS equipment is discussed in section 3.11.

Protection of the ECCS from missiles is discussed in section 3.5. Protection of the ECCS against dynamic effects associated with rupture of piping is described in section 3.6. Protection from flooding is also discussed in section 3.4, paragraph 3.6.1.3, and appendix 3F.

The elevated temperature of the containment emergency sump solution during recirculation is well within the design temperature of all ECCS components. In addition, consideration has been given to the potential for corrosion of various types of metals exposed to the fluid conditions prevalent immediately after the accident or during long-term recirculation operations.

6.3.2 SYSTEM DESIGN

The emergency core cooling system (ECCS) components are designed such that a minimum of three accumulators, one charging pump, one safety injection (SI) pump, one residual heat removal (RHR) pump, and one RHR heat exchanger, together with their associated valves and piping, will ensure adequate core cooling in the event of a design basis accident (DBA). The redundant onsite emergency diesels ensure adequate emergency power to at least one train of electrically operated components in the event that a loss of offsite power occurs simultaneously with an accident, even assuming a single failure in the emergency power system.

6.3.2.1 Schematic Piping and Instrumentation Diagrams

Piping and instrumentation diagrams of the ECCS are shown in drawings 1X4DB116-1, 2X4DB116-1, 1X4DB116-2, 2X4DB116-2, 1X4DB119, 2X4DB119, 1X4DB120, 2X4DB120, 1X4DB121, 2X4DB121, 1X4DB122 and 2X4DB122. Process flow diagrams of the ECCS are shown in figure 6.3.2-1. Pertinent design and operating parameters for the components of the ECCS are given in table 6.3.2-1. The codes and standards to which the individual components of the ECCS are designed are listed in table 3.2.2-1.

Process flow diagrams (figure 6.3.2-1) are developed for illustrative purposes only and are not intended to represent the flowrates and temperatures used in the various accident analyses. For the cold leg injection mode, they illustrate best estimate system performance based on maximum ECCS safeguards (both trains of ECCS components) operating. The system operation conditions presented are based on the assumption that the reactor coolant system (RCS) is fully depressurized and is in equilibrium with the containment at 0 psig.

Flowrates to the RCS are provided in chapter 15, where appropriate, for the accident analyses. The accident analyses flowrates are developed from certified pump performance curves and calculated system resistances based on plant piping layouts. Minimum ECCS safeguards flowrates are determined by degrading the minimum composite pump performance curves by an amount representing design margins of the pumps and also by uniformly degrading those resulting curves by 5 percent (reference 1) and, to maximize the spill, the injection line with the lowest resistance is assumed to spill to either the containment or to the RCS. Refer to paragraph 15.6.5.2 for details.

The component interlocks used in different modes of system operation are listed below.

A. The SI signal is interlocked with the following components and initiates the indicated action:

	Component	Action
1.	Centrifugal charging pumps	Start
2.	Refueling water storage tank (RWST) suction valves to charging pumps	Open
3.	Discharge header (parallel) valves for Unit 1	Open

boron injection tank (BIT) and Unit 2 CVCS charging pumps high head cold leg injection valves.

4.	Normal charging path valves	Close
5.	Charging pump miniflow valves	Close
6.	Charging pump alternate miniflow valves	Open[(a)]
7.	SI pumps	Start
8.	RHR pumps	Start
9.	Any closed accumulator isolation valve	Open
10.	Volume control tank outlet isolation valves	Close

B. Switchover from injection mode to recirculation involves the following interlocks:

1. The RHR suction valves from the containment emergency sumps open when two out of four RWST level transmitters indicate a low-low level in conjunction with an SI signal.
2. The SI pump and charging pump suction isolation valves from the RHR pump discharge can be opened provided that the SI pump miniflow and charging pump alternate miniflow isolation valves have been closed.

6.3.2.2 Equipment and Component Descriptions

The component design and operating conditions are specified as the most severe conditions to which each respective component is exposed during either normal plant operation or during operation of the ECCS. For each component these conditions are considered in relation to the code to which it is designed. By designing the components in accordance with applicable codes and with due consideration for the design and operating conditions, the fundamental assurance of structural integrity of the ECCS components is maintained. Components of the ECCS are designed to withstand the appropriate seismic loadings in accordance with their safety class as given in table 3.2.2-1. Specific equipment parameters are given in table 6.3.2-1.

A gas accumulation monitoring and trending process for the Vogtle Unit 1 and 2 ECCS and containment spray systems has been established to meet the requirements of NRC Generic Letter 2008-01.

A discussion of the major mechanical components of the ECCS follows.

6.3.2.2.1 Accumulators

The accumulators are pressure vessels partially filled with borated water and pressurized with nitrogen gas. During normal operation each accumulator is isolated from the RCS by two check

[a] Valves are enabled by the SI signal and will open or close based on centrifugal charging pump discharge pressure.

valves in series. Should the RCS pressure fall below the accumulator pressure, the check valves open and borated water is forced into the RCS. One accumulator is attached to each of the cold legs of the RCS. Mechanical operation of the swing-disc check valves is the only action required to open the injection path from the accumulators to the core via the cold legs.

Connections are provided for remotely adjusting the level and boron concentration of the borated water in each accumulator during normal plant operation as required. Accumulator water level may be adjusted either by draining to the RWST or by pumping borated water from the RWST to the accumulator. Samples of the solution in the accumulators are taken periodically to check boron concentration.

Accumulator pressure is provided by a supply of nitrogen gas and can be adjusted as required during normal plant operation. However, the accumulators are normally isolated from this nitrogen supply. Gas relief valves on the accumulators protect them from pressures in excess of design pressure. The redundant capability to vent the accumulators is described in paragraph 5.4.7.2.3.

The accumulators are located within the containment but outside the secondary shield wall which protects them from missiles.

Accumulator level and pressure are monitored by indicators and alarms. The operator can take action as required to maintain plant operation within the requirements of the technical specification covering accumulator operability.

6.3.2.2.2 Boron Injection Tank (Unit 1 only)

The BIT will contain anywhere from 0 to 2600 ppm borated water solution (the upper limit being the same as the borated water solution in the refueling water storage tank (RWST) and is connected to the discharge of the centrifugal charging pumps. Upon actuation of the SI signal, the charging pumps provide the pressure to inject the borated water solution from the RWST into the RCS when the isolation valves associated with the BIT discharge path open automatically.

6.3.2.2.3 Deleted

6.3.2.2.4 Residual Heat Removal Pumps

In the event of an accident, the RHR pumps are started automatically on receipt of an SI signal. The RHR pumps take suction from the RWST during the injection phase and are automatically realigned to the containment emergency sump during the recirculation phase, although manual action is required to close the suction path from the RWST. Each RHR pump is a single-stage, vertical position, centrifugal pump.

A minimum flow bypass line is provided on each pump discharge to recirculate and return the pump discharge fluid to the pump suction should these pumps be started with the RCS pressure above their shutoff head. Once flow is established to the RCS, each pump bypass line automatically closes. The minimum flow bypass lines prevent deadheading of the pumps and permit pump testing during normal operation.

The RHR pumps are discussed further in subsection 5.4.7. A pump performance curve is given in figure 6.3.2-2.

The net positive suction head (NPSH) of the RHR pumps was evaluated for normal plant cooldown operation and for both the injection and recirculation modes of operation for the DBA.

The recirculation mode of operation gives the limiting NPSH requirement for the RHR pumps, and the NPSH available as determined from the following equation:

$$NPSH_{available} = h_{ambient\ pressure} + h_{static\ head} - h_{vapor} - h_{system\ pressure\ losses}$$

To evaluate the adequacy of the available NPSH, the debris generation from a high energy line break within the containment and the resultant impact on the RHR pump performance was evaluated using the following.

A. The minimum flood level inside the containment, based on the RWST water discharged during injection and switchover to recirculation, plus the water volume of 3 accumulator tanks, is at el 177 ft 0 in. The minimum flood level also takes into account the flooding of the reactor cavity/incore instrument tunnel via the penetrations in the primary shield.

B. The RHR pump suction elevation is 124 ft 3 ¾ in.

C. The calculated maximum line losses which include losses through pipe fittings, valves, entrance and exit are 17.0 ft at 4500 gal/min, RHR pump runout flow.

D. The quantity of insulation debris generated by the double-ended rupture of the RCS hot leg at its connection to the steam generator, the limiting case, is 432.0 ft^3.

E. For one RHR and one CS pump train operation, the volume of debris transported and deposited on one RHR sump screen is 58.4% of the total debris. This is based on the ratio of individual pump flowrate to total recirculation flowrate.

F. The maximum RHR emergency sump screen head loss, assuming 58.4% of the total debris generated is evenly deposited on the trash guard-screen cage, is 16.65 ft.

G. The vapor pressure of the pump liquid is assumed to be in equilibrium with the containment ambient pressure (i.e., no credit is taken for subcooling of the sump fluid) for sump fluid temperatures of equal to or greater than 211°F.

$$h_{containment\ ambient\ pressure} = h_{vapor\ pressure}$$

The NPSH equation for the recirculation mode when suction is taken from the containment emergency sump becomes:

$$NPSH_{available} = h_{static\ head} + h_{static\ head} - h_{line\ loss} - h_{sump\ screen\ loss}$$

Using the above values, the calculated RHR pump available NPSH is 26.6 ft. The RHR pump required NPSH at 4500 gal/min is 19 ft. Therefore adequate available NPSH margin is provided for proper RHR pump operation.

6.3.2.2.5 Centrifugal Charging Pumps

In the event of an accident, the centrifugal charging pumps are started automatically on receipt of an SI signal and are automatically aligned to take suction from the RWST during the injection phase. During recirculation, suction is provided from the RHR pump discharge.

These high-head pumps deliver flow to the RCS at the prevailing RCS pressure. Each centrifugal charging pump is a multistage diffuser design, barrel-type casing with vertical suction and discharge nozzles.

A minimum flow bypass line is provided on each pump discharge to prevent pump deadheading and to permit pump testing during power operations. Each minimum flow bypass line contains an isolation valve that closes automatically upon receipt of an SI signal. A third isolation valve is provided in the common header downstream of the two individual pump minimum flow lines. An alternate minimum flow line is provided for each pump to prevent pump deadheading should RCS pressure rise following isolation of the normal minimum flow lines. An isolation valve in each of these lines is enabled by the SI signal and opens upon receipt of a high pressure signal from a pressure switch connected to the centrifugal charging pump discharge. When the isolation valve opens, flow will be discharged to the RWST. Both isolation valves in each alternate minimum flow line are closed from the control room as part of the ECCS realignment from the injection to the recirculation mode.

A pump performance curve for the centrifugal charging pumps is given in figure 6.3.2-3.

6.3.2.2.6 Safety Injection Pumps

Two SI pumps are provided. Each pump is a multistage, diffuser design, split-case centrifugal pump with side suction and side discharge.

In the event of an accident, the SI pumps are started automatically on receipt of an SI signal.

These pumps deliver water to the RCS from the RWST during the injection phase and from the containment emergency sump via the RHR pumps during the recirculation phase.

A minimum flow bypass line is provided on each pump discharge to recirculate flow to the RWST in the event that the pumps are started with the RCS pressure above pump shutoff head. This line also permits pump testing during normal plant operation. Two parallel valves in series, with a third downstream in a common header, are provided for isolation of the minimum flow lines.

These valves are manually closed from the control room as part of the ECCS realignment from the injection to the recirculation mode. A pump performance curve is presented in figure 6.3.2-4.

The NPSH for the SI and charging pumps was evaluated for both the injection and recirculation modes of operation for the DBA. The end of the injection mode of operation gives the limiting NPSH available (minimum static head). The NPSH available was determined from the elevation head and vapor pressure of the water in the RWST, which is at atmospheric pressure, and from the pressure drop in the suction piping from the tank to the pumps.

The NPSH evaluation for the charging and SI pumps from the RWST was based on all safeguards pumps operating, with the pump being analyzed at its runout flowrate.

When a predetermined low RWST level is reached, the SI and charging pumps are manually aligned to take suction from the RHR pump discharge headers. The NPSH requirements of these pumps are therefore satisfied by the discharge head of the RHR pumps during the recirculation mode of system operation.

6.3.2.2.7 Deleted

6.3.2.2.8 Residual Heat Removal Heat Exchangers

The RHR heat exchangers are conventional shell and U-tube type units. During normal cooldown operation, the RHR pumps recirculate reactor coolant through the tube side while component cooling water flows through the shell side. During ECCS recirculation operation, water from the containment emergency sump flows through the tube side. The tubes are seal welded to the tube sheet.

The RHR heat exchangers are discussed further in subsection 5.4.7.

6.3.2.2.9 Refueling Water Storage Tank

The RWST provides a source of water for the SI, containment spray, centrifugal charging, and RHR pumps following an accident. The tank volume is sufficiently large to supply the injection phase water requirements. The RWST is a cylindrical, reinforced concrete tank with a stainless steel liner. The walls and roof slab are sufficiently thick to protect the tank against tornado-generated missiles. The tank is designed to Seismic Category 1 requirements. The tank outlet is designed to prevent vortex propagation to the pump suction lines. The outlet is also screened to prevent the passage of particles larger than 1/8 in. in diameter. A recirculation heater is provided to maintain a minimum water temperature of 50°F, and external connections to the tank are heat traced to ensure their operability. The RWST provides a sufficient volume of water to adequately account for instrument error, working allowance, transfer allowance, single failure, and unusable volume. RWST sizing is shown in figure 6.3.2-5.

The discussion which follows addresses the adequacy of the RWST volume, including the shortest times available for ECCS injection and switchover.

6.3.2.2.9.1 Injection Mode Allowance. The SI mode of ECCS operation consists of the ECCS pumps (charging pumps, SI pumps, and RHR pumps) and the containment spray pumps taking suction from the RWST and delivering to the RCS and containment, respectively. In order to analyze the shortest time available for injection mode operation, the following conservative bases are established:

A. The RWST volume available for injection mode operation is approximately 455,635 gal.

B. Containment and RCS pressures are conservatively assumed to be 0 psig to maximize flow out of the RWST.

C. Flow out of the RWST during the injection mode includes conservative allowances for two pumps of each type operating at the following flowrates:

SI pump	450 gal/min per pump
Charging pump	450 gal/min per pump
RHR pump	3700 gal/min per pump
Spray pump	3200 gal/min per pump

Total RWST outflow during injection mode operation is 15,600 gal/min.

Based on a minimum available RWST volume of 435,522 gal for injection mode operation, and the maximum total flowrate out of the RWST, the shortest injection mode operation time is approximately 28 min.

6.3.2.2.9.2 Transfer Allowance. During the injection mode of ECCS operation, the operator monitors the RWST level and containment emergency sump level in anticipation of switchover. Upon receipt of the RWST low-low level alarm, the operator is required to initiate the manual operations to complete switchover in a timely manner.

The switchover from injection to cold leg recirculation is initiated automatically upon receipt of the RWST low-low level trip signal and is completed via timely operator action at the main control board. Switchover is initiated via automatic opening of the containment emergency sump isolation valves (HV-8811 A and B). This automatic action aligns the suction of the RHR pumps to the containment emergency sump to ensure continued availability of a suction source. Manual actions of table 6.3.2-7 must be performed following switchover initiation prior to loss of the RWST transfer allowance to ensure that all pumps are protected with suction flow available from the containment emergency sump. The manual actions required to complete switchover are shown in table 6.3.2-7. The switchover procedure is structured so that the operator simultaneously switches both trains from injection to recirculation, repositioning functionally similar switches as part of the same procedural steps.

The time available for switchover is dependent on the flowrate out of the RWST as the switchover manual actions are performed. As valves are repositioned, the flowrate out of the RWST is reduced in magnitude. In order to analyze the shortest time available for switchover, the following conservative bases are established:

A. The RWST transfer allowance available for ECCS manual switchover is approximately 135,000 gal. The RWST volume available for containment spray system (CSS) manual switchover is approximately 62,000 gal.

B. Approximately 20,000 gal below the minimum submergence level but above the bottom of the RWST is provided for vortex prevention and is not included in the above volumes.

C. Containment and RCS pressures for large break conditions are conservatively assumed to be 0 psig. Thus, no credit is taken for the reduction in RWST outflow that results with higher containment and RCS pressures following a large break.

   The same conservative assumption is made for the small break conditions (except that RCS pressure is assumed to be greater than RHR pump shutoff head resulting in no RHR pump flow to the RCS for small break conditions).

D. Pumped flowrates are assumed to be constant during switchover and include the following conservative flowrate allowances assuming two pumps of each type are operating:

SI pump	450 gal/min per pump
Charging pump	450 gal/min per pump
RHR pump	3700 gal/min per pump
Spray pump	3200 gal/min per pump

E. Flowrate out of the RWST for the worst ECCS single failure condition is determined assuming one of the RWST isolation valves (HV-8812 A or B) fails to close on demand. This single failure maximizes RWST outflow during

switchover. Flowrates out of the RWST assume no operator corrective action to mitigate the single failure; i.e., stop the affected RHR pump and close the appropriate sump isolation valves.

6.3.2.2.10 Valves

Design features employed to minimize valve leakage include:

A. Where possible, packless valves are used.

B. Other valves which are normally open, except check valves and those which perform a control function, are provided with backseats to limit stem leakage.

C. Normally closed globe valves are installed with recirculation fluid pressure under the seat to prevent stem leakage of recirculated (radioactive) water.

D. Relief valves are enclosed; i.e., they are provided with a closed bonnet.

E. Live-load packing is used on some valves including valves equipped with valve stem leakoff connections to maintain a constant force on the gland follower as the packing ages and compresses.

6.3.2.2.11 Motor-Operated Valves

Design parameters for motor-operated valves used in the ECCS are given in table 6.3.2-1.

The seating design of all motor-operated valves is of the Crane flexible wedge design. This design releases the mechanical holding force during the first increment of travel so that the motor operator works only against the frictional component of the hydraulic unbalance on the disc and the packing box friction. The disc is guided throughout the full disc travel to prevent chattering and to provide ease of gate movement. The seating surfaces are hard faced to prevent galling and to reduce wear.

Where a gasket is employed for the body to bonnet joint, it is either a fully trapped, controlled compression, spiral wound gasket with provisions for seal welding or it is of the pressure seal design with provisions for seal welding. The valve stuffing boxes are designed with a reduced packing configuration and the deletion of the valve stem leakoff line, or lantern ring leakoff connection with a set of packing below and above the lantern ring.

6.3.2.2.12 Manual Globe, Gate, and Check Valves

Gate valves employ a wedge design and are straight through. The wedge is either split or solid. All gate valves have backseat and outside screw and yoke.

Globe valves, "T" and "Y" style, are full ported with outside screw and yoke construction.

Check valves are spring-loaded lift piston types for sizes 2 in. and smaller, swing type for sizes 2 1/2 in. to 4 in., and tilting disc type for sizes 4 in. and larger.

Stainless steel check valves have no penetration welds other than the inlet, outlet, and bonnet. The check hinge is serviced through the bonnet.

The stem packing and gasket of the stainless steel manual globe and gate valves are similar to those described above for motor-operated valves. Carbon steel manual valves are employed to

pass nonradioactive fluids only and, therefore, do not contain the double packing and seal weld provisions.

6.3.2.2.13 Accumulator Check Valves (Swing-Disc)

The accumulator check valve is designed with a low-pressure drop configuration with all operating parts contained within the body.

Design considerations and analyses, which ensure that leakage across the check valves located in each accumulator injection line does not impair accumulator availability, are as follows:

A. During normal operation, the check valves are in the closed position with a nominal differential pressure across the disc of approximately 1650 psi. Since the valves remain in this position except for testing or when called upon to open following an accident and are, therefore, not subject to the abuse of flowing operation or impact loads caused by sudden flow reversal and seating, they do not experience significant wear of the moving parts and are expected to function with minimal backleakage. This backleakage can be checked via the test connection, as described in subsection 6.3.4.

B. When the RCS is being pressurized during the normal plant heatup operation, the check valves are tested for leakage at a test pressure of greater than 350 psig. This test confirms the seating of the disc and whether there has been an increase in the leakage since the last test.

C. The experience derived from the check valves employed in the emergency injection systems indicates that the system is reliable and workable. This is substantiated by the satisfactory experience obtained from operation of the Robert Emmett Ginna plant and subsequent plants where the usage of check valves is identical to VEGP.

D. The accumulators can accept some inleakage from the RCS without affecting availability. Continuous inleakage would require, however, that the accumulator water volume and boron concentration be adjusted periodically to meet Technical Specification requirements.

6.3.2.2.14 Relief Valves

Relief valves are installed in various sections of the ECCS to protect lines which have a lower design pressure than the RCS. The valve stem and spring adjustment assembly are isolated from the system fluids by a bellows seal between the valve disc and spindle. The closed bonnet provides an additional barrier for enclosure of the relief valves. Table 6.3.2-2 lists the ECCS relief valves with their capacities and setpoints.

ECCS piping has been reviewed to identify those lengths of piping which are isolated by normally closed valves and which do not have pressure relief protection in the piping section between the valves. These sections include:

1. Between valves 1205-U4-021 and 1205-U4-022 in the residual heat removal system cross-connection line to the chemical and volume control system. This piping is outside containment, is not an ECCS flowpath, and is not needed to achieve safety-grade cold shutdown.

2. Portions of ECCS test lines. This piping is inside containment, is not an ECCS accident mitigation flowpath, and is not needed to achieve safety-grade cold shutdown.
3. Containment spray system sump suction lines, between valves HV-9002A-9003A and HV-9002B-9003B. This piping is the recirculation flowpath outside of containment.
4. Between valves 1205-U6-027 and 1205-U4-226 on the RHR pump discharge to the RWST. This piping is outside the containment and is not in the ECCS flow path and is not required for safety-grade cold shutdown.
5. Piping vents, drains, test connections, etc., typically have two closed valves or one closed valve and a blind flange.
6. Check valve test lines have sections that are isolated by two normally closed valves.

The piping vents, drains, test corrections, and check valve lines have design pressure/temperature conditions compatible with the process piping to which they connect. Thus valve leakage will not function to overpressurize the isolated piping sections and pressure relief provisions are not required.

6.3.2.2.15 Butterfly Valves

Each main RHR line has an air-operated butterfly valve which is normally open and is designed to fail in the open position. The actuator is arranged such that air pressure on the diaphragm overcomes the spring force, causing the linkage to move the butterfly to the closed position. Upon loss of air pressure, the spring returns the butterfly to the open position. These valves are left in the full-open position during normal operation to maximize flow from this system to the RCS during the injection mode of the ECCS operation. These valves are used during normal RHR system operation to control cooldown flowrate.

Each RHR heat exchanger bypass line has an air-operated butterfly valve which is normally closed and is designed to fail closed. These valves are used during normal cooldown to control RCS cooldown rates.

Other air-operated valves in the ECCS along with their failure positions and position indications are provided in table 6.3.2-8.

6.3.2.2.16 Motor-Operated Valves Control

Remotely operated valves for the injection mode which are under manual control; i.e., valves which normally are in their ready position and do not receive an SI signal, have their positions indicated on a common portion of the control board. If a component is out of its proper position, its monitor light indicates this on the control panel. At any time during operation when one of these valves is not in the ready position for injection, this condition is shown visually on the board, and an audible alarm is sounded in the control room.

Spurious movement of a motor-operated valve due to an electrical fault in the motor actuation circuitry, coincident with a loss-of-coolant accident (LOCA), has been analyzed (WCAP-8966) and found to be an acceptably low probability event. In addition, power lockout in accordance with Branch Technical Position ICSB-18 is provided for those valves whose spurious movement could result in degraded ECCS performance. Power lockout is accomplished by providing a

control power isolation switch for each of these valves on the main control board. Further details on power lockout are provided in paragraph 8.3.1.1.11. Table 6.3.2-3 provides a listing of the motor-operated isolation valves in the ECCS, showing interlocks, automatic features, and position indication.

Periodic visual inspection and operability testing of the motor-operated valves in the ECCS ensure that there is no potential for impairment of valve operability due to boric acid crystallization which could result from valve stem leakage.

In addition, the location of all motor-operated valves within the containment has been examined to identify any motor operators which may be submerged following a postulated LOCA. Based on a maximum post-LOCA flood level of el 181 ft 4 in., the only potentially submerged valves are the accumulator discharge valves, HV-8808 A, B, C, and D. These valves are positioned prior to startup and then have power removed from the motor operator. These valves are not required to change position after a LOCA. Therefore, the flooding of these motor operators and any resultant postulated failure does not present any problems for the short or long-term ECCS operations, containment isolation, or any other safety-related function.

6.3.2.2.17 Manual Valves

The 24-in. manually operated gate valve on the suction line from the RWST is the only manual valve which, if mispositioned, could totally interrupt ECCS flow. For this reason and in accordance with Regulatory Guide 1.47, the valve is locked open, administratively controlled, and provided with redundant system level input to the bypassed/inoperable status panel in the control room.

Manual valves are generally used as maintenance isolation valves or throttling valves. When used for these functions they are under administrative control, which requires them to be locked in the correct position. Manual valves used as maintenance isolation valves occur on the SI and RHR pump discharges. They are located so that no single valve can isolate both trains of ECCS equipment. Manual valves are also used as throttling valves in the injection branch paths of the high-head ECCS pumps.

To preclude the possibility of ECCS degradation due to valve mispositioning, branch line connections such as vent and drain lines, test connections, pressure points, flow element test points, flush connections, local sample points, and bypass lines are provided with double isolation or sealed barriers. The isolation is provided by one of the following methods: two valves in a series; a single valve with a screwed cap or blind flange; a single locked closed valve; or a blind flange. These valves are under administrative control.

6.3.2.2.18 Accumulator Motor-Operated Valve Control

As part of the plant shutdown procedures, the operator is required to close these valves. This prevents a loss of accumulator water inventory to the RCS and is done after the RCS has been depressurized below 1000 psig. The redundant pressure and level alarms on each accumulator function to alert the operator to close these valves, if any are inadvertently left open. Power is locked out after the valves are closed.

During plant startup, Technical Specifications require these valves be open with power removed in mode 3 when the RCS pressure exceeds 1000 psig. Monitor lights in conjunction with an audible alarm alert the operator should any of these valves be left inadvertently closed. The audible alarm is enabled once the RCS pressure increases beyond the SI unblock setpoint. Power is locked out after these valves are opened.

The accumulator isolation valves are not required to move for the accumulators to perform their safety function during power operation or in a post-accident situation. For a discussion of limiting conditions for operation and surveillance requirements of these valves, refer to the Technical Specifications.

The accumulator isolation valves receive an SI signal to ensure that they are open in the event of an accident occurring during inservice inspection testing.

For further discussion of the instrumentation associated with these valves, refer to subsections 6.3.5, 7.3.1, and 7.6.4.

6.3.2.3 Applicable Codes and Standards

Applicable codes and standards for the ECCS are discussed in section 3.2.

6.3.2.4 Material Specifications and Compatibility

Materials employed for engineered safety feature components are discussed in subsection 6.1.1. Materials for ECCS components are selected to meet the applicable material requirements of the codes in table 3.2.2-1 and the following additional requirements:

A. All parts of components in contact with borated water are fabricated of or clad with austenitic stainless steel or equivalent corrosion resistant material.

B. All parts of components in contact (internal) with containment emergency sump solution during recirculation are fabricated of austenitic stainless steel or equivalent corrosion resistant material.

C. Valve seating surfaces are hard faced with Stellite No. 6 or equivalent to prevent galling and to reduce wear.

D. Valve stem materials are selected for their corrosion resistance, high-tensile properties and their resistance to surface scoring by the packing.

Table 6.3.2-4 summarizes the materials employed for ECCS components.

6.3.2.5 System Reliability

Reliability of the ECCS is considered in all aspects of the system from initial design to periodic testing of the components during plant operation. The ECCS is a two-train, fully redundant, safety-related system. The system has been designed and proven by analysis to withstand any credible single active failure during injection or active or passive failure during recirculation and to satisfy the performance requirements. Two trains of pumps, heat exchangers, and flowpaths are provided for redundance, since only one train is required to satisfy the performance requirements. The initiating signals for the ECCS are derived from independent sources as measured from process; e.g., pressurizer low pressure or environmental (containment high pressure) variables. Redundant as well as functionally independent variables are measured to initiate ECCS operation. Each train is physically separated and protected where necessary so that a single event cannot initiate a common failure. Power sources for the ECCS are divided into two independent trains supplied from the emergency buses from offsite power. Sufficient diesel generating capacity is maintained onsite to provide required power to the emergency buses if offsite power is not available. The diesel generators and their auxiliary systems are completely independent and each supplies power to one of the two ECCS trains.

The preoperational testing program ensures that the systems as designed and constructed meet the functional requirements as calculated in design. The ECCS is designed with the ability for online testing of most components so the availability and operational status can be readily determined. In addition to the above, the integrity of the ECCS is ensured through examination of critical components during the routine inservice inspection.

The reliability program extends to the procurement of ECCS components such that only designs which have been proven by past use in similar applications are acceptable for use. The procurement quality assurance program is discussed in chapter 17.

The ECCS is a two-train, fully redundant, engineered safety feature. During the long-term cooling period following a LOCA, ECCS flowpaths may be separated into two subsystems or trains, either of which can provide minimum core cooling functions. Due to this concept, either of the two subsystems can be isolated and removed from service if maintenance is required on any ECCS component.

Each compartment is provided with radiation shielding such that access may be allowed to compartments for maintenance during the recirculation phase. To obtain access to a given compartment, limited pump(s) in that compartment would be stopped and the lines flushed with water from the RWST. Provisions for washing down the floors and walls are provided to reduce contamination in the event that a leak occurs in a compartment during the recirculation phase. Ventilation is provided to facilitate access for maintenance.

During the long-term period, the RCS is depressurized to containment ambient pressure following the LOCA. During this period, the heat generated in the reactor core is in the form of residual decay heat; consequently, the RHR pumps provide the required flow; i.e., from the containment sump through the residual heat exchangers and into the reactor core, to perform reactor core residual decay heat removal.

The piping and valves associated with the pumps are arranged so that the system can be drained and flushed prior to maintenance. To meet this requirement, manual valves are provided with extended reach rods so that they can be operated from a position external to the pump compartments.

Proper initial fill and venting of the ECCS ensures that water hammer does not occur in ECCS lines. In addition, the head of water provided by the RWST further ensures the lines remain full and water hammer concerns do not develop. High point vents in the ECCS lines are provided to ensure means for proper venting of lines and pumps. Fill and venting procedures for the ECCS ensure removal of air from the system to prevent the possibility of a water hammer if injection flow is initiated. The RWST location/configuration ensures that the Technical Specification limit for the RWST low water level is above the ECCS high point required to maintain water solid ECCS lines.

Further, the existence of high point vents and the positive head of water provides means by which the operator can confirm water solid ECCS lines.

6.3.2.5.1 Active Failure Criteria

An active failure is the failure of a powered component such as a piece of mechanical equipment, a component of the electrical supply system, or instrumentation and control equipment to act on command to perform its design function. Examples include the failure of a motor-operated valve to move to its correct position, the failure of an electrical breaker or relay to respond, the failure of a pump, fan, or diesel generator to start, etc.

The failure mode and effects analysis (FMEA), provided in table 6.3.2-5, demonstrates the ability of the ECCS to withstand any single active failure. The analysis illustrates that the ECCS can sustain an active failure in either the short or long term and still meet the required level of performance for core cooling.

Since the short-term operation of the active components of the ECCS following a steam line rupture or a steam generator tube rupture is similar to that following a LOCA, the same analysis is applicable and the ECCS can sustain the failure of any single active component and still meet the level of performance for the addition of shutdown reactivity.

Portions of the ECCS are also relied upon to provide boration and makeup during a safety grade cold shutdown. The capability of the ECCS to sustain an active failure and still perform in conjunction with other systems of the cold shutdown design is presented in subsection 5.4.7.

6.3.2.5.2 Passive Failure Criteria

A passive failure is the structural failure of a static component which limits the component's effectiveness in carrying out its design function. Examples include cracks in pipes, sprung flanges, valve packing leaks, or pump seal failures.

A single passive failure analysis is presented in table 6.3.2-6. It demonstrates that the ECCS can sustain a single passive failure during the long-term phase and still retain an intact flowpath to the core to supply sufficient flow to keep the core covered and to effect the removal of decay heat. The procedure followed to establish the alternate flowpath also isolates the component which failed.

The following philosophy provides for necessary redundancy in component and system arrangement to meet the intent of the general design criterion on single failure as it specifically applies to failure of passive components in the ECCS. Thus, for the long term, the system design is based on accepting either a passive or an active failure, assuming no failures in the short term.

A. Redundancy of Flowpaths and Components for Long-Term Emergency Core Cooling

   In design of the ECCS, the following criteria are utilized:

   1. During the long-term cooling period following an accident, the emergency core cooling flowpaths are separable into two subsystems, either of which can provide minimum core cooling functions and return spilled water from the floor of the containment back to the RCS.
   2. Either of the two subsystems can be isolated and removed from service in the event of a leak outside the containment.
   3. Adequate redundancy of check valves is provided to tolerate failure of a check valve during the long term as a passive component.
   4. Should one of these two subsystems be isolated in this long-term period, the other subsystem remains operable.
   5. Provisions are also made in the design to detect leakage from components outside the containment, collect this leakage, and provide for maintenance of the affected equipment.

   For the long-term emergency core cooling function, adequate core cooling capacity exists with one flowpath removed from service.

B. Subsequent Leakage from Components in Safeguards Systems

With respect to piping and mechanical equipment outside the containment, considering the provisions for visual inspection and leak detection, leaks can be detected before they propagate to major proportions. A review of the equipment in the system indicates that the largest sudden leak potential would be the sudden failure of a pump shaft seal. Evaluation of the leakrate assuming only the presence of a seal retention ring around the pump shaft showed flows less than 50 gal/min would result. Piping leaks, valve packing leaks, or flange gasket leaks have been of a nature to build up slowly with time and are considered less severe than the pump seal failure.

Larger leaks in the ECCS are prevented by the following:

1. The piping is classified in accordance with American National Standards Institute (ANSI) Safety Class 2 and receives the American Society of Mechanical Engineers (ASME) Class 2 quality assurance program associated with this safety class.
2. The piping, equipment, and supports are designed to ANSI Safety Class 2, Seismic Category 1 requirements, permitting no loss of function for the safe shutdown earthquake.
3. The system piping is located within a controlled area on the plant site.
4. The piping system receives periodic pressure tests and is accessible for periodic visual inspection.
5. The piping is austenitic stainless steel which, due to its ductility, can withstand severe distortion without failure.

The design of the auxiliary building and related equipment is based upon handling of leaks up to a maximum of 50 gal/min. Subsection 9.3.3 describes the design features provided to detect and isolate such leaks in the ECCS flowpath within 30 min.

6.3.2.5.3 Lag Times

Lag times for initiation and operation of the ECCS are limited by pump startup time and consequential loading sequence of these motors onto the safeguard buses. Most valves are normally in the position conducive to safety; therefore, valve operation time is not considered for these valves. If there is no loss of offsite power, all pump motors and valve motors are loaded immediately onto the safeguards buses according to the sequencer. The charging pumps and all valves are applied to the buses in 0.5 s, the SI pumps in 5.5 s, and the RHR pumps in 10.5 s. Safeguards pumps are capable of obtaining operating speed and rated flow within 4 s of receipt of the start signal. In the case of loss of offsite power, the diesel generator is designed for a 9.5-s delay to start and to obtain operating speed and voltage prior to the safeguards pumps and valves being sequenced onto the safeguards buses, which is less than the time assumed in the accident analysis. These lag times refer to the time after initiation of the SI signal.

6.3.2.5.4 Potential Boron Precipitation

Boron precipitation in the reactor vessel can be prevented by a backflush of ECCS water through the core to terminate boiloff and the resulting increase in boron concentration of the

water remaining in the reactor vessel. This is accomplished by the switchover from cold leg to hot leg recirculation at about 7.5 h following an accident. In addition to preventing boron precipitation by backflushing the core, hot leg recirculation provides subcooled water to terminate boiloff.

Three flowpaths, each with sufficient capacity to prevent precipitation, are available for hot leg recirculation of containment emergency sump water. Each SI pump can discharge to two hot legs with suction taken from the RHR pump discharge, either directly or indirectly via the charging pump cross connect. Each SI pump flow path provides an independent train of hot leg recirculation flow. In addition, the RHR pump of each train can also be aligned to deliver flow directly to two hot legs via a common hot leg recirculation header, through train B valve HV-8840 to provide the third flow path. Normal operator response is to align both RHR pumps to the hot legs through HV-8840 and both SI pumps to the hot legs. Sufficient flow to prevent precipitation only requires one SI pump aligned to the hot legs.

Loss of one pump or one flowpath or one complete train (including train B) will not prevent hot leg recirculation since redundant methods are available for use.

6.3.2.5.5 Safety Grade Cold Shutdown Function

During a safety-grade cold shutdown, the ECCS is relied upon to provide one of the two redundant flowpaths for boration and makeup. The BIT high-head injection header (Unit 1) and the CVCS charging pump high head cold leg injection header (Unit 2) provide this function. The redundant flowpath is the normal charging header, which is part of the chemical and volume control system (CVCS). Two independent subsystems, each consisting of a charging pump and the associated valves and piping, are provided and are powered by redundant emergency buses in a manner that ensures that at least one subsystem is always operable. A solenoid valve provided in each subsystem and located in the CVCS ensures that the remote throttling capability necessary for RCS inventory control and shutdown is available. Provisions are also included in the ECCS design to ensure that the accumulators can be either isolated or vented so that RCS depressurization can be accomplished. Details of the cold shutdown design are discussed in subsection 5.4.7. A failure mode and effects analysis for safety-grade cold shutdown operations is provided in table 6.3.2-9.

6.3.2.6 Protection Provisions

The provisions taken to protect the system from damage that might result from dynamic effects are discussed in section 3.6. The provisions taken to protect the system from missiles are discussed in section 3.5. The provisions to protect the system from seismic damage are discussed in sections 3.7.N, 3.9.N, and 3.10.N. Thermal stresses on the RCS are discussed in section 5.2.

6.3.2.7 Provisions for Performance Testing

Test lines are provided for performance testing of the ECCS as well as individual components. These test lines and instrumentation are shown in drawings 1X4DB119, 2X4DB119, 1X4DB120 and 1X4DB121. All pumps have miniflow lines for use in testing operability. Additional information on testing can be found in paragraph 6.3.4.2.

6.3.2.8 Manual Actions

The ECCS is automatically actuated by those accidents identified in subsection 6.3.1. Following actuation, the ECCS continues to operate in the injection mode until its operation is terminated by the operator or until its operation is switched to the recirculation mode. During the injection mode no manual actions are required for proper operation of the ECCS. For the loss-of-secondary-coolant accident and the tube rupture accident, the operator should stabilize plant conditions and terminate ECCS operation after satisfying the criteria for ECCS termination. For the LOCA, the operator may not be able to terminate ECCS operation and may have to initiate manual actions to align the ECCS for the recirculation mode. The following discussion addresses the limited manual actions that are required of the operator to realign the system for the cold leg recirculation mode of operation and, after approximately 7.5 h, for the hot leg recirculation mode of operation. These actions are delineated in table 6.3.2-7. Operator action (both short term and long term) required for the various modes of ECCS operation to mitigate the consequences of a LOCA or steam line break, as well as other accident conditions, is presented in the emergency operating procedures. These procedures discuss the alarms/indications available to the operator to lead him to take the appropriate actions.

The switchover from the injection mode to recirculation mode is initiated automatically and completed manually by operator action from the main control room. Protection logic is provided to automatically open the two containment emergency sump isolation valves when two out of four RWST level channels indicate a low-low level in conjunction with an SI signal. This automatic action aligns the two RHR pumps to take suction from the containment emergency sump. The RHR pumps continue to operate during this automatic switchover from the injection mode to the recirculation mode.

The two charging pumps and the two SI pumps continue to take suction from the RWST until manual operator action is taken to align these pumps in series with the RHR pumps.

The RWST level protection logic consists of four level channels with each level channel assigned to a separate process control protection set. Four RWST level transmitters provide level signals to corresponding normally deenergized level channel bistables. Each level channel bistable is energized on receipt of an RWST level signal less than the low-low level setpoint. A two-out-of-four coincident logic is utilized in both protection cabinets A and B to ensure a trip signal in the event that two of the four level channel bistables are energized. This trip signal, in conjunction with the SI signal, provides the actuation signal to automatically open the corresponding containment emergency sump isolation valves.

The RWST low-low level signal is also alarmed to inform the operator to initiate the manual action required to realign the RHR, charging, and SI pumps for the recirculation mode.

Following the automatic and manual switchover action, the two RHR pumps are aligned to take suction from the containment emergency sump and deliver directly to the RCS cold legs. The train A RHR pump is also aligned to provide suction to the two charging pumps which are aligned to deliver directly to the RCS cold legs. The train B RHR pump is also aligned to provide suction to the two SI pumps which are aligned to deliver directly to the RCS cold legs. As part of the manual switchover procedure, the suctions of the SI and charging pumps are cross-connected so that one RHR pump can deliver flow to the RCS and the SI and charging pumps in the event of the failure of the second RHR pump. The switchover would begin at least 19 min after the initiation of ECCS operation. The RWST is sized to allow the operator approximately 45 min after the initiation of ECCS injection before completing the ECCS and CSS switchover operation.

The switchover from the cold leg recirculation mode to the hot leg recirculation mode requires further manual actions. The charging pumps are not realigned and they continue to deliver to the RCS cold legs.

See section 7.5 for process information available to the operator in the control room following an accident.

Time delay following a low-low RWST signal, in which the operator's failure to act has no adverse effects, is greater than 10 min.

The consequences of the operator failing to act altogether are loss of the SI, charging, and containment spray pumps.

6.3.2.9 References

1. Westinghouse Letter GP-15580 to McCoy, C. K., dated April 3, 1992.
2. Deleted.

6.3.3 PERFORMANCE EVALUATION

The accidents identified in subsection 6.3.1 result in emergency core cooling system (ECCS) actuation and are mitigated within acceptance criteria by ECCS operation. For the purpose of evaluation in chapter 15, the accidents that result in ECCS actuation are categorized as follows:

A. Increase in Heat Removal by the Secondary System
   1. Inadvertent opening of a steam generator power- operated atmospheric steam relief or safety valve.
   2. Steam system piping failure.

B. Decrease in Heat Removal by the Secondary System
   1. Feedwater system piping failure.

C. Decrease in Reactor Coolant System (RCS) Inventory
   1. Steam generator tube rupture.
   2. Loss-of-coolant accident (LOCA) from a spectrum of postulated RCS piping failures.
   3. Loss of coolant due to a rod cluster control assembly ejection accident. This type of accident is enveloped by the RCS piping failures.

Each of these accidents results in generation of a safety injection (SI) signal and ECCS operation. The SI signal can be generated by any of the following:

- Pressurizer low pressure.
- Steam line low pressure.
- Containment high pressure.
- Manual actuation.

In addition to initiating ECCS operation, the SI signal initiates other safeguards automatic actions, including reactor trip, auxiliary feedwater system initiation, feedwater isolation, and containment isolation.

Upon receipt of an SI signal, the actions in paragraph 6.3.2.1.A are automatically initiated and the ECCS is aligned to operate in the injection mode. The charging pumps are aligned to deliver borated water from the refueling water storage tank to the RCS cold legs. The SI pumps, residual heat removal (RHR) pumps, and the accumulators are aligned to deliver to the RCS cold legs should RCS pressure drop below their respective pump shutoff heads or tank static pressure, respectively. The ECCS pumps and the accumulator flowrates vary depending on the type of accident and its characteristic pressure transient.

6.3.3.1 Increase in Heat Removal by the Secondary System

A number of events have been postulated which could result in an increase in heat removal from the RCS by the secondary system. Detailed analyses of these events are presented in section 15.1. For those events which result in ECCS actuation, the following summarizes ECCS performance:

A. Inadvertent Opening of a Steam Generator Relief or Safety Valve

The most severe core conditions resulting from an accidental depressurization of the main steam system are associated with an inadvertent opening of a single steam dump, relief, or safety valve. Refer to subsection 15.1.4 for a detailed description of this accident, including acceptance criteria and analytical results.

For this accident, the ECCS is actuated upon generation of an SI signal and the charging pumps function to inject the borated water solution from the RWST into the RCS cold legs. Although the borated water solution does not provide sufficient negative reactivity to maintain the reactor below criticality, the core is ultimately shut down by the solution and the departure from nucleate boiling (DNB) design basis is met. The charging pump flow also functions to increase RCS inventory and to repressurize the RCS. For this accident, the RCS does not depressurize sufficiently to permit the SI pumps, RHR pumps, or accumulators to deliver to the RCS. Subsequent to stabilizing plant conditions and satisfying ECCS termination criteria, the operator terminates ECCS operation and initiates plant shutdown operations.

B. Steam System Pipe Failure

The most severe core conditions resulting from a steam system piping failure are associated with a double-ended rupture of a main steam line which occurs at zero power. Effects of smaller piping failures at higher power levels are bounded by the double-ended rupture at zero power. Refer to subsection 15.1.5 for a detailed description of this accident, including acceptance criteria and analytical results.

For this accident, the ECCS functions as described in paragraph 6.3.3.1.A for the inadvertent opening of a steam generator relief or safety valve. However, this piping failure constitutes a more severe cooldown transient. The negative reactivity provided by operation of the charging pumps is not sufficient to prevent the reactor from returning to criticality during the transient. However, the core is ultimately shut down by the borated water solution, and the DNB design basis is met.

6.3.3.2 Decrease in Heat Removal by the Secondary System

A number of events have been postulated which could result in a decrease in heat removal from the RCS by the secondary system. Detailed analyses of these events are presented in section 15.2. For those events which result in ECCS actuation, the following summarizes ECCS performance:

A. Feedwater System Pipe Failure

The most severe core conditions resulting from a feedwater system piping failure are associated with a double-ended rupture of a feed line at full power. Depending on break size and power level, a feedwater system pipe failure could cause either an RCS cooldown transient or RCS heatup transient. Only the RCS heatup transient is evaluated as a feedwater system pipe failure, since the spectrum of cooldown transients is bounded by the steam system pipe failure analyses. The heatup transient effects of smaller piping failures at reduced power levels are bounded by the double-ended feed line rupture at full power. Refer to subsection 15.2.8 for a detailed description of this accident, including acceptance criteria and analytical results.

For this accident, the ECCS is actuated upon generation of an SI signal and the charging pumps inject the borated water solution from the RWST into the RCS cold legs. The charging pump flow functions to increase RCS inventory to ensure that sufficient inventory exists to keep the core covered with water. Since the accident is characterized by a heatup transient, the borated water solution from the RWST is not required and is not taken credit for in the analysis to control core reactivity. The RCS does not depressurize to permit the SI pumps, RHR pumps, or accumulators to deliver to the RCS. Subsequent to stabilizing plant conditions and satisfying ECCS termination criteria, the operator terminates ECCS operation and initiates plant shutdown operations.

6.3.3.3 Decrease in RCS Inventory

A number of events have been postulated which could result in a decrease in RCS inventory. Detailed analyses of these events are presented in section 15.6. For those events which result in ECCS actuation, the following summarize ECCS performance:

A. Steam Generator Tube Rupture

Although a steam generator tube rupture is an accident which results in a decrease in RCS inventory, severe core conditions are not associated with a steam generator tube rupture. The accident analyzed is a complete severance of a single steam generator tube that occurs at power with the reactor coolant contaminated with fission products, corresponding to continuous operation with a limited amount of defective fuel rods. Effects of smaller breaks are bounded by the complete severance. Refer to subsection 15.6.3 for a detailed description of this accident, including acceptance criteria and analytical results.

For this accident, the ECCS is actuated upon generation of an SI signal and the charging pumps inject the borated water solution from the RWST into the RCS cold legs. The charging pump flow functions to replace RCS inventory that is being lost through the ruptured steam generator tube, provide a heat sink which helps absorb decay heat, and repressurize the RCS. Subsequent to stabilizing

plant conditions and satisfying ECCS termination criteria, the operator terminates ECCS operation and initiates plant shutdown operations.

B. Loss-of-Coolant Accident

A LOCA is defined as a rupture of the RCS piping or branch piping which results in a decrease in RCS inventory that exceeds the flow capability of the normal makeup system. Ruptures which result in break flow within the capability of the normal makeup system will not result in decreasing RCS pressure and ECCS actuation. The maximum break size for which the normal makeup system can maintain RCS pressure is obtained by comparing the calculated flow from the RCS through the postulated break against the charging system makeup flow capacity when aligned for maximum charging at normal RCS pressure. Makeup flow from two charging pumps is adequate to sustain pressurizer pressure at approximately 2210 psig for a break through a 0.375-in. diameter hole. This break results in a loss of approximately 17.5 lbm/s. For breaks less than a 0.375-in. diameter hole, the normal makeup system can maintain RCS pressure and permit the operator to execute an orderly shutdown.

For the purpose of evaluation, the spectrum of postulated piping breaks in the RCS is divided into major pipe breaks (large break) and minor pipe breaks (small breaks). The large break is defined as a rupture with a total cross-sectional area equal to or greater than 1 ft^2. The small break is defined as a rupture with a total cross-sectional area less than 1 ft^2 but larger than the maximum break size for which the normal makeup system can maintain RCS pressure as described above. Refer to subsection 15.6.5 for a detailed description of this accident, including acceptance criteria and analytical results.

For this accident, the ECCS is actuated upon receipt of an SI signal. Once actuated, the ECCS mitigates the spectrum of LOCA accidents, but its performance varies depending on the LOCA transient. The charging pumps function to immediately inject borated water from the refueling water storage tank. The SI pumps and RHR pumps function to start delivering borated water from the refueling water storage tank when the RCS depressurizes to approximately 1500 psia and 200 psia, respectively. The accumulators begin to inject when the RCS depressurizes to approximately 600 psia. During the LOCA transient, flow to the RCS is dependent on the RCS pressure transient. The ECCS water injected into the RCS provides for heat transfer from the core, prevents excessive core clad temperatures, and eventually accomplishes core reflood (large break) or core recovery (small break). The LOCA analyses do not take credit for the boron content of the injected water.

Following completion of core reflood (large break) or core recovery (small break), the ECCS continues to supply water to the RCS for long-term cooling. After the water level in the refueling water storage tank reaches the low-low level setpoint, switchover to cold leg recirculation is initiated automatically and completed by manual operator action as discussed in paragraph 6.3.2.8. This permits continued cooling of the core by recirculation of the spilled water in the containment emergency sumps. At approximately 7.5 h after initiation of the LOCA, the ECCS is manually realigned in the hot leg recirculation mode to control boric acid concentration in the reactor vessel.

Figure 6.3.2-1 provides process flow diagrams which illustrate ECCS performance for the various modes of system operation.

6.3.3.4 Use of Dual Function Components

The ECCS contains components which have no other operating function and components which are shared with other systems. Components in each category are as follows:

A. Components of the ECCS which perform no other function are:

1. One accumulator for each loop which discharges borated water into its respective cold leg of the reactor coolant loop piping.
2. Two SI pumps, which supply borated water for core cooling to the RCS.
3. Deleted.
4. One BIT (Unit 1 only).
5. Deleted.
6. Associated piping, valves, and instrumentation.

B. Components which also have a normal operating function are as follows:

1. Two RHR pumps and two residual heat exchangers which are normally used for decay heat removal during the latter stages of normal plant shutdown and when the reactor is held at cold shutdown for maintenance or refueling conditions. During all other plant operating modes, they are aligned to perform the low-head injection function.
2. Two charging pumps which are normally aligned for charging service with suction from the volume control tank. As a part of the chemical and volume control system, the normal operation of these pumps is discussed in subsection 9.3.4.
3. One refueling water storage tank which is used to fill the refueling canal for refueling operations and to provide makeup to the spent fuel pool. However, during all other plant operating periods it is aligned to the suction of the SI pumps. It is also aligned to the suction of the RHR pumps during plant thermal power operating periods. The charging pumps are automatically aligned to the suction of the refueling water storage tank upon receipt of an SI signal or a volume control tank low level signal.

An evaluation of all components required for operation of the ECCS demonstrates that either:

A. The component is not shared with other systems.

B. If the component is shared with other systems, it is either aligned or not aligned during normal plant operation to perform its accident mitigation function.

If not aligned to perform its accident mitigation function, two valves in parallel are provided to align the system for injection, and two valves in series are provided to isolate portions of the system not utilized for injection. These valves are automatically actuated by the SI signal.

Table 6.3.3-1 provides a shared function evaluation that indicates the alignment of components during normal operation and the realignment required to perform the accident function.

In all cases of component operation, SI has the priority usage such that an SI signal will override all other signals and start or align systems for injection.

6.3.3.5 Limits on System Parameters

The analyses show that the design basis performance characteristic of the ECCS is adequate to meet the requirements for core cooling following an accident with the minimum engineered safety features equipment operating. In order to ensure this capability in the event of the simultaneous failure to operate any single active component, technical specifications are established for reactor operation.

Normal operating status of ECCS components is given in table 6.3.3-2.

The ECCS components are available whenever the coolant energy is high and the reactor is critical. During low-temperature physics tests there is a negligible amount of stored energy in the coolant and low decay heat; therefore, an accident comparable in severity to accidents occurring at operating conditions is not possible and ECCS components are not required. The possibility of a LOCA during startup and shutdown has been considered. It has been demonstrated by analysis that a LOCA at startup or shutdown is bounded by the LOCA analyses described in chapter 15, even though various ECCS components are intentionally disabled in the shutdown mode. Portions of the ECCS may be temporarily realigned for check valve leakage testing prior to startup in Mode 3 by closing either HV-8809A or HV-8809B. Conservative analyses have been performed for ECCS configurations more restrictive than those established during check valve testing. The conservative analyses assumed that the SI actuation on low pressurizer pressure was blocked and that flow was available from one RHR pump, one safety injection pump, and one charging pump (no accumulator injection). The results demonstrate that the minimum ECCS flow requirements during Mode 3 are met and that the Chapter 15 LOCA analyses remain limiting.

The principal system parameters and the number of components which may be out of operation in test, quantities and concentrations of coolant available, and allowable time in a degraded status are provided in the Technical Specifications. If efforts to restore the operable status of the ECCS are not accomplished within Technical Specification requirements, the plant is required to be placed in a lower operational mode; i.e., hot standby to hot shutdown, hot shutdown to cold shutdown, etc.

6.3.4 TESTS AND INSPECTIONS

6.3.4.1 Emergency Core Cooling System Performance Tests

Conformance with the recommendations of Regulatory Guide 1.79 is discussed in section 1.9.

6.3.4.1.1 Preoperational Test Program at Ambient Conditions

Preoperational testing of the emergency core cooling system (ECCS) can be conducted during the testing of the reactor coolant system (RCS) following flushing and hydrostatic testing, with the system cold and the reactor vessel head removed. Provision should be made for excess water to drain into the refueling canal. The ECCS must be aligned for normal power operation. The operation of the emergency diesels can be tested in conjunction with the ECCS. This test should provide information including the following facets:

- Satisfactory safety injection (SI) signal generation and transmission.

- Proper operation of the emergency diesel generators, including sequential load pickup.
- Valve operating times.
- Pump starting times.
- Pump delivery rates at runout conditions (one point on the operating curve).

6.3.4.1.2 Components

6.3.4.1.2.1 Pumps. Separate flow tests of the pumps in the ECCS are conducted during preoperational testing (with the reactor vessel head off) to check capability for sustained operation. The charging, SI, and residual heat removal (RHR) pumps discharge into the reactor vessel through the injection lines, with the overflow from the reactor vessel passing into the refueling canal. Each pump is tested separately with water drawn from the refueling water storage tank. Data are taken to determine pump head and flow at this time. Pumps are then run on miniflow circuits and data taken to determine a second point on the head-flow characteristic curve.

6.3.4.1.2.2 Accumulators. Each accumulator is filled with water from the refueling water storage tank and pressurized with the motor-operated valve on the discharge line closed. Then the valve is opened and the accumulator allowed to discharge into the reactor vessel as part of the operational startup testing with the reactor cold and the vessel head off.

6.3.4.2 Reliability Tests and Inspections

Routine periodic testing of the ECCS components and all necessary support systems at power is planned. Valves which operate after a loss-of-coolant accident are operated through a complete cycle; pumps are operated individually on either their miniflow lines, or on recirculation back to the RWST except the charging pumps which are tested by their normal charging function. If such testing indicates a need for corrective maintenance, the redundancy of equipment in these systems permits such maintenance to be performed without shutting down under certain conditions. These conditions include considerations such as the period within which the component should be restored to service and the capability of the remaining equipment to provide the minimum required level of performance during such a period.

The series check valves between the accumulator and the RCS are tested to verify that each of the series check valves can independently sustain differential pressure across its disc and also verify that the valve is in its closed position. The required periodic tests are performed after each refueling just prior to plant startup, after the RCS has been pressurized.

Where series pairs of check valves form the high-pressure to low-pressure isolation barrier between the RCS and ECCS piping outside the reactor containment, periodic testing of these check valves is performed to provide assurance that certain postulated failure modes will not result in a loss of coolant from the low-pressure system outside containment with a simultaneous loss of ECCS pumping capacity.

The ECCS test line subsystem provides the capability for determination of the integrity of the pressure boundary formed by series check valves. The tests performed verify that each of the series check valves can independently sustain differential pressure across its disc and also verify that the valve is in its closed position. The required periodic tests are performed after each refueling just prior to plant startup, after the RCS has been pressurized.

For some series check valve tests, the ECCS may be temporarily realigned for a brief period of time. Due to redundancy of ECCS equipment, the minimum ECCS flow delivery requirements continue to be met.

Lines in which the series check valves are to be tested are the SI pump cold and hot leg injection lines and the RHR pump cold and hot leg injection lines.

To implement the periodic component testing requirements, technical specifications have been established. During periodic system testing, a visual inspection of pump seals, valve packings, flanged connections, and relief valves is made to detect leakage. Inservice inspection provides further confirmation that no significant deterioration is occurring in the ECCS pressure boundary.

Design measures have been taken to ensure that the following testing can be performed:

A. Active components may be tested periodically for operability (e.g., pumps on miniflow, certain valves, etc.).

B. An integrated system actuation test can be performed when the plant is cooled down and the RHR system is in operation. The ECCS is arranged so that no flow is introduced into the RCS for this test. Detailed discussion of SI signal testing provisions is provided in section 7.2.

C. An initial flow test of the full operational sequence can be performed.

The design features which ensure this test capability are specifically:

A. Power sources are provided to permit individual actuation of each active component of the ECCS.

B. The SI pumps can be tested periodically during plant operation using the miniflow recirculation lines provided.

C. The RHR pumps are used every time the RHR system is put into operation. They can also be tested periodically when the plant is at power using either the miniflow recirculation lines, or recirculation line back to the RWST.

D. The centrifugal charging pumps are either normally in use for charging service or can be tested periodically on miniflow.

E. Remotely operated valves can be exercised during routine plant maintenance.

F. Level and pressure instrumentation is provided for each accumulator tank for continuous monitoring of these parameters during plant operation.

G. Flow from each accumulator tank can be directed through a test line to verify isolation valve position. The test line can be used, when the RCS is pressurized, to ascertain backleakage through the accumulator check valves.

H. A flow indicator is provided in the charging pump, SI pump, and RHR pump headers. Pressure instrumentation is also provided in these lines.

I. An integrated system test can be performed when the plant is cooled down and the RHR system is in operation. This test does not introduce flow into the RCS but does demonstrate the operation of the valves, pump circuit breakers, and

automatic circuitry, including diesel starting and the automatic loading of ECCS components on the diesels (by simultaneously simulating a loss of offsite power to the vital electrical buses).

The Technical Specifications specify requirements for test frequency, acceptability of testing, and measured parameters. The ECCS components and systems are designed to meet the intent of the American Society of Mechanical Engineers Code, Section XI, for inservice inspection.

6.3.5 INSTRUMENTATION REQUIREMENTS

Instrumentation and associated analog and logic channels employed for initiation of emergency core cooling system (ECCS) operation is discussed in section 7.3. This section describes the instrumentation employed for monitoring ECCS components during normal plant operation and also ECCS post-accident operation. All alarms are annunciated in the control room.

6.3.5.1 Deleted

6.3.5.1.1 Deleted

6.3.5.1.2 Residual Heat Exchanger Temperature

The fluid temperature at both the inlet and the outlet of each residual heat exchanger is recorded on the integrated plant computer. There is also a locally mounted temperature indicator at the outlet of each residual heat exchanger.

6.3.5.1.3 Deleted

6.3.5.2 Pressure Indication

6.3.5.2.1 Deleted

6.3.5.2.2 Charging Pump Suction and Discharge Pressure

There is local pressure indication at the suction and discharge of each charging pump.

6.3.5.2.3 Safety Injection Pump Suction Pressure

There is a locally mounted pressure indicator at the suction of each safety injection pump.

6.3.5.2.4 Safety Injection Pump Discharge Pressure

Safety injection pump discharge pressure for each pump is indicated in the control room.

6.3.5.2.5 Accumulator Pressure

Duplicate pressure channels are installed on each accumulator. Pressure indication in the control room and high- and low-pressure alarms are provided by each channel.

6.3.5.2.6 Test Line Pressure

A local pressure indicator, used to check for proper seating of the accumulator check valves between the injection lines and the reactor coolant system, is installed on the leakage test line.

6.3.5.2.7 Residual Heat Removal Pump Suction Pressure

Local pressure indication is provided at the inlet to each residual heat removal pump.

6.3.5.2.8 Residual Heat Removal Pump Discharge Pressure

Residual heat removal discharge pressure for each pump is indicated in the control room. A high-pressure alarm is actuated by each channel.

6.3.5.3 Flow Indication

6.3.5.3.1 Deleted

6.3.5.3.2 Charging Pump Injection Flow

Charging pump injection header flow to the reactor cold legs is indicated in the control room.

6.3.5.3.3 Safety Injection Pump Header Flow

Safety injection pump flow for each pump is indicated in the control room.

6.3.5.3.4 Safety Injection Pump Minimum Flow

A local flow indicator is installed in the safety injection pump minimum flow line.

6.3.5.3.5 Test Line Flow

A local flow indicator is provided in the accumulator test line.

6.3.5.3.6 Residual Heat Removal Pump Flow

The flow from each residual heat removal pump is indicated in the control room. These instruments also control the residual heat removal bypass valves, maintaining constant return flow to the reactor coolant system during normal cooldown.

6.3.5.3.7 Residual Heat Removal Pump Minimum Flow

A flowmeter installed in the discharge of each residual heat removal pump provides control for the valve located in the pump miniflow line.

6.3.5.4 Level Indication

6.3.5.4.1 Refueling Water Storage Tank Level

Four level indicator channels, which indicate in the control room, are provided for the refueling water storage tank. Each channel is provided with a high, low, low-low, and empty level alarm. The high level alarm is provided to protect against possible overflow of the refueling water storage tank. The low level alarm is provided to ensure that a sufficient volume of water is always available in the refueling water storage tank in conformance with the Technical Specifications. The refueling water storage tank low level signal isolates the sludge mixing system, which also is used to maintain temperature from the tank at a 50°F minimum. The low-low level alarm alerts the operator to realign the ECCS from the injection to the recirculation mode following an accident and automatically opens the RHR containment emergency sump isolation valves. The empty level alarm alerts the operator to initiate manual actions to realign the containment spray pumps from the injection mode to the recirculation mode. (Upon receipt of the empty alarm, the RWST contents are nearly exhausted; the remaining volume is only sufficient to allow manual switchover of the containment spray pumps.) It is not expected that any ECCS pumps will be taking suction from the RWST when the empty level alarm is reached. Any ECCS pumps taking suction from the RWST when the empty level alarm is received should be secured and the steps necessary to realign their suction source to the containment sump should be completed prior to restarting the pumps. Each channel also provides level indication in the main control room.

Refer to subsection 6.3.2 and figure 6.3.2-5 for additional information on RWST sizing and alarm setpoints for ECCS and containment spray pump suction switchover from injection mode to recirculation mode.

6.3.5.4.2 Accumulator Level

Duplicate level channels are provided for each accumulator. Both channels provide indication in the control room and actuate high- and low-level alarms.

6.3.5.4.3 Deleted

6.3.5.5 Operating Status Indication

6.3.5.5.1 Pumps

The operating status of ECCS pumps is indicated on the control board by red (running) and green (stopped) lights that are integral with the pump switch assembly for each pump. Pump operating status is also indicated by monitor lights which are grouped in a common portion of the control board. The operating status of each ECCS pump is indicated by its monitor light which is dark (stopped) or bright (running) depending on pump operating status.

6.3.5.5.2 Valves

The position of ECCS valves is indicated on the control board by red (open) and green (closed) lights that are integral with the valve switch assembly for each valve. Valve position is also indicated by monitor lights which are grouped in a common portion of the control board. The position of each valve is indicated by its monitor light. For the centrifugal charging pump alternate miniflow valves (HV-8508A and HV-8508B), the monitor panel lights indicate that the valves are in the enabled mode; therefore, valve position is indicated only at the handswitches. The position indication lights for motor-operated valves are controlled by motor operator limit switches. For air-operated valves, these lights are controlled by stem-mounted limit switches.

The accumulator motor-operated valves have additional position indication. For each valve, an alarm annunciator point is activated by both a motor operator limit switch and by a stem-mounted limit switch whenever an accumulator valve is not fully open for any reason with the system at pressure. (The pressure at which the safety injection block is unblocked is approximately 1900 psig.) A separate annunciator point is used for each accumulator valve. The alarm activated by the stem-mounted limit switch will be recycled at approximately 1-h intervals to remind the operator of the improper valve lineup, until corrective action is taken.

TABLE 6.3.2-1 (SHEET 1 OF 3)

EMERGENCY CORE COOLING SYSTEM
COMPONENT PARAMETERS

Accumulators	
Number	4
Design pressure (psig)	700
Design temperature (°F)	300
Operating temperature (°F)	60-120
Normal operating pressure (psig)	650
Total volume (ft^3)	1350 each
Nominal water volume (ft^3)	900 each
Nominal volume N gas (ft^3)	400 each
Boron concentration, nominal (ppm)	1900-2600
Centrifugal charging pumps (See figure 6.3.2-3.)	
Number	2
Design pressure (psig)	2800
Design temperature (°F)	300
Design flow (gal/min)	150
Design head (ft)	5800
Maximum flow (gal/min)	555
Design head at maximum flow (ft)	1400
Design head at shutoff (ft)	6200
Motor rating (hp)	600
Required NPSH at maximum flow (ft)	(See figure 6.3.2-3.)
Available NPSH at maximum flow (ft) from RWST	78
Discharge orifice (1FO-10118 & 1FO-10122)	(See drawing 1X6AH02-300000 for sizing)
Discharge orifice (2FO-10122 & 2FO-10123)	(See drawing 2X6AH02-30000 for sizing)
SI pumps (See figure 6.3.2-4.)	
Number	2
Design pressure (psig)	1750
Design temperature (°F)	300
Design flow (gal/min)	425
Design head (ft)	2680
Maximum flow (gal/min)	660
Design head at maximum flow (ft)	1660
Design head at shutoff (ft)	3545
Motor rating (hp)	450
Required NPSH at maximum flow (ft)	(See figure 6.3.2-4.)

TABLE 6.3.2-1 (SHEET 2 OF 3)

Available NPSH at maximum flow (ft) from RWST	59

RHR pumps (See figure 6.3.2-2.)

Number	2
Design pressure (psig)	600
Design temperature (°F)	400
Design flow (gal/min)	3000
Design head (ft)	375
Maximum flow (gal/min)	4500
Design head at maximum flow (ft)	325
Design head at shutoff (ft)	450
Motor rating (hp)	400
Required NPSH at maximum flow (ft)	(See figure 6.3.2-2.)
Available NPSH at maximum flow (ft)	
From RWST	92
From emergency sumps	34.7

Residual heat exchangers

(See subsection 5.4.7 for design parameters.)

Boron injection tank (Unit 1 only)

Number	1
Total volume (gal)	900
Usable volume at operating conditions, solution (gal)	900
Boron concentration, nominal (ppm)	0-2600
Design pressure (psig)	2735
Operating pressure (psig)	2684
Design temperature	300
Operating temperature	Ambient
Heaters	Determinated

TABLE 6.3.2-1 (SHEET 3 OF 3)

Motor-operated valves stroke times are provided in table 6.3.2-3.

Refueling water storage tank

Number	1
Total volume, nominal (gal)	715,000
Boron concentration (ppm)	2400-2600
Operating pressure	Atmospheric
Operating temperature	Ambient, 50°F minimum
Heating system	
Number of heaters	1
Heater capacity (kw)	50
Number of pumps	1
Pump capacity (gal/min)	200
Number of eductors	9

TABLE 6.3.2-2

EMERGENCY CORE COOLING SYSTEM RELIEF VALVE DATA

Description	Fluid Discharged	Fluid Inlet Temperature Normal (°F)	Set Pressure (psig)	Backpressure Constant (psig)	Maximum Total Backpressure (psig)	Capacity
N_2 supply to accumulators	Nitrogen	120	700	0	0	1500 sf /min
SI pump discharge	Dilute H_3BO_3	120	1750	0 to 15	50	20 gal/min
RHR pump SI line	Dilute H_3BO_3	120	600	0 to 15	50	20 gal/min
SI pumps suction header	Dilute H_3BO_3	100	220	0 to 15	50	25 gal/min
Accumulator to containment	Nitrogen	120	700	0	0	1500 sf /min

TABLE 6.3.2-3 (SHEET 1 OF 3)

MOTOR-OPERATED ISOLATION VALVES IN THE EMERGENCY CORE COOLING SYSTEM

Location	Valve Identification	Interlocks	Automatic Features[a]	Position Indication[b]	Alarms
Accumulator isolation valves	HV-8808 A,B,C,D	SI signal, RCS pressure> unblock.	Opens on SI signal if closed and RCS pressure > unblock.	MCB	Yes-out of position
SI pump suction from RWST	HV-8806, HV-8923 A,B	None	None [c]	MCB	Yes-out of position
RHR suction from RWST	HV-8812 A,B	Cannot be opened unless corresponding sump valve closed and RHR discharge to SI or charging pumps closed.	None[c]	MCB	Yes-out of position
RHR discharge to SI/charging pump suction	HV-8804 A,B	Cannot be opened unless SI pump miniflow isolated, charging pump alternate miniflow isolated, RHR suction from RCS isolated, and corresponding sump valve open.	None[c]	MCB	Yes-out of position
SI hot leg injection	HV-8802 A,B[d]	None	None	MCB	Yes-out of position
RHR hot leg injection	HV-8840[d]	None	None	MCB	Yes-out of position
Containment emergency sump isolation valve	HV-8811 A,B[d]	Cannot be opened in normal operation unless RHR suction valves from RWST and RCS closed.	Opens on RWST low-low with SI signal.	MCB	Yes-out of position
CVCS suction from RWST	LV-112 D,E	SI signal	Opens on SI signal or VCT low-low level 15 s[c][e]	MCB	None
CVCS normal suction	LV-112 B,C	SI signal	Closes on SI signal or VCT low-low level if CVCS suction valves from RWST open 10 s[e]	MCB	Yes-out of position
SI pump to cold leg	HV-8835	None	None	MCB	Yes-out of position

TABLE 6.3.2-3 (SHEET 2 OF 3)

Location	Valve Identification	Interlocks	Automatic Features[a]	Position Indication[b]	Alarms
CVCS pump discharge	HV-8105, HV-8106, HV-8116	SI signal	Closes on SI signal. 10 s[f] for HV-8116 17 s[i] for HV-8105, HV-8106	MCB	None
BIT suction (Unit 1 only)	HV-8803 A,B	None	None[g]	None	None
BIT discharge (Unit 1) and CVCS charging pump high head cold leg	HV-8801 A,B	SI signal	Opens on SI signal.	MCB[b]	Yes-out of position
Charging pump/SI pump crossover crossover	HV-8807 A,B, HV-8924	None	None[c]	MCB	Yes-out of position
RHR to RCS cold legs	HV-8809 A,B	None	None	MCB	Yes-out of position
SI pump miniflow	HV-8813, HV-8814, HV-8920	Cannot be opened unless RHR discharge to SI and charging pumps closed.	None[c]	MCB	Yes-out of position
RHR cross-connect	HV-8716 A,B[h]	None	None[c]	MCB	Yes-out of position
SI pump cross-connect	HV-8821 A,B	None	None	MCB	Yes-out of position
Charging pump normal miniflow	HV-8110, HV-8111 A,B	SI signal	Closes on SI signal. 15 s	MCB	Yes-out of position
Charging pump suction	HV-8471 A,B	None	None	MCB	Yes-out of position
Charging pump discharge	HV-8485 A,B, HV-8438	None	None	MCB	Yes-out of position

TABLE 6.3.2-3 (SHEET 3 OF 3)

Location	Valve Identification	Interlocks	Automatic Features[a]	Position Indication[b]	Alarms
Charging pump alternate miniflow	HV-8508 A,B	Cannot be opened by operator unless volume control tank discharge valves closed and RHR discharge to SI and charging pumps closed.	Enabled on SI signal and will open or close based on centrifugal charging pump discharge pressure.	MCB	Yes-incorrect mode
	HV-8509 A,B	Cannot be opened unless RHR discharge to SI and charging pumps closed.	None[c]	MCB	Yes-out of position
RHR pump miniflow	FV-610, 611	None	Open if pump discharge flow is less than 824 gpm at 350°F, 780 gpm at 100°F and close when the flow exceeds 1944 gpm at 350°F, 1841 gpm at 100°F (10 s).	MCB	None

a. Times are maximum motor-operated valve stroke times that are significant to safety analysis/evaluations. No time is indicated where stroke time was irrelevant to safety analyses/evaluations.
b. MCB - main control board.
c. Vogtle FSAR table 6.3.2-7 provides the switchover sequence from the post-accident cold leg injection mode to the cold leg recirculation mode of operation. The times provided in FSAR table 6.3.2-7, in conjunction with the RWST outflow, are used to verify that there is sufficient volume between the RWST Lo-Lo and empty alarms to complete the switchover sequence. Changes to valve stroke time should be evaluated for impact on available RWST volume.
d. Valve disk is provided with bonnet vent on containment side of disk.
e. Valves LCV-112D, E open automatically on an SI signal. Valves LCV-112B, C begin to close when LCV-112D, E reach full open. The safety analyses assume that the SI flow path is unavailable until both sets of valves reach their final positions.
f. The analysis assumes ECCS flow at 25 s. The operating time for the closure of the normal discharge valves and opening the injection valves should not be significantly different; therefore, the nominal value of 10 s was specified for these valves.
g. Valve is normally locked open, with power removed at component. Valve is operated manually for maintenance only.
h. Valve disk is provided with bonnet vent to RHR pump side.
i. The analysis assumes ECCS flow at 25 s. The limiting operation is the alignment of the charging pump suction to the RWST from the VCT. The normal discharge valves and the injection valves can have stroke times up to 25 s before affecting the time it takes to establish safety injection. These valves should have similar stroke times to reduced charging pump runout. Closure of the normal discharge valves is 17 s and the injection valves is 17 s, and the opening time of the injection valves is 17 s.

TABLE 6.3.2-4 (SHEET 1 OF 2)

MATERIALS EMPLOYED FOR
EMERGENCY CORE COOLING SYSTEM COMPONENTS

Component	Material
Accumulators	Carbon steel clad with austenitic stainless steel
Boron injection tank (Unit 1 only)	Austenitic stainless steel
Boron injection surge tank[(a)] (Unit 1 only)	Austenitic stainless steel
Pumps	
Centrifugal charging	Austenitic stainless steel
Safety injection	Austenitic stainless steel
Residual heat removal	Austenitic stainless steel
Residual heat exchangers	
Shell	Carbon steel
Shell end cap	Carbon steel
Tubes	Austenitic stainless steel
Channel	Austenitic stainless steel
Channel cover	Austenitic stainless steel
Tube sheet	Austenitic stainless steel
Valves	
Motor-operated valves containing radioactive fluids	
Pressure containing parts	Austenitic stainless steel or equivalent corrosion resistant material
Body-to-bonnet bolting and nuts	Low alloy steel
Seating surfaces	Stellite No. 6 or equivalent corrosion resistant material
Stems	Austenitic stainless steel or 17-4 pH stainless

TABLE 6.3.2-4 (SHEET 2 OF 2)

Component	Material
Motor-operated valves, containing nonradioactive, boron-free fluids	
Body, bonnet, and flange	Carbon steel
Stems	Corrosion resistance steel
Diaphragm valves	Austenitic stainless steel
Accumulator check valves	
Parts contacting borated water	Austenitic stainless steel
Clapper arm shaft	17-4 pH stainless
Relief valves	
Stainless steel bodies	Stainless steel
Carbon steel bodies	Carbon steel
All nozzles, discs, spindles, and guides	Austenitic stainless steel
Bonnets for stainless steel valves without balancing bellows	Stainless steel or plated carbon steel
All other bonnets	Carbon steel
Piping	
All piping in contact with borated water	Austenitic stainless steel

a. The Unit 1 boron injection tank recirculation system is physically isolated from the ECCS flowpath.

TABLE 6.3.2-5 (SHEET 1 OF 9)

EMERGENCY CORE COOLING SYSTEM - SAFEGUARDS
OPERATIONS - FAILURE MODES AND EFFECTS ANALYSIS

Component[a]	Failure Mode	Function	Effect on System Operation	Failure Detection Method[b]	Remarks
1. Motor-operated gate valve LV-112B (LV-112C analogous)	Fails to close on demand.	Provides isolation of fluid discharge from the volume control tank (VCT) to the suction of charging pumps.	Failure reduces redundancy of providing tank discharge isolation. Negligible effect on system operation. Alternate isolation valve LV-112C (LV-112B) provides backup tank discharge isolation.	Valve open/close position indication and valve close position monitor light and alarm for group monitoring of components at main control board (MCB).	Valve is electrically interlocked with isolation valve LV-112D (LV-112E) and the instrumentation that monitors fluid level of the VCT. Valve closes upon receipt of an SI signal or upon receipt of a VCT low water level signal providing that isolation valve LV-112D (LV-112E) is at full open position.
2. Motor-operated gate valve LV-112D (LV-112E analogous)	Fails to open on demand.	Provides isolation of fluid discharge from the RWST to the suction of charging pumps and an electrical interlock to the closing of isolation valve LV-112B (LV-112C).	Failure reduces redundancy of providing fluid flow from RWST to suction of charging pumps. Negligible effect on system operation. Alternate isolation valve LV-112E (LV-112D) opens to provide backup flowpath.	Valve open/close position indication and valve open position monitor light at MCB.	Valve is electrically interlocked with the instrumentation that monitors fluid level of the VCT. Valve opens upon receipt of an SI signal or upon receipt of a VCT low water level signal.
	Fails to close on demand.		Failure reduces redundancy of providing isolation of fluid discharged from residual heat exchanger 1 to RWST. No immediate effect on system operation during recirculation. Alternate isolation check valve 1208-U4-189 in common line from RWST provides backup tank isolation.		

TABLE 6.3.2-5 (SHEET 2 of 9)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Method[b]	Remarks
3. Centrifugal charging pump train A (pump train B analogous)	Fails to deliver working fluid.	Provides fluid flow of emergency coolant through the BIT for Unit 1 and the CVCS charging pump high head cold leg injection for Unit 2 to the RCS at the prevailing incident RCS pressure.	Failure reduces redundancy of providing emergency coolant to the RCS at high RCS pressures. Fluid flow from charging train A (train B) will be lost. Minimum flow requirements for high-head SI will be met by charging train B (train A).	Charging pump discharge header pressure and flow indication at MCB. Open/close pump switchgear circuit breaker indication on MCB. Circuit breaker close position monitor light for group monitoring of component at MCB. Common breaker trip alarm at MCB.	One pump may be used for normal charging of RCS during plant operation. Both pumps start upon receipt of an SI signal.
4. Motor-operated globe valve HV-8110	Fails to close on demand.	Provides isolation of fluid flow from the charging pump discharge header to the seal water heat exchanger via minimum flow bypass line.	Failure reduces redundancy of providing isolation of charging pump miniflow line. Negligible effect on system operation. Alternate isolation valves HV-8111A and HV-8111B in individual charging pump minimum flow bypass lines provide backup miniflow line isolation.	Same as item 1.	Valve closes upon receipt of an SI signal.
5. Motor-operated globe valve HV-8111A (HV-8111B analogous)	Fails to close on demand.	Provides isolation of fluid flow from charging train A (train B) to the seal water heat exchanger flow via minimum bypass line.	Failure reduces redundancy of providing isolation of charging pump miniflow line. Negligible effect on system operation. Alternate isolation valve 8110 provides backup miniflow line isolation.	Same as item 1.	Valve closes upon receipt of an SI signal.
6. Motor-operated gate valve HV-8105 (HV-8106 analogous)	Fails to close on demand.	Provides isolation of fluid flow from the charging pump discharge header to the chemical and volume control system (CVCS) normal charging line to the RCS.	Failure reduces redundancy of providing isolation of charging pump discharge to normal charging line. Negligible effect on system operation. Alternate isolation valve HV-8106 (HV-8105) provides backup normal charging line isolation.	Same as item 1 except no valve close monitor alarm for group monitoring.	Same as item 4.

TABLE 6.3.2-5 SHEET 3 of 9)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Method[(b)]	Remarks
7. Motor-operated gate valve HV-8801A (HV-8801B analogous)	Fails to open on demand.	Provides isolation of fluid discharge from the BIT for Unit 1 and the CVCS charging pumps for Unit 2 to high head injection header connected to the cold legs.	Failure reduces redundancy of providing fluid flow from BIT for Unit 1 and CVCS charging pumps for Unit 2 to high head injection header feeding the cold legs. Negligible effect on system operation. Alternate isolation valve HV-8801B (HV-8801A) opens to provide backup flowpath to header.	Valve open/close position indication and valve open position monitor light and alarm for group monitoring of components at MCB.	Valve opens upon receipt of an SI signal.
8. Motor-operated gate valve FV-610 (FV-611 analogous)	Fails open.	Provides regulation of fluid flow through miniflow bypass line to suction of train A (train B) to protect against overheating of the pump and loss of discharge flow from the pump.	Failure reduces working fluid delivered to RCS from RHR train A (train B). Minimum flow requirements will be met by RHR train B (train A) and SI and charging pumps.	Same as item 1.	Valves are regulated by signals from flow transmitter located in each pump discharge header. The control valves open when a RHR pump discharge flow is less than approximately 824 gpm at 350°F, 780 gpm at100°F and close when the flow exceeds approximately 1944 gpm at 350°F, 1841 gpm at 100°F.
	Fails closed.		Failure results in an insufficient fluid flow through RHR train A (train B) pump for a small LOCA or steam break resulting in possible pump damage. Minimum flow requirements will be met by RHR train B (train A) and SI and charging pumps delivering coolant fluid to RCS.		
9. RHR pump train A (train B pump analogous)	Fails to deliver working fluid.	Provides fluid flow of emergency coolant to the RCS when the incident RCS loop pressure drops below shutoff head of pump and provides long term recirculation capability for core cooling following the injection phase of LOCA.	Failure reduces redundancy of providing emergency coolant to the RCS at low RCS pressure. Fluid flow from RHR pump train A (train B) will be lost. Minimum flow requirement will be met by RHR pump train B (train A).	Same as that stated for item 3 except RHR pump discharge pressure and flow indication at MCB.	The RHR pumps are used to deliver reactor coolant through the residual heat exchanger to meet the plant cooldown requirements and are used during cooldown and startup operation. The RHR pumps start upon receipt of an SI signal.

TABLE 6.3.2-5 (SHEET 4 of 9)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Method[(b)]	Remarks
10. Motor-operated gate valve HV-8811A (HV-8811B analogous)	Fails to open on demand.	Provides isolation of fluid discharge from containment emergency sump to suction line of RHR train A (train B).	Failure reduces redundancy of providing fluid flow from the containment emergency sump to the RCS. RHR train A (train B) not available for recirculation. Minimum flow requirements will be met by RHR train B (train A) through opening of isolation valve HV-8811B (HV-8811A). Negligible effect on system operation.	Same as item 7.	Valves open automatically on receipt of a 2/4 RWST lolo level signal in coincidence with SI signal being present (i.e., latched in.) Valve is electrically interlocked from being remotely opened from MCB.
11. Motor-operated gate valve HV-8812A (HV-8812B analogous)	Fails to close on demand.	Provides isolation of fluid discharge from the RWST to suction line of RHR train A (train B).	Failure reduces redundancy of providing RWST isolation from suction line of RHR train A (train B). Negligible effect on system operation. A series check valve 1205-U4-001(1205-U4-002) provides backup isolation against fluid flow from the suction of RHR train A (train B) to the RWST.	Same as item 1.	
12. Motor-operated gate valve HV-8716A (HV-8716B analogous)	Fails to close on demand.	Controls the RHR system resistance to prevent RHR pump runout by blocking or opening flowpaths Provides separation between two independent flowpaths outside containment during cold leg recirculation. Directs LHSI flow to hot legs during hot leg recirculation.	Failure reduces redundancy to prevent excessive RHR pump runout during cold leg recirculation. No effect on system operation. Isolation valve HV-8716B (HV-8716A) provides backup isolation to limit RHR pump runout flow.	Same as item 1.	During the first 11 h of long-term core cooling phase incident recovery RHR, SI, and charging pumps are aligned for injection into cold legs of RCS coolant loops. After 11 h, RHR and SI pumps are aligned by operator for recirculation flow into the hot legs.
	Fails to open on demand.		Failure reduces redundancy of providing fluid flow from RHR pumps for injection into hot legs of RCS loops. Minimum flow requirements will be met by opening of isolation valve HV-8716B (HV-8716A) and flow from RHR train B (train A).		Hot leg RCS coolant loop recirculation from at least SI pump required to prevent boron precipitation during longterm core cooling.

TABLE 6.3.2-5 (SHEET 5 of 9)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Method[(b)]	Remarks
13. Motor-operated gate valve HV-8809A (HV-8809B analogous)	Fails to close on demand.	Provides isolation of fluid flow from RHR train A (train B) to cold leg injection header of RCS coolant loops.	Failure reduces flow of recirculation coolant to hot legs of RCS coolant loops from RHR train A (train B) Minimum flow requirements to hot leg of RCS coolant loops will be met by delivery of coolant from RHR train B (train A) and SI pumps to the hot legs.	Same as item 1.	
14. Motor-operated gate valve HV-8840	Fails to open on demand.	Provides isolation of fluid flow from RHR pumps to hot leg injection header of RCS coolant loops.	Failure prevents fluid fluid flow from RHR pumps directly to hot leg injection header of RCS coolant loops. Minimum flow requirements to hot legs of RCS coolant loop will be met by delivery of coolant from either of the two SI pumps, thus maintaining redundant hot leg recirculation capability.	Same as item 7.	Same as item 12.
	Fails to close on demand.		Failure reduces redundancy of providing isolation of recirculation of fluid into hot legs of RCS coolant loops by RHR pumps. Negligible effect on recirculation into cold legs of RCS coolant loops. Alternate fluid flow isolation provided by closing of isolation valves HV-8716A and HV-8716B.		
15. Motor-operated gate valve HV-8804A	Fails to open on demand.	Provides isolation of fluid flow from RHR train A via RHR heat exchanger A to suction line of charging pumps.	Failure reduces redundancy of providing flow to the suction of the charging pumps from the RHR pumps. No effect on system operation. Charging pumps will be provided suction head by RHR train B via opening valve HV-8804B and the high-head SI suction crosstie via opening valve HV-8807A or HV-8807B.	Same as item 7.	

TABLE 6.3.2-5 (SHEET 6 of 9)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Method[(b)]	Remarks
16. Motor-operated gate valve HV-8804B	Fails to open on demand.	Provides isolation of fluid flow from RHR train B via RHR heat exchanger B to suction line of SI pumps.	Failure reduces redundancy of providing flow to the suction of the SI pumps from the RHR pumps. No effect on system operation. SI pumps will be provided suction head by RHR train A via opening valve HV-8804A and the HHSI suction crosstie via opening valve HV-8807A or HV-8807B.	Same as item 7.	
17. Motor-operated gate valve HV-8807A (HV-8807B analogous)	Fails to open on demand.	Provides fluid flow between the suction of the charging pumps and the SI pumps.	Failure reduces redundancy of providing long term recirculation fluid between the suction of the charging pumps and the SI pumps. Negligible effect on system operation. Suction fluid flow provided by opening alternate isolation valve HV-8807B (HV-8807A).	Same as item 7.	
18. Motor-operated gate valve HV-8924	Fails to close on demand.	Provides isolation barrier to separate the suction of the charging pumps and SI pumps in the event of a single passive failure which occurs in the recirculation mode.	No effect on system operation. Isolation barrier is provided by closing of alternate isolation valves HV-8807A and HV-8807B.	Same as item 1.	The normal operating position of the valve during recirculation is open.
19. Motor-operated gate valve HV-8835	Fails to close on demand.	Provides isolation of fluid flow from SI pumps discharge line to cold legs of RCS coolant loops.	Failure reduces redundancy of providing flow isolation of SI pump flow to cold coolant loops. No effect on safety for system operation. Alternate isolation valves HV-8821A and HV-8821B in discharge crosstie line between SI pumps provide backup isolation against flow of coolant to cold legs.	Same as item 1.	

TABLE 6.3.2-5 (SHEET 7 of 9)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Method[(b)]	Remarks
20. Motor-operated gate valve HV-8802A (HV-8802B analogous)	Fails to open on demand.	Provides isolation of fluid flow from SI train A (train B) discharge line to hot legs of RCS coolant loops.	Failure reduces redundancy of providing fluid flow from SI pumps to hot legs of RCS coolant loops. Minimum flow requirements will be met by SI train B (train A) and RHR pump flow to hot legs of RCS coolant loops.	Same as item 7.	Same as item 12. Valve is positioned open by operator for recirculation into hot legs 1 and 4 (hot legs 2 and 3) of RCS coolant loops.
21. Motor-operated gate valve HV-8821A(HV-8821B analogous)	Fails to close on demand.	Directs SI flow to cold legs during cold leg recirculation. Provides separation between two independent flowpath outside containment during hot leg recirculation isolation to separate SI flowpaths.	Failure reduces redundancy to provide independent SI flowpaths during hot leg recirculation. No effect on system operation. Valve HV-8821B (HV-8821A) provides backup	Same as item 1.	Same as item 12.
22. SI train A (train B analogous)	Fails to deliver working fluid.	Provides fluid flow of emergency coolant to the RCS when the RCS loop pressure drops below shutoff head of pump and provides long-term recirculation capability for core cooling following the injection phase of LOCA.	Failure reduces redundancy of providing emergency coolant to the RCS at high RCS pressure. Fluid flow from SI pump train A (train B) will be lost. Minimum flow requirements for high-head SI will be met by SI pump train B (train A) and charging pumps.	Same as stated for item 3 except SI pump discharge pressure and flow indication at MCB.	The SI pumps start upon receipt of an SI signal.
23. Motor-operated globe valve HV-8813	Fails to close on demand.	Provides isolation of fluid flow from the SI pump discharge header to the RWST.	Failure reduces redundancy of providing isolation of SI pumps miniflow line isolation from RWST. No effect on safety for system operation. Alternate isolation valves HV-8814 and HV-8920 in each SI pump miniflow line provide backup isolation.	Same as item 1.	Valve is electrically interlocked with isolation valves HV-8804A and HV-8804B and may not be opened unless these valves are closed.

TABLE 6.3.2-5 (SHEET 8 of 9)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Method[(b)]	Remarks
24. Motor-operated globe valve HV-8814 (HV-8920 analogous)	Fails to close on demand.	Provides isolation of fluid flow from SI pump train A (train B) to the RWST.	Failure reduces redundancy of providing isolation of SI pump train A (train B) miniflow isolation from RWST. No effect on safety for system operation. Alternate isolation valve HV-8813 in miniflow header provides backup isolation.	Same as item 1.	Same as item 23.
25. Motor-operated gate valve HV-8806	Fails to close on demand.	Provides isolation of fluid discharge from the RWST to suction line of SI pumps.	Failure reduces redundancy of providing isolation of SI pump suction to RWST. No effect on safety for system operation. Alternate check isolation valve1204-U4-090 provides backup isolation.	Same as item 1.	
26. Motor-operated gate valve HV-8923A (HV-8923B analogous)	Fails to close on demand.	Provides isolation barrier to form two independent SI pump flowpaths in the event of a single passive failure.	No effect on system operation. Isolation barrier is provided by closing of alternate isolation valve HV-8923B (HV-8923A).	Same as item 1.	The normal operating position of the valve during recirculation is open.
27. Motor-operated globe valve HV-8508A (HV-8508B analogous)	Fails to open on demand.	Provides alternate miniflow path for charging pump train A (train B) following isolation of normal miniflow line.	Failure prevents use of alternate miniflow line following receipt of SI signal. Charging train A (train B) degradation may occur if RCS pressure then increases to pump shutoff head. High-head SI injection flow will be provided by charging pump train B (train A) and SI pumps.	Valve open/close position indication at valve handswitch. Valve enabled condition at monitor light box and alarm for group monitoring of components at MCB.	Valve enabled by SI signal and opens or closes based on centrifugal charging pump discharge pressure.
	Fails to close on demand.		Failure reduces redundancy of providing isolation of charging pump alternate miniflow line. Alternate isolation valve HV-8509B (HV-8509A) provides backup miniflow line isolation.		Valve is closed by the operator during the switch-over from injection to recirculation.

TABLE 6.3.2-5 (SHEET 9 of 9)

Component	Failure Mode	Function	Effect on System Operation	Failure Detection Method[b]	Remarks
28. Motor-operated globe valve HV-8509A (HV-8509B analogous)	Fails to close on demand.		Failure reduces redundancy of providing isolation of charging pump train B (train A) alternate miniflow line. Alternate isolation valve HV-8508B (HV-8508A) provides backup miniflow line isolation.	Same as item 1.	Same as item 27.

a. Components 1 through 6, 27, and 28 are components of the CVCS that perform an ECCS safeguards function. Components 8, 9, and 12 are components of the RHR system that perform an ECCS safeguards function.

b. As part of plant operation, periodic tests, surveillance inspections, and instrumentation calibrations are made to monitor equipment and performance. Failures may be detected during such monitoring of equipment in addition to detection methods noted.

TABLE 6.3.2-6

EMERGENCY CORE COOLING SYSTEM RECIRCULATION PIPING PASSIVE FAILURE ANALYSIS
LONG TERM PHASE

Flowpath	Indication of Loss of Flowpath	Alternate Flowpath
Low-head recirculation		
From containment sump to low-head injection header via the RHR pumps and the residual heat exchangers.	Accumulation of water in a RHR pump compartment or auxiliary building sump.	Via the independent, identical low-head flowpath utilizing the second residual heat exchanger and RHR pump.
High-head recirculation		
From containment sump to the high-head injection header via RHR pump, heat exchanger, and the high-head SI pumps.	Accumulation of water in a RHR pump compartment or the auxiliary building sump or SI or charging pump compartment.	From containment sump to the high-head SI headers via alternate RHR pump, RHR heat exchanger, and SI or charging pump.

TABLE 6.3.2-7 (SHEET 1 OF 4)

SEQUENCE OF SWITCHOVER OPERATIONS

Switchover from Cold Leg Injection to Cold Leg Recirculation

During the cold leg injection mode and prior to the receipt of the RWST low-low level alarm, the operator is to:

- Verify that all ECCS pumps are operating and are delivering flow to the RCS cold legs.
- Monitor the RWST and the containment emergency sump levels.
- Verify that all CSS pumps are operating.

The RWST low-low level signal automatically initiates opening of the containment emergency sump isolation valves (HV-8811A and B). Upon receipt of the RWST low-low level signal, the operator is required to perform manual switchover steps to complete the switchover in an orderly and timely manner and in the proper sequence. The list of manual switchover steps stated below summarizes the significant procedural steps performed. Upon completion of the switchover steps, the operator will verify proper operation and alignment of all ECCS and CSS components.

From the assumed allowances and flow rates, the minimum time available to complete the ECCS switchover, isolation, and verification (steps 1 through 8) is approximately 11.1 minutes from the time the RWST low-low level alarm is actuated. The available switchover time is calculated considering the worst-case single active failure, which is not isolating one RHR pump from the RWST.

From the assumed allowances and flow rates, the minimum time available to complete the containment spray switchover and verification (steps 9 through 12) is approximately 6 minutes from the time the RWST empty level alarm is actuated. The available switchover time is calculated considering the worst-case single active failure, which is not isolating one RHR pump from the RWST.

TABLE 6.3.2-7 (SHEET 2 OF 4)

Switchover Steps[a]

The steps 1 through 6 manual actions function to align the suction of the RHR pumps to the containment emergency sump and to align the suction of the charging and SI pumps to the discharge of the RHR pumps, thereby assuring an available suction source for all ECCS pumps. The steps 7 and 8 manual actions provide redundant isolation of the RWST from the recirculation fluid. In the cold leg recirculation alignment, both RHR pumps, both SI pumps, and both charging pumps are delivering to the RCS cold legs.

Step 1: When each containment emergency sump isolation valve has reached the full open position, take action to close the corresponding RWST to RHR pump suction isolation valve (HV-8812 A and B). The maximum time allowed for operator actions prior to and including completion of this step is 6.5 minutes from receipt of the RWST low-low level alarm.

Step 2: Close the three SI pump miniflow valves (HV-8813, HV-8814, and HV-8920).

Step 3: Close the two isolation valves in each charging pump's alternate miniflow line (HV-8508 A and B, HV-8509 A and B).

Step 4: Close the two valves in the crossover line downstream of the RHR heat exchangers (HV-8716 A and B).

Step 5: Open the two parallel valves in the common suction lines between the charging pump suction and the SI pump suction (HV-8807 A and B).

Step 6: Open each valve from each RHR pump discharge line to the charging pump suction and to the SI pump suction (HV-8804 A and B, respectively).

Step 7: Close the two parallel valves in the line from the RWST to the charging pump suction (LV-112D and E).

Step 8: Restore power to and close the valve in the common line from the RWST to both SI pumps (HV-8806).

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a. The operator actions for switchover from injection to cold leg recirculation and CSS switchover are not to be interrupted until all of the steps in the switchover are completed; however, corrective actions for any components failures during the switchover procedure will be performed following completion of the switchover procedure.

TABLE 6.3.2-7 (SHEET 3 OF 4)

Following ECCS realignment from injection to recirculation and upon receipt of an RWST empty level alarm, the spray pumps' suctions are remote manually transferred to the containment emergency sumps. The steps 9 through 12 manual actions provide for this alignment and isolation of the RWST from the recirculation fluid. Upon completion of step 12, the CSS is aligned for recirculation mode of operation, with both CSS pumps taking suction from the containment emergency sumps and delivering flow to the containment spray ring headers.

Step 9: Open the containment emergency sump isolation valves in train A of the CSS (HV-9002A and HV-9003A).

Step 10: When the containment emergency sump isolation valves have reached the full open position, take action to close the corresponding RWST to CSS pump suction isolation valve (HV-9017A).

Step 11: Open the containment emergency sump isolation valves in train B of the CSS (HV-9002B and HV-9003B).

Step 12: When the containment emergency sump isolation valves have reached the full open position, take action to close the corresponding RWST to CSS pump suction isolation valve (HV-9017B).

TABLE 6.3.2-7 (SHEET 4 OF 4)

Switchover from Cold Leg Recirculation to Hot Leg Recirculation

At approximately 7.5 h after the accident, hot leg recirculation shall be initiated. The manual operator switchover steps stated below are normally used to perform the switchover operation from the cold leg recirculation mode to the hot leg recirculation mode. Upon completion of the switchover steps, both RHR pumps are delivering from the containment emergency sumps directly to the RCS hot legs and are also delivering to the suction of the SI and charging pumps. Both SI pumps are delivering to the RCS hot legs and both charging pumps are delivering to the RCS cold legs. The CSS is not affected by the switchover to the hot leg recirculation procedure.

Switchover Steps

Step 1: Close the RHR pump discharge cold leg header isolation valves (HV-8809 A and B).

Step 2: Open the RHR pump discharge crossover isolation valves (HV-8716 A and B).

Step 3: Open the RHR pump discharge hot leg header isolation valve (HV-8840).

Step 4: Stop SI train A pump.

Step 5: Close the corresponding SI pump discharge crossover header isolation valve (HV-8821 A).

Step 6: Open the corresponding SI pump discharge hot leg header isolation valve (HV-8802 A).

Step 7: Restart SI train A pump.

Step 8: Stop SI train B pump.

Step 9: Close the corresponding SI pump discharge crossover isolation valve (HV-8821 B).

Step 10: Open the corresponding SI pump discharge hot leg header isolation valve (HV-8802 B).

Step 11: Restart SI train B pump.

Step 12: Close the SI pump discharge cold leg header isolation valve (HV-8835).

TABLE 6.3.2-8

EMERGENCY CORE COOLING SYSTEM AIR-OPERATED VALVES[a]

Valve Location Number	Correct Position Following Safeguards Actuation	Failure Position	Automatic Positioning Signal	Position Indication Red/Green	Position Indication Monitor Lights
HV-8843	C	FC	CI-A	Yes	Yes
HV-8882	C	FC	--	Yes	--
HV-8964	C	FC	CI-A	Yes	Yes
HV-8871	C	FC	CI-A	Yes	Yes
HV-8888	C	FC	CI-A	Yes	Yes
HV-8879 A,B,C,D	C	FC	--	Yes	--
HV-8877 A,B,C,D	C	FC	--	Yes	--
HV-8878 A,B,C,D	C	FC	--	Yes	--
HV-8880	C	FC	CI-A	Yes	Yes
HV-8889 A,B,C,D	C	FC	--	Yes	--
HV-8823	C	FC	CI-A	Yes	Yes
HV-8824	C	FC	CI-A	Yes	Yes
HV-8825	C	FC	CI-A	Yes	Yes
HV-8881	C	FC	CI-A	Yes	Yes
HV-8890 A,B	C	FC	CI-A	Yes	Yes
FV-618	C	FC	--	No	--
FV-619	C	FC	--	No	--
HV-606	O	FO	--	No[b]	Yes
HV-607	O	FO	--	No[b]	Yes

a. Abbreviations:

FC - fails closed.
FO - fails open.
C - closed.
O - open.
SI - safety injection.
CI-A --containment isolation phase A.

b. Position indication by percent valve opening.

TABLE 6.3.2-9 (SHEET 1 OF 5)

FAILURE MODES AND EFFECTS ANALYSIS FOR SAFETY GRADE COLD SHUTDOWN OPERATONS

Component[a]	Failure Mode	Function	Effect on System Operation	Failure Detection Methods[c]	Remarks
1. Motor-operated gate valve HV-8812A (HV-8812B analogous).	Fails to close on demand.	Provides isolation of fluid from the RWST to suction of RHR pump 1 during cooldown operation.	No effect on safety for system operation. Plant cooldown requirements are met by reactor coolant flow from hot leg loop 4 flowing through train B of RHRS; however, time required to reduce RCS temperature is extended.	Valve open/closed position indication at CB and valve (closed) monitor light and alarm at CB.	Valve is normally open to align RHRS for ECCS operation during plant power operation and load follow. Valve must be closed during plant cooldown to satisfy electrical interlock to permit valves HV-8701A and B (HV-8702A, B) to be opened.
2. Centrifugal charging pump 1 (pump 2 analogous).	Fails to deliver working fluid.	Provides fluid flow of borated water from the BAT or RWST to the RCS.	Failure reduces redundancy of providing borated water to the RCS at high RCS pressures. Fluid flow from charging pump1 is lost. Minimum flow requirements for boration and makeup are met by charging pump 2.	Charging pump discharge header pressure and flow indication at CB. Open/close pump switchgear circuit breaker indication on CB. Circuit breaker close position monitor light for group monitoring of component at CB. Common breaker trip alarm at CB.	The charging pumps provide boration and makeup flow to the RCS during safety grade cold shutdown operations.
3. Motor-operated gate valve LCV-112B (LCV-112C analogous).	Fails to close on demand	Provides isolation of fluid discharge from the VCT to the suction of charging pumps.	Failure reduces redundancy of providing VCT discharge isolation. Negligible effect on safety for system operation. Alternate isolation valve provides backup tank discharge isolation.	Same as item 1.	The charging pumps suction is isolated from the VCT and aligned to the BAT (for boration) or RWST (for boration and makeup) during safety grade cold shutdown operations.
4. Motor-operated gate valve LCV-112D (LCV-112E analogous).	Fails to open on demand.	Provides isolation of fluid discharge from the RWST to the suction of charging pumps.	Failure reduces redundancy of providing fluid flow from RWST to suction of charging pumps. Negligible effect on safety for system operation. Alternate isolation valve opens to provide backup flowpath to suction of charging pumps. This path is also the alternate to HV-8104 for boration during safety grade cold shutdown operations (see item 18).	Valves open/close position indication at CB and valve (open) monitor light and alarm at CB.	The charging pumps suction is aligned to the RSWT for boration and makeup to the RCS during safety grade cold shutdown operations.
5. Motor-operated gate valve HV-8803A (HV-8803B analogous) (Unit 1 only).	N/A	Provides isolation of fluid flow from charging pump discharge header to the inlet of the BIT.	N/A	N/A	Electric power supply has been disconnected. Hand switches have been removed.

TABLE 6.3.2-9 (SHEET 2 OF 5)

Component[a]	Failure Mode	Function	Effect on System Operation	Failure Detection Methods[c]	Remarks
6. Solenoid-operated globe valve HCV-190A (HCV-190B analogous).	Fails to open on demand.	Provides control of fluid flow from charging pump to RCS during plant boration and makeup.	Failure reduces redundancy of controlling boration and makeup flow to the RCS. Negligible effect on safety for system operation. Alternate control valve HCV-190B flow from charging pump.	Valve position indication at CB; and charging pump discharge header flow indication at CB.	
7. Motor-operated globe valve HV-8116.	Fails to open on demand.	Provides isolation of fluid flow from charging pump discharge header to RCS through valve HCV-190A.	Failure reduces redundancy of providing boration flow to the RCS. Negligible effect on safety for system operation. Boration flow provided by charging pump through valve HCV-190B.	Same as item 4.	Same as item 5.
8. Solenoid-operated globe valve HV-8095A (HV-8095B analogous).	a. Fails to open on demand.	Provides isolation of fluid flow from the RV head to the PRT.	a. Failure reduces redundancy of providing flow from the RV head to the PRT. Neglig ble effect on safety for system operation. RV head letdown flow provided by parallel head letdown path through alternate isolation valve.	Valve open/close position, indication at CB; and RV head letdown high temperature indication and alarm at CB.	The RV head letdown path to the PRT provides fluid flow out of the RCS to accommodate boration flow into the RCS.
	b. Fails to close on demand.		b. Failure reduces redundancy of isolating flow from the RV head to the PRT. Neglig ble effect on safety for system operation. RV head letdown flow isolation provided by alternate series isolation valve.		
9. Solenoid-operated globe valve HV-8096-A (HV-8096B analogous).	a. Fails to open on demand.	Same as item 8.	a. Same as item 8.a.	Same as item 8.	Same as item 8.
	b. Fails to close on demand.		b. Same as item 8.b.		
10. Solenoid-operated globe valve HCV-442A (HCV-442B analogous).	Fails to open on demand.	Same as item 8.	Same as item 8.a.	Valve position indication at CB; RV letdown temperature indication at CB.	Same as item 8.

TABLE 6.3.2-9 (SHEET 3 OF 5)

Component[a]	Failure Mode	Function	Effect on System Operation	Failure Detection Methods[c]	Remarks
11. Solenoid- operated power-operated relief valve PCV-455A (PCV-456 analogous).	a. Fails to open on demand.	Provides isolation of fluid flow from pressurizer to PRT.	a. Failure reduces redundancy of providing flow from pressurizer to PRT. Negligible effect on safety for system operation. Pressurizer vent flow provided by a parallel pressurizer vent path through alternate isolation valves.	Valve open/close position indication at CB; Pressurizer power-operated relief valve outlet temperature indication at CB.	Pressurizer vent path to the PRT provides fluid flow out of the RCS to permit RCS depressurization to RHRS initiation conditions
	b. Fails to close on demand.		b. Failure reduces redundancy of isolating flow from the pressurizer to the PRT. Negligible effect on safety for system operation. Pressurizer vent flow isolation provided by alternate series isolation valve.		
12. Motor-operated gate valve HV-8000A (HV-8000B analogous).	Fails to close on demand.	Same as item 11.	Same as item 11.b except pressurizer vent flow isolation provided by alternate series isolation valve.	Same as item 11.	Same as item 11.
13. Motor-operated gate valve HV-8808A (HV-8808B, HV-8808C, and HV-8808D analogous).	Fails to close on demand.	Provides isolation of fluid flow from accumulator 1 to the RCS.	Failure prevents isolation of accumulator 1 from the RCS. Negligible effect on safety for system operation. Accumulator 1 is depressurized by opening vent isolation valves.	Valve open/closed position indication at CB, valve (closed) monitor light and alarm at CB; and accumulator pressure indication and low alarm at CB.	Accumulators are isolated or vented during plant cooldown to not affect RCS depressurization.
14. Solenoid-operated globe valve HV-8875A (HV-8875B, HV-8875C, and HV-8875D analogous).	Fails to open on demand.	Provides venting of nitrogen gas from accumulator 1 to containment.	Failure reduces redundancy for venting accumulator 1 to containment. No effect on safety for system operation. Accumulator 1 can be vented by opening vent valves HV-8875E and HCV-943A, or isolated from the RCS by closing isolation valve HV-8808A.	Valve open/closed position indication at CB and accumulator pressure indication and low alarm at CB.	Same as item 13.

TABLE 6.3.2-9 (SHEET 4 OF 5)

Component[a]	Failure Mode	Function	Effect on System Operation	Failure Detection Methods[c]	Remarks
15. Solenoid-operated globe valve HV-8875E (HV-8875F, HV-8875G, and HV-8875H analogous).	Fails to open on demand.	Same as item 14.	Failure reduces redundancy for venting accumulator 1 to containment. No effect on safety for system operation. Accumulator 1 can be vented by opening vent valves HV-8875A and HCV-943A, or isolated from the RCS by closing isolation valve HV-8808A.	Same as item 14.	Same as item 13.
16. Solenoid-operated globe valve HCV-943A (HCV-943B analogous).	Fails to open on demand.	Provides venting of nitrogen gas from accumulators to containment.	Failure reduces redundancy for venting accumulators to containment. No effect on safety for system operation. Accumulators can be vented by opening vent valve HCV-943B or isolated from RCS by closing isolation valves HV-8808A, B, C, and D.	Valve position indication at CB and accumulator pressure indication and low alarm at CB.	Same as item 13.
17. Boric acid transfer pump 1 (pump 2 analogous).	Fails to deliver working fluid.	Provides fluid flow of concentrated boric acid from BAT to charging pump suction.	Failure reduces redundancy of providing concentrated boric acid to charging pump suction. Fluid flow from boric acid transfer pump 1 is lost. Minimum flow requirements for boration is met by boric acid transfer pump 2.	Pump motor start relay position indication (open) at CB and local pump discharge pressure indication.	The boric acid transfer pumps provide boration flow to the charging pumps suction during safety grade cold shutdown operations.
18. Motor-operated globe valve HV-8104.	Fails to open on demand.	Provides isolation of fluid flow from either boric acid transfer pump to charging pump suction.	Failure reduces redundancy of providing concentrated boric acid to charging pump suction. Negligible effect on safety for system operation Concentrated boric acid provided to charging pump suction through valves LV-112D or LV-112E from the RWST.	Valve open/close position indication at CB; and boration flow indication at CB.	The charging pumps' suction is aligned to the BAT pumps for A-train boration of the RCS during safety grade cold shutdown operations. Alternate A and B train paths are aligned through valves LV-112D or LV-112E from the RWST.

TABLE 6.3.2-9 (SHEET 5 OF 5)

a. Components 1, 5, and 13 through 16 are components of the ECCS that perform a safety-grade cold shutdown function. Components 2 through 4, 6, 7, 17 and 18 are components of the CVCS that perform a safety-grade cold shutdown function. Components 8 through 12 are components of the RCS that perform a safety-grade cold shutdown function.

b. List of acronyms and abbreviations.

Auto	-	Automatic.
BAT	-	Boric acid tank.
BIT	-	Boron injection tank (Unit 1 only).
CB	-	Main control board.
CVCS	-	Chemical and volume control system.
ECCS	-	Emergency core cooling system.
HELB	-	High-energy line break.
MELB	-	Moderate-energy line break.
PRT	-	Pressurizer relief tank.
RC	-	Reactor coolant.
RCS	-	Reactor coolant system.
RHR	-	Residual heat removal.
RHRS	-	Residual heat removal system.
RWST	-	Refueling water storage tank.
RV	-	Reactor vessel.
SI	-	Safety injection.
VCT	-	Volume control tank.

c. As part of plant operation, periodic tests, surveillance inspections, and instrument calibrations are made to monitor equipment and performance. Failures may be detected during such monitoring of equipment in addition to detection methods noted.

TABLE 6.3.3-1

EMERGENCY CORE COOLING SYSTEM SHARED FUNCTIONS EVALUATION

Component	Normal Operating Arrangement	Accident Arrangement
Refueling water storage tank	Lined up to suction of SI and RHR pumps.	Lined up to suction of centrifugal charging, SI, and RHR pumps.
Charging pumps	Lined up for charging service suction from volume control tank; discharge via normal charging line.	Suction from refueling water storage tank; discharge lined up to cold legs of reactor coolant piping through high head header. Valves for realignment meet single failure criteria.
RHR pumps	Lined up to cold legs of reactor coolant piping.	Lined up to cold legs of reactor coolant piping.
Residual heat exchangers	Lined up to cold legs of reactor coolant piping.	Lined up to cold legs of reactor coolant piping.

TABLE 6.3.3-2

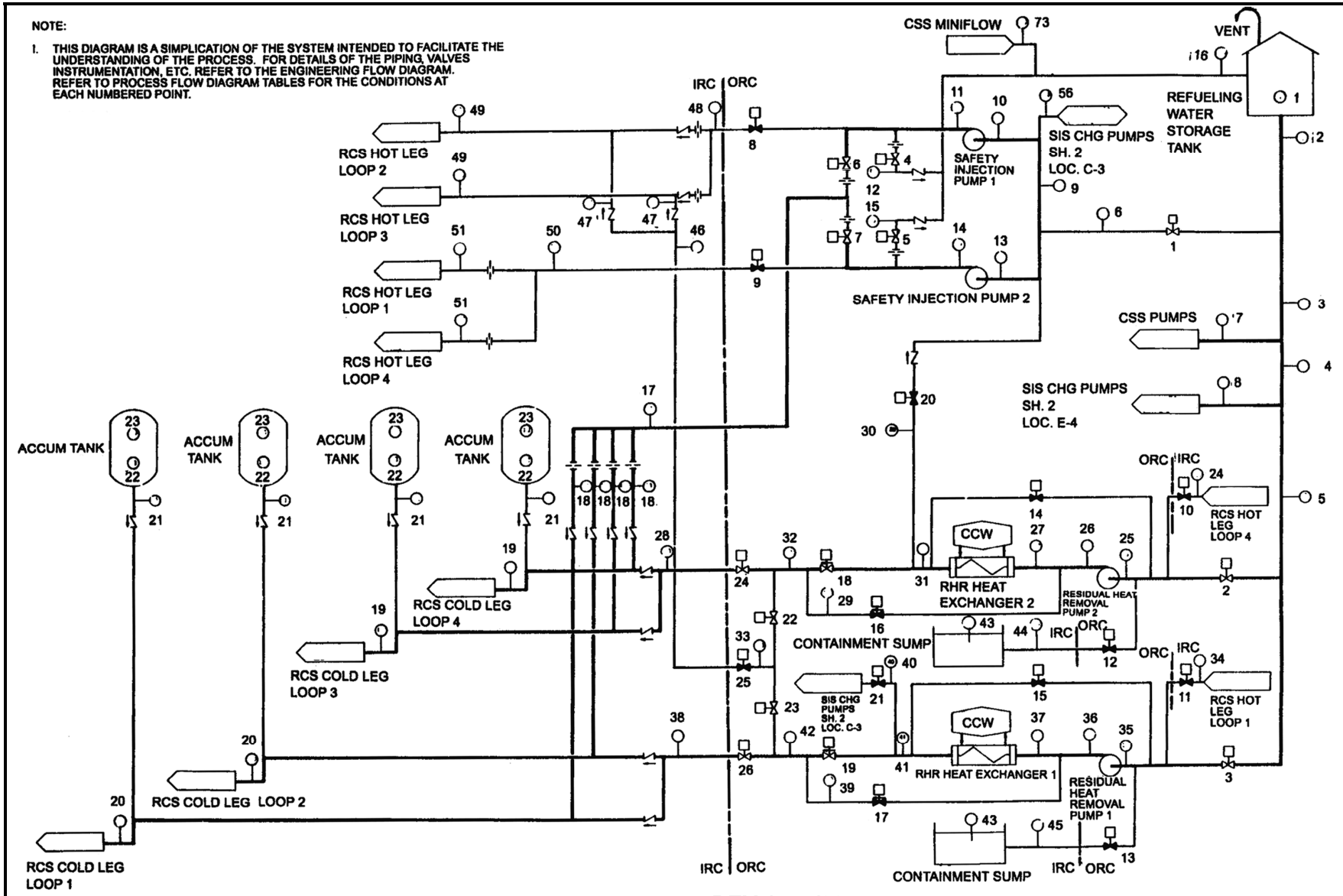
NORMAL OPERATING STATUS OF EMERGENCY CORE COOLING SYSTEM COMPONENTS FOR CORE COOLING

Number of SI pumps operable	2
Number of charging pumps operable	2
Number of RHR pumps operable	2
Number of residual heat exchangers operable	2
Refueling water storage tank volume, nominal (gal)	715,000
Boron concentration in refueling water storage tank (ppm)	2400-2600
Boron concentration in accumulators (ppm)	1900-2600
Boron concentration in BIT (ppm) (Unit 1 only)	0-2600[(a)]
Number of accumulators operable	4
Minimum accumulator pressure (psig)	617
Nominal accumulator water volume (ft^3)	900
System valves, interlocks, and piping required for the above components which are operable	All

a. No credit taken in accident analyses.

NOTE:

1. THIS DIAGRAM IS A SIMPLICATION OF THE SYSTEM INTENDED TO FACILITATE THE UNDERSTANDING OF THE PROCESS. FOR DETAILS OF THE PIPING, VALVES INSTRUMENTATION, ETC. REFER TO THE ENGINEERING FLOW DIAGRAM. REFER TO PROCESS FLOW DIAGRAM TABLES FOR THE CONDITIONS AT EACH NUMBERED POINT.



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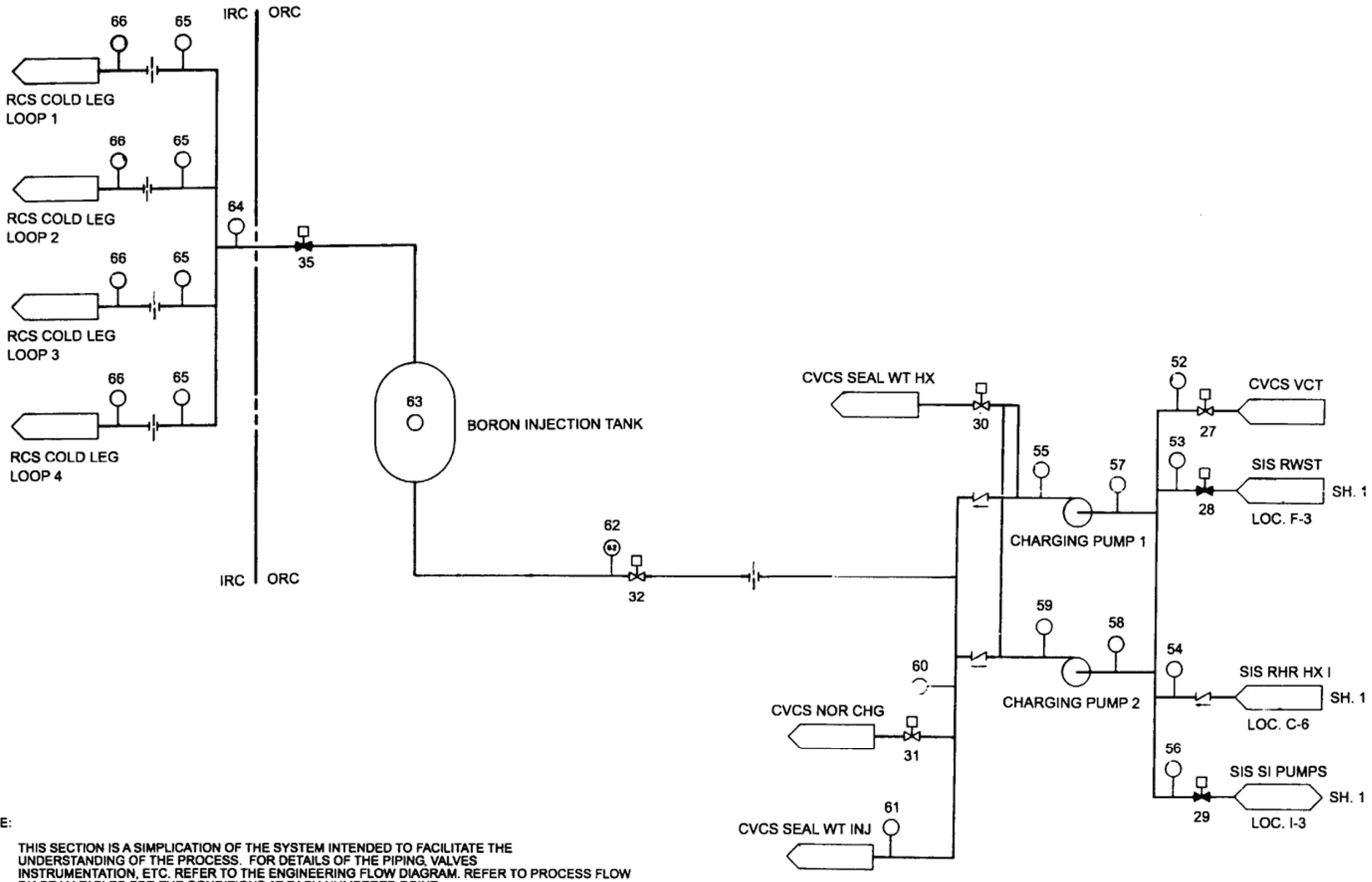
UNIT 1 AND UNIT 2

EMERGENCY CORE COOLING SYSTEM
PROCESS FLOW DIAGRAM

FIGURE 6.3.2–1 (SHEET 1 OF 22)



NOTE:

1. THIS SECTION IS A SIMPLICATION OF THE SYSTEM INTENDED TO FACILITATE THE UNDERSTANDING OF THE PROCESS. FOR DETAILS OF THE PIPING, VALVES INSTRUMENTATION, ETC. REFER TO THE ENGINEERING FLOW DIAGRAM. REFER TO PROCESS FLOW DIAGRAM TABLES FOR THE CONDITIONS AT EACH NUMBERED POINT.

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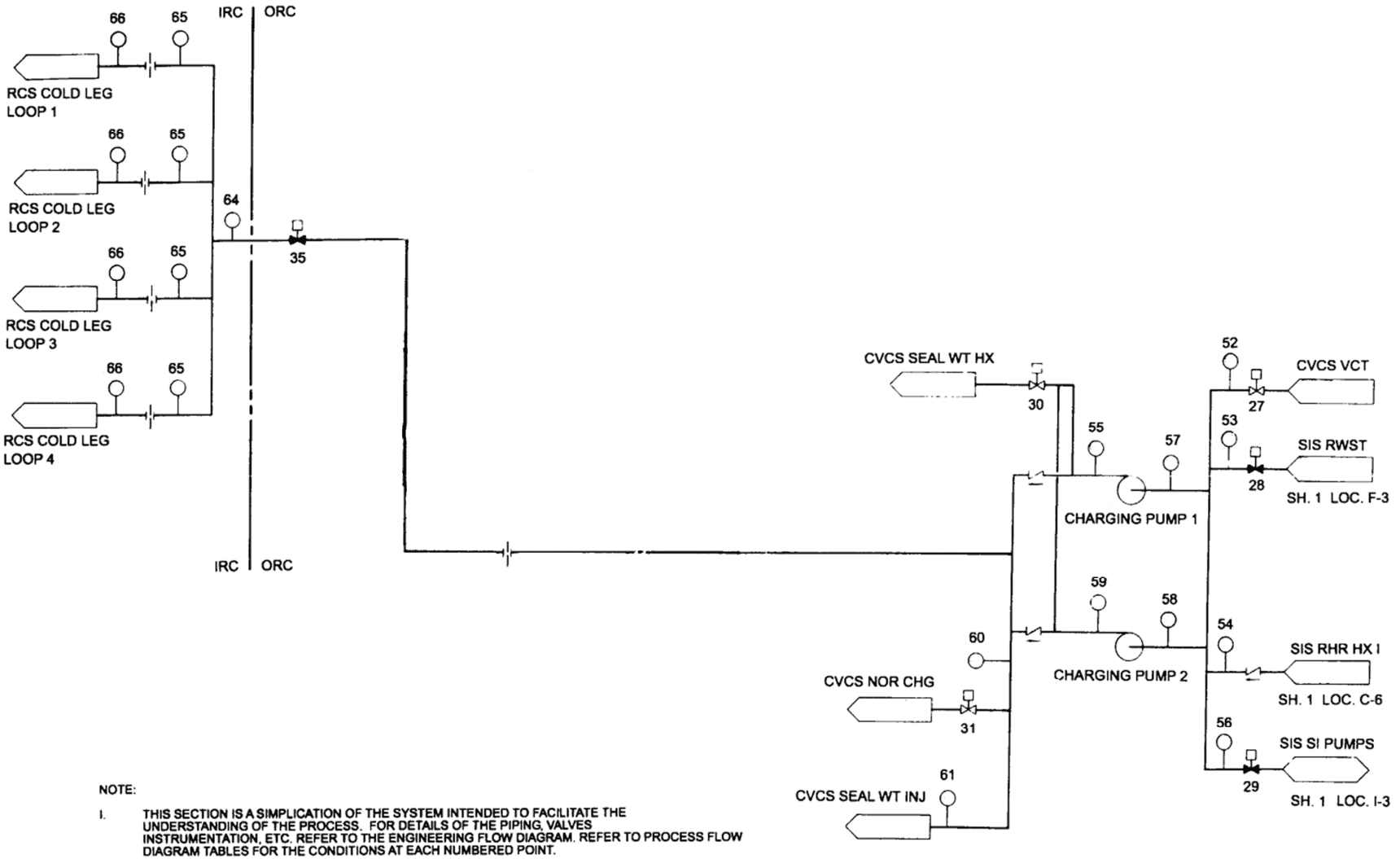
EMERGENCY CORE COOLING SYSTEM
PROCESS FLOW DIAGRAM (UNIT 1)

FIGURE 6.3.2–1 (SHEET 2 OF 22)



NOTE:

1. THIS SECTION IS A SIMPLICATION OF THE SYSTEM INTENDED TO FACILITATE THE UNDERSTANDING OF THE PROCESS. FOR DETAILS OF THE PIPING, VALVES INSTRUMENTATION, ETC. REFER TO THE ENGINEERING FLOW DIAGRAM. REFER TO PROCESS FLOW DIAGRAM TABLES FOR THE CONDITIONS AT EACH NUMBERED POINT.

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UNIT 1 AND UNIT 2

EMERGENCY CORE COOLING SYSTEM
PROCESS FLOW DIAGRAM (UNIT 2)

FIGURE 6.3.2–1 (SHEET 3 OF 22)

MODES OF OPERATION

MODE A - INJECTION

This mode presents the process conditions for the case of maximum safeguards, i.e., all pumps operating, following accumulator delivery. Two RHR pumps, two SI pumps, and two charging pumps operate, taking suction from the RWST and delivering to the reactor through the cold leg connections. Note that the flow from each pump is less than its maximum runout since the pump discharge piping is shared by the two pumps of each subsystem. Note also that the SI pump branch connections to the residual heat removal lines are close to their discharge into the accumulator lines, thereby minimizing any increase in the RHR branch line head loss due to the combined flows of the RHR and SI pumps.

MODE B - COLD LEG RECIRCULATION

This mode presents the process conditions for the case of cold leg recirculation assuming RHR pump train B, SI pumps A and B, and charging pumps A and B operating. It is assumed that the spray pumps have emptied the RWST at this time.

In this mode the safeguards pumps operate in series, with only the RHR pump capable of taking suction from the containment sump. The recirculated coolant is then delivered by the RHR pump to both of the SI pumps which deliver to the reactor through their cold leg connections and to both of the charging pumps which deliver to the reactor through their cold leg connections. The RHR pump also delivers flow directly to the reactor through two cold legs since the RHR discharge cross-connect valves are closed when making the transfer from injection to recirculation.

MODE C - HOT LEG RECIRCULATION

This mode presents the process conditions for the case of hot leg recirculation assuming train A RHR pump, charging pumps A and B, and SI pumps A and B operating.

In this mode, the safeguards pumps again operate in series with only the RHR pump taking suction from the containment emergency sump. The recirculated coolant is then delivered by the RHR pump to both of the charging pumps which continue to deliver to the reactor through their cold leg connections and to both of the SI pumps which also deliver to the reactor through their hot leg connections. The RHR pump also delivers directly to the reactor through two hot leg connections.

VALVE ALIGNMENT CHART

Valve	Injection (Mode A)	Cold Leg Recirculation (Mode B)	Hot Leg Recirculation (Mode C)
HV-8806	Open	Closed	Closed
HV-8812B	Open	Closed	Closed
HV-8812A	Open	Closed	Closed
HV-8814	Open	Closed	Closed
HV-8920	Open	Closed	Closed
HV-8821A	Open	Open	Closed
HV-8821B	Open	Open	Closed
HV-8802A	Closed	Closed	Open
HV-8802B	Closed	Closed	Open
HV-8702A	Closed	Closed	Closed
HV-8701A	Closed	Closed	Closed
HV-8811B	Closed	Open	Open
HV-8811A	Closed	Open	Open
FV-0611	Closed	Closed	Closed
FV-0610	Closed	Closed	Closed
FV-0619	Closed	Closed	Closed
FV-0618	Closed	Closed	Closed
HV-0607	Open	Open	Open
HV-0606	Open	Open	Open
HV-8804B	Closed	Open	Open
HV-8804A	Closed	Open	Open
HV-8716B	Open	Closed	Closed (a)
HV-8716A	Open	Closed	Open
HV-8809B	Open	Open	Closed
HV-8840	Closed	Closed	Open
HV-8809A	Open	Open	Closed
LV-112C	Closed	Closed	Closed
LV-112D/E	Open	Closed	Closed
HV-8924	Open	Open	Open
HV-8110	Closed	Closed	Closed
HV-8106	Closed	Closed	Closed
HV-8803A/B(Unit 1 only)	Open	Open	Open
HV-8801A/B	Open	Open	Open

a. Valve HV-8716B is normally open during the hot leg recirculation mode. However, for the "mode C" scenario, the B-train RHR is assumed to be faulted. Therefore, valve HV-8716B would be closed to separate the two trains of RHR.

MODE A - INJECTION PHASE
(RUNOUT CONDITIONS FOLLOWING ACCUMULATOR DELIVERY)

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)[a]	Volume (gal)
1	Refueling water	Atmospheric tank	100	-	715,000
2	Refueling water	-	100	15,600	
3	Refueling water	13 psia	100	14,660	-
4	Refueling water	-	100	7920	-
5	Refueling water	-	100	7000	-
6	Refueling water	11 psia	100	940	-
7	Refueling water	10 psia	100	6740[b]	
8	Refueling water	10 psia	100	920	-
9	Refueling water	10 psia	100	470	-
10	Refueling water	10 psia	100	470	-
11	Refueling water	1325	100	470	-
12	Refueling water	25	100	38	-
13	Refueling water	10 psia	100	470	-
14	Refueling water	1325	100	470	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
15	Refueling water	25	100	38	-
16	Refueling water	-	100	76	-
17	Refueling water	1250	100	864	-
18	Refueling water	105	100	216	-
19	Refueling water	-	100	1966	-
20	Refueling water	-	100	1966	-
21	Nitrogen	0	100	0	-
22	Nitrogen	0	100	0	850
23	Nitrogen	0	100	0	500
24	Reactor coolant	-	120	0	-
25	Refueling water	~ 0	100	3500	-
26	Refueling water	175	100	3500	
27	Refueling water	175	100	3500	-
28	Refueling water	150	100	3500	-
29	Refueling water	160	100	0	-
30	Refueling water	160	100	0	-
31	Refueling water	160	100	3500	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
32	Refueling water	160	100	3500	-
33	Refueling water	160	100	0	-
34	Reactor coolant	-	120	0	-
35	Refueling water	~ 0	100	3500	-
36	Refueling water	175	100	3500	-
37	Refueling water	175	100	3500	-
38	Refueling water	150	100	3500	-
39	Refueling water	160	100	0	-
40	Refueling water	160	120	0	-
41	Refueling water	160	100	3500	-
42	Refueling water	160	100	3500	-
43	Recirc-ulating coolant	Containment pressure	220	0	-
44	Recirc-ulating coolant	Containment pressure	120	0	-
45	Recirc-ulating coolant	Containment pressure	120	0	-
46	Refueling water	Low pressure	100	0	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
47	Refueling water	Low pressure	100	0	-
48	Refueling water	Low pressure	100	0	-
49	Refueling water	Low pressure	100	0	-
50	Refueling water	Low pressure	100	0	-
51	Refueling water	Low pressure	100	0	-
52	Refueling water	Low pressure	100	0	-
53	Refueling water	10 psia	100	920	-
54	Refueling water	-	100	0	-
55	Refueling water	1620	100	460	-
56	Refueling water	-	100	0	-
57	Refueling water	10 psia	100	460	-
58	Refueling water	10 psia	100	460	-
59	Refueling water	1620	100	460	-
60	Refueling water	1610	100	104	-
61	Refueling water	1610	100	104	-
62	Refueling water	1500	100	816	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
63(Unit 1 only)	Refueling water	-	100	816	900
64	Refueling water	1400	100	816	-
65	Refueling water	1170	100	204	-
66	Refueling water	40	100	204	-

a. At reference conditions 100°F and 0 psig.

b. Estimated spray pump flow.

MODE B - COLD LEG RECIRCULATION
(RHR PUMP 2 Operating)

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)[a]	Volume (gal)
1	Refueling water	Atmospheric tank	100	-	-
2	Refueling water	-	100	0	-
3	Refueling water	-	100	0	-
4	Refueling water	-	100	0	-
5	Refueling water	-	100	0	-
6	Recirc-ulating coolant	-	182	0	-
7	Refueling water	-	100	0	-
8	Refueling water	-	100	0	-
9	Recirc-ulating coolant	~ 35	182	1350	-
10	Recirc-ulating coolant	~ 35	185	430	-
11	Recirc-ulating coolant	1325	182	430	-
12	Refueling water	-	100	0	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
13	Recirc-ulating coolant	~ 35	182	430	-
14	Recirc-ulating coolant	1325	182	430	-
15	Refueling water	-	100	0	-
16	Refueling water	-	100	0	-
17	Recirc-ulating coolant	1250	182	860	-
18	Recirc-ulating coolant	105	182	215	-
19	Recirc-ulating coolant	-	182	1975	-
20	Recirc-ulating coolant	-	182	215	-
21	Nitrogen	0	Ambient	0	-
22	Nitrogen	0	Ambient	0	850(b)
23	Nitrogen	0	Ambient	0	500
24	Recirc-ulating coolant	-	212	0	-
25	Recirc-ulating coolant	10	212	4300	-
26	Recirc-ulating coolant	170	212	4300	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
27	Recirc-ulating coolant	-	212	4300	-
28	Recircu-ulating coolant	130	182	2520	-
29	Recirc-ulating coolant	140	182	0	-
30	Recircu-ulating coolant	150	182	1780	-
31	Recirc-ulating coolant	160	182	4300	-
32	Recirc-ulating coolant	140	182	2520	-
33	Recirc-ulating coolant	0	182	0	-
34	Recirc-ulating coolant	-	212	0	-
35	Refueling water	-	100	0	-
36	Refueling water	-	100	0	-
37	Refueling water	-	100	0	-
38	Refueling water	-	100	0	-
39	Refueling water	-	100	0	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
40	Refueling water	-	100	0	-
41	Refueling water	-	100	0	-
42	Refueling water	-	100	0	-
43	Recirc-ulating coolant	Containment pressure	212	0	~540,000
44	Recirc-ulating coolant	Containment pressure	212	4300	-
45	Recirc-ulating coolant	Containment pressure	212	0	-
46	Refueling water	Low pressure	100	0	-
47	Refueling water	Low pressure	100	0	-
48	Refueling water	Low pressure	100	0	-
49	Refueling water	Low pressure	100	0	-
50	Refueling water	Low pressure	100	0	-
51	Refueling water	Low pressure	100	0	-
52	Recirc-ulating coolant	-	182	0	-
53	Recirc-ulating coolant	-	182	0	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
54	Recirc-ulating coolant	-	182	0	-
55	Recirc-ulating coolant	1620	182	460	-
56	Recirc-ulating coolant	30	182	920	-
57	Recirc-ulating coolant	30	182	460	-
58	Recirc-ulating coolant	30	182	460	-
59	Recir-ulating coolant	1620	182	460	-
60	Recirc-ulating coolant	1610	182	104	-
61	Recirc-ulating coolant	1610	182	104	-
62	Recirc-ulating coolant	1500	182	816	-
63	Recirc-ulating coolant (Unit 1 only)	-	182	816	-
64	Recirc-ulating coolant	1400	182	816	-
65	Recirc-ulating coolant	1170	182	204	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
66	Recirc-ulating coolant	40	182	204	-

a. At reference conditions 212°F and 0 psig.

b. Minimum water volume at operating conditions.

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EMERGENCY CORE COOLING SYSTEM
PROCESS FLOW DIAGRAM

FIGURE 6.3.2–1 (SHEET 16 OF 22)

MODE C - HOT LEG RECIRCULATION
(RHR PUMP NO. 1 OPERATING)

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)[a]	Volume (gal)
1	Refueling water	Atmospheric tank	100	-	-
2	Refueling water	-	100	0	-
3	Refueling water	-	100	0	-
4	Refueling water	-	100	0	-
5	Refueling water	-	100	0	-
6	Recirc-ulating coolant	-	182	0	-
7	Refueling water	-	100	0	-
8	Refueling water	-	100	0	-
9	Recirc-ulating coolant	25	182	660	-
10	Recirc-ulating coolant	25	182	660	-
11	Recirc-ulating water	805	182	660	-
12	Refueling water	-	100	0	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
13	Recirc-ulating coolant	25	182	660	-
14	Recirc-ulating coolant	805	182	660	-
15	Refueling water	-	100	0	-
16	Refueling water	-	100	0	-
17	Recirc-ulating coolant	0	182	0	-
18	Recirc-ulating coolant	-	182	0	-
19	Recirc-ulating coolant	-	182	0	-
20	Recirc-ulating coolant	-	182	0	-
21	Nitrogen	-	Ambient	0	-
22	Nitrogen	0	Ambient	0	850(b)
23	Nitrogen	0	Ambient	0	500
24	Recirc-ulating coolant	-	212	0	-
25	Recirc-ulating coolant	-	< 212	0	-
26	Recirc-ulating coolant	-	< 212	0	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
27	Recirc-ulating coolant	-	< 212	0	-
28	Recirc-ulating coolant	-	< 182	0	-
29	Recirc-ulating coolant	-	< 182	0	-
30	Recirc-ulating coolant	-	< 182	0	-
31	Recirc-ulating coolant	-	< 182	0	-
32	Recirc-ulating coolant	-	< 182	0	-
33	Recirc-ulating coolant	50	182	2060	-
34	Recirc-ulating coolant	-	212	0	-
35	Recirc-ulating coolant	12	212	4300	-
36	Recirc-ulating coolant	140	212	4300	-
37	Recirc-ulating coolant	-	212	4300	-
38	Recirc-ulating coolant	-	< 182	0	-

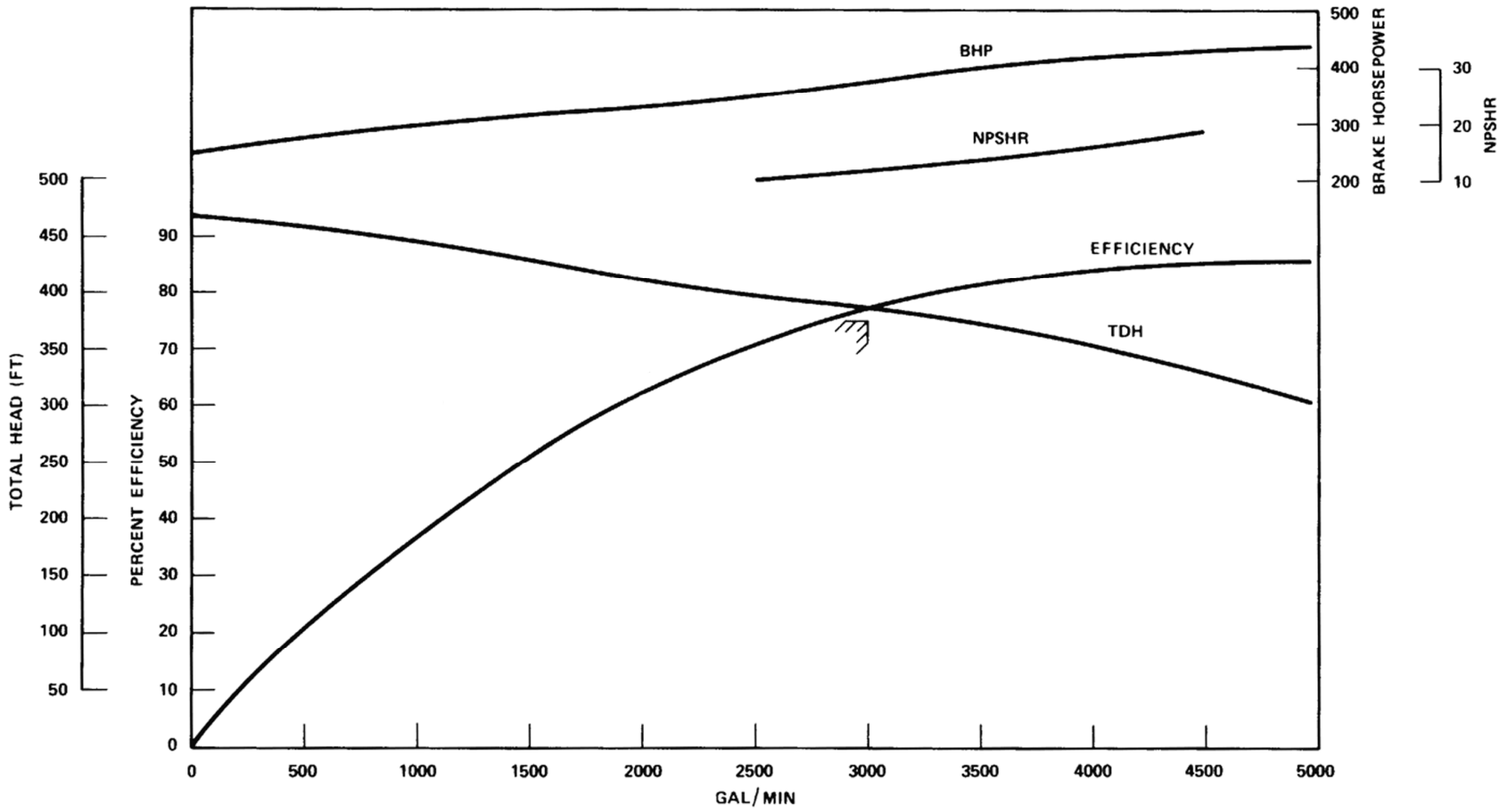
Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
39	Recirc-ulating coolant	120	182	0	-
40	Recirc-ulating coolant	130	182	2240	-
41	Recirc-ulating coolant	135	182	4300	-
42	Recirc-ulating coolant	120	182	2060	-
43	Recirc-ulating coolant	Containment pressure	212	-	~540,000
44	Recirc-ulating coolant	Containment pressure	212	0	-
45	Recirc-ulating coolant	Containment pressure	212	4300	-
46	Recirc-ulating coolant	10	182	2060	-
47	Recirc-ulating coolant	10	182	1030	-
48	Recirc-ulating coolant	700	182	660	-
49	Recirc-ulating coolant	-	182	1360	-
50	Recirc-ulating coolant	700	182	660	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
51	Recirc-ulating coolant	-	182	330	-
52	Recirc-ulating coolant	-	182	0	-
53	Recirc-ulating coolant	-	182	0	-
54	Recirc-ulating coolant	-	182	2240	-
55	Recirc-ulating coolant	1620	182	460	-
56	Recirc-ulating coolant	35	182	1320	-
57	Recirc-ulating coolant	35	182	460	-
58	Recirc-ulating coolant	35	182	460	-
59	Recirc-ulating coolant	1620	182	460	-
60	Recirc-ulating coolant	1620	182	104	-
61	Recirc-ulating coolant	1620	182	104	-
62	Recirc-ulating coolant	1500	182	816	-

Location	Fluid	Pressure (psig)	Temperature (°F)	Flow (gal/min)	Volume (gal)
63(Unit 1 only)	Recirc-ulating coolant	-	182	816	-
64	Recirc-ulating coolant	1400	182	816	-
65	Recirc-ulating coolant	1170	182	204	-
66	Recirc-ulating coolant	40	182	204	-

a. At reference conditions 212°F and 0 psig.

b. Minimum water volume at operating conditions.



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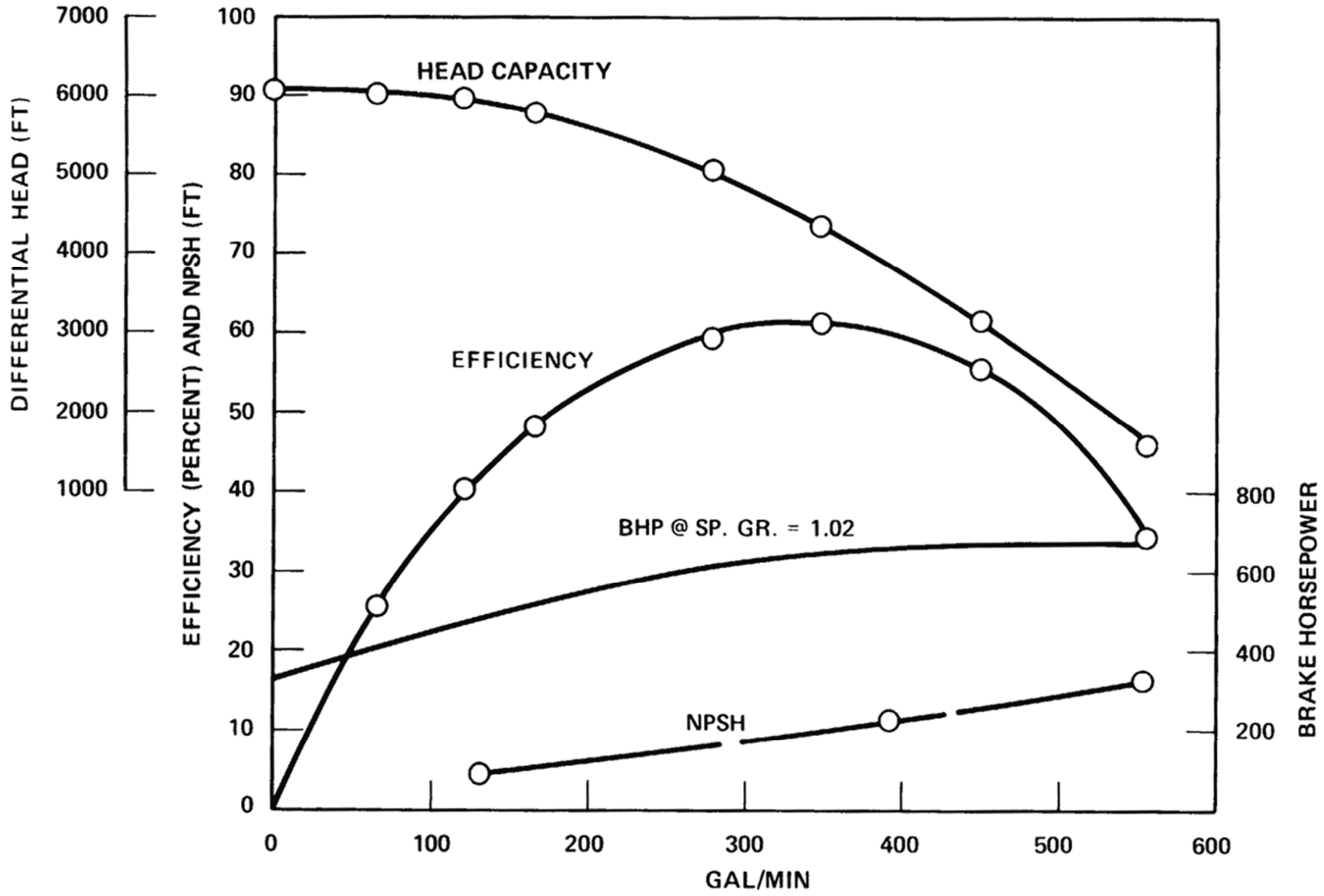
PERFORMANCE CURVES
RESIDUAL HEAT REMOVAL PUMPS

FIGURE 6.3.2–2



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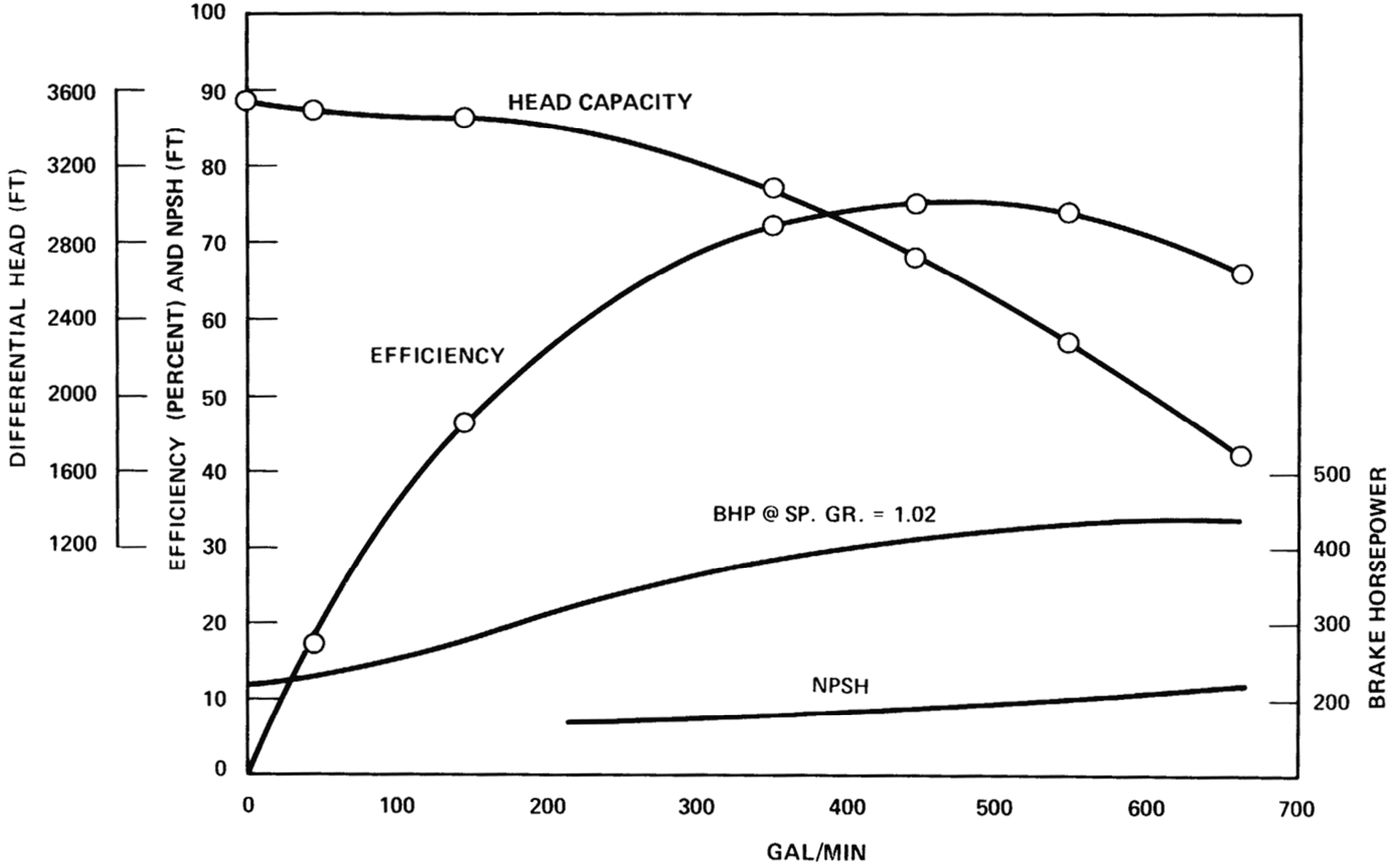
PERFORMANCE CURVES
CENTRIFUGAL CHARGING PUMPS

FIGURE 6.3.2–3



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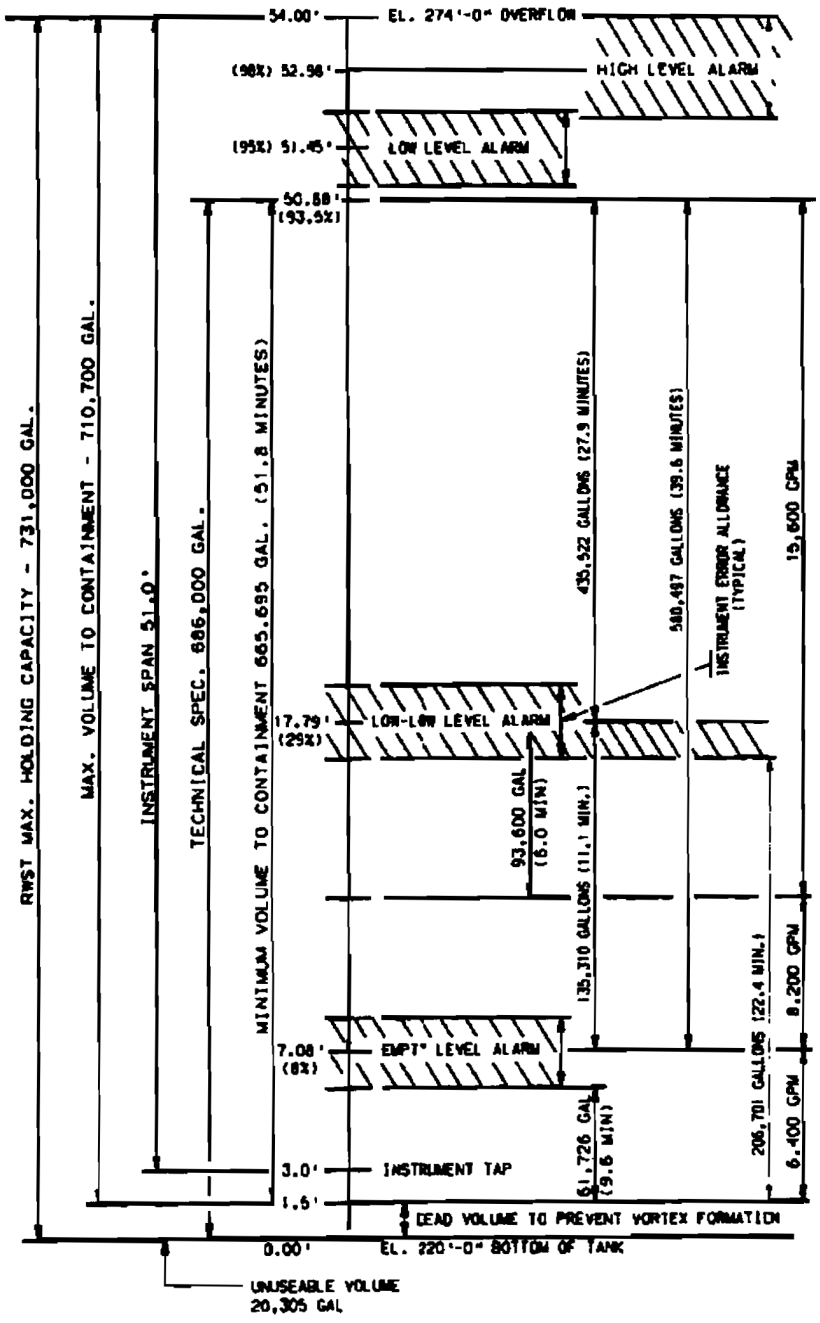
PERFORMANCE CURVES
SAFETY INJECTION PUMPS

FIGURE 6.3.2–4



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UNIT 1 AND UNIT 2

RWST SIZING

FIGURE 6.3.2–5

6.4 HABITABILITY SYSTEMS

The control room habitability systems include missile protection; radiation shielding; radiation monitoring; and smoke detection capability; air filtration, adsorption, and pressurization; and air-conditioning, lighting, personnel support, and fire protection equipment. (Refer also to section 3.1 for a discussion on conformance with 10 CFR 50, Appendix A, General Design Criterion 19.)

The heating, ventilation, and air-conditioning (HVAC) equipment discussed in this section is also discussed in subsection 9.4.1 which is directed toward normal use of the equipment. This section only addresses emergency service requirements and the response and operation of control room HVAC equipment under emergency conditions. Other equipment and systems are described only as necessary to define their connection with control room habitability. Reference is made to other sections as appropriate.

6.4.1 DESIGN BASES

The safety design bases for the control room habitability systems are as follows.

The habitability systems provide coverage for the control room envelope defined in paragraph 6.4.2.1.

The control room emergency ventilation and air-conditioning system is capable of maintaining the control room atmosphere in a condition suitable for prolonged occupancy throughout the duration of any one of the postulated accidents discussed in chapter 15.

The control room emergency ventilation and air-conditioning system is capable of maintaining an environment suitable for sustained occupancy for a five-person minimum, with higher occupancy levels for shorter periods of time.

Food, water, medical supplies, and sanitary facilities are provided for a minimum sustained control room occupancy of five persons for 5 days. The control room will have approximately five hundred 130-mg potassium iodide tablets.

The radiation exposure of control room personnel through the duration of any one of the postulated limiting faults discussed in chapter 15 does not exceed the limits set by 10 CFR 50, Appendix A, General Design Criterion 19.

The habitability systems provide the capability to detect and protect control room personnel from smoke, and airborne radioactivity.

Respiratory, eye, and skin protection is provided for emergency use within the control room envelope.

The control room emergency HVAC system is capable of automatic and manual transfer from its normal operating mode to the emergency modes. Smoke and radiation detectors, and control equipment are provided at plant locations as necessary to ensure the appropriate operation of the system.

A single active failure of any component of the control room emergency HVAC system, assuming a loss of offsite power, does not impair the ability of the system to function. Each train of the control room HVAC system is connected to a separate and independent Class 1E power supply.

The control room emergency HVAC system is designed to remain functional during and after a safe shutdown earthquake. All airducts and their supports above the control room suspended ceiling, as well as the ceiling itself, are Seismic Category 1.

The control room normal HVAC system is described in subsection 9.4.1.

Protection of the habitability systems in the control room from wind and tornado effects is discussed in section 3.3. Flood design is discussed in section 3.4. Missile protection is discussed in section 3.5. Protection against dynamic effects associated with the postulated rupture of piping is discussed in section 3.6. Environmental design is discussed in section 3.11. The fire protection system is discussed in subsection 9.5.1. The fire hazard analysis is discussed in appendix 9A. The control room ventilation isolation is described in subsection 7.3.6. The design of the control room habitability system meets the intent of Regulatory Guides 1.52, 1.78, and 1.95 as discussed in section 1.9.

6.4.2 SYSTEM DESIGN

6.4.2.1 Definition of the Control Room Envelope

The areas, equipment, and materials to which the control room operator requires access during an emergency are shown in drawing AX1D11A04. Those spaces requiring continuous or frequent operator occupancy are also shown in drawing AX1D11A04. A description of shielding required to maintain habitability of the control room during the course of postulated accidents is provided in section 12.3.

6.4.2.2 Ventilation System Design

6.4.2.2.1 General Description

Subsection 9.4.1 contains an overall description of the control room heating, ventilation, and air-conditioning (HVAC) system. The system is shown schematically in drawings AX4DB206-1, AX4DB206-2 and AX4DB206-3. Drawing AX6DD310 shows the plant layout, including the location of potential radiological and onsite toxic chemical release points with respect to the control room air intakes. Elevation and plan drawings with descriptions providing building dimensions and locations are in section 1.2. Potential sources of toxic gas and radiological releases are discussed in subsection 2.2.3.

The volume of the habitability zone served by the HVAC system in the emergency mode or the isolation mode is approximately 161,500 ft^3.

Environmental design criteria for the air purification system are based on the most limiting conditions resulting from any of the postulated design basis accidents (DBAs) and on their duration, in accordance with Regulatory Guide 1.52, as discussed in section 1.9.

Two redundant and physically separated air handling unit trains with a moisture eliminator, an electric preheater, high-efficiency particulate air (HEPA) filters, and charcoal adsorbers are provided for each unit to process intake airflow and recirculated airflow in the combined control room. Each of the two redundant units belongs to a different safety train. Performance characteristics and parameters for major components are listed in table 6.4.2-1. The seismic

and quality classifications of components, instrumentation, and ducting are given in table 3.2.2-1.

6.4.2.2.2 Component Description

Each essential air handling unit contains a fan, a moisture eliminator, an electric heating coil, an upstream HEPA filter, an activated charcoal filter, a downstream HEPA filter, and a cooling coil. Pneumatic and/or motor-operated dampers are provided for system isolation purposes.

The control room emergency HVAC system is shared by both Units 1 and 2. The two units' control rooms are partially separated by a partition. Areas in the vicinity of the shift technical advisor work station, and the balance of plant (BOP) and nuclear steam supply system (NSSS) control panels provide access between the Unit 1 and Unit 2 areas of the control room as shown in drawing AX1D11A04. The air ducts serving the control room are a common system connected to the Units 1 and 2 safety-related air handling units. Four safety-related air handling units are available to serve the control room envelope.

A. Filter Unit Housings

The filter unit housings are Seismic Category 1 and are made of carbon steel. Each housing is provided with a service access door, explosion-proof light, filter test connections, connections for pressure gauges, and floor drains. The housings are of all-welded construction. Filter unit housings are designed in accordance with Regulatory Guide 1.52, as discussed in section 1.9.

B. Moisture Separator

The moisture separator is a two-stage unit using louvers followed by relatively coarse glass fiber or galvanized steel pads. The moisture separator will remove 99 percent of 5 to 10 μm diameter droplets without any visible carryover. The moisture separator is designed and qualified in accordance with Regulatory Guide 1.52, as discussed in section 1.9.

C. Electric Heaters

The electric heater is sized to reduce the relative humidity of the airstream to 70 percent from as high as 100 percent. Heating elements are a finned tubular type, with 80-percent nickel/20-percent chromium resistance wire embedded in insulation inside a monel sheath. Fins are of monel and are permanently attached to the tubes. The elements are supported in a type 304 stainless steel casing. The electric heaters are designed in accordance with American National Standards Institute N509, as discussed in paragraph 1.9.52.2.

D. HEPA Filters

HEPA filter elements are of pleated fiberglass with aluminum separator design, measure 24 x 24 x 11.5 in., and are capable of handling a nominal flowrate of 1000 ft^3/min each. The filter medium is cased in stainless steel, has face guards on both sides, and is water and fire resistant. HEPA filter elements are manufactured and tested prior to installation in accordance with MIL-F-51068D. The filter element minimum acceptance criterion is removal of 99.97 percent of 0.3-mm thermal-generated, monodispersed dioctyl phthalate particles.

E. Carbon Adsorbers

The carbon adsorbers for the emergency air handling units are of the bulk type, are 4 in. deep, and have an all-welded design. The carbon adsorbers are a rechargeable type. Minimum air residence time in the carbon is 0.5 s at a nominal face velocity of 40 ft/min. An 8 x 16 mesh of impregnated, activated charcoal is used in each filter.

F. Cooling Coil

The cooling coils are of nonferrous construction with copper fins mechanically bonded to seamless 90-percent copper/10-percent nickel tubing. Coils are arranged for counterflow operation using chilled water. The tube bundle is enclosed in a stainless steel frame. Coils are arranged for horizontal airflow and are provided with inlet and outlet piping, vent, and drain connections. The chilled water system is discussed in subsection 9.2.9. The cooling coil is Seismic Category 1 and American Society of Mechanical Engineers Section III, Class 3.

G. Emergency Filtration Train Fans

The emergency filtration train fans are Seismic Category 1 and are capable of delivering 25,000-ft^3/min flowrate with all filters at their design pressure drop. Fans are chosen with a steeply rising pressure-flow characteristic to maintain a reasonable constant airflow over the full filter train life. Fan and motor materials are suitable for operation under the environmental conditions associated with the postulated DBA, in conformance with Regulatory Guide 1.52, as discussed in section 1.9. The emergency ventilation recirculation rate for each train is balanced for a flow of 19,000 ft^3/min.

H. Control Room Return Fan

The emergency control room return fans are disabled and abandoned in place as their function is not required.

I. Ductwork and Dampers

The system ductwork and dampers are Seismic Category 1 and are designed in accordance with Regulatory Guide 1.52. Ductwork is redundant where required to provide functional support to active components in meeting the single active failure criteria. Leaktight ductwork and bubbletight isolation dampers are provided, where required, to isolate the system from unfiltered outside air.

In general conformance with Position C4 of Regulatory Guide 1.52 as discussed in section 1.9, accessibility and adequate working space for maintenance and testing operations are provided in the design and layout of the air purification system equipment.

J. Control Room Access Doors

To minimize inleakage, the control room access doors are equipped with self-closing devices that shut the doors automatically following the passage of personnel. Two sets of doors with a vestibule between, acting as an airlock, are provided at each of the entrances to the combined control room and associated spaces.

A separate doorway is provided to access the record storage area which, while not part of the control room proper, is within the control room HVAC envelope. The door is equipped with a self-closing device to shut the door automatically

following the passage of personnel and can only be accessed by passing through an air lock (two-door vestibule) or another door in series.

K. Isolation Dampers

System isolation dampers are capable of automatically closing in 6 s after receipt of an actuation signal, as verified by manufacturer testing. The isolation dampers are factory tested as bubbletight dampers for zero leakage.

L. Radiation Detectors

Redundant radiation detectors are installed in the control room ventilation outside air intake plenum. Each unit is responsive to gaseous activity at concentrations as low as 10-6 $\mu Ci/cm^3$ of Xe-133. Airborne particulate and iodine activities are also detected. The detectors are described in section 11.5.

M. Smoke Detectors

Redundant smoke detectors are installed in each control room ventilation outside air intake (a total of four detectors). These detectors indicate the presence of smoke entering the control room envelope from outside. Each smoke detector actuates an alarm in the control room on the HVAC control panel. Redundant smoke detectors are also installed inside the control room envelope. These smoke detectors detect smoke inside the control room envelope and actuate an alarm in the control room on the fire protection panel.

N. Breathing Apparatus

Self-contained portable breathing equipment with air bottles is stored within the habitability area of the control room. The quantity available is sufficient to allow manning of five people for 6 h each for each individual (30 h).

The remainder of the system; i.e., supply/recirculation fans, exhaust fans, ductwork, and dampers, are components that function during normal operation and are described in subsection 9.4.1.

O. Backdraft Dampers

Backdraft dampers in the outside air and return ductwork are provided to prevent excessive backflow in faulted conditions. See Failure Modes and Effects Analysis, table 6.4.4-1, for further description.

6.4.2.3 Leaktightness (HISTORICAL)

The exfiltration and infiltration analyses are performed using the methods and assumptions given in American Society of Heating, Refrigerating, and Air-Conditioning Engineers Handbook of Fundamentals and Regulatory Guide 1.78 and "Conventional Buildings for Reactor Containment," published by Atomics International, Catalog No. NAA-SR-10100, dated June 15, 1965. The leakage rates were calculated using the following equations:

A. Penetrations and Doors

$Q = AP + BP^{1/2}$

where:

Q = leakage rate per unit leak path (ft^3/min).

P = differential pressure (in. WG).

A and B = coefficients from test data.

B. Dampers

Leaktightness is determined from actual test data on dampers.

The leak paths considered are ductwork, piping, and electrical penetrations; dampers and doors; and construction joints and materials.

Table 6.4.2-2 provides a listing of leakage data and total leakage rates for potential leak paths. For analysis of exfiltration from the pressurized control room envelope, a positive 1/8-in. WG pressure differential is considered for all leak paths resulting in a total outleakage of 1500 ft^3/min (at emergency conditions). For analysis of infiltration to the unpressurized control room envelope, a negative 1/8-in. WG pressure differential is considered for all leak paths resulting in a total inleakage of 750 ft^3/min. The control room envelope is pressurized during normal operation. The normal outside air supply is designed to pressurize the control room to 1/8 in. WG and is sized to deliver up to 3000-ft^3/min flowrate into the control room during the normal mode of operation. Based on the rate of outleakage, this flowrate is adequate to maintain a 1/8-in. positive pressure in the control room envelope during normal operation.

6.4.2.4 Interaction with Other Zones and Pressurized Equipment

The outside air intake duct is located such that:

A. It is protected from the effects of a main steam line break.

B. It minimizes the introduction of airborne radioactive material from unit release points.

C. It minimizes the introduction of diesel generator exhaust and other noxious gases.

The probability of radioactive material, noxious gases, or steam being transferred directly into the control room from adjacent areas and buildings other than through the outside air duct is minimized by the following design arrangements and considerations.

A. The control room is maintained at 1/8-in. WG pressure above atmospheric to prevent infiltration of air during normal conditions. The volume of the control room and other space protected by the habitability system is approximately 161,500 ft^3. The outside air supply of 3000 ft^3/min for the normal mode ensures pressurization of the area in excess of 1/8 in. WG so that all flow of air through the potential leakage paths, doors, ductwork, filtration units, and cable penetrations is outward and not inward. The outside air intake is through the plenum system at el 281 ft 0 in. The inlet to the plenum is through openings at the upper part of the building above the roof. The plenum system is designed as a Seismic Category 1 structure, which is an integral part of the building structure.

   The two air intakes are located at the southeast and southwest corners of the control building. There is no direct horizontal path from any sources of radioactivity, noxious gases, or steam to the air intake.

B. The normal releases from the auxiliary, fuel handling, and containment buildings are exhausted through an elevated stack atop the containments. This precludes any direct transfer of contaminants to the control room intake.

C. The control room consists of two air spaces separated partially by a suspended ceiling. The upper air space contains cable penetrations (sealed) from the upper

cable spreading room above, Seismic Category 1 duct hangers, Seismic Category 1 tray hangers, Seismic Category 1 ceiling hangers, recessed light fixture enclosures (with power connections), and the Seismic Category 1 HVAC air ducts. There is no leakage path from any of these attachments nor penetrations in the 8-in. floor slab to the cable spreading room above the control room. The suspended ceiling is not sealed from the lower air space containing the control room equipment.

D. The floor of the control room contains sealed cable penetrations from the cable spreading area below the control room. There is, therefore, no leakage path from the lower cable spreading room through the control room floor into the control room.

E. There are three doorways into the combined control room:

1. At the northwest corner of the control room.
2. In the south wall of the control room.
3. At the east wall of the control room.

The doorways to the control room are each arranged with two sets of doors acting as an airlock. The doors are provided with seals to reduce leakage and to maintain pressurization.

A separate doorway is provided to access the record storage area which, while not part of the control room proper, is within the control room HVAC envelope. The door is equipped with a self-closing device to shut the door automatically following the passage of personnel and can only be accessed by passing through an air lock (two-door vestibule) or another door in series.

F. The ductwork for the essential HVAC system for the control room under accident conditions is separated from connections to other areas or to the normal operating HVAC air handling units by two Seismic Category 1, bubbletight dampers independently actuated and powered by the two engineered safety features trains. Each isolation damper automatically closes when an emergency air handling unit is started in the corresponding safety train. The emergency air handling units start automatically on either of the following signals: safety injection, or high radiation levels in the outside air intake. Under emergency conditions, filters are used for the makeup air supply to the essential HVAC system, in conformance with Regulatory Guide 1.52, as discussed in section 1.9.

G. When the control room is isolated but not pressurized, the air leakage into the control room is no greater than 750 ft^3/min from all pathways, based on a 1/8-in. WG differential. This amounts to approximately 0.3-h^{-1} air change. The infiltration is distributed as shown in table 6.4.2-2. (HISTORICAL)

H. Halon 1301 bottles provide fire protection capability for the shutdown panels, records storage, and plant and technical support center computer areas.

6.4.2.5 Shielding Design

The design basis loss-of-coolant accident (LOCA) dictates the shielding requirements for the control room. Control room shielding design bases are discussed in section 12.3. Descriptions of the design basis LOCA source terms, control room shielding parameters, and evaluation of DBA doses to control room personnel are presented in section 15.6.

The control room and its location in the plant, identifying distances and shield thicknesses with respect to radiation sources discussed in section 15.6, are shown on drawing AX6DD404.

6.4.2.6 Reference

1. "Testing of Nuclear Air-Cleaning Systems," ANSI/ASME N510-1980.

6.4.3 SYSTEM OPERATIONAL PROCEDURES

The control room normal and emergency airflow schematic is shown in drawings AX4DB206-1, AX4DB206-2, AX4DB206-3, AX4DB255-1, AX4DB255-3, and AX4DB256-1.

6.4.3.1 Normal Mode

Control room heating, ventilation, and air-conditioning (HVAC) system operation in the normal mode is described in subsection 9.4.1.

6.4.3.2 Emergency Mode

A safety injection signal or the detection of high radiation levels in the control room outside air intake shall cause the initiation of the control room isolation (CRI) signal. The CRI signal causes the activation of the emergency air filtration systems for the control room and the onsite technical support center. The control room system logic permits only a single train to start through lead/lag logic. In addition, the CRI closes the isolation dampers between the normal and emergency systems. Lead/lag control logic permits only one train of fans per unit to operate during the emergency mode. The logic provides automatic start of a standby (lag) fan if the lead fan fails, such that no more than two fans total (for two units) are running. The control room normal air handling units will automatically trip as the isolation dampers close. If one train from each unit's control room HVAC is simultaneously activated, the control room operator may stop one train manually by first resetting the CRI signal for the selected train and then stopping the selected train's HVAC system. During this mode of operation, conference room, kitchen, and toilet exhaust ducts are also isolated through automatic closure of the isolation dampers on the receipt of the CRI signal. Subsection 7.3.6 provides a discussion of the actuation of the CRI signal and the operation of the emergency HVAC system.

6.4.3.3 Smoke Removal Mode

This operation mode is provided to remove smoke from the control room envelope by exhausting smoke-contaminated air to the atmosphere while introducing 100-percent outside air as dilutant makeup.

When there is smoke inside the control room, interior smoke detectors are actuated and sound an alarm in the control room. The operator analyzes the situation, and then manually opens or closes the appropriate dampers to start the smoke removal mode of operation.

In this mode of operation, 100-percent outside air from the intake plenum at el 281 ft 0 in. is supplied by a normal air handling unit which purges the control room. The control room return

dampers are isolated, the purge exhaust damper is opened, and the return/exhaust fan exhausts the air by discharging it to the outside at el 302 ft 0 in.

When there is smoke outside the control room, the smoke detectors in the outside air intake plenum actuate the annunciator alarms in the control room. The operator then analyzes the situation on the HVAC panel and, if necessary, actuates the isolation mode by manually generating a CRI signal and closing the outside air intake dampers. In the isolation mode of operation the emergency air filtration system runs in recirculation without outside air. The air from the control room is continually recirculated, cooled, and filtered by the emergency filtration units.

6.4.4 DESIGN EVALUATIONS

6.4.4.1 Radiological Protection

The effects of potential radiological accidents are analyzed in chapter 15. The radiological protection afforded to the operators in the event of an accident is described in subsections 6.4.2, 12.3.2, 12.3.3, and 12.3.4 and in section 11.5.

6.4.4.2 Toxic Gas Protection

Control room protection from the effects of toxic gases is in accordance with Regulatory Guide 1.78 as discussed in subsection 2.2.3. The analysis of potential sources for toxic gases is presented in subsection 2.2.3. The analysis of onsite and offsite sources for toxic gases is presented in paragraphs 2.2.3.1.4.1 through 2.2.3.1.4.3. These sources are analyzed deterministically and it is shown that either the 8-h toxicity limit is not exceeded in the control room, or that there is at least 2-min between detection and reaching the short-term toxicity limit, such that the operators have time to put on breathing apparatus. These results are shown in table 2.2.3-20.

6.4.4.3 Implementation of Design Bases

Control room habitability system components discussed in paragraph 6.4.2.2.2 are arranged in redundant safety-related ventilation trains as shown in drawings AX4DB206-1, AX4DB206-2, AX4DB206-3, AX4DB256-1, 1X4DB257-1, AX4DB266-2, 2X4DB266-1, AX4DB376, AX4DB269, and AX4DB270. The location of components and ductwork within the control room envelope ensures an adequate supply of filtered air to all areas requiring access as shown in drawing AX1D11A04.

By using chilled water cooling coils, the control room essential air-conditioning system maintains the temperature between 70°F and 85°F and the relative humidity below 50 percent. The control room pressure is maintained at least 1/8 in. WG above atmospheric pressure during normal operation. The control room emergency air-conditioning system maintains the same temperature and humidity conditions when operating in the emergency and isolation modes.

The control room air-conditioning system is capable of removing sensible and latent heat loads including consideration of equipment heat loads and minimum personnel occupancy requirements. The transfer to emergency or isolation operation mode does not create a hazard for CO_2 buildup. In case of emergency operation, there is a supply of outside air of 1500 ft^3/min

and the long term equilibrium for CO_2 will remain below one part per thousand for a five-person occupancy. In case of isolation mode operation, where the control room is sealed, the critical level of 3 percent would be reached in 5 days for an occupancy of five persons. The technical support center will provide an additional habitable location to relieve crowding in the control room as discussed in paragraphs 9.4.1.8 and 9.5.10.2.

Food, water, medical supplies, and sanitary facilities are provided for a minimum occupancy of five persons for 5 days. Storage locations provided ensure that the above supplies will not be contaminated as a result of postulated accidents. The supply of food and water is sufficient for a prolonged occupancy because outside supplies can be provided within the 5-day interval.

The control room air purification system and shielding designs are based on the most limiting design basis assumptions contained in Regulatory Guide 1.4. Automatic transfer of the control room from the normal heating, ventilation, and air-conditioning (HVAC) system to the emergency system is accomplished upon receipt of a control room isolation signal which is generated on receipt of either the high-radiation signal from the outside air intake duct radiation detector or the safety injection actuation signal. Transfer to the emergency system also may be manually initiated from the control room. Refer to subsection 7.3.6 for a discussion of the actuation logic.

The airborne fission product source term in the reactor containment following the postulated loss-of-coolant accident (LOCA) is assumed to leak from the containment at a rate of 0.2 percent per day for the first 24 h after the accident and 0.1 percent per day thereafter. The concentration of radioactivity, which is postulated to surround the control room after the postulated accident, is evaluated as a function of the fission product decay constants, the containment spray system effectiveness, the containment leak rate, and the meteorological conditions assumed to occur. The assessment of the amount of radioactivity within the control room takes into consideration the flowrate through the control room outside air intake, the effectiveness of the control room air purification system, the radiological decay of fission products, and the exfiltration rate from the control room.

Air within the control room is recirculated continuously through the emergency air-conditioning units, which contain upstream high-efficiency particulate air (HEPA) filters, charcoal adsorbers, downstream HEPA filters, cooling coil, and fan, to control the room temperature and airborne radioactivity. The outside air required for pressurization is mixed with the return air before it enters the filtration unit. During the emergency mode of operation, the control room HVAC is designed to pressurize the control room to 1/8-in. WG pressure to prevent unfiltered inleakage.

Doses to control room personnel resulting from a postulated LOCA are presented in section 15.6. A detailed discussion of the calculational models is given in appendix 15A. Air leaks have been taken into account in the calculations for ingress and egress losses in conformance with Regulatory Guide 1.78.

Control room shielding design, based on the most limiting design basis LOCA fission product release, is discussed in section 12.3 and is evaluated in chapter 15.

As discussed and evaluated in subsection 9.5.1, the use of noncombustible construction and heat- and flame-resistant materials throughout the plant minimizes the likelihood of fire and consequential fouling of the control room atmosphere.

A supply of protective clothing, respirators, and self-contained breathing apparatus adequate for at least five persons is stored at specified locations within the control room envelope. Five persons is the design basis operating shift crew size for operation as described in section 13.1.

To protect against high airborne radioactivity inside the control room, the control room HVAC system is automatically transferred from the normal mode to the emergency mode of operation

upon receipt of a control room outside air intake high radiation signal. Transfer of the system to emergency or isolation modes may also be initiated manually from the control room.

The filtration and cooling functions of the control room HVAC system may be performed fully even if the capability of the system is reduced by a single active component failure within the system or its supporting systems. Should the lead air filtration unit fail, the lag (redundant) train will provide the required cooling and air filtration. Should an excessive pressure drop develop across one filter train, the operator manually starts the redundant train to provide the required cooling and filtration. Normally open isolation dampers are arranged in series, so that the failure of one damper to shut upon transfer to the emergency mode will not prevent isolation. There are two emergency diesel generators for each unit. If one of the emergency diesel generators fails to start and assume its load, the control room emergency ventilation system equipment powered by the other diesel generator will provide the required cooling and air filtration.

A failure modes and effects analysis is provided in table 6.4.4-1.

Refer to subsection 7.3.6 for control room emergency filtration system actuation logic.

6.4.5 TESTING AND INSPECTION

A detailed program of preoperational and postoperational testing requirements to ensure continued system capability is implemented prior to station operation. Emphasis is placed on tests and inspections essential to a determination that performance criteria and operational capability are achieved and maintained.

Acceptance testing is performed to verify the unfiltered inleakage for a pressurized control room (emergency mode) is less than or equal to 120 cfm and the unfiltered makeup rate is less than or equal to 3000 cfm with the normal control room HVAC in service. Measurement of the control room envelope (CRE) pressure with respect to areas adjacent to the CRE, at designated locations, is performed in assessing the CRE boundary on a periodic basis. Obtaining and trending pressure data provides additional assurance that significant degradation of the CRE boundary will not go undetected between CRE inleakage determinations.

The control room isolation capability and the ability to process outside air through one of the two high-efficiency filter trains are tested periodically. The filtration trains are tested periodically by standard methods in general conformance with Regulatory Guide 1.52, as discussed in section 1.9.

Performance of fans is initially verified in accordance with Air Moving and Conditioning Association Standard 210[(1)] at the maximum anticipated system pressure drop. Ductwork is tested for leakage during installation. Testing and inspection of the control room normal outside air intake duct radiation monitors is discussed in section 11.5.

6.4.5.1 Preoperational Testing

High-efficiency filter air (HEPA) filter elements are tested individually prior to installation to verify an efficiency of 99.97 percent with a thermally generated, monodispersed, 0.3-μm dioctyl phthalate penetration aerosol. HEPA filter banks are tested in place prior to operation and periodically thereafter in conformance with American National Standards Institute (ANSI) N510 and conform with Position C.5.b of Regulatory Guide 1.52, as discussed in section 1.9.

The original and replacement batches of impregnated, activated carbon are batch tested in accordance with Regulatory Guide 1.52, as discussed in section 1.9, prior to loading into the adsorber section. The carbon adsorber section is filled with carbon in a manner which ensures a uniform packing density and minimizes dusting. The adsorber section is leak tested for gaseous halogenated hydrocarbon refrigerant prior to operation and periodically thereafter to verify a minimum of 99.95 percent retention, in conformance with ANSI N510, as discussed in paragraph 1.9.52.2. In addition, a periodic laboratory test of a representative sample of the impregnated, activated carbon is performed to verify iodine removal efficiencies in accordance with Position C.6 and Table 2 of Regulatory Guide 1.52, as discussed in section 1.9, for the assigned decontamination efficiency and bed depth.

Design and testing of filtration systems is consistent with the recommendations of Regulatory Guide 1.52, as discussed in section 1.9.

The emergency mode of the control room emergency ventilation system will undergo an acceptance test to verify that the system maintains a 1/8-in. WG positive pressure, with respect to adjacent areas, in the emergency zone.

Testing performed on the control room emergency ventilation systems is discussed in chapter 14.

6.4.5.2 Inservice Testing

Inservice testing of the control room essential heating, ventilation, and air-conditioning system is conducted in accordance with the surveillance requirements specified in the Technical Specifications.

Portable equipment such as air samplers, personnel dosimeters, and other radiation analysis equipment applicable to control room habitability is tested and inspected periodically, as noted in section 12.5.

Inplace leak testing of the HEPA filters is conducted initially prior to operation and at least every 18 months thereafter to confirm that the allowable penetration as defined in Regulatory Guide 1.52 is not exceeded. The inplace testing will be performed to the requirements of ANSI N510, as discussed in paragraph 1.9.52.2.

Carbon adsorbers are leak tested with a gaseous halogenated hydrocarbon refrigerant to ensure that bypass leakage is equal to or less than the allowable leakage required by Regulatory Guide 1.52, as discussed in section 1.9. Leak testing is performed in accordance with ANSI 510, as discussed in paragraph 1.9.52.2, prior to operation and periodically when HEPA testing is conducted.

6.4.5.3 Reference

1. "Laboratory Methods of Testing Fans for Rating Purposes," Air Moving and Conditioning Association Standard 210, 1974.

6.4.6 INSTRUMENTATION REQUIREMENT

The indications in the control room to monitor the heating, ventilation, and air-conditioning (HVAC) systems are listed in table 6.4.6-1.

Instrumentation required for actuation of the control room essential HVAC system is discussed in paragraph 6.4.2.2.2 and in subsection 7.3.6. The control room ventilation logic diagram is shown in drawings AX5DN020-4, AX5DN020-5, AX5DN020-6, AX5DN020-8, AX5DN020-10, AX5DN031-1, AX5DN031-2, AX5DN031-4, AX5DN032-1, AX5DN032-3, AX5DN034-2, and AX5DN037-1.

Details of the radiation monitors used to provide the control room indication actuation signal for the control room essential ventilation system are given in section 11.5.

The instrumentation is designed as Seismic Category 1. A description of initiating circuits, logic interlocks, periodic testing requirements, and redundancy of instrumentation relating to control room habitability is provided in subsection 7.3.6.

TABLE 6.4.2-1 (SHEET 1 OF 2)

PERFORMANCE CHARACTERISTICS OF MAJOR SYSTEM COMPONENTS

Control Building Control Room Filter Units

Quantity	4
System components	
Supply fan	
Type	Centrifugal
Capacity (ft^3/min) (maximum)	25,000
Static pressure (in. WG)	14
Motor (hp)	125
Charcoal absorber	
Efficiency (%)	99 at 70% relative humidity (for elemental and organic iodines)
Face velocity (ft^3/min)	40
Residence time (s/4-in. bed depth)	0.5
Nominal size (Tyler mesh)	8 x 16
HEPA filters	
Filter element	Pleated fiberglass
Size (in.)	24 x 24 x 12
Efficiency (%)	99.97 on 0.3 μm and larger
Capacity for size indicated (ft^3/min)	1000
Moisture eliminator	
Separator element	Fiberglass or galvanized steel
Efficiency (%)	99% for 5 to 10 μm droplets
Electric heater	
Heater element	80% Ni/20% Cr
Heating capacity (kW)	118
Cooling coil	
Cooling capacity (Btu/h)	1.09×10^6
Entering water temperature (°F)	44
Leaving water temperature (°F)	56
Chilled waterflow (gal/min)	175

TABLE 6.4.2-1 (SHEET 2 OF 2)

Control Building Control Room Return Air Fan

Type	Vane axial
Capacity (ft^3/min)	24,800
Static pressure (in. WG)	2.5
Motor (hp)	20

(NOTE: The return air fans are disabled and abandoned in place as their function is not required.

(HISTORICAL)
TABLE 6.4.2-2

LEAKAGE DATA AND LEAKAGE RATES

Leak Path	*Inleakage Rate at 1/8 in. WG (ft^3/min)*	*Outleakage Rate at 1/8 in. WG (ft^3/min)*
Concrete walls and floors	*2*	*5*
Ducts, piping, and electrical penetrations	*559*	*500*
Dampers	*0*	*10*
Doors	*125*	*125*
Ductwork	*20*	*15*
Supply fans and filtration units	*24*	*675*
Return fan	*4*	*150*
Supply fan enclosure	*16*	*20*
TOTAL	*750*	*1500*

TABLE 6.4.4-1 (SHEET 1 OF 14)

CONTROL ROOM EMERGENCY HVAC SYSTEM FAILURE MODES AND EFFECTS ANALYSIS

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
1.	1HV12146 air-operated on-off damper, normally open/fail closed (NO/FC) (supply side)	Remain open to allow flow of air, and closed on CRI so that EFU will provide HVAC	A	Inadvertent closed	Position indicating lights	None. Damper can be manually opened to provide HVAC in normal mode.	
			B, C	Fail to close	Position indicating lights	None. Redundant damper (item 2) available.	
			D	Inadvertent closed	Position indicating lights	None. Dampers can be manually opened to remove smoke.	
2.	1HV12147 air-operated on-off damper, NO/FC (supply side)	Remain open to allow flow of air, and closed on CRI so that EFU will provide HVAC	A	Inadvertent closed	Position indicating lights	None. Damper can be manually opened to provide HVAC in normal mode.	
			B, C	Fail to close	Position indicating lights	None. Redundant damper (item 1) available.	
			D	Inadvertent closed	Position indicating lights	None. Damper can be manually opened to remove smoke.	
3.	1HV12148 air-operated on-off damper, NO/FC (return side)	Remain open to allow flow of air, and closed on CRI so that EFU will provide HVAC	A	Inadvertent closed	Position indicating lights	None. Damper can be manually opened to provide HVAC in normal mode.	
			B, C	Fail to close	Position indicating lights	None. Redundant damper (item 4) available.	
			D	Inadvertent closed	Position indicating lights	None. Damper can be manually opened to remove smoke.	

TABLE 6.4.4-1 (SHEET 2 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
4.	1HV12149 air-operated on-off damper, NO/FC (return side)	Remain open to allow flow of air, and closed on CRI so that EFU will provide HVAC	A	Inadvertent closed	Position indicating lights	None. Damper can be manually opened to provide HVAC in normal mode.	
			B, C	Fail to close	Position indicating lights	None. Redundant damper (item 3) available	
			D	Inadvertent closed	Position indicating lights	None. Damper can be manually opened to remove smoke.	
5.	No. 16 breaker on 1ABA 480-V MCC, 1E Bus, for item 7, normally closed (NC)	Provide continuity and protection to damper motor (item 7)	A	Inadvertent open	Position indicating lights; motor control center (MCC) alarm	None. Loss of intake air from Unit 1. Intake air from Unit 2 available.	
			B	Inadvertent open	Position indicating lights; MCC alarm	None. No loss of EFU.	
			C	Inadvertent open	Position indicating lights; MCC alarm	None. Open damper will close.	
6.	No. 16 motor starter for item 7, NC	Provide continuity to damper (item 7)	A	Inadvertent open	Position indicating lights	None. Loss of intake air from Unit 1. Intake air from Unit 2 available.	
			B	Inadvertent open	Position indicating lights	None. No loss of EFU.	
			C	Fail to open	Position indicating lights	None. Dampers (item10) available.	
7.	1HV12114 motor-operated on-off damper, NO	Remain open to allow flow of air on NU and CRI modes	A	Inadvertent closed	Position indicating lights	None. Loss of intake air from Unit 1. Intake air from Unit 2 available.	
			B	Inadvertent closed	Position indicating lights	None. No loss of EFU. Loss of intake air from Unit 1. Intake air from Unit 2 available to maintain positive pressure.	
			C	Fail to close	Position indicating lights	None. Damper (item 10) is already closed.	

TABLE 6.4.4-1 (SHEET 3 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
8.	No. 16 breakers on 1BBA 480-V MCC, 1E Bus, for item 10, NC	Provide continuity and protection to damper motor (item 10)	A	Inadvertent open	Position indicating lights; MCC alarm	None. Loss of intake air from Unit 1. Intake air from Unit 2 available.	
			B	Inadvertent open	Position indicating lights; MCC alarm	None. No loss of EFU.	
			C	Inadvertent open	Position indicating lights; MCC alarm	None. Open damper will close.	
9.	No. 16 motor starter for item 10, NC	Provide continuity to 1HV12115 (item 10)	A	Inadvertent open	Position indicating lights	None. Loss of intake air from Unit 1. Intake air from Unit 2 available.	
			B	Inadvertent open	Position indicating lights	None. No loss of EFU.	
			C	Fail to open	Position indicating lights	None. Dampers (item 7) available.	
10.	1HV12115 motor-operated on-off damper, NO	Remain open to allow flow of air on NU and CRI modes	A	Inadvertent closed	Position indicating lights	None. Loss of intake air from Unit 1. Intake air from Unit 2 available.	
			B	Inadvertent closed	Position indicating lights	None. No loss of EFU. Loss of intake air from Unit 1. Intake air from Unit 2 available to maintain positive pressure.	
			C	Fail to close	Position indicating lights	None. Dampers (item 7) is already closed.	
11.	HV12152 air-operated on-off damper, NO/FC	Remain open to allow flow of air during normal mode, and closed on CRI so that EFU will provide HVAC	A	Inadvertent closed	Flow alarm, low; position indicating lights	None. Damper can be manually opened to provide HVAC in normal mode.	Common to Units 1 and 2
			B, C	Fail to close	Position indicating lights	None. Redundant damper (item 12) available.	

TABLE 6.4.4-1 (SHEET 4 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
			D	Inadvertent closed	Position indicating lights	None. Smoke mode intake available.	
12.	HV12153 air-operated on-off damper, NO/FC	Remain open to allow flow of air, and closed on CRI so that EFU will provide HVAC	A	Inadvertent closed	Flow alarm, low; position indicating lights	None. Damper can be manually opened to provide HVAC in normal mode.	Common to Units 1 and 2
			B, C	Fail to close	Position indicating lights	None. Redundant damper (item 11) available.	
			D	Inadvertent closed	Position indicating lights	None. Smoke mode intake available.	
13.	DELETED						
14.	DELETED.						
15.	DELETED.						
16.	DELETED.						
17.	DELETED.						
18.	DELETED.						
19.	Breakers, 480-V switchgear, 1E Bus, for item 20, No. 4 breaker on 1AB05	Provide continuity and protection to fan motor (item 20)	A	Inadvertent closed	Motor indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Motor indicating lights; flow alarm, low	Loss of train A. Train B available.	

TABLE 6.4.4-1 (SHEET 5 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
20.	Control building control room filter unit fan motor, normally deenergized (ND) 1-1531-N7-001-M01	Provide motive power to circulate air	A	N/A	N/A	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to operate fan motor	Flow alarm, low; motor indicating lights	Loss of train A. Train B available.	
21.	Breakers, 480-V switchgear, 1E Bus, for item 22, No. 4 breaker on 1BB07	Provide continuity and protection to fan motor item 22	A	Inadvertent closed	Motor indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Motor indicating lights; flow alarm, low	None. Loss of train B. Train A available.	
22.	Control building control room filter unit fan motor, ND 1-1531-N7-002-M01	Provide motor power to circulate air	A	N/A	N/A	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to operate fan motor	Flow alarm, low; motor indicating lights	None. Loss of train B. Train A available.	
23.	Breakers, 480-V switchgear, 1E Bus, for item 24 No. 06 breaker on 1AB05	Provide continuity and protection to heater (item 24)	A	Inadvertent closed	Heater indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Heater indicating lights; flow alarm, low	None. Loss of train A. Train B available.	
24.	Control building control room electrical heater, ND 1-1531-N7-001-H01	Provide heat, and reduce relative humidity and extract moisture	A	N/A	N/A	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to operate	Moisture alarm; temperature indicating lights	None. Loss of train A. Train B available.	

TABLE 6.4.4-1 (SHEET 6 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
25.	Breaker 480-V switchgear, 1E Bus, for item 26, NO No. 6 breaker on 1BB07	Provide continuity and protection to heater (item 26)	A	Inadvertent closed	Heater indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Heater indicating lights; flow alarm, low	None. Loss of train B. Train B available.	
26.	Control building control room electrical heater, ND 1-1531-N7-002-H01	Provide heat, and reduce relative humidity and extract moisture	A	N/A	N/A	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to operate	Moisture alarm; temperature indicating lights	None. Loss of train B. Train A available.	
27.	Breakers, 480-V MCC, 1E Bus, for item 29, NC No. 23 breaker on 1ABA	Provide continuity and protection to damper (item 29)	A, D	Inadvertent open	Position indicating lights; MCC alarm	None. Normally closed damper will remain closed.	
			B, C	Inadvertent open	Flow alarm, low; position indicating lights; MCC alarm	None. Loss of train A. Train B available.	
28.	Motor starter for item 29, NC No. 23 motor starter on 1ABA	Provide continuity to damper (item 29)	A, D	Inadvertent closed	Position indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Flow alarm, low; position indicating lights	None. Loss of train A. Train B available.	

TABLE 6.4.4-1 (SHEET 7 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
29.	Motor-operated on-off damper, NC 1HV12128 on 1ABA	Remain close on normal mode, and opens on CRI	A, D	Inadvertent open	Position indicating lights	None. EFU not required. NUs provide HVAC. In smoke mode, close the damper manually.	
			B, C	Fail to open	Flow alarm, low; position indicating lights	None; Loss of train A. Train B available.	
30.	Breaker, 480-V MCC, 1E Bus, for item 32, NC No. 23 breaker on 1BBA	Provide continuity and protection to damper (item 32)	A, D	Inadvertent open	Position indicating lights; MCC alarm	None. Normally closed damper will remain closed.	
			B, C	Inadvertent open	Flow alarm, low; position indicating lights; MCC alarm	None. Loss of train B. Train A available.	
31.	Motor starter for item 32, NC No. 23 motor starter on 1BBA	Provide continuity to damper (item 32)	A, D	Inadvertent closed	Position indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Flow alarm, low; position indicating lights	None. Loss of train B. Train A available.	
32.	Motor-operated on/off damper, NC 1HV12129 on 1BBA	Remain closed on normal mode, and open on CRI	A, D	Inadvertent open	Position indicating lights	None. EFU not required. NUs provide HVAC. In smoke mode, close the damper manually.	
			B, C	Fail to open	Flow alarm, low; position indicating lights	None. Loss of train B. Train A available.	
33.	Breakers, 480-V MCC, 1E Bus, for item 35, NC No. 6 breaker on 1ABA[(c)]	Provide continuity and protection to fan motor (item 35)	A, D	Inadvertent open	MCC alarm; motor indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Inadvertent open	Flow alarm, low; position indicating lights; MCC alarm	None. Loss of train A. Train B available.	

TABLE 6.4.4-1 (SHEET 8 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
34.	Motor starter for item 35, NO No. 6 motor starter on 1ABA[(c)]	Provide continuity and protection to fan motor (item 35)	A, D	Inadvertent closed	Motor indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Flow alarm, low; motor indicating lights	None. Loss of train A. Train B available.	
35.	Control building control room return air fan motor, ND 1-1531-B7-005-M01[(c)]	Provide motive power to circulate air	A, D	N/A	N/A	None. EFU not required.	
			B, C	Fail to operate fan motor	Flow alarm, low; motor indicating lights	None. Loss of train A. Train B available.	
36.	Breaker, 480-V MCC, 1E Bus, for item 38, NC No. 6 breaker on 1BBA[(c)]	Provide continuity and protection to fan motor (item 38)	A, D	Inadvertent open	MCC alarm; motor indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Inadvertent open	Flow alarm, low; position indicating lights; MCC alarm	None. Loss of train B. Train A available.	
37.	Motor starter for item 38, NO No. 6 motor starter on 1BBA[(c)]	Provide continuity and protection to fan motor (item 38)	A, D	Inadvertent closed	Motor indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Flow alarm, low; motor indicating lights	None. Loss of train B. Train A available.	
38.	Control building control room return air fan motor, ND 1-1531-B7-006-M01[(c)]	Provide motor power to circulate air	A, D	N/A	N/A	None. EFU not required.	
			B, C	Fail to operate	Flow alarm, low; motor indicating lights	None. Loss of train B. Train A available.	
39.	Breaker, 480-V MCC, 1E Bus, for item 41, NC No. 21 breaker on 1ABA	Provide continuity and protection to damper (item 41)	A, D	Inadvertent open	Position indicating lights; MCC alarm	None. Normally closed damper will remain closed.	

TABLE 6.4.4-1 (SHEET 9 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
			B, C	Inadvertent open	Flow alarm, low; position indicating lights; MCC alarm	None. Loss of train A. Train B available.	
40.	Motor starter for item 41, NO No. 21 motor starter on 1ABA	Provide continuity to damper (item 41)	A, D	Inadvertent close	Position indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Flow alarm, low; position indicating lights	None. Loss of train A. Train B available.	
41.	Motor-operated on-off damper, NC 1HV12130 damper on 1ABA	Remain closed on normal mode, and open on CRI	A, D	Inadvertent open	Position indicating lights	None. EFU not required. NUs provide HVAC. In smoke mode, damper can be closed manually.	
			B, C	Fail to open	Flow alarm, low; position indicating lights	None. Loss of train A. Train B available.	
42.	Breaker, 480-V MCC, 1E Bus, for item 44, NC No. 21 breaker on 1BBA	Provide continuity and protection to damper (item 44)	A, D	Inadvertent open	Position indicating lights; MCC alarm	None. Normally closed damper will remain closed.	
			B, C	Inadvertent open	Flow alarm, low; position indicating lights; MCC alarm	None. Loss of train B. Train A available.	
43.	Motor starter for item 44, NO No. 21 motor starter on 1BBA	Provide continuity to damper item 44	A, D	Inadvertent close	Position indicating lights	None. EFU not required. NUs provide HVAC.	
			B, C	Fail to close	Flow alarm, low; position indicating lights	None. Loss of train B. Train A available.	
44.	Motor-operated on-off damper, NC 1HV12131 on 1BB1	Remain closed on normal mode and opens on CRI	A, D	Inadvertent open	Position indicating lights	None. EFU not required. NUs provide HVAC. In smoke mode, damper can be closed manually.	
			B, C	Fail to open	Flow alarm, low; position indicating lights	None. Loss of train B. Train A available.	

TABLE 6.4.4-1 (SHEET 10 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
45.	Breakers, 480-V MCC, 1E Bus, for item 47, NC No. 14 breaker on 1ABA	Provide continuity and protection to fan motor item 47	A, B, C, D	Inadvertent open	MCC alarm; motor indicating lights; flow alarm, low	None. Loss of train A. Train B available.	
46.	Motor starter for item 47, NC No. 14 motor starter on 1ABA	Provide continuity to fan motor, item 47	A, B, C, D	Inadvertent open	Flow alarm, low; motor indicating lights	None. Loss of train A. Train B available.	
47.	Control building control room engineered safety features (ESF) chiller room exhaust fan motor, normally energized (NE) 1-1531-B7-002-M01	Provide motive power, to exhaust air	A, B, C, D	Fail to operate	Flow alarm, low; motor indicating lights	None. Loss of train A. Train B available.	
48.	Breaker 480-V MCC, 1E Bus, for item 50, NC No. 14 breaker on 1BBA	Provide continuity protection to fan motor item 50	A, B, C, D	Inadvertent open	MCC alarm; motor indicating lights; flow alarm, low	None. Loss of train B. Train A available.	
49.	Motor starter for item 50, NC No. 14 motor starter on 1BBA	Provide continuity to fan motor (item 50)	A, B, C, D	Inadvertent open	Flow alarm, low; motor indicating lights	None. Loss of train B. Train A available.	
50.	Control building control room ESF chiller room exhaust fan motor NE 1-1531-B7-004-M01	Provide motor power to exhaust air	A, B, C, D	Fail to operate	Flow alarm, low; motor indicating lights	None. Loss of train B. Train A available.	
51.	Fan, fan shaft, bearing, filter, damper, etc., for air filtration unit 1-1531-N7-001-000	Provide circulation filtration and control of air flow	B, C	Mechanical failure	Flow alarm, low; pressure differential alarm, high; temperature alarm, high	None. Loss of train A. Train B available.	
52.	Fan, fan shaft, bearing, filter, damper, etc., for air filtration unit 1-1531-N7-002-000	Provide circulation filtration and control of air flow	B, C	Mechanical failure	Flow alarm, low; pressure differential alarm, high; temperature alarm, high	None. Loss of train B. Train A available.	

TABLE 6.4.4-1 (SHEET 11 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
53.	Cooling coil for air filtration unit 1-1531-N7-001-000	Provide cooling and heat removal in the area	B, C	Leakage in cooling coil	Water flow alarm, low; temperature alarm, low	None. Loss of train A. Train B available.	
54.	Cooling coil for air filtration unit 1-1531-N7-002-000	Provide cooling and heat removal in the area	B, C	Leakage in cooling coil	Water flow alarm, low; temperature alarm, high	None. Loss of Train B. Train A available.	
55.	HV12162 air-operated on-off damper NO/FC	Remain open to allow flow of air in normal and smoke modes, and close on CRI so that EFU will provide HVAC	A	Inadvertent closed	Position indicating lights	None. Damper can be manually opened.	Common to Units 1 and 2
			B, C	Fail to close	Position indicating lights	None. Item 56 available.	
			D	Inadvertent closed	Position indicating lights	None. Damper can be manually opened	
56.	HV12163 air-operated on-off dampers NO/FC	Remain open to allow flow of air in normal and smoke modes, and close on CRI so that EFU will provide HVAC	A	Inadvertent closed	Position indicating lights	None. Damper can be manually opened.	Common to Units 1 and 2
			B, C	Fail to close	Position indicating lights	None. Item 55 available.	
			D	Inadvertent closed	Position indicating lights	None. Damper can be manually opened.	
57.	1-1531-B7-002-000 fan, fan shaft bearing, motor, etc.	Provide motive power to circulate air	B, C, D	Mechanical failure	Flow alarm, low; temperature alarm, high	None. Loss of train A. Train B available.	

TABLE 6.4.4-1 (SHEET 12 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
58.	1-1531-B7-004-000 fan, fan shaft, bearing, motor, etc.	Provide motive power to circulate air	B, C, D	Mechanical failure	Flow alarm, low; temperature alarm, high	None. Loss of train B. Train A available.	
59.	DELETED.						
60.	DELETED.						
61.	Smoke monitor 1AE12167	Monitor smoke in intake air and alarms at high smoke concentration and isolates intake air	C	Fail to give smoke alarm at high smoke concentration	Smoke alarm high on smoke monitor 1AE12166	None. Automatically isolate Unit 1 side intake by closing dampers 1HV12114 and 1HV12115. Use Unit 2 air intake.	If smoke concentration is also high on Unit 2 air intake, go on recirculation mode with no outside air intake
				False alarm	No alarm on smoke monitor 1AE12166	None. Smoke concentration is not high.	
62.	Smoke monitor 1AE12166	Monitor smoke in intake air and alarms at high smoke concentration and isolates intake air	C	Fail to give smoke alarm at high smoke concentration	Smoke alarm high on smoke monitor 1AE12167	None. Automatically isolate Unit 1 side intake by closing dampers 1HV12114 and 1HV12115. Use Unit 2 air intake.	If smoke concentration is also high in Unit 2 air intake go on recirculation mode with no outside air intake
				False alarm	No alarm on smoke monitor 1AE12167	None. Smoke concentration is not high.	
63.	Radiation monitor 1RE12117	Monitor radiation in intake air and alarms at high radiation level	C	Fail to give radiation alarm at high radiation	Radiation alarm high on radiation monitor 1RE12116	None. Use EFU to filter iodine. Item 64 available also.	
				False alarm	No alarm on radiation monitor 1RE12116	None. Radiation level is not high.	

TABLE 6.4.4-1 (SHEET 13 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
64.	Radiation monitor 1RE12116	Monitor radiation in intake air, and alarms at high radiation level	C	Fail to give radiation alarm at high radiation	Radiation alarm high on radiation monitor 1RE12117	None. EFU to filter iodine. Item 63 available also.	
				False alarm	No alarm on radiation monitor 1RE12117	None. Radiation level is not high.	
65.	1-1531-D7-103 backdraft damper (return duct)	Open with air flow in return direction and close on opposite flow	A&D	N/A	N/A	N/A	
			B&C	One blade fails to open	None	None. Flow variation is within allowable tolerances. Control room +VE pressure will be maintained.	
66.	1-1531-D7-104 backdraft damper (outside air)	Open with air flow in outside air supply direction and close on opposite flow	A&C&D	N/A	N/A	N/A	
			B	One blade fails to open	None	None. Flow variation is within allowable tolerances. Control room +VE pressure will be maintained.	
67.	1-1531-D7-105 backdraft damper (return duct)	Same as Item 65					
68.	1-1531-D7-106 backdraft damper (outside air)	Same as Item 66					

TABLE 6.4.4-1 (SHEET 14 OF 14)

Item No.	Description of Component	Safety Function	Plant Operating Mode	Failure Mode(s)	Method of Failure Detection	Failure Effect on System Safety Function Capability	General Remarks
69.	Flow switch FSL-12045	Monitor supply air flow of filter unit 1-1531-N7-002 to prevent starting of 1-1531-N7-001; or upon sensing low flow, to provide permissive start of filter unit 1-1531-N7-001	A&D	N/A	N/A	N/A	
			B&C	Fail to detect air flow; no power	Lights for both units 1-1531-N7-001 1-1531-N7-002 will run until N7-002[(b)]	None. Flow switch FSL-12045 will detect no flow in 1-1531-N7-002 supply duct and thus start 1-1531-N7-001.	
70.	Flow switch FSL-12046	Monitor supply air flow of filter unit 1-1531-N7-001 to stop lead filtration unit 1-1531-N7-002 upon sensing flow	A&D	N/A	N/A	N/A	
			B&C	Fail to detect air flow; no power	None	None. Flow indicator FI-12192 will indicate 1-1531-N7-002 is running.	
				False flow indication	None	None, Lead/lag logic will start 1-1531-N7-001	

a. Plant operating modes are as follows:

A - Normal mode: HVAC normal units (NU) operating; outside and recirculation supply air; positive room pressure relative to the atmosphere.
B - Emergency mode: HVAC emergency filtration units (EFU) operating; outside and recirculation supply air; positive room pressure relative to the atmosphere. *Trips after FSL-12046 sensing flow on N7-001 unit.
C - Isolation mode: EFU operating; recirculation only; zero pressure differential; outside smoke.
D - Smoke purge mode: HVAC NU operating; outside air only; negative room pressure relative to the atmosphere (smoke inside control room).

b. Trips after FSL-12046 sensing flow on N7-001 unit.

c. The return air fans are disabled and abandoned in place.

TABLE 6.4.6-1

CONTROL ROOM HVAC INDICATIONS AND ALARMS

Control room differential pressure (high or low alarm)

Control room area radiation (indication and high alarm)

Control room smoke (high alarm)

Smoke in control room intake (high alarm)

Radiation level in control room intake (indication and high alarm)

Fan operating status

Isolation damper position

Differential pressure across first HEPA filter (indication and high alarm)

Differential pressure across total filter unit (indication and high alarm)

Moisture content downstream of the moisture eliminator (indication and high alarm)

Temperature in charcoal filter (high alarm, high high alarm)

Temperature of filter unit upstream and downstream of the charcoal filter (indication)

Airflow rate at filter unit outlet (indication and high or low alarm)

6.5 FISSION PRODUCT REMOVAL AND CONTROL SYSTEMS

Several plant features serve to reduce or limit the release of fission products following a postulated loss-of-coolant accident (LOCA) or fuel handling accident. This section provides a discussion of the function of the emergency filter systems (subsection 6.5.1) and the containment spray system (subsection 6.5.2) to mitigate the consequences of an accident. The design of each of these engineered safety features (ESF) is discussed in other referenced sections. Chapter 15 addresses the radiological consequences of postulated accidents and demonstrates the adequacy of the fission product removal and control systems.

Other sections provide the design bases and safety evaluations which demonstrate that the design and construction of these systems is commensurate with acceptable practices for ESF. This includes, but is not limited to, ensuring redundancy, isolation from nonsafety-related portions, seismic classification, conformance with Regulatory Guide 1.52 (section 1.9), suitability of material for the intended service, Class 1E power supply from onsite or offsite sources, qualification testing, and capability for inspection and testing.

6.5.1 ENGINEERED SAFETY FEATURES FILTER SYSTEMS

The ESF filter systems include the control room heating, ventilation, and air-conditioning (HVAC) system discussed in section 6.4 and subsection 9.4.1, the fuel handling building post-accident exhaust system discussed in subsection 9.4.2, and the piping penetration filter exhaust system discussed in subsection 9.4.3. The performance of the systems under postulated accident conditions is discussed in sections 15.6 and 15.7. The post-LOCA purge exhaust system is nonsafety related; it is discussed in subsection 6.2.5.

6.5.1.1 Design Bases

The ESF filter systems are designed to accomplish the following:

A. The control room HVAC system ensures that the radiation exposures to operating personnel in the control room resulting from a design basis accident, as discussed in chapter 15, are within the guideline values of 10 CFR 50, Appendix A, General Design Criterion (GDC) 19.

B. The piping penetration filter exhaust system is designed to maintain the filtration unit rooms at - 1/4 in. WG with respect to atmosphere. This condition ensures that the piping penetration areas which contain post-LOCA recirculation components are maintained at a negative pressure with respect to adjacent areas to prevent uncontrolled exfiltration of potentially contaminated air and to minimize release of airborne radioactivity to the outside atmosphere resulting from containment and penetration area leakage under accident conditions. The piping penetration filter exhaust system ensures that the offsite radiation exposures resulting from the postulated post-LOCA leakage in recirculation piping, as discussed in subsection 15.6.5, are within the guideline values of 10 CFR 100. It also ensures that the emergency core cooling system and containment spray pump rooms can be purged to allow access for repair and maintenance of the equipment.

   The drain isolation valves for ESF equipment rooms and negative pressure boundary areas are procured as project class 313 but are installed as project

class 414. These valves are required to be locked closed to ensure validity of the offsite dose exposure analysis as tabulated in table 15.6.5-6.

C. The fuel handling building post-accident exhaust system is designed to maintain a slightly negative pressure within the fuel handling building following a fuel handling accident to minimize release of airborne radioactivity to the outside atmosphere. The post-accident exhaust system filter will reduce doses, but is not required to ensure that the offsite radiation exposures and exposures to operating personnel in the control room resulting from a postulated fuel handling accident in the fuel handling building, are within the guideline values of 10 CFR 100 and 10 CFR 50, Appendix A, GDC 19, respectively. As discussed in subsection 15.7.4, no credit is taken for filter operation.

D. The failure of any active component in a filtration system, assuming loss of offsite power, cannot impair the ability of the system to perform its safety function.

E. The ESF filter systems are designed to remain intact and functional in the event of a safe shutdown earthquake.

F. The ESF filter systems are designed to be consistent with the recommendations of Regulatory Guide 1.52, as discussed in section 1.9.

The design bases for sizing the filters, fans, and associated ductwork are discussed in subsections 9.4.1, 9.4.2, 9.4.3, and 9.4.5.

6.5.1.2 System Design

6.5.1.2.1 General System Description

The control room emergency ventilation and air-conditioning system is described in section 6.4 and subsection 9.4.1. The piping penetration filter exhaust system is described in subsection 9.4.3. The fuel handling building post-accident cleanup system is described in subsection 9.4.2. Flow diagrams for each system are shown in the appropriate subsections.

6.5.1.2.2 Component Description

Each ESF filter train consists of a moisture separator, a heating coil, an upstream high-efficiency particulate air (HEPA) filter, a charcoal adsorber with fire detection temperature sensors, and a downstream HEPA filter. The filtration trains are connected to fans with direct drive motors, associated ductwork, and controls. Specific component design parameters are provided in table 6.5.1-1.

The filter housing design provides adequate space for filter maintenance and inspection. The housing is fitted with the necessary ports for testing. Pipe, cable, and conduit penetrations are sealed to minimize leakage. Access doors are marine-type, bulkhead doors with gas-tight seals.

The charcoal adsorber portion of each filter train is provided with a fire detection system and a water spray system to allow flooding of the charcoal bed to prevent bed ignition from radioactivity-induced heat. Fire protection systems for the carbon adsorbers are discussed in subsection 9.5.1.

The electric heaters provided in the control room emergency ventilation and air conditioning system air filtration units are designed to reduce the relative humidity of the entering air stream mixture to 70 percent from as high as 100 percent.

Relative humidity is maintained by use of a moisture controller which, as relative humidity approaches 70 percent, modulates the electric heater to gradually raise the air temperature, thus lowering the relative humidity.

The electric heaters in the piping penetration filter exhaust system and the fuel handling building post-accident exhaust system reduce relative humidity, thereby improving adsorber efficiency. However, the heaters are not required to maintain the relative humidity of the entering air stream mixture to less than 70 percent. Although a moisture controller is provided to modulate the electric heaters as relative humidity approaches 70 percent, no credit is taken for humidity control in these systems.

6.5.1.2.3 System Operation

In the event of a LOCA, the piping penetration filter exhaust system functions to limit and reduce the potential release of fission products due to ECCS recirculation line and component leakage. Specific details of system operation following a LOCA are provided in subsections 9.4.3 and 9.4.5.

In the event of a fuel handling accident in the fuel handling building, the emergency exhaust system functions to reduce the fission product release from that building. Specific details of system operation following a fuel handling accident are provided in subsection 9.4.2.

In the event of high radiation levels in the control room outside air intake, the control room emergency HVAC systems provide the control room with a filtered supply of air. Specific details of system operation following a LOCA are discussed in section 6.4 and subsection 9.4.1.

6.5.1.3 Design Evaluation

A. The performance capability of the control room emergency filters and the design of individual components which ensure the capability to perform the safety function are discussed in section 6.4. Control room doses resulting from postulated radiological accidents are given in section 15.6. These doses are within the guideline values of 10 CFR 50, Appendix A, GDC 19.

B. Component descriptions and safety evaluation for the piping penetration exhaust filters are provided in subsection 9.4.3. Dose analyses of post-LOCA leakage in recirculation piping is discussed in subsection 15.6.5. Offsite radiation exposures and control room doses resulting from this leakage are within the guideline values of 10 CFR 100 and 10 CFR 50, Appendix A, GDC 19, respectively.

C. Component descriptions and safety evaluation for the fuel handling building post-accident exhaust system are provided in subsection 9.4.2. Dose analyses of postulated fuel handling accidents are discussed in subsection 15.7.4. Offsite radiation exposures and control room doses resulting from these accidents are within the guideline values of 10 CFR 100 and 10 CFR 50, Appendix A, GDC 19, respectively.

D. The control room HVAC system, the piping penetration filter exhaust system, and the fuel handling building post-accident exhaust system each consist of two

independent and redundant filtration trains with respect to active components. Should any active component in one train fail, filtration can be performed by the other train, which is powered from a separate Class 1E electrical bus. Failure modes and effects analyses are provided in section 6.4 and subsections 9.4.2 and 9.4.3.

E. The ESF filter systems are designed to Seismic Category 1 requirements. The components and supporting structures of any system, piece of equipment, or structure that is not Seismic Category 1, and whose collapse could result in loss of safety function of the ESF filter systems through either impact or flooding, have been evaluated to determine that they will not collapse when subjected to seismic loading.

F. The ESF filter systems are designed and constructed to be consistent with the recommendations of Regulatory Guide 1.52 as discussed in section 1.9.

G. The ESF filters have been conservatively analyzed to determine the bed temperature under worst-case accident conditions. The analysis demonstrated that the temperature of the charcoal beds will remain below the 300°F limit recommended by ANSI N509-1980. Thus, no cooling mechanism is supplied for the filters.

6.5.1.4 Tests and Inspections

6.5.1.4.1 Preoperational Testing

The HEPA filters are manufactured and tested prior to installation in accordance with MIL-F-51068. The HEPA filter banks are tested in place prior to operation to verify a minimum of 99.95-percent retention (99.0 percent for FHB filters) with a dioctyl phthalate aerosol (DOP) in accordance with American National Standards Institute (ANSI) N510 and conforms with Position C.5.b of Regulatory Guide 1.52, as discussed in section 1.9.

The original and replacement batches of impregnated, activated carbon are batch tested in accordance with Regulatory Guide 1.52, as discussed in section 1.9, prior to loading into the adsorber section. Tests include particle size distribution, hardness, density, moisture content, impregnant content, ash content, impregnant leachout, and elemental iodine and methyl iodine removal efficiencies at postulated accident conditions. The charcoal adsorber is leak tested prior to operation to verify a minimum of 99.95-percent retention. In addition, a laboratory test of a representative sample of the impregnated, activated charcoal is performed to verify iodine removal efficiencies in accordance with Position C.6 and table 2 of Regulatory Guide 1.52, as discussed in section 1.9.

Air filtration units are acceptance tested in accordance with ANSI N510, as discussed in section 1.9.

Design and testing of ESF filtration systems are consistent with the recommendations of Regulatory Guide 1.52, as discussed in section 1.9.

Fans are factory tested in accordance with standards of the Air Moving and Conditioning Association Standard 210.(2)

Moisture separators are tested in accordance with ANSI N509, paragraph 5.4, as discussed in subsection 1.9.52, and capable of removing at least 99 percent of the entrained moisture in an airstream.

The drain design and the accessibility of components and provisions for maintenance are in accordance with ANSI N509, as discussed in section 1.9.

6.5.1.4.2 Inservice Testing

Inservice testing of the ESF filtration systems is conducted in accordance with the surveillance requirements given in the Technical Specifications and Technical Requirements Manual, as applicable.

6.5.1.5 Instrumentation Requirements

Controls and instrumentation for the control room HVAC, piping penetration filter exhaust, and fuel handling building post-accident exhaust systems are discussed in section 7.3 and subsections 9.4.1, 9.4.2, and 9.4.3. Instrumentation is summarized in table 6.5.1-2. Each system is designed to function automatically upon receipt of an ESF actuation system signal. Fans can also be controlled from the control room. The status of the ESF filter systems equipment is displayed in the control room during both normal and accident operations. Instrumentation is consistent with the recommendations of Regulatory Guide 1.52, as discussed in section 1.9.

6.5.1.6 Materials

The materials of construction used in or on the filter systems are given in table 6.5.1-3. Each of the materials is compatible with the normal and accident environments postulated in the control room, the fuel handling building, and the penetration areas.

6.5.1.7 Standard Review Plan Evaluation

The design of the ESF atmospheric cleanup systems conforms to the majority of the items delineated in table 6.5.1-1. The requirements of table 6.5.1-1 and nonconformative items are listed below with the VEGP design rationale:

A. A prefilter ahead of the pre-HEPA is required for the removal of submicron particles.

   The demister provided for the ESF filtration systems has the same filtration efficiency as a prefilter in removing submicron particles in the airstream. Therefore, the demister serves the same function as a prefilter in preserving the life of the HEPA filter. Regulatory Guide 1.52, paragraph C.2.(2) permits demisters to fulfill the prefilter function.

B. Local and control room temperature readout and alarm signals are required for the air entering the electric heater.

   The basic purpose of providing an electric heater in the airstream is to lower the relative humidity of the air before it reaches the charcoal filters, which are less

efficient at high relative humidity. Moisture entering the HEPA filters is monitored with a high alarm provided in the control room. In addition, temperature indication upstream and downstream of the charcoal filters is also provided in the control room.

For clarification, air that passes through the ESF filtration systems is room air conditioned to the design temperatures which range from 40°F to 104°F.

C. Local pressure drop indication and high alarm signal are required for the second HEPA (post-HEPA).

A local high alarm of the pressure drop across the second HEPA filter is not provided, since the filter equipment rooms are potentially high radiation areas under post-accident conditions. Dust loading would result in a high pressure drop across the first HEPA filter, which is alarmed, before it would cause high pressure drop across the second HEPA filter.

Charcoal used in the charcoal adsorbers meets the specifications of ANSI N509; therefore, only a small quantity of charcoal fines would be expected to accumulate on the second HEPA filter. Since the ESF filtration systems do not operate during normal operation, periodic maintenance and surveillance in accordance with the Technical Specifications and Technical Requirements Manual will ensure filter pressure drops are in normal operating ranges.

D. Local and control room hand switch and status indication is required for the deluge valves.

The ESF filtration units charcoal filters are protected by manual deluge systems with automatic detection. Unit 1/Common filtration units have the deluge system water supply positively isolated by removal of a pipe spool piece. Unit 2 filtration units are provided with a manual fire hose connection.

E. No local indication (or continuous recording in the control room) is provided for unit inlet or outlet flow.

System flow is indicated and high-flow or low-flow is alarmed in the control room, which is a continuously manned location. Local indication is not provided since filter equipment rooms are potentially high radiation areas under post-accident conditions.

F. No local high alarm signal is provided of the pressure drop across the prefilter (demister in the VEGP design).

Demister pressure drop is indicated locally. Standard Review Plan table 6.5.1-1 does not require an alarm, which is listed as optional.

G. Local status indication is provided for the electric heater.

H. No local indication, high alarm, and low alarm signals are provided, and no high alarm, low alarm, and trip alarm signals are provided in the control room for a temperature sensor located between the heater and the first HEPA filter.

Temperature downstream of the first HEPA filter is monitored in lieu of the upstream temperature. The downstream temperature, which is indicated in the control room, is representative of the upstream temperature. The temperature sensor was located on the downstream side of the HEPA filter to minimize the influence of thermal radiation from the heaters. Local indication and alarm is not

provided since equipment rooms are potentially high radiation areas under post-accident conditions.

In addition to the above indication, temperature of the charcoal bed is also monitored. A two-stage high-temperature alarm is provided in the control room. Although no low temperature alarm is provided in the control room, a high humidity alarm is available which provides direct indication of high humidity rather than the indirect indication a low temperature alarm would provide.

The heater unit also has two temperature sensing elements that will automatically shut down the heater. The heater is automatically shut down when the temperature in the airstream exceeds approximately 210°F. As the temperature is reduced, the cutout will automatically reset. If the temperature in the airstream should exceed approximately 260°F, the heater is automatically shut down. As the temperature is reduced, the cutout will automatically reset. Although no high alarm or trip alarm signals are provided in the control room for a temperature sensor between the heater and the first HEPA filter, the above described automatic cutout features of the heater unit, the temperature indication in the control room for the temperature downstream of the first HEPA filter, and the high alarm and high-high alarm in the control room for the charcoal bed temperature assure that the ESF atmosphere cleanup systems perform their design safety functions.

I. No local high alarm signal is provided, and no recorded indication is provided in the control room of the pressure drop across the first HEPA filter.

A local high alarm of the pressure drop across the first HEPA filter is not provided, since the filter equipment rooms are potentially high radiation areas under post-accident conditions.

Although no recorded indication is provided in the control room of the pressure drop across the first HEPA filter, indication and a high pressure drop alarm across the first HEPA filter is provided in the control room.

A high pressure drop across the entire filtration unit is also alarmed in the control room. This provides the necessary information to access the effectiveness of the first HEPA filter.

J. No local two-stage high alarm signal is provided for a temperature sensor located between the adsorber and the second HEPA filter.

A local two-stage high alarm of the temperature between the charcoal adsorber and the second HEPA filter is not provided, since the filter equipment rooms are potentially high radiation areas under post-accident conditions.

A two-stage high alarm signal is provided in the control room for charcoal bed temperature as described in "H" above.

6.5.1.8 References

1. "Filter, Particulate, High Efficiency, Fire Resistant," MIL-F-51068D.

2. "Laboratory Methods of Testing Fans for Rating Purposes," Air Moving and Conditioning Association Standard 210, 1974.

6.5.2 CONTAINMENT SPRAY SYSTEMS (FISSION PRODUCT REMOVAL)

The containment spray system (CSS) is an engineered safety features (ESF) system that is operated, following a postulated loss-of-coolant accident (LOCA), to reduce the pressure, temperature, and iodine activity of the containment atmosphere. To accomplish these aims, the system sprays subcooled solution into the containment through nozzles suspended from the containment dome.

Radioiodines are of primary concern in the radiological consequence evaluation of a LOCA. A primary function of the spray system is the absorption of elemental iodine.

To enhance the spray's iodine absorption capacity, the spray recirculation solution is adjusted to an alkaline pH which promotes the hydrolysis of iodine to nonvolatile forms.

The mechanical components of the CSS are described in subsection 6.2.2. The actuation times for the CSS are event dependent and are described in the individual accident analyses.

6.5.2.1 Design Bases (For Fission Product Removal)

The CSS provides a spray injection solution with a pH of approximately 4.5 and a final recirculation sump solution with a pH in the range of 7.5 to 10.5.

Fifty percent of the core equilibrium iodine inventory is assumed to be released to the containment atmosphere and available for leakage to the environment. The iodine released from the core is assumed to consist of 91% elemental, 4% organic, and 5% particulates.(1)

The CSS reduces the elemental iodine inventory in the containment atmosphere until a decontamination factor (DF) of 21.4 is achieved. The CSS also removes particulates from the containment atmosphere. The extent to which credit is taken for particulate removal is discussed in paragraph 6.5.2.3.

Additional design bases associated with the heat removal capability of the CSS are presented in subsection 6.2.2.

6.5.2.2 System Design (For Fission Product Removal)

The containment spray pumps and headers described in subsection 6.2.2 are utilized for fission product removal.

Crystalline trisodium phosphate (TSP) is utilized to raise the pH of the ECCS solution in the containment sump since the initial pH of the boric acid ECCS solution will be approximately 4.5. The TSP will be stored in baskets located in the post-LOCA flooded region of the containment building. As the initial spray solution and subsequently the recirculation solution comes in contact with the TSP, the TSP will dissolve, raising the pH of the sump solution to an equilibrium value in the range of 7.5 to 10.5.

This ensures that elemental iodine will be retained in the sump and will not re-evolve from the spray droplets during the recirculation phase.

The mechanical components and operation of the spray additive system are described in detail in subsection 6.2.2. There are two independent trains of spray nozzles located in the containment dome region. Each train consists of three ring headers. There is a total of 342 spray nozzles. The spray nozzle locations and orientations are shown in drawings 1X4DL4A01, 1X4DL4D01, and 1X4DL4D09.

The spray nozzles used are SPRACO model 1713A. The nozzles are the ramp bottom, swirl chamber type with a discharge orifice diameter of 3/8 in. The nozzles cannot clog since there are no internal parts, such as swirl vanes, and the maximum particle size entrained in the spray flow is limited to < 3.8 in. in diameter (see section 6.1.2) by the emergency sump screen perforations. Spray drop size distribution is presented in figure 6.5.2-1 and is discussed in detail in reference 2. The spray system operating modes, water sources, and initiation signals are described in paragraph 6.2.2.2. The system component design parameters are presented in table 6.2.2-4.

Approximately 78 percent of the net free containment volume is sprayed. This volume consists of the containment volume above the operating deck and the regions below the operating deck impinged directly by the spray. The unsprayed containment regions are described in table 6.5.2-1. The mixing rate between the sprayed and unsprayed containment volumes is assumed to be the minimum safeguards containment cooler air flowrate.

6.5.2.3 Design Evaluation

Containment spray system performance is evaluated using a spray model developed by Westinghouse. The model considers the effects of spray drop size distribution, droplet coalescence, gas and liquid phase mass transfer resistance, drop trajectories, and condensation of steam on the drops.

The elemental iodine removal capability of the containment spray is described in terms of the individual spray droplets. The behavior of the aggregate spray is related to the behavior of the individual drops, by means of a drop size distribution function.

An advantage to using this microscopic approach is the ability to derive the model from first principles. Thus, the model is free of any scaling factors which would be required to extrapolate laboratory test data to a full-size reactor containment.

A conservative spray removal coefficient as shown in table 6.5.2-2 is used based on data provided in references 5 and 6. The elemental iodine removal analysis assumes that the containment spray system is operating at minimum capacity (one of two spray pumps) and that the emergency core cooling system (ECCS) is operating at maximum capacity. Although the CSS will operate for a minimum of 2 h, no spray removal of elemental iodine is assumed after a DF of 21.4 is achieved for the containment atmosphere.

The particulate washout coefficient is calculated by the method outlined in reference 5. Removal of particulate iodine is assumed for the duration of the spray period.

The deposition removal coefficient for elemental iodine is determined based on guidance of reference 5. Deposition is assumed to occur only on galvanized surfaces and surfaces coated with zinc based paint or epoxy paint. Deposition removal of iodine is assumed to continue at the initial rate until a DF of 50 is achieved for the containment atmosphere and then at a reduced rate until a DF of 200 is achieved.

The spray system performance was evaluated at the peak containment pressure postulated to occur following a LOCA. Presented in figure 6.5.2-2 is the spray reduction factor as a function of post-LOCA containment saturation temperature. In the determination of operating deck coverage, the reduction factor was applied to both the spray throw distance and the spray envelope diameter.

The spray nozzle capacity curve is presented in figure 6.5.2-3. In those cases where operation of containment spray has been initiated and a LOCA is indicated, the containment will be

sprayed continuously for a minimum of 2 hours. The transfer from spray injection to spray recirculation will be made manually.

Table 6.5.2-2 lists the input parameters and results of the spray iodine removal analysis.

6.5.2.4 Tests and Inspection

Refer to subsection 6.2.2 for a description of provisions for testing the CSS.

6.5.2.5 Instrumentation Requirements

Refer to subsection 6.2.2 for a description of the CSS instrumentation.

6.5.2.6 Materials

Refer to subsection 6.2.2 for a discussion of the CSS materials chemistry.

6.5.2.7 Standard Review Plan Evaluation

The VEGP is equipped with a semiautomatic switchover from injection to recirculation modes.

The ECCS switchover from injection to recirculation is semiautomatic. The sump suction valves open automatically on low refueling water storage tank level. The valve changes to switch suctions of the safety injection and charging pumps to the residual heat removal discharge and the isolation of the suctions from the refueling water storage tank are manual. Thus the switchover is semiautomatic. The CSS switchover from injection to recirculation is entirely manual. The CSS is designed to continuously deliver spray to containment and is capable of continuing for a time equal to or greater than the duration of the initial 2-h period following the accident.

6.5.2.8 References

1. U.S. Nuclear Regulatory Commission, "Containment Spray as a Fission Product Cleanup System," Standard Review Plan 6.5.2.
2. Sanford, M. O., "SPRACO Model-1713A Nozzle Spray Drop-Size Distribution," WCAP-8258-R1 (Nonproprietary).
3. Somers, E. V., and Sanford, M. O., "Iodine Removal by Spray in the Joseph M. Farley Station Containment," WCAP-8376 (Nonproprietary).
4. Deleted.
5. U.S. Nuclear Regulatory Commission, "Technological Bases for Models of Spray Washout of Airborne Contaminants in Containment Vessels," NUREG/CR-0009.
6. U.S. Atomic Energy Commission, "A Review of Mathematical Models for Predicting Spray Removal of Fission Products in Reactor Containment Vessels," WASH-1329, June 15, 1974.

6.5.3 FISSION PRODUCT CONTROL SYSTEMS

6.5.3.1 Primary Containment

The containment consists of a prestressed, post-tensioned, reinforced concrete structure with cylindrical walls, hemispherical dome, and base slab lined with welded 1/4-in. carbon steel liner plate, which forms a continuous, leaktight membrane. Details of the containment structural design are discussed in section 3.8. Layout drawings of the containment structure and the related items are given in the general arrangement drawings of section 1.2.

The containment walls, liner plate, penetrations, and isolation valves function to limit the release of radioactive materials subsequent to postulated accidents, such that the resulting offsite doses are less than the guideline values of 10 CFR 100. Containment parameters affecting fission product release accident analyses are given in table 6.5.3-1.

Long-term containment pressure and temperature response to the design basis accident are shown in section 6.2. Relative to this time period, the spray system is operated to reduce iodine concentrations and containment atmospheric temperature and pressure commencing with system initiation, as discussed in section 6.2, and ending when containment pressure has returned to normal.

The containment minipurge system may be operated for personnel access to the containment when the reactor is at power, as discussed in subsection 9.4.6. For this reason, the radiological assessment of a loss-of-coolant accident assumes that the minipurge valves are open at the initiation of the event. However, the minipurge valves receive automatic signals to shut from diverse parameters. The valves are designed to close within 5 s. (See the description in subsection 6.2.4.)

The fission product removal capability of the containment spray system is discussed in subsection 6.5.2.

Redundant, safety-related hydrogen recombiners are provided in the containment as the primary means of controlling post-accident hydrogen concentrations. A hydrogen purge system is provided for backup hydrogen control. Although use of the hydrogen purge system is not expected for post-accident hydrogen control, offsite dose analyses assuming the operation of the hydrogen purge system have been performed to determine its incremental contribution on the radiological doses. This analysis is provided in chapter 15.

Containment combustible gas control systems are discussed in detail in subsection 6.2.5.

6.5.3.2 Secondary Containment

This paragraph is not applicable to the VEGP.

TABLE 6.5.1-1 (SHEET 1 OF 3)

ESF FILTER SYSTEM DESIGN PARAMETERS
(FOR UNIT 1 OR 2)

Control Room Emergency Filter System	
Quantity	2 (one on standby)
Capacity (ft^3/min)	25,000
HEPA Filters	
Number of stages	2 (one upstream and one downstream of charcoal filter)
Cell size	24 in. x 24 in. x 12 in.
Pressure drop	
Clean (in. WG)	1.0
Loaded (in. WG)	2.0
Efficiency	99.97% for 0.3-μm particles
Charcoal Filter	
Bed depth (in.)	4
Face velocity (ft/min)	40
Average residence time (s)	0.25 per 2-in. bed depth
Filter media	Impregnated coconut shell
Decontamination efficiency	99% at 70% relative humidity (for elemental and organic iodines)
Filter capacity	2.5 mg of total iodine per gram of activated carbon
Moisture Eliminator	
Eliminator media	Spun glass fiber or galvanized steel
Maximum pressure drop (in. WG)	1.0
Efficiency	99% of 5 to 10 μm diameter droplets
Heating Coil	
Heating capacity (kW)	118
Heating element	Finned tubular
Heating coil	80% Ni/20% Cr
Fan	
Quantity	1
Type	Centrifugal
Static press (in. WG)	14
Motor (hp)	125
Cooling Coils	
Cooling capacity (Btu/h)	1.09×10^6
Air entering temperature (°F)	82 dry bulb, 65 wet bulb
Air exiting temperature (°F)	50.5 dry bulb, 50 wet bulb

TABLE 6.5.1-1 (SHEET 2 OF 3)

Water entering temperature (°F)	44
Water exiting temperature (°F)	56
Piping Penetration Filter System	
Quantity	2 (one on standby)
Capacity (ft^3/min)	15,500
HEPA Filters	
Number of stages	2 (one upstream and one downstream of charcoal filter)
Cell size	24 in. x 24 in. x 12 in.
Pressure drop	
Clean (in. WG)	1.0
Loaded (in. WG)	2.0
Efficiency	99.97% for 0.3-μm particles
Charcoal Filter	
Bed depth (in.)	4
Face velocity (ft/min)	40
Average residence time (s)	0.25 per 2-in. bed depth
Filter media	Impregnated coconut shell
Decontamination efficiency	90% elemental iodine, 30% organic iodine, at 95% relative humidity
Filter capacity	2.5 mg of total iodine per gram of activated carbon
Eliminator media	Spun glass fiber or galvanized steel
Maximum pressure drop (in. WG)	1.0
Efficiency	99% of 5 to 10 μm diameter droplets
Heating Coil	
Heating capacity (kW)	80
Heating element	Finned tubular
Heating material	80% Ni/20% Cr
Fan	
Quantity	1
Type	Vane axial
Static pressure (in. WG)	16
Motor (hp)	75
Fuel Handling Building Post-Accident Filter System (shared by both units)	
Quantity	2 (one on standby)
Capacity (ft^3/min)	5000

TABLE 6.5.1-1 (SHEET 3 OF 3)

HEPA Filters	
Number of stages	2 (one upstream and one downstream of charcoal filter)
Cell size	24 in. x 24 in. x 12 in.
Resistance	
Clean (in. WG)	1.0
Loaded (in. WG)	2.0
Efficiency	99.97% for 0.3-μm particles
Charcoal Filters	
Bed depth (in.)	4.0
Face velocity (ft/min)	40
Average residence time (s)	0.25 per 2-in. bed depth
Filter media	Impregnated coconut shell
Decontamination efficiency	90% elemental iodine, 30% organic iodine, at 95% relative humidity
Filter capacity	2.5 mg of total iodine per gram of activated carbon
Eliminator media	Spun glass fiber or galvanized steel
Maximum pressure drop (in. WG)	1.0
Efficiency	99% of 5 to 10 μm diameter droplets
Heating Coil	
Heating capacity (kW)	20
Heating element	Finned tubular
Heating coil material	80% Ni, 20% Cr
Fan	
Quantity	1
Type	Vane axial
Static pressure (in. WG)	14
Motor (hp)	40

TABLE 6.5.1-2 (SHEET 1 OF 2)

ESF FILTER SYSTEM INSTRUMENTATION
(TYPICAL FOR EACH FILTER TRAIN)

Sensing Location	Local	Control Room
Unit outlet flowrate	---	Indication; alarm[(a)]
Moisture separator pressure drop	Indication	---
Space between heater and first HEPA filter moisture content	---	Indication; alarm[(a)]
First HEPA filter pressure drop	Indication	Indication; alarm[(a)]
Space between first HEPA and charcoal filter temperature	---	Indication
Charcoal filter temperature	---	Alarm;[(a)] high-high alarm
Space between charcoal filter and second HEPA temperature	---	Indication
Second HEPA filter pressure drop	Indication	---
Fan	---	Hand switch status indication
Damper	---	Hand switch status indication
Deluge valves	---	Status indication
Electric Heater	Indication	---

TABLE 6.5.1-2 (SHEET 2 OF 2)

Sensing Location	Local	Control Room
Total system pressure drop	Indication	Indication alarm[(a)]

a. There is one trouble alarm for each filter train. The alarm is actuated by high or low filter outlet flow, high pressure drop across first HEPA filter, high charcoal filter temperature, high pressure drop across total filter system, or high moisture content in space between heater and first HEPA filter.

TABLE 6.5.1-3 (SHEET 1 OF 3)

ESF FILTER SYSTEM MATERIALS

(Control Room Emergency Air-Conditioning Units)

Component	Material/ Chemical Composition	Estimated Quantity per Housing (lb)
Filter housing	ASTM A-36	-
Moisture eliminators		
Eliminator medium	Spun fiberglass or galvanized steel	6
Holding frame	304 SS	445
Total assembly	304 SS	1004
HEPA filters		
Filter medium	Glass fiber with 5% binder	48 total
Separator	Aluminum foil	83 total
Holding frames	304 SS; ASTM A-240	829 total
Charcoal filters		
Filter media	Impregnated, activated coconut shell charcoal	8905
Holding frames	304 SS; ASTM A-240	9364
Electric heater		
Element	304 SS; ASTM A-240	252
Casing	304 SS; ASTM A-240	283
Cooling coils		4625 dry; 5208 wet
Coils	Copper; ASTM B-152; UNS-C11000	2950
Fins and header	Copper-nickel ASME SB-111; UNS-C70600	92; 1086
Casing	304 SS; ASTM A-240	497
Exhaust fans		
Housing	Carbon steel; ASTM A-36	610 [(a)]
Blades	Ex-Ten 50; ASTM A-607	60

TABLE 6.5.1-3 (SHEET 2 OF 3)

(Piping Penetration Room Filtration Units)

Composition	Material/ Chemical Component	Estimated Quantity per Housing (lb)
Filter housing	ASTM A-36	-
Moisture eliminators		
Eliminator medium	Spun fiberglass or galvanized steel	
Holding frame	304 SS	254
Total assembly	ASTM A-240	1127
HEPA filters		
Filter medium	Glass fiber with 5% binder	128 total
Separator	Aluminum foil	221 total
Holding frames	304 SS; ASTM A-240	2211 total
Charcoal filters		
Filter media	Impregnated, activated coconut shell charcoal	5446
Holding frames	304 SS; ASTM A-240	5408
Electric heater		
Element	304 SS; ASTM A-240	210
Casing	304 SS; ASTM A-240	200
Exhaust fans		
Housing	ASTM A-283 grade D	294[(b)]
Blades	Aluminum, ASTM B-108	21 total[(c)]

TABLE 6.5.1-3 (SHEET 3 OF 3)

(Fuel Handling Building Post-Accident Cleanup Units)

Composition	Material/ Chemical Component	Estimated Quantity per Housing (lb)
Filter housing	ASTM A-36	-
Moisture eliminators		
Eliminator medium	Spun fiberglass or galvanized steel	3
Holding frame	304 SS	114
Total assembly	304 SS	592
HEPA filters		
Filter medium	Glass fiber with 5% binder	48 total
Separators	Aluminum foil	83 total
Holding frames	304 SS; ASTM A-240	829 total
Charcoal filters		
Filter media	Impregnated, activated coconut shell charcoal	1918
Holding frames	304 SS; ASTM A-240	3019
Electric heater		
Element	304 SS; ASTM A-240	63
Casing	304 SS; ASTM A-240	187
Exhaust fans		
Housing	ASTM A-283 grade D	148[b]
Blades	Aluminum; ASTM B-108	5 total[c]

---

a. Housing weights consist of shell material only and do not include stiffening or roll shapes.

b. Housing weights consist of outer casing and flanges.

c. Blade weight includes only blades and no studs.

TABLE 6.5.2-1

CONTAINMENT UNSPRAYED REGIONS

Unsprayed Containment Net Free Volume above Operating Level	Volume (ft^3)
Sheltered volume by steam generator compartment concrete	6900
Pressurizer compartment net free volume	200
Sheltered volume by stairs, No. 1 and No. 2	8300
Sheltered volume by containment cooling units	300
Sheltered volume by auxiliary containment cooling units	8800
Sheltered volume by containment building preaccess filtration unit	33600
Volume underneath polar crane girders	10800
Total unsprayed net free volume above operating deck	68900
Unsprayed Containment Volume below Operating Level	
Total net free volume below operating level	603200
Refueling canal	42500
Total unsprayed volume below operating level	560700
Total unsprayed containment net free volume above and below operating level	629600
Total net free containment volume	2932000
Percentage of unsprayed containment volume	21.5
Percentage of sprayed containment volume	78.5

TABLE 6.5.2-2 (SHEET 1 OF 2)

INPUT PARAMETERS AND RESULTS OF
SPRAY IODINE REMOVAL ANALYSIS

Total containment free volume (ft^3)	2.93×10^6
Unsprayed containment free volume (%)	21.5
Area coverage at the operating deck (%)	87
Mixing rate between sprayed and unsprayed volumes (ft^3/min)	87,000
Containment model	Two region
Minimum vertical distance to operating deck from lowest spray header (ft)	134
Net spray flowrate per train, injection phase (gal/min)	2500
Number of spray pumps operating	1
Minimum spray solution pH	
Injection phase	4.5
Recirculation phase	7.5
Partition factor between liquid and gas phases	40
Average spray drop diameter (μm)	1240
Elemental iodine spray removal coefficient (h^{-1})	10 (DF ≤ 21.4) 0 (DF > 21.4)
Particulate iodine spray removal coefficient (h^{-1})	4.19 (DF ≤ 50) 0.419 (DF > 50)
Duration of spray phase (h)	2

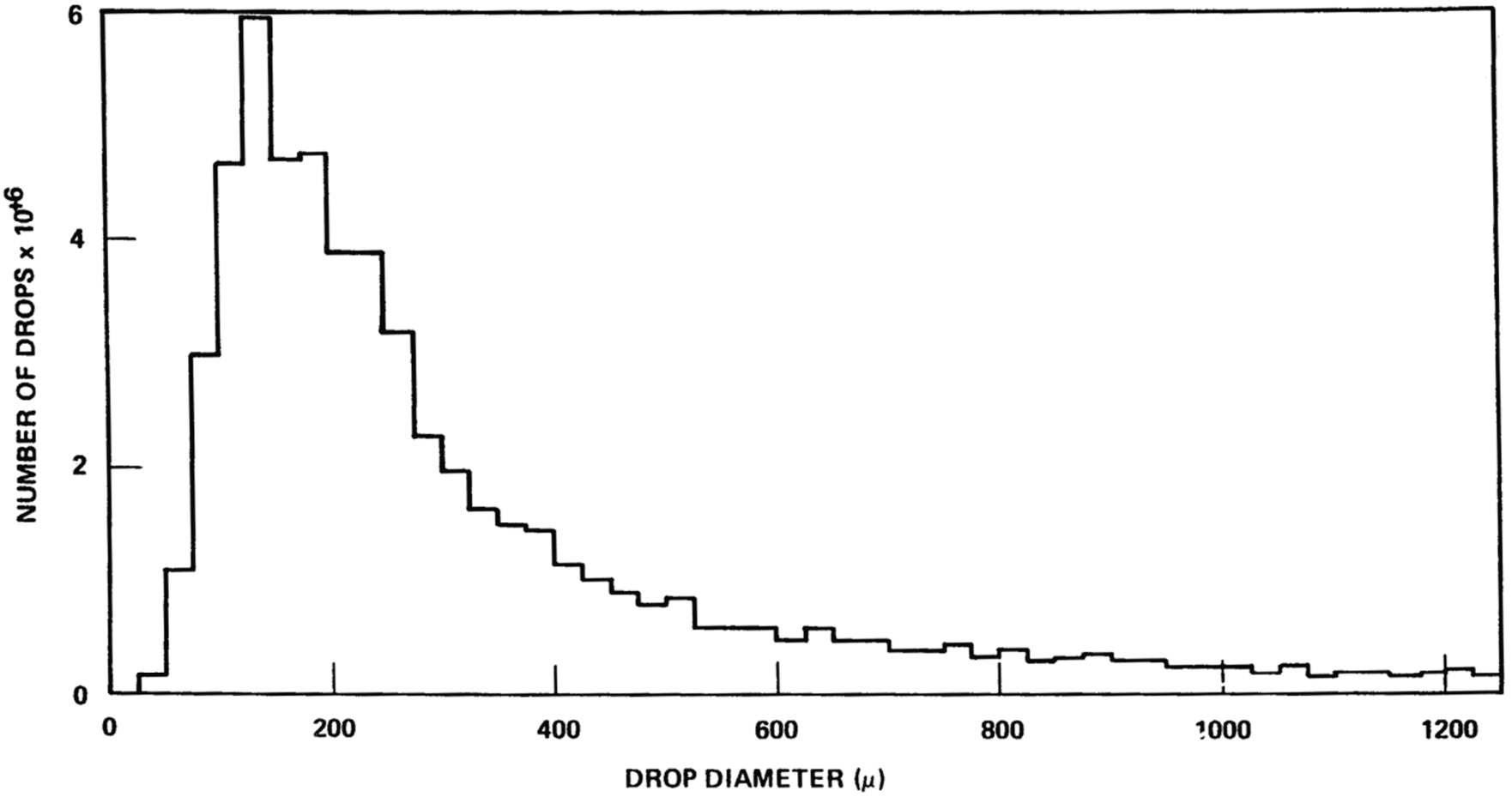
TABLE 6.5.2-2 (SHEET 2 OF 2)

Elemental iodine wall deposition coefficient (h^{-1})	4.76 (DF ≤ 200) 0 (DF > 200)
Area in containment subject to iodine deposition, i.e., coated with epoxy paint, zinc based paint or galvanized (ft^2)	7.94×10^5
Average iodine deposition mass transfer coefficient (m/h)	4.9

TABLE 6.5.3-1

PRIMARY CONTAINMENT OPERATION
FOLLOWING A DESIGN BASIS ACCIDENT

Type of structure	Reinforced concrete cylindrical containment with hemispherical dome. Interior wall lined with 1/4-in. liner plate.
Containment free volume	2.75×10^6 ft^3
Containment leak rate	0.2 v/o per day, 0-24 h 0.1 v/o per day, 1-30 days



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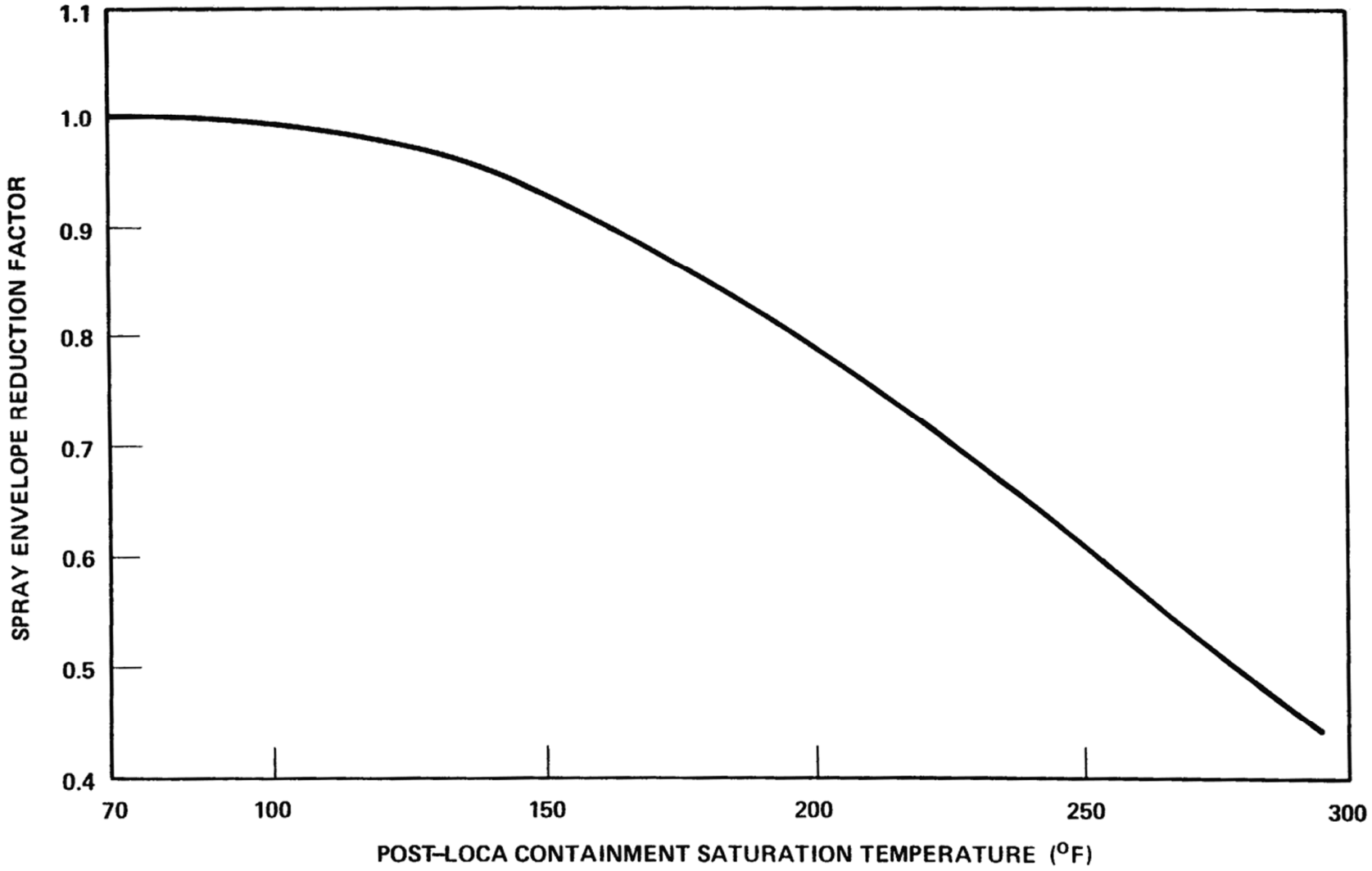
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

SPATIAL DROP–SIZE DISTRIBUTION

FIGURE 6.5.2–1



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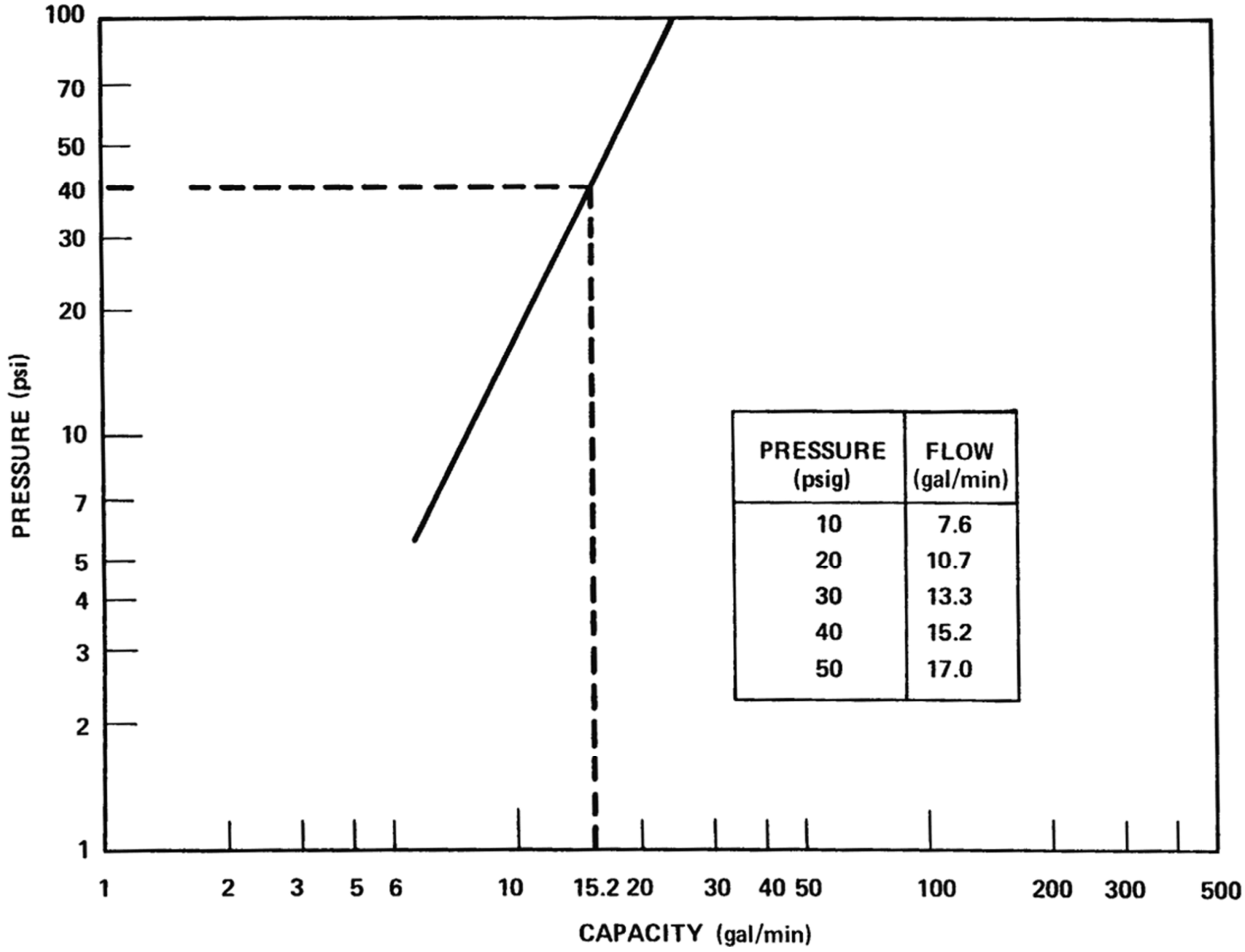
SPRAY ENVELOPE REDUCTION FACTOR

FIGURE 6.5.2–2



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CAPACITY CURVE SPRACO 1713A NOZZLE

FIGURE 6.5.2–3

6.6 INSERVICE INSPECTION OF CLASS 2 AND 3 COMPONENTS[a]

6.6.1 COMPONENTS SUBJECT TO EXAMINATION

Inservice inspection and testing of Class 2 and 3 pressure-retaining components such as vessels, piping, pumps, valves, bolting, and supports shall be performed in accordance with Section XI of the American Society of Mechanical Engineers (ASME) Code including subsections IWC and IWD and any applicable addenda of the code in accordance with 10 CFR 50.55a(g) (specific edition and any applicable addenda of the code will be delineated in each program). The testing of pumps and valves is discussed in subsection 3.9.6. Class 1 component examinations are addressed in subsection 5.2.4. Certain exceptions to the above requirements may be taken whenever specific written relief is granted by the Nuclear Regulatory Commission (NRC) in accordance with 10 CFR 50.55a(g)(6)(i).

The preservice inspection program requirements for each unit were completed prior to the commercial operation date for each of the respective units. The preservice inspection program for Unit 1 complied with the ASME Code, Section XI, 1980 Edition including addenda through Winter 1980, except that reactor pressure vessel examinations were performed using the 1980 Edition including addenda through Winter 1981. The preservice inspection program for Unit 2 complied with the ASME Code, Section XI, 1983 Edition including addenda through Summer 1983, except that reactor pressure vessel examinations were performed using the 1980 Edition including addenda through Winter 1981. Certain preservice inspection requirements of the ASME Code, Section XI were determined to be impractical and relief requests were granted by the NRC pursuant to 10 CFR 50.55a(g) (i). The relief requests were supported by information pursuant to 10 CFR 50.55a(a) (3). In addition, the preservice inspection program included a volumetric examination of a minimum of 8 percent of the Class 2 piping welds in the engineered safety systems.

The inservice inspection program and inservice test program were submitted to the NRC prior to commercial operation. These programs comply with applicable inservice inspection provisions of 10 CFR 50.55a(g) and the NRC guidelines attached as an appendix to section 121.0 of review questions entitled, "Guidance for Preparing Preservice and Inservice Inspection Programs and Relief Requests Pursuant to 10 CFR 50.55a(g)." Where compliance with code requirements is not practical, relief requests have been submitted to the NRC for review and approval. The inservice programs will detail the areas subject to examination and method, extent, and frequency of examinations. Additionally, component supports and snubber testing requirements are included in the inspection programs.

6.6.2 ACCESSIBILITY

The physical arrangement of components was designed to allow personnel and equipment access to the extent practical to perform the inservice inspection examinations. Removable insulation was provided on those piping systems requiring volumetric and surface inspection. Removable hangers and pipe whip restraints are provided as necessary and practical to facilitate inservice inspection. Working platforms were provided in areas requiring inspection and servicing of pumps and valves. Temporary or permanent platforms, scaffolding, and ladders were provided to facilitate access to piping welds.

[a] The Inservice Inspection Program is credited as a license renewal aging management program (see subsection 19.2.13).

An inservice inspection design review was undertaken to identify exceptions to the access requirements of the code with subsequent design modifications and/or inspection technique development to ensure code compliance to the extent practical. Additional exceptions may be identified and reported to the Nuclear Regulatory Commission after plant operation, as specified in 10 CFR 50.55a(g)(5)(iv).

Space is provided to handle and store insulation, structural members, shielding, and other material related to the inspection. Suitable hoists and other handling equipment, lighting, and sources of power for inspection equipment were installed at appropriate locations.

6.6.3 EXAMINATION TECHNIQUES AND PROCEDURES

The visual, surface, and volumetric examination techniques and procedures are in accordance with the requirements of American Society of Mechanical Engineers Code, Section XI, subarticle IWA-2200. Where compliance with code requirements is not practical, relief requests and proposed alternatives have been submitted to the NRC for review and approval. SNC will apply the code cases listed in the latest revision of Regulatory Guide 1.147 endorsed by the NRC in 10 CFR 50.55a on a case-by-case basis as the need arises during inservice inspection. Code cases, which are determined as necessary to accomplish inservice inspection activities, will be used.

The liquid penetrant or magnetic particle methods are used for surface examinations. Radiography or ultrasonic methods, whether manual or remote, are used for volumetric examinations.

The reportable indications and data compilation format provide for comparison of data from subsequent examinations.

6.6.4 INSPECTION INTERVALS

Inspection intervals are as defined in subarticle IWA-2400 of American Society of Mechanical Engineers (ASME) Code, Section XI. The periods within each inspection interval may be extended by as much as 1 year to permit weld inspections to be concurrent with plant outages. It is intended that inservice examinations be performed during normal plant outages such as refueling shutdowns or maintenance shutdowns occurring during the inspection interval.

6.6.5 EXAMINATION CATEGORIES AND REQUIREMENTS

Examination categories are in accordance with subsection IWC and table IWC-2500 of American Society of Mechanical Engineers (ASME) Code, Section XI, and the methods used comply with table IWC-2500 for Class 2 components. The examination categories of Class 3 components and the methods used comply with subsection IWD. The preservice examination of Class 2 and 3 components was in accordance with the requirements of IWC-2200 and IWD-2100, respectively.

6.6.6 EVALUATION OF EXAMINATION RESULTS

Examination results are evaluated per IWA-3000, IWC-3000, and IWD-3000 of American Society of Mechanical Engineers (ASME) Code, Section XI. Repair procedures are in accordance with IWC-4000 and IWD-4000. If the guidelines of IWC-4000 and IWD-4000 are inappropriate for the components, then the guidelines of IWA-4000 apply.

6.6.7 SYSTEM PRESSURE TESTS

System pressure tests comply with IWA-5000, IWC-5000 and IWD-5000 of American Society of Mechanical Engineers (ASME) Code, Section XI, for Class 2 and 3 components.

6.6.8 AUGMENTED INSERVICE INSPECTION TO PROTECT AGAINST POSTULATED PIPING FAILURES

An augmented inservice inspection program is provided for high-energy fluid systems piping between containment isolation valves or where no isolation valve is used inside containment, between the first rigid pipe connection to the containment penetration or the first pipe whip restraint inside containment and the outside isolation valve.

This program includes 100 percent volumetric examination of welds in the affected piping during each inspection interval and will be conducted in accordance with American Society of Mechanical Engineers (ASME) Code, Section XI, and covers the high-energy fluid systems described in subsections 3.6.1 and 3.6.2.